

Krishna's
B.C.A.

MATHEMATICS-I

As per U.P. UNIFIED Syllabus (*w.e.f.* 2011–2012)

(*for B.C.A. Ist Sem. Students of all colleges affiliated to Universities in Uttar Pradesh*)

By

J.P. Chauhan

M.Sc. (Mathematics)

C.C.S. University, Meerut

Sharad Kumar

M.Sc. M.Phil., (N.E.T.)

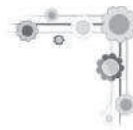
Dep't. of Mathematics

B.S.A. (P.G.) College., Mathura (U.P.)



KRISHNA Prakashan Media (P) Ltd.

KRISHNA HOUSE, 11, Shivaji Road, Meerut-250 001 (U.P.), India



Jai Shri Radhey Shyam

Dedicated
to
Lord
Krishna

Author & Publishers



— *Preface to the Fourth Revised Edition* —

- ▶ The present book **Mathematics-I** has been written according to the latest **U.P. Unified Syllabus** w.e.f. 2011-2012, keeping in view the requirements of the students of B.C.A. 1st Semester of all universities in Uttar Pradesh (U.P.).
- ▶ I feel great pleasure in bringing out Fourth revised edition of this book.
- ▶ Efforts have been made to move the types and correct the mismatched cross references. I hope there are none now.
- ▶ Sequencing of topics in some of the chapters has been altered to make the development of the subject matter more consistent.
- ▶ I have expanded on the contents by adding 150 more solved examples.
- ▶ The reference of last year question papers of all U.P. Universities (Unified Syllabus) for B.C.A. examination have been mentioned at concerned places.
- ▶ In view of the above mentioned features, it is expected that this new edition will prove more useful to the readers.
- ▶ Any constructive comments for improving the contents of this volume would be warmly appreciated.

— *J.P. Chauhan*

Preface to the First Edition

Introduction

Mathematics-I for B.C.A. has been written to fulfil the needs of B.C.A. students. It is needless to say that the Determinants, Limits and Continuity, Vectors and Matrices forms the basis of Mathematical analysis and is an indispensable and powerful tool for Mathematical. Detailed chapters of Gamma and Beta function, differentiation and integration gives the better outline to acquire Mathematics knowledge.

Features

Some prominent of this book are:

Accessibility: Each chapter of the book has been written so that is easily accessible. The basic idea of each topic is carefully and clearly explained.

Prerequisites: Prerequisite topics of each chapter are listed in the contents of the book and a basic understanding to those topics will suffice for the study of that particular chapter.

Exercise sets: Each chapter includes a comprehensive exercise set containing routine, intermediate and challenging exercises so that the learners may have sufficient practice in the subject. Solutions to all exercises are given at the end of each chapter.

References: References from previous years question papers of different regions have been added where appropriate as an indicator to important topics.

Goals

The goal of this book is to present a wide variety of interesting applications of B.C.A. Mathematics in a flexible and accessible format. The book entitled "**B.C.A. Mathematics-I**" has been written to meet requirements of B.C.A. courses of all Indian Universities on this subject. It covers topics form Determinants, Limit and Continuity, Eigen Values and Vectors Differentiation and Integration, Vector Analysis and Matrices. This book contains illustrative examples, theorems with proofs and comprehensive exercises that are designed to reinforce the learning of Discrete Mathematics, logic and computibility.

Throughout the book, the subject matter has been dealt in such a simple, logical & systematic way that students will not find any difficulty to understand it.

— J.P. Chauhan

[illegible]

Syllabus

U.P. UNIFIED (w.e.f. 2011-12)

B.C.A. Mathematics-I

BCA-S105

L	T	P	C
4	0	0	4

Unit-I

Determinants:

Definition, Minors, Cofactors, Properties of Determinants MATRICES: Definition, Types of Matrices, Addition, Subtraction, Scalar Multiplication and Multiplication of Matrices, Adjoint, Inverse, Cramer's Rule, Rank of Matrix Dependence of Vectors, Eigen Vectors of a Matrix, Caley-Hamilton Theorem (without proof).

Unit-II

Limits & Continuity:

Limit at a Point, Properties of Limit, Computation of Limits of Various Types of Functions, Continuity at a Point, Continuity Over an Interval, Intermediate Value Theorem, Type of Discontinuities.

Unit-III

Differentiation:

Derivative, Derivatives of Sum, Differences, Product & Quotients, Chain Rule, Derivatives of Composite Functions, Logarithmic Differentiation, Rolle's Theorem, Mean Value Theorem, Expansion of Functions (Maclaurin's & Taylor's), Indeterminant Forms, L' Hospital's Rule, Maxima & Minima, Curve Tracing, Successive Differentiation & Leibnitz Theorem.

Unit-IV

Integration:

Integral as Limit of Sum, Fundamental Theorem of Calculus (without proof.), Indefinite Integrals, Methods of Integration Substitution, By parts, Partial Fractions, Reduction Formulae for Trigonometric Functions, Gamma and Beta Functions (definition).

Unit-V

Vector Algebra:

Definition of a Vector in 2 and 3 Dimensions; Double and Triple Scalar and Vector Product and physical interpretation of area and volume.

Brief Contents

<i>Dedication</i>	(iii)
<i>Preface to the Fourth Revised Edition</i>	(iv)
<i>Preface to the First Edition</i>	(v)
<i>Acknowledgement</i>	(vi)
<i>Syllabus</i>	(vii)
<i>Brief Contents</i>	(viii)
<i>Detailed Contents</i>	(ix–xii)

Unit-I

Chapter 1. Determinants	(01 – 32)
Chapter 2. Algebra of Matrices and Type of Matrices	(33 – 82)
Chapter 3. Rank of Matrix	(83 – 108)
Chapter 4. Eigen Values, Eigen Vectors and Cayley-Hamilton Theorem ..	(109 – 136)

Unit-II

Chapter 5. Limits and Continuity	(137 – 190)
---	-------------

Unit-III

Chapter 6. Differentiation	(191 – 262)
Chapter 7. Rolles and Mean Value Theorems	(263 – 298)
Chapter 8. Expansion of Functions	(299 – 310)
Chapter 9. Indeterminant Forms	(311 – 338)
Chapter 10. Maxima and Minima	(339 – 376)
Chapter 11. Curve Tracing	(377 – 408)
Chapter 12. Successive Differentiation and Leibnitz Theorem	(409 – 440)

Unit-IV

Chapter 13. Indefinite Integral	(441 – 518)
Chapter 14. Definite Integral	(519 – 538)
Chapter 15. Reduction Formulae	(539 – 564)
Chapter 16. Gamma and Beta Functions	(565 – 582)

Unit-V

Chapter 17. Vector Algebra	(583 – 664)
---	-------------

Detailed Contents

Unit-I

Chapter 1: Determinants(1 – 32)

- 1.1 Determinant 03
- 1.2 Product of Two Determinants of the Same Order 11
- 1.3 Solution of Linear Equations (Cramer's Rule) 25

Chapter 2: Algebra of Matrices and Type of Matrices(33 – 82)

- 2.1 Matrices 33
- 2.2 Addition of Matrices 48
- 2.3 Multiplication of Matrices 52
- 2.4 Inverse of a Matrix 64
- 2.5 Solution of Linear Equations by the use of Inverse of a Matrix 77

Chapter 3: Rank of Matrix(83 – 108)

- 3.1 Rank of a Matrix 83
- 3.2 Elementary Transformations

Chapter 4: Eigen Values, Eigen Vectors and Cayley-Hamilton Theorem..(109–136)

- 4.1 Eigen Value & Eigen Vector of Matrix 109
- 4.2 Certain Relations between Eigen Values and Eigen Vectors 111
 - Some Important Theorems 111
- 4.3 Nature of the Eigen Values of Special Types of Matrices 112
- 4.4 The Process of Finding the Eigen Values & Eigen Vector of Matrix 114

Unit-II

Chapter 5: Limits and Continuity(137 – 190)

- 5.1 Concept of Limits 139
- 5.2 Continuity 157
- 5.3 Discontinuous Functions 160

Unit-III

Chapter 6: Differentiation(191 – 262)

- 6.1 Definition of Differential Coefficient 193
- 6.2 Differential Coefficients of Standard Function 196
- 6.3 Differentiation of a Function Represented Parametrically 197
- 6.4 Facts Towards Derivatives of Composite Functions 215
- 6.5 Derivatives of Inverse Functions 219
- 6.6 Logarithmic Differentiation 241
- 6.7 Derivative of an Implicit Function 249

Chapter 7: Rolles and Mean Value Theorems(263 – 298)

- 7.1 Rolle's Theorem 263
- 7.2 Lagrange's Mean Value Theorem 275
- 7.3 Cauchy's Mean Value Theorem 292

Chapter 8: Expansion of Functions(299 – 310)

- 8.1 Maclaurin's Theorem 299
- 8.2 Taylor's Theorem 300
- 8.3 Some Important Expansions 301

Chapter 9: Indeterminate Forms(311 – 338)

- 9.1 Indeterminate Forms 311

Chapter 10: Maxima and Minima(339 – 376)

- 10.1 Maximum and Minimum Values of a Function 339
- 10.2 Local Maxima and Local Minima 339
- 10.3 Extreme Values 340
- 10.4 Tests for Finding Maxima or Minima 340
- 10.5 Application of Maxima and Minima to Practical Problems 356

Chapter 11: Curve Tracing(377 – 408)

- 11.1 Tracing of Curves 377

Chapter 12: Successive Differentiation and Leibnitz Theorem(409 – 440)

- 12.1 Successive Differentiation 409
- 12.2 n^{th} Derivative of Some Standard Functions 413
- 12.3 Leibnitz's Theorem 424
- 12.4 n^{th} Differential Coefficient for $(x = 0)$ 433

Unit-IV

Chapter 13: Indefinite Integral(441 – 518)

13.1	Integration	443
13.2	Methods of Integration	451
13.3	Some Standard Results	460
13.4	Some Standard Result	466
13.5	Some Special Integral	472
13.6	Some Important Substitution	473
13.3	Integration (Continued)	495

Chapter 14: Definite Integral(519 – 538)

14.1	Fundamental Theorems of Integrals Calculus	519
------	--	-----

Chapter 15: Reduction Formulae(539 – 564)

15.1	Reduction Formulae	539
15.2	Reduction Formulae for $\int \sin^n x \, dx$ and $\int \cos^n x \, dx$	539
15.3	Walli's Formula	540
15.4	Reduction Formula for $\int \sin^m x \cos^n x \, dx$	541
15.5	To Find the Value of the Definite Integral $\int_0^{p/2} \sin^m x \cos^n x \, dx$ when m and n are Positive Integers	544
15.6	Gamma Function	545
15.7	For any Positive Integer	546
15.8	Reduction Formulae for $\int \tan^n x \, dx$ and $\int \cot^n x \, dx$	554
15.9	Reduction Formulae for $\int \sec^n x \, dx$ and $\int \operatorname{cosec}^n x \, dx$	554
15.10	Reduction Formulae for $x^n \sin mx$ and $x^n \cos mx$	557
15.11	Reduction Formulae for $x \sin^n x$ and $x \cos^n x$	558
15.12	Reduction Formulae for $\int \cos^m x \sin nx \, dx$	558

Chapter 16: Gamma and Beta Functions(565 – 582)

16.1	Beta Function	565
16.2	An Important Property of Beta Function	565
16.3	Other Forms of Beta Function	566
16.4	Evaluation of Beta Function	567
16.5	Gamma Function	568

16.6	Relation between Beta and Gamma Function	570
16.7	Duplication Formula	572

Unit-V

Chapter 17: Vector Algebra	(583 – 664)	
17.1	Vector Algebra	585
17.2	Laws of Addition of Vector	586
17.3	Components of a Vector	588
17.4	Scalar Multiplication	589
17.5	Properties of Scalar Product	594
17.6	Scalar Product in Terms of Rectangular Components	598
17.7	Vector (or Cross) Product of Vectors	622
17.8	Properties of Vector Product	623
17.9	Product of Three Vectors	649

○○○

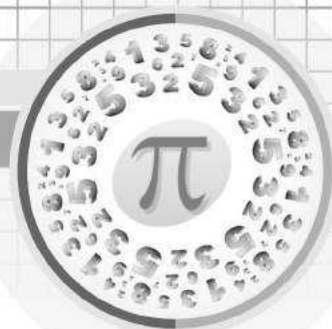
UNIT

1

1. Determinants
2. Algebra of Matrices and Types of Matrices
3. Rank of Matrix
4. Eigen Values, Eigen Vectors and Cayley-Hamilton Theorem



Unit-1



Chapter

1

Determinants

1.1 Determinant

Consider the two linear equations in x

$$a_1x + b_1 = 0 \quad \dots(1)$$

$$a_2x + b_2 = 0 \quad \dots(2)$$

From (1),
$$x = -\frac{b_1}{a_1}$$

Put the value of x in (2), we get

$$a_2 \left(-\frac{b_1}{a_1} \right) + b_2 = 0$$

or
$$a_1b_2 - a_2b_1 = 0 \quad \dots(3)$$

From (1) and (2) by eliminating x , we find

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 0 \quad \dots(4)$$

The two rows of a_1, b_1 and a_2, b_2 enclosed by two vertical bars is called a **determinant** of second order

$$\begin{array}{ccc}
 a_1 & & b_1 \\
 a_2 & \text{and} & b_2 \\
 \text{column 1} & & \text{column 2} \\
 \text{Row 1} & \rightarrow & \left| \begin{array}{cc} a_1 & b_1 \end{array} \right| \\
 \text{Row 2} & \rightarrow & \left| \begin{array}{cc} a_2 & b_2 \end{array} \right|
 \end{array}$$

Each quantity a_1, b_1, a_2, b_2 is a column or in a row is called **element** or a **constituent**.

From (3) and (4), we know that

$$\left| \begin{array}{cc} a_1 & b_1 \\ a_2 & b_2 \end{array} \right| = a_1 b_2 - a_2 b_1$$

$a_1 b_2 - a_2 b_1$ is called the **expansion of the determinant** of $\left| \begin{array}{cc} a_1 & b_1 \\ a_2 & b_2 \end{array} \right|$.

Illustrative Examples

Example 1: Expand the determinant $\left| \begin{array}{cc} 4 & 6 \\ 2 & 5 \end{array} \right|$.

Solution: $\left| \begin{array}{cc} 4 & 6 \\ 2 & 5 \end{array} \right| = 4 \times 5 - 2 \times 6 = 20 - 12 = 8.$

Example 2: Expand the determinant $\left| \begin{array}{cc} 8 & 5 \\ 3 & 1 \end{array} \right|$.

Solution: $\left| \begin{array}{cc} 8 & 5 \\ 3 & 1 \end{array} \right| = 8 \times 1 - 3 \times 5 = 8 - 15 = -7.$

1.1.1 Determinant of Order 3

The symbol $\Delta = \left| \begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array} \right|$

is called a **Determinant of Order 3** and its value is the number

$$a_{11} \left| \begin{array}{cc} a_{22} & a_{23} \\ a_{32} & a_{33} \end{array} \right| - a_{12} \left| \begin{array}{cc} a_{21} & a_{23} \\ a_{31} & a_{33} \end{array} \right| + a_{13} \left| \begin{array}{cc} a_{21} & a_{22} \\ a_{31} & a_{32} \end{array} \right|$$

This is called the **expansion of the determinant** along its first row.

A determinant of order 3 has three rows and three columns. It has $3 \times 3 = 9$ elements.

1.1.2 Determinants of Order 4

The symbol $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$

is called a **Determinant of Order 4** and its value is the number.

$$a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \\ a_{41} & a_{42} & a_{44} \end{vmatrix} - a_{14} \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{vmatrix}$$

This is the expansion of Δ along its first row. The determinant of order 4 has 4 rows and 4 columns. We can expand it along any of its rows or columns as is the case of a determinant of order 3. Proceeding in the same fashion we can evaluate the determinant of order greater than 4.

1.1.3 Minors and Cofactors

Minors: Consider the determinant $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

If we leave the row and the column passing through the element a_{ij} , then second order determinant thus obtained is called the **minor** of the element a_{ij} . We shall denote the minor of the element a_{ij} by M_{ij} .

For Example:

The minor of the element $a_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} = M_{21}$.

The minor of the element $a_{32} = \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} = M_{32}$ and so on.

In terms of the notation of minors, if we expand Δ along the first row then

$$\begin{aligned} \Delta &= (-1)^{1+1} a_{11} M_{11} + (-1)^{1+2} a_{12} M_{12} + (-1)^{1+3} a_{13} M_{13} \\ &= a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13}. \end{aligned}$$

Cofactors: The minor M_{ij} multiplied by $(-1)^{i+j}$ is called the **cofactor** of the element a_{ij} .

We shall denote the cofactor of an element by the corresponding capital letter.

The cofactor of $a_{ij} = A_{ij} = (-1)^{i+j} M_{ij}$.

For Example:

The cofactor of the element $a_{21} = A_{21} = (-1)^{2+1} M_{21} = - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$

The cofactor of the element $a_{32} = A_{32} = (-1)^{3+2} M_{32} = - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$

Thus, the cofactor of any element $a_{ij} = (-1)^{i+j} \times$ the determinant obtained by leaving the row and column passing through that element.

In terms of notation of cofactors, we have

$$\Delta = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

or

$$\Delta = a_{31}A_{31} + a_{32}A_{32} + a_{33}A_{33} \text{ and so on.}$$

Therefore, in a determinant the sum of the products of the elements of any row or column with the corresponding cofactors is equal to the value of the determinant.

Example 3: Find

(i) Minors

(ii) Cofactors of the elements of the first row of the determinant $\begin{vmatrix} 2 & 3 & 5 \\ 4 & 1 & 0 \\ 6 & 2 & 7 \end{vmatrix}$.

Solution: (i) The minor of the element 2 is

$$\begin{vmatrix} 2 & 3 & 5 \\ 4 & 1 & 0 \\ 6 & 2 & 7 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 2 & 7 \end{vmatrix} = 1 \times 7 - 0 \times 2 = 7 - 0 = 7$$

The minor of the element 3 is

$$\begin{vmatrix} 2 & 3 & 5 \\ 4 & 1 & 0 \\ 6 & 2 & 7 \end{vmatrix} = \begin{vmatrix} 4 & 0 \\ 6 & 7 \end{vmatrix} = 4 \times 7 - 0 \times 6 = 28 - 0 = 28$$

The minor of the element 5 is

$$\begin{vmatrix} 2 & 3 & 5 \\ 4 & 1 & 0 \\ 6 & 2 & 7 \end{vmatrix} = \begin{vmatrix} 4 & 1 \\ 6 & 2 \end{vmatrix} = 4 \times 2 - 6 \times 1 = 8 - 6 = 2$$

(ii) The cofactor of 2 $= (-1)^{1+1} 7 = +7$

The cofactor of 3 $= (-1)^{1+2} (28) = -28$

The cofactor of 5 $= (-1)^{1+3} (2) = +2$.

Example 4: Find the order and value of the following determinants.

$$(i) \begin{vmatrix} 4 & -3 \\ 2 & -1 \end{vmatrix}$$

$$(ii) \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -3 \\ 2 & -2 & 1 \end{vmatrix}$$

Solution: (i) The determinant $\begin{vmatrix} 4 & -3 \\ 2 & -1 \end{vmatrix}$ has 2 rows and 2 columns, so the order of this determinant is 2.

$$\text{Let } \Delta = \begin{vmatrix} 4 & -3 \\ 2 & -1 \end{vmatrix} = (4)(-1) - (-3)(2) = -4 + 6 = 2.$$

(ii) The determinant $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -3 \\ 2 & -2 & 1 \end{vmatrix}$ has 3 rows and 3 columns, so the order of this determinant is 3.

$$\text{Let } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -3 \\ 2 & -2 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & -3 \\ -2 & 1 \end{vmatrix} - 1 \cdot \begin{vmatrix} 2 & -3 \\ 2 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & 1 \\ 2 & -2 \end{vmatrix}$$

(Expanding the determinant along the first row)

$$= 1 \{ (1) \cdot (1) - (-2) (-3) \} - 1 \cdot \{ (2) (1) - (2) (-3) \} + 1 \{ (2) (-2) - (2) (1) \} \\ = (1 - 6) - (2 + 6) + (-4 - 2) = -5 - 8 - 6 = -19.$$

Example 5: Expand the determinant $\begin{vmatrix} 6 & 2 & 3 \\ 2 & 3 & 5 \\ 4 & 2 & 1 \end{vmatrix}$.

Solution: Expanding the determinant along the first row, we get

$$\begin{vmatrix} 6 & 2 & 3 \\ 2 & 3 & 5 \\ 4 & 2 & 1 \end{vmatrix} = 6 (\text{cofactor of } 6) - 2 (\text{cofactor of } 2) + 3 (\text{cofactor of } 3) \\ = 6 (3 \times 1 - 5 \times 2) - 2 (2 \times 1 - 4 \times 5) + 3 (2 \times 2 - 3 \times 4) \\ = 6 (3 - 10) - 2 (2 - 20) + 3 (4 - 12) \\ = 6 (-7) - 2 (-18) + 3 (-8) = -42 + 36 - 24 = -30.$$

Example 6: Expand the fourth order determinant

$$|A| = \begin{vmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 \\ 2 & 0 & 1 & 3 \\ 1 & 2 & 1 & 0 \end{vmatrix}$$

Solution: Expanding $|A|$ along the first row, we get

$$\begin{aligned}
 |A| &= 0 \cdot (-1)^{1+1} \begin{vmatrix} 0 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 1 & 0 \end{vmatrix} + 1 \cdot (-1)^{1+2} \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 3 \\ 1 & 1 & 0 \end{vmatrix} \\
 &\quad + 2 \cdot (-1)^{1+3} \begin{vmatrix} 1 & 0 & 0 \\ 2 & 0 & 3 \\ 1 & 2 & 0 \end{vmatrix} + 3 \cdot (-1)^{1+4} \begin{vmatrix} 1 & 0 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 1 \end{vmatrix} \\
 |A| &= 0 - \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 3 \\ 1 & 1 & 0 \end{vmatrix} + 2 \begin{vmatrix} 1 & 0 & 0 \\ 2 & 0 & 3 \\ 1 & 2 & 0 \end{vmatrix} - 3 \begin{vmatrix} 1 & 0 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 1 \end{vmatrix} \quad \dots(i)
 \end{aligned}$$

Now, $\begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 3 \\ 1 & 1 & 0 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & 3 \\ 1 & 0 \end{vmatrix} - 2 \begin{vmatrix} 2 & 3 \\ 1 & 0 \end{vmatrix} + 0 \cdot \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} = 1(0-3) - 2(0-3) + 0 \cdot (2-1) = 3.$

$$\begin{vmatrix} 1 & 0 & 0 \\ 2 & 0 & 3 \\ 1 & 2 & 0 \end{vmatrix} = 1 \begin{vmatrix} 0 & 3 \\ 2 & 0 \end{vmatrix} - 0 \begin{vmatrix} 2 & 3 \\ 1 & 0 \end{vmatrix} + 0 \begin{vmatrix} 2 & 0 \\ 1 & 2 \end{vmatrix} = 1(0-6) = -6$$

$$\begin{vmatrix} 1 & 0 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 1 \end{vmatrix} = 1 \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} - 0 \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 2 & 0 \\ 1 & 2 \end{vmatrix} = 1(0-2) - 0 + 2(4-0) = 6$$

Putting these values in (i), we get

$$|A| = 0 - 3 + 2(-6) - 3(6) = -3 - 12 - 18 = -33.$$

Rules of Sarrus

(For third order determinants only)

After writing the determinant, repeat the first two columns as below

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 & a_1 & b_1 \\ a_2 & b_2 & c_2 & a_2 & b_2 \\ a_3 & b_3 & c_3 & a_3 & b_3 \end{vmatrix}$$

$= (a_1 b_2 c_3 + b_1 c_2 a_3 + c_1 a_2 b_3) + (-c_1 b_2 a_3 - a_1 c_2 b_3 - b_1 a_2 c_3)$

Example 7: Expand the determinant $\Delta = \begin{vmatrix} 2 & 3 & 4 \\ 1 & 5 & 3 \\ 3 & 0 & 5 \end{vmatrix}$ by Rule of Sarrus.

Solution:

$$\begin{vmatrix} 2 & 3 & 4 \\ 1 & 5 & 3 \\ 3 & 0 & 5 \end{vmatrix} = (2 \times 5 \times 5) + (3 \times 3 \times 3) + (4 \times 1 \times 0) - [(4 \times 5 \times 3) + (2 \times 3 \times 0) + (3 \times 1 \times 5)]$$

$$= (50 + 27 + 0) - (60 + 0 + 15) = 77 - 75 = 2.$$



Comprehensive Exercise 1.1

Expand the following by Rule of sarrus

$$1. \begin{vmatrix} 1 & 4 & 2 \\ 2 & 5 & 3 \\ 3 & 6 & 4 \end{vmatrix}$$

$$2. \begin{vmatrix} 6 & 3 & 7 \\ 32 & 13 & 37 \\ 10 & 4 & 11 \end{vmatrix}$$

$$3. \begin{vmatrix} 3 & 2 & -4 \\ 5 & 1 & -1 \\ -2 & 6 & 7 \end{vmatrix}$$



Answers

1.	0	2.	10	3.	-155
----	---	----	----	----	------

1.1.4 Determinant of a Square Matrix

Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order n . Then the determinant given below is called the determinant of the matrix A and is denoted by $|A|$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, \quad |A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

Matrix Determinant

Since the determinant is always square, therefore, only square matrices can have determinant.

Example 8: Find the value of the determinant of the matrix, $A = \begin{bmatrix} 2 & 0 & 5 \\ 3 & 1 & 1 \\ 1 & 4 & 6 \end{bmatrix}$.

Solution: We have,

$$|A| = \begin{vmatrix} 2 & 0 & 5 \\ 3 & 1 & 1 \\ 1 & 4 & 6 \end{vmatrix} = 2 \begin{vmatrix} 1 & 1 \\ 4 & 6 \end{vmatrix} - 0 \begin{vmatrix} 3 & 1 \\ 1 & 6 \end{vmatrix} + 5 \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix}$$

$$= 2(6 - 4) - 0 + 5(12 - 1) = 4 + 55 = 59.$$

1.1.5 Properties of Determinants

- The value of a determinant does not change when rows and columns are interchanged

$$i.e., \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

2. If any two rows (or two columns) of a determinant are inter-changed the value of the determinant is multiplied by -1 .

$$\text{i.e., } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = - \begin{vmatrix} a_3 & b_3 & c_3 \\ a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \end{vmatrix}$$

3. If any row or column of a determinant Δ be passed over n rows or columns the resulting determinant will be $(-1)^n \Delta$.
4. If all the elements of any row (or any column) of a determinant are multiplied by the same number k , the value of the new determinant is k times the value of the given determinant.

$$\text{i.e., } \begin{vmatrix} a_1 & b_1 & c_1 \\ ka_2 & kb_2 & kc_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = k \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

5. If A be an n -rowed square matrix and k be any scalar then

$$|kA| = k^n |A|.$$

6. If all the elements of any row (or any column) of a determinant are zero, the value of the determinant is zero

$$\text{i.e., } \begin{vmatrix} 0 & 0 & 0 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0 \text{ or } \begin{vmatrix} 0 & a_2 & a_3 \\ 0 & b_2 & b_3 \\ 0 & c_2 & c_3 \end{vmatrix} = 0$$

7. If two rows (or two columns) of a determinant are identical, the value of the determinant is zero.

$$\text{i.e., } \begin{vmatrix} a_1 & a_1 & c_1 \\ a_2 & a_2 & c_2 \\ a_3 & a_3 & c_3 \end{vmatrix} = 0 \text{ or } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

8. In a determinant the sum of the products of the elements of any row (or any column) with the cofactors of the corresponding elements of any other row (column) is zero.
9. If in a determinant each element in any row (or column) consists of the sum of two terms then the determinant can be expressed as the sum of two determinants of the same order.

$$\text{i.e., } \begin{vmatrix} a_1 + a_1 & b_1 & c_1 \\ a_2 + a_2 & b_2 & c_2 \\ a_3 + a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

10. If the elements of a row (or column) of a determinant are added m times the corresponding elements of another row (or column), the value of the determinant thus obtained is equal to the value of the original determinant.

$$\text{i.e., Let } \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ and } \Delta' = \begin{vmatrix} a_1 + mb_1 & b_1 & c_1 \\ a_2 + mb_2 & b_2 & c_2 \\ a_3 + mb_3 & b_3 & c_3 \end{vmatrix}$$

be the determinant obtained by adding to the elements of first column of Δ m times the elements of the second column.

1.2 Product of Two Determinants of the Same Order

1.2.1 Rule for the Multiplication of two Determinants of the Third Order

Let
$$\Delta_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ and } \Delta_2 = \begin{vmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{vmatrix}$$

be two determinants of the third order.

Then
$$\Delta_1 \cdot \Delta_2 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \begin{vmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_1\alpha_1 + b_1\alpha_2 + c_1\alpha_3 & a_1\beta_1 + b_1\beta_2 + c_1\beta_3 & a_1\gamma_1 + b_1\gamma_2 + c_1\gamma_3 \\ a_2\alpha_1 + b_2\alpha_2 + c_2\alpha_3 & a_2\beta_1 + b_2\beta_2 + c_2\beta_3 & a_2\gamma_1 + b_2\gamma_2 + c_2\gamma_3 \\ a_3\alpha_1 + b_3\alpha_2 + c_3\alpha_3 & a_3\beta_1 + b_3\beta_2 + c_3\beta_3 & a_3\gamma_1 + b_3\gamma_2 + c_3\gamma_3 \end{vmatrix}$$

This is the row by column multiplication rule.

Note: The value of the determinant does not change by interchanging the rows and columns. Therefore, while writing down the product of two determinants of the same order, we can also follow the row-by-row multiplication rule or the column-by-row multiplication rule or column-by-column multiplication rule.

Working rule: In finding the value of a determinant we always try to make the elements as zero at maximum number of places in any particular row (or column) and then to expand the determinant along the row (or column).

The first, second and third rows of a determinant are denoted by R_1, R_2, R_3 respectively and columns by C_1, C_2, C_3 etc. The addition of R_i and m times R_j is written as $R_i = R_i + mR_j$.

Here R_i changes while R_j will remain as it is. The interchanging of R_i and R_j denoted as $R_i (\leftrightarrow) R_j$.

Example 9: Evaluate, $\Delta = \begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$ [B.C.A. (Kanpur) 2006]

where ω is one of the imaginary cube roots of unity.

Solution: Applying $C_1 = C_1 + C_2 + C_3$

$$\begin{vmatrix} 1 + \omega + \omega^2 & \omega & \omega^2 \\ 1 + \omega + \omega^2 & \omega^2 & 1 \\ 1 + \omega + \omega^2 & 1 & \omega \end{vmatrix} = \begin{vmatrix} 0 & \omega & \omega^2 \\ 0 & \omega^2 & 1 \\ 0 & 1 & \omega \end{vmatrix} = 0 \quad (\because 1 + \omega + \omega^2 = 0)$$

Example 10: Solve the equation $\begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$. [B.C.A. (Allahabad) 2010]

Solution: Applying $R_1 = R_1 + R_2 + R_3$

$$\begin{vmatrix} 3x-2 & 3x-2 & 3x-2 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$$

Taking common $(3x-2)$ from R_1 , we find

$$(3x-2) \begin{vmatrix} 1 & 1 & 1 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$$

Again applying $C_2 = C_2 - C_1$, $C_3 = C_3 - C_1$

$$(3x-2) \begin{vmatrix} 1 & 0 & 0 \\ 3 & 3x-11 & 0 \\ 3 & 0 & 3x-11 \end{vmatrix} = 0$$

Expanding the determinant from the first row, we get

$$(3x-2)(3x-11)^2 = 0$$

$$\Rightarrow x = \frac{2}{3} \text{ and } x = \frac{11}{3}, \frac{11}{3}$$

Example 11: Show that $\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$. [B.C.A. (Bundelkhand) 2007]

Solution: Multiplying R_1, R_2 and R_3 by a, b and c respectively, we find

$$= \frac{1}{abc} \begin{vmatrix} abc & a^2 & a^3 \\ abc & b^2 & b^3 \\ abc & c^2 & c^3 \end{vmatrix}$$

Taking common abc from C_1 , we find

$$= \frac{abc}{abc} \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

Example 12: Show that

$$\Delta = \begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = 2(a+b+c)^3.$$

Solution: Applying $R_1 = R_1 - R_2, R_2 = R_2 - R_3$

$$\Delta = \begin{vmatrix} a+b+c & -(a+b+c) & 0 \\ 0 & b+c+a & -(b+c+a) \\ c & a & c+a+2b \end{vmatrix}$$

Taking common $(a+b+c)$ from R_1 and R_2

$$= (a+b+c)^2 \begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ c & a & c+a+2b \end{vmatrix}$$

Again applying $C_2 = C_2 + C_1$

$$= (a+b+c)^2 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ c & a+c & a+c+2b \end{vmatrix}$$

Expanding the determinant along the first row, we get

$$\begin{aligned} \Delta &= (a+b+c)^2 \begin{vmatrix} 1 & -1 \\ a+c & c+a+2b \end{vmatrix} \\ &= (a+b+c)^2 (c+a+2b+a+c) \\ &= (a+b+c)^2 (2a+2b+2c) \\ &= 2(a+b+c)^3 \end{aligned}$$

Example 13: Show that, $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$.

Solution: Applying $R_1 = R_1 + R_2 + R_3$

$$\begin{aligned} &= \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} \\ &= (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} \\ &= (a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ 2b & b-c-a-2b & 2b-2b \\ 2c & 2c-2c & c-a-b-2c \end{vmatrix} \\ &= (a+b+c)^3 \quad (C_2 = C_2 - C_1, C_3 = C_3 - C_1) \end{aligned}$$

Example 14: If ω be a cube root of unity, prove that $(a + b\omega + c\omega^2)$ is factor of the determinant

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

and show that the determinant is equal to $-(a^3 + b^3 + c^3 - 3abc)$.

Solution: $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \begin{vmatrix} a + b\omega + c\omega^2 & b & c \\ b + c\omega + a\omega^2 & c & a \\ c + a\omega + b\omega^2 & a & b \end{vmatrix}$ (By applying $C_1 = C_1 + \omega C_2 + \omega^2 C_3$)

$$= \begin{vmatrix} a + b\omega + c\omega^2 & b & c \\ \omega^2(b\omega + c\omega^2 + a) & c & a \\ \omega(\omega^2 c + a + b\omega) & a & b \end{vmatrix} \quad (\omega^3 = 1)$$

$$= (a + b\omega + c\omega^2) \begin{vmatrix} 1 & b & c \\ \omega^2 & c & a \\ \omega & a & b \end{vmatrix}$$

$$= (a + b\omega + c\omega^2) \begin{vmatrix} 1 + \omega + \omega^2 & a + b + c & a + b + c \\ \omega^2 & c & a \\ \omega & a & b \end{vmatrix} \quad (R_1 = R_1 + R_2 + R_3)$$

$$= (a + b\omega + c\omega^2) \begin{vmatrix} 0 & a + b + c & a + b + c \\ \omega^2 & c & a \\ \omega & a & b \end{vmatrix}$$

$$= (a + b\omega + c\omega^2) (a + b + c) \begin{vmatrix} 0 & 1 & 1 \\ \omega^2 & c & a \\ \omega & a & b \end{vmatrix} \quad (C_2 = C_2 - C_3)$$

$$= (a + b\omega + c\omega^2) (a + b + c) \begin{vmatrix} 0 & 0 & 1 \\ \omega^2 & c - a & a \\ \omega & a - b & b \end{vmatrix}$$

$$= (a + b\omega + c\omega^2) (a + b + c) (a\omega^2 - b\omega^2 - c\omega + a\omega)$$

$$= (a + b\omega + c\omega^2) (a + b + c) (a(\omega^2 + \omega) - b\omega^2 - c\omega)$$

$$= (a + b + c) (a + b\omega + c\omega^2) (-a - b\omega^2 - c\omega)$$

$$= -(a + b + c) (a + b\omega + c\omega^2) (a + b\omega^2 + c\omega)$$

$$= -(a + b + c) (a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= -(a^3 + b^3 + c^3 - 3abc).$$

Example 15: If x, y and z are all different and if
$$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

Prove that $xyz = -1$.

Solution: We have,
$$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

Expressing the given determinant into two determinant, we get

$$\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$$

Taking x, y, z common from R_1, R_2, R_3 of IInd determinant, we get

$$\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$$\Rightarrow (-1)^2 \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$$\Rightarrow (1 + xyz) \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

$$\Rightarrow 1 + xyz = 0 \text{ or } xyz = -1.$$

Example 16: Prove that
$$\begin{vmatrix} a^2 & a & bc \\ b^2 & b & ca \\ c^2 & c & ab \end{vmatrix} = - \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}.$$

Solution: We have,
$$\Delta = \begin{vmatrix} a^2 & a & bc \\ b^2 & b & ca \\ c^2 & c & ab \end{vmatrix}$$

Multiplying R_1, R_2, R_3 by a, b and c respectively, we get

$$\begin{aligned}
 &= \frac{1}{abc} \begin{vmatrix} a^3 & a^2 & abc \\ b^3 & b^2 & abc \\ c^3 & c^2 & abc \end{vmatrix} = \frac{abc}{abc} \begin{vmatrix} a^3 & a^2 & 1 \\ b^3 & b^2 & 1 \\ c^3 & c^2 & 1 \end{vmatrix} \\
 &= \begin{vmatrix} a^3 & a^2 & 1 \\ b^3 & b^2 & 1 \\ c^3 & c^2 & 1 \end{vmatrix} \quad (C_1 \longleftrightarrow C_3) \\
 &= - \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = - \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} \text{ (By interchanging row and column).}
 \end{aligned}$$

Example 17: Show that $x = -(a + b + c)$ is one root of the equation

$$\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0 \text{ and solve the equation completely.}$$

Solution: Applying $C_1 = C_1 + C_2 + C_3$, we find

$$\begin{vmatrix} x+a+b+c & b & c \\ x+a+b+c & x+c & a \\ x+a+b+c & a & x+b \end{vmatrix} = 0$$

$$\text{or} \quad (x+a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & x+c & a \\ 1 & a & x+b \end{vmatrix} = 0$$

$$\text{or} \quad (x+a+b+c) \begin{vmatrix} 1 & b & c \\ 0 & x-b+c & a-c \\ 0 & a-b & x+b-c \end{vmatrix} = 0 \quad (R_2 = R_2 - R_1, R_3 = R_3 - R_1)$$

On expanding the first column

$$(x+a+b+c) [(x-b+c)(x+b-c) - (a-b)(a-c)] = 0$$

$$(x+a+b+c) [x^2 - (b-c)^2 - (a^2 - ac - ab + bc)] = 0$$

$$(x+a+b+c) [x^2 - b^2 - c^2 + 2bc - a^2 + ac + ab - bc] = 0$$

$$(x+a+b+c) (x^2 - b^2 - c^2 - a^2 + ab + bc + ca) = 0$$

$$\Rightarrow \text{either } x+a+b+c=0 \quad \text{or} \quad x^2 - a^2 - b^2 - c^2 + ab + bc + ca = 0$$

$$\Rightarrow x = -(a+b+c)$$

$$\text{or} \quad x = \pm \sqrt{a^2 + b^2 + c^2 - ab - bc - ca}$$

Example 18: Show that

$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3.$$

Solution: L.H.S. = $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$

Applying $C_1 = C_1 - bC_3, C_2 = C_2 + aC_3$

$$= \begin{vmatrix} 1+a^2+b^2 & 0 & -2b \\ 0 & 1+a^2+b^2 & 2a \\ b(1+a^2+b^2) & -a(1+a^2+b^2) & 1-a^2-b^2 \end{vmatrix}$$

$$= (1+a^2+b^2)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ b & -a & 1-a^2-b^2 \end{vmatrix}$$

Applying $R_3 = R_3 - bR_1$, we get

$$= (1+a^2+b^2)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ 0 & -a & 1-a^2+b^2 \end{vmatrix}$$

Expanding along the first column, we get

$$\begin{aligned} &= (1+a^2+b^2)^2 [(1-a^2+b^2) + 2a^2] \\ &= (1+a^2+b^2)^2 (1+a^2+b^2) \\ &= (1+a^2+b^2)^3 \end{aligned}$$

Example 19: Show that

$$\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$$

Solution: L.H.S. = $\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}$

Applying $C_1 = C_1 + C_2 + C_3$, we get

$$= \begin{vmatrix} a+b+c & b & c \\ 0 & b-c & c-a \\ 2(a+b+c) & c+a & a+b \end{vmatrix}$$

Taking common $(a + b + c)$ from C_1 , we find

$$= (a + b + c) \begin{vmatrix} 1 & b & c \\ 0 & b - c & c - a \\ 2 & c + a & a + b \end{vmatrix}$$

Now, applying $R_3 = R_3 - 2R_1$

$$= (a + b + c) \begin{vmatrix} 1 & b & c \\ 0 & b - c & c - a \\ 0 & c + a - 2b & a + b - 2c \end{vmatrix}$$

Expanding the determinant along first column, we find

$$\begin{aligned} &= (a + b + c) \begin{vmatrix} b - c & c - a \\ c + a - 2b & a + b - 2c \end{vmatrix} \\ &= (a + b + c) [(b - c)(a + b - 2c) - (c - a)(c + a - 2b)] \\ &= (a + b + c) [ab + b^2 - 2bc - ac - bc + 2c^2 - c^2 - ac + 2bc + ac + a^2 - 2ab] \\ &= (a + b + c) (a^2 + b^2 + c^2 - ab - bc - ca) \\ &= a^3 + b^3 + c^3 - 3abc \end{aligned}$$

Example 20: Prove that

$$\Delta = \begin{vmatrix} b + c & a + b & a \\ c + a & b + c & b \\ a + b & c + a & c \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$$

Solution: Applying $C_1 = C_1 + C_3$, we get

$$\Delta = \begin{vmatrix} a + b + c & a + b & a \\ a + b + c & b + c & b \\ a + b + c & c + a & c \end{vmatrix} = (a + b + c) \begin{vmatrix} 1 & a + b & a \\ 1 & b + c & b \\ 1 & c + a & c \end{vmatrix}$$

Applying $R_2 = R_2 - R_1, R_3 = R_3 - R_1$

$$\begin{aligned} &= (a + b + c) \begin{vmatrix} 1 & a + b & a \\ 0 & c - a & b - a \\ 0 & c - b & c - a \end{vmatrix} \\ &= (a + b + c) \begin{vmatrix} c - a & b - a \\ c - b & c - a \end{vmatrix} \\ &= (a + b + c) [(c - a)^2 - (c - b)(b - a)] \\ &= (a + b + c) [c^2 + a^2 - 2ac - bc + ac + b^2 - ab] \\ &= (a + b + c) [a^2 + b^2 + c^2 - ab - bc - ca] \\ &= a^3 + b^3 + c^3 - 3abc \end{aligned}$$

Example 21: Show that

$$\begin{vmatrix} a^2 & bc & c^2 + ca \\ a^2 + ab & b^2 & ca \\ ab & b^2 + bc & c^2 \end{vmatrix} = 4a^2b^2c^2.$$

Solution: We have

$$\begin{vmatrix} a^2 & bc & c^2 + ca \\ a^2 + ab & b^2 & ca \\ ab & b^2 + bc & c^2 \end{vmatrix}$$

Taking common a, b and c from C_1, C_2 and C_3 , we find

$$= abc \begin{vmatrix} a & c & c + a \\ a + b & b & a \\ b & b + c & c \end{vmatrix}$$

Applying $C_3 = C_3 - (C_1 + C_2)$

$$\begin{aligned} &= abc \begin{vmatrix} a & c & c + a - a - c \\ a + b & b & a - a - b - b \\ b & b + c & c - b - b - c \end{vmatrix} = abc \begin{vmatrix} a & c & 0 \\ a + b & b & -2b \\ b & b + c & -2b \end{vmatrix} \\ &= (abc)(-2b) \begin{vmatrix} a & c & 0 \\ a + b & b & 1 \\ b & b + c & 1 \end{vmatrix} \end{aligned}$$

Now, applying $R_3 = R_3 - R_2$

$$= -2ab^2c \begin{vmatrix} a & c & 0 \\ a + b & b & 1 \\ -a & c & 0 \end{vmatrix} = 2ab^2c \begin{vmatrix} a & c \\ -a & c \end{vmatrix} = 2ab^2c(2ac) = 4a^2b^2c^2.$$

Example 22: Show that $\begin{vmatrix} 1 & a & b + c \\ 1 & b & c + a \\ 1 & c & a + b \end{vmatrix} = 0$

Solution: Let $\Delta = \begin{vmatrix} 1 & a & b + c \\ 1 & b & c + a \\ 1 & c & a + b \end{vmatrix}$

Applying $C_3 \rightarrow C_3 + C_2$, we get

$$\Delta = \begin{vmatrix} 1 & a & a + b + c \\ 1 & b & b + c + a \\ 1 & c & a + b + c \end{vmatrix} = (a + b + c) \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} = 0, \text{ since } C_1 \text{ and } C_3 \text{ are identical.}$$

Taking $a + b + c$ common from C_3

Example 23: Without expanding, show that

$$\begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix} = 0.$$

Solution: Let $\Delta = \begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix}$

Multiplying the 1st, 2nd and 3rd rows by a, b, c respectively, we get

$$\Delta = \frac{1}{abc} \begin{vmatrix} ab^2c^2 & abc & ab+ac \\ bc^2a^2 & bca & bc+ba \\ ca^2b^2 & cab & ca+cb \end{vmatrix} = \frac{(abc)^2}{abc} \begin{vmatrix} bc & 1 & ab+ac \\ ca & 1 & bc+ba \\ ab & 1 & ca+cb \end{vmatrix}$$

Taking abc common from 1st and 2nd columns

$$\begin{aligned} &= abc \begin{vmatrix} bc & 1 & ab+bc+ca \\ ca & 1 & ab+bc+ca \\ ab & 1 & ab+bc+ca \end{vmatrix}, \text{ applying } C_3 \rightarrow C_3 + C_1 \\ &= abc (ab+bc+ca) \begin{vmatrix} bc & 1 & 1 \\ ca & 1 & 1 \\ ab & 1 & 1 \end{vmatrix} \end{aligned}$$

Taking $(ab+bc+ca)$ common from third column

$$= 0, \text{ since } C_2 \text{ and } C_3 \text{ are identical.}$$

Example 24: Show that

$$\begin{vmatrix} b-c & c-a & a-b \\ c-a & a-b & b-c \\ a-b & b-c & c-a \end{vmatrix} = 0.$$

Solution: Let $\Delta = \begin{vmatrix} b-c & c-a & a-b \\ c-a & a-b & b-c \\ a-b & b-c & c-a \end{vmatrix}$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$, we get

$$\Delta = \begin{vmatrix} 0 & 0 & 0 \\ c-a & a-b & b-c \\ a-b & b-c & c-a \end{vmatrix} = 0, \text{ since all elements of } R_1 \text{ are zero.}$$

Example 25: Prove that

$$\Delta = \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) = abc + ab + bc + ca.$$

[B.C.A. (Lucknow) 2007]

Solution: Taking a, b, c common from first, second and third columns respectively, we get

$$\Delta = abc \begin{vmatrix} \frac{1}{a} + 1 & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{a} & \frac{1}{b} + 1 & \frac{1}{c} \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} + 1 \end{vmatrix} = abc \begin{vmatrix} 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & \frac{1}{b} & \frac{1}{c} \\ 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & \frac{1}{b} + 1 & \frac{1}{c} \\ 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & \frac{1}{b} & \frac{1}{c} + 1 \end{vmatrix},$$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$

$$= abc \begin{vmatrix} 1 & \frac{1}{b} & \frac{1}{c} \\ 1 & \frac{1}{b} + 1 & \frac{1}{c} \\ 1 & \frac{1}{b} & \frac{1}{c} + 1 \end{vmatrix}$$

Taking $\left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$ common from C_1

$$= abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \begin{vmatrix} 1 & \frac{1}{b} & \frac{1}{c} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix},$$

Applying $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$= abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) = abc + ab + bc + ca.$$

Example 26: Prove that,

$$\Delta = \begin{vmatrix} x-y-z & 2x & 2x \\ 2y & y-x-z & 2y \\ 2z & 2z & z-x-y \end{vmatrix} = (x+y+z)^3.$$

Solution: Applying $R_1 \rightarrow R_1 + R_2 + R_3$, we get

$$\begin{aligned} \Delta &= \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ 2y & y-x-z & 2y \\ 2z & 2z & z-x-y \end{vmatrix} \\ &= (x+y+z) \begin{vmatrix} 1 & 1 & 1 \\ 2y & y-x-z & 2y \\ 2z & 2z & z-x-y \end{vmatrix} \\ &= (x+y+z) \begin{vmatrix} 1 & 0 & 0 \\ 2y & -(x+y+z) & 0 \\ 2z & 0 & -(x+y+z) \end{vmatrix} = (x+y+z)^3 \end{aligned}$$

[Applying $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$]

Example 27: Solve the equation

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0.$$

Solution: Applying $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$, we get

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ -2 & -6 & -12 \\ -6 & -24 & -60 \end{vmatrix} = 0 \text{ or } \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 1 & 3 & 6 \\ 1 & 4 & 10 \end{vmatrix} = 0$$

Expanding the determinant along the first row, we get

$$(x-2)6 - (2x-3)4 + (3x-4)1 = 0$$

$$6x - 12 - 8x + 12 + 3x - 4 = 0$$

or

$$x - 4 = 0 \text{ or } x = 4.$$



Comprehensive Exercise 1.2

1. Evaluate the following determinants:

(i) $\begin{vmatrix} a & b \\ -b & a \end{vmatrix}$

(ii) $\begin{vmatrix} x+y & 1 \\ x^3 & x^2 - xy + y^2 \end{vmatrix}$

2. Evaluate to each of the following determinants:

(i) $\begin{vmatrix} 1 & 3 & -1 \\ -2 & 1 & 2 \\ 4 & 5 & 1 \end{vmatrix}$

(ii) $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ x & y & z \end{vmatrix}$

3. Show that $\begin{vmatrix} \sin 10^\circ & -\cos 10^\circ \\ \sin 80^\circ & \cos 80^\circ \end{vmatrix} = 1$.

4. Find the cofactors of each element of the following determinants:

(i) $\begin{vmatrix} 5 & 7 \\ 3 & -1 \end{vmatrix}$

(ii) $\begin{vmatrix} 0 & 1 & 2 \\ 1 & 4 & 1 \\ 5 & 3 & 7 \end{vmatrix}$

(iii) $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$

5. Using the method of cofactors find the value of determinant

$$\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$$

6. Evaluate the following determinants:

$$\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}.$$

$$7. \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \quad 8. \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} \quad 9. \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \beta + \gamma & \gamma + \alpha & \alpha + \beta \end{vmatrix}$$

10. Without expanding the determinant, show that $\left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$ is a factor of the

$$\text{determinant } \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}.$$

11. Using properties of determinants, show that

$$\begin{vmatrix} a+b+2c & a & b \\ c & 2a+b+c & b \\ c & a & a+2b+c \end{vmatrix} = 2(a+b+c)^3$$

12. Using properties of determinants, prove the following results:

$$(i) \begin{vmatrix} x+9 & x & x \\ x & x+9 & x \\ x & x & x+9 \end{vmatrix} = 243(x+3)$$

$$(ii) \begin{vmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix} = 16(3x+4)$$

$$(iii) \begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$$

$$(iv) \begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

13. Prove the following identities

$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz(x-y)(y-z)(z-x).$$

$$14. \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c).$$

$$15. \begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix} = 2(a+b)(b+c)(c+a).$$

$$16. \begin{vmatrix} 1 & b+c & b^2+c^2 \\ 1 & c+a & c^2+a^2 \\ 1 & a+b & a^2+b^2 \end{vmatrix} = (a-b)(b-c)(c-a).$$

$$17. \begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2$$

Solve the Following Equations:

$$18. \begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$$

$$19. \begin{vmatrix} x+a & b & c \\ a & x+b & c \\ a & b & x+c \end{vmatrix} = 0$$

$$20. \begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$$



Answers

1.	(i) $a^2 + b^2$ (ii) y^3	2.	(i) 35 (ii) $3x - z$
4.	(i) $A_{11} = -1, A_{12} = -3, A_{21} = -7, A_{22} = 5$		
	(ii) $A_{11} = 25, A_{12} = -2, A_{13} = -17, A_{21} = -1, A_{22} = -10,$ $A_{23} = 5, A_{31} = -7, A_{32} = 2, A_{33} = -1$		
	(iii) $A_{11} = a(b^2 - c^2), A_{12} = a(c - b), A_{13} = c - b, A_{21} = b(c^2 - a^2),$ $A_{22} = b(a - c), A_{23} = a - c, A_{31} = c(a^2 - b^2), A_{32} = c(b - a),$ $A_{33} = b - a$		
5.	0	6.	0
7.	$abc + 2fgh - af^2 - bg^2 - ch^2$	8.	$(a-b)(b-c)(c-a)$
9.	0	18.	1, 1, -9
19.	$x = 0, -(a+b+c)$	20.	$x = \frac{2}{3}, \frac{11}{3}, \frac{11}{3}$

1.3 Solution of Linear Equations (Cramer's Rule)

Solution of Simultaneous Linear Equations by Determinants (Cramer's rule)–

Let us solve the following equation:

$$\begin{aligned}a_1x + b_1y + c_1z &= d_1 \\a_2x + b_2y + c_2z &= d_2 \\a_3x + b_3y + c_3z &= d_3\end{aligned}$$

Let

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

or

$$x \Delta = \begin{vmatrix} a_1x & b_1 & c_1 \\ a_2x & b_2 & c_2 \\ a_3x & b_3 & c_3 \end{vmatrix}$$

$$[C_1 = C_1 + yC_2 + zC_3]$$

$\Rightarrow$

$$x \Delta = \begin{vmatrix} a_1x + b_1y + c_1z & b_1 & c_1 \\ a_2x + b_2y + c_2z & b_2 & c_2 \\ a_3x + b_3y + c_3z & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} = \Delta_1$$

$\Rightarrow$

$$\frac{x}{\Delta_1} = \frac{1}{\Delta}$$

Similarly

$$\frac{y}{\Delta_2} = \frac{1}{\Delta}, \quad \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$\Rightarrow$

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$\Rightarrow$

$$x = \frac{\Delta_1}{\Delta}, \quad y = \frac{\Delta_2}{\Delta}, \quad z = \frac{\Delta_3}{\Delta}$$

Note: If $\Delta = 0$ then the system is not consistent

Example 28: Solve $x - 2y + 3z = 6$

$$2x + 4y + z = 17$$

$$3x + 2y + 9z = 2$$

Using Cramer's rule.

Solution:

$$x - 2y + 3z = 6$$

$$2x + 4y + z = 17$$

$$3x + 2y + 9z = 2$$

Let

$$\Delta = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 4 & 1 \\ 3 & 2 & 9 \end{vmatrix} = 1(36 - 2) + 2(-3 + 18) + 3(4 - 12) = 40$$

Thus, $\Delta \neq 0$ and therefore, the system has unique solution given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

where $\Delta_1 = \begin{vmatrix} 6 & -2 & 3 \\ 17 & 4 & 1 \\ 2 & 2 & 9 \end{vmatrix} = 6(36-2) + 2(153-2) + 3(34-8) = 584$

$$\Delta_2 = \begin{vmatrix} 1 & 6 & 3 \\ 2 & 17 & 1 \\ 3 & 2 & 9 \end{vmatrix} = 1(153-2) - 6(18-3) + 3(4-51) = -80$$

$$\Delta_3 = \begin{vmatrix} 1 & -2 & 6 \\ 2 & 4 & 17 \\ 3 & 2 & 2 \end{vmatrix} = 1(8-34) + 2(4-51) + 6(4-12) = -168$$

$$\therefore x_1 = \frac{\Delta_1}{\Delta} = \frac{584}{40} = \frac{73}{5}, \quad y_1 = \frac{\Delta_2}{\Delta} = \frac{-80}{40} = -2, \quad z_1 = \frac{\Delta_3}{\Delta} = \frac{-168}{40} = -\frac{21}{5}$$

Example 29: Use Cramer's rule to solve the following system of equations

$$3x + y - z = 1, \quad 5x + 2y + 3z = 2, \quad 8x + 3y + z = 3.$$

Solution: $3x + y - z = 1$

$$5x + 2y + 3z = 2$$

$$8x + 3y + z = 3$$

Let $\Delta = \begin{vmatrix} 3 & 1 & -1 \\ 5 & 2 & 3 \\ 8 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -1 \\ 14 & 5 & 3 \\ 11 & 4 & 1 \end{vmatrix} = -1 \begin{vmatrix} 14 & 5 \\ 11 & 4 \end{vmatrix} = -(56-55) = -1$

$$[C_1 \rightarrow C_1 + 3C_3, C_2 \rightarrow C_2 + C_3]$$

Thus, $\Delta \neq 0$ and therefore, the system has a unique solution given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$$\Rightarrow \frac{x}{\begin{vmatrix} 1 & 1 & -1 \\ 2 & 2 & 3 \\ 3 & 3 & 1 \end{vmatrix}} = \frac{y}{\begin{vmatrix} 3 & 1 & -1 \\ 5 & 2 & 3 \\ 8 & 3 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 3 & 1 & 1 \\ 5 & 2 & 2 \\ 8 & 3 & 3 \end{vmatrix}} = \frac{1}{\begin{vmatrix} 3 & 1 & -1 \\ 5 & 2 & 3 \\ 8 & 3 & 1 \end{vmatrix}}$$

$$\Delta_1 = \begin{vmatrix} 1 & 1 & -1 \\ 2 & 2 & 3 \\ 3 & 3 & 1 \end{vmatrix} = 0$$

$$[C_1 \rightarrow C_1 + 3C_3, C_2 \rightarrow C_2 + C_3]$$

$$\Delta_2 = \begin{vmatrix} 3 & 1 & -1 \\ 5 & 2 & 3 \\ 8 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -1 \\ 14 & 5 & 3 \\ 11 & 4 & 1 \end{vmatrix} = -1 \begin{vmatrix} 14 & 5 \\ 11 & 4 \end{vmatrix} = -(56-55) = -1$$

$$\Delta_3 = \begin{vmatrix} 3 & 1 & 1 \\ 5 & 2 & 2 \\ 8 & 3 & 3 \end{vmatrix} = 0$$

The solution is given by $\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$

$$\frac{x}{0} = \frac{y}{-1} = \frac{z}{0} = \frac{1}{-1}$$

Or

Hence, $x = 0, y = 1, z = 0$

Example 30: Use Cramer's rule to solve the following equations:

$$x - y - z = 1; \quad y - z - x = 1, \quad z - x - y = 1$$

Solution: $x - y - z = 1$

$$-x + y - z = 1$$

$$-x - y + z = 1$$

$$[C_2 \rightarrow C_2 + C_1, C_3 \rightarrow C_3 + C_1]$$

$$\text{Let } \Delta = \begin{vmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ -1 & 0 & -2 \\ -1 & -2 & 0 \end{vmatrix} = 1 \begin{vmatrix} 0 & -2 \\ -2 & 0 \end{vmatrix} = -4$$

Thus, $\Delta \neq 0$ and therefore the system has a unique solution given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$$\begin{vmatrix} x & & \\ 1 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} y & & \\ 1 & 1 & -1 \\ -1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} z & & \\ 1 & -1 & 1 \\ -1 & 1 & 1 \\ -1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & & \\ 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{vmatrix}$$

$$[C_2 \rightarrow C_2 + C_1, C_3 \rightarrow C_3 + C_1]$$

$$\Delta_1 = \begin{vmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} = 1 \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 1(4 - 0) = 4$$

$$[C_1 \rightarrow C_1 + C_3, C_2 \rightarrow C_2 + C_3]$$

$$\Delta_2 = \begin{vmatrix} 1 & 1 & -1 \\ -1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -1 \\ -2 & 0 & -1 \\ 0 & 2 & 1 \end{vmatrix} = -1 \begin{vmatrix} -2 & 0 \\ 0 & 2 \end{vmatrix} = -(-4 - 0) = 4$$

$$\Delta_3 = \begin{vmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ -1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & -1 & 0 \\ 0 & 1 & 2 \\ -2 & -1 & 0 \end{vmatrix} = 1 \begin{vmatrix} 0 & 2 \\ -2 & 0 \end{vmatrix} = (0 + 4) = 4$$

The solution is given by $\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$

$$\text{or } \frac{x}{4} = \frac{y}{4} = \frac{z}{4} = -\frac{1}{4}$$

Hence, $x = -1, y = -1, z = -1$.

Example 31: Apply Cramer's rule and solve

$$3x + 5y - 7z = 13$$

$$4x + y - 12z = 6$$

$$2x + 9y - 3z = 20$$

Solution: $3x + 5y - 7z = 13$

$$4x + y - 12z = 6$$

$$2x + 9y - 3z = 20$$

$$[C_1 \rightarrow C_1 - 4C_2; C_3 \rightarrow C_3 + 12C_2]$$

Let
$$\Delta = \begin{vmatrix} 3 & 5 & -7 \\ 4 & 1 & -12 \\ 2 & 9 & -3 \end{vmatrix} = \begin{vmatrix} -17 & 5 & 53 \\ 0 & 1 & 0 \\ -34 & 9 & 105 \end{vmatrix} = 1 \begin{vmatrix} -17 & 53 \\ -34 & 105 \end{vmatrix}$$

$$= -17 \begin{vmatrix} 1 & 53 \\ 2 & 105 \end{vmatrix} = -17(105 - 106) = 17$$

Thus, $\Delta \neq 0$ and therefore the system has a unique solution given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$$\begin{array}{c|c|c|c} x & y & z & 1 \\ \hline \begin{vmatrix} 13 & 5 & -7 \\ 6 & 1 & -12 \\ 20 & 9 & -3 \end{vmatrix} & \begin{vmatrix} 3 & 13 & -7 \\ 4 & 6 & -12 \\ 2 & 20 & -3 \end{vmatrix} & \begin{vmatrix} 3 & 5 & 13 \\ 4 & 1 & 6 \\ 2 & 9 & 20 \end{vmatrix} & \begin{vmatrix} 3 & 5 & -7 \\ 4 & 1 & -12 \\ 2 & 9 & -3 \end{vmatrix} \end{array}$$

$$[C_1 \rightarrow C_1 - 6C_2, C_3 \rightarrow C_3 + 12C_2]$$

$$\Delta = \begin{vmatrix} 13 & 5 & -7 \\ 6 & 1 & -12 \\ 20 & 9 & -3 \end{vmatrix} = \begin{vmatrix} -17 & 5 & 53 \\ 0 & 1 & 0 \\ -34 & 9 & 105 \end{vmatrix} = \begin{vmatrix} -17 & 53 \\ -34 & 105 \end{vmatrix}$$

$$= -17 \begin{vmatrix} 1 & 53 \\ 2 & 105 \end{vmatrix} = -17(105 - 106) = 17$$

Expanding the determinant along first row

$$\Delta_2 = \begin{vmatrix} 3 & 13 & -7 \\ 4 & 6 & -12 \\ 2 & 20 & -3 \end{vmatrix} = 3 \begin{vmatrix} 6 & -12 \\ 20 & -3 \end{vmatrix} - 13 \begin{vmatrix} 4 & -12 \\ 2 & -3 \end{vmatrix} - 7 \begin{vmatrix} 4 & 6 \\ 2 & 20 \end{vmatrix}$$

$$= 3(-18 + 240) - 13(-12 + 24) - 7(80 - 12)$$

$$= 3(222) - 13(12) - 7(68) = 666 - 156 - 476 = 34$$

$$[C_1 \rightarrow C_1 - 4C_2, C_3 \rightarrow C_3 - 6C_2]$$

$$\Delta_3 = \begin{vmatrix} 3 & 5 & 13 \\ 4 & 1 & 6 \\ 2 & 9 & 20 \end{vmatrix} = \begin{vmatrix} -17 & 5 & -17 \\ 0 & 1 & 0 \\ -34 & 9 & -34 \end{vmatrix} = \begin{vmatrix} -17 & -17 \\ -34 & -34 \end{vmatrix} = 0$$

The solution is given by $\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$

$$\frac{x}{17} = \frac{y}{34} = \frac{z}{0} = \frac{1}{17}$$

Hence,

$$x = 1, y = 2, z = 0$$

Example 32: Solve by Cramer's rule the following equations

$$x + y + z = 2, \quad x + 2y + 3z = 1, \quad 3x + y - 5z = 4$$

Solution: $x + y + z = 2$

$$x + 2y + 3z = 1$$

$$3x + y - 5z = 4$$

Let
$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & -5 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 2 \\ 3 & -2 & -8 \end{vmatrix} \quad \begin{matrix} c_2 \rightarrow c_2 - c_1 \\ c_3 \rightarrow c_3 - c_1 \end{matrix}$$

$$= 1 \begin{vmatrix} 1 & 2 \\ -2 & -8 \end{vmatrix} = -8 + 4 = -4$$

Thus, $\Delta \neq 0$ and therefore the system has a unique solution given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 4 & 1 & -5 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 1 & 3 \\ 3 & 4 & -5 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 3 & 1 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & -5 \end{vmatrix}$$

$$\Delta_1 = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 3 \\ 4 & 1 & -5 \end{vmatrix} = -16$$

$$[C_2 \rightarrow C_2 - 2C_1, C_3 \rightarrow C_3 - C_1]$$

$$\Delta_2 = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 1 & 3 \\ 3 & 4 & -5 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & -1 & 2 \\ 3 & -2 & -8 \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ -2 & -8 \end{vmatrix} = 8 + 4 = 12$$

$$[C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - 2C_1]$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 3 & 1 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & -1 \\ 3 & -2 & -2 \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ -2 & -2 \end{vmatrix} = 2 - 2 = -4$$

The solution is given by $\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$

or
$$\frac{x}{-16} = \frac{y}{12} = \frac{z}{-4} = -\frac{1}{4}$$

Hence, $x = 4, y = -3, z = 1$.

Example 33: Solve by Cramer's rule

$$2x + 3y - 3z = 0, \quad 5x - 2y + 2z = 19, \quad x + 7y - 5z = 5$$

Solution: $2x + 3y - 3z = 0$

$$5x - 2y + 2z = 19$$

$$x + 7y - 5z = 5$$

Let
$$\Delta = \begin{vmatrix} 2 & 3 & -3 \\ 5 & -2 & 2 \\ 1 & 7 & -5 \end{vmatrix} = \begin{vmatrix} 2 & -11 & 7 \\ 5 & -37 & 27 \\ 1 & 0 & 0 \end{vmatrix} \quad \begin{matrix} c_2 \rightarrow c_2 - 7c_1, c_3 \rightarrow c_3 + 5c_1 \end{matrix}$$

$$= \begin{vmatrix} -11 & 7 \\ -37 & 27 \end{vmatrix} = -38$$

Thus, $\Delta \neq 0$ and therefore the system has a unique solution given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$$\begin{vmatrix} 0 & 3 & -3 \\ 19 & -2 & 2 \\ 5 & 7 & -5 \end{vmatrix} = \begin{vmatrix} 2 & 0 & -3 \\ 5 & 19 & 2 \\ 1 & 5 & -5 \end{vmatrix} = \begin{vmatrix} 2 & 3 & 0 \\ 5 & -2 & 19 \\ 1 & 7 & 5 \end{vmatrix} = \begin{vmatrix} 2 & 3 & -3 \\ 5 & -2 & 2 \\ 1 & 7 & -5 \end{vmatrix}$$

$$[C_2 \rightarrow C_2 + C_3]$$

$$\Delta_1 = \begin{vmatrix} 0 & 3 & -3 \\ 19 & -2 & 2 \\ 5 & 7 & -5 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -3 \\ 19 & 0 & 2 \\ 5 & 2 & -5 \end{vmatrix} = -3 \begin{vmatrix} 19 & 0 \\ 5 & 2 \end{vmatrix}$$

$$= -3(38 - 0) = -114$$

$$[C_2 \rightarrow C_2 - 5C_1, C_3 \rightarrow C_3 + 5C_1]$$

$$\Delta_2 = \begin{vmatrix} 2 & 0 & -3 \\ 5 & 19 & 2 \\ 1 & 5 & -5 \end{vmatrix} = \begin{vmatrix} 2 & -10 & 7 \\ 5 & -6 & 27 \\ 1 & 0 & 0 \end{vmatrix} = 1 \begin{vmatrix} -10 & 7 \\ -6 & 27 \end{vmatrix}$$

$$= 1(-270 + 42) = -228$$

$$[C_2 \rightarrow C_2 - 7C_1, C_3 \rightarrow C_3 - 5C_1]$$

$$\Delta_3 = \begin{vmatrix} 2 & 3 & 0 \\ 5 & -2 & 19 \\ 1 & 7 & 5 \end{vmatrix} = \begin{vmatrix} 2 & -11 & -10 \\ 5 & -37 & -6 \\ 1 & 0 & 0 \end{vmatrix} = 1 \begin{vmatrix} -11 & -10 \\ -37 & -6 \end{vmatrix}$$

$$= 66 - 370 = -304$$

The solution is given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

or

$$\frac{x}{-114} = \frac{y}{-228} = \frac{z}{-304} = \frac{1}{-38}$$

Hence,

$$x = 3, \quad y = 6, \quad z = 8.$$

Example 34: Solve the following system of linear equations:

$$x + y + z = 1$$

$$2x + 3y + z = 4$$

$$4x + 9y + z = 16$$

Solution: Let

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 4 & 9 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & -1 \\ 4 & 5 & -3 \end{vmatrix} \quad \begin{array}{l} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{array}$$

$$= \begin{vmatrix} 1 & -1 \\ 5 & -3 \end{vmatrix} = (-3 + 5) = 2$$

Thus, $\Delta \neq 0$ and therefore, the system has a unique solution given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$$\begin{array}{c} x \qquad \qquad y \qquad \qquad z \qquad \qquad 1 \\ \left| \begin{array}{ccc} 1 & 1 & 1 \\ 4 & 3 & 1 \\ 16 & 9 & 1 \end{array} \right| = \left| \begin{array}{ccc} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 4 & 16 & 1 \end{array} \right| = \left| \begin{array}{ccc} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 4 & 9 & 16 \end{array} \right| = \left| \begin{array}{ccc} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 4 & 9 & 1 \end{array} \right| \end{array}$$

$$[C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1]$$

$$\Delta_1 = \left| \begin{array}{ccc} 1 & 1 & 1 \\ 4 & 3 & 1 \\ 16 & 9 & 1 \end{array} \right| = \left| \begin{array}{ccc} 1 & 0 & 0 \\ 4 & -1 & -3 \\ 16 & -7 & -15 \end{array} \right| = 1 \left| \begin{array}{cc} -1 & -3 \\ -7 & -15 \end{array} \right|$$

$$= 15 - 21 = -6$$

$$[C_3 \rightarrow C_3 - C_1, C_2 \rightarrow C_2 - C_1]$$

$$\Delta_2 = \left| \begin{array}{ccc} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 4 & 16 & 1 \end{array} \right| = \left| \begin{array}{ccc} 1 & 0 & 0 \\ 2 & 2 & -1 \\ 4 & 12 & -3 \end{array} \right| = 1 \left| \begin{array}{cc} 2 & -1 \\ -12 & -3 \end{array} \right|$$

$$= (-6 + 12) = 6$$

$$[C_3 \rightarrow C_3 - C_1, C_2 \rightarrow C_2 - C_1]$$

$$\Delta_3 = \left| \begin{array}{ccc} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 4 & 9 & 16 \end{array} \right| = \left| \begin{array}{ccc} 1 & 0 & 0 \\ 2 & 1 & 2 \\ 4 & 5 & 12 \end{array} \right| = 1 \left| \begin{array}{cc} 1 & 2 \\ 5 & 12 \end{array} \right| = (12 - 10) = 2$$

The solution is given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$$\frac{x}{-6} = \frac{y}{6} = \frac{z}{2} = \frac{1}{2}$$

Hence,

$$x = -3, \quad y = 3, \quad z = 1.$$

Example 35: Using Cramer's rule solve the following system of equations:

$$2x - 3y + z = 7$$

$$2x + y - z = 1$$

$$4y + 3z = -11$$

[B.C.A (Agra) 2002]

Solution: We have

$$2x - 3y + z = 7 \quad \dots(1)$$

$$2x + y - z = 1 \quad \dots(2)$$

$$4y + 3z = -11 \quad \dots(3)$$

Let

$$\Delta = \left| \begin{array}{ccc} 2 & -3 & 1 \\ 2 & 1 & -1 \\ 0 & 4 & 3 \end{array} \right| = \left| \begin{array}{ccc} 2 & -3 & 1 \\ 0 & 4 & -2 \\ 0 & 4 & 3 \end{array} \right| = 2(12 + 8) = 40$$

Thus, $\Delta \neq 0$ and therefore the given system has a unique solution, given by

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$$i.e., \quad \Delta_1 = \begin{vmatrix} 7 & -3 & 1 \\ 1 & 1 & -1 \\ -11 & 4 & 3 \end{vmatrix} = 7(3+4) + 3(3-11) + 1(4+11) = 40$$

$$\Delta_2 = \begin{vmatrix} 2 & 7 & 1 \\ 2 & 1 & -1 \\ 0 & -11 & 3 \end{vmatrix} = \begin{vmatrix} 2 & 7 & 1 \\ 0 & -6 & -2 \\ 0 & -11 & 3 \end{vmatrix} = 2(-18-22) = -80$$

$$(R_2 = R_2 - R_1)$$

$$\Delta_3 = \begin{vmatrix} 2 & -3 & 7 \\ 2 & 1 & 1 \\ 0 & 4 & -11 \end{vmatrix} = 2(-11-4) + 3(-22-0) + 7(8-0) = -40$$

$$\text{Hence,} \quad x = \frac{40}{40} = 1, \quad y = -\frac{80}{40} = -2, \quad z = \frac{-40}{40} = -1$$

$$i.e., \quad x = 1, \quad y = -2, \quad z = -1.$$



Comprehensive Exercise 1.3

1. Solve the following system of equations using determinants only

$$x + 2y + 5z = 23$$

$$3x + y + 4z = 26$$

$$6x + y + 7z = 47$$

2. Solve the equations (by Cramer's rule):

$$x + y + z = 1$$

$$3x + 5y + 6z = 4$$

$$9x + 2y - 36z = 17$$

[B.C.A. (Purvanchal) 2006]

3. Using determinants, solve the following system of equations

$$x - y = 3$$

$$2x + 3y + 4z = 17$$

$$y + 2z = 7$$

4. Apply Cramer's rule to solve the following equations

$$x + y + z = -1$$

$$x + 2y + 2z = -4$$

$$x + 3y + yz = -6$$

5. Solve the following equations

$$x + 9y - z = 4$$

$$2x + 7y + 3z = 7$$

$$3x + 10y + 4z = 9$$

[B.C.A. (Garhwal) 2010]



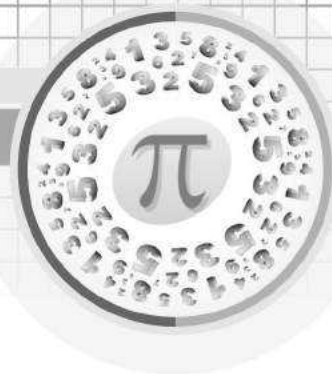
Answers

1.	$x = 4, y = 2, z = 3$	2.	$x = \frac{1}{3}, y = 1, z = \frac{-1}{3}$	3.	$x = -1, y = 2, z = 4$
4.	$x = 1, y = -1, z = -1$	5.	$x = -3, y = 1, z = 2$		





Unit-1



Chapter 2

Algebra of Matrices and Type of Matrices

2.1 Matrices

Let us consider a set of simultaneous equations

$$x + 2y + 3z + 4t = 0$$

$$4x + 6y + 5z + 7t = 0$$

$$3x + 4y + 2z + 6t = 0$$

Now, we write down the coefficients of x, y, z and t of the above equations and enclose

them within brackets and we get $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 6 & 5 & 7 \\ 3 & 4 & 2 & 6 \end{bmatrix}$

The above system of numbers, arranged in a rectangular array in rows and columns and bounded by the brackets is an example of a **matrix**. The horizontal lines are called **rows** and vertical lines are called **columns** of the matrix.

There are 3 rows and 4 columns in this matrix. The numbers 1, 2, 3, 4..... constituents of this matrix are called **elements**. It is termed as 3×4 matrix, to be read as [3 by 4 matrix]. In the double subscripts of an element, the first subscript denotes the row and the second subscript stands for the column in which the elements lies, a_{ij} lies in the i^{th} row and j^{th} column.

Thus, $a_{24} = 7$. There is a lot of difference between a matrix and a determinant. The value of a determinant is a number. Matrix is not a number. It has no numerical value. Matrix gives some information. As elements in the above matrix are the coefficients of x, y, z and t . Here, there are few examples of the matrix.

$$A = \begin{bmatrix} a & b & c & d \\ 1 & 2 & 3 & 4 \end{bmatrix}_{2 \times 4} \quad B = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix}_{3 \times 3}$$

An $m \times n$ matrix in general form is written as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

Matrix A is also represented in short form $A = [a_{ij}]$

2.1.1 Type of Matrices

(i) **Row Matrix:** If a matrix has a only one row and any number of columns, it is called a row matrix *eg.*, $[1 \ 2 \ 3 \ 4]_{1 \times 4}$ is a row matrix of the type 1×4 .

(ii) **Column Matrix:** A matrix, having one column and any number of rows is called a column matrix *eg.*, $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is a column matrix of the type 3×1 .

(iii) **Null Matrix or Zero Matrix:** Any matrix, in which all the elements are zeros, is called zero matrix or null matrix. [B.C.A. (Meerut) 2008]

eg., $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2 \times 3}$ is zero matrix of the type 2×3 . It is usually denoted by 0.

(iv) **Square Matrix:** A matrix, in which the number of rows is equal to the number of columns, is called a square matrix. *eg.*, $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$. [B.C.A. (Meerut) 2009]

The element on the principal diagonals are 1, 4. In general the elements, $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ are called the diagonal elements and the lines along which they lie is called the principal diagonal of the matrix.

(v) **Diagonal Matrix:** A square matrix is called a diagonal matrix, if all its non-diagonal elements are zero. [B.C.A. (Meerut) 2008, 2009]

eg., $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

Thus, Diagonal matrix can be written as diagonal (1, 3, 4). Thus, $A = [a_{ij}]_{m \times n}$ is diagonal matrix if $a_{ij} = 0$ for $i \neq j$.

(vi) **Scalar Matrix:** A square matrix $A = [a_{ij}]$ is said to be a scalar matrix if

[B.C.A. (Meerut) 2009, 2010]

- (a) all its non-diagonal elements are zero.
- (b) all its diagonal elements are equal.

$$\text{Thus, } a_{ij} = \begin{cases} o & \text{when } i \neq j \\ k & \text{when } i = j \end{cases} \text{ for example } \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

This scalar matrix can be written as diagonal (2,2,2).

(vii) **Unit or identity Matrix:** A diagonal matrix is called a unit matrix if all the diagonal elements are 1 and non-diagonal elements are zero

[B.C.A. (Meerut) 2008, 2009]

$$\text{e.g., } I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

A square matrix is said to be a unit matrix or identity matrix if

- (a) all its non-diagonal elements are zero.
- (b) all its diagonal elements are each equal to 1.

$$\text{Thus, } A = [a_{ij}] \text{ is said to be a unit matrix if } a_{ij} = \begin{cases} 0 & \text{when } i \neq j \\ 1 & \text{when } i = j \end{cases}$$

It is denoted by I , I_n is a unit matrix of order n , in unit matrix $A = [a_{ij}]$, $a_{ii} = 1$ when $i = j$ and $a_{ij} = 0$ when $i \neq j$.

(viii) **Transpose of Matrix:** If in a given matrix A , we interchange the rows and the corresponding columns, the new matrix obtained is called the transpose of the matrix A and is denoted by A' or A^T .

$$\text{For Example: If } A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad A' = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$$

The transpose of a matrix $A = [a_{ij}]_{m \times n}$ is defined as the matrix $A' = [b_{ij}]_{n \times m}$ where $b_{ij} = a_{ji}$, in other words the $(i, j)^{\text{th}}$ element of A is the $(j, i)^{\text{th}}$ element of A'

(ix) **Symmetric Matrix:** A square matrix $A = [a_{ij}]$ is said to be symmetric if its (i, j) th element = the element (j, i) th of $[a_{ij}]$ in short $a_{ij} = a_{ji}$

Hence, $A' = A$

[B.C.A. (Meerut) 2009]

$$\text{For Example: } \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}; \begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & -2 \\ 4 & -2 & 0 \end{bmatrix}$$

- (x) **Skew symmetric Matrix:** A square matrix is called a skew symmetric matrix if $a_{ij} = -a_{ji}$ and all the diagonal elements are zero [B.C.A. (Meerut) 2008]

For Example:
$$\begin{bmatrix} 0 & -h & g \\ h & 0 & -f \\ -g & f & 0 \end{bmatrix}$$

Hence, $A' = -A$

- (xi) **Upper Triangular Matrix:** A square matrix, all of whose elements below the principal diagonal are zero, is called an upper triangular matrix.

Thus, $A = [a_{ij}]_{n \times n}$ is an upper triangular matrix if $a_{ij} = 0$ for $i > j$.

For Example:
$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

- (xii) **Lower triangular Matrix:** A square matrix all of whose elements above the principal diagonal are zero, is called a **lower triangular** matrix *e.g.*, thus, $A = [a_{ij}]_{n \times n}$ is a lower triangular matrix if $a_{ij} = 0$ for $i < j$.

For Example:
$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$$

- (xiii) **Orthogonal Matrix:** A square matrix is called an **orthogonal matrix**, if the product of the matrix A and its transpose matrix A' is an identity matrix, *e.g.*, $A.A' = I$.

if $|A| = \pm 1$, matrix is proper.

- (xiv) **Conjugate of a Matrix:** A matrix obtained by replacing the elements by their corresponding conjugate elements, is called **conjugate** of a matrix and denoted by $\bar{A}$. For Example

if $A = \begin{bmatrix} 1+i & 2-3i & 4 \\ 7+2i & -i & 3-2i \end{bmatrix}$ then $\bar{A} = \begin{bmatrix} 1-i & 2+3i & 4 \\ 7-2i & i & 3+2i \end{bmatrix}$

- (xv) **Matrix A^θ :** Transpose of the conjugate of a matrix A is A^θ . For example,

Let $A = \begin{bmatrix} 2+i & 3 & i \\ 1-i & 1+2i & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2-i & 3 & -i \\ 1+i & 1-2i & 0 \end{bmatrix}$ conjugate of matrix

$$A^\theta = \begin{bmatrix} 2-i & 1+i \\ 3 & 1-2i \\ -i & 0 \end{bmatrix}$$

(xvi) **Unitary Matrix:** A square matrix A is said to be unitary matrix, if $A^\theta \cdot A = I$.

For Example: $A = \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$

(xvii) **Hermitian Matrix:** A square matrix $A = (a_{ij})_{n \times n}$ is called Hermitian matrix, if every (i, j) the elements of A is equal to the conjugate complex (j, i) the element of A . In other words $a_{ij} = \bar{a}_{ji}$

For Example: $\begin{bmatrix} i & 2+3i & 3+i \\ 2-3i & 2 & 1-2i \\ 3-i & 1+2i & 5 \end{bmatrix}$

Necessary and sufficient conditions for a matrix to be Hermitian is that $A = A^\theta$ i.e., conjugate transpose of A .

or $A = (\bar{A})'$

(xviii) **Skew Hermitian Matrix:** A square matrix $A = (a_{ij})_{n \times n}$ is said to be Skew Hermitian matrix if every (i, j) the element of A is equal to negative conjugate complex of (j, i) the element of A .

If other words $a_{ij} = -\bar{a}_{ji}$

For Example: $\begin{bmatrix} i & 2-3i & 4+5i \\ -(2+3i) & 0 & 2i \\ -(4-5i) & 2i & -3i \end{bmatrix}$

The necessary and sufficient conditions for a matrix A to be Skew Hermitian matrix is that $A^\theta = -A$ or $(\bar{A})' = -A$.

(xix) **Idempotent Matrix:** A matrix such that $A^2 = A$ is called idempotent matrix

For Example: $A = \begin{bmatrix} 2 & -2 & 4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \Rightarrow A^2 = A. A = \begin{bmatrix} 2 & -2 & 4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A$

(xx) **Periodic Matrix:** A matrix A is called a periodic matrix if $A^{k+1} = A$, where k is a +ve integer. If k is the least +ve integer for which $A^{k+1} = A$, then k is said to be the period of A . If we choose $k = 1$, we get $A^2 = A$ and we call it to be idempotent matrix.

(xxi) **Nilpotent Matrix:** A matrix is called a Nilpotent matrix. If $A^k = O$ (null matrix) where k is a +ve integer. If however k is the least +ve integer for which $A^k = O$, then k is the index of the nilpotent matrix,

For Example: $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$

A is a nilpotent matrix whose index is 2.

(xxii) **Involuntary Matrix:** A matrix is called an involuntary matrix if $A^2 = I$ (Identity matrix). Since $I^2 = I$ always.

$\therefore$ Unit matrix is an involuntary matrix.

(xxiii) **Equal Matrix:** Two matrices $A = (a_{ij})$ and $B = (b_{ij})$ are said to be equal matrices if

(a) They are of the same order.

(b) The elements in the corresponding position are equal i.e., $a_{ij} = b_{ij}$, for each i and j .

Example: Find the value of x, y, z and a which satisfies the matrix equation

$$\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4a-6 \end{bmatrix} = \begin{bmatrix} 10 & -17 \\ 13 & -12a \end{bmatrix}$$

Solution: As the given matrices are equal, so their corresponding elements are also equal,

$$\begin{cases} x+3 = 10 \\ 2y+x = -17 \\ z-1 = 13 \\ 4a-6 = 12a \end{cases} \Rightarrow x=7, y=-12, z=14, a=-3/4$$

(xxiv) **Sub-Matrix:** A matrix which is obtained from a given matrix by deleting any number of rows and number of columns is called sub-matrix of the given matrix.

For Example: The sub-matrix of $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ is $\begin{bmatrix} a & b \\ d & e \end{bmatrix}$

(xxv) **Non singular and singular matrix:** A matrix A is said to be non singular if $|A| \neq 0$, if $|A| = 0$ then A is said to be singular matrix.

[B.C.A. (Meerut) 2009]

Illustrative Examples

Example 1: Show that if A is an orthogonal matrix then its determinant, $|A| = 1$.

Solution: $AA' = I$, since A is orthogonal matrix taking determinant on both sides, we have

$$|AA'| = |I|$$

$$|A| |A'| = 1, \quad |A'| = |A|$$

$$\Rightarrow |A| |A| = 1$$

$$\Rightarrow |A|^2 = 1$$

$$\Rightarrow |A| = 1$$

Example 2: If A is orthogonal matrix show that A' is also orthogonal.

Solution: If A is an orthogonal matrix, then

$$AA' = A'A = I$$

$$\Rightarrow (AA')' = (A'A)' = I, \quad I' = I$$

$$\Rightarrow (A')' A' = (A)' (A')' = I$$

Hence, A' is orthogonal. Transpose of an orthogonal matrix is also orthogonal.

Example 3: If A is an orthogonal matrix, show that A^{-1} is also orthogonal.

Solution: A is an orthogonal matrix

$$\Rightarrow AA' = A'A = I$$

Taking inverse on both sides

$$(AA')^{-1} = (A'A)^{-1} = I$$

$$\Rightarrow (A')^{-1}(A)^{-1} = (A)^{-1}(A')^{-1} = I$$

$$\Rightarrow (A^{-1})' A^{-1} = A^{-1} (A^{-1})' = I$$

Hence, A^{-1} is orthogonal.

Example 4: If A and B are two orthogonal matrices, show that AB is also orthogonal matrix.

Solution: $AA' = A'A = I$ as A is orthogonal matrix

$$BB' = B'B = I \text{ as } B \text{ is orthogonal matrix}$$

$$(AB)(AB)' = (AB)(B'A') = A(BB')A' = (AI)A' = AA' = I$$

$$\text{Similarly, } (AB)'(AB) = (B'A')(AB) = B'(A'A)B = B'IB = B'B = I$$

$$\therefore (AB)(AB)' = (AB)'(AB) = I$$

$\Rightarrow AB$ is orthogonal matrix.

Example 5: Define an orthogonal matrix and prove that the matrix

$$\begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \text{ is orthogonal.}$$

Solution: A square matrix is said to be orthogonal if $AA' = I$

Let $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ and $A' = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$

$$AA' = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & -\cos \alpha \sin \alpha + \sin \alpha \cos \alpha \\ -\sin \alpha \cos \alpha + \cos \alpha \sin \alpha & \sin^2 \alpha + \cos^2 \alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Hence, A is an orthogonal matrix.

Example 6: Show that $A = \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$ is orthogonal.

[B.C.A. (Meerut) 2007]

Solution: We have $A = \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$, $A' = \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$

$$\Rightarrow AA' = \frac{1}{9} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$\Rightarrow AA' = I$$

$\Rightarrow A$ is orthogonal matrix

Example 7: Determine the value of α, β, γ , when $\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal.

Solution: Let, $A = \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$, $A' = \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix}$

If A is orthogonal matrix then $AA' = I$

$$\Rightarrow \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix} \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

or
$$\begin{bmatrix} 4\beta^2 + \gamma^2 & 2\beta^2 - \gamma^2 & -2\beta^2 + \gamma^2 \\ 2\beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 \\ -2\beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equating the corresponding elements, we have

$$\begin{cases} 4\beta^2 + \gamma^2 = 1 \\ 2\beta^2 - \gamma^2 = 0 \end{cases} \Rightarrow \beta = \pm \frac{1}{\sqrt{6}}, \gamma = \pm \frac{1}{\sqrt{3}}$$

$$\alpha^2 + \beta + \gamma^2 = 1$$

as
$$\beta = \pm \frac{1}{\sqrt{6}}, \gamma = \pm \frac{1}{\sqrt{3}}, \alpha = \pm \frac{1}{\sqrt{2}}.$$

Example 8: Prove that every square matrix can be expressed in one and only one way as a sum of a symmetric and a skew symmetric matrix.

[B.C.A. (Rohilkhand) 2003, B.C.A. (Meerut) 2010]

Solution: Let A be a square matrix. It can be expressed as

$$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A') = P + Q$$

where
$$P = \frac{1}{2}(A + A') \text{ and } Q = \frac{1}{2}(A - A')$$

Now, we have to show that P is symmetric matrix and Q is a skew symmetric matrix

$$P' = \frac{1}{2}(A + A')' = \frac{1}{2}(A' + (A')') = \frac{1}{2}(A' + A) = P$$

$$Q = \frac{1}{2}(A - A')' = \frac{1}{2}(A' - (A')') = \frac{1}{2}(A' - A) = -\frac{1}{2}(A - A') = -Q$$

Hence, $A = P + Q$, where P is a symmetric matrix and Q is a skew symmetric matrix.

Uniqueness: Let $A = R + S$ be another expression of A where R is symmetric matrix and S is a skew symmetric.

$$R' = R, S' = -S$$

Now
$$A' = (R + S)' = R' + S' = R - S$$

$$\Rightarrow R = \frac{1}{2}(A + A') = P \text{ and } S = \frac{1}{2}(A - A') = Q$$

So
$$A = R + S = P + Q$$

Hence, $A = P + Q$ is the unique expression, where P is a symmetric matrix and Q is a skew symmetric matrix.

Example 9: Express the matrix $A = \begin{bmatrix} 4 & 2 & -3 \\ 1 & 3 & -6 \\ -5 & 0 & -7 \end{bmatrix}$ as the sum of a symmetric and a skew

symmetric matrix.

Solution:

$$A = \begin{bmatrix} 4 & 2 & -3 \\ 1 & 3 & -6 \\ -5 & 0 & -7 \end{bmatrix}, A' = \begin{bmatrix} 4 & 1 & -5 \\ 2 & 3 & 0 \\ -3 & -6 & -7 \end{bmatrix}$$

$$A + A' = \begin{bmatrix} 4+4 & 2+1 & -3-5 \\ 1+2 & 3+3 & -6+0 \\ -5-3 & 0-6 & -7-7 \end{bmatrix} = \begin{bmatrix} 8 & 3 & -8 \\ 3 & 6 & -6 \\ -8 & -6 & -14 \end{bmatrix} \quad \dots(1)$$

$$A - A' = \begin{bmatrix} 4-4 & 2-1 & -3+5 \\ 1-2 & 3-3 & -6-0 \\ -5+3 & 0+6 & -7+7 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -6 \\ -2 & 6 & 0 \end{bmatrix} \quad \dots(2)$$

Adding (1) and (2), we have

$$2A = \begin{bmatrix} 8 & 3 & -8 \\ 3 & 6 & -6 \\ -8 & -6 & -14 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -6 \\ -2 & 6 & 0 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 4 & \frac{3}{2} & -4 \\ \frac{3}{2} & 3 & -3 \\ -4 & -3 & -7 \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{2} & 1 \\ -\frac{1}{2} & 0 & -3 \\ -1 & 3 & 0 \end{bmatrix}$$

[Symmetric matrix] [Skew-Symmetric matrix]

Example 10: Prove that matrix $A = \begin{bmatrix} 1 & 1-i & 2 \\ 1+i & 3 & i \\ 2 & -i & 0 \end{bmatrix}$ is Hermitian.

Solution: We have $A = \begin{bmatrix} 1 & 1-i & 2 \\ 1+i & 3 & i \\ 2 & -i & 0 \end{bmatrix}$ and $\bar{A} = \begin{bmatrix} 1 & 1+i & 2 \\ 1-i & 3 & -i \\ 2 & i & 0 \end{bmatrix}$

$$(\bar{A})' = \begin{bmatrix} 1 & 1-i & 2 \\ 1+i & 3 & i \\ 2 & -i & 0 \end{bmatrix} \Rightarrow A^{\theta} = A$$

$\Rightarrow A$ is Hermitian matrix

Example 11: Show that $A = \begin{bmatrix} i & 3+2i & -2-i \\ -3+2i & 0 & 3-4i \\ 2-i & -3-4i & -2i \end{bmatrix}$ is Skew Hermitian matrix.

Solution:

$$\bar{A} = \begin{bmatrix} -i & 3-2i & -2+i \\ -3-2i & 0 & 3+4i \\ 2+i & -3+4i & 2i \end{bmatrix}$$

$$(\bar{A})' = \begin{bmatrix} -i & -3-2i & 2+i \\ 3-2i & 0 & -3+4i \\ -2+i & 3+4i & 2i \end{bmatrix} = - \begin{bmatrix} i & 3+2i & -2-i \\ -3+2i & 0 & 3-4i \\ 2-i & -3-4i & -2i \end{bmatrix} = -A$$

$$\Rightarrow A^\theta = -A \Rightarrow A \text{ is Skew Hermitian matrix.}$$

Example 12: Show that every square matrix can be expressed as $R + iS$ uniquely where R and S are Hermitian matrices. [B.C.A. (Lucknow) 2010]

Solution: Let A be any square matrix. It can be rewritten as

$$A = \frac{1}{2}(A + A^\theta) + i \left[\frac{1}{2i}(A - A^\theta) \right] = R + iS$$

$$\text{Where } R = \frac{1}{2}(A + A^\theta), S = \frac{1}{2i}(A - A^\theta).$$

Now, we have to show that R and S are Hermitian matrices.

$$R^\theta = \frac{1}{2}(A + A^\theta)^\theta = \frac{1}{2}(A^\theta + (A^\theta)^\theta) = \frac{1}{2}(A^\theta + A) = \frac{1}{2}(A + A^\theta) = R$$

Thus R is a Hermitian matrix

$$\begin{aligned} \text{Now } S^\theta &= \left[\frac{1}{2i}(A - A^\theta) \right]^\theta = \frac{-1}{2i}(A - A^\theta)^\theta = \frac{-1}{2i}(A^\theta - (A^\theta)^\theta) = \frac{-1}{2i}(A^\theta - A) \\ &= \frac{1}{2i}(A - A^\theta) = S \end{aligned}$$

Thus S is Hermitian matrix.

Hence, $A = R + iS$, where R and S are Hermitian matrices

Now, we have to show its **uniqueness**.

Let $A = P + iQ$ be another expression, where P and Q are Hermitian matrices i.e.,

$$P^\theta = P, Q^\theta = Q.$$

$$\text{Then, } A^\theta = (P + iQ)^\theta = P^\theta + (iQ)^\theta = P^\theta - iQ^\theta = P - iQ$$

$$A = P + iQ \text{ and } A^\theta = P - iQ$$

$$\Rightarrow P = \frac{1}{2}(A + A^\theta) = R \text{ and } Q = \frac{1}{2i}(A - A^\theta) = S.$$

Hence, $A = R + iS$ is the unique expression, where R and S are Hermitian matrices.

Example 13: Express the matrix. $A = \begin{bmatrix} 1+i & 2 & 5-5i \\ 2i & 2+i & 4+2i \\ -1+i & -4 & 7 \end{bmatrix}$ as the sum of Hermitian

and Skew Hermitian matrix.

$$\text{Solution: Given that } A = \begin{bmatrix} 1+i & 2 & 5-5i \\ 2i & 2+i & 4+2i \\ -1+i & -4 & 7 \end{bmatrix} \quad \dots(1)$$

$$\Rightarrow \bar{A} = \begin{bmatrix} 1-i & 2 & 5+5i \\ -2i & 2-i & 4-2i \\ -1-i & -4 & 7 \end{bmatrix}$$

$$A^\theta = \begin{bmatrix} 1-i & -2i & -1-i \\ 2 & 2-i & -4 \\ 5+5i & 4-2i & 7 \end{bmatrix} = (\bar{A})' \quad \dots(2)$$

$$(1) + (2) \Rightarrow A + A^\theta = \begin{bmatrix} 2 & 2-2i & 4-6i \\ 2+2i & 4 & 2i \\ 4+6i & -2i & 14 \end{bmatrix}$$

$$\text{Let } R = \frac{1}{2}(A + A^\theta) = \begin{bmatrix} 1 & 1-i & 2-3i \\ 1+i & 2 & i \\ 2+3i & -i & 7 \end{bmatrix} \quad \dots(3)$$

$$(1) - (2) \Rightarrow A - A^\theta = \begin{bmatrix} 2i & 2+2i & 6-4i \\ -2+2i & 2i & 8+2i \\ -6-4i & -8+2i & 0 \end{bmatrix}$$

$$\Rightarrow \frac{1}{2}(A - A^\theta) = \begin{bmatrix} i & 1+i & 3-2i \\ -1+i & i & 4+i \\ -3-2i & -4+i & 0 \end{bmatrix} \quad \dots(4)$$

Hence, from (3) and (4) we find

$$A = \begin{bmatrix} 1 & 1-i & 2-3i \\ 1+i & 2 & i \\ 2+3i & -i & 7 \end{bmatrix} + \begin{bmatrix} i & 1+i & 3-2i \\ -1+i & i & 4+i \\ -3-2i & -4+i & 0 \end{bmatrix}$$

[Hermitian matrix] [Skew Hermitian matrix]

Example 14: Prove that the matrix $\frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ is unitary

Solution: Let $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \Rightarrow A^\theta [\bar{A}]^1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$

$$A^\theta A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1+(1+i) & (1+i)-(1+i) \\ (1-i)-(1-i) & (1+i)+1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Hence, A is unitary matrix.

Example 15: If A and B are two unitary matrices, show that AB is unitary matrix.

[B.C.A. (Meerut) 2000]

Solution: We have $AA^\theta = A^\theta A = I$

$$BB^\theta = B^\theta B = I$$

as A and B are unitary matrix

$$\text{Now } (AB)(AB)^\theta = (AB)(B^\theta A^\theta) = A(BB^\theta)A^\theta = AIA^\theta = AA^\theta = I$$

$$\text{Again } (A^\theta B^\theta)(AB) = (B^\theta A^\theta)(AB) = B^\theta(A^\theta A)B = B^\theta(I)B = B^\theta B = I.$$

Hence, AB is unitary matrix.

Example 16: If A is a unitary matrix, show that A^{-1} is also unitary.

Solution: Since A is unitary matrix then

$$AA^\theta = A^\theta A = I$$

$$\Rightarrow (AA^\theta)^{-1} = (A^\theta A)^{-1} = I \quad (\text{by taking inverse})$$

$$\Rightarrow (A^{-1})^\theta A = A^{-1}(A^{-1})^\theta = I \Rightarrow A^{-1} \text{ is unitary matrix.}$$

Example 17: Show that the matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is Idempotent

Solution:

$$A^2 = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A$$

$$\therefore A^2 = A \Rightarrow \text{is Idempotent.}$$

Example 18: If B is Idempotent matrix, show that $A = (I - B)$ is also Idempotent.

Solution: Since B is Idempotent matrix then $B^2 = B$.

$$\text{Now } A^2 = (I - B)^2 = (I - B)(I - B) = I - IB - BI + B^2 = I - B - B + B^2$$

$$= I - 2B + B = I - 2B + B = I - B = A$$

$\Rightarrow A$ is a Idempotent matrix.

Example 19: Show that the matrix $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ is Involutory

[B.C.A. (Rohilkhand) 2004]

Solution:

$$A^2 = AA = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Example 20: Show that A is involutory, if and only if $(I + A)(I - A) = 0$.

Solution: Let A be an involutory matrix of order n

$$\text{Then } A^2 = I \Rightarrow I - A^2 = 0 \Rightarrow I^2 - A^2 = 0 \Rightarrow (I - A)(I + A) = 0.$$

Conversely if $(I + A)(I - A) = 0$

$$\Rightarrow I^2 - IA + AI - A^2 = 0$$

$$\Rightarrow I^2 - A + A - A^2 = 0$$

$$\Rightarrow I^2 - A^2 = 0$$

$$\Rightarrow A^2 = I$$

Example 21: Show that the matrix $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is a nilpotent matrix

Solution:
$$A^2 = A \cdot A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

Hence, A^2 is nilpotent matrix of order 2.



Comprehensive Exercise 2.1

1. Show that the matrix $\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{-2}{3} \\ \frac{-2}{3} & \frac{2}{3} & \frac{-1}{3} \end{bmatrix}$ is orthogonal matrix.

[B.C.A. (Meerut) 2006]

2. Verify that the matrix $A = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ is orthogonal.

3. Define orthogonal matrix and show that

$$A = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \text{ is orthogonal.}$$

4. Express the matrix $\begin{bmatrix} 1 & 0 & 5 & 3 \\ -2 & 1 & 6 & 1 \\ 3 & 2 & 7 & 1 \\ 4 & -4 & -2 & 0 \end{bmatrix}$ as the sum of a symmetric and a skew symmetric matrix.

5. If A and B be two symmetric matrices of same order, show that AB is symmetric if $AB = BA$.
[B.C.A. (Meerut) 2003]

6. Which of the following matrices are symmetric?

(a) $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 3 & 4 \\ 3 & 5 & -1 \\ 4 & -1 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 4 & 8 \\ 6 & 2 & 6 \\ 4 & 6 & 2 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1 & 2 \\ 3 & 0 & 4 \\ 5 & 6 & 0 \end{bmatrix}$

7. Which of the following matrices are skew symmetric ?

(a) $\begin{bmatrix} -1 & 2 & -3 \\ -2 & -1 & -4 \\ 3 & 4 & -1 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & 4 \\ 3 & -4 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & 7 \\ -3 & -7 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & a & b \\ -a & 0 & -c \\ -b & c & 0 \end{bmatrix}$

8. Define with examples (i) symmetric and skew symmetric matrix (ii) orthogonal matrix.
[B.C.A (Meerut) 2004]

9. Which of the following matrices are Hermitian and Skew Hermitian?

(a) $\begin{bmatrix} 1 & 2+i & 3-i \\ 2+i & 2 & 4-i \\ 3+i & 4+i & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 4 & 2-i & 5+2i \\ 2+i & 1 & 2-5i \\ 5-2i & 2+5i & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 2i & -3 & 4 \\ 3 & 3i & -5 \\ -4 & 5 & 4i \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1-i & 2+3i \\ -1-i & 0 & 6i \\ -2+3i & 6i & 0 \end{bmatrix}$

10. Show that the matrix $A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ -i & -1 \end{bmatrix}$ is unitary.

11. Prove that the following matrix is unitary $\begin{bmatrix} \frac{1}{2}(1+i) & \frac{1}{2}(-1+i) \\ \frac{1}{2}(1+i) & \frac{1}{2}(1-i) \end{bmatrix}$

12. Show that matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is involuntary matrix.

13. Show that the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is nilpotent and find its index.

14. Show that the matrix $A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & 4 \end{bmatrix}$ is a nilpotent matrix.

15. Distinguish between matrix and determinant with example. [B.C.A (Delhi) 2008]

16. Distinguish between the following:

- (i) Scalar matrix and unit matrix
- (ii) Square matrix and non-singular matrix
- (iii) Diagonal matrix and symmetric matrix

[B.C.A. (Meerut) 2009]



Answers

6.	(a), (b)	7.	(c), (d)
9.	(b) is Hermitian, (c), (d) be Skew Hermitian matrices.		
13.	3 is least positive integer such that $A^3 = 0$. Hence the index of the given matrix is 3.		

2.2 Addition of Matrices

Let $A = (a_{ij})_{m \times n}$ $B = (b_{ij})_{m \times n}$ be two matrices.

The sum $A + B$ of A and B is defined as $A + B = (c_{ij})$ where $c_{ij} = a_{ij} + b_{ij}$ for each i and j .

Remark: Two matrices can be added if they are of the same order otherwise not.

$$\begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 2+1 & 4+0 \\ 5+2 & 6+4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 7 & 10 \end{bmatrix}$$

2.2.1 Properties of Matrix Addition

Let $A = (a_{ij})_{m \times n}$, $B = (b_{ij})_{m \times n}$ be the matrices of the same order $m \times n$.

(i) **Commutative Law** : $A + B = B + A$

Proof: $A + B = [a_{ij}]_{m \times n} + [b_{ij}]_{m \times n}$

$$= [a_{ij} + b_{ij}]_{m \times n} \text{ (Addition of two matrices)}$$

$$= [b_{ij} + a_{ij}]_{m \times n} \text{ (commutative law of addition of numbers)}$$

$$= [b_{ij}]_{m \times n} + [a_{ij}]_{m \times n} \text{ (Addition of two numbers)}$$

$$= B + A$$

(ii) **Associative Law:** Let A, B and C three matrices each of the same order $m \times n$, then $(A + B) + C = A + (B + C)$.

Proof: Let $A = (a_{ij})_{m \times n}$, $B = (b_{ij})_{m \times n}$, $C = (c_{ij})_{m \times n}$

$$\begin{aligned}
 \text{Then } (A+B)+C &= \left[(a_{ij})_{m \times n} + (b_{ij})_{m \times n} \right] + (c_{ij})_{m \times n} \\
 &= [a_{ij} + b_{ij}]_{m \times n} + [c_{ij}]_{m \times n} && \text{(Addition of A and B)} \\
 &= [(a_{ij} + b_{ij}) + c_{ij}]_{m \times n} \\
 &= [a_{ij} + (b_{ij} + c_{ij})]_{m \times n} && \text{(Associative law of Addition of numbers)} \\
 &= [a_{ij}]_{m \times n} + (b_{ij} + c_{ij})_{m \times n} \\
 &= A + (B + C)
 \end{aligned}$$

(iii) **Existence of additive identity:** Let $A = [a_{ij}]$ be any $m \times n$ matrix and O be the $m \times n$ null matrix, then

$$A + O = A = O + A$$

Proof: $A + O = [a_{ij} + O]_{m \times n} = [a_{ij}]_{m \times n} = A$

$$O + A = [O + a_{ij}]_{m \times n} = [a_{ij}]_{m \times n} = A$$

Hence, $A + O = A = O + A$

(iv) **Existence of additive inverse:** Let $A = [a_{ij}]$ be $m \times n$ matrix.

There exists another $m \times n$ matrix B such that

$$A + B = O = B + A$$

where O is the $m \times n$ null matrix.

Here, matrix B is called the additive inverse of the matrix A or negative of A .

Also the (i,j) th element of B is $-a_{ij}$ if $A = [a_{ij}]$

(v) **Cancellation Law:** Let A, B and C be three matrices of the same order $m \times n$.

$$A + B = A + C \Rightarrow B = C \quad \text{(Left cancellation law)}$$

$$B + A = C + A \Rightarrow B = C \quad \text{(Right cancellation law)}$$

Proof: Given $A + B = A + C$

Adding $-A$ to both sides we have

$$-A + (A + B) = -A + (A + C)$$

$$(-A + A) + B = (-A + A) + C \quad \text{(Associative law of matrix addition)}$$

$$O + B = O + C$$

$$B = C$$

Similarly the right cancellation law.

Theorem: Prove that the transpose of the sum of two matrices is the sum of their transposes i.e., $(A + B)' = A' + B'$

Proof: Let $A = [a_{ij}]$, $B = [b_{ij}]$ be $m \times n$ matrices, since A and B are both $m \times n$ matrices, therefore $A + B$ exists and is a $m \times n$ matrix. Hence $(A + B)'$ is an $n \times m$ matrix.

Now, A' and B' are both $n \times m$ matrices, so that $A' + B'$ is also $n \times m$ matrix. Thus the matrices $(A + B)$ and $A' + B'$ are comparable, each being $n \times m$.

(i, j) the element of $(A + B)' = (j, i)$ the element of $(A + B)$

$$= a_{ji} + b_{ji}$$

$$= (j, i)\text{th element of } A + (j, i)\text{th element of } B.$$

$$= (i, j)\text{th element of } A' + (i, j)\text{th element of } B'.$$

$$= (i, j)\text{th element of } (A' + B')$$

Thus, the matrices $(A + B)'$ and $A' + B'$ are comparable and their corresponding elements are equal.

Hence, $(A + B)' = A' + B'$

Example 21: If $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ 1 & -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 & 2 \\ -1 & 0 & 1 \\ 0 & 0 & 3 \end{bmatrix}$ show that $A + B = B + A$

Solution:

$$\begin{aligned} A + B &= \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ 1 & -1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 3 & 2 \\ -1 & 0 & 1 \\ 0 & 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2+1 & 3+3 & 4+2 \\ 1-1 & 2+0 & 3+1 \\ 1+0 & -1+0 & 2+3 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 6 \\ 0 & 2 & 4 \\ 1 & -1 & 5 \end{bmatrix} \quad \dots(1) \end{aligned}$$

$$\begin{aligned} B + A &= \begin{bmatrix} 1 & 3 & 2 \\ -1 & 0 & 1 \\ 0 & 0 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ 1 & -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 1+2 & 3+3 & 2+4 \\ -1+1 & 0+2 & 1+3 \\ 0+1 & 0-1 & 3+2 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 6 \\ 0 & 2 & 4 \\ 1 & -1 & 5 \end{bmatrix} \quad \dots(2) \end{aligned}$$

From (1) and (2) $A + B = B + A$

(vi) **Scalar Multiple of a matrix:** Let A be a matrix and k be a number. Then kA is defined as the matrix each element of which is k times the corresponding elements of the matrix.

For Example $-4 \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} -4 & -8 & -12 \\ -16 & -20 & -24 \end{bmatrix}$

In symbols if $A = [a_{ij}]$, then $kA = [ka_{ij}]$, where k is a scalar.

Example 22: If $A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 2 & 0 \\ 3 & 4 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 6 \\ 2 & 4 & 0 \\ 3 & 1 & -1 \end{bmatrix}$ verify $2(A+B) = 2A + 2B$

Solution:

$$A+B = \begin{bmatrix} 2+1 & 3+0 & -1+6 \\ 1+2 & 2+4 & 0+0 \\ 3+3 & 4+1 & 1-1 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 5 \\ 3 & 6 & 0 \\ 6 & 5 & 0 \end{bmatrix}$$

$$2(A+B) = \begin{bmatrix} 2 \times 3 & 2 \times 3 & 2 \times 5 \\ 2 \times 3 & 2 \times 6 & 2 \times 0 \\ 2 \times 6 & 2 \times 5 & 2 \times 0 \end{bmatrix} = \begin{bmatrix} 6 & 6 & 10 \\ 6 & 12 & 0 \\ 12 & 10 & 0 \end{bmatrix} \quad \dots(1)$$

$$2A = 2 \begin{bmatrix} 2 & 3 & -1 \\ 1 & 2 & 0 \\ 3 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 2 \times 2 & 2 \times 3 & 2 \times -1 \\ 2 \times 1 & 2 \times 2 & 2 \times 0 \\ 2 \times 3 & 2 \times 4 & 2 \times 1 \end{bmatrix} = \begin{bmatrix} 4 & 6 & -2 \\ 2 & 4 & 0 \\ 6 & 8 & 2 \end{bmatrix}$$

$$2B = 2 \begin{bmatrix} 1 & 0 & 6 \\ 2 & 4 & 0 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 \times 1 & 2 \times 0 & 2 \times 6 \\ 2 \times 2 & 2 \times 4 & 2 \times 0 \\ 2 \times 3 & 2 \times 1 & 2 \times -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 12 \\ 4 & 8 & 0 \\ 6 & 2 & -2 \end{bmatrix}$$

$$\begin{aligned} 2A+2B &= \begin{bmatrix} 4 & 6 & -2 \\ 2 & 4 & 0 \\ 6 & 8 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 12 \\ 4 & 8 & 0 \\ 6 & 2 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 4+2 & 6+0 & -2+12 \\ 2+4 & 4+8 & 0+0 \\ 6+6 & 8+2 & 2-2 \end{bmatrix} = \begin{bmatrix} 6 & 6 & 10 \\ 6 & 12 & 0 \\ 12 & 10 & 0 \end{bmatrix} \quad \dots(2) \end{aligned}$$

From (1) and (2) we find, $2(A+B) = 2A + 2B$



Comprehensive Exercise 2.2

1. If $A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ 2 & 2 \\ 3 & 1 \end{bmatrix}$, find $3A - 2B$.

2. If $A = \begin{bmatrix} 2 & 4 & 1 \\ 1 & 5 & 0 \\ 0 & -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 & 5 \\ -2 & 2 & 6 \\ -1 & 3 & 7 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 2 & 3 \\ 1 & 3 & 2 \\ 1 & 4 & 1 \end{bmatrix}$,

verify that $A + (B + C) = (A + B) + C$.

3. Given $3 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$ find x, y and z .



Answers

1. $\begin{bmatrix} 7 & 6 \\ 2 & -1 \\ -3 & -2 \end{bmatrix}$

3. $x = 2, y = 4, z = 1, w = 3$

2.3 Multiplication of Matrices

Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times p}$ be two matrices such that the number of columns in A is equal to the number of rows in B . Two such matrices are said to be comfortable for multiplication.

Then $C = [c_{ij}]_{m \times p}$ such that

$$C_{ik} = \sum_{j=1}^n a_{ij} b_{jk} = a_{i1} b_{1k} + a_{i2} b_{2k} + \dots + a_{in} b_{nk}$$

is said to be the product of the matrices A and B i.e. $AB = C$

A is called as **prefactor** and B is said to be **postfactor** in the product AB and multiplication is row by column.

Example 23: If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 0 \\ 2 & -1 \end{bmatrix}$

Obtain the Product AB and explain why BA is not defined.

Solution: The number of columns in A is 3 and the number of rows in B is also 3 therefore the product AB is defined.

$$\begin{aligned} &= \begin{bmatrix} 0 \times 1 + 1 \times (-1) + 2 \times 2 & 0 \times (-2) + 1 \times 0 + 2 \times (-1) \\ 1 \times 1 + 2 \times (-1) + 3 \times 2 & 1 \times (-2) + 2 \times 0 + 3 \times (-1) \\ 2 \times 1 + 3 \times (-1) + 4 \times 2 & 2 \times (-2) + 3 \times 0 + 4 \times (-1) \end{bmatrix} \\ &= \begin{bmatrix} 0 - 1 + 4 & 0 + 0 - 2 \\ 1 - 2 + 6 & -2 + 0 - 3 \\ 2 - 3 + 8 & -4 + 0 - 4 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 5 & -5 \\ 7 & -8 \end{bmatrix} \end{aligned}$$

Since the number of columns 2 of B is not equal to the number of rows 3 of A , BA is not defined.

2.3.1 Properties of Matrix Multiplication

(i) Matrix Multiplication is not necessarily Commutative

$$AB \neq BA$$

[B.C.A. (Meerut) 2001]

Proof: There are three cases.

Case I: Let the matrix A be $m \times n$ and the matrix B be $n \times k$. The product BA does not exist. Since the number of columns of B is not equal to the number of rows in A , BA is not comfortable.

Case II: Let the matrix A be $m \times n$ and the matrix B be $n \times m$. Then matrix AB and matrix BA exist, but matrix AB is $m \times m$ while matrix BA is $n \times n$.

$$\therefore AB \neq BA$$

Case III : Let the matrix A be $n \times n$ and the matrix B be $n \times n$. The matrix AB and the matrix BA exist but AB and BA may or may not be equal. For example

$$\text{Let } A = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$\text{Then } AB = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 \times 1 + 1 \times 1 & 2 \times 2 + 1 \times 1 \\ 3 \times 1 + 0 \times 1 & 3 \times 2 + 0 \times 1 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 3 & 6 \end{bmatrix}^n$$

$$\text{and } BA = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 1 \times 2 + 2 \times 3 & 1 \times 1 + 2 \times 0 \\ 1 \times 2 + 1 \times 3 & 1 \times 1 + 1 \times 0 \end{bmatrix} = \begin{bmatrix} 8 & 1 \\ 5 & 1 \end{bmatrix}$$

$$AB \neq BA$$

(ii) Associative law of matrix multiplication

$$A(BC) = (AB)C$$

where A, B and C are matrices comfortable for products.

Proof: Let $A = (a_{ij})_{m \times n}$, $B = (b_{ij})_{n \times p}$, $C = (c_{ij})_{p \times q}$

$$\text{Let } AB = (d_{ij})_{m \times p}, d_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

$$BC = (e_{ij})_{n \times q}, e_{ij} = \sum_{k=1}^p b_{ik} c_{kj}$$

The order of $A(BC)$ and $(AB)C$ is $m \times q$

$$(i, j) \text{ the element of } A(BC) = \sum_{k=1}^n a_{ik} e_{kj}.$$

$$\begin{aligned} &= \sum_{k=1}^n a_{ik} \left(\sum_{l=1}^p b_{lk} c_{lj} \right) = \sum_{k=1}^p \left(\sum_{l=1}^n (a_{il} b_{lk}) \right) c_{kj} \\ &= \sum_{k=1}^p d_{ik} c_{kj} = (i, j)^{th} \text{ element of } (AB)C \end{aligned}$$

$$\text{Hence, } A(BC) = (AB)C.$$

(iii) Distributive law of multiplication over addition

$$A(B + C) = AB + AC$$

Proof: Let $A = (a_{ij})_{m \times n}$, $B = (b_{ij})_{n \times p}$, $C = (c_{ij})_{n \times p}$

So, AB, AC are of each order $m \times p$. The order of $AB + AC$ is also $m \times p$.

As $B + C$ is of order $n \times p$, so $A(B + C)$ is of the order $m \times p$.

$$\begin{aligned} (i, j)\text{th element of } A(B + C) &= \sum_{k=1}^n a_{ik} (b_{kj} + c_{kj}) \\ &= \sum_{k=1}^n (a_{ik} b_{kj} + a_{ik} c_{kj}) = \sum_{k=1}^n a_{ik} b_{kj} + \sum_{k=1}^n a_{ik} c_{kj} \\ &= (i, j)^{\text{th}} \text{ element of } AB + (i, j)^{\text{th}} \text{ element of } AC \\ &= (i, j)^{\text{th}} \text{ element of } (AB + AC) \end{aligned}$$

Hence, $A(B + C) = AB + AC$.

Theorem: The transpose of the product of two matrices is the product in reverse order of their transpose i.e.,

$$(AB)' = B' A'$$

Proof: Let $A = (a_{ij})_{m \times n}$, $B = (b_{ij})_{n \times p}$ be two matrices of the type $m \times n$ and $n \times p$ respectively.

Then $A' = (c_{ij})_{n \times m}$, $c_{ij} = a_{ji}$ and $B' = (d_{ij})_{p \times n}$, $d_{ij} = b_{ji}$

we know that the order of AB is $m \times p$

The order of $(AB)'$ is $p \times m$. The order of $B' A'$ is $p \times m$

The (i, j) th element of $(AB)' = (j, i)$ th element of AB

$$\begin{aligned} &= \sum_{k=1}^n a_{jk} b_{ki} = \sum_{k=1}^n c_{kj} d_{ik} \\ &= \sum_{k=1}^n d_{ik} c_{kj} = (i, j)\text{th element of } B' \cdot A' \end{aligned}$$

Hence, $(AB)' = B' A'$



Illustrative Examples

Multiplication of Matrix

Example 1: If A, B are square matrices, such that $A^2 = A$ and $B^2 = B$ show that $(AB)^2 = AB$, if A, B commute.

[B.C.A. (Meerut) 2001]

Solution:

$$(AB)^2 = (AB)(AB) = A(BA)B \quad [AB = BA]$$

$$= A(AB)B = A^2 B^2 = AB \quad [A^2 = A, B^2 = B]$$

Example 2: If $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$, find AB and show that $AB \neq BA$

[B.C.A. (Meerut) 2001, 2003]

Solution:

$$AB = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-8+6 & 3-10+3 \\ -8+8+10 & -12+10+5 \end{bmatrix} = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix} \quad \dots(1)$$

$$BA = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 2-12 & -4+6 & 6+15 \\ 4-20 & -8+10 & 12+15 \\ 2-4 & -4+2 & 6+5 \end{bmatrix}$$

$$= \begin{bmatrix} -10 & 2 & 21 \\ -16 & 2 & 27 \\ -2 & -2 & 11 \end{bmatrix} \quad \dots(2)$$

From (1) and (2) $AB \neq BA$.

Example 3: If $A = \begin{bmatrix} 0 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 0 \end{bmatrix}$ and I be a identify matrix prove that

$$(I + A) = (I - A) \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

[B.C.A. (Delhi) 2004]

Solution:

$$(I + A) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 0 \end{bmatrix} = \begin{bmatrix} 1 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 1 \end{bmatrix} \quad \dots(1)$$

$$(I - A) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 0 \end{bmatrix} = \begin{bmatrix} 1 & \tan \frac{\theta}{2} \\ -\tan \frac{\theta}{2} & 1 \end{bmatrix}$$

$$(I - A) \begin{bmatrix} \cos \theta - \sin \theta \\ \sin \theta - \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & \tan \theta / 2 \\ -\tan \theta / 2 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta - \sin \theta \\ \sin \theta - \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta + \tan (\theta / 2) \sin \theta & -\sin \theta + \cos \theta \tan (\theta / 2) \\ -\tan (\theta / 2) \cos \theta + \sin \theta & \tan (\theta / 2) \sin \theta + \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta + \frac{\sin (\theta / 2)}{\cos (\theta / 2)} \sin \theta & -\sin \theta + \cos \theta \frac{\sin (\theta / 2)}{\cos (\theta / 2)} \\ -\frac{\sin (\theta / 2)}{\cos (\theta / 2)} \cos \theta + \sin \theta & \frac{\sin (\theta / 2)}{\cos (\theta / 2)} \sin \theta + \cos \theta \end{bmatrix}$$

$$= \frac{1}{\cos (\theta / 2)} \begin{bmatrix} \cos \theta \cos (\theta / 2) + \sin (\theta / 2) \sin \theta & -\sin \theta \cos (\theta / 2) + \cos \theta \sin (\theta / 2) \\ -\sin (\theta / 2) \cos \theta + \sin \theta \cos (\theta / 2) & \sin (\theta / 2) \sin \theta + \cos \theta \cos (\theta / 2) \end{bmatrix}$$

$$\begin{aligned}
&= \frac{1}{\cos \theta/2} \begin{bmatrix} \cos \left(\theta - \frac{\theta}{2} \right) & -\sin \left(\theta - \frac{\theta}{2} \right) \\ \sin \left(\theta - \frac{\theta}{2} \right) & \cos \left(\theta - \frac{\theta}{2} \right) \end{bmatrix} \\
&= \begin{bmatrix} \frac{\cos \theta/2}{\cos \theta/2} & \frac{-\sin \theta/2}{\cos \theta/2} \\ \frac{\sin \theta/2}{\cos \theta/2} & \frac{\cos \theta/2}{\cos \theta/2} \end{bmatrix} = \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \quad \dots(2)
\end{aligned}$$

From (1) and (2)

$$(I + A) = (I - A) \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Example 4: Show that

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta/2 \\ -\tan \theta/2 & 1 \end{bmatrix}^{-1}$$

[B.C.A. (Kanpur) 2005]

Solution: We have $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & \tan \theta/2 \\ -\tan \theta/2 & 1 \end{bmatrix}$

$$= \begin{bmatrix} \cos \theta + \sin \theta \tan \theta/2 & \cos \theta \tan \theta/2 - \sin \theta \\ \sin \theta - \cos \theta \tan \theta/2 & \sin \theta \tan \theta/2 + \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta + \sin \theta \frac{\sin \theta/2}{\cos \theta/2} & \cos \theta \frac{\sin \theta/2}{\cos \theta/2} - \sin \theta \\ \sin \theta - \cos \theta \frac{\sin \theta/2}{\cos \theta/2} & \sin \theta \frac{\sin \theta/2}{\cos \theta/2} + \cos \theta \end{bmatrix}$$

$$= \frac{1}{\cos \theta/2} \begin{bmatrix} \cos \theta \cos \theta/2 + \sin \theta \sin \theta/2 & \cos \theta \sin \theta/2 - \sin \theta \cos \theta/2 \\ \sin \theta \cos \theta/2 - \cos \theta \sin \theta/2 & \sin \theta \sin \theta/2 + \cos \theta \cos \theta/2 \end{bmatrix}$$

$$= \frac{1}{\cos \theta/2} \begin{bmatrix} \cos (\theta - \theta/2) & -\sin (\theta - \theta/2) \\ \sin (\theta - \theta/2) & \cos (\theta - \theta/2) \end{bmatrix}$$

$$= \frac{1}{\cos \theta/2} \begin{bmatrix} \cos \theta/2 & -\sin \theta/2 \\ \sin \theta/2 & \cos \theta/2 \end{bmatrix} = \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix}$$

or

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta/2 \\ -\tan \theta/2 & 1 \end{bmatrix}^{-1}$$

Example 5: If $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}$, find AA' and show that AA' is symmetric matrix.

Solution: $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}$ then $A' = \begin{bmatrix} 3 & 0 \\ 1 & 1 \\ -1 & 2 \end{bmatrix}$

$$AA' = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 1 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 9+1+1 & 0+1-2 \\ 0+1-2 & 0+1+4 \end{bmatrix} = \begin{bmatrix} 11 & -1 \\ -1 & 5 \end{bmatrix}$$

Here $a_{12} = a_{21} = -1$

Hence, AA' is symmetric matrix

Example 6: Show that the matrix $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ is a solution of the equation

$A^2 - 5A + 7I_2 = O$, where I_2 is 2×2 unit matrix.

[B.C.A. (Meerut) 2000]

Solution: $A^2 = A \cdot A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 9-1 & 3+2 \\ -3-2 & -1+4 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} \quad \dots(1)$

$$-5A = -5 \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -15 & -5 \\ 5 & -10 \end{bmatrix} \quad \dots(2)$$

$$7I_2 = 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \quad \dots(3)$$

Adding (1), (2) and (3) we get,

$$\begin{aligned} A^2 - 5A + 7I_2 &= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} + \begin{bmatrix} -15 & -5 \\ 5 & -10 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 8-15+7 & 5-5+0 \\ -5+5+0 & 3-10+7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \end{aligned}$$

Hence A is solution of $A^2 - 5A + 7I_2 = O$

Example 7: If $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ prove that $(2I + 3E)^2 = 8I + 36E$.

[B.C.A. (Delhi) 2006]

Solution: $2I = 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ and $3E = 3 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$

$$(2I + 3E) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$$

$$(2I + 3E)^2 = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 12 \\ 0 & 4 \end{bmatrix}$$

$$(2I + 3E)^3 = \begin{bmatrix} 4 & 12 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 36 \\ 0 & 8 \end{bmatrix} \quad \dots(1)$$

$$8I + 36E = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} + \begin{bmatrix} 0 & 36 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 8 & 36 \\ 0 & 8 \end{bmatrix} \quad \dots(2)$$

From (1) and (2) we have,

$$(2I + 3E)^3 = 8I + 36E$$

Example 8: If $A = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$ and

$$(A+B)^2 = A^2 + B^2 + 2AB, \text{ find } a \text{ and } b.$$

Solution: Given $(A+B)^2 = A^2 + B^2 + 2AB$

$$\Rightarrow (A+B)(A+B) = (A^2 + B^2) + 2AB$$

$$\Rightarrow AA + AB + BA + BB = A^2 + B^2 + 2AB$$

$$\Rightarrow A^2 + AB + BA + B^2 = A^2 + B^2 + 2AB$$

$$\Rightarrow BA = AB.$$

$$\Rightarrow \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a+2 & -a+1 \\ b-2 & -b-1 \end{bmatrix} = \begin{bmatrix} a-b & 1+1 \\ 2a+b & 2-1 \end{bmatrix}$$

$$\Rightarrow a+2 = a-b, \quad -a+1 = 2 \Rightarrow a = -1$$

$$\text{and } b-2 = 2a+b, \quad -b-1 = 1 \Rightarrow b = -2$$

$$\therefore a = -1, \quad b = -2$$

Example 9: If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ show that $A^k = \begin{bmatrix} 1+2k & -4k \\ k & 1-2k \end{bmatrix}$ where k is any positive integer.

[B.C.A. (Meerut) 2005, 2006, 2007]

Solution:

$$\begin{aligned} A^2 &= A \cdot A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 9-4 & -12+4 \\ 3-1 & -4+1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 1+2(2) & -4(2) \\ (2) & 1-2(2) \end{bmatrix} \end{aligned}$$

Thus, theorem is true for $k = 2$

Now assume that the theorem is true for $k = n$

$$\begin{aligned} A^n &= \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix} \\ A^{n+1} &= \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 3+6n-4n & -4-8n+4n \\ 3n+1-2n & -4n-1+2n \end{bmatrix} \\ &= \begin{bmatrix} 3+2n & -4-4n \\ n+1 & -2n-1 \end{bmatrix} = \begin{bmatrix} 1+2(n+1) & -4(n+1) \\ (n+1) & 1-2(n+1) \end{bmatrix} \end{aligned}$$

This theorem is true for $k = n+1$ if it is true for $k = n$.

But it is true for $k = 2$. Hence, it is true for all values of k .

Example 10: If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ prove that $A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ where n is a positive integer.

[B.C.A. (Delhi) 2002, 2007]

Solution:
$$A^2 = A \cdot A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Thus, the theorem is true for $n = 2$

Now assume that the theorem is true for n

$$A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$$

$$A^{n+1} = A^n A = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & n+1 \\ 0 & 1 \end{bmatrix}$$

Thus, theorem is also true for $(n+1)$ if it is true for n . But it is true for $n=2$.

Hence, by law of induction it is true for all positive integral values of n .

Example 11: If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ prove that

$$A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$$

[B.C.A. (Meerut) 2002, 2008]

Solution:
$$A^2 = A \cdot A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \alpha - \sin^2 \alpha & \cos \alpha \sin \alpha + \sin \alpha \cos \alpha \\ -\sin \alpha \cos \alpha - \cos \alpha \sin \alpha & -\sin^2 \alpha + \cos^2 \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$$

Thus, the theorem is true for $n = 2$

Now assume that the theorem is true for n

So
$$A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$$

$$A^{n+1} = A^n \cdot A = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos n\alpha \cos \alpha - \sin n\alpha \sin \alpha & \cos n\alpha \sin \alpha + \sin n\alpha \cos \alpha \\ -\sin n\alpha \cos \alpha - \cos n\alpha \sin \alpha & -\sin n\alpha \sin \alpha + \cos n\alpha \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos(n\alpha + \alpha) & \sin(n\alpha + \alpha) \\ -\sin(n\alpha + \alpha) & \cos(n\alpha + \alpha) \end{bmatrix}$$

$$= \begin{bmatrix} \cos(n+1)\alpha & \sin(n+1)\alpha \\ -\sin(n+1)\alpha & \cos(n+1)\alpha \end{bmatrix}$$

Thus, theorem is also true for $n+1$ if it is true for n .

Hence, it is true for all value of n .

Example 12: If $A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$ then prove that

(i) $A_{(\alpha)} A_{(-\alpha)} = I$ (ii) $(A_\alpha A_\beta) = A_{(\alpha + \beta)}$ [B.C.A (Rohilkhand) 2002, 2005]

Solution:

$$A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix},$$

$$A_{(-\alpha)} = \begin{bmatrix} \cos(-\alpha) & -\sin(-\alpha) \\ \sin(-\alpha) & \cos(-\alpha) \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$\begin{aligned} \text{(i)} \quad A_{(\alpha)} A_{(-\alpha)} &= \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & \sin \alpha \cos \alpha - \cos \alpha \sin \alpha \\ \sin \alpha \cos \alpha - \sin \alpha \cos \alpha & \sin^2 \alpha + \cos^2 \alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad A_\alpha \cdot A_\beta &= \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix} \\ &= \begin{bmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & -\cos \alpha \sin \beta - \sin \alpha \cos \beta \\ \sin \alpha \cos \beta + \cos \alpha \sin \beta & -\sin \alpha \sin \beta + \cos \alpha \cos \beta \end{bmatrix} \\ &= \begin{bmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix} = A_{(\alpha + \beta)} \end{aligned}$$

Example 13: If $A = \begin{bmatrix} 1 & 1 & -1 \\ -2 & 3 & -4 \\ 3 & -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$,

prove that $AB = O$ but $BA \neq O$

Solution:

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 1 & -1 \\ -2 & 3 & -4 \\ 3 & -2 & 3 \end{bmatrix} \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix} \\ &= \begin{bmatrix} -1+6-5 & -2+12-10 & -1+6-5 \\ 2+18-20 & 4+36-40 & 2+18-20 \\ -3-12+15 & -6-24+30 & -3-12+15 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \\ BA &= \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ -2 & 3 & -4 \\ 3 & -2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} -1+4-3 & -1-6+2 & 1+8-3 \\ 6-24+18 & 6+36-12 & -6-48+18 \\ 5-20+15 & 5+30-10 & -5-40+15 \end{bmatrix} = \begin{bmatrix} 0 & -5 & 6 \\ 0 & 30 & -36 \\ 0 & 25 & -30 \end{bmatrix} = O \end{aligned}$$

Example 14: Find three matrices A, B and C such that $B \neq C$ but $AB = AC$.

[B.C.A. (Rohilkhand) 2005]

Solution: Let $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix}$,

Thus, $AB = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}$

And $AC = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}$

Hence, $AB = AC$ but $B \neq C$.

Example 15: Show that $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ satisfies the equation $A^2 - 4A - 5I = O$, where I

is the identity matrix and O denotes the zero matrix. Hence or otherwise find the inverse of A .

[B.C.A. (Meerut) 2001]

Solution: $A^2 = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$

$$-4A = -4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} -4 & -8 & -8 \\ -8 & -4 & -8 \\ -8 & -8 & -4 \end{bmatrix}$$

$$-5I = -5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -5 \end{bmatrix}$$

$$A^2 - 4A - 5I = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} + \begin{bmatrix} -4 & -8 & -8 \\ -8 & -4 & -8 \\ -8 & -8 & -4 \end{bmatrix} + \begin{bmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 9-4-5 & 8-8+0 & 8-8+0 \\ 8-8+0 & 9-4-5 & 8-8+0 \\ 8-8+0 & 8-8+0 & 9-4-5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

$\therefore A^2 - 4A - 5I = O$ multiplying A^{-1} on both sides

$$A - 4I - 5A^{-1} = O \text{ or } 5A^{-1} = A - 4I$$

or
$$5A^{-1} = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-4 & 2-0 & 2-0 \\ 2-0 & 1-4 & 2-0 \\ 2-0 & 2-0 & 1-4 \end{bmatrix} = \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

Hence,
$$A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

Example 16: If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, prove that $A^3 = A^{-1}$ [B.C.A. (Rohilkhand) 2003]

Solution:
$$A^2 = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix}$$

$$A^4 = A^2 \cdot A^2 = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix} \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$\therefore A^4 = I$ multiplying A^{-1} on both sides

$$A^3 = A^{-1}$$

Example 17: Find the product of the following matrices.

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

[B.C.A. (Kanpur) 2004]

Solution:
$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} = \begin{bmatrix} ax+hy+gz & hx+by+fz & gx+fy+cz \end{bmatrix}$$

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} ax+hy+gz & hx+by+fz & gx+fy+cz \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= ax^2 + hxy + gxz + hxy + by^2 + fyz + gxz + fyz + cz^2$$

$$= ax^2 + by^2 + cz^2 + 2hxy + 2fyz + 2gzx$$

Example 18: Show that $A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$ satisfies the equation $x^2 - 3x - 7 = O$. Also find

A^{-1} .

[B.C.A. (Meerut) 2009]

Solution: We have $x^2 - 3x - 7 = O$, put $x = A$, then

$$A^2 - 3A - 7I = O \quad \dots(1)$$

$$\text{Now, } A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}, A^2 = A.A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 25-3 & 15-6 \\ -5+2 & -3+4 \end{bmatrix} = \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{Then, } A^2 - 3A - 7I &= \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix} - 3 \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} - 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix} + \begin{bmatrix} -15 & -9 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \end{aligned}$$

Now to find A^{-1} , we have $A^2 - 3A - 7I = O$

$$\Rightarrow -7I = -A^2 + 3A$$

$$\Rightarrow -7A^{-1} = -A + 3I$$

$$\Rightarrow -A^{-1} = \frac{1}{7} \{-A + 3I\}$$

$$\Rightarrow -A^{-1} = \frac{1}{7} \left\{ \begin{bmatrix} -5 & -3 \\ 1 & 2 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

$$= \frac{1}{7} \left\{ \begin{bmatrix} -2 & -3 \\ 1 & 5 \end{bmatrix} \right\}$$

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

Example 19: If $A = \begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix}$ and $B = [3 \ 2 \ -2]$ verify that $(A \ B)^T = B^T A^T$ where A^T

denotes the transpose of A .

[B.C.A. (Meerut) 2009]

Solution:

$$AB = \begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix} [3 \ 2 \ -2] = [3 \ -10 \ -14]$$

$$(AB)^T = \begin{bmatrix} 3 \\ -10 \\ -14 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} 3 \\ 2 \\ -2 \end{bmatrix} [1 \ -5 \ 7] = [3 \ -10 \ -14]$$

$$\therefore (AB)^T = B^T A^T$$



Comprehensive Exercise 2.3

1. If $A = \begin{bmatrix} 1 & 2 \\ -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$, $C = \begin{bmatrix} -3 & 1 \\ 2 & 0 \end{bmatrix}$, verify

$$(AB)C = A(BC) \text{ and } A(B+C) = AB + AC.$$

2. If $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, choose α and β so that $(\alpha I + \beta A)^2 = A$.

3. If $A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 3 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -2 & 1 \\ 0 & -1 & 4 \\ 5 & 2 & -1 \end{bmatrix}$ find AB .

4. Evaluate $A^2 - 3A + 9I$ if $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$

5. If $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 1 \\ 0 & 0 & 2 \\ 4 & -3 & 2 \end{bmatrix}$ verify.

$$(A+B)^2 = A^2 + AB + BA + B^2$$

6. Prove that the product of two matrices.

$$\begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{bmatrix} \text{ and } \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$$

is zero when θ and ϕ differ by odd multiple of $\pi/2$.



Answers

2.	$\alpha = \beta = \pm \frac{1}{\sqrt{2}}$	3.	$AB = \begin{bmatrix} 28 & 6 & 4 \\ -23 & -9 & 15 \end{bmatrix}$	4.	$\begin{bmatrix} -6 & 1 & 2 \\ 5 & 4 & 4 \\ 2 & 8 & -3 \end{bmatrix}$
----	---	----	--	----	---

2.4 Inverse of a Matrix

2.4.1 Adjoint of a Square Matrix

If $A = [a_{ij}]_{n \times n}$ be any square matrix, then the matrix $B = (b_{ij})_{n \times n}$ such that $b_{ij} = A_{ji}$ where A_{ji} is the cofactor of a_{ji} in the determinant $|A|$ is called the **adjoint of A** and is denoted by $\text{Adj}(A)$.

Thus, the adjoint of a matrix A is the transpose of the matrix formed by cofactor of A , i.e., if

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

$$\text{Matrix formed by cofactors} = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \dots & \dots & \dots & \dots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{bmatrix}$$

Then $\text{Adj}(A) = \text{Transpose of the matrix formed by cofactors}$

$$= \begin{bmatrix} A_{11} & A_{21} & \dots & A_{n1} \\ A_{12} & A_{22} & \dots & A_{n2} \\ \dots & \dots & \dots & \dots \\ A_{1n} & A_{2n} & \dots & A_{nn} \end{bmatrix}$$

Theorems

Theorem 1: If $A = [a_{ij}]$ be $n \times n$ matrix, then $A.(\text{Adj } A) = (\text{Adj } A).A = |A|I$ where I is $n \times n$ identity matrix.

Proof:

$$A = (a_{ij})_{n \times n}$$

Let

$$\text{Adj}(A) = [b_{ij}]_{n \times n}$$

Then $b_{ij} = c_{ji} = \text{cofactor of } a_{ji}$ in $|A|$ since the matrices A and $\text{Adj } A$ are both $n \times n$ therefore both the products $A(\text{Adj } A)$ and $(\text{Adj } A).A$ exist

$$\text{The } (i, j) \text{ the element of } A.(\text{Adj } A) = \sum_{k=1}^n a_{ik} b_{kj}$$

$$= \sum_{k=1}^n a_{ik} c_{jk} = \begin{cases} |A| & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

All the elements of $A.(\text{Adj } A)$ along the principal diagonal ($i = j$) are equal to $|A|$ and the non-diagonal ($i \neq j$) elements are all equal to zero.

$$\begin{aligned} \text{Therefore, } A.(\text{Adj } A) &= \begin{bmatrix} |A| & 0 & 0 & \dots & 0 \\ 0 & |A| & 0 & \dots & 0 \\ 0 & 0 & |A| & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & |A| \end{bmatrix} \\ &= |A| \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix} = |A|I_n \end{aligned}$$

Similarly, the (i, j) the element of $(\text{Adj } A) \cdot A$

$$= \sum_{k=1}^n b_{ik} a_{kj} = \sum_{k=1}^n c_{ki} a_{kj} = \begin{cases} |A| & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Therefore, $(\text{Adj } A) \cdot A = A \cdot (\text{Adj } A) = |A| I_n$

Theorem 2: For a Square matrix A of order n , prove that

$$|(\text{Adj } A)| = (\det A)^{n-1}.$$

Solution: We know that

$$|A \cdot \text{Adj } A| = \begin{vmatrix} |A| & 0 & 0 & \dots & 0 \\ 0 & |A| & 0 & \dots & 0 \\ 0 & 0 & |A| & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & |A| \end{vmatrix} = |A|^n$$

$$|A| |\text{Adj } A| = |A|^n \quad (|AB| = |A| |B|)$$

Dividing both sides by $|A|$, we have

$$|\text{Adj } A| = |A|^{n-1}$$

Theorem 3: Show that, for any square matrix of order $n \times n$

$$\text{Adj}(A^T) = (\text{Adj } A)^T$$

Solution: Let A be a square matrix of order n , then both $\text{Adj}(A^T)$ and $(\text{Adj } A)^T$ are also square matrices of order n .

$$\begin{aligned} (i, j)\text{th element of } (\text{Adj } A)^T &= (j, i) \text{th element of } \text{Adj } A \\ &= \text{cofactor of } (i, j) \text{th element of } |A| \\ &= \text{cofactor of } (j, i) \text{th element of } |A^T| \\ &= (i, j) \text{th element of } \text{Adj}(A^T) \end{aligned}$$

Hence, $(\text{Adj } A)^T = \text{Adj}(A^T)$

2.4.2 Invertible Matrix

Let A be any n rowed square matrix. Then a matrix B if it exists such that

$$AB = BA = I$$

is called **inverse of A** . The inverse of A is denoted by A^{-1}

Note: The inverse of matrix A is obtain by

$$A^{-1} = \frac{\text{Adj}(A)}{|A|}$$

Theorems

Theorem 1 : Show that the necessary and sufficient condition for square matrix to possess an inverse is that it must be non singular.

[B.C.A (Meerut) 2004]

Solution: The necessary condition. Let A be an n rowed square matrix and let B be its inverse.

$$AB = I$$

$$\Rightarrow |AB| = |I| = 1$$

$$|A| |B| = 1$$

$$[|AB| = |A| |B|]$$

$$\therefore |A| \neq 0$$

The sufficient condition

$$\text{If } |A| \neq 0$$

$$\text{Let, } B = \frac{1}{|A|} (\text{Adj } A)$$

$$AB = A \left[\frac{1}{|A|} (\text{Adj } A) \right]$$

$$= \frac{1}{|A|} (A \text{ Adj } A)$$

$$= \frac{1}{|A|} |A| I = I$$

$$\text{Similarly, } BA = \left[\frac{1}{|A|} \text{Adj } A \right] A = \frac{1}{|A|} (\text{Adj } A) \cdot A$$

$$= \frac{1}{|A|} |A| I = I$$

$$\text{Thus, } AB = BA = I$$

Hence, the matrix A is invertible and B is the inverse of A .

Theorem 2: Prove that the inverse of every square matrix, if it exists, is unique.

[B.C.A. (Delhi) 2003]

Solution: Let B and C be two inverse matrices of a square matrix A .

Since B is an inverse of A , therefore

$$AB = BA = I \quad \dots(1)$$

Again since C is an inverse of A , therefore

$$AC = CA = I \quad \dots(2)$$

From (1), we find that

$$C(AB) = CI = C \quad \dots(3)$$

Also from (2), we find that

$$(CA) B = IB = B \quad \dots(4)$$

or

$$C (AB) = B$$

From (3) and (4), it follows that

$$B = C$$

Theorem 3: If A, B are two non singular matrices of order n , then prove that AB is also a non-singular matrix and

$$(AB)^{-1} = B^{-1} A^{-1}$$

[B.C.A. (Meerut) 2004]

Solution: Since A and B are n rowed invertible matrices, therefore A^{-1} and B^{-1} are both n rowed square matrices, and consequently $B^{-1} A^{-1}$ is also an n rowed square matrix. Let us denote $B^{-1} A^{-1}$ by C . To show that C is the inverse of the n rowed square matrix AB .

We have to verify that

$$C (AB) = (AB) C = I$$

Now,

$$\begin{aligned} C (AB) &= (B^{-1} A^{-1}) (AB) \\ &= B^{-1} (A^{-1} A) B = B^{-1} I B = B^{-1} B = I \\ (AB) C &= (AB) (B^{-1} A^{-1}) \\ &= A (BB^{-1}) A^{-1} = A I A^{-1} = AA^{-1} = I \end{aligned}$$

Thus, we find that $C (AB) = (AB) C = I$

Hence, it follows that AB is invertible and that C is inverse of AB .

i.e. $(AB)^{-1} = B^{-1} A^{-1}$

Theorem 4: If A be an n rowed non singular matrix, prove that $(A^T)^{-1} = (A^{-1})^T$ where A^T is the transpose of A .

Solution: We know that $AA^{-1} = A^{-1}A = I$

Taking transpose of both sides

$$(AA^{-1})^T = (A^{-1}A)^T = (I)^T$$

By reversal law for transposes, we have

$$(A^{-1})^T A^T = A^T (A^{-1})^T = I \quad \dots(1)$$

Equation (1) shows that $(A^{-1})^T$ is the inverse of A^T

Hence, $(A^T)^{-1} = (A^{-1})^T \quad (|A'| = |A| \neq 0)$

The inverse of the transpose of a matrix is the transpose of the inverse.

Illustrative Examples

Example 1: Find the adjoint of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$ [B.C.A. (Meerut) 2009]

Solution: The minors of the elements of the first row of the determinant

$$\begin{aligned} |A| \text{ are } & \begin{vmatrix} 2 & -3 \\ -1 & 3 \end{vmatrix}, \begin{vmatrix} 1 & -3 \\ 2 & 3 \end{vmatrix}, \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} \\ & = (2 \times 3) - (-1 \times -3), (1 \times 3) - (2 \times -3), (1 \times -1) - (2 \times 2) \\ & = 3, 9, -5 \end{aligned}$$

The minors of the elements of the second row of the determinant $|A|$ are

$$\begin{aligned} & \begin{vmatrix} 1 & 1 \\ -1 & 3 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} \\ & = (1 \times 3) - (-1 \times 1), (1 \times 3) - (2 \times 1), (1 \times -1) - (2 \times 1) \\ & = 4, 1, -3 \end{aligned}$$

The minors of the elements of the third row of the determinant $|A|$ are

$$\begin{aligned} & \begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 1 & -3 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} \\ & = (1 \times -3) - (1 \times 2), (1 \times -3) - (1 \times 1), (1 \times 2) - (1 \times 1) \\ & = -5, -4, 1 \end{aligned}$$

The matrix of the cofactors of $|A|$

$$= \begin{bmatrix} 3 & -9 & -5 \\ -4 & 1 & 3 \\ -5 & 4 & 1 \end{bmatrix} \quad [\text{cofactor} = (-1)^{\text{row} + \text{column}} \text{ minor}]$$

On transposing the matrix of cofactors we get adjoint A .

$$= \begin{bmatrix} 3 & -4 & -5 \\ -9 & 1 & 4 \\ -5 & 3 & 1 \end{bmatrix}$$

Example 2: Find the inverse of the matrix

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & 2 \end{bmatrix}$$

Solution:

$$|A| = \begin{vmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -1 \\ 2 & 3 & 1 \\ 9 & 4 & 2 \end{vmatrix} \quad [C_1 \rightarrow C_1 + 2C_3, C_2 \rightarrow C_2 + C_3]$$

$$\begin{aligned}
 &= -1 \begin{vmatrix} 2 & 3 \\ 9 & 4 \end{vmatrix} \text{expanding the determinant along the first row.} \\
 &= -(8 - 27) = 19
 \end{aligned}$$

Since, $|A| \neq 0$, therefore A^{-1} exists.

Now the cofactors of the elements of the first row of $|A|$ are

$$\begin{vmatrix} 2 & 1 \\ 2 & 2 \end{vmatrix}, - \begin{vmatrix} 0 & 1 \\ 5 & 2 \end{vmatrix}, \begin{vmatrix} 0 & 2 \\ 5 & 2 \end{vmatrix} \text{ i.e. } 2, 5, -10 \text{ respectively.}$$

The cofactors of the elements of the second row of $|A|$ are

$$- \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix}, \begin{vmatrix} 2 & -1 \\ 5 & 2 \end{vmatrix}, - \begin{vmatrix} 2 & 1 \\ 5 & 2 \end{vmatrix} \text{ i.e. } -4, 9, 1 \text{ respectively.}$$

The cofactors of the elements of the third row of $|A|$ are

$$\begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix}, - \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix}, \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} \text{ i.e. } 3, -2, 4 \text{ respectively.}$$

$$\text{Matrix of cofactors} = \begin{bmatrix} 2 & 5 & -10 \\ -4 & 9 & 1 \\ 3 & -2 & 4 \end{bmatrix}$$

Therefore the $\text{Adj}(A)$ = the transpose of the matrix of cofactors

$$\therefore \text{Adj}(A) = \begin{bmatrix} 2 & -4 & 3 \\ 5 & 9 & -2 \\ -10 & 1 & 4 \end{bmatrix}$$

$$\text{Now, } A^{-1} = \frac{1}{|A|} \text{Adj.}(A) \text{ and here } |A| = 19$$

$$\therefore A^{-1} = \frac{1}{19} \begin{bmatrix} 2 & -4 & 3 \\ 5 & 9 & -2 \\ -10 & 1 & 4 \end{bmatrix} = \begin{bmatrix} \frac{2}{19} & -\frac{4}{19} & \frac{3}{19} \\ \frac{5}{19} & \frac{9}{19} & -\frac{2}{19} \\ -\frac{10}{19} & \frac{1}{19} & \frac{4}{19} \end{bmatrix}$$

Example 3: Find the adjoint and inverse of the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 3 & -1 \end{bmatrix}$

Solution:

$$\begin{aligned}
 |A| &= \begin{vmatrix} 2 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 3 & -1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 2 & -1 \end{vmatrix} [C_1 \rightarrow C_1 + 2C_3, C_2 \rightarrow C_2 + C_3] \\
 &= -1 \begin{vmatrix} 0 & 1 \\ -1 & 2 \end{vmatrix} \text{expanding the determinant along the first row}
 \end{aligned}$$

$$= -1$$

Since $|A| \neq 0$, therefore A^{-1} exists.

Now the cofactors of the elements of the first row of $|A|$ are

$$\begin{vmatrix} 1 & 0 \\ 3 & -1 \end{vmatrix}, -\begin{vmatrix} 0 & 0 \\ 1 & -1 \end{vmatrix}, \begin{vmatrix} 0 & 1 \\ 1 & 3 \end{vmatrix} \text{ i.e. } -1, 0, -1 \text{ respectively.}$$

The cofactors of the elements of the second row of $|A|$ are

$$-\begin{vmatrix} 1 & -1 \\ 3 & -1 \end{vmatrix}, \begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix}, -\begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} \text{ i.e. } -2, -1, -5 \text{ respectively.}$$

The cofactors of the elements of the third row of $|A|$ are

$$\begin{vmatrix} 1 & -1 \\ 1 & 0 \end{vmatrix}, -\begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix}, \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} \text{ i.e. } 1, 0, 2 \text{ respectively.}$$

$$\text{Matrix of cofactors} = \begin{bmatrix} -1 & 0 & -1 \\ -2 & -1 & -5 \\ 1 & 0 & 2 \end{bmatrix}$$

Therefore, $\text{Adj}(A)$ = the transpose of the matrix of cofactors

$$\text{Adj}(A) = \begin{bmatrix} -1 & -2 & 1 \\ 0 & -1 & 0 \\ -1 & -5 & 2 \end{bmatrix}$$

$$\text{Now } A^{-1} = \frac{1}{|A|} \text{Adj}(A) \text{ and here } |A| = -1$$

$$\therefore A^{-1} = -1 \begin{bmatrix} -1 & -2 & 1 \\ 0 & -1 & 0 \\ -1 & -5 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 1 & 5 & -2 \end{bmatrix}$$

Example 4: Obtain the reciprocal matrix of the matrix

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & -6 & -7 \end{bmatrix}$$

Solution:

$$|A| = \begin{vmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & -6 & -7 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 3 & 4 & 8 \\ 0 & -6 & -7 \end{vmatrix} [C_3 \rightarrow C_3 + C_1]$$

$$= 1 \begin{vmatrix} 4 & 8 \\ -6 & -7 \end{vmatrix} \text{ expanding the determinant along the first row.}$$

$$= 20$$

Since $|A| \neq 0$, therefore A^{-1} exists.

Now the cofactors of the elements of the first row of $|A|$ are

$$\begin{vmatrix} 4 & 5 \\ -6 & -7 \end{vmatrix}, -\begin{vmatrix} 3 & 5 \\ 0 & -7 \end{vmatrix}, \begin{vmatrix} 3 & 4 \\ 0 & -6 \end{vmatrix} \text{ i.e. } 2, 21-18 \text{ respectively.}$$

The cofactors of the elements of the second row of $|A|$ are

$$-\begin{vmatrix} 0 & -1 \\ -6 & -7 \end{vmatrix}, \begin{vmatrix} 1 & -1 \\ 0 & -7 \end{vmatrix}, -\begin{vmatrix} 1 & 0 \\ 0 & -6 \end{vmatrix} \text{ i.e. } 6, -7, 6 \text{ respectively.}$$

The cofactors of the elements of the third row of $|A|$ are

$$\begin{vmatrix} 0 & -1 \\ 4 & 5 \end{vmatrix}, -\begin{vmatrix} 1 & -1 \\ 3 & 5 \end{vmatrix}, \begin{vmatrix} 1 & 0 \\ 3 & 4 \end{vmatrix} \text{ i.e., } 4, -8, 4 \text{ respectively.}$$

The matrix of cofactors =
$$\begin{bmatrix} 2 & 21 & -18 \\ 6 & -7 & 6 \\ 4 & -8 & 4 \end{bmatrix}$$

Therefore, $\text{Adj}(A)$ = the transpose of the matrix of the cofactors

$$\text{Adj}(A) = \begin{bmatrix} 2 & 6 & 4 \\ 21 & -7 & -8 \\ -18 & 6 & 4 \end{bmatrix}$$

Now
$$A^{-1} = \frac{1}{|A|} \text{Adj}(A) \text{ and here } |A| = 20$$

$$\therefore A^{-1} = \frac{1}{20} \begin{bmatrix} 2 & 6 & 4 \\ 21 & -7 & -8 \\ -18 & 6 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{10} & \frac{3}{10} & \frac{1}{5} \\ \frac{21}{20} & -\frac{7}{20} & -\frac{2}{5} \\ -\frac{9}{10} & \frac{3}{10} & \frac{1}{5} \end{bmatrix}$$

Example 5: Find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 6 & 7 & 9 \end{bmatrix}$$

[B.C.A. (Meerut) 2007]

Solution:

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 6 & 7 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 4 & -3 & -6 \\ 6 & -5 & -9 \end{vmatrix} & \begin{matrix} [C_2 \rightarrow C_2 - 2C_1 \\ C_3 \rightarrow C_3 - 3C_1] \end{matrix} \\ &= 1 \begin{vmatrix} -3 & -6 \\ -5 & -9 \end{vmatrix} \text{ expanding the determinant along the first row} \\ &= -3 \end{aligned}$$

Since $|A| \neq 0$, therefore A^{-1} exists.

Now the cofactors of the elements of the first row of $|A|$ are

$$\begin{vmatrix} 5 & 6 \\ 7 & 9 \end{vmatrix}, -\begin{vmatrix} 4 & 6 \\ 6 & 9 \end{vmatrix}, \begin{vmatrix} 4 & 5 \\ 6 & 7 \end{vmatrix} \text{ i.e. } 3, 0, -2 \text{ respectively.}$$

The cofactors of the elements of the second row of $|A|$ are

$$-\begin{vmatrix} 2 & 3 \\ 7 & 9 \end{vmatrix}, -\begin{vmatrix} 1 & 3 \\ 6 & 9 \end{vmatrix}, -\begin{vmatrix} 1 & 2 \\ 6 & 7 \end{vmatrix} \text{ i.e. } 3, -9, 5 \text{ respectively.}$$

The cofactors of the elements of the third row of $|A|$ are

$$\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix}, -\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix}, \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} \text{ i.e. } -3, 6, -3 \text{ respectively.}$$

The matrix of the cofactors =
$$\begin{bmatrix} 3 & 0 & -2 \\ 3 & -9 & 5 \\ -3 & 6 & -3 \end{bmatrix}$$

Therefore, the $\text{Adj}(A)$ = the transpose of the matrix of the cofactors

$$\text{Adj}(A) = \begin{bmatrix} 3 & 3 & -3 \\ 0 & -9 & 6 \\ -2 & 5 & -3 \end{bmatrix}$$

Now
$$A^{-1} = \frac{1}{|A|} \text{Adj}(A) \text{ here } |A| = -3$$

$$\therefore A^{-1} = -\frac{1}{3} \begin{bmatrix} 3 & 3 & -3 \\ 0 & -9 & 6 \\ -2 & 5 & -3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 3 & -2 \\ \frac{2}{3} & -\frac{5}{3} & 1 \end{bmatrix}$$

Example 6: Find the inverse of the following matrix:

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

[B.C.A. (Meerut) 2008]

Solution:

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} & \begin{matrix} [C_2 \rightarrow C_2 - 3C_1 \\ C_3 \rightarrow C_3 - 3C_1] \end{matrix} \\ &= 1 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \end{aligned}$$

Since $|A| \neq 0$, therefore A^{-1} exists.

Now the cofactors of the elements of the first row of $|A|$ are

$$\begin{vmatrix} 4 & 3 \\ 3 & 4 \end{vmatrix}, -\begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix}, \begin{vmatrix} 1 & 4 \\ 1 & 3 \end{vmatrix} \text{ i.e. } 7, -1, -1 \text{ respectively.}$$

The cofactors of the elements of the second row of $|A|$ are

$$-\begin{vmatrix} 3 & 3 \\ 3 & 4 \end{vmatrix}, \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix}, -\begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix} \text{ i.e. } -3, 1, 0 \text{ respectively.}$$

The cofactors of the elements of the third row of $|A|$ are

$$\begin{vmatrix} 3 & 3 \\ 4 & 3 \end{vmatrix}, -\begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix}, \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} \text{ i.e. } -3, 0, 1 \text{ respectively.}$$

$$\text{Matrix of cofactors} = \begin{bmatrix} 7 & -1 & -1 \\ -3 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

Therefore, the $\text{Adj}(A)$ = the transpose of the matrix of cofactors

$$\text{Adj}(A) = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Now $A^{-1} = \frac{1}{|A|} \text{Adj}(A) \text{ and } |A| = 1$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Example 7: Find the inverse of the matrix

$$A = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 6 & 2 \\ -3 & 2 & -1 \end{bmatrix}$$

Solution:

$$\begin{aligned} |A| &= \begin{vmatrix} 2 & 1 & 4 \\ 3 & 6 & 2 \\ -3 & 2 & -1 \end{vmatrix} = \begin{vmatrix} 0 & 1 & 0 \\ -9 & 6 & -22 \\ -7 & 2 & -9 \end{vmatrix} \begin{matrix} [C_1 \rightarrow C_1 - 2C_2 \\ C_3 \rightarrow C_3 - 4C_2] \end{matrix} \\ &= -1 \begin{vmatrix} -9 & -22 \\ -7 & -9 \end{vmatrix} = -(81 - 154) = 73 \end{aligned}$$

Since $|A| \neq 0$, therefore A^{-1} exists.

Now the cofactors of the elements of the first row of $|A|$ are

$$\begin{vmatrix} 6 & 2 \\ 2 & -1 \end{vmatrix}, -\begin{vmatrix} 3 & 2 \\ -3 & -1 \end{vmatrix}, \begin{vmatrix} 3 & 6 \\ -3 & 2 \end{vmatrix} \text{ i.e. } -10, -3, 24 \text{ respectively.}$$

Now the cofactors of the elements of the second row of $|A|$ are

$$-\begin{vmatrix} 1 & 4 \\ 2 & -1 \end{vmatrix}, \begin{vmatrix} 2 & 4 \\ -3 & -1 \end{vmatrix}, -\begin{vmatrix} 2 & 1 \\ -3 & 2 \end{vmatrix} \text{ i.e. } 9, 10, -7 \text{ respectively.}$$

Now the cofactors of the elements of the third row of $|A|$ are

$$\begin{vmatrix} 1 & 4 \\ 6 & 2 \end{vmatrix}, -\begin{vmatrix} 2 & 4 \\ 3 & 2 \end{vmatrix}, \begin{vmatrix} 2 & 1 \\ 3 & 6 \end{vmatrix} \text{ i.e. } -22, 8, 9 \text{ respectively.}$$

$$\text{The matrix of the cofactors} = \begin{bmatrix} -10 & -3 & 24 \\ 9 & 10 & -7 \\ -22 & 8 & 9 \end{bmatrix}$$

Therefore, the $\text{Adj}(A)$ = the transpose of the matrix of cofactors

$$= \begin{bmatrix} -10 & 9 & -22 \\ -3 & 10 & 8 \\ 24 & -7 & 9 \end{bmatrix}$$

But

$$A^{-1} = \frac{1}{|A|} \text{Adj } A \text{ and } |A| = 73.$$

$$A^{-1} = \frac{1}{73} \begin{bmatrix} -10 & 9 & -22 \\ -3 & 10 & 8 \\ 24 & -7 & 9 \end{bmatrix}$$

Example 8: Find the inverse of

$$A = \begin{bmatrix} a + ib & c + id \\ -c + id & a - ib \end{bmatrix}$$

where $a^2 + b^2 + c^2 + d^2 = 1$

Solution:

$$|A| = \begin{vmatrix} a + ib & c + id \\ -c + id & a - ib \end{vmatrix} = (a + ib)(a - ib) - (-c + id)(c + id) = a^2 + b^2 + (c^2 + d^2) = 1$$

Since $|A| \neq 0$, therefore A^{-1} exists.

Now the cofactor of the elements of the first row of $|A|$ are $(a - ib)$, $-(-c + id)$

The cofactors of the elements of the second row of $|A|$ are $-(c + id)$, $(a + ib)$

$$\text{The matrix of cofactors} = \begin{bmatrix} a - ib & c - id \\ -c - id & a + ib \end{bmatrix}$$

Therefore the $\text{Adj}(A)$ = the transpose of the matrix of cofactors

$$= \begin{bmatrix} a - ib & -c - id \\ c - id & a + ib \end{bmatrix}$$

Therefore $A^{-1} = \frac{1}{|A|} \text{Adj}(A)$ here $|A| = 1$

$$A^{-1} = \begin{bmatrix} a - ib & -c - id \\ c - id & a + ib \end{bmatrix}$$

Example 9: Find the inverse of the matrix

$$= \begin{bmatrix} 3 & -10 & -1 \\ -2 & 8 & 2 \\ 2 & -4 & -2 \end{bmatrix}$$

[B.C.A. (Meerut) 2004]

Solution:

$$\text{The given matrix is } A = \begin{bmatrix} 3 & -10 & -1 \\ -2 & 8 & 2 \\ 2 & -4 & -2 \end{bmatrix}$$

$$|A| = 3(-16 + 8) + 10(4 - 4) - 1(8 - 16) = -24 + 8 = -16$$

Now cofactor of A

$$C_{11} = 8(-2) - 2(-4) = -8$$

$$C_{12} = 0 \quad C_{13} = -8$$

$$C_{21} = -16 \quad C_{22} = -4, \quad C_{23} = -8$$

$$C_{31} = -12, \quad C_{32} = -4, \quad C_{33} = 4$$

$$\text{Matrix of cofactor of } A = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

$\text{Adj}(A) = \text{transpose of cofactors of } A$

$$= \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} = \begin{bmatrix} -8 & -16 & -12 \\ 0 & -4 & -4 \\ -8 & -8 & 4 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{\text{Adj}(A)}{|A|} = -\frac{1}{16} \begin{bmatrix} -8 & -16 & -12 \\ 0 & -4 & -4 \\ -8 & -8 & 4 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \frac{1}{2} & 1 & \frac{3}{4} \\ 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$$



Comprehensive Exercise 2.4

Find the inverse of matrix

$$1. \quad A = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

[B.C.A. (Meerut) 2005]

$$2. \quad A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$3. \quad A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

[B.C.A. (Meerut) 2009]

$$4. \quad A = \begin{bmatrix} 9 & 5 & 6 \\ 7 & -1 & 8 \\ 3 & 4 & 2 \end{bmatrix}$$

$$5. \quad A = \begin{bmatrix} 2 & 2 & -1 \\ -2 & 3 & 1 \\ 4 & 2 & -1 \end{bmatrix}$$

6. $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix}$

[B.C.A. (Delhi) 2007]



Answers

1.	$\frac{1}{5} \begin{bmatrix} 2 & 2 & -3 \\ -3 & 2 & 2 \\ 2 & -3 & 2 \end{bmatrix}$	2.	$\frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$
3.	$\begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$	4.	$-\frac{1}{70} \begin{bmatrix} -34 & 14 & 46 \\ 10 & 0 & -30 \\ 31 & -21 & -44 \end{bmatrix}$
5.	$\frac{1}{10} \begin{bmatrix} -5 & 0 & 5 \\ 2 & 2 & 0 \\ -16 & 4 & 2 \end{bmatrix}$	6.	Inverse of matrix does not exist. Since the given matrix is singular.

25 Solution of Linear Equations by the use of Inverse of a Matrix

Consider a system of linear equations.

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n = b_3$$

$$\dots\dots\dots$$

$$\dots\dots\dots$$

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n = b_n$$

We can express these equations as a single matrix equation:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \dots \\ b_n \end{bmatrix}$$

$$AX = B$$

Let $|A| \neq 0$ so that A^{-1} exists uniquely

Pre multiplying both sides of $AX = B$ by A^{-1} we get

$$A^{-1}(AX) = A^{-1}B$$

$$(A^{-1}A)X = A^{-1}B$$

$$IX = A^{-1}B$$

$$X = A^{-1}B$$

Hence, $X = A^{-1}B$ is the unique solution of $AX = B$, $|A| \neq 0$.

Remark: If $|A| = 0$, then the above method fails. The equation $AX = B$ can be solved with the help of elementary row transformations.

Illustrative Examples

Example 1: Solve $x - 3y + z = -1$, $2x + y - 4z = -1$, $6x - 7y + 8z = 7$

Solution: The given equations may be written as a single matrix equation.

$$\begin{bmatrix} 1 & -3 & 1 \\ 2 & 1 & -4 \\ 6 & -7 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 7 \end{bmatrix}$$

$$AX = B$$

$$|A| = \begin{vmatrix} 1 & -3 & 1 \\ 2 & 1 & -4 \\ 6 & -7 & 8 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 7 & -6 \\ 6 & 11 & 2 \end{vmatrix} = \begin{vmatrix} 7 & -6 \\ 11 & 2 \end{vmatrix} = 14 + 66 = 80$$

$$[C_2 = C_2 + 3C_1, C_3 = C_3 - C_1]$$

Since $|A| \neq 0$, therefore A^{-1} exists and the solution can be written

$$X = A^{-1}B \quad \dots(1)$$

The cofactors of the elements of the first row of $|A|$ are

$$\begin{vmatrix} 1 & -4 \\ -7 & 8 \end{vmatrix}, -\begin{vmatrix} 2 & -4 \\ 6 & 8 \end{vmatrix}, \begin{vmatrix} 2 & 1 \\ 6 & -7 \end{vmatrix} \text{ i.e. } -20, -40, -20 \text{ respectively.}$$

The cofactors the elements of the second row of $|A|$ are

$$-\begin{vmatrix} -3 & 1 \\ -7 & 8 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 6 & 8 \end{vmatrix}, -\begin{vmatrix} 1 & -3 \\ 6 & -7 \end{vmatrix} \text{ i.e. } 17, 2, -11 \text{ respectively.}$$

The cofactors of the elements of the third row of $|A|$ are

$$\begin{vmatrix} -3 & 1 \\ 1 & -4 \end{vmatrix}, -\begin{vmatrix} 1 & 1 \\ 2 & -4 \end{vmatrix}, \begin{vmatrix} 1 & -3 \\ 2 & 1 \end{vmatrix} \text{ i.e. } 11, 6, 7 \text{ respectively.}$$

$$\text{Matrix of cofactors} \begin{bmatrix} -20 & -40 & -20 \\ 17 & 2 & -11 \\ 11 & 6 & 7 \end{bmatrix}$$

$$\text{Adj } (A) = \begin{bmatrix} -20 & 17 & 11 \\ -40 & 2 & 6 \\ -20 & -11 & 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{Adj}(A) = \frac{1}{80} \begin{bmatrix} -20 & 17 & 11 \\ -40 & 2 & 6 \\ -20 & -11 & 7 \end{bmatrix}$$

Now, substituting the values of XA^{-1} and B in (1), we get

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{80} \begin{bmatrix} -20 & 17 & 11 \\ -40 & 2 & 6 \\ -20 & -11 & 7 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \\ 7 \end{bmatrix} \\ &= \frac{1}{80} \begin{bmatrix} 20 - 17 + 77 \\ 40 - 2 + 42 \\ 20 + 11 + 49 \end{bmatrix} = \frac{1}{80} \begin{bmatrix} 80 \\ 80 \\ 80 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

Hence, $x = 1, y = 1, z = 1$ is the required solution.

Example 2: Solve the simultaneous equations.

$$\begin{aligned} x + y + z &= 3 \\ x + 2y + 3z &= 4 \\ x + 4y + 9z &= 6 \end{aligned}$$

Solution: The given equations may be written as a single matrix equation:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}$$

$$AX = B$$

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 8 \end{vmatrix} [C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1] \\ &= 1 \begin{vmatrix} 1 & 2 \\ 3 & 8 \end{vmatrix} = 8 - 6 = 2 \end{aligned}$$

Since $|A| \neq 0$, therefore A^{-1} exists and the solution can be written as

$$X = A^{-1}B \quad \dots(1)$$

The cofactors of the elements of the first row of $|A|$ are

$$\begin{vmatrix} 2 & 3 \\ 4 & 9 \end{vmatrix}, -\begin{vmatrix} 1 & 3 \\ 1 & 9 \end{vmatrix}, \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} \text{ i.e. } 6, -6, 2 \text{ respectively.}$$

The cofactors of the elements of the second row of $|A|$ are

$$-\begin{vmatrix} 1 & 1 \\ 4 & 9 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 1 & 9 \end{vmatrix}, -\begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} \text{ i.e. } -5, 8, -3 \text{ respectively.}$$

The cofactors of the elements of the third row of

$$|A| \text{ are } \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}, -\begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} \text{ i.e. } 1, -2, 1 \text{ respectively.}$$

$$\text{Matrix of cofactors} = \begin{bmatrix} 6 & -6 & 2 \\ -5 & 8 & -3 \\ 1 & -2 & 1 \end{bmatrix}, \text{Adj}(A) = \begin{bmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{bmatrix}$$

$$\text{and } A^{-1} = \frac{1}{|A|} \text{Adj}(A) = \frac{1}{2} \begin{bmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{bmatrix}$$

Now substituting the values of X , A^{-1} and B in (1), we get

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 18 - 20 + 6 \\ -18 + 32 - 12 \\ 6 - 12 + 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow x = 2, y = 1, z = 0.$$

Example 3: Find the inverse of the matrix

$$A = \begin{vmatrix} 1 & 2 & 3 \\ 1 & -2 & 3 \\ 1 & 2 & -3 \end{vmatrix}$$

and hence solve the equation:

$$x + 2y + 3z = 11$$

$$x - 2y + 3z = 3$$

$$x + 2y - 3z = -1$$

$$\begin{aligned} \text{Solution: } |A| &= \begin{vmatrix} 1 & 2 & 3 \\ 1 & -2 & 3 \\ 1 & 2 & -3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & -4 & 0 \\ 1 & 0 & -6 \end{vmatrix} \quad [C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1] \\ &= 1 \begin{vmatrix} -4 & 0 \\ 0 & -6 \end{vmatrix} = 24 \neq 0 \end{aligned}$$

Hence, A^{-1} exists.

$$\text{Matrix of cofactors} = \begin{bmatrix} 0 & 6 & 4 \\ 12 & -6 & 0 \\ 12 & 0 & -4 \end{bmatrix} \text{ and } |A| = 12 + 12 = 24$$

$$\text{Adj}(A) = \begin{bmatrix} 0 & 12 & 12 \\ 6 & -6 & 0 \\ 4 & 0 & -4 \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{24} \begin{bmatrix} 0 & 12 & 12 \\ 6 & -6 & 0 \\ 4 & 0 & -4 \end{bmatrix}$$

The given system of equations can be written in the single matrix equations

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & 3 \\ 1 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 3 \\ -1 \end{bmatrix}$$

$$AX = B, \Rightarrow X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{24} \begin{bmatrix} 0 & 12 & 12 \\ 6 & -6 & 0 \\ 4 & 0 & -4 \end{bmatrix} \begin{bmatrix} 11 \\ 3 \\ -1 \end{bmatrix}$$

$$= \frac{1}{24} \begin{bmatrix} 0 + 36 - 12 \\ 66 - 18 - 0 \\ 44 + 0 + 4 \end{bmatrix} = \frac{1}{24} \begin{bmatrix} 24 \\ 48 \\ 48 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$x = 1, y = 2, z = 2$$



Comprehensive Exercise 2.5

Solve the following equations with the help of matrices.

1. $3x + y + 2z = 3, 2x - 3y - z = -3, x + 2y + z = 4$
2. $x + y + z = 6, x - y + 2z = 5, 3x + y + z = 8$
3. $9x + 4y + 3z = -1, 5x + y + 2z = 1, 7x + 3y + 4z = 1$



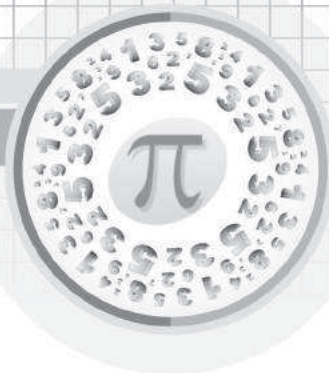
Answers

1.	$x = 1, y = 2, z = -1$	2.	$x = 1, y = 2, z = 3$	3.	$x = 0, y = -1, z = 1$
----	------------------------	----	-----------------------	----	------------------------





Unit-1



Chapter

3

Rank of Matrix

3.1 Rank of a Matrix

Minor: If from matrix $A_{m \times n}$, $m-p$ rows and $n-p$ columns are removed, the remaining square sub of p rows and p column is left. The determinant of a square sub matrix $p \times p$ is called a minor of A of order p .

For Example: In the matrix $A = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 2 & 3 & 1 & 0 \\ 4 & 1 & 3 & 6 \\ 8 & 1 & 2 & 0 \end{bmatrix}$

Every element of the matrix is the minor of order 1.

$$\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}, \begin{vmatrix} 3 & 6 \\ 2 & 0 \end{vmatrix}, \begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix}, \begin{vmatrix} 0 & 4 \\ 1 & 0 \end{vmatrix}, \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} \text{ etc.}$$

are minors of order 2.

$$\begin{vmatrix} 3 & 1 & 0 \\ 1 & 3 & 6 \\ 1 & 2 & 0 \end{vmatrix}, \begin{vmatrix} 2 & 1 & 0 \\ 4 & 3 & 6 \\ 8 & 2 & 0 \end{vmatrix}, \begin{vmatrix} 2 & 3 & 1 \\ 4 & 1 & 3 \\ 8 & 1 & 2 \end{vmatrix} \text{ etc.}$$

are the minors of order 3.

Definition: A positive integer r is said to be the rank of a non zero matrix A , if

- I. there exists at least one minor in A of order which is not zero
- II. every minor in A of order greater than r is zero.

it is written as $\rho(A) = r$

The rank of a zero matrix is defined to be zero.

For Example

1. Rank of $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = 2$ since $\begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 5 \neq 0$

2. Rank of $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 3$ since $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \neq 0$



Illustrative Examples

Example 1: Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}$

Solution: $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}$

$$|A| = 1(1 - 6) - 2(3 - 4) + 3(9 - 2) = -5 + 2 + 21 = 18 \neq 0$$

A minor of order 3 is not zero. Therefore,

$$\text{the rank of } A \geq 3 \quad \dots(1)$$

Since A does not possess any 4 rowed minor therefore

$$\text{Rank of } A \leq 3 \quad \dots(2)$$

From (1) and (2), the rank of $A = 3$

Example 2: Find the rank of the matrix $A = \begin{bmatrix} 4 & 1 & 3 & 8 \\ 6 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$

Solution: $|A| = \begin{vmatrix} 4 & 1 & 3 & 8 \\ 6 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{vmatrix}$

$$R_3 \rightarrow R_3 - R_2, R_4 \rightarrow R_4 - R_3$$

$$= \begin{vmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 6 & 1 & 3 & 8 \\ 6 & 1 & 3 & 8 \end{vmatrix} \text{ as } R_1, R_3 \text{ and } R_4 \text{ are identical.}$$

$$= 0$$

$$\text{A minor of order 3} = \begin{vmatrix} 6 & 1 & 3 \\ 4 & 2 & 6 \\ 10 & 3 & 9 \end{vmatrix} = \begin{vmatrix} 6 & 1 & 3 \\ 4 & 2 & 6 \\ 6 & 1 & 3 \end{vmatrix} = 0$$

$$R_3 \rightarrow R_3 - R_2$$

Since R_1 and R_3 are identical.

Similarly, we can prove that all the minors of order 3 are zero.

$$\text{Now, a minor of order 2} = \begin{vmatrix} 4 & 2 \\ 6 & 1 \end{vmatrix} = 4 - 12 = -8 \neq 0$$

Hence, the rank of the given matrix = 2.

Example 3: Find the rank of matrix A .

$$A = \begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{bmatrix}$$

when a, b, c all are real.

[B.C.A. (Meerut) 2001]

Solution:

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^3 & b^3-a^3 & c^3-a^3 \end{vmatrix}$$

$$[C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1]$$

$$= \begin{vmatrix} b-a & c-a \\ b^3-a^3 & c^3-a^3 \end{vmatrix}$$

$$= (b-a)(c-a) \begin{vmatrix} 1 & 1 \\ b^2+ab+a^2 & c^2+ac+a^2 \end{vmatrix}$$

$$= (b-a)(c-a)[c^2+ac+a^2-b^2-ab-a^2]$$

$$= (b-a)(c-a)(c^2+ac-b^2-ab)$$

$$= (b-a)(c-a)[(c+b)(c-b)+a(c-b)]$$

$$= (b-a)(c-a)(c-b)(c+b+a)$$

$$= (a-b)(b-c)(c-a)(a+b+c)$$

Case I: If $a = b = c$, then

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ a & a & a \\ a^3 & a^3 & a^3 \end{vmatrix} = 0$$

And all the minors of order 2 of A are also zero. But the elements of A are not zero. Hence, in this case the rank of $A = 1$.

Case II: $a = b \neq c$

Then $|A| = \begin{vmatrix} 1 & 1 & 1 \\ a & a & c \\ a^2 & a^2 & c^2 \end{vmatrix} = 0$ as c_1, c_2 are identical.

Also A has a minor $\begin{vmatrix} 1 & 1 \\ a & c \end{vmatrix} = c - a \neq 0 \quad \because a \neq c$

Hence, the rank of $A = 2$.

Case III: a, b, c are all different but $a + b + c = 0$, we know that $|A| = (a - b)(b - c)(c - a)$
 $(a + b + c) = 0$. Also A has a minor of order 2

$$\begin{vmatrix} 1 & 1 \\ b & c \end{vmatrix} = c - b \neq 0 \quad \because b \neq c$$

Hence, the rank of A is 2.

Case IV: a, b, c are all different and $a + b + c \neq 0$, we know that

$$|A| = (a - b)(b - c)(c - a)(a + b + c) \neq 0$$

The minor of order 3 of A is non zero

$$\text{Rank of } A \geq 3 \quad \dots(1)$$

Also A has no minor of order 4

$$\text{Rank of } A \leq 3 \quad \dots(2)$$

From (1) and (2), we get

$$\text{Rank of } A = 3$$

Example 4: Show that the points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are collinear if the rank of the matrix.

$$A = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \text{ is less than 3.}$$

Solution: Here, the rank of A is less than 3. Therefore, the minor of order 3 of the matrix $A = 0$.

$$A = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} = 0 \quad \dots(1)$$

Let the vertices of the triangle ABC be (x_1, y_1) , (x_2, y_2) and (x_3, y_3) .

$$\text{Area of the triangle } ABC = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0 \quad \text{From (1)}$$

Since the area of the triangle ABC is zero, so its vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are collinear.

Example 5: Under what conditions the rank of the matrix is 3?

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 2 & 1 & 2 \\ 1 & 0 & x \end{bmatrix}$$

Solution: Here, the rank of matrix $A = 3$.

Therefore, the minor of order 3 of matrix $A \neq 0$.

$$\begin{vmatrix} 2 & 4 & 2 \\ 2 & 1 & 2 \\ 1 & 0 & x \end{vmatrix} \neq 0, \begin{vmatrix} 2 & 0 & 0 \\ 2 & -3 & 0 \\ 1 & -2 & x-1 \end{vmatrix} \neq 0 \quad [C_3 \rightarrow C_2 - 2C_1, C_3 \rightarrow C_3 - C_1]$$

$$2(-3x + 3 - 0) \neq 0$$

(expanding the determinants with respect to the first row)

$$\Rightarrow -3x + 3 \neq 0$$

$$\Rightarrow 3x \neq 3 \Rightarrow x \neq 1$$

Example 6: Find the values of a so that $\text{rank } \rho(A) < 3$, where A is the matrix

$$A = \begin{bmatrix} 3a-8 & 3 & 3 \\ 3 & 3a-8 & 3 \\ 3 & 3 & 3a-8 \end{bmatrix}$$

Solution: Here, the rank of matrix $A < 3$. Therefore, the minor of order 3 of $A = 0$.

$$\begin{vmatrix} 3a-8 & 3 & 3 \\ 3 & 3a-8 & 3 \\ 3 & 3 & 3a-8 \end{vmatrix} = 0$$

$$\begin{vmatrix} 3a-8+3+3 & 3a-8+3+3 & 3a-8+3+3 \\ 3 & 3a-8 & 3 \\ 3 & 3 & 3a-8 \end{vmatrix} = 0$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\Rightarrow \begin{vmatrix} 3a-2 & 3a-2 & 3a-2 \\ 3 & 3a-8 & 3 \\ 3 & 3 & 3a-8 \end{vmatrix} = 0$$

$$\Rightarrow (3a-2) \begin{vmatrix} 1 & 1 & 1 \\ 3 & 3a-8 & 3 \\ 3 & 3 & 3a-8 \end{vmatrix} = 0$$

$$(3a-2) \begin{vmatrix} 1 & 0 & 0 \\ 3 & 3a-11 & 0 \\ 3 & 0 & 3a-11 \end{vmatrix} = 0 \quad \begin{array}{l} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{array}$$

$$\Rightarrow (3a-2)(3a-11)(3a-11) = 0$$

$$a = \frac{2}{3}, \frac{11}{3}$$

Example 7: Find the rank of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$$

[B.C.A. (Meerut) 2006, 2009]

Solution: We have

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{by } R_4 = R_4 - R_3 - R_2 - R_1$$

$$\Rightarrow A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & -3 & 2 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{by } R_2 = R_2 - 2R_1, R_3 = R_3 - 3R_1$$

Hence, the rank of $A = 3$.

$$\therefore \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & -3 \\ 0 & -4 & -8 \end{vmatrix} = -12 \neq 0$$



Comprehensive Exercise 3.1

Find the rank of

1. $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{bmatrix}$

2. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 7 & 8 & 10 \end{bmatrix}$

[B.C.A. (Meerut) 2004]

3. $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$

4. $A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 3 & 4 & 3 & 0 \\ -1 & 0 & 2 & -8 \end{bmatrix}$

[B.C.A. (Delhi) 2003, 05, B.C.A.; (Rohilkhand) 2005]

5. $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 4 & 6 & 2 \\ 1 & 2 & 3 & 2 \end{bmatrix}$

[B.C.A. (Meerut) 2001]



Answers

1.	3	2.	2	3.	3	4.	3	5.	2
----	---	----	---	----	---	----	---	----	---

3.2 Elementary Transformations

Any one of the following operations on a matrix is called an elementary transformation.

1. Interchanging any two rows (or columns). This transformation is indicated by $R_i \leftrightarrow R_j$ or $C_i \leftrightarrow C_j$
2. Multiplication of the elements of any row (or column) by a non-zero scalar quantity K and indicated as $R_i \rightarrow K R_i$.
3. Addition of constant multiple of the elements of any row to the corresponding element of any other row, indicated as $R_i \rightarrow R_i + K R_j$.

Two matrices are said to be equivalent if one is obtained from the other by elementary transformations. The symbol $\sim$ is used for equivalence.

Definition: A matrix obtained from a unit matrix by subjecting it to any of the elementary transformation is called elementary matrix.

For example: $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad R_3 \leftrightarrow R_1$

$$I \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \quad R_3 \leftrightarrow 5R_3$$

$$I \sim \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_2 \rightarrow R_2 + 3R_1$$

$$I \sim \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad C_1 \leftrightarrow C_3$$

Illustrative Examples

Example 1: Reduce the following matrix to upper triangular form :

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 1 & 2 \end{bmatrix}$$

Solution: **Upper Triangular Matrix.** If in a square matrix all the elements below the principal diagonal are zero, the matrix is called an upper triangular matrix.

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 1 & 2 \end{bmatrix} &\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & -5 & -7 \end{bmatrix} & \begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \end{aligned} \\ &\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix} & R_3 \rightarrow R_3 + 5R_2 \end{aligned}$$

Example 2: Transform $\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix}$ into a unit matrix.

Solution: $\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & -2 & 4 \\ 0 & -1 & -5 \end{bmatrix} \quad \begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \end{aligned}$

$$\sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & -2 \\ 0 & -1 & -5 \end{bmatrix}$$

$$R_2 \rightarrow -\frac{1}{2} R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & -7 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 3 R_2$$

$$R_3 \rightarrow R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow -\frac{1}{7} R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 9 R_3$$

$$R_3 \rightarrow R_2 + 2 R_3$$

Example 3: Use the method of elementary row transformation to compute the inverse of

$$\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

[B.C.A. (Meerut) 2004]

Solution: Let $A = I \quad A \Rightarrow \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 1 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & -1 & -9 \\ 0 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - 2 R_1$$

$$R_3 \rightarrow R_3 + R_1$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & -1 & -9 \\ 0 & 0 & -21 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & 3 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 + 3 R_2$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 9 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ \frac{5}{21} & -\frac{1}{7} & -\frac{1}{21} \end{bmatrix} A$$

$$R_2 \rightarrow -R_2$$

$$R_3 \rightarrow -\frac{1}{21} R_3$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{4}{21} & \frac{5}{7} & \frac{5}{21} \\ -\frac{1}{7} & \frac{2}{7} & \frac{3}{7} \\ \frac{5}{21} & -\frac{1}{7} & -\frac{1}{21} \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 5 R_3$$

$$R_2 = R_2 - 9 R_3$$

$$I = PA \Rightarrow P = A^{-1}$$

$$A^{-1} = \begin{bmatrix} \frac{2}{21} & \frac{1}{7} & -\frac{13}{21} \\ -\frac{1}{7} & \frac{2}{7} & \frac{3}{7} \\ \frac{5}{21} & -\frac{1}{7} & -\frac{1}{21} \end{bmatrix}$$

$$R_1 = R_1 - 2R_2$$

Example 4: Using elementary row operations show that the matrix

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

is invertible. Find A^{-1} .

[B.C.A. (Delhi) 2003, 2007]

Solution: Let $A = I A$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} A$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & \frac{3}{2} & 2 \\ 0 & 0 & 1 & \frac{4}{3} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix} A$$

$$R_2 \rightarrow \frac{1}{2} R_2$$

$$R_3 \rightarrow \frac{1}{3} R_3$$

$$R_4 \rightarrow \frac{1}{4} R_4$$

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & \frac{3}{2} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - 4R_4$$

$$R_2 \rightarrow R_2 - 2R_4$$

$$R_3 \rightarrow R_3 - \frac{4}{3}R_4$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - 3R_3$$

$$R_2 \rightarrow R_2 - \frac{3}{2}R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - 2R_2$$

$$I = PA \Rightarrow P = A^{-1}$$

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix}$$



Comprehensive Exercise 3.2

Reduce the following matrices to upper triangular form.

1. $\begin{bmatrix} 1 & 3 & 4 \\ 2 & 4 & 6 \\ 3 & 5 & 9 \end{bmatrix}$

Find the inverse matrices of the following matrices.

2. $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 1 \\ -1 & 1 & 1 \end{bmatrix}$



Answers

1.	$\begin{bmatrix} 1 & 3 & 4 \\ 0 & -2 & -2 \\ 0 & 0 & 1 \end{bmatrix}$	2.	$\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$	3.	$\frac{1}{21} \begin{bmatrix} 2 & 3 & -13 \\ -3 & 6 & 9 \\ 5 & -3 & -1 \end{bmatrix}$
----	---	----	---	----	---

3.2.1 Echelon Form of a Matrix

1. All the non-zero rows of A , if any precede the zero rows.
2. The number of zero preceeding the first non-zero element in a row is less than the number of such zero in the succeeding row.
3. The first non-zero element in a row is unity, then it is in the Echelon form.

Illustrative Examples

Example 1: A matrix in the Echelon form $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 3 & 5 & 1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

1. Here, there are three non-zero rows which precedes the fourth zero row.
2. Number of zeros in R_2, R_3, R_4 are 1, 2, 5 which are in ascending order.
3. The first non-zero element is unity.

Here there are three non-zero rows so the rank is 3.

Note: To obtain the rank of a matrix we reduce it to upper triangular matrix and we count the number of non-zero rows, the rank of the matrix is equal to the number of non-zero rows.

Example 2: Find the rank of the matrix $\begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & 2 \\ -1 & 2 & 2 \end{bmatrix}$

Solution: Let $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & 2 \\ -1 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} -1 & 2 & 2 \\ 3 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix} \quad R_1 \leftrightarrow R_3$

$$\sim \begin{bmatrix} -1 & 2 & 2 \\ 0 & 7 & 8 \\ 0 & 7 & 8 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 + 3R_1 \\ R_3 \rightarrow R_3 + 2R_1 \end{array}$$

$$\sim \begin{bmatrix} -1 & 2 & 2 \\ 0 & 7 & 8 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & \frac{8}{7} \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} R_1 \rightarrow -R_1 \\ R_2 \rightarrow \frac{1}{7}R_2 \end{array}$$

The last equivalent matrix is in Echelon form. The number of non-zero rows in this matrix = 2

Rank of A = Number of non-zero rows = 2

Example 3: Find the rank of the matrix $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 4 & 5 & 2 \\ 2 & 3 & 4 & 0 \end{bmatrix}$

Solution:

$$\begin{aligned}
 A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 4 & 5 & 2 \\ 2 & 3 & 4 & 0 \end{bmatrix} &\sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & -2 \end{bmatrix} & \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array} \\
 &\approx \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & -1 \end{bmatrix} & R_3 \rightarrow R_3 - R_2
 \end{aligned}$$

Rank of A = Number of non-zero rows = 3

Example 4: Reduce the matrix $A = \begin{bmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{bmatrix}$ to the triangular form and hence,

find the rank.

[M.C.A. (Meerut) 2002, 2008]

$$\begin{aligned}
 \text{Solution: } A = \begin{bmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{bmatrix} &\sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 5 & 3 & 14 & 4 \end{bmatrix} & R_1 \leftrightarrow R_3, R_3 \rightarrow R_3 - 5R_1 \\
 &\sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 8 & 4 & 4 \end{bmatrix} & R_3 \rightarrow R_3 - 8R_2 \\
 &\sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & \frac{1}{3} \end{bmatrix} & R_3 \rightarrow -\frac{1}{12} R_3
 \end{aligned}$$

 $\Rightarrow$ Rank of $A = 3$

Example 5: Find the rank of the matrix: $A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 3 & 2 & 5 & 1 \\ 0 & 4 & 4 & -4 \end{bmatrix}$

$$\begin{aligned}
 \text{Solution: } A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 3 & 2 & 5 & 1 \\ 0 & 4 & 4 & -4 \end{bmatrix} &\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 2 & -2 \\ 0 & 4 & 4 & -4 \end{bmatrix} & R_2 \rightarrow R_2 - 3R_1 \\
 &\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 2 & -2 \\ 0 & 4 & 4 & -4 \end{bmatrix} & R_3 \rightarrow R_3 - 2R_2 \\
 &\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} & R_2 \rightarrow \frac{1}{2} R_2
 \end{aligned}$$

 $\Rightarrow$ Rank of $A = 2$

Example 6: Find the rank of matrix:

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -4 & 7 \\ -1 & -2 & -1 & -2 \end{bmatrix}$$

Solution: Let

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -4 & 7 \\ -1 & -2 & -1 & -2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & -2 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 + R_1$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow -\frac{1}{2}R_2$$

The last equivalent matrix is in the Echelon form. The number of non-zero rows in the matrix is 2. Therefore, the rank of the matrix A is 2.

Example 7: Find the value of the matrix

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$$

Solution: Let

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 8 & 5 & 0 \\ 0 & -2 & 1 & -8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{5}{8} & 0 \\ 0 & -2 & 1 & -8 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{8}R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{5}{8} & 0 \\ 0 & 0 & \frac{9}{4} & -8 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{5}{8} & 0 \\ 0 & 0 & 1 & -\frac{32}{9} \end{bmatrix} \quad R_3 \rightarrow \frac{4}{9}R_3$$

The last equivalent matrix is in the Echelon form. The number of non-zero rows in the matrix are three. Therefore, the rank of A is 3.

Example 8: Find the rank of the matrix :

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & -2 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & 0 & 3 \end{bmatrix}$$

using elementary row transformation.

Solution: Let

$$\begin{aligned} A &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & -2 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & 0 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & -2 & -5 & 0 \\ 0 & 0 & -3 & 0 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1 \\ R_4 \rightarrow R_4 - 3R_1 \end{array} \\ &\sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & 0 & -8 & 0 \\ 0 & 0 & -3 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 + R_2 \\ &\sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & 0 & -8 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_4 \rightarrow R_4 - \frac{3}{8}R_3 \\ &\sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & -3 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \left(R_3 - \frac{1}{8}R_3 \right) \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -\frac{3}{2} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_2 \rightarrow \frac{1}{2}R_2 \end{aligned}$$

The last equivalent matrix is in Echelon form. The number of non-zero rows in this matrix is 3. Therefore its rank is 3.

Example 9: Find the rank of the matrix:

$$\begin{bmatrix} 6 & 1 & 3 & 8 \\ 16 & 4 & 12 & 15 \\ 5 & 3 & 3 & 4 \\ 4 & 2 & 6 & -1 \end{bmatrix}$$

Solution:

$$A = \begin{bmatrix} 1 & 6 & 3 & 8 \\ 4 & 16 & 12 & 15 \\ 3 & 5 & 3 & 4 \\ 2 & 4 & 6 & -1 \end{bmatrix} \quad C_1 \leftrightarrow C_2$$

$$\begin{aligned}
 & \sim \begin{bmatrix} 1 & 6 & 3 & 8 \\ 0 & -8 & 0 & -17 \\ 0 & -13 & -6 & -20 \\ 0 & -8 & 0 & -17 \end{bmatrix} & \begin{aligned} R_2 &\rightarrow R_2 - 4R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \\ R_4 &\rightarrow R_4 - 2R_1 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 6 & 3 & 8 \\ 0 & -8 & 0 & -17 \\ 0 & 0 & -6 & \frac{61}{8} \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{aligned} R_3 &\rightarrow R_3 - \frac{13}{8}R_2 \\ R_4 &\rightarrow R_4 - R_2 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 6 & 3 & 8 \\ 0 & 1 & 0 & \frac{17}{8} \\ 0 & 0 & 1 & -\frac{61}{48} \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{aligned} R_2 &\rightarrow -\frac{1}{8}R_2 \\ R_3 &\rightarrow -\frac{1}{6}R_3 \end{aligned}
 \end{aligned}$$

The last equivalent matrix is in Echelon form. The number of non-zero rows in this matrix is 3. Hence, the rank of the given matrix = 3.

Example 10: Find the rank of:

$$\begin{bmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{bmatrix}$$

Solution: The given matrix is

$$\begin{aligned}
 A &= \begin{bmatrix} 1 & 4 & 9 & 16 \\ 4 & 9 & 16 & 25 \\ 9 & 16 & 25 & 36 \\ 16 & 25 & 36 & 49 \end{bmatrix} \\
 & \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & -7 & -20 & -39 \\ 9 & -20 & -56 & -108 \\ 16 & -39 & -108 & -207 \end{bmatrix} & \begin{aligned} C_2 &\rightarrow C_2 - 4C_1 \\ C_3 &\rightarrow C_3 - 9C_1 \\ C_4 &\rightarrow C_4 - 16C_1 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & -20 & -39 \\ 0 & -20 & -56 & -108 \\ 0 & -39 & -108 & -207 \end{bmatrix} & \begin{aligned} R_2 &\rightarrow R_2 - 4R_1 \\ R_3 &\rightarrow R_3 - 9R_1 \\ R_4 &\rightarrow R_4 - 16R_1 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & -20 & 1 \\ 0 & -20 & -56 & 4 \\ 0 & -39 & -108 & 9 \end{bmatrix} & C_4 \rightarrow C_4 - 2C_3
 \end{aligned}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 1 & 1 \\ 0 & -20 & 4 & 4 \\ 0 & -39 & 9 & 9 \end{bmatrix} \quad C_3 \rightarrow C_3 - 3C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 1 & 0 \\ 0 & -20 & 4 & 0 \\ 0 & -39 & 9 & 0 \end{bmatrix} = B \quad C_4 \rightarrow C_4 - C_3$$

$$\Rightarrow |B| = 0 \quad \rho(A) < 4$$

Consider a minor of order 3.

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & -7 & 1 \\ 0 & -20 & 4 \end{vmatrix} = 1(-28 + 20) = -8 \neq 0$$

Hence, Rank of $A = 3$.

Example 11: Find the rank of the matrix:

$$\begin{bmatrix} 3 & -1 & 1 & 3 \\ -1 & -4 & -2 & -7 \\ 2 & 1 & 3 & 0 \\ -1 & -2 & 3 & 0 \end{bmatrix}$$

Solution:

$$A = \begin{bmatrix} -1 & -4 & -2 & -7 \\ 3 & -1 & 1 & 3 \\ 2 & 1 & 3 & 0 \\ -1 & -2 & 3 & 0 \end{bmatrix} \quad R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} -1 & -4 & -2 & -7 \\ 0 & -13 & -5 & -18 \\ 0 & -7 & -1 & -14 \\ 0 & 2 & 5 & 7 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 + 3R_1 \\ R_3 \rightarrow R_3 + 2R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 4 & 2 & 7 \\ 0 & 2 & 5 & 7 \\ 0 & -7 & -1 & -14 \\ 0 & -13 & -5 & -18 \end{bmatrix} \quad R_2 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 4 & 2 & 7 \\ 0 & 1 & \frac{5}{2} & \frac{7}{2} \\ 0 & -7 & -1 & -14 \\ 0 & -13 & -5 & -18 \end{bmatrix} \quad R_2 \rightarrow \frac{1}{2}R_2$$

$$\begin{aligned}
 & \sim \begin{bmatrix} 1 & 4 & 2 & 7 \\ 0 & 1 & \frac{5}{2} & \frac{7}{2} \\ 0 & 0 & \frac{33}{2} & -\frac{21}{2} \\ 0 & 0 & \frac{55}{2} & \frac{55}{2} \end{bmatrix} & \begin{aligned} R_3 &\rightarrow R_3 + 7R_2 \\ R_4 &\rightarrow R_4 + 13R_2 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 4 & 2 & 7 \\ 0 & 1 & \frac{5}{2} & \frac{7}{2} \\ 0 & 0 & 1 & -\frac{7}{11} \\ 0 & 0 & 1 & 1 \end{bmatrix} & \begin{aligned} R_3 &\rightarrow \frac{2}{33} R_3 \\ R_4 &\rightarrow \frac{2}{55} R_4 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 4 & 2 & 7 \\ 0 & 1 & \frac{5}{2} & \frac{7}{2} \\ 0 & 0 & 1 & -\frac{7}{11} \\ 0 & 0 & 0 & \frac{11}{18} \end{bmatrix} & R_4 \rightarrow R_4 - R_3 \\
 & \sim \begin{bmatrix} 1 & 4 & 2 & 7 \\ 0 & 1 & \frac{5}{2} & \frac{7}{2} \\ 0 & 0 & 1 & -\frac{7}{11} \\ 0 & 0 & 0 & 1 \end{bmatrix} & R_4 \rightarrow \frac{11}{18} R_4
 \end{aligned}$$

The last equivalent matrix is in the Echelon form. The number of non-zero rows is 4. Therefore, the rank of $A = 4$.

Example 12: Find the rank of the matrix A , where :

$$A = \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$$

Solution:

$$A = \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$$

$$\sim \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_4 \rightarrow R_4 - (R_1 + R_3)$$

$$\sim \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 - (R_1 + R_2)$$

$$\begin{aligned}
 & \sim \begin{bmatrix} 4 & 2 & 6 & -1 \\ 6 & 1 & 3 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & R_1 \leftrightarrow R_2 \\
 & \sim \begin{bmatrix} 4 & 2 & 6 & -1 \\ 0 & -2 & -6 & \frac{19}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & R_2 \rightarrow R_2 - \frac{3}{2}R_1 \\
 & \sim \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{4} \\ 0 & 1 & 3 & -\frac{19}{4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{aligned} R_1 &\rightarrow \frac{1}{4}R_1 \\ R_2 &\rightarrow 0 - \frac{1}{2}R_2 \end{aligned}
 \end{aligned}$$

The last equivalent matrix is in the Echelon form. The number of non-zero rows in the matrix is 2. Therefore, the rank of A is 2.

Example 13: If $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$ find its rank:

Solution:

$$\begin{aligned}
 A &= \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix} \\
 & \sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 5 & -4 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix} & \begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - R_1 \\ R_4 &\rightarrow R_4 + R_1 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 1 & -\frac{4}{5} \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix} & R_2 \rightarrow \frac{1}{5}R_2 \\
 & \sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 1 & -\frac{4}{5} \\ 0 & 0 & 0 & \frac{16}{5} \\ 0 & 0 & 0 & 1 \end{bmatrix} & \begin{aligned} R_3 &\rightarrow R_3 - 4R_2 \\ R_4 &\rightarrow R_4 - 5R_2 \end{aligned}
 \end{aligned}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 1 & -\frac{4}{5} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_3 \rightarrow \frac{5}{16} R_3$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 1 & -\frac{4}{5} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_4 \rightarrow R_4 - R_3$$

The last equivalent matrix is in the Echelon form. The number of non-zero rows is 3. The rank of A is 3.

3.2.2 Normal Form of a Matrix

A matrix A can be reduced by elementary row and column transformation into one of the following equivalent matrix.

$$\text{I. } \begin{bmatrix} I_n & O \\ O & O \end{bmatrix} \quad \text{II. } \begin{bmatrix} I_n \\ O \end{bmatrix} \quad \text{III. } [I_n \ O] \quad \text{IV. } [I_n]$$

where I_n is $n \times n$ identity matrix and O is null matrix of any order.

The above forms are known as normal or canonical form of A .

Then rank of A is n .



Illustrative Examples

Example 1: Reduce the following matrix to the normal form and hence, find its rank:

$$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Solution: Let

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\begin{aligned}
 & \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix} & \begin{aligned} R_3 &\rightarrow R_3 - 3R_1 \\ R_4 &\rightarrow R_4 - R_2 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{aligned} R_3 &= R_3 - R_2 \\ R_4 &= R_4 - R_2 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{aligned} C_4 &\rightarrow C_4 - C_1 + C_2 \\ C_3 &\rightarrow C_3 - C_1 + 3C_2 \end{aligned} \\
 & \sim \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}
 \end{aligned}$$

The last equivalent matrix is in the normal form. The rank of A is 2.

Example 2: Find the rank of:

$$A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 3 & 0 & 3 \\ 1 & -2 & -3 & -3 \\ 1 & 1 & 2 & 3 \end{bmatrix}$$

by reducing it to the normal form by means of elementary transformation.

Solution: We have the matrix $A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 3 & 0 & 3 \\ 1 & -2 & -3 & -3 \\ 1 & 1 & 2 & 3 \end{bmatrix}$

$$\begin{aligned}
 & \sim \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 2 & -2 & 0 \\ 0 & -3 & -5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{aligned} R_2 &\rightarrow R_2 - R_1 \\ R_3 &\rightarrow R_3 - R_1 \\ R_4 &\rightarrow R_4 - R_1 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & -1 & 0 \\ 0 & -3 & -5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} & R_2 \rightarrow \frac{1}{2} R_2 \\
 & \sim \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -8 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} & R_3 \rightarrow R_3 + 3R_2
 \end{aligned}$$

$$\begin{aligned}
 & \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -8 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{aligned} C_2 &\rightarrow C_2 - C_1 \\ C_3 &\rightarrow C_3 - 2C_1 \\ C_4 &\rightarrow C_4 - 3C_1 \end{aligned} \\
 & \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -8 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} & C_3 \rightarrow C_3 + C_2 \\
 & \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{aligned} C_3 &\rightarrow -\frac{1}{8}C_3 \\ C_4 &\rightarrow -\frac{6}{8}C_3 \end{aligned}
 \end{aligned}$$

Thus, the matrix A is equivalent to the matrix $\begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$ hence, the rank of $A = 3$.

Example 3: Reduce the matrix $A = \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 2 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$ to the normal form and hence, find

its rank:

Solution:

$$\begin{aligned}
 A &= \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 2 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix} \\
 & \sim \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 2 & 4 & 1 \\ 0 & 4 & 4 & 1 \\ 0 & 6 & 8 & 2 \end{bmatrix} & \begin{aligned} R_3 &\rightarrow R_3 - R_1 \\ R_4 &\rightarrow R_4 - R_1 \end{aligned} \\
 & \sim \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 2 & 4 & 1 \\ 0 & 4 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} & R_4 \rightarrow R_4 - (R_2 + R_3) \\
 & \sim \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 2 & 4 & 1 \\ 0 & 0 & -4 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} & R_3 \rightarrow R_3 - 2R_2 \\
 & \sim \begin{bmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & 2 \\ 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{aligned} R_1 &\rightarrow \frac{1}{2}R_2 \\ R_2 &\rightarrow \frac{1}{2}R_2 \\ R_3 &\rightarrow -\frac{1}{4}R_3 \end{aligned}
 \end{aligned}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_2 \rightarrow C_2 + \frac{1}{2}C_1$$

$$C_3 \rightarrow C_3 - \frac{3}{2}C_1$$

$$C_4 \rightarrow C_4 - 2C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - 2C_2$$

$$C_4 \rightarrow C_4 - \frac{1}{2}C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_4 \rightarrow C_4 - \frac{1}{4}C_3$$

$$\sim \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

The last equivalent matrix is in the normal form. The rank of A is 3.

Example 4: Find two non-singular matrices P and Q such that PAQ is in the normal form where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix}$$

Also find the rank of the matrix A

Solution:

$$A = I_3 AI_3$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We have to reduce the matrix on the left hand side to the normal form by elementary row operations. Every E-row operations will also be applied on prefactor I_3 on the right hand side and every E-column operations to the post factor I_3 on the right hand side.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -2 \\ 0 & -2 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & -2 \\ 0 & -2 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - C_1$$

$$C_3 \rightarrow C_3 - C_1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ -3 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{2}R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ -2 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 + 2R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ -2 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix} = PAQ$$

where

$$P = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ -2 & -1 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

Since $A \sim \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$, therefore rank $A = 2$.

Example 5: Reduce the matrix: $A = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 1 & 3 \end{bmatrix}$ to its normal form and hence,

determine its rank.

Solution: We have

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 1 \\ 3 & 1 & 1 & 3 \end{bmatrix} \quad R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & -8 & -9 \end{bmatrix} \quad R_3 \rightarrow R_3 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & -4 \end{bmatrix} \quad R_3 \rightarrow R_3 + 5R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & -4 \end{bmatrix} \quad \begin{array}{l} C_2 \rightarrow C_2 - 2C_1 \\ C_3 \rightarrow C_3 - 3C_1 \\ C_4 \rightarrow C_4 - 4C_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & -4 \end{bmatrix} \quad \begin{array}{l} C_3 \rightarrow C_3 - 2C_2 \\ C_4 \rightarrow C_4 - C_2 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \end{bmatrix} \quad \begin{array}{l} C_4 \rightarrow C_4 + 2 C_3 \\ \\ C_3 \rightarrow \frac{1}{2} C_3 \\ C_4 \rightarrow C_4 + 2 C_3 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Thus, the matrix A is equivalent to the matrix $[I_3, 0]$. Hence, the rank of A is 3.



Comprehensive Exercise 3.3

Find the rank of the following matrices:

$$1. \begin{bmatrix} 3 & 2 & 5 & 7 & 12 \\ 1 & 1 & 2 & 3 & 5 \\ 3 & 3 & 6 & 9 & 15 \end{bmatrix} \quad 2. \begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix} \quad 3. \begin{bmatrix} 0 & c & -b & d' \\ -c & 0 & a & b' \\ b & -a & 0 & c' \\ -d' & -b' & c' & 0 \end{bmatrix}$$

where $ad' + bb' + cc' = 0$ and a, b, c are all positive numbers.

$$4. \begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 3 & 0 & 3 \\ 1 & -2 & -3 & -3 \\ 1 & 1 & 2 & 3 \end{bmatrix} \quad 5. \begin{bmatrix} 1 & 1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix} \quad 6. A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$$

$$7. A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 4 & 6 & 2 \\ 1 & 2 & 3 & 2 \end{bmatrix}$$

[B.C.A. (Meerut) 2001]

$$8. A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$$

[B.C.A. (Meerut) 2006, 2007]



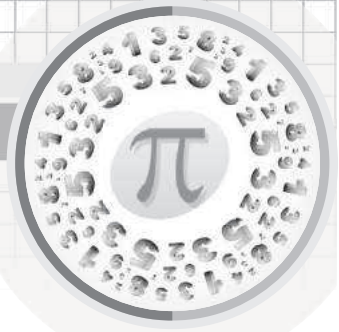
Answers

1.	2	2.	3	3.	2	4.	3
5.	4	6.	3	7.	3	8.	2





Unit-1



Chapter

4

Eigen Values, Eigen Vectors & Cayley-Hamilton Theorem

4.1 Eigen Value & Eigen Vector of Matrix

4.1.1 Eigen Value

Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order n over the field F . An element λ in F is called an **eigen value** of A if $|(A - \lambda I)| = 0$, where I is the unit matrix of order n

Eigen Vector. Let λ be an eigen value of an n -square matrix A , then a non-zero matrix X of the type $n \times 1$ such that $AX = \lambda X$ is called an **eigen vector** of A corresponding to the eigen value λ .

4.1.2 Characteristic Roots

If A be an n -square matrix, then the problem to determine the scalar λ and the non-zero vectors X which simultaneously satisfy the equation $AX = \lambda X$ is known as **Characteristic value** or roots.

Eigen values are some times also called **proper values** characteristic values or spectral values similarly eigen vectors are called proper vectors, Characteristic vectors or spectral vectors.

4.1.3 Characteristic Value Problem

If A be n -square matrix, then, the problem to determine the scalar λ and the non-zero vectors X which simultaneously satisfy the equation.

$$AX = \lambda X \quad \dots(1)$$

is known as characteristic value problem

$$\text{If } X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{(n \times 1)} \quad \text{and} \quad A = [a_{ij}]_{(n \times n)} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}_{(n \times n)}$$

The equation (1) in n -unknown can be obtained in the following system of homogeneous linear equations.

$$\left. \begin{aligned} (a_{11} - \lambda)x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\ a_{21}x_1 + (a_{22} - \lambda)x_2 + \dots + a_{2n}x_n &= 0 \\ \dots & \dots \dots \dots \dots \dots \\ \dots & \dots \dots \dots \dots \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + (a_{nn} - \lambda)x_n &= 0 \end{aligned} \right\} \quad \dots(2)$$

(2) can be written in matrix form, as

$$\begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \cdots & \vdots \\ \vdots & \vdots & \cdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & \cdots & a_{nn} - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix} = 0$$

$$\Rightarrow (A - \lambda I) X = 0 \quad \dots(3)$$

where I is the unit matrix of order n . The equation (3) has a non-trivial solution X if and only if

$$|A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix} = 0 \quad \dots(4)$$

4.1.4 Characteristic Matrix

The matrix $A - \lambda I$ is called the characteristic matrix.

4.1.5 Characteristic Polynomial

The expansion of the determinant $|A - \lambda I|$ yield a polynomial $f(\lambda)$ of degree n in λ which is called characteristic polynomial of the matrix A .

4.1.6 Characteristic Equation

$$|A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix} = 0$$

is called the characteristic equation of A and its roots $\lambda_1, \lambda_2, \dots, \lambda_n$ are called the characteristic roots or eigen values of A or latent roots.

4.1.7 Spectrum

The set of all characteristic roots of A is called **spectrum**

4.1.8 Characteristic Vector

If $\lambda = \lambda_i$ be any characteristic root (eigen value) of A , then for this value of λ , equation (3) is satisfied and hence the equation $(A - \lambda_i I) X = 0$ has non-trivial solution for X . Every such non-trivial solution is called a characteristic vector or eigen vector of A .

4.2 Certain Relations between Eigen Values and Eigen Vectors



Some Important Theorems

Theorem 1: λ is an eigen value of a matrix A if and only if there exists a non-zero vector X such that

$$AX = \lambda X$$

Proof: Let λ is an eigen value of matrix A . Then $(A - \lambda I) X = 0$ and the matrix $(A - \lambda I)$ is singular

Therefore $(A - \lambda I) X = 0$ i.e. $AX = \lambda X$

Conversely: Let there exists a non-zero vector X such that $AX = \lambda X$, i.e. $(A - \lambda I) X = 0$ since the matrix equation (1) have a non-zero solution therefore the coefficient matrix $(A - \lambda I)$ must be non-singular i.e.

$$|A - \lambda I| = 0$$

Hence λ is an eigen value of matrix A

Theorem 2: If X is an eigen value of a matrix A , corresponding to the eigen value λ , then kX is also an eigen vector of A corresponding to the same eigen value λ . Hence k is any non-zero scalar.

Proof: Let $X \neq 0$, then

$$AX = \lambda X \quad \dots(1)$$

If k is any non-zero scalar $kX \neq 0$

$$\begin{aligned} A(kX) &= k(AX) = k(\lambda X) & [\text{From (1)}] \\ &= \lambda(kX) \end{aligned}$$

Now, kX is non-zero vector, such that

$$A(kX) = \lambda(kX)$$

Hence kX is eigen vector of A corresponding eigen value λ .

4.3 Nature of the Eigen Values of Special Types of Matrices

Theorem 3: The eigen values of a Hermitian matrix are real.

Proof: Let A be Hermitian matrix λ an eigen value λ and X be eigen vector, then

$$AX = \lambda X \quad \dots(1)$$

Pre-multiplying both sides of (1) by X^* , we get

$$X^* \cdot AX = \lambda X^* X \quad \dots(2)$$

Taking transposed conjugate or tranjugate on both sides of (2), we get

$$(X^* \cdot AX)^* = (\lambda X^* X)^*$$

$$\Rightarrow X^* A^* (X^*)^* = \bar{\lambda} X^* (X^*)^* \quad [\because (AB)^* = B^* A^*]$$

$$\Rightarrow X^* AX = \bar{\lambda} X^* X \quad \dots(3)$$

$$[\because (X^*)^* = X \text{ and } A^* = A, \text{ because } A \text{ is a Hermitian matrix}]$$

Now, from 2 and 3

$$\lambda X^* X = \bar{\lambda} X^* X \Rightarrow (\lambda - \bar{\lambda}) X^* X = 0$$

But

$$X \neq 0 \text{ therefore } X^* X \neq 0$$

Now

$$\lambda - \bar{\lambda} = 0. \text{ Hence } \lambda = \bar{\lambda} \text{ } \lambda \text{ is real}$$

Cor 1: The eigen value of a real symmetric are all real.

Proof: Let A is a real symmetric matrix, then

$$\overline{A'} = A' = A \Rightarrow A^* = A$$

Hence, A is Hermitian matrix

Thus a real symmetric is Hermitian and hence the result.

Cor 2: The eigen value of a Skew-Hermitian matrix are either pure imaginary or zero.

Proof: Let A is a skew-Hermitian matrix. Then iA is Hermitian, let λ be an eigen value of A , then

$$AX = \lambda X \Rightarrow (iA)X = (i\lambda)X$$

$i\lambda$ is an eigen value of iA which is Hermitian. Hence $i\lambda$ is real.

Therefore either λ must be zero or pure imaginary.

Cor 3: The eigen value of a real symmetric are either pure imaginary or zero.

Proof: Since every real symmetric matrix is skew-Hermitian, so that required result.

Theorem 4: The eigen value of a unitary matrix are of unit modulus.

Proof: Let A be unitary matrix. Then

$$A^* A = I$$

Suppose λ be an eigen value of A . Then

$$AX = \lambda X \quad \dots(1)$$

Taking transposed conjugate or tranjugate on both sides

$$(AX)^* = (\lambda X)^*$$

$$X^* \cdot A^* = \bar{\lambda} X^* \quad [\because (AB)^* = B^* A^*] \quad \dots(2)$$

Now, from (1) and (2), we get

$$\Rightarrow (X^* A^*)(AX) = \bar{\lambda} \lambda X^* X$$

$$\Rightarrow X^* (A^* A) X = \bar{\lambda} \lambda X^* X$$

$$X^* I X = \bar{\lambda} \lambda X^* X$$

$$[\because A \text{ is unitary matrix} \Rightarrow A^* A = I]$$

$$\Rightarrow X^* X = \bar{\lambda} \lambda X^* X$$

$$\Rightarrow (\bar{\lambda} \lambda - 1) X^* X = 0 \quad \dots(3)$$

Since, $X^* X \neq 0$, therefore (3) gives

$$\Rightarrow \bar{\lambda} \lambda = 1 \quad \Rightarrow |\lambda|^2 = 1$$

Cor 4: The eigen value of an orthogonal matrix are of unit modulus.

We know that if the elements of a unitary matrix A are all real, then A is said to be orthogonal matrix. Hence the result.

Theorem 5: The eigen value of idempotent matrix are zero or one.

Proof: Let A be an idempotent matrix. Then $A^2 = A$, let λ be eigen value of A and X a corresponding eigen vector. Then $AX = \lambda X$

$$\begin{aligned}
 \text{Now} \quad & AX = \lambda X \Rightarrow A(AX) = A(\lambda X) = \lambda(AX) \\
 \Rightarrow & (AA)X = \lambda(\lambda X) \quad [\because AX = \lambda X] \\
 \Rightarrow & A^2 X = \lambda^2 X \\
 \Rightarrow & AX = \lambda^2 X \quad [\because A^2 = A] \\
 \Rightarrow & \lambda X = \lambda^2 X \\
 \Rightarrow & (\lambda - \lambda^2) X = 0 \\
 \Rightarrow & \lambda - \lambda^2 = 0 \quad [\because X \neq 0] \\
 \Rightarrow & \lambda(1 - \lambda) = 0 \\
 \Rightarrow & \lambda = 0 \quad \text{or} \quad \lambda = 1
 \end{aligned}$$

Hence eigen values of an idempotent matrix are zero or one.

Theorem 6: The eigen values of involutory matrix are ± 1 .

Proof: Let A be involutory matrix. Then $A^2 = I$ where I is unit matrix. Then rest proof is similar to theorem 5.

4.4 The Process of Finding the Eigen Values & Eigen Vector of Matrix

Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order n . First write the characteristic equation i.e. $|A - \lambda I| = 0$. This equation will be of degree n in λ . So it will have n roots. These n roots will give us the eigen values of matrix A . If λ_1 is an eigen value of A , then the vector corresponding eigen vectors of A will be given by non-zero vector

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

satisfying the equation

$$AX = \lambda_1 X \quad \text{or} \quad (A - \lambda_1 I)X = 0$$

Illustrative Examples

Example 1: Determine the eigen (Characteristic) values of the matrix $A = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$

Solution: Here A characteristic matrix

$$= A - \lambda I = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3-\lambda & 2 \\ -1 & 0-\lambda \end{bmatrix}$$

Therefore A is characteristic polonomic

$$\begin{aligned} &= |A - \lambda I| = \begin{vmatrix} 3-\lambda & 2 \\ -1 & -\lambda \end{vmatrix} \\ &= (3-\lambda)(-\lambda) - 2 \times (-1) \\ &= \lambda^2 - 3\lambda + 2 \end{aligned}$$

$\therefore A$ is characteristic equation

$$|A - \lambda I| = 0$$

$$\Rightarrow \lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda = 1, 2$$

Hence, eigen values of A is (1) and (2)

Example 2: Determine the eigen values of matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{bmatrix}$$

Solution: A is characteristic matrix

$$\begin{aligned} &= A - \lambda I = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0-\lambda & 1 & 2 \\ 1 & 0-\lambda & -1 \\ 2 & -1 & 0-\lambda \end{bmatrix} \end{aligned}$$

Thus the characteristic polynomial of $A = |A - \lambda I|$

$$\begin{aligned} &= \begin{vmatrix} -\lambda & 1 & 2 \\ 1 & -\lambda & -1 \\ 2 & -1 & -\lambda \end{vmatrix} \\ &= -\lambda (\lambda^2 - 1) - 1(-\lambda + 2) + 2(-1 + 2\lambda) \\ &= -\lambda^3 + 6\lambda - 4 \end{aligned}$$

∴ The characteristic equation of A is

$$|A - \lambda I| = 0 \Rightarrow \lambda^3 - 6\lambda + 4 = 0$$

$$\Rightarrow (\lambda - 2)(\lambda^2 + 2\lambda - 2) = 0$$

$$\Rightarrow \lambda = 2, -1 \pm \sqrt{3}$$

Hence, the eigen values of matrix A are $2, -1 \pm \sqrt{3}$

Example 3: Show that matrix $A = \begin{bmatrix} 0 & h & g \\ h & 0 & f \\ g & f & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & f & h \\ f & 0 & g \\ h & g & 0 \end{bmatrix}$, $C = \begin{bmatrix} 0 & g & f \\ g & 0 & h \\ f & h & 0 \end{bmatrix}$

having the same characteristics $\lambda^3 + \lambda(f^2 + g^2 + h^2) - 2fgh = 0$

Solution: The characteristics equation of A is

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 0 - \lambda & h & g \\ h & 0 - \lambda & f \\ g & f & 0 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda(\lambda^2 - f^2) - h(-h\lambda - gf) + g(fh + g\lambda) = 0$$

$$\Rightarrow \lambda^3 - \lambda(f^2 + g^2 + h^2) - 2fgh = 0 \quad \dots(1)$$

Similarly, we can see that this equation is also characteristic roots for B and C matrix.

Example 4: Determine the eigen values and the corresponding eigen vectors of the matrix

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$$

Solution: The characteristic equation of matrix A is

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 4 - \lambda & 2 \\ 3 & 3 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (4 - \lambda)(3 - \lambda) - 6 = 0$$

$$\Rightarrow \lambda^2 - 7\lambda + 6 = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 6) = 0$$

The roots of this equation is $\lambda_1 = 1, \lambda_2 = 6$, hence eigen value or characteristic roots of A is 1 and 6.

The eigen vector $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ of matrix A corresponding eigen value $\lambda_1 = 1$ is obtain by

$$(A - I)X = 0$$

$$\Rightarrow \begin{bmatrix} 4-1 & 2 \\ 3 & 3-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad [\text{by } R_2 \rightarrow R_2 - R_1]$$

The coefficient matrix of these equations is of rank 1. Therefore these equations have $2 - 1 = 1$ linearly independent solution. Thus there is only one linearly independent eigen vector corresponding to eigen value $\lambda_1 = 1$. These equation can be written as $3x_1 + 2x_2 = 0$

$$\Rightarrow x_1 = 2, x_2 = -3. \text{ Therefore } X_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix} \text{ is eigen vector of } A \text{ when eigen value } \lambda_1 = 1$$

Similarly eigen vector of A when eigen value $\lambda_2 = 6$ is given by

$$|A - 6I| = 0$$

$$\Rightarrow \begin{bmatrix} 4-6 & 2 \\ 3 & 3-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & 2 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -2x_1 + 2x_2 = 0, 3x_1 - 3x_2 = 0 \Rightarrow x_1 = x_2$$

If we have $x_1 = 1, x_2 = 1$ then $X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, be eigen vector corresponding eigen value $\lambda_2 = 6$

Example 5: Determine the eigen values and the corresponding eigen vectors of the

matrix
$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Also write the corresponding eigen values.

Solution: The characteristic equation of matrix A is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (8-\lambda)\{(7-\lambda)(3-\lambda)-16\} + 6\{-6(3-\lambda)+8\} + 2\{24-2(7-\lambda)\} = 0$$

$$\Rightarrow \lambda^3 - 18\lambda^2 + 45\lambda = 0$$

$$\Rightarrow \lambda(\lambda-3)(\lambda-15)=0$$

$$\Rightarrow \lambda=0,3,15$$

Hence eigen value of A is 0,3 and 15

The eigen vector corresponding to eigen value 0 are given by non-zero solution of the equation

$$(A-0I)X=0 \Rightarrow \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & -4 & 3 \\ -6 & 7 & -4 \\ 8 & -6 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [R_1 \leftrightarrow R_3]$$

$$\Rightarrow \begin{bmatrix} 2 & -4 & 3 \\ 0 & -5 & 5 \\ 0 & 10 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [R_1 \rightarrow R_2 + 3R_1, R_3 \rightarrow R_3 - 4R_1]$$

$$\Rightarrow \begin{bmatrix} 2 & -4 & 3 \\ 0 & -5 & 5 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [R_3 \rightarrow R_3 + 2R_2]$$

The co-efficient matrix of these equation is of rank 2. Therefore these equation have $3-2=1$.

linearly independent solution. Thus there is only one linearly independent vector corresponding to the eigen value 0, these equations can be written as:

$$2x_1 - 4x_2 + 3x_3 = 0 \quad \dots(1)$$

$$-5x_2 + 5x_3 = 0 \quad \dots(2)$$

From (2) we get $x_2 = x_3$, Let us take $x_2 = 1, x_3 = 1$ so that from (1) $x_1 = \frac{1}{2}$.

Therefore $X_1 = \left[\frac{1}{2}, 1, 1\right]^T$ is an eigen vector of A when eigen value is 0. If K_1 is any non-zero scalar, then $K_1 X_1$ is also an eigen vector of A corresponding eigen value 0.

The eigen vectors of A corresponding to the eigen value 3 are given by the non-zero solution of the equation.

$$(A-3I)X=0$$

$$\Rightarrow \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & -2 & -2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [R_1 \rightarrow R_1 + R_2]$$

$$\Rightarrow \begin{bmatrix} -1 & -2 & -2 \\ 0 & 16 & 8 \\ 0 & -8 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [R_2 \rightarrow R_2 - 6R_1, R_3 \rightarrow R_3 + 2R_1]$$

$$\Rightarrow \begin{bmatrix} -1 & -2 & -2 \\ 0 & 16 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \left[R_3 \rightarrow R_3 + \frac{1}{2} R_2 \right]$$

The coefficient matrix of these equations is of rank 2. Therefore these equations have $3 - 2 = 1$ linearly independent solution. These equations can be written as

$$-x_1 - 2x_2 - 2x_3 = 0 \quad \dots(3)$$

$$16x_2 + 8x_3 = 0 \quad \dots(4)$$

From (4) $x_2 = -\frac{1}{2}x_3$ Now take $x_3 = 4, x_2 = -2$

From(3) we get $x_1 = -4$. Therefore $X_2 = [-4, -2, 4]'$, is an eigen vector of A corresponding the eigen value 3.

Every non-zero multiple of X_2 i.e. $K_2 X_2$ where K_2 is any non-zero scalar, is an eigen vector of A corresponding to eigen value 3. The eigen vectors of A corresponding to eigen value 15 are given by the non-zero solution of the equation.

$$(A - 15I)X = 0$$

$$\Rightarrow \begin{bmatrix} 8-15 & -6 & 2 \\ -6 & 7-15 & -4 \\ 2 & -4 & 3-15 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & 2 & 6 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [R_1 \rightarrow R_1 - R_2]$$

$$\Rightarrow \begin{bmatrix} -1 & 2 & 6 \\ 0 & -20 & -40 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} R_2 \rightarrow R_2 - 6R_1 \\ R_3 \rightarrow R_3 + 2R_1 \end{bmatrix}$$

The co-efficient matrix of these equations is of rank 2. Therefore these equations have $3 - 2 = 1$ linearly independent solution. These equation can be written as

$$-x_1 + 2x_2 + 6x_3 = 0 \quad \dots(5)$$

$$-20x_2 - 40x_3 = 0 \quad \dots(6)$$

From (6) $x_2 = -2x_3$, Let $x_3 = 1$, $x_2 = -2$ then from (5) $x_1 = 2$ therefore

$x_3[2, -2, 1]'$ is an eigen vector of A corresponding to the eigen value 15. If K_3 is any non-zero scalar, then $K_3 X_3$ is also an eigen vector of A corresponding to eigen value 15.

Example 6: Find all the eigen values and corresponding eigen vectors of the matrix
or Determine values and vector of the following matrix.

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

Solution: The characteristic equation of matrix A

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)[(3-\lambda)(2-\lambda)-2] - 2[1 \times (2-\lambda) - 1] + 1[2 - 1 \times (3-\lambda)] = 0$$

$$\Rightarrow \lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

$$\Rightarrow (\lambda - 5)(\lambda - 1)^2 = 0$$

$$\Rightarrow \lambda = 5, 1, 1$$

$$\Rightarrow \lambda_1 = 5, \lambda_2 = 1, \lambda_3 = 1$$

The eigen vectors $X = [x_1, x_2, x_3]'$ of A corresponding to eigen value $\lambda_1 = 5$ are given by the non-zero solution of the equation

$$(A - 5I)X = 0 \Rightarrow \begin{bmatrix} 2-5 & 2 & 1 \\ 1 & 3-5 & 1 \\ 1 & 2 & 2-5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -3 & 2 & 1 \\ 1 & -2 & 1 \\ 1 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -3x_1 + 2x_2 + x_3 = 0$$

$$x_1 - 2x_2 + x_3 = 0$$

$$x_1 + 2x_2 - 3x_3 = 0$$

These solution have a non-trivial solution given by

$$\frac{x_1}{2+2} = \frac{x_2}{1+3} = \frac{x_3}{6-2}$$

$$\Rightarrow x_1 = x_2 = x_3 = K_1, \text{ Let}$$

The given vectors corresponding to eigen value $\lambda_1 = 5$ is $\begin{bmatrix} K_1 \\ K_1 \\ K_1 \end{bmatrix}$. For eigen vectors

corresponding to the given value $\lambda_2 = 1$, we get

$$(A - I)X = 0$$

$$\Rightarrow \begin{bmatrix} 2 & -1 & 2 & 1 \\ 1 & 3 & -1 & 1 \\ 1 & 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow x_1 + 2x_2 + x_3 = 0$$

Hence, we obtain linearly independent solutions given by

$$x_1 = 2, x_2 = -1, x_3 = 0$$

$$x_1 = 1, x_2 = 0, x_3 = -1$$

Thus the eigen vectors corresponding to eigen value $\lambda_2 = 1$ is all linear combinations of

vectors $\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ which are obtained by the following relation

$$K_2 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + K_3 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2K_2 + K_3 \\ -K_2 \\ -K_3 \end{bmatrix}$$

Example 7: Determine the eigen values and the corresponding vectors of the matrix

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

Also find the corresponding eigen spaces.

Solution: The characteristic equation of the matrix A is

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 3 - \lambda & 2 & 4 \\ 2 & 0 - \lambda & 2 \\ 4 & 2 & 3 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (3 - \lambda)\{(-\lambda) \times (3 - \lambda) - 4\} - 2\{2 \times (3 - \lambda) - 8\} + 4\{4 - 4 \times (-\lambda)\} = 0$$

$$\Rightarrow -\lambda^3 + 6\lambda^2 + 15\lambda + 8 = 0$$

$$\Rightarrow (\lambda + 1)^2(\lambda - 8) = 0$$

$$\Rightarrow \lambda = 8, -1, -1$$

Hence eigen values of A are $\lambda_1 = 8, \lambda_2 = -1$ with algebraic multiplicity 2

Now for $\lambda_1 = 8$

$$(A - 8I)X = 0$$

$$\Rightarrow \begin{bmatrix} 3-8 & 2 & 4 \\ 2 & 0-8 & 2 \\ 4 & 2 & 3-8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -5 & 2 & 4 \\ 2 & -8 & 2 \\ 4 & 2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -5 & 2 & 4 \\ -18 & 0 & 18 \\ 9 & 0 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} R_2 \rightarrow R_2 + 4R_1 \\ R_3 \rightarrow R_3 - R_1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -5 & 2 & 4 \\ -1 & 0 & 1 \\ 9 & 0 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [R_2 \rightarrow R_2 (1/18)]$$

$$\Rightarrow \begin{bmatrix} -5 & 2 & 4 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad [R_3 \rightarrow R_3 + 9R_2]$$

$$\Rightarrow -5x_1 + 2x_2 + 4x_3 = 0$$

and $-x_1 + x_3 = 0$

$$x_1 = x_3$$

$x_1 = x_3$ and $x_3 = 2x_2$. Then eigen vector is $w X_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

For $\lambda_2 = -1$, we have

$$(A + I)X = 0$$

$$\Rightarrow \begin{bmatrix} 4 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow 2x_1 + x_2 + 2x_3 = 0 \text{ or } x_2 = -2x_1 - 2x_3$$

If we take $x_1 = 1, x_3 = 0$ then $x_2 = -2$ and if we take $x_1 = 0, x_3 = 1$ then $x_2 = -2$, Hence

$$X_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

are two linearly independent solutions by this equation therefore X_2 and X_3 are two L.I. eigen vectors of A when eigen value is -1 . If K_1, K_2 are non-zero scalars, then $K_1 X_2 + K_2 X_3, A$ gives all the eigen vectors of A when eigen value is -1

Example 8: Show that the matrices A and A' have the same eigen values.

Solution: We know that

$$(A - \lambda I)' = A' - \lambda I' = A' - \lambda I \quad [\because I' = I]$$

$$\therefore |(A - \lambda I)'| = |A' - \lambda I|$$

$$\Rightarrow |A - \lambda I| = |A' - \lambda I| \quad [\because |B'| = |B|]$$

$$\Rightarrow |A - \lambda I| = 0 \text{ iff } |A' - \lambda I| = 0$$

i.e. λ is an eigen value of A is and only if λ is an eigen value of A' .

Example 9: Show that 0 is an eigen value of a matrix if and only if the matrix is singular.

Solution: Since 0 is eigen value of $A \Rightarrow \lambda = 0$, and $|A - \lambda I| = 0 \Rightarrow |A| = 0$

$\Rightarrow A$ is singular

Conversely: If A is singular, then $|A| = 0$

$$\Rightarrow \lambda = 0 \text{ satisfies the equation } |A - \lambda I| = 0$$

$$\Rightarrow 0 \text{ is eigen value of } A$$

Example 10: Prove that the scalar λ is an eigen value of a matrix A iff $(A - \lambda I)$ is singular.

Solution: We have λ is eigen value of matrix A

$\Rightarrow \lambda$ satisfies the characteristic equation

$$\Rightarrow |A - \lambda I| = 0$$

$$\Rightarrow (A - \lambda I) \text{ is singular}$$

Conversely: If $(A - \lambda I)$ is singular matrix then there exists a non-zero vector X such that

$$(A - \lambda I)X = 0$$

$$\Rightarrow AX - \lambda IX = 0$$

$$\begin{aligned} \Rightarrow & (AX - \lambda X) = 0 \\ \Rightarrow & AX = \lambda X \\ \Rightarrow & \lambda \text{ is an eigen value of matrix } A. \end{aligned}$$

Example 11: If A is non-singular matrix, prove that A^{-1} are reciprocals of the eigen values of A .

Solution: Let X be an eigen values of A and X be a corresponding eigen vector, then

$$\begin{aligned} & AX = \lambda X \\ \Rightarrow & A^{-1}(AX) = A^{-1}(\lambda X) \\ \Rightarrow & (A^{-1}A)X = \lambda(A^{-1}X) \\ \Rightarrow & X = \lambda(A^{-1}X) \quad [\because AA^{-1} = I] \\ \Rightarrow & \frac{1}{\lambda} X = A^{-1}X \quad [\because A \text{ is non-singular} \Rightarrow \lambda \neq 0] \\ \Rightarrow & A^{-1}X = \frac{1}{\lambda} X \end{aligned}$$

$\frac{1}{\lambda}$ is eigen value of A^{-1} and X is a corresponding eigen vector.

Conversely: Let K is a eigen value of A^{-1}

$$\begin{aligned} \Rightarrow & A^{-1} \text{ is non-singular} \\ \Rightarrow & \frac{1}{K} \text{ is an eigen value of } (A^{-1})^{-1} \\ & \frac{1}{K} \text{ is an eigen value of } A \quad [\because (A^{-1})^{-1} = A] \end{aligned}$$

Thus each eigen value of A^{-1} is equal to the reciprocal of same equation value of A .

Example 12: Show that if λ is an eigen-value of matrix A , then $K + \lambda$ is an eigen value of the matrix $A + KI$.

Solution: Let λ be an eigen-value of A and X a corresponding eigen vector of a matrix A , then $X \neq 0$ is such that

$$\begin{aligned} & AX = \lambda X \quad \dots(1) \\ (A + KI)X &= AX + KIX \\ &= \lambda X + KX \quad \text{From eqn. (1)} \\ &= (\lambda + K)X \quad \dots(2) \end{aligned}$$

Since $X \neq 0$ therefore from (2) we see that $\lambda + K$ is an eigen-value of matrix $A + KI$ and X is corresponding eigen vector.

Example 13: Prove that if eigen value of A are $\lambda_1, \lambda_2, \dots, \lambda_n$, then eigen value of A^2 are $\lambda_1^2, \lambda_2^2, \dots, \lambda_n^2$

Solution: Let λ be an eigen value of the matrix A , then there exists $X \neq 0$ such that

$$AX = \lambda X$$

$$\Rightarrow A(AX) = A(\lambda X)$$

$$\Rightarrow A^2 X = \lambda(AX)$$

$$\Rightarrow A^2 X = \lambda(\lambda X) \quad [\because AX = \lambda X]$$

$$\Rightarrow A^2 X = \lambda^2 X \quad \dots(1)$$

Since X is a non-zero vector, therefore from (1) it is clear that λ^2 is an eigen value of the matrix. Hence if $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigen values of A , then $\lambda_1^2, \lambda_2^2, \dots, \lambda_n^2$ are eigen-values of the matrix A .

Example 14: Show that the eigen values of the matrix $A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$ are ± 1 . Find the corresponding eigen vectors.

Solution: The Characteristic equation of A is

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{bmatrix} \cos \theta - \lambda & \sin \theta \\ \sin \theta & -\cos \theta - \lambda \end{bmatrix} = 0$$

$$\Rightarrow -(\cos \theta - \lambda)(\cos \theta + \lambda) - \sin^2 \theta = 0$$

$$\Rightarrow \cos^2 \theta - \lambda^2 + \sin^2 \theta = 0$$

$$\Rightarrow \lambda^2 - 1 = 0 \Rightarrow \lambda^2 = 1$$

$$\Rightarrow \lambda = \pm 1$$

Now $\lambda = 1$ the matrix of A is

$$(A - \lambda I)X = 0$$

$$\Rightarrow (A - I)X = 0$$

$$\Rightarrow \begin{bmatrix} \cos \theta - 1 & \sin \theta \\ \sin \theta & -\cos \theta - 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 \sin^2 \theta / 2 & 2 \sin \theta / 2 \cos \theta / 2 \\ 2 \sin \theta / 2 \cos \theta / 2 & -2 \cos^2 \theta / 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -\sin \theta / 2 & \cos \theta / 2 \\ \sin \theta / 2 & -\cos \theta / 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \left[R_1 \rightarrow R_2 \left(\frac{1}{2 \sin (\theta / 2)} \right), R_2 \rightarrow R_2 \left(\frac{1}{2 \cos (\theta / 2)} \right) \right]$$

$$\Rightarrow \begin{bmatrix} -\sin \theta / 2 & \cos \theta / 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} [R_2 \rightarrow R_2 + R_1]$$

$$\Rightarrow -\sin \theta / 2 \cdot x_1 + \cos \theta / 2 \cdot x_2 = 0$$

This equation is satisfied if $x_1 = \cos \theta / 2, x_2 = \sin \theta / 2$

Hence: $\lambda = 1$ the corresponding eigen vector is $\begin{bmatrix} \cos(\theta / 2) \\ \sin(\theta / 2) \end{bmatrix}$

Similarly, for $\lambda = -1$, the corresponding eigen vector is $\begin{bmatrix} \sin \theta / 2 \\ -\cos(\theta / 2) \end{bmatrix}$



Comprehensive Exercise 4.1

(Eigen value and Eigen Vectors)

1. Find the characteristic equation of the following matrices

(i) $\begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$ (ii) $\begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & -1 & 2 \end{bmatrix}$

2. Determine the eigen values and the corresponding eigen vectors of the following matrices

(i) $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ (iii) $\begin{bmatrix} 0 & 3 \\ 2 & -1 \end{bmatrix}$

3. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

4. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$
5. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$
6. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 3 & 6 & 6 \\ 0 & 2 & 0 \\ -3 & -12 & -6 \end{bmatrix}$
7. Show that the eigen values of the matrix $A = \begin{bmatrix} b & c & a \\ c & a & b \\ a & b & c \end{bmatrix}$ are
 $a + b + c, \pm \sqrt{a^2 + b^2 + c^2 - ab - bc - ca}.$



Answers

1. (i) $\lambda^2 - 8\lambda + 16 = 0$, (ii) $\lambda^2 + 3 = 0$ (iii) $\lambda^2 - 12 = 0$ (iv) $\lambda = 1, 2, 3$
2. (i) $\lambda = i, -i, \begin{bmatrix} i \\ 1 \end{bmatrix}, \begin{bmatrix} -i \\ 1 \end{bmatrix}$ (ii) $\lambda = 4, -1, \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ (iii) $\lambda = -3, 2, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix}$
3. $\lambda = 0, 0, 3, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ 4. $\lambda = 2, 2, 2, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ 5. $\lambda = 1, 2, 2, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$
6. $\lambda = 0, 2, -3, \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 12 \\ -5 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

4.4.1 Cayley-Hamilton Theorem

Theorem 7: Every square matrix satisfies its own characteristic equation

or Every matrix is a zero of its characteristic polynomial.

Example 15: Find the characteristic equation of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix}$ and verify

that it satisfied by A and hence of obtain A^{-1}

Solution: Here, $A - \lambda I = \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0-\lambda & 1 & 2 \\ 2 & -3-\lambda & 0 \\ 1 & 1 & -1-\lambda \end{bmatrix}$

$\therefore$ The characteristic equation of A is

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 0-\lambda & 1 & 2 \\ 2 & -3-\lambda & 0 \\ 1 & 1 & -1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda\{(-3-\lambda)(-1-\lambda)-0\} - 1\{2(-1-\lambda)-0\} + 2\{2-1(-3-\lambda)\} = 0$$

$$\Rightarrow -\lambda^3 + 4\lambda^2 - \lambda + 12 = 0$$

$$\Rightarrow \lambda^3 + 4\lambda^2 - \lambda - 12 = 0 \quad \dots(1)$$

Now we shall verify that A satisfies its own characteristic equation (1) that is

$$A^3 + 4A^2 - A - 12I = 0 \quad \dots(2)$$

We know that

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

$$A^2 = A \cdot A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0+2+2 & 0-3+2 & 0+0-2 \\ 0-6-0 & 2+9+0 & 4+0+0 \\ 0+2-1 & 1-3-1 & 2+0+1 \end{bmatrix} = \begin{bmatrix} 4 & -1 & -2 \\ -6 & 11 & 4 \\ 1 & -3 & 3 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 4 & -1 & -2 \\ -6 & 11 & 4 \\ 1 & -3 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0-2-2 & 4+3-2 & 8+0+2 \\ 0+22+4 & -6-33+4 & -12+0-4 \\ 0-6+3 & 1+9+3 & 2+0-3 \end{bmatrix} = \begin{bmatrix} -4 & 5 & 10 \\ 26 & -35 & -16 \\ -3 & 13 & -1 \end{bmatrix}$$

Therefore

$$A^3 + 4A^2 - A - 12I = \begin{bmatrix} -4 & 5 & 10 \\ 26 & -35 & -16 \\ -3 & 13 & -1 \end{bmatrix} + 4 \begin{bmatrix} 4 & -1 & -2 \\ -6 & 11 & 4 \\ 1 & -3 & 3 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix} - 12 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} -4+16-12 & 5-4-1 & 10-8-2 \\ 26-24-2 & -35+44+3-12 & -16+16 \\ -3+4-1 & 13-12-1 & -1+12+1-12 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O
\end{aligned}$$

Multiplying (2) by A^{-1} , we find

$$A^2 + 4A - I - 12A^{-1} = O$$

$$\therefore A^{-1} = \frac{1}{12}(A^2 + 4A - I)$$

$$\begin{aligned}
\text{Now, } A^2 + 4A - I &= \begin{bmatrix} 4 & -1 & -2 \\ -6 & 11 & 4 \\ 1 & -3 & 3 \end{bmatrix} + 4 \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 4+0-1 & -1+4-0 & -2+8-0 \\ -6+8-0 & 11-12-1 & 4+0-0 \\ 1+4-0 & -3+4-0 & 3-4-1 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 3 & 6 \\ 2 & -2 & 4 \\ 5 & 1 & -2 \end{bmatrix}
\end{aligned}$$

$$\therefore A^{-1} = \frac{1}{12}(A^2 + 4A - I) = \frac{1}{12} \begin{bmatrix} 3 & 3 & 6 \\ 2 & -2 & 4 \\ 5 & 1 & -2 \end{bmatrix}$$

Example 16: Obtain the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ and verify

that it is satisfied by A and hence find A^{-1}

Solution: Here, $|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 2 \\ 0 & 2-\lambda & 1 \\ 2 & 0 & 3-\lambda \end{vmatrix}$

$$\begin{aligned}
&= (1-\lambda)\{(2-\lambda)(3-\lambda)-0\} + 2\{0-2(2-\lambda)\} \\
&= (2-\lambda)\{(1-\lambda)(3-\lambda)-4\} \\
&= (2-\lambda)[\lambda^2 - 4\lambda - 1] \\
&= -(\lambda^3 - 6\lambda^2 + 7\lambda + 2)
\end{aligned}$$

$\therefore$ The characteristic equation of A is $|A - \lambda I| = 0$

$$\Rightarrow \lambda^3 + 6\lambda^2 + 7\lambda + 2 = 0 \quad \dots(1)$$

Verification of Cayley-Hamilton theorem of a matrix A .

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

$$A^2 = A \cdot A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix}$$

Now

$$A^3 - 6A^2 + 7A + 2I$$

$$= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - 6 \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} + 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - \begin{bmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 14 \\ 0 & 14 & 7 \\ 14 & 0 & 21 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{bmatrix} - \begin{bmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{bmatrix}$$

$$A^3 - 6A^2 + 7A + 2I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \quad \dots(2)$$

Hence, Cayley-Hamilton theorem is verified.

Multiplying (2) by A^{-1} , we find

$$A^2 - 6A + 7I + 2A^{-1} = O$$

$$\therefore A^{-1} = -\frac{1}{2}(A^2 - 6A + 7I)$$

$$= -\frac{1}{2} \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} - \frac{7}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -3 & 0 & 2 \\ -1 & 1/2 & 1/2 \\ 2 & 0 & -1 \end{bmatrix}$$

Example 17: Show that the matrix $A = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{bmatrix}$ satisfies Cayley-Hamilton theorem.

Hence find A^{-1}

Solution: The characteristic equation of A is

$$\begin{aligned} |A - \lambda I| &= 0 \\ \Rightarrow \begin{vmatrix} 0-\lambda & 0 & 1 \\ 3 & 1-\lambda & 0 \\ 2 & 1 & 4-\lambda \end{vmatrix} &= 0 \\ \Rightarrow (-\lambda) \{ (1-\lambda)(4-\lambda) - 0 \} + 1 \{ 3 - 2(1-\lambda) \} &= 0 \\ \Rightarrow -\lambda^3 + 5\lambda^2 - 2\lambda + 1 &= 0 \\ \Rightarrow \lambda^3 - 5\lambda^2 + 2\lambda - 1 &= 0 \end{aligned} \quad \dots(1)$$

Now we shall show A satisfies Cayley-Hamilton theorem

i.e. $A^3 - 5A^2 + 2A - I = O$

Here $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{bmatrix}$

$\therefore A^2 = A \cdot A = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 1 & 3 \\ 11 & 5 & 18 \end{bmatrix}$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 1 & 3 \\ 11 & 5 & 18 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 5 & 18 \\ 9 & 4 & 15 \\ 51 & 23 & 83 \end{bmatrix}$$

Now $A^3 - 5A^2 + 2A - I = \begin{bmatrix} 11 & 5 & 18 \\ 9 & 4 & 15 \\ 51 & 23 & 83 \end{bmatrix} - 5 \begin{bmatrix} 2 & 1 & 4 \\ 3 & 1 & 3 \\ 11 & 5 & 18 \end{bmatrix} + 2 \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 11 & 5 & 20 \\ 15 & 6 & 15 \\ 55 & 25 & 91 \end{bmatrix} - \begin{bmatrix} 11 & 5 & 20 \\ 15 & 6 & 15 \\ 55 & 25 & 91 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

Hence, A satisfies Cayley-Hamilton theorem.

Example 18: Show that the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ satisfies its own characteristic

equation. Hence find A^{-1}

Solution: The characteristic equation of A is

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 0 \\ 2 & -1-\lambda & 0 \\ 0 & 0 & -1-\lambda \end{vmatrix}$$

$$= (-1-\lambda)\{(1-\lambda)(-1-\lambda)-4\}$$

$$= -(\lambda+1)(\lambda^2-5) = -(\lambda^3 + \lambda^2 - 5\lambda - 5)$$

Hence, the characteristic equation of matrix A is

$$|A - \lambda I| = 0$$

$$\Rightarrow \lambda^3 + \lambda^2 - 5\lambda - 5 = 0$$

Now we shall verify that

$$A^3 + A^2 - 5A - 5I = O$$

Here

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$A^2 = A \cdot A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$A^3 = A^2 \cdot A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 0 \\ 10 & -5 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{aligned}
 \therefore A^3 + A^2 - 5A - 5I &= \begin{bmatrix} 5 & 10 & 0 \\ 10 & -5 & 0 \\ 0 & 0 & -1 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 10 & 10 & 0 \\ 10 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 10 & 10 & 0 \\ 10 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O
 \end{aligned}$$

Multiplying (2) by A^{-1} , we find

$$A^2 + A - 5I - 5A^{-1} = O$$

$$\Rightarrow A^{-1} = \frac{1}{5}(A^2 + A - 5I)$$

$$\begin{aligned}
 \text{Now } A^2 + A - 5I &= \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} + \begin{bmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -5 \end{bmatrix} \\
 &= \begin{bmatrix} 5+1-5 & 0+2+0 & 0+0+0 \\ 0+2+0 & 5-1-5 & 0+0+0 \\ 0+0+0 & 0+0+0 & 1-1-5 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -5 \end{bmatrix}
 \end{aligned}$$

$$\text{Thus } A^{-1} = \frac{1}{5} \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -5 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 1/5 & 2/5 & 0 \\ 2/5 & -1/5 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Example 19: Find the characteristic equation of matrix

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \text{ and verify that it is satisfied by } A \text{ and hence obtain } A^{-1}$$

Solution: The characteristic equation of matrix A is

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)\{(2-\lambda)^2 - 1\} + 1\{(-1)(2-\lambda) + 1\} + 1\{1 - (2-\lambda)\} = 0$$

$$\Rightarrow (2-\lambda)(3-4\lambda+\lambda^2) + (\lambda-1) + (\lambda-1) = 0$$

$$\Rightarrow -\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

$$\text{Now to find } A^3 - 6A^2 + 9A - 4I = O \quad \dots(1)$$

We know that
$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$A^2 = A \cdot A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

$$\begin{aligned} \text{Thus } A^3 - 6A^2 + 9A - 4I &= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - 6 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} + 9 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \\ &\quad - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

Multiplying (1) A^{-1} , we get

$$A^2 - 6A + 9I - 4A^{-1} = O$$

$$\Rightarrow A^{-1} = \frac{1}{4} [A^2 - 6A + 9I]$$

$$\text{Now } A^2 - 6A + 9I = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} + \begin{bmatrix} -12 & 6 & -6 \\ 6 & -12 & 6 \\ -6 & 6 & -12 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$



Comprehensive Exercise 4.2

1. Find the eigen values of matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{bmatrix}$ verify that matrix A satisfies its

characteristic equation.

2. Show that the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ satisfies Cayley-Hamilton theorem. Also

determine the eigen values and corresponding eigen vectors of matrix A .

3. Verify that the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ satisfies its own characteristic equation.

Is it true for every square matrix ? State the theorem that applies here.

4. Verify Cayley-Hamilton theorem for matrix $A = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ -2 & 1 & 4 \end{bmatrix}$ hence or otherwise

evaluate A^{-1}

5. Verify Cayley-Hamilton $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix}$, hence or otherwise evaluate A^{-1}

6. What is Cayley-Hamilton theorem?



Answers

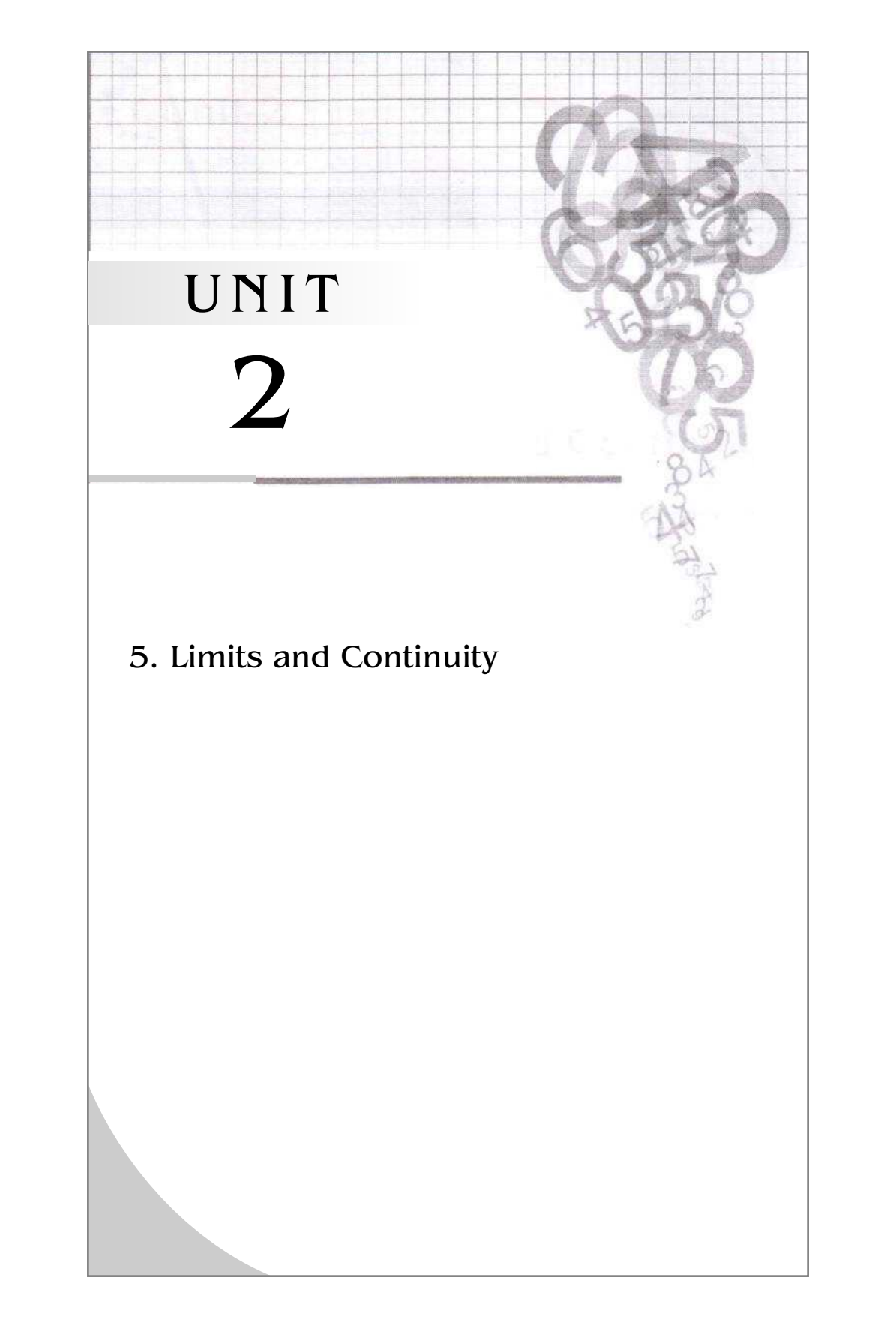
1. $1, -4, 7$
2. Eigen values are $\lambda = 5, 1$ 1 eigen vectores corresponding to the eigen value $\lambda = 5$ is $[1, 1, 1]$ corresponding to the eigen value $\lambda = 1$ there are two L.I. vectores $[1, 0, -1]$ and $[1, 1, -3]$

3. Yes, $A^{-1} = \begin{bmatrix} 1/5 & 2/5 & 0 \\ 2/5 & -1/5 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

4. $A^{-1} = \frac{1}{5} \begin{bmatrix} 4 & 1 & -1 \\ -12 & 2 & 3 \\ 5 & 0 & 0 \end{bmatrix}$

5. Yes, $A^{-1} = \frac{1}{40} \begin{bmatrix} -4 & 11 & -5 \\ -4 & 1 & 25 \\ 8 & -2 & -10 \end{bmatrix}$



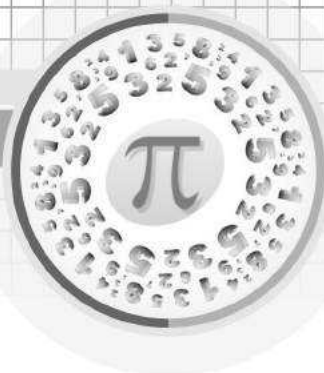


UNIT 2

5. Limits and Continuity



Unit-2



Chapter

5

Limits and Continuity

5.1 Concept of Limits

Consider the real function $f(x) = \frac{x^2 - 16}{x - 4}$. This function is not defined at $x = 4$, because the value of the function at $x = 4$ is $\frac{0}{0}$ which is meaningless.

Now suppose that x is not exactly 4 but close to 4 (*i.e.* $x = 3.9999$ or $x = 4.0001$ say) so that $x - 4 \neq 0$ and we shall have

$$f(x) = \frac{(x - 4)(x + 4)}{x - 4} = x + 4$$

It follows that when x is close to 4, $f(x)$ becomes close to 8. In other words, as x tends to 4 either from left or from right the difference between $f(x)$ and 8.

$$\begin{aligned} \text{i.e. } f(x) - 8 &= \frac{(x - 4)(x + 4)}{(x - 4)} - 8 \\ &= \frac{(x - 4)^2}{x - 4} = x - 4 \end{aligned}$$

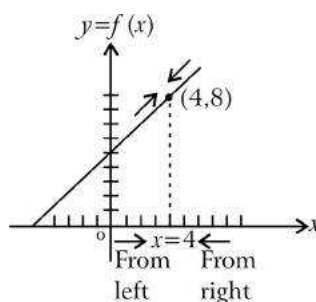


Fig. 5.1

tends to zero. This is the intuitive concept of limits. Clearly, the concept of limits can somehow extend the definition of a function to that point, where it is not defined. This necessitates the study of the behavior of a function for those values of x that are very close

to some particular value, where the function is not defined. We now define. If a real valued function $f(x)$ of a real variable x gets arbitrarily close to some number ' l ' as x gets close to some constant a (from either side of a), without ever being equal to ' a ' then we say that the limit of $f(x)$ is ' l ' as x tends to ' a ' and write it as

$$\lim_{x \rightarrow a} f(x) = l$$

5.1.1 Definition of Limit

[B.C.A. (Agra) 2006; B.C.A. (Meerut) 2004, 2008]

A function $f(x)$ has a limit ' l ' as x tends to ' a ' if for each number $\epsilon > 0$, however small, a positive number δ can be found such that $|f(x) - l| < \epsilon$ whenever $|x - a| < \delta$. In symbols, we then write $\lim_{x \rightarrow a} f(x) = l$.

i.e., $l - \epsilon < f(x) < l + \epsilon$ for all those values of x (except at $x = a$) which lies between $a - \delta$ to $a + \delta$ (i.e., $a - \delta < x < a + \delta$)

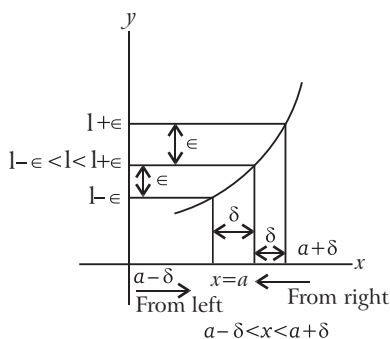


Fig. 5.2

In short we can say: $\lim_{x \rightarrow a} f(x) = l$ iff given $\epsilon > 0$, $\exists \delta > 0$ such that

$$|f(x) - l| < \epsilon \text{ whenever } 0 < |x - a| < \delta$$

5.1.2 Left Hand and Right Hand Limits

[B.C.A. (Bundelkhand) 2008]

- (i) **Left hand limit:** A function $f(x)$ is said to tend to ' l ' as x tends to ' a ' if for each number $\epsilon > 0$, there exist a number $\delta > 0$, such that

$$|f(x) - l| < \epsilon \quad \text{whenever } a - \delta < x < a$$

So that

$$l - \epsilon < f(x) < l + \epsilon \text{ whenever } a - \delta < x < a$$

The limit in this case is called left hand limit (L.H.L.) and denoted by $f(a - 0)$.

Thus

$$f(a - 0) = \lim_{x \rightarrow a - 0} f(x)$$

- (ii) **Right hand limit:** A function $f(x)$ is said to tend to ' l ' as x tends to ' a ' if for each number $\epsilon > 0$, there exist a number $\delta > 0$, such that

$$|f(x) - l| < \epsilon \quad \text{whenever } a < x < a + \delta$$

So that

$$l - \epsilon < f(x) < l + \epsilon \text{ whenever } a < x < a + \delta$$

The limit in this case is called right hand limit (R.H.L.) and denoted by $f(a + 0)$.

Thus

$$f(a + 0) = \lim_{x \rightarrow a + 0} f(x)$$

(iii) Existence of a limit at a point**[B.C.A (Meerut) 2004]**

A function $f(x)$ is said to tend to a limit ' l ' as x tends to ' a ' if both the left and right hand limits exist and are equal to ' l ' i.e. if $f(a-0) = f(a+0) = l$ then

$$\lim_{x \rightarrow a} f(x) = l$$

Note-

How to find the left hand and right hand limits :

- (i) To find $f(a-0)$ or $\lim_{x \rightarrow a-0} f(x)$

We first put $x = a - h$ in $f(x)$ and then take the limit as $h \rightarrow 0$.

$$\text{Thus } f(a-0) = \lim_{x \rightarrow a-0} f(x) = \lim_{h \rightarrow 0} f(a-h)$$

- (ii) To find $f(a+0)$ or $\lim_{x \rightarrow a+0} f(x)$

We first put $x = a + h$ in $f(x)$ and then take the limit $h \rightarrow 0$.

$$\text{Thus } f(a+0) = \lim_{x \rightarrow a+0} f(x) = \lim_{h \rightarrow 0} f(a+h)$$

The limit of the function $f(x)$ at $x = a$ does not exist if either one or both (L.H.L. and R.H.L.) limits do not exist. Even if both limits exist but are not equal in value then the limit of the function $f(x)$ at $x = a$ does not exist.

5.1.3 Useful Formulae in Finding the Limits of Certain Functions

If $f(x)$ and $g(x)$ are two real functions and k is any constant, then

(i) $\lim_{x \rightarrow a} k = k$

(ii) $\lim_{x \rightarrow a} \{k f(x)\} = k \lim_{x \rightarrow a} f(x)$

(iii) $\lim_{x \rightarrow a} \sqrt{f(x)} = \sqrt{\lim_{x \rightarrow a} f(x)}$

(iv) $\lim_{x \rightarrow a} [f(x)]^{m/n} = \left[\lim_{x \rightarrow a} f(x) \right]^{m/n}$

(v) $\lim_{x \rightarrow a} [f\{g(x)\}] = \lim_{x \rightarrow b} f(x)$ where $b = \lim_{x \rightarrow a} g(x)$

(vi) $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

(vii) If n is an integer or fractions then

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \text{ provided that } a > 0$$

(viii) If m and n are rational numbers, then

$$\lim_{x \rightarrow a} \frac{x^m - a^m}{x^n - a^n} = \frac{m}{n} a^{m-n} \text{ provided that } a > 0$$

(ix) $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$

- (x) $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$
- (xi) $\lim_{x \rightarrow 0} \log |x| = -\infty$
- (xii) $\lim_{x \rightarrow 0} \sin \frac{1}{x} = a$ finite quantity lying between -1 and 1 .



Illustrative Examples

(Limits)

Example 1: Do the following limits exist? If yes find them:

- (a) $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$ [B.C.A. (Meerut) 2005, 2011; B.C.A. (Rohilkhand) 2007, 2010]
- (b) $\lim_{x \rightarrow 0} \frac{1}{1 + e^{1/x}}$ or $\lim_{x \rightarrow 0} \frac{1}{1 + e^{-1/x}}$ [B.C.A. (Meerut) 2008]
- (c) $\lim_{x \rightarrow 0} 2^{1/(x-1)}$
- (d) $\lim_{x \rightarrow 1} \sin \frac{1}{x-1}$ [B.C.A. (Kanpur) 2006]
- (e) $\lim_{x \rightarrow 0} \frac{e^{1/x}}{e^{1/x} + 1}$ [B.C.A. (Meerut) 2006; B.C.A. (Rohikhand) 2000, 2005, 2008]
- (f) $\lim_{x \rightarrow 1} f(x)$ where $f(x) = \begin{cases} 3x-2 & \text{when } x < 1 \\ 4x^2-3x & \text{when } x > 1 \end{cases}$ [B.C.A. (Bundelkhand) 2007]
- (g) $\lim_{x \rightarrow 0} \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}}$ does not exist [B.C.A. (Meerut) 2007]

Solution:

- (a) $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$, taking $f(x) = x \sin \frac{1}{x}$
- $$\begin{aligned} \lim_{h \rightarrow 0} f(0-h) &= \text{L. H. L.} = f(0-0) \\ &= \lim_{h \rightarrow 0} (0-h) \sin \frac{1}{0-h} \\ &= \lim_{h \rightarrow 0} h \sin \frac{1}{h} = \lim_{h \rightarrow 0} (h) \lim_{h \rightarrow 0} \sin \frac{1}{h} \\ &= 0 \times a \text{ finite quantity between } -1 \text{ and } 1 \\ &= 0 \end{aligned}$$

$$\lim_{h \rightarrow 0} f(0+h) = \text{R. H. L.} = f(0+0) = \lim_{h \rightarrow 0} (0+h) \sin \frac{1}{0+h}$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} h \sin \frac{1}{h} = \lim_{h \rightarrow 0} (h) \lim_{h \rightarrow 0} \sin \frac{1}{h} \\
 &= 0 \times \text{a finite quantity between } -1 \text{ and } 1 \\
 &= 0
 \end{aligned}$$

$$f(0-h) = (0+h) \text{ i.e. L.H.L.} = \text{R.H.L.}$$

$$\therefore \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

$$(b) \quad \lim_{x \rightarrow 0} \frac{1}{1+e^{1/x}} \text{ taking } f(x) = \frac{1}{1+e^{1/x}}$$

$$\begin{aligned}
 \lim_{h \rightarrow 0} f(0-h) &= \text{L. H. L.} = f(0-0) = \lim_{h \rightarrow 0} \frac{1}{1+e^{1/(0-h)}} = \lim_{h \rightarrow 0} \frac{1}{1+e^{-1/h}} \\
 &= \frac{1}{1+e^{-\infty}} = \frac{1}{1+0} = 1 \because e^{-\infty} = 0
 \end{aligned}$$

$$\begin{aligned}
 \lim_{h \rightarrow 0} f(0+h) &= \text{R. H. L.} = f(0+0) \\
 &= \lim_{h \rightarrow 0} \frac{1}{1+e^{1/(0+h)}} \\
 &= \frac{1}{1+e^{\infty}} = \frac{1}{1+\infty} = \frac{1}{\infty} = 0 \quad \because e^{\infty} = \infty
 \end{aligned}$$

$$\text{i.e. L.H.L.} \neq \text{R.H.L.}$$

Therefore limit does not exist.

$$(c) \quad \lim_{x \rightarrow 1} 2^{1/(x-1)} \text{ taking } f(x) = 2^{1/(x-1)}$$

$$\begin{aligned}
 \lim_{h \rightarrow 0} f(1-h) &= \text{L. H. L.} = f(1-0) = \lim_{h \rightarrow 0} 2^{1/(1-h-1)} \\
 &= \lim_{h \rightarrow 0} 2^{-1/h} = 2^{-1/0} = 2^{-\infty} = \frac{1}{2^{\infty}} = \frac{1}{\infty} = 0
 \end{aligned}$$

$$\lim_{h \rightarrow 0} f(1+h) = \text{R. H. L.} = f(1+0) = \lim_{h \rightarrow 0} 2^{1/(1+h)-1} = \lim_{h \rightarrow 0} 2^{1/h} = 2^{1/0} = 2^{\infty} = \infty$$

Since L.H.L. $\neq$ R.H.L.

$$\therefore \lim_{x \rightarrow 1} 2^{1/(x-1)} \text{ does not exist.}$$

$$(d) \quad \lim_{x \rightarrow 1} \sin \left(\frac{1}{x-1} \right), \text{ taking } f(x) = \sin \left(\frac{1}{x-1} \right)$$

$$\begin{aligned}
 \lim_{h \rightarrow 0} f(1-h) &= \text{L. H. L.} = f(1-0) = \lim_{h \rightarrow 0} \sin \left(\frac{1}{1-h-1} \right) \\
 &= \lim_{h \rightarrow 0} \sin \left(-\frac{1}{h} \right) = - \lim_{h \rightarrow 0} \sin \frac{1}{h}
 \end{aligned}$$

Now as $h \rightarrow 0$, $\sin \frac{1}{h}$ is finite and oscillates between -1 to 1 , so it does not tend to any unique and definite value as $h \rightarrow 0$. Hence L.H.L. does not exist.

Similarly, $\lim_{h \rightarrow 0} f(1+h) = \text{R. H. L.} = f(1+0)$

$$= \lim_{h \rightarrow 0} \sin \frac{1}{1+h-1} = \lim_{h \rightarrow 0} \sin \frac{1}{h} = \text{value oscillates between } -1 \text{ and } 1$$

$\therefore$ R.H.L. does not exist.

Thus $\lim_{x \rightarrow 1} \sin \frac{1}{x-1}$ does not exist.

(e) $\lim_{x \rightarrow 0} \frac{e^{1/x}}{e^{1/x} + 1}$ Let $f(x) = \frac{e^{1/x}}{e^{1/x} + 1}$

$$\begin{aligned} \lim_{h \rightarrow 0} f(0-h) &= \text{L. H. L.} = f(0-0) = \lim_{h \rightarrow 0} \frac{e^{1/0-h}}{e^{1/0-h} + 1} = \lim_{h \rightarrow 0} \frac{e^{-1/h}}{e^{-1/h} + 1} \\ &= \frac{e^{-1/0}}{e^{-1/0} + 1} = \frac{e^{-\infty}}{e^{-\infty} + 1} = \frac{0}{0+1} = 0 \quad \because e^{-\infty} = 0 \end{aligned}$$

$$\lim_{h \rightarrow 0} f(0+h) = \text{R. H. L.} = f(0+0) = \lim_{h \rightarrow 0} \frac{e^{1/0+h}}{e^{1/0+h} + 1}$$

$$= \lim_{h \rightarrow 0} \frac{e^{1/h}}{e^{1/h} + 1} \quad \text{from } \frac{\infty}{\infty}$$

$$= \lim_{h \rightarrow 0} \frac{1}{1 + e^{-1/h}} = \frac{1}{1 + e^{-\infty}} = \frac{1}{1+0} = 1$$

[divide the denominator and numerator by $e^{1/h}$]

Since L.H.L. $\neq$ R.H.L.

$\lim_{x \rightarrow 0} \frac{e^{1/x}}{e^{1/x} + 1}$ does not exist.

(f) $\lim_{x \rightarrow 1} f(x)$ where $f(x) = \begin{cases} 3x-2 & \text{when } x < 1 \\ 4x^2-3x & \text{when } x > 1 \end{cases}$

$$\begin{aligned} \lim_{h \rightarrow 0} f(1-h) &= \text{L. H. L.} = f(1-0) = \lim_{h \rightarrow 0} 3(1-h) - 2 \quad \because 1-h < 1 \\ &= \lim_{h \rightarrow 0} (3-3h) - 2 = 1 \end{aligned}$$

$$\begin{aligned} \lim_{h \rightarrow 0} f(1+h) &= \text{R. H. L.} = f(1+0) = \lim_{h \rightarrow 0} 4(1+h)^2 - 3(1+h) \quad \because 1+h > 1 \\ &= \lim_{h \rightarrow 0} (4h^2 + 5h + 1) = 1 \end{aligned}$$

$\therefore$ L.H.L. = R.H.L. = 1

Hence, $\lim_{x \rightarrow 1} f(x) = 1$

Example 2: Prove that $\lim_{x \rightarrow 1} [x]$ does not exist.

Solution: Let $f(x) = [x]$

$$\lim_{h \rightarrow 0} f(1-h) = \text{L.H.L.} = f(1-0) = \lim_{h \rightarrow 0} [1-h] = \lim_{h \rightarrow 0} 0 = 0$$

$$\lim_{h \rightarrow 0} f(1+h) = \text{R.H.L.} = f(1+0) = \lim_{h \rightarrow 0} [1+h] = \lim_{h \rightarrow 0} 1 = 1$$

Since L.H.L. $\neq$ R.H.L., therefore $\lim_{x \rightarrow 1} [x]$ does not exist.

Example 3: Prove that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Solution: L.H.L. $= f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{|0-h|}{0-h} = \lim_{h \rightarrow 0} \frac{h}{-h} = \lim_{h \rightarrow 0} (-1) = -1$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{|0+h|}{0+h} = \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} (1) = 1$$

$f(0+0) \neq f(0-0)$ i.e., R.H.L. $\neq$ L.H.L.

Therefore limit of a function $f(x) = \frac{|x|}{x}$ does not exist at $x=0$.

Example 4: Evaluate $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$

Solution: Put $x = \frac{1}{y}$ so that when $x \rightarrow \infty$, $y \rightarrow 0$ and thus

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sin x}{x} &= \lim_{y \rightarrow 0} \frac{\sin(1/y)}{1/y} = \lim_{y \rightarrow 0} y \sin \frac{1}{y} = \lim_{y \rightarrow 0} y \lim_{y \rightarrow 0} \sin \frac{1}{y} \\ &= 0 \times \text{finite value oscillate between } -1 \text{ and } 1 = 0 \end{aligned}$$

Thus $\lim_{y \rightarrow 0} y \sin \frac{1}{y} = 0$ or $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$

Example 5: If $\lim_{x \rightarrow 2} \frac{x^n - 2^n}{x - 2} = 80$ find the value of n

Solution: We have

$$\lim_{x \rightarrow 2} \frac{x^n - 2^n}{x - 2} = n(2)^{n-1} = 80 = 5(2)^4 = 5(2)^{5-1}$$

$$\Rightarrow n = 5$$

Example 6: If $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2}$ find the value of k .

Solution: We have

$$\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{x^4 - 1^4}{x - 1} = 4(1)^{4-1} = 4$$

and
$$\lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2} = \lim_{x \rightarrow k} \frac{\frac{x^3 - k^3}{x - k}}{\frac{x^2 - k^2}{x - k}} = \frac{\lim_{x \rightarrow k} \left(\frac{x^3 - k^3}{x - k} \right)}{\lim_{x \rightarrow k} \left(\frac{x^2 - k^2}{x - k} \right)} = \frac{3(k)^{3-1}}{2(k)^{2-1}} = \frac{3}{2} k$$

Therefore we have
$$\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2} \Rightarrow 4 = \frac{3}{2} k \Rightarrow k = \frac{8}{3}$$

Example 7: Evaluate
$$\lim_{x \rightarrow 0} \frac{(1+x)^6 - 1}{(1+x)^2 - 1}$$

Solution: We have
$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(1+x)^6 - 1}{(1+x)^2 - 1} &= \lim_{x \rightarrow 0} \frac{(1 + 6x + 15x^2 + \dots) - 1}{(1 + 2x + x^2) - 1} \\ &= \lim_{x \rightarrow 0} \frac{6x + 15x^2 + \dots}{2x + x^2} = \lim_{x \rightarrow 0} \frac{6 + 15x + \dots}{2 + x} = \frac{6}{2} = 3 \end{aligned}$$

Example 8: Evaluate
$$\lim_{x \rightarrow 0} \frac{1 - \cos 6x}{x}$$

Solution:
$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos 6x}{x} &= \lim_{x \rightarrow 0} \frac{2 \sin^2 3x}{x} \\ &= \lim_{x \rightarrow 0} \left(6 \times \frac{\sin 3x}{3x} \times \sin 3x \right) \\ &= 6 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \lim_{x \rightarrow 0} \sin 3x = 6 \times 1 \times 0 = 0 \end{aligned}$$

Example 9: Evaluate
$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

Solution:
$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} &= \lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{x^3 \cos x} \\ &= \lim_{x \rightarrow 0} \frac{\sin x \left(2 \sin^2 \frac{x}{2} \right)}{x^3 \cos x} \\ &= \lim_{x \rightarrow 0} 2 \left(\frac{\sin x}{x} \right) \left(\frac{\sin x/2}{x/2} \right)^2 \left(\frac{1}{4 \cos x} \right) \\ &= \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \lim_{x \rightarrow 0} \left(\frac{\sin x/2}{x/2} \right)^2 \lim_{x \rightarrow 0} \left(\frac{1}{\cos x} \right) \\ &= \frac{1}{2} \times 1 \times (1)^2 \times 1 = \frac{1}{2} \end{aligned}$$

Example 10: Evaluate $\lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{(\pi - 2x)^2}$

Solution: Put $\pi - 2x = 2t$ or $\pi/2 - x = t$

So that when $x \rightarrow \frac{\pi}{2}$, $t = 0$, and we have

$$\begin{aligned} \lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{(\pi - 2x)^2} &= \lim_{t \rightarrow 0} \frac{1 + \cos(\pi - 2t)}{4t^2} \\ &= \lim_{t \rightarrow 0} \frac{1 - \cos 2t}{4t^2} = \lim_{t \rightarrow 0} \frac{2\sin^2 t}{4t^2} = \frac{1}{2} \lim_{t \rightarrow 0} \left(\frac{\sin t}{t} \right)^2 = \frac{1}{2} \times 1 = \frac{1}{2} \end{aligned}$$

Example 11: Evaluate $\lim_{x \rightarrow \pi/4} \frac{1 - \tan x}{x - \pi/4}$

Solution: Put $x - \frac{\pi}{4} = t$ or $x = \frac{\pi}{4} + t$ so that when $x \rightarrow \frac{\pi}{4}$, $t \rightarrow 0$

and we have

$$\begin{aligned} \lim_{x \rightarrow \pi/4} \frac{1 - \tan x}{x - \pi/4} &= \lim_{t \rightarrow 0} \frac{1 - \tan(\pi/4 + t)}{t} \\ &= \lim_{t \rightarrow 0} \frac{1}{t} \left[1 - \frac{1 + \tan t}{1 - \tan t} \right] = \lim_{t \rightarrow 0} \frac{-2 \tan t}{t(1 - \tan t)} \\ &= \lim_{t \rightarrow 0} \left(\frac{-2}{1 - \tan t} \times \frac{\tan t}{t} \right) \\ &= \lim_{t \rightarrow 0} \frac{-2}{1 - \tan t} \times \lim_{t \rightarrow 0} \frac{\tan t}{t} = -\frac{2}{1} \times 1 = -2 \end{aligned}$$

Example 12: Evaluate the limit of the function $f(x)$ at $x = 2$ defined by

$$f(x) = \frac{|x - 2|}{x - 2}$$

Solution: We have $f(x) = \frac{|x - 2|}{x - 2}$

$$\text{R.H.L.} = f(2+0) = \lim_{h \rightarrow 0} f(2+h) = \lim_{h \rightarrow 0} \frac{|2+h-2|}{2+h-2} = \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} (1) = 1$$

$$\text{L.H.L.} = f(2-0) = \lim_{h \rightarrow 0} f(2-h) = \lim_{h \rightarrow 0} \frac{|2-h-2|}{2-h-2} = \lim_{h \rightarrow 0} \frac{|-h|}{-h} = \lim_{h \rightarrow 0} \frac{h}{-h} = \lim_{h \rightarrow 0} (-1) = -1$$

$$\text{L.H.L.} \neq \text{R.H.L. i.e. } f(2-0) \neq f(2+0)$$

Hence, $\lim_{x \rightarrow 2} \frac{|x - 2|}{x - 2}$ does not exist.

Example 13: Using the definition of limit, prove that

$$(i) \lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a} = 2a \quad (ii) \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0 \quad (iii) \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$

Solution:

(i) Here $f(x) = \frac{x^2 - a^2}{x - a}, x \neq a$

We must show that for any $\epsilon > 0, \exists \delta > 0$ s.t.

$$|f(x) - 2a| < \epsilon \text{ for } 0 < |x - a| < \delta$$

$$\begin{aligned} \text{Now } \left| \frac{x^2 - a^2}{x - a} - 2a \right| &= \left| \frac{x^2 - a^2 - 2ax + 2a^2}{x - a} \right| = \left| \frac{x^2 - 2ax + a^2}{x - a} \right| \\ &= \left| \frac{(x - a)^2}{(x - a)} \right| = |x - a| \end{aligned}$$

$$\therefore |f(x) - 2a| < \epsilon \text{ whenever } 0 < |x - a| < \epsilon$$

Choosing $\delta = \epsilon$

$$|f(x) - 2a| < \epsilon \text{ whenever } 0 < |x - a| < \delta$$

$$\text{Hence } \lim_{x \rightarrow 0} \frac{x^2 - a^2}{x - a} = 2a$$

(ii) Here $f(x) = x \sin \frac{1}{x}$

We must show that any $\epsilon > 0, \exists \delta > 0$ s.t.

$$|f(x) - 0| = \left| x \sin \frac{1}{x} - 0 \right| = |x| \left| \sin \frac{1}{x} \right| < \epsilon \text{ whenever } 0 < |x - 0| < \epsilon$$

$$\text{Now } |f(x) - 0| = |x| \left| \sin \frac{1}{x} \right| \leq |x| \text{ because } \left| \sin \frac{1}{x} \right| \leq 1$$

$$|f(x) - 0| < \epsilon \text{ whenever } 0 < |x| < \epsilon, \text{ Choosing } \delta = \epsilon$$

$$|f(x) - 0| < \epsilon \text{ whenever } 0 < |x| < \delta$$

$$\text{Hence } \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

(iii) Here $f(x) = x^2 \sin \frac{1}{x}$

We have to show that $|f(x) - 0| < \epsilon$ whenever $0 < |x - 0| < \delta$

$$\text{Now } |f(x) - 0| = \left| x^2 \sin \frac{1}{x} - 0 \right| = |x^2| \left| \sin \frac{1}{x} \right| \leq |x^2|$$

$$(\text{because } \left| \sin \frac{1}{x} \right| \leq 1)$$

$$\therefore |f(x) - 0| < \epsilon \text{ whenever } 0 < |x^2| < \epsilon$$

$$\Rightarrow |f(x) - 0| < \epsilon \text{ whenever } 0 < |x| \leq \sqrt{\epsilon}$$

Choosing $\delta = \sqrt{\epsilon}$, then we have

$$|f(x) - 0| < \epsilon \text{ whenever } 0 < |x| < \delta$$

$$\text{Hence } \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$

Example 14: Let $f(x) = \frac{x^2 + 2}{x^2 + 1}$, then given $\epsilon > 0$,

Find a real number $\delta > 0$ such that $|f(x) - 2| < \epsilon$ for $0 < |x| < \delta$

Solution: $|f(x) - 2| = \left| \frac{x^2 + 2}{x^2 + 1} - 2 \right|$
 $= \left| \frac{x^2 + 2 - 2x^2 - 2}{x^2 + 1} \right| < \epsilon$

or $\left| \frac{-x^2}{x^2 + 1} \right| < \epsilon \quad (\because |-x| = |x|)$

or $\frac{x^2}{x^2 + 1} < \epsilon \quad (\because \frac{x^2}{x^2 + 1} \geq 0)$

or $x^2 < (x^2 + 1)\epsilon \quad (\because x^2 + 1 > 0)$

or $(1 - \epsilon)x^2 < \epsilon$

or $x^2 < \frac{\epsilon}{1 - \epsilon}$

or $|x| < \sqrt{\frac{\epsilon}{1 - \epsilon}}$

Choosing $\delta = \sqrt{\frac{\epsilon}{1 - \epsilon}}$ where $0 < \epsilon < 1$

We have $|f(x) - 2| < \epsilon$ for $0 < |x| < \delta$

So $\delta = \sqrt{\frac{\epsilon}{1 - \epsilon}}$

Example 15: Let $f(x) = \frac{1}{x}$, $x \neq 0$. Prove that from definition (ϵ, δ) that $\lim_{x \rightarrow 2} f(x) = \frac{1}{2}$

Solution: To prove that $\lim_{x \rightarrow 2} f(x) = \frac{1}{2}$ we have to show that for any $\epsilon > 0$, we can find $\delta > 0$ s.t.

$$|f(x) - \frac{1}{2}| < \epsilon \text{ when } 0 < |x - 2| < \delta$$

Now $\left| f(x) - \frac{1}{2} \right| = \left| \frac{1}{x} - \frac{1}{2} \right| = \left| \frac{2 - x}{2x} \right| = \frac{|2 - x|}{2|x|} \quad \dots(1)$

Choosing $\delta \leq 1$ and $0 < |x - 2| < \delta$, we have

$$0 < |x - 2| < 1$$

$$\Rightarrow |x - 2| > 0 \text{ and } |x - 2| < 1$$

$$\Rightarrow x \neq 2 \text{ and } 2 - 1 < x < 2 + 1$$

$$\Rightarrow x \neq 2 \text{ and } 1 < x < 3$$

$$\Rightarrow x \neq 2 \text{ and } 1 > \frac{1}{x} > \frac{1}{3}$$

$$\Rightarrow x \neq 2 \text{ and } \frac{1}{3} < \frac{1}{x} < 1$$

$$\Rightarrow x \neq 2 \text{ and } \frac{1}{|x|} < 1 \left[\because \frac{1}{x} > \frac{1}{3} > 0 \quad \therefore \frac{1}{x} = \frac{1}{|x|} \right]$$

$$\text{From (i)} \quad \left| f(x) - \frac{1}{2} \right| = \frac{|x-2|}{2} \cdot \frac{1}{|x|} < \frac{\delta}{2} \cdot 1$$

Let us choose δ such that $\frac{\delta}{2} < \epsilon$ i.e. $\delta < 2\epsilon$

Also $\delta \leq 1$ $\therefore$ Choosing $\delta = \min \{1, 2\epsilon\}$, we have

$$\left| f(x) - \frac{1}{2} \right| < \frac{\delta}{2} < \epsilon \text{ when } 0 < |x-2| < \delta$$

$$\therefore \lim_{x \rightarrow 2} f(x) = \frac{1}{2}$$

Example 16: Which one of the following limit exists

$$(i) \quad \lim_{x \rightarrow 1} \frac{1}{(x-1)}$$

$$(ii) \quad \lim_{x \rightarrow 0} \frac{\sin |x|}{|x|}$$

$$(iii) \quad \lim_{x \rightarrow 0} \frac{|x|}{x}$$

[B.C.A. (Meerut) 2001]

$$(iv) \quad \lim_{x \rightarrow 2} \sin \left(\frac{1}{x-2} \right)$$

[B.C.A. (Meerut) 2004]

Solution:

$$(i) \quad \text{Given function is } \lim_{x \rightarrow 1} \left(\frac{1}{x-1} \right) = \frac{1}{0} = \infty.$$

Hence limit does not exist.

$$(ii) \quad \lim_{x \rightarrow 0} \frac{\sin(x)}{x}, \text{ we know } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

So therefore $\lim_{x \rightarrow 0} \frac{\sin |x|}{|x|} = 1$ and so it exists

$$(iii) \quad \lim_{x \rightarrow 0} \frac{|x|}{x} \text{ where } |x| = \begin{cases} x & , \quad x \geq 0 \\ -x & , \quad x < 0 \end{cases}$$

in both cases the limit will exist.

$$(iv) \quad \lim_{x \rightarrow 2} \sin \left(\frac{1}{x-2} \right) = \sin \left(\frac{1}{0} \right) = \sin(\infty) = \text{does not exist.}$$

Example 17: If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} g(x)$ does not exist, can $\lim_{x \rightarrow a} [f(x) + g(x)]$ exist? Prove your assertion.

Solution: If $\lim_{x \rightarrow a} f(x)$ exists, let $\lim_{x \rightarrow a} f(x) = l$

$$\Rightarrow f(a-0) = f(a+0) = l$$

$\therefore \lim_{x \rightarrow a} g(x)$ does not exist, let

$$g(a-0) = m_1 \text{ and } g(a+0) = m_2 \text{ where } m_1 \neq m_2$$

$$\text{Now } (f+g)(a-0) = f(a-0) + g(a-0) = l + m_1$$

$$(f+g)(a+0) = f(a+0) + g(a+0) = l + m_2$$

$$\text{Since } l + m_1 \neq l + m_2$$

$\therefore \lim_{x \rightarrow a} [f(x) + g(x)]$ does not exist.

Example 18: If $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} f(x)g(x)$ both exist, then does it follow that

$\lim_{x \rightarrow a} g(x)$ exists?

Solution: Let $f(x) = x$ and $g(x) = \frac{|x|}{x}, x \neq 0$

$$f(x)g(x) = |x|$$

$$\text{Now } \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x = 0 \text{ exists}$$

$$\text{and } \lim_{x \rightarrow 0} f(x)g(x) = \lim_{x \rightarrow 0} |x| = 0 \text{ exists}$$

$$\text{But } g(0-0) = \lim_{h \rightarrow 0} g(0-h) = \lim_{h \rightarrow 0} \frac{|0-h|}{0-h} = -1$$

$$g(0+0) = \lim_{h \rightarrow 0} g(0+h) = \lim_{h \rightarrow 0} \frac{|0+h|}{0+h} = 1$$

$$g(0+0) \neq g(0-0) \text{ so } \lim_{h \rightarrow 0} g(x) \text{ does not exist.}$$

Thus, $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} f(x)g(x)$ both exist but it is not necessary that $\lim_{x \rightarrow a} g(x)$ exists.

Example 19: If $\lim_{x \rightarrow a} f(x) = l$ then show that $\lim_{x \rightarrow a} |f(x)| = |l|$. Is its converse true?

Solution: $\lim_{x \rightarrow a} f(x) = l \Rightarrow$ for any given $\epsilon > 0$, $\exists \delta > 0$

$$\text{Such that } |f(x) - l| < \epsilon \text{ whenever } 0 < |x - a| < \delta \quad \dots(1)$$

$$\text{Since } |a - b| \geq ||a| - |b||$$

$$\therefore |f(x) - l| \geq ||f(x)| - |l||$$

$$\Rightarrow ||f(x)| - |l|| \leq |f(x) - l| < \epsilon \text{ whenever } 0 < |x - a| < \delta \quad [\text{using (i)}]$$

$$\Rightarrow \lim_{x \rightarrow a} |f(x)| = |l|$$

The converse of this statement is not always true.

For example, consider $f(x) = \begin{cases} -1 & \text{if } x < a \\ 1 & \text{if } x \geq a \end{cases}$

$$f(a+0) = \lim_{h \rightarrow 0} f(a+h) = \lim_{h \rightarrow 0} 1 = 1$$

$$f(a-0) = \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} (-1) = -1$$

$$f(a+0) \neq f(a-0)$$

$\Rightarrow \lim_{x \rightarrow a} f(x)$ does not exist.

But $|f(x)| = 1 \quad \forall x$

$\Rightarrow \lim_{x \rightarrow a} |f(x)| = \lim_{x \rightarrow a} 1 = 1$ exists

Example 20: If $f(x) \leq g(x) \leq h(x)$ and $\lim_{x \rightarrow a} f(x) = l = \lim_{x \rightarrow a} h(x)$ then prove that $\lim_{x \rightarrow a} g(x)$

exists and is equal to l .

Solution: If $\lim_{x \rightarrow a} f(x) = l = \lim_{x \rightarrow a} h(x)$

$\Rightarrow$ given $\epsilon > 0$, then exist $\delta_1, \delta_2 > 0$ such that

$$|f(x) - l| < \epsilon \quad \text{for } 0 < |x - a| < \delta_1$$

and $|h(x) - l| < \epsilon \quad \text{for } 0 < |x - a| < \delta_2$

$\Rightarrow l - \epsilon < f(x) < l + \epsilon \quad \text{for } 0 < |x - a| < \delta_1$

and $l - \epsilon < h(x) < l + \epsilon \quad \text{for } 0 < |x - a| < \delta_2$

Let $\delta = \min \{\delta_1, \delta_2\}$, then

$$l - \epsilon < f(x) < l + \epsilon \text{ and } l - \epsilon < h(x) < l + \epsilon \text{ for } 0 < |x - a| < \delta \quad \dots(1)$$

$$\text{Also given } f(x) \leq g(x) \leq h(x) \quad \dots(2)$$

From (1) and (2)

$$l - \epsilon < f(x) \leq g(x) \leq h(x) < l + \epsilon \text{ for } 0 < |x - a| < \delta$$

$\Rightarrow l - \epsilon < g(x) < l + \epsilon \quad \text{for } 0 < |x - a| < \delta$

$\Rightarrow |g(x) - l| < \epsilon \quad \text{for } 0 < |x - a| < \delta$

$\Rightarrow \lim_{x \rightarrow a} g(x) = l$

Example 21: Show that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} f(x-a)$

Solution: Let $\lim_{x \rightarrow a} f(x-a) = l$ then we have to show that $\lim_{x \rightarrow a} f(x) = l$.

If $\lim_{x \rightarrow a} f(x-a) = l$ then by definition of limit for any $\epsilon > 0$ there exists $\delta > 0$ such that.

$$|f(x-a) - l| < \epsilon \quad \text{whenever } 0 < |x-a| < \delta$$

Put $x-a = y$, we get

$$|f(y) - l| < \epsilon \quad \text{whenever } 0 < |y| < \delta$$

$\Rightarrow |f(y) - l| < \epsilon \quad \text{whenever } 0 < |y-0| < \delta$

$\Rightarrow \lim_{y \rightarrow 0} f(y) = l$

or $\lim_{x \rightarrow 0} f(x) = l$

Example 22: Evaluate $\lim_{x \rightarrow a} \frac{|\sin x|}{x}$

Solution: If $f(x) = \frac{|\sin x|}{x}$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{|\sin(0+h)|}{0+h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad [\because h > 0 \text{ so } \sin h > 0]$$

$$f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{|\sin(0-h)|}{0-h}$$

$$= \lim_{h \rightarrow 0} \frac{|-\sin h|}{-h} = \lim_{h \rightarrow 0} \frac{\sin h}{-h} = - \lim_{h \rightarrow 0} \frac{\sin h}{h} = -1$$

$f(0+0) \neq f(0-0)$ therefore $\lim_{x \rightarrow 0} \frac{|\sin x|}{x}$ does not exist.

Example 23: Evaluate $\lim_{x \rightarrow 0} \frac{x - |x|}{x}$

Solution: Let $f(x) = \frac{x - |x|}{x}$

$$f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{(0+h) - |0+h|}{(0+h)}$$

$$= \lim_{h \rightarrow 0} \frac{h-h}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = \lim_{h \rightarrow 0} 0 = 0$$

$$f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{(0-h) - |0-h|}{(0-h)}$$

$$= \lim_{h \rightarrow 0} \frac{-h-h}{-h} \quad (\text{because } |-h| = h)$$

$$= \lim_{h \rightarrow 0} \frac{2h}{h} = \lim_{h \rightarrow 0} 2 = 2$$

$$f(0+0) \neq f(0-0)$$

So $\lim_{x \rightarrow 0} \frac{x - |x|}{x}$ does not exist.

Example 24: Let $f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ -x & \text{if } x \text{ is irrational} \end{cases}$

Show that $\lim_{x \rightarrow a} f(x)$ exists only when $a = 0$.

Solution:

Case I: When a is a non-zero rational number.

$$\text{L.H.L.} = f(a-0) = \lim_{h \rightarrow 0} f(a-h) = \begin{cases} \lim_{h \rightarrow 0} (a-h) = a & \text{if } a-h \text{ is rational} \\ \lim_{h \rightarrow 0} -(a-h) = -a & \text{if } a-h \text{ is irrational} \end{cases}$$

i.e. $f(a-0)$ is not unique.

Therefore $f(a-0)$ does not exist and hence $\lim_{x \rightarrow a} f(x)$ does not exist.

Case II: When $a = 0$

$$\begin{aligned} f(0-0) &= \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(-h) \\ &= \begin{cases} \lim_{h \rightarrow 0} (-h) = 0 & \text{if } -h \text{ is rational} \\ \lim_{h \rightarrow 0} h = 0 & \text{if } -h \text{ is irrational} \end{cases} \end{aligned}$$

i.e. $f(0-0) = 0$

$$\text{R.H.L. } f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} f(h) = \begin{cases} \lim_{h \rightarrow 0} h = 0 & \text{if } h \text{ is rational} \\ \lim_{h \rightarrow 0} (-h) = 0 & \text{if } h \text{ is irrational} \end{cases}$$

i.e. $f(0+0) = 0$

$$f(0-0) = f(0+0) = 0 \text{ hence } \lim_{x \rightarrow a} f(x) = 0$$

Case III: When a is irrational number.

$$f(a-0) = \lim_{h \rightarrow 0} f(a-h) = \begin{cases} \lim_{h \rightarrow 0} (a-h) = a & \text{if } a-h \text{ is rational} \\ \lim_{h \rightarrow 0} -(a-h) = -a & \text{if } a-h \text{ is irrational} \end{cases}$$

$f(a-0) = a$ or $-a$ i.e. is not unique.

$\therefore f(a-0)$ does not exist and hence $\lim_{x \rightarrow a} f(x)$ does not exist.

Example 25: Show that $\lim_{x \rightarrow a} \sin\left(\frac{1}{x-a}\right)$ does not exist.

[B.C.A. (Meerut) 2004, 2008]

Solution: Let $f(x) = \sin\left(\frac{1}{x-a}\right) = \sin\left(\frac{1}{y}\right)$

where $x-a = y$ so that $y \rightarrow 0$ as $x \rightarrow a$ we know that $\lim_{y \rightarrow 0} \sin\left(\frac{1}{y}\right)$ does not exist.

Hence, $\lim_{x \rightarrow a} \sin\left(\frac{1}{x-a}\right)$ does not exist.



Comprehensive Exercise 5.1

(Limits)

1. Evaluate the following:

$$(i) \quad \lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}$$

$$(ii) \quad \lim_{x \rightarrow 2} \frac{2x^2 - 8}{x - 2}$$

[B.C.A. (Meerut) 2005]

$$(iii) \quad \lim_{x \rightarrow 2} \frac{x^2 + 3x + 2}{x - 2}$$

$$(iv) \quad \lim_{x \rightarrow 0} \frac{(1+x)^{1/3} - (1-x)^{1/3}}{x}$$

$$(v) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$(vi) \quad \lim_{x \rightarrow 0} (1+x)^{1/x}$$

$$(vii) \quad \lim_{x \rightarrow 0} \frac{a^x - 1}{x}$$

$$(viii) \quad \lim_{x \rightarrow 0} \frac{1}{x} e^{1/x}$$

[B.C.A (Meerut) 2006]

$$(ix) \quad \lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x}$$

$$(x) \quad \lim_{x \rightarrow a} \frac{x^m - a^m}{x - a}$$

$$(xi) \quad \lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}$$

$$(xii) \quad \lim_{x \rightarrow 0} \frac{xe^{1/x}}{1 + e^{1/x}}$$

[B.C.A. (Meerut) 2007]

2. Discuss the existence of the limit of the function f defined as

$$f(x) = \begin{cases} 1, & \text{if } x < 1 \\ 2 - x, & \text{if } 1 < x < 2 \\ 2, & \text{if } x \geq 2 \end{cases}$$

at $x = 1$ and $x = 2$

3. Let $f(x) = \begin{cases} x, & \text{if } 0 \leq x < 1 \\ 3 - x, & \text{if } 1 \leq x \leq 2 \end{cases}$

Show that $f(1+0) = 2$. Does the limit of $f(x)$ at $x = 1$ exist? Give reasons for your answers.

4. If $f(x) = \frac{\sin [x]}{[x]}$, $[x] \neq 0$ and $f(x) = 0, [x] = 0$ where $[x]$ denoted the greatest integer less than or equal to x , then find $\lim_{x \rightarrow 0} f(x)$.

5. Evaluate $\lim_{x \rightarrow \infty} \frac{(3x-1)(4x-2)}{(x+8)(x-1)}$

6. Establish the existence of the limit of the function $f(x)$ defined as

$$f(x) = \begin{cases} x, & \text{if } 0 \leq x < \frac{1}{2} \\ 0, & \text{if } x = \frac{1}{2} \\ x-1, & \text{if } \frac{1}{2} < x \leq 1 \end{cases}$$

7. Find the limit as $x \rightarrow 0$ of the function $f(x)$ defined as

$$f(x) = \begin{cases} 1, & \text{if } x \neq 0 \\ 2, & \text{if } x = 0 \end{cases}$$

8. Use the formula $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a$ to find $\lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{(x+1)} - 1}$

9. If f is defined on R as

$$f(x) = \begin{cases} 2, & \text{if } x \text{ is irrational} \\ 1, & \text{if } x \text{ is rational} \end{cases}$$

Prove that $\lim_{x \rightarrow a} f(x)$ does not exist for any $a \in R$.

[B.C.A. (Rohtak) 2006, 2009, 2011]

10. Give an example to show that $\lim_{x \rightarrow a} f(x)$ may exist even when the function is not defined for $x = a$.

11. If $x \rightarrow 0$, then does the limit of the following function f exist or not ?

$$f(x) = \begin{cases} x, & \text{if } x < 0 \\ 1, & \text{if } x = 0 \\ x^2, & \text{if } x > 0 \end{cases}$$

12. If $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n$, then prove that $\lim_{x \rightarrow a} f(x) = f(a)$.

[B.C.A. (Garhwal) 2011]

13. If f is defined on R as

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ 1, & \text{if } x \text{ is rational} \end{cases}$$

Prove that $\lim_{x \rightarrow a} f(x)$ does not exist for any $a \in R$.

14. If $\lim_{x \rightarrow a} f(x) = \pm \infty$, then $\lim_{x \rightarrow a} \frac{1}{f(x)} = 0$

15. Evaluate $\lim_{x \rightarrow 0} \sin \frac{1}{x}$

[B.C.A. (Kanpur) 2010]

16. Evaluate the following limits

(i) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$

[B.C.A. (Meerut) 2002]

(ii) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$

[B.C.A. (Meerut) 2002]

(iii) $\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{x}$

[B.C.A. (Meerut) 2006]

(iv) $\lim_{x \rightarrow 2} \left\{ \frac{2x^2 - 8}{x - 2} \right\}$

[B.C.A. (Meerut) 2005]



Answers

1.	(i) $\frac{3}{2}$	(ii) 8	(iii) Limits does not exists.	(iv) $\frac{2}{3}$
	(v) 1	(vi) e	(vii) $\log a$	(viii) Limit does not exists.
	(ix) n	(x) ma^{m-1}	(xi) Limit does not exists.	(xii) 0
2.	Limit exists at $x = 1$ and equal to 1, at $x = 2$ limit does not exists			
3.	$f(1 - 0) = 1$ so limit does not exists.			
4.	Limit does not exists.			
5.	12;	6.	Limit does not exist.	
7.	limit exists and equal to 1.	8.	$2 \log 2$;	
10.	$f(x) = \frac{x^2 - a^2}{x - a}$	11.	Limit exists and equal to zero.	
15.	Limit des not exists			
16.	(i) $\log_e a$	(ii) $\frac{1}{2}$	(iii) 1	(iv) 8

5.2 Continuity

5.2.1 Definition

[B.C.A. (Meerut) 2002, 2003, 2006, 2009]

A function $f(x)$ is said to be continuous at $x = a$ if given $\epsilon > 0$ however small, we can find a number $\delta > 0$ such that

$$|f(x) - f(a)| < \epsilon \quad \text{when} \quad |x - a| < \delta$$

Remark 1: $|f(x) - f(a)| < \epsilon \Rightarrow f(a) - \epsilon < f(x) < f(a) + \epsilon$

$$\text{and} \quad |x - a| < \delta \Rightarrow a - \delta < x < a + \delta$$

Remark 2: The value of δ depends upon the choice of ϵ .

Remark 3: If a function is continuous at every point of its domain of definition, then it is said to be an everywhere continuous functions.

5.2.2 Continuity at a Point

A function $y = f(x)$ is said to be continuous at a point $x = a$ if the following conditions are satisfied.

(i) $f(x)$ is defined at $x = a$ i.e. $f(a)$ exists.

(ii) $\lim_{x \rightarrow a} f(x)$ exists i.e. $f(a+0) = f(a-0)$

(iii) $\lim_{x \rightarrow a} f(x) = f(a)$

If any one of the above conditions are not satisfied, then function is said to be discontinuous at $x = a$.

5.2.3 Continuity in Open Interval $]a, b[$

A function $y = f(x)$ is said to be continuous in an open interval $]a, b[$ if it is continuous at every point of the interval $]a, b[$.

5.2.4 Continuity in Closed Interval $[a, b]$

A function $f(x)$ is said to be continuous in $[a, b]$, if it satisfies the following :

(i) $f(x)$ is continuous at $x = c$ where $a < c < b$

$$\text{i.e.} \quad f(c+0) = f(c-0) = f(c)$$

$$\text{or} \quad \lim_{x \rightarrow c} f(x) = f(c)$$

(ii) $f(x)$ is continuous at $x = a$ from the right.

$$\text{i.e.} \quad f(a+0) = f(a)$$

(iii) $f(x)$ is continuous at $x = b$ from the left

$$\text{i.e.} \quad f(b-0) = f(b)$$

It may be noted that when we are talking about the continuity of a real function without mentioning the domain of the function, it means that the function is continuous at every point of the domain. Roughly speaking, a function is continuous if there is no break or jump in the graph of this function.

5.2.5 Continuity from the Left and Continuity from the Right

If a function $f(x)$ is defined on an interval I and $a \in I$, we say that $f(x)$ is continuous from the left at a if $f(a-0)$ exists and equal to $f(a)$. Similarly if $f(x)$ is said to be continuous from the right at $x = a$ if $f(a+0)$ exists and equal to $f(a)$.

Working formula for testing the continuity of a function

Let $y = f(x)$ be a given function and we have to test the continuity at $x = a$ then proved as follows:

(i) Find $f(a+0)$ i.e. $\lim_{h \rightarrow 0} f(a+h)$ (R.H.L.), if exists then

(ii) Find $f(a-0)$ i.e. $\lim_{h \rightarrow 0} f(a-h)$ (L.H.L.), if exists then

(iii) Find $f(a)$ i.e. value of the function at $x = a$ if exists

The function will be continuous at $x = a$ if

$$f(a+0) = f(a-0) = f(a) \text{ or } \lim_{x \rightarrow a} f(x) = f(a)$$

If any one of the above conditions are not satisfied then the function is said to be discontinuous at $x = a$.

Following are some of the frequently occurring functions which are also continuous.

5.2.6 Constant Function

If $f(x) = c$ where c is a constant and $c \in R$ then graph is a line parallel to x -axis at a distance c from x -axis and thus there is no break in the graph, i.e. constant function is a continuous function.

5.2.7 Identity Function

The identity function $f(x) = x$ is a continuous function because the graph of this function is a line which makes an angle 45° with the x -axis and passes through the origin.

5.2.8 Sine Function

If $f(x) = \sin x$ and $x \in R$, is continuous on R . If $c \in R$ then $f(c) = \sin c$

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \sin x = \sin c$$

$$\text{i.e. } \lim_{x \rightarrow c} f(x) = f(c) = \sin c$$

So $\sin x$ continuous at $x = c \in R$

i.e., $\sin x$ is continuous on R .

5.2.9 Polynomial Function

[B.C.A. (Meerut) 2003]

Let $f(x) = a_0x^n + a_1x^{n-1} + \dots + a_n$ where $x \in R$ is continuous on R .

If $c \in R$ then $f(c) = a_0c^n + a_1c^{n-1} + \dots + a_n$

$$\text{Also } \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} [a_0x^n + a_1x^{n-1} + \dots + a_n] = a_0c^n + a_1c^{n-1} + \dots + a_n$$

$$\text{i.e. } \lim_{x \rightarrow c} f(x) = f(c) \text{ where } c \in R$$

Hence every polynomial is continuous on R .

5.2.10 Rational Function

If $f(x) = \frac{h(x)}{g(x)}$ where $h(x)$ and $g(x)$ both are polynomials is also a continuous function on R .

$$\text{If } c \in R \text{ then } f(c) = \frac{h(c)}{g(c)}$$

$$\text{Also } \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \left[\frac{h(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow c} h(x)}{\lim_{x \rightarrow c} g(x)} = \frac{h(c)}{g(c)}$$

i.e. $\lim_{x \rightarrow c} f(x) = \frac{h(c)}{g(c)}$ provided $g(c) \neq 0$

Then $\lim_{x \rightarrow c} f(x) = f(c) = \frac{h(c)}{g(c)} \quad \forall c \in R$

Therefore $f(x)$ is continuous on R .

- (i) The functions $\cos x, e^x$ and a^x are continuous everywhere.
- (ii) The functions $\tan^{-1} x$ and $\cot^{-1} x$ are continuous everywhere.
- (iii) The functions $\sin^{-1} x$ and $\cos^{-1} x$ are continuous in $[-1, 1]$
- (iv) The functions $\sec^{-1} x$ and $\operatorname{cosec}^{-1} x$ are continuous in $]-\infty, -1] \cup [1, \infty[$

5.2.11 Properties of Continuous Functions

- (a) If $f(x)$ is continuous at $x = a$ and $g(x)$ is another continuous functions at $x = a$ then the composite function $(g \circ f)(x)$ is also continuous at $x = a$. For example, if $f(x) = x^2 + 1$ and $g(x) = x + 1$ are two continuous functions on an interval I then $(g \circ f)(x) = x^2 + 2$ is also continuous function.

- (b) If $f(x)$ is continuous function and $f(x) \neq 0$ then $\frac{1}{f(x)}$ is also continuous function.

For example, if $f(x) = x^4$ is continuous, then $\frac{1}{x^4}$ is also continuous except at $x = 0$.

- (c) A continuous function on an interval has a least and greatest value.

5.3 Discontinuous Functions

A function $f(x)$ is said to be discontinuous at a point 'a' if it not continuous at that point. The point 'a' is called the point of discontinuity.

5.3.1 Types of Discontinuity

5.3.2 Removable Discontinuity

If a function $f(x)$ at $x = a$ is such that

- (i) $f(a+0)$ and $f(a-0)$ both exists and equal i.e. $\lim_{x \rightarrow a} f(x)$ exists.
- (ii) The function is not defined at $x = a$

or $\lim_{x \rightarrow a} f(x) \neq f(a)$

Then this type of continuity is called removable discontinuity because if the function can be defined suitable at $x = a$ so that the new function becomes continuous at $x = a$.

Example: Is it possible to define $f(x)$ in a way that extends

$$f(x) = \frac{x^2 + x - 6}{x^2 - 4}$$

to be continuous at $x = 2$? If so, what value should $f(x)$ have?

Solution: For $f(x)$ to be continuous at $x = 2$ we must have $\lim_{x \rightarrow 2} f(x) = f(2)$

$$\text{Now } \lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{(x-2)(x+3)}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{x+3}{x+2} = \frac{5}{4}$$

$$\text{Now } f(2) = \frac{4+2-6}{4-0} = \frac{0}{4} \text{ (Meaningless)}$$

i.e. $f(2)$ does not exist. So function is discontinuous at $x = 2$.

If we take function such that

$$f(x) = \begin{cases} \frac{x^2 + x - 6}{x^2 - 4} & \text{if } x \neq 2 \\ \frac{5}{4} & \text{if } x = 2 \end{cases}$$

Then this function is continuous at $x = 2$ because $\lim_{x \rightarrow 2} f(x) = f(2)$

This type of discontinuity is called **removable discontinuity**.

5.3.3 Infinite Discontinuity

This type of discontinuity occurs when neither the limit exists at $x = a$ nor the function is defined there. In other way we can say that a function $f(x)$ is said to have an infinite discontinuity at $x = a$ if $f(a+0)$ or $f(a-0)$ is $+\infty$ or $-\infty$.

Such type of discontinuities cannot be removed.

5.3.4 Discontinuity of First Kind

If (i) $f(x)$ is defined at $x = a$ and

(ii) $f(a+0)$ and $f(a-0)$ exist but are not equal *i.e.* $f(a+0) \neq f(a-0)$

Then function $f(x)$ has discontinuity of first kind.

5.3.5 Discontinuity of Second Kind

If (i) The function $f(x)$ is defined at $x = a$ and

(ii) Neither $f(a+0)$ nor $f(a-0)$ exists at $x = a$

Then the function $f(x)$ has discontinuity of second kind.

5.3.6 Mixed Discontinuity

If (i) The function $f(x)$ is defined at $x = a$ and

(ii) only one limit $f(a+0)$ or $f(a-0)$ exists

Then $f(x)$ is said to have mixed discontinuity at $x = a$.



Some Important Theorems

One-Sided Limits and Continuity

Theorem 1: Let $f(x)$ be a function defined on an interval I , and $a \in I$ then, $f(x)$ is continuous at $x = a$, iff, $f(a+0)$, and $f(a-0)$ both exists and $f(a-0) = f(a+0) = f(a)$.

[B.C.A. (Kurukshetra) 2011]

Proof : Since $f(x)$ is defined on I and $a \in I$, therefore $f(a)$ exists.

Let us suppose that $f(x)$ is continuous at $x = a$, then by definition $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} f(x) = f(a)$, $\lim_{x \rightarrow a} f(x)$ exists means $f(a+0) = f(a-0)$

i.e. $\lim_{x \rightarrow a} f(x) = f(a) \Rightarrow f(a+0) = f(a-0) = f(a)$

Conversely, If $f(a+0)$ and $f(a-0)$ exists and

$$f(a+0) = f(a-0) = f(a)$$

$$\Rightarrow \lim_{x \rightarrow a} f(x) = f(a)$$

This shows that functions $f(x)$ is continuous at $x = a$.

Algebra of Continuous Functions

Theorem 2: If $f(x)$ and $g(x)$ be two functions defined on an interval I , $a \in I$ and c is any real number. If $f(x)$ and $g(x)$ are continuous at $x = a$, then so also are :

$$(i) f + g \quad (ii) cf \quad (iii) fg \quad (iv) f/g, \text{ provided } g(a) \neq 0$$

[B.C.A (Rohilkhand) 2003, 2008, 2011]

Proof: We have $f(x)$ and $g(x)$ are continuous at $x = a$,

$$\therefore \lim_{x \rightarrow a} f(x) = f(a) \quad \dots(1)$$

$$\lim_{x \rightarrow a} g(x) = g(a) \quad \dots(2)$$

(i) We know that

$$(f + g)(x) = f(x) + g(x), \quad x \in I$$

and in particular

$$(f + g)(a) = f(a) + g(a)$$

$$\begin{aligned} \text{Now } \lim_{x \rightarrow a} (f + g)(x) &= \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) \\ &= f(a) + g(a) = (f + g)(a) \quad [\text{From (1) and (2)}] \end{aligned}$$

$\therefore f + g$ is continuous at $x = a$

(ii) We have $(cf)(x) = cf(x), \quad x \in I$

$$\text{In particular } (cf)(a) = cf(a)$$

$$\lim_{x \rightarrow a} (cf)(x) = \lim_{x \rightarrow a} cf(x) = c \lim_{x \rightarrow a} f(x) = cf(a) = (cf)(a)$$

$\therefore cf$ is continuous at $x = a$

(iii) We know that

$$(fg)(x) = f(x)g(x), \quad x \in I$$

In particular, $(fg)(a) = f(a)g(a)$

$$\begin{aligned} \lim_{x \rightarrow a} (fg)(x) &= \lim_{x \rightarrow a} f(x)g(x) = \lim_{x \rightarrow a} f(x) \lim_{x \rightarrow a} g(x) \\ &= f(a)g(a) = fg(a) \end{aligned}$$

$\therefore fg$ is continuous at $x = a$

(iv) We know that

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \quad x \in I, g(x) \neq 0$$

and in particular $\left(\frac{f}{g}\right)(a) = \frac{f(a)}{g(a)}, g(a) \neq 0$

$$\text{Now, } \lim_{x \rightarrow a} \left(\frac{f}{g}\right)(x) = \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{f(a)}{g(a)} = \left(\frac{f}{g}\right)(a)$$

provided $g(a) \neq 0$

$\therefore \frac{f}{g}$ is continuous at $x = a$

Corollary I: Let f be a function defined on an interval I , and $a \in I$. If f and g be continuous at a , then so is $f - g$.

$$(f - g)(x) = [f + (-g)](x) = f(x) + (-g)(x), \quad x \in I$$

If $x = a$ then $(f - g)(a) = [f + (-g)](a) = f(a) + (-g)(a) = f(a) - g(a)$

$$\begin{aligned} \text{Now } \lim_{x \rightarrow a} (f - g)(x) &= \lim_{x \rightarrow a} [f + (-g)](x) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} [-g(x)] \\ &= \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = f(a) - g(a) = (f - g)(a) \end{aligned}$$

$\therefore (f - g)$ is continuous at $x = a$.

Theorem 3: Let f be the function defined on an interval I and $a \in I$. If f is continuous at a then also $|f|$ is continuous at a .

Proof: Since f is continuous at a then by definition of continuity.

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Also by definition

$$|f|(x) = |f(x)|, \quad x \in I$$

In particular

$$|f|(a) = |f(a)|$$

$$\text{Now,} \quad \lim_{x \rightarrow a} |f|(x) = \lim_{x \rightarrow a} |f(x)| = \left| \lim_{x \rightarrow a} f(x) \right| = |f(a)| = |f|(a)$$

$\therefore |f|$ is continuous at $x = a$

Note- The converse of the above theorem is not true. Consider a function $f(x)$ on R defined by

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ -1 & \text{if } x \text{ is irrational} \end{cases}$$

Then $|f|(x) = 1 \quad \forall x \in R$

$|f|$ is continuous being a constant function but $f(x)$ is not continuous. Hence $|f|$ is continuous but $f(x)$ is not continuous.

Theorem 4: Let f be a continuous function defined on an interval I , $a \in I$ and g be a function defined on an interval J , such that $f(I) \subseteq J$. If f is continuous at a and g be continuous at $f(a)$, then $(g \circ f)$ is continuous at a .

Proof: Since, f is continuous at $a \in I$, then $\lim_{x \rightarrow a} f(x) = f(a)$

Also g is continuous at $f(a) \in J$, then $\lim_{y \rightarrow f(a)} g(y) = g\{f(a)\}$

By definition of composite of mappings.

$$g \circ f(x) = g[f(x)] \quad \text{where } x \in I$$

For $x = a$ we have

$$g \circ f(a) = g[f(a)]$$

Let $\langle a_n \rangle$ be any sequence in I , such that $\lim_{n \rightarrow \infty} a_n = a$

Since $\lim_{n \rightarrow a} f(x) = f(a)$

$$\therefore \lim_{n \rightarrow \infty} f(a_n) = f(a)$$

we have $f(I) \subseteq J$ and $\langle f(a_n) \rangle$ is a sequence in J . Also we have $\lim_{y \rightarrow f(a)} g(y) = g\{f(a)\}$.

$$\therefore \lim_{n \rightarrow \infty} g[f(a_n)] = g[f(a)]$$

$$\text{or} \quad \lim_{n \rightarrow \infty} g \circ f(a_n) = g \circ f(a)$$

This is true for every sequence $\langle a_n \rangle$ in I , that converges to a , therefore

$$\lim_{x \rightarrow a} (g \circ f)(x) = g \circ f(a)$$

Hence $g \circ f$ is continuous at point a .

Theorem 5: Prove that a function defined on an interval I , is continuous at a point $c \in I$ if and only if for every sequence $\langle x_n \rangle$ converging to c , we have

$$\lim_{n \rightarrow \infty} f(x_n) = f(c) \quad \text{[B.C.A. (Meerut) 2007]}$$

Proof: Let $f(x)$ be continuous at c and let $\langle x_n \rangle$ be a sequence in I such that

$$\lim_{n \rightarrow \infty} x_n = c$$

Since $f(x)$ is continuous at $x = c$ then for a given $\varepsilon > 0$, there exists a corresponding number $\delta > 0$ such that

$$|f(x) - f(c)| < \varepsilon \quad \text{whenever} \quad |x - c| < \delta \quad \dots(1)$$

Also, since $\lim_{n \rightarrow \infty} x_n = c$, then there exists a positive integer m such that

$$|x_n - c| < \delta \quad \text{whenever} \quad n \geq m \quad \dots(2)$$

Letting $x = x_n$ in (1) we get

$$|f(x_n) - f(c)| < \varepsilon \quad \text{whenever} \quad |x_n - c| < \delta \quad \dots(3)$$

From (2) and (3), we get

$$|f(x_n) - f(c)| < \varepsilon, \text{ whenever } n \geq m$$

$$\therefore \lim_{n \rightarrow \infty} f(x_n) = f(c)$$

Conversely: Suppose for every sequence $\langle x_n \rangle$ in I converging to c , we have

$$\lim_{n \rightarrow \infty} f(x_n) = f(c)$$

Then we have to show that $f(x)$ is continuous at $x = c$.

The converse is proved by negation contradiction method.

Let $f(x)$ be not continuous at $x = c$ then there exist $\varepsilon > 0$. s.t., for every $\delta > 0, |x - c| < \delta \Rightarrow |f(x) - f(c)| \geq \varepsilon$.

Let us choose $\delta = \frac{1}{n}$ and thus get $a < x_n >$ which is such that

$$|x_n - c| < \frac{1}{n} \quad \text{but} \quad |f(x_n) - f(c)| \geq \varepsilon$$

This means $\lim_{n \rightarrow \infty} x_n = c$ but $\lim_{n \rightarrow \infty} f(x_n) \neq f(c)$

which is contradiction to the fact that $\lim_{n \rightarrow \infty} f(x_n) = f(c)$. Therefore, our supposition is false. Hence, $f(x)$ is continuous at $x = c$.



Illustrative Examples

(Continuity)

Example 26: Discuss the continuity and discontinuity of the following functions:

(i) $f(x) = x + \frac{1}{x}$

(ii) $f(x) = \sin x$

(iii) $f(x) = \cos\left(\frac{1}{x}\right)$ when $x \neq 0$, $f(0) = 0$

[B.C.A. (Kanpur) 2006]

(iv) $f(x) = \frac{\sin x}{x}$ when $x \neq 0$ and $f(0) = 0$

[B.C.A. (Rohilkhand) 2009]

(v) $f(x) = \frac{e^{1/x}}{1 + e^{1/x}}$ when $x \neq 0$ and $f(0) = 0$

[B.C.A. (Meerut) 2006]

$$(vi) \quad f(x) = \frac{xe^{1/x}}{1+e^{1/x}} + \sin\left(\frac{1}{x}\right) \text{ when } x \neq 0 \text{ and } f(0) = 0$$

$$(vii) \quad f(x) = \frac{\sin^{-1} x}{x} \text{ when } x \neq 0 \text{ and } f(0) = 1$$

[B.C.A. (Agra) 2003]

Solution: (i) $f(x) = x + \frac{1}{x}$ is defined at every point $a \in R$ except at $a = 0$, therefore we discuss continuity of this function in two cases (i) when $a \neq 0$ and (ii) when $a = 0$.

Case I: If $0 \neq a \in R$

We have
$$f(a) = a + \frac{1}{a}$$

and
$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \left(x + \frac{1}{x} \right) = a + \frac{1}{a}$$

i.e.
$$\lim_{x \rightarrow a} f(x) = f(a)$$

$\therefore f(x)$ is continuous when $0 \neq a \in R$

Case II: When $a = 0$ the function is not defined and therefore we need to check R.H.L. and L.H.L.

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \left\{ (0+h) + \frac{1}{0+h} \right\} = \lim_{h \rightarrow 0} \left\{ h + \frac{1}{h} \right\} = 0 + \infty = \infty$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0-h)$$

$$= \lim_{h \rightarrow 0} \left\{ (0-h) + \frac{1}{0-h} \right\} = \lim_{h \rightarrow 0} \left(-h - \frac{1}{h} \right) = (-0) - \frac{1}{0} = 0 - \infty = -\infty$$

R.H.L. $\neq$ L.H.L., therefore function is not continuous at $a = 0$ and the function has discontinuity of second kind.

(ii) $f(x) = \sin x$ this function is defined for all real numbers.

If $a \in R$ then $f(a) = \sin a$

and
$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \sin x = \sin a$$

i.e.,
$$\lim_{x \rightarrow a} f(x) = f(a)$$

Therefore continuous $\forall a \in R$.

(iii) $f(x) = \cos\left(\frac{1}{x}\right)$ when $x \neq 0$ and $f(0) = 0$

The function is defined for $0 \neq a \in R$ but not defined at $a = 0$. Although $f(0) = 0$ given, we use this value only to check the continuity of the function at $a = 0$.

Case I: When $a \neq 0$, $f(a) = \cos \frac{1}{a}$

and
$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \cos\left(\frac{1}{x}\right) = \cos \frac{1}{a}$$

i.e. $\lim_{x \rightarrow a} f(x) = f(a)$, therefore function is continuous at $0 \neq a \in R$.

Case II: When $a = 0$

In this case the function is not defined at $a = 0$ so we need to check R.H.L. and L.H.L.

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \cos\left(\frac{1}{0+h}\right) = \lim_{h \rightarrow 0} \cos \frac{1}{h}$$

But $\cos \frac{1}{h}$ oscillates between -1 and $+1$ when $h \rightarrow 0$ i.e. we can not get a unique value, therefore R.H.L. does not exist.

$$\begin{aligned} \text{L.H.L.} &= f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \cos\left(\frac{1}{0-h}\right) = \lim_{h \rightarrow 0} \cos\left(-\frac{1}{h}\right) \\ &= \lim_{h \rightarrow 0} \cos\left(\frac{1}{h}\right) \text{ does not exist due to the reason given in case of R.H.L.} \end{aligned}$$

Neither R.H.L. nor L.H.L. exists, therefore function has discontinuity of second kind. When $x = 0$.

$$(iv) \quad f(x) = \frac{\sin x}{x} \text{ when } x \neq 0 \text{ and } f(0) = 0$$

The function $f(x)$ is defined for $0 \neq a \in R$ but undefined for $x = 0$ or $a = 0$. Now we have to discuss the continuity of this function in two cases (i) when $a \neq 0$ and (ii) when $a = 0$.

Case I: When $a \neq a \in R$.

$$f(a) = \frac{\sin a}{a}$$

$$\text{and} \quad \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \frac{\sin x}{x} = \frac{\sin a}{a}$$

$$\text{i.e.} \quad \lim_{x \rightarrow a} f(x) = f(a)$$

Therefore function is continuous at $x = a$ where $a \neq 0$.

Case II: When $a = 0$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{\sin(0+h)}{0+h} = \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \left[\because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right]$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{\sin(0-h)}{0-h} = \lim_{h \rightarrow 0} \left\{ \frac{-\sin h}{-h} \right\} = \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$$

$$\text{i.e.} \quad \text{R.H.L.} = \text{L.H.L.} = f(0) = 1$$

Therefore function is continuous at $x = 0$ or $a = 0$.

Hence, $f(x)$ continuous for all real numbers.

$$(v) \quad f(x) = \frac{e^{1/x}}{1+e^{1/x}} \text{ for } x \neq 0 \text{ and } f(0) = 0$$

(i) We know that $e^\infty = \infty$ and $e^{-\infty} = 0$, therefore if $x \neq 0$ or $a \neq 0$ where $a \in \mathbb{R}$ the function is defined and $f(a) = \frac{e^{1/a}}{1 + e^{1/a}}$

Now
$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \frac{e^{1/x}}{1 + e^{1/x}} = \frac{e^{1/a}}{1 + e^{1/a}}$$

i.e.,
$$\lim_{x \rightarrow a} f(x) = f(a) = \frac{e^{1/a}}{1 + e^{1/a}}$$

Therefore function is continuous when $a \rightarrow \mathbb{R}$ and $a \neq 0$.

(ii) When $a = 0$ the value of the function is not defined. We have $f(0) = 0$ only to test the continuity.

$$\begin{aligned} \text{R.H.L.} &= f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{e^{1/0+h}}{1 + e^{1/0+h}} = \lim_{h \rightarrow 0} \frac{e^{1/h}}{1 + e^{1/h}} \\ &= \lim_{h \rightarrow 0} \frac{e^{1/h}}{e^{1/h}} \left[\frac{1}{\frac{1}{e^{1/h}} + 1} \right] = \lim_{h \rightarrow 0} \frac{1}{1 + e^{-1/h}} = \frac{1}{1 + e^{-\infty}} = \frac{1}{1+0} = 1 \end{aligned}$$

$$\begin{aligned} \text{L.H.L.} &= f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{e^{1/0-h}}{1 + e^{1/0-h}} \\ &= \lim_{h \rightarrow 0} \frac{e^{-1/h}}{1 + e^{-1/h}} = \frac{e^{-\infty}}{1 + e^{-\infty}} = \frac{0}{1+0} = \frac{0}{1} = 0 \end{aligned}$$

R.H.L. $\neq$ L.H.L so function is discontinuous at $x = 0$ and we have

$$f(0-0) = f(0) \neq f(0+0)$$

This shows that $f(x)$ has an ordinary discontinuity at $x = 0$ and there is a jump at $x = 0$.

(vi) $f(x) = x \frac{e^{1/x}}{1 + e^{1/x}} + \sin \frac{1}{x}$ when $x \neq 0$ and $f(0) = 0$

(i) The given function is defined for $x = a$ where $0 \neq a \in \mathbb{R}$

$$f(a) = \frac{ae^{1/a}}{1 + e^{1/a}} + \sin \frac{1}{a}$$

and
$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \left(\frac{xe^{1/x}}{1 + e^{1/x}} + \sin \frac{1}{x} \right) = \frac{ae^{1/a}}{1 + e^{1/a}} + \sin \frac{1}{a}$$

$$\lim_{x \rightarrow 0} f(x) = f(a)$$

Therefore, function is continuous when x is a non-zero real number

(ii) $f(x)$ is not defined when $x = 0$ and $a = 0$, therefore in this case we have to test R.H.L. and L.H.L.

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \left[\frac{(0+h)e^{1/0+h}}{1 + e^{1/0+h}} + \sin \frac{1}{0+h} \right]$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{h e^{1/h}}{1 + e^{1/h}} + \lim_{h \rightarrow 0} \sin \left(\frac{1}{h} \right) \\
&= \lim_{h \rightarrow 0} \frac{e^{1/h}}{e^{1/h}} \left[\frac{h}{\frac{1}{e^{1/h}} + 1} \right] + \lim_{h \rightarrow 0} \sin \left(\frac{1}{h} \right) \\
&= \lim_{h \rightarrow 0} \frac{h}{e^{-1/h} + 1} + \lim_{h \rightarrow 0} \sin \left(\frac{1}{h} \right) = \frac{0}{e^{-\infty} + 1} + \lim_{h \rightarrow 0} \sin \left(\frac{1}{h} \right) \\
&= \frac{0}{0+1} + \text{value oscillates between } -1 \text{ and } 1 \\
&= 0 + \text{limit does not exist.} \\
&= \text{limit does not exist.}
\end{aligned}$$

$$\begin{aligned}
f(0-0) &= \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \left[\frac{(0-h) e^{1/0-h}}{1 + e^{1/0-h}} + \sin \frac{1}{0-h} \right] \\
&= \lim_{h \rightarrow 0} \frac{-h e^{1/-h}}{1 + e^{1/-h}} + \lim_{h \rightarrow 0} \sin \left(-\frac{1}{h} \right) \\
&= \frac{-0 \times e^{-\infty}}{1 + e^{-\infty}} - \lim_{h \rightarrow 0} \sin \left(\frac{1}{h} \right) \\
&= \frac{-0 \times 0}{1+0} - \lim_{h \rightarrow 0} \sin \left(\frac{1}{h} \right) = 0 - \lim_{h \rightarrow 0} \sin \left(\frac{1}{h} \right) \\
&= 0 - \text{value oscillates between } -1 \text{ and } 1.
\end{aligned}$$

i.e. $f(0-0)$ does not exist.

$f(0+0)$ and $f(0-0)$ both do not exist, therefore function is not continuous at $x=0$ and the discontinuity is of second kind.

(vii) $f(x) = \frac{\sin^{-1} x}{x}$ when $x \neq 0$ and $f(0) = 1$

Case I: When $x = a \neq 0$ real number, then

$$f(a) = \frac{\sin^{-1} a}{a}$$

and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \frac{\sin^{-1} x}{x} = \frac{\sin^{-1} a}{a}$$

$$\lim_{x \rightarrow a} f(x) = f(a) = \frac{\sin^{-1} a}{a}$$

So function is continuous when $x = a \neq 0$ real number.

Case II: When $x = a = 0$

$$\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} \text{ if we put } \sin^{-1} x = \theta \Rightarrow x = \sin \theta$$

$$\therefore \frac{\sin^{-1} x}{x} = \frac{\theta}{\sin \theta} \text{ and if } x \rightarrow 0 \text{ then } \theta \rightarrow 0$$

$$\text{Now } \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1 \left[\text{because } \lim_{x \rightarrow 0} \frac{x}{\sin x} = 1 \right]$$

i.e. function is continuous at $x = 0$ also.

Hence function is continuous for all real numbers.

Example 27: Examine for continuity at the origin of the functions

$$(i) \quad f(x) = \begin{cases} \frac{x-|x|}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \quad [\text{B.C.A. (Meerut) 2008, 2011}]$$

$$(ii) \quad f(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \quad [\text{B.C.A. (Meerut) 2001}]$$

$$(iii) \quad f(x) = \begin{cases} \frac{x}{|x|} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases} \quad [\text{B.C.A. (Purvanchal) 20011; B.C.A (Meerut) 2001}]$$

Solution:

$$(i) \quad f(x) = \frac{x-|x|}{x} \text{ if } x \neq 0 \text{ and } f(0) = 1$$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{(0+h)-|0+h|}{0+h}$$

$$= \lim_{h \rightarrow 0} \frac{h-h}{h} = \lim_{h \rightarrow 0} \left(\frac{0}{h} \right) = \lim_{h \rightarrow 0} 0 = 0$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{(0-h)-|0-h|}{(0-h)}$$

$$= \lim_{h \rightarrow 0} \frac{-h-h}{-h} = \lim_{h \rightarrow 0} \frac{2h}{h} = \lim_{h \rightarrow 0} 2 = 2, \text{ because } |-h| = |h| = h$$

L.H.L. $\neq$ R.H.L.

$\therefore f(x)$ is not continuous at $x = 0$

or $f(x)$ is discontinuous at origin.

$$(ii) \quad f(x) = \frac{|x|}{x} \text{ if } x \neq 0 \text{ and } f(0) = 1$$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{|0+h|}{(0+h)} = \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} 1 = 1$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{|0-h|}{(0-h)} = \lim_{h \rightarrow 0} \frac{h}{-h} = \lim_{h \rightarrow 0} (-1) = -1$$

R.H.L. $\neq$ L.H.L.

$\therefore$ Function is not continuous at $x = 0$

or function $f(x)$ is discontinuous at origin.

$$(iii) \quad f(x) = \frac{x}{|x|} \text{ if } x \neq 0 \text{ and } f(0) = 0$$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} (0+h) = \lim_{h \rightarrow 0} \frac{0+h}{|0+h|} = \lim_{h \rightarrow 0} \left(\frac{h}{h} \right) = \lim_{h \rightarrow 0} (1) = 1$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} (0-h) = \lim_{h \rightarrow 0} \frac{0-h}{|0-h|} = \lim_{h \rightarrow 0} \frac{-h}{h} = \lim_{h \rightarrow 0} (-1) = -1$$

$\therefore$ Function is not continuous at $x=0$ or function is discontinuous at origin.

Example 28: Discuss the continuity at $x=0, 1, 2$ of the function

$$f(x) = \begin{cases} -x^2 & \text{if } x \leq 0 \\ 5x-4 & \text{if } 0 < x \leq 1 \\ 4x^2-3x & \text{if } 1 < x < 2 \\ 3x+4 & \text{if } x \geq 2 \end{cases}$$

Solution: Discussion at $x=0$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \{5(0+h)-4\} = -4$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \{-(0-h)^2\} = \lim_{h \rightarrow 0} (-h^2) = 0$$

$$\text{R.H.L.} \neq \text{L.H.L.}$$

$\therefore$ Function is not continuous at $x=0$.

Discussion at $x=1$

$$\begin{aligned} \text{R.H.L.} &= f(1+0) = \lim_{h \rightarrow 0} (1+h) = \lim_{h \rightarrow 0} \{4(1+h)^2 - 3(1+h)\} \\ &= \lim_{h \rightarrow 0} [4(1+2h+h^2) - 3(1+h)] = \lim_{h \rightarrow 0} (1+5h+4h^2) = 1 \end{aligned}$$

$$\text{L.H.L.} = f(1-0) = \lim_{h \rightarrow 0} (1-h) = \lim_{h \rightarrow 0} \{5(1-h)-4\} = \lim_{h \rightarrow 0} (1-5h) = 1$$

$$\text{Also} \quad f(1) = 5 \times 1 - 4 = 1 \text{ because for } x=1 \text{ we have } f(x) = 5x-4$$

$$\text{We have } \text{R.H.L.} = \text{L.H.L.} = f(1) = 1$$

$\therefore$ Function is continuous at $x=1$

Discussion at $x=2$

$$\text{R.H.L.} = f(2+0) = \lim_{h \rightarrow 0} f(2+h) = \lim_{h \rightarrow 0} \{3(2+h)+4\} = 10$$

$$\begin{aligned} \text{L.H.L.} &= f(2-0) = \lim_{h \rightarrow 0} f(2-h) = \lim_{h \rightarrow 0} \{4(2-h)^2 - 3(2-h)\} \\ &= \lim_{h \rightarrow 0} \{4(4+h^2-4h) - 6+3h\} \\ &= \lim_{h \rightarrow 0} (10-13h+4h^2) = 10 \end{aligned}$$

For $x=2$ we have $f(x) = 3x+4$ therefore

$$f(2) = 3 \times 2 + 4 = 10$$

we have $\text{R.H.L.} = \text{L.H.L.} = f(2) = 10$

$\therefore$ Function is continuous at $x=2$.

Example 29: Discuss the continuity of the following functions at the points indicated against them:

(i) $f(x) = 2^{1/x}$ when $x \neq 0$ and $f(0) = 0$ at $x = 0$ [B.C.A. (Bhopal) 2011]

(ii) $f(x) = \frac{\sin^2 ax}{x^2}$, when $x \neq 0$ and $f(0) = 1$ at $x = 0$ [B.C.A. (Meerut) 2008]

(iii) $f(x) = (1 + 2x)^{1/x}$ when $x \neq 0$ and $f(0) = e^2$ at $x = 0$ [B.C.A. (Meerut) 2004, 2006]

(iv) $f(x) = \frac{\sin 2x}{x}$ when $x \neq 0$ and $f(0) = 1$ at $x = 0$ [B.C.A. (Meerut) 2002]

Solution:

(i) We have $f(x) = 2^{1/x}$ when $x \neq 0$ and $f(0) = 0$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} 2^{1/0+h} = \lim_{h \rightarrow 0} 2^{1/h} = 2^\infty = \infty$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} 2^{1/-h} = \lim_{h \rightarrow 0} 2^{-(1/h)} = 2^{-\infty} = 0$$

$$f(0+0) \neq f(0-0)$$

Therefore function is discontinuous at $x = 0$.

(ii) $f(x) = \frac{\sin^2 ax}{x^2}$, when $x \neq 0$ and $f(0) = 1$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h)$$

$$= \lim_{h \rightarrow 0} \frac{[\sin a(0+h)]^2}{(0+h)^2} = \lim_{h \rightarrow 0} \frac{\sin^2 ah}{h^2} = \lim_{h \rightarrow 0} a^2 \left\{ \frac{\sin^2 ah}{a^2 h^2} \right\}$$

$$= a^2 \lim_{h \rightarrow 0} \left(\frac{\sin ah}{ah} \right)^2 = a^2 \left\{ \lim_{h \rightarrow 0} \frac{\sin ah}{ah} \right\}^2 = a^2 (1)^2 = a^2$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{[\sin a(0-h)]^2}{(0-h)^2}$$

$$= \lim_{h \rightarrow 0} \frac{[-\sin ah]^2}{h^2} = \lim_{h \rightarrow 0} a^2 \left\{ \frac{\sin^2 ah}{a^2 h^2} \right\}$$

$$= \lim_{h \rightarrow 0} a^2 \left(\frac{\sin ah}{ah} \right)^2 = a^2 \left\{ \lim_{h \rightarrow 0} \frac{\sin ah}{ah} \right\}^2 = a^2 (1)^2 = a^2$$

$$f(0+0) = f(0-0) \neq f(0)$$

$\therefore$ Function is discontinuous at $x = 0$

(iii) $f(x) = (1 + 2x)^{1/x}$ when $x \neq 0$ and $f(0) = e^2$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} [1 + 2(0+h)]^{1/0+h}$$

$$= \lim_{h \rightarrow 0} (1+2h)^{1/h} = \lim_{h \rightarrow 0} [(1+2h)^{1/2h}]^2 = e^2$$

$$\left[\text{because } \lim_{y \rightarrow 0} (1+y)^{1/y} = e \right]$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} [1+2(0-h)]^{1/0-h}$$

$$= \lim_{h \rightarrow 0} (1-2h)^{1/-h} = \lim_{h \rightarrow 0} [(1-2h)^{-1/2h}]^2 = e^2$$

$$\left[\text{because } \lim_{y \rightarrow 0} (1-y)^{1/y} = e \right]$$

$$f(0+0) = f(0-0) = f(0) = e^2$$

Therefore function is continuous at $x=0$.

(iv) $f(x) = \frac{\sin 2x}{x}$ when $x \neq 0$ and $f(0) = 1$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \frac{\sin 2(0+h)}{(0+h)} = \lim_{h \rightarrow 0} 2 \left[\frac{\sin 2h}{2h} \right]$$

$$= 2 \lim_{h \rightarrow 0} \left(\frac{\sin 2h}{2h} \right) = 2 \times 1 = 2$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{\sin 2(0-h)}{(0-h)} = \lim_{h \rightarrow 0} \frac{-\sin 2h}{-h}$$

$$= \lim_{h \rightarrow 0} 2 \left(\frac{\sin 2h}{2h} \right) = 2 \lim_{h \rightarrow 0} \frac{\sin 2h}{2h} = 2 \times 1 = 2$$

$$\therefore \quad \text{L.H.L.} = \text{R.H.L.} \neq f(0)$$

$\Rightarrow f(x)$ is discontinuous at $x=0$

Example 30: Test for continuity the function defined by

$$f(x) = \begin{cases} \left(\frac{x^2}{a} \right) - a & \text{when } x < a \\ a - \left(\frac{a^2}{x} \right) & \text{when } x > a \\ 0 & \text{when } x = a \end{cases}$$

at $x = a$

[B.C.A. (Rohilkhand) 2001, 2007;
B.C.A. (Delhi) 2007, 2011]

Solution:

$$\text{R.H.L.} = f(a+0) = \lim_{h \rightarrow 0} f(a+h) = \lim_{h \rightarrow 0} \left[a - \frac{a^2}{a+h} \right] \left[\text{because } x = a+h > a \right]$$

$$= \lim_{h \rightarrow 0} \left[\frac{a^2 + ah - a^2}{a+h} \right] = \lim_{h \rightarrow 0} \left[\frac{ah}{a+h} \right] = \frac{0}{a+0} = 0$$

$$\begin{aligned}
 \text{L.H.L.} &= f(a-0) = \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} \left\{ \frac{(a-h)^2}{a} - a \right\} \quad [\text{because } x = a-h < a] \\
 &= \lim_{h \rightarrow 0} \left[\frac{a^2 - 2ah + h^2 - a^2}{ah} \right] = \lim_{h \rightarrow 0} \frac{h^2 - 2ah}{a} \\
 &= \frac{0-0}{a} = \frac{0}{a} = 0 \\
 f(a+0) &= f(a-0) = f(0) = 0
 \end{aligned}$$

Hence $f(x)$ is continuous at $x = a$.

Example 31: Test the following function for continuity at $x = 0$

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & \text{when } x \neq 0 \\ x, & \text{when } x = 0 \end{cases}$$

[B.C.A. (Delhi) 2010; B.C.A. (Meerut) 2001, 2010]

Also show that it is continuous for all real values of x .

Solution: R.H.L. at $x = 0$

$$\begin{aligned}
 f(0+0) &= \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} (0+h) \sin \frac{1}{0+h} \\
 &= \lim_{h \rightarrow 0} h \sin \frac{1}{h} = \lim_{h \rightarrow 0} h \lim_{h \rightarrow 0} \sin \frac{1}{h} \\
 &= 0 \times \text{value oscillates between } -1 \text{ and } 1 = 0
 \end{aligned}$$

L.H.L. at $x = 0$

$$\begin{aligned}
 f(0-0) &= \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} (0-h) \sin \frac{1}{0-h} \\
 &= \lim_{h \rightarrow 0} (-h) \sin \left(-\frac{1}{h} \right) = \lim_{h \rightarrow 0} (-h) \left\{ -\sin \frac{1}{h} \right\} \\
 &= \lim_{h \rightarrow 0} h \sin \frac{1}{h} = \lim_{h \rightarrow 0} h \lim_{h \rightarrow 0} \sin \frac{1}{h} \\
 &= 0 \times \text{finite quantity lying between } -1 \text{ and } 1 = 0
 \end{aligned}$$

Also $f(0) = 0$

Therefore $f(0+0) = f(0-0) = f(0) = 0$

Hence the function is continuous at $x = 0$

Let $x = a$ where a is non-zero real number, then

$$\begin{aligned}
 f(a) &= a \sin \frac{1}{a} \\
 f(a+0) &= \lim_{h \rightarrow 0} f(a+h) = \lim_{h \rightarrow 0} (a+h) \sin \frac{1}{a+h} = a \sin \frac{1}{a} \\
 f(a-0) &= \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} (a-h) \sin \frac{1}{a-h} = a \sin \frac{1}{a} \\
 f(a+0) &= f(a-0) = f(a) = a \sin \frac{1}{a}
 \end{aligned}$$

Hence the function is continuous for all real numbers.

Example 32: Show that $f(x) = [x]$ is not continuous at $x = 1$

[B.C.A. (Kurukshetra) 2009; B.C.A. (Delhi) 2008;
B.C.A. (Avadh) 2007; B.C.A. (Meerut) 2001]

Solution: We have $f(x) = [x]$ i.e.

$$f(x) = \begin{cases} 0 & \text{when } 0 < x < 1 \\ 1 & \text{when } 1 < x < 2 \\ 1 & \text{when } x = 1 \end{cases}$$

because $[x] = \text{greatest integer } \leq x$

$$\text{R.H.L.} = f(1+0) = \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} 1 = 1 \quad \because 1 < 1+h < 2$$

$$\text{L.H.L.} = f(1-0) = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} 0 = 0 \quad \because 0 < 1-h < 1$$

$$f(0+0) \neq f(0-0)$$

Therefore function is not continuous at $x = 1$.

Example 33: Draw the graph of function

$$f(x) = |x-1| + 2|x-2| + 3|x-3| \text{ in } [0, 4]$$

and discuss the continuity in this interval.

[B.C.A. (Meerut) 2002, 2005]

Solution:

$$f(0) = |0-1| + 2|0-2| + 3|0-3| = 1 + 4 + 9 = 14$$

$$f(1) = |1-1| + 2|1-2| + 3|1-3| = 0 + 2 + 6 = 8$$

$$f(2) = |2-1| + 2|2-2| + 3|2-3| = 1 + 0 + 3 = 4$$

$$f(3) = |3-1| + 2|3-2| + 3|3-3| = 2 + 2 + 0 = 4$$

$$f(4) = |4-1| + 2|4-2| + 3|4-3| = 3 + 4 + 3 = 10$$

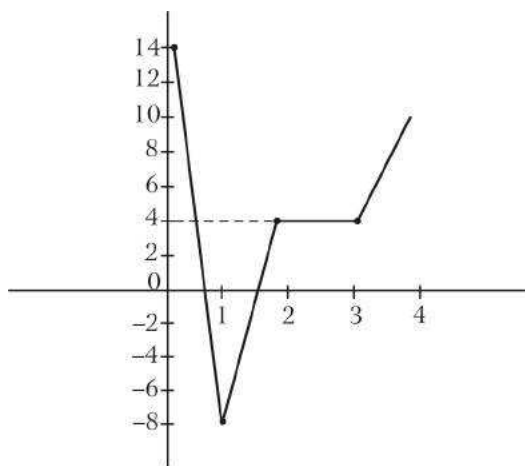


Fig. 5.3

Example 34: Show that the function $f(x) = |x| + |x-1|$ is continuous at $x=0$ and $x=1$.

[B.C.A (Kanpur) 2003]

Solution: The function $f(x)$ can be redescribed as follows:

$$f(x) = \begin{cases} 1-2x & \text{if } x \leq 0 \\ 1 & \text{if } 0 < x < 1 \\ 2x-1 & \text{if } x \geq 1 \end{cases}$$

we have

$$|x| = x \text{ when } x > 0$$

$$|x| = -x \text{ when } x < 0$$

$$|x| = 0 \text{ when } x = 0$$

(i) when $x \leq 0$ then

$$|x| + |x-1| = -x + \{-(x-1)\} \\ = 1-2x$$

(ii) when $0 < x < 1$

$$|x| + |x-1| = x + \{-(x-1)\} = 1$$

(iii) When $x \geq 1$

$$|x| + |x-1| = x + (x-1) = 2x-1$$

(i) At $x=0$

$$\text{R.H.L.} = f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} 1 = 1$$

$$\text{L.H.L.} = f(0-0) = \lim_{h \rightarrow 0} f(0-h)$$

$$= \lim_{h \rightarrow 0} f(1-2(0-h)) = \lim_{h \rightarrow 0} (1+2h) = 1$$

$$f(0+0) = f(0-0) = f(0) = 1, \text{ because } f(0) = 1.$$

(ii) At $x=1$

$$\text{R.H.L.} = f(1+0) = \lim_{h \rightarrow 0} f(1+h)$$

$$= \lim_{h \rightarrow 0} \{2(1+h)-1\} = \lim_{h \rightarrow 0} (1+2h) = 1$$

$$\text{L.H.L.} = f(1-0) = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} (1) = 1$$

$$f(1-0) = f(1+0) = f(1) = 1$$

Hence, function is continuous at $x=0$ and $x=1$.

Example 35: Examine the continuity at $x=0$ of the function defined below for different values of k .

$$f(x) = \frac{\sin 3x}{x} \text{ when } x \neq 0, \text{ and } f(x) = k \text{ when } x = 0$$

[B.C.A. (Delhi) 2005, 2008, 2011]

Solution: We know that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

$$\therefore \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \lim_{x \rightarrow 0} 3 \left(\frac{\sin 3x}{3x} \right) = 3 \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \right) = 3 \times 1 = 3$$

Also $f(0) = k$ (given)

If $k = 3$ then $\lim_{x \rightarrow 0} f(x) = f(0) = 3$

$\therefore f(x)$ is continuous at $x = 0$

If $k \neq 3$ then $\lim_{x \rightarrow 0} f(x) \neq f(0)$

$\therefore f(x)$ is discontinuous at $x = 0$

Example 36: Let a function f be defined on R by

$$f(x) = \begin{cases} 1 & \text{where } x \text{ is rational} \\ -1 & \text{where } x \text{ is irrational} \end{cases}$$

Show that $f(x)$ is discontinuous everywhere.

Solution: Let a be any real number. Then a is either rational or irrational.

(i) Let a be any rational number so that $f(a) = 1$

$$\text{R.H.L} = f(a+0) = \lim_{h \rightarrow 0} f(a+h)$$

We know that $a+h$ is a *nbd* of a and every *nbd* of a rational number contains infinitely many irrationals so $f(a+h)$ may be irrational number so that

$$f(a+h) = -1 \text{ therefore}$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} f(a+h) = \lim_{h \rightarrow 0} (-1) = -1$$

$$\text{i.e., } f(a+0) \neq f(a)$$

$$\Rightarrow f(x) \text{ is not continuous at } x = a$$

(ii) Let a be any irrational number so that $f(a) = -1$

$$\text{R.H.L.} = f(a+0) = \lim_{h \rightarrow 0} f(a+h)$$

We have $a+h$ is a *nbd* of a and every *nbd* of irrational number contains infinitely many rationals so that

$$f(a+h) = 1 \text{ therefore}$$

$$\text{R.H.L.} = f(a+0) = \lim_{h \rightarrow 0} f(a+h) = \lim_{h \rightarrow 0} (1) = 1$$

$$\text{i.e., } f(a) \neq f(a+0)$$

$$\Rightarrow f(x) \text{ is not continuous at } x = a.$$

Hence f is discontinuous everywhere.

Example 37: Discuss the continuity of the function $f(x) = [x]$ at the points $\frac{1}{2}$ and 1 .

where $[x]$ denote the greatest integers $\leq x$.

Solution: At $x = \frac{1}{2}$

$$f\left(\frac{1}{2}+0\right) = \lim_{h \rightarrow 0} f\left(\frac{1}{2}+h\right) = \lim_{h \rightarrow 0} \left[\frac{1}{2}+h\right] = \lim_{h \rightarrow 0} (0) = 0$$

$$\therefore 0 < \frac{1}{2} + h < 1$$

$$f\left(\frac{1}{2}-0\right) = \lim_{h \rightarrow 0} f\left(\frac{1}{2}-h\right) = \lim_{h \rightarrow 0} \left[\frac{1}{2}-h\right] = \lim_{h \rightarrow 0} (0) = 0$$

$$\therefore 0 < \frac{1}{2}-h < 1$$

Now
$$f\left(\frac{1}{2}\right) = \left[\frac{1}{2}\right] = 0$$

$$\therefore f\left(\frac{1}{2}+0\right) = f\left(\frac{1}{2}-0\right) = 0 = f\left(\frac{1}{2}\right)$$

i.e. function is continuous at $x = \frac{1}{2}$

At $x = 1$

$$f(1-0) = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} [1-h] = \lim_{h \rightarrow 0} (0) = 0$$

$$\therefore 0 < 1-h < 1$$

$$f(1+0) = \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} [1+h] = \lim_{h \rightarrow 0} (1) = 1$$

$$\therefore 1 < 1+h < 2$$

i.e.
$$f(1+0) \neq f(1-0)$$

$\therefore$ Function is discontinuous at $x = 1$

Example 38: Show that the function $f(x) = x - [x]$, where x is a positive variable and $[x]$ denotes the integral part of x is discontinuous for all integral values of x and continuous for all others. Draw its graph. [B.C.A. (Meerut) 2007]

Solution: If n is any integer then

$$f(x) = \begin{cases} x - (n-1) & \text{when } n-1 < x < n \\ 0 & \text{when } x = n \\ x - n & \text{when } n < x < n+1 \end{cases}$$

Case I: When n is an integer

$$\begin{aligned} f(n+0) &= \lim_{h \rightarrow 0} f(n+h) = \lim_{h \rightarrow 0} (n+h) - [n+h] \\ &= \lim_{h \rightarrow 0} \{(n+h) - n\} = \lim_{h \rightarrow 0} h = 0 \end{aligned}$$

$$\therefore n < n+h < n+1$$

$$\begin{aligned} f(n-0) &= \lim_{h \rightarrow 0} f(n-h) = \lim_{h \rightarrow 0} \{(n-h) - [n-h]\} \\ &= \lim_{h \rightarrow 0} \{(n-h) - (n-1)\} = \lim_{h \rightarrow 0} (1-h) = 1 \end{aligned}$$

$$\therefore n-1 < n-h < n$$

$$f(n+0) \neq f(n-0)$$

i.e. function is discontinuous for all $x = n$

Case II: When $n < x < n+1$ where $n = 0, 1, 2, 3, \dots$

If $0 < x < 1$ then $[x] = 0$ and $f(x) = x - [x] = x - 0$

also if $x = 1$ then $[x] = 1$ and $f(x) = 1 - 1 = 0$

If $1 < x < 2$ then $[x] = 1$ and $f(x) = x - [x] = x - 1$

also If $x = 2$ then $[x] = 2$ and $f(x) = 2 - 2 = 0$

also if $2 < x < 3$ then $[x] = 2$ and $f(x) = x - [x] = x - 2$ and so on

Now graph of the function is

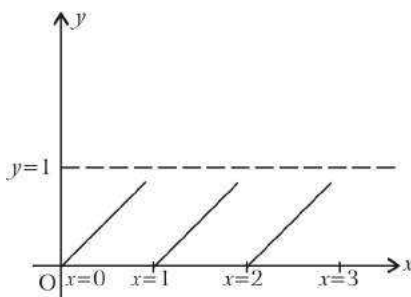


Fig. 5.4

The graph when $0 \leq x \leq 1$, $1 \leq x \leq 2$, $2 \leq x \leq 3$ and so on is discontinuous at $x = 1, 2, 3, \dots$ and continuous for all other values of x .

Example 39: Show that the function f defined as

$$f(x) = \begin{cases} 0 & \text{when } x = 0 \\ \frac{1}{2} - x & \text{when } 0 < x < \frac{1}{2} \\ \frac{1}{2} & \text{when } x = \frac{1}{2} \\ \frac{3}{2} - x & \text{when } \frac{1}{2} < x < 1 \\ 1 & \text{when } x = 1 \end{cases}$$

has three points of discontinuity which you are required to find. Also draw the graph of the function.

[B.C.A. (Delhi) 2008; B.C.A. (Rohilkhand) 2005]

Solution: We know that any polynomial is continuous at each point, therefore $\frac{1}{2} - x$ when $0 < x < \frac{1}{2}$ and $\frac{3}{2} - x$ when $\frac{1}{2} < x < 1$ are both polynomial so continuous at each point.

Thus we have only to require to find the continuity as $x = 0, \frac{1}{2}$ and 1 .

(i) When $x = 0$ we have $f(0) = 0$

$$f(0+0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} \left\{ \frac{1}{2} - (0+h) \right\} = \frac{1}{2}$$

i.e. $f(0) \neq f(0+0)$ so function is not continuous at $x = 0$

(ii) When $x = \frac{1}{2}$ we gave $f\left(\frac{1}{2}\right) = \frac{1}{2}$

$$f\left(\frac{1}{2}+0\right) = \lim_{h \rightarrow 0} f\left(\frac{1}{2}+h\right) = \lim_{h \rightarrow 0} \left\{ \frac{3}{2} - \left(\frac{1}{2}+h\right) \right\} = 1$$

$$f\left(\frac{1}{2}-0\right) = \lim_{h \rightarrow 0} f\left(\frac{1}{2}-h\right) = \lim_{h \rightarrow 0} \left\{ \frac{1}{2} - \left(\frac{1}{2}-h\right) \right\} = 0$$

$$\text{i.e. } f\left(\frac{1}{2}+0\right) \neq f\left(\frac{1}{2}-0\right) \neq f\left(\frac{1}{2}\right)$$

i.e. function is discontinuous at $x = \frac{1}{2}$

(iii) When $x = 1$ we have $f(x) = 1$, i.e. $f(1) = 1$

$$f(1-0) = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} \left\{ \frac{3}{2} - (1-h) \right\} = \frac{1}{2}$$

i.e. $f(1) \neq f(1-0)$ and therefore $f(x)$ is discontinuous at $x = 1$.

Hence the function is discontinuous at $x = 0, \frac{1}{2}$ and 1 .

The graph of the function consists of the point $(0,0)$; the segment of the line $y = \frac{1}{2} - x, 0 < x < \frac{1}{2}$. The point $\left(\frac{1}{2}, \frac{1}{2}\right)$; the segment of the line $y = \frac{3}{2} - x, \frac{1}{2} < x < 1$; and the point $(1,1)$.

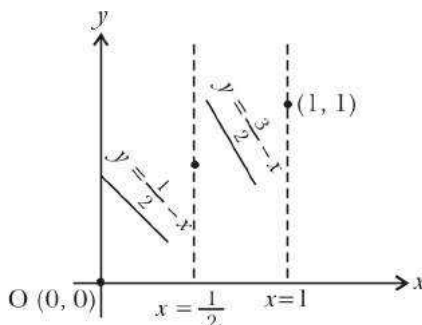


Fig. 5.5

Example 40: Prove that the function $y = E(x)$, where $E(x)$ denotes the integral part of x is discontinuous for all integers. Also draw the graph.

Solution: We have $E(x)$ is the integral part of x i.e. $E(x) = [x]$ denotes the greatest integer not greater than x . Thus we have

$$E(x) = \begin{cases} n-1 & \text{if } n-1 \leq x < n \\ n & \text{if } n \leq x < n+1 \\ n+1 & \text{if } n+1 \leq x < n+2 \end{cases}$$

and so on where n is an integer.

Now if $x = n$ where $n = 0, \pm 1, \pm 2, \pm 3, \dots$

Then $E(x) = n$ or $E(n) = n$

$$E(n-0) = \lim_{h \rightarrow 0} E(n-h) = \lim_{h \rightarrow 0} (n-1) = n-1$$

$$E(n+0) = \lim_{h \rightarrow 0} E(n+h) = \lim_{h \rightarrow 0} (n) = n$$

i.e. $E(n+0) \neq E(n-0)$,

Therefore function is discontinuous when $x = n$ is an integer.

Evidently it is continuous for all other values of x .

In order to draw the graph let us find the value of the function $y = E(x)$ for $n = 0, \pm 1, \pm 2, \pm 3, \pm 4, \dots$ we get

$y = E(x) = -4$	if	$-4 \leq x < -3$
$y = E(x) = -3$	if	$-3 \leq x < -2$
$y = E(x) = -2$	if	$-2 \leq x < -1$
$y = E(x) = -1$	if	$-1 \leq x < 0$
$y = E(x) = 0$	if	$0 \leq x < 1$
$y = E(x) = 1$	if	$1 \leq x < 2$
$y = E(x) = 2$	if	$2 \leq x < 3$
$y = E(x) = 3$	if	$3 \leq x < 4$
$y = E(x) = 4$	if	$4 \leq x < 5$ and so on.

The graph of the function is

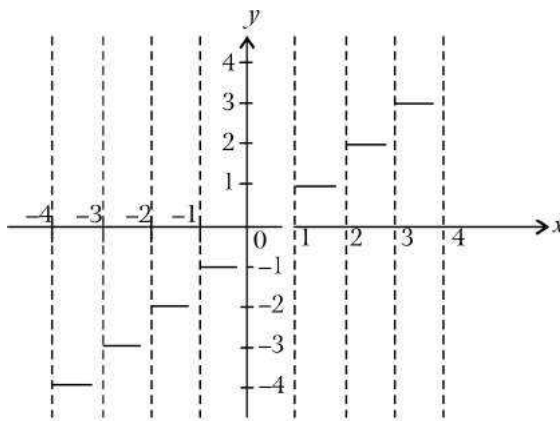


Fig. 5.6

Example 41: Is the function $f(x) = \frac{3x + 4 \tan x}{x}$ continuous at $x = 0$. If not, how may be function be defined to make it continuous at this point.

Solution: $f(x) = \frac{3x + 4 \tan x}{x} = 3 + 4 \frac{\tan x}{x}$

$$\lim_{h \rightarrow 0} f(x) = \lim_{h \rightarrow 0} 3 + 4 \lim_{h \rightarrow 0} \frac{\tan x}{x} = 3 + 4 \times 1 = 7$$

∴ Limit of the function at $x=0$ exists and equal to 7.

$$f(0) = \frac{3 \times 0 + 4 \times 0}{0} = \frac{0}{0}$$

which is an undefined quantity *i.e.* $f(0)$ does not exist.

Hence $f(x)$ is discontinuous at $x=0$

Now if we define the function $f(x)$ as below:

$$f(x) = \begin{cases} \frac{3x+4 \tan x}{x} & \text{when } x \neq 0 \\ 7 & \text{when } x = 0 \end{cases}$$

Then clearly $\lim_{x \rightarrow 0} f(x) = f(0) = 7$ and so $f(x)$ will be continuous at $x = 7$.

Example 42: Draw the graph of the function

$$f(x) = |x-1| + |x-2|$$

in the interval $[0, 3]$ and discuss the continuity in that interval. [B.C.A. (Meerut) 2006]

Solution: We have by definition of absolute value of x

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$\begin{aligned} \therefore f(x) &= 1-x+2-x = 3-2x, & \text{when } x \leq 1 \\ f(x) &= x-1+2-x = 1, & \text{when } 1 \leq x \leq 2 \\ f(x) &= x-1+x-2 = 2x-3, & \text{when } x \geq 2 \end{aligned}$$

Thus, the graph consists of the segments of the three straight line $y = 3-2x$, $y = 1$ and $y = 2x-3$ corresponding to the intervals $[0,1]$, $[1,2]$, $[2,3]$ respectively. The graph of the function for the interval $[0,3]$ is as given in the figure. The graph shows that the function is continuous throughout the interval.

$$\therefore f(x) = \begin{cases} 3-2x, & \text{when } x \leq 1 \\ 1, & \text{when } 1 \leq x \leq 2 \\ 2x-3, & \text{when } x \geq 2 \end{cases}$$

Test the continuity at $x = 1$

$$\text{L.H.L. at } (x=1) = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} \{3-2(1-h)\} = 1$$

$$\text{R.H.L. at } (x=1) = \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} (1) = 1$$

and $f(1) = 1$

$$\therefore \text{L.H.L.} = \text{R.H.L.} = f(1) = 1$$

Hence function is continuous at $x = 1$

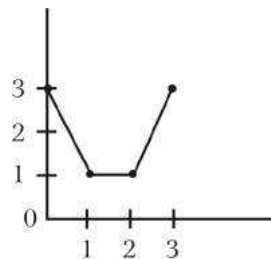


Fig. 5.7

Test the continuity at $(x = 2)$

$$\text{L.H.L. at } (x = 2) = \lim_{h \rightarrow 0} f(2 - h) = \lim_{h \rightarrow 0} (1) = 1$$

$$\text{R.H.L. at } (x = 2) = \lim_{h \rightarrow 0} f(2 + h) = \lim_{h \rightarrow 0} \{2(2 + h) - 3\} = 1$$

and

$$f(2) = 1$$

$$\text{L.H.L.} = \text{R.H.L.} = f(2) = 1$$

Hence function is continuous at $x = 2$.

Example 43: Show that a polynomial function is continuous at each point.

[B.C.A. (Meerut) 2003]

Solution: Let $f(x) = a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n$ be a polynomial in x of degree n ,

where $a_0, a_1, \dots, a_n$ are real numbers and $a_0 \neq 0$,

Here we shall show that $f(x)$ is continuous for all $x \in R$.

Let c be any real number, then

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} [a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n] \\ &= a_0 \lim_{x \rightarrow c} x^n + a_1 \lim_{x \rightarrow c} x^{n-1} + \dots + a_{n-1} \lim_{x \rightarrow c} x + \lim_{x \rightarrow c} a_n \\ &= a_0c^n + a_1c^{n-1} + \dots + a_{n-1}c + a_n = f(c) \left[\because \lim_{x \rightarrow c} x = c \right] \end{aligned}$$

Since $\lim_{x \rightarrow c} f(x) = f(c)$, hence, $f(x)$ is continuous at $x = c$.

Thus, $f(x)$ is continuous at every real number c and so $f(x)$ is continuous for all $x \in R$.

Example 44: Discuss the continuity of the function

$$f(x) = \begin{cases} x \log \sin^2 x, & \text{for } x \neq 0, \\ 0, & \text{for } x = 0, \end{cases}$$

[B.C.A. (Garhwal) 2006, 2011; B.C.A. (Meerut) 2003]

Solution: Let $0 \neq c \in R$ then, we have

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} (x \log \sin^2 x) = c \log \sin^2 c = f(c)$$

Therefore, $f(x)$ is continuous $\forall 0 \neq x \in R$,

Now, we check continuity at $x = 0$. We have $f(0) = 0$

Now

$$\begin{aligned} f(0-0) &= \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} (-h) \\ &= \lim_{h \rightarrow 0} (-h) \log \sin^2 (-h) = \lim_{h \rightarrow 0} (-h) \log \sin^2 h \end{aligned}$$

[it is indeterminate form $0 \times \infty$]

$$= - \lim_{h \rightarrow 0} \frac{\log(\sin^2 h)}{(1/h)}, \quad \left[\text{Form } \frac{\infty}{\infty} \right] \text{ Use L. Hospital's rule}$$

$$= - \lim_{h \rightarrow 0} \frac{\left(\frac{1}{\sin^2 h} \cdot 2 \sin h \cos h \right)}{(-1/h^2)} = \lim_{h \rightarrow 0} \frac{2h^2}{\tan h} \quad \left[\text{Form } \frac{0}{0} \right]$$

$$= \lim_{h \rightarrow 0} \frac{4h}{\sec^2 h} = 0. \quad [\text{Again by L. Hospital's rule}]$$

and

$$\begin{aligned} f(0+0) &= \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} (h) \\ &= \lim_{h \rightarrow 0} h \log(\sin^2 h) \quad [\text{Form } 0 \times \infty] \end{aligned}$$

$$= \lim_{h \rightarrow 0} \frac{\log(\sin^2 h)}{(1/h)} \quad \left[\text{Form } \frac{\infty}{\infty} \right] \quad \text{Use L. Hospital's rule}$$

$$= \lim_{h \rightarrow 0} \frac{\left(\frac{1}{\sin^2 h} \times 2 \sin h \cos h \right)}{(-1/h^2)} = \lim_{h \rightarrow 0} \left[\frac{-2h^2}{\tan h} \right] \quad \left(\frac{0}{0} \right)$$

$$= \lim_{h \rightarrow 0} \left[\frac{-4h}{\sec^2 h} \right] = 0$$

Since $f(0+0) = f(0) = f(0-0)$, therefore the function $f(x)$ is continuous at $x=0$. Hence $f(x)$ is continuous for all value of $x \in R$.

Example 45: Examine the continuity of the function $f(x)$ defined by

$$f(x) = \begin{cases} 5x-4, & \text{if } 0 < x \leq 1 \\ 4x^3-3x, & \text{if } 1 < x < 2 \end{cases} \text{ at } x=1 \quad [\text{B.C.A. (Meerut) 2005}]$$

Solution: We have $f(1) = 5 \cdot 1 - 4 = 1$

$$\text{L.H.L. at } (x=1) = f(1-0) = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} \{5(1-h)-4\} = \lim_{h \rightarrow 0} \{1-5h\} = 1$$

$$\text{R.H.L. at } (x=1) = f(1+0) = \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} \{4(1+h)^3 - 3(1+h)\}$$

$$= \{4(1+0)^3 - 3(1+0)\} = 4 - 3 = 1$$

$$\therefore f(1-0) = f(1) = f(1+0)$$

$$\Rightarrow f(x) \text{ is continuous at } x=1$$

Example 46: If $f(x) = \begin{cases} -x, & \text{if } x \leq 0 \\ x, & \text{if } 0 < x < 1 \\ 2-x, & \text{if } x \geq 1 \end{cases}$

Examine the continuity of the function at $x=0, 1$

[B.C.A (Rohilkhand) 2007, 2009]

Solution: We have

$$\text{L.H.L. at } (x=0) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} [-(-h)] = 0$$

$$\text{R.H.L. at } (x=0) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} (h) = 0$$

But $f(0) = 0$

$$\text{L.H.L.} = \text{R.H.L} = f(0) = 0$$

Therefore, $f(x)$ is continuous at $x = 0$

Also at $x = 1$,

$$\text{R.H.L. at } (x=1) = f(1+0) = \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} 2 - (1+h) = 1$$

$$\text{L.H.L. at } (x=1) = f(1-0) = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} (1-h) = 1$$

$$\therefore f(1-0) = f(1+0) = 1 = f(1) \quad [f(1) = 2 - 1 = 1]$$

$$\Rightarrow f(x) \text{ is continuous at } x = 1$$

Example 47: Show that the function defined by

$$f(x) = \begin{cases} \frac{x^2 - 9}{x - 3} & , \quad x \neq 3 \\ 6 & , \quad x = 3 \end{cases}$$

is continuous at $x = 3$

[B.C.A. (Meerut) 2008, 2010]

Solution: We have $f(3) = 6$

$$\begin{aligned} \text{L.H.L. at } (x=3) &= f(3-0) = \lim_{h \rightarrow 0} f(3-h) = \lim_{h \rightarrow 0} \frac{(3-h)^2 - 9}{3-h-3} \\ &= \lim_{h \rightarrow 0} \left\{ \frac{3^2 + h^2 - 6h - 9}{-h} \right\} = \lim_{h \rightarrow 0} \frac{h(h-6)}{(-h)} = \lim_{h \rightarrow 0} (6-h) = 6 \end{aligned}$$

$$\begin{aligned} \text{R.H.L. at } (x=3) &= \lim_{h \rightarrow 0} f(3+h) = \lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{3+h-3} \\ &= \lim_{h \rightarrow 0} \frac{3^2 + h^2 + 6h - 9}{3+h-3} = \lim_{h \rightarrow 0} \frac{h(h+6)}{h} = 6 \end{aligned}$$

$$\text{Since } f(3-0) = f(3+0) = f(3) = 6$$

Therefore, $f(x)$ is continuous at $x = 3$.



Comprehensive Exercise 5.2

(Continuity)

1. Discuss the continuity and discontinuity of the following functions:

(i) $f(x) = x^2 + 2x$

(ii) $f(x) = e^{-\frac{1}{x}}$

(iii) $f(x) = \sin\left(\frac{1}{x}\right)$ if $x \neq 0$ and $f(0) = 0$

(iv) $f(x) = \begin{cases} \sin x \cos \frac{1}{x} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$

(v) $f(x) = \begin{cases} \frac{1}{1-e^{-\frac{1}{x}}} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$

(vi) $f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$

(vii) $f(x) = \frac{1}{1-e^{1/x}}, x \neq 0$ and $f(0) = 0$

[B.C.A. (Meerut) 2004]

2. Show that the function $f(x) = \frac{1}{x-a} \sin \frac{1}{x-a}$ is discontinuous at $x = a$.

3. Examine continuity and discontinuity of the following function at $x = 0$:

(i) $f(x) = \frac{\sin x}{x}, x \neq 0$ and $f(0) = 1$

(ii) $f(x) = x \cos \frac{1}{x}, x \neq 0$ and $f(0) = 0$

(iii) $f(x) = \begin{cases} \frac{xe^{1/x}}{1+e^{1/x}} & \text{when } x \neq 0 \\ 1 & \text{when } x = 0 \end{cases}$

(iv) $f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$

[B.C.A. (Meerut) 2004]

(v) $f(x) = \begin{cases} \frac{e^{1/x} \sin\left(\frac{1}{x}\right)}{1+e^{1/x}} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$

(vi) $f(x) = [x] - [-x]$ where $[x]$ denotes the integral part of x .

(vii) $f(x) = |x|$

[B.C.A. (Meerut) 2004]

4. Is the function $f(x) = 2x^2 + 3x - 1$ continuous at $x = 2$?

5. Prove that the function $f(x)$ defined by

$$f(x) = \begin{cases} \frac{1}{2}, & \text{If } x \text{ is rational} \\ \frac{1}{3}, & \text{If } x \text{ is irrational} \end{cases}$$

is discontinuous everywhere.

6. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = \frac{1}{x-4}$.

Draw the graph and discuss the type of discontinuity which the function $f(x)$ has $]-\infty, \infty[$.

7. Examine the discontinuity of the function

$$f(x) = \begin{cases} \frac{x^2 - 4x + 3}{x^2 - 1} & \text{for } x \neq 1 \\ 2 & \text{for } x = 1 \end{cases}$$

at $x = 1$.

8. Show that the function $f(x) = [x] + [-x]$ has removable discontinuity for integral values of x .

9. Find the values of a, b and c for which the function

$$f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{when } x < 0 \\ c & \text{when } x = 0 \\ \frac{(x + bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{when } x > 0 \end{cases}$$

is continuous at $x = 0$

10. Show that the function

$$f(x) = \begin{cases} \frac{1}{x-a} \operatorname{cosec}\left(\frac{1}{x-a}\right) & \text{when } x \neq a \\ 0 & \text{when } x = a \end{cases}$$

is discontinuous at $x = a$

[B.C.A. (Meerut) 2006, 2011]

11. Show that the function

$$f(x) = \begin{cases} 1+x & \text{when } x \leq 2 \\ 5-x & \text{when } x \geq 2 \end{cases}$$

is continuous at $x = 2$

12. Discuss the continuity of the function

$$f(x) = \begin{cases} x^2 & \text{if } x < -2 \\ 4 & \text{if } -2 \leq x \leq 2 \\ x^2 & \text{if } x > 2 \end{cases}$$

13. Show that the function

$$f(x) = \frac{x^2}{1+x^2} + \frac{x^2}{(1+x^2)^2} + \frac{x^2}{(1+x^2)^3} + \frac{x^2}{(1+x^2)^4} + \dots \infty$$

is not continuous at $x = 0$

14. Show that the function

$$f(x) = \begin{cases} \frac{1}{x} \sin \frac{1}{x} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$$

is discontinuous at $x = 0$

15. A function is defined on $[0, 1]$ is given by

$$f(x) = \begin{cases} x, & \text{If } x \text{ is rational} \\ 1-x, & \text{If } x \text{ is irrational} \end{cases}$$

Show that the function is continuous only at $x = \frac{1}{2}$

16. Give an example of a function $f(x)$ such that f is not continuous but $|f|$ is continuous. Show that f^2 can be continuous when f is not.
17. Give an example each of the following:
- (i) The function which possesses a limit $x = 1$ but it is not defined at $x = 1$
 - (ii) The function which is neither has a limit at $x = 1$ nor defined at $x = 1$
18. Show that the function

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x^2} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

in the closed interval $[0, 1]$ is continuous.

19. Define the continuity of a function on an interval. Discuss the continuity of function.

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

[B.C.A. (Meerut) 2003]

20. Prove that a polynomial function is continuous at each point.

[B.C.A. (Meerut) 2003]

21. Describe different kinds of Discontinuities of a function by given one example of each.

[B.C.A. (Meerut) 2002]



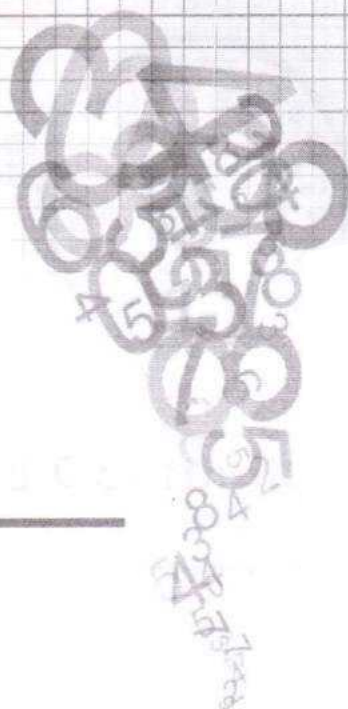
Answers

1.	(i)	Continuous everywhere	(ii)	Discontinuous at $x = 0$
	(iii)	Discontinuous at $x = 0$	(iv)	Continuous everywhere
	(v)	Discontinuous at $x = 0$	(vi)	Discontinuous at $x = 0$
	(vii)	Discontinuous at $x = 0$, only		
3.	(i)	Continuous at $x = 0$	(ii)	Continuous
	(iii)	Discontinuous at $x = 0$	(iv)	Discontinuous at $x = 0$
	(v)	Discontinuous at $x = 0$	(vi)	Discontinuous at $x = 0$
	(vii)	Continuous at $x = 0$		
4.	Yes			
5.	Discontinuous everywhere			
6.	Discontinuous only at $x = 4$			
7.	Discontinuous only at $x = 4$			
9.	$a = -\frac{3}{2}, c = \frac{1}{2}$ and $b \in R - \{0\}$			
12.	Continuous at $x = 2$ and -2			
19.	Continuous everywhere			



UNIT

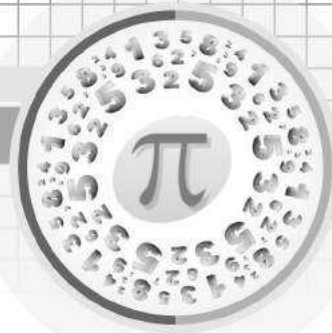
3



6. Differentiation
7. Rolle's and Mean Value Theorems
8. Expansion of Functions
9. Indeterminant Forms
10. Maxima and Minima
11. Curve Tracing
12. Successive Differentiation and Leibnitz Theorem



Unit-3



Chapter

6

Differentiation

6.1 Definition of Differential Coefficient

Let $y = f(x)$ be a function then the differential coefficient of y w.r.t x is

$$\frac{dy}{dx} \text{ or } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

differential coefficient of y w.r.t ' x ' at $x = a$ is

$$\left(\frac{dy}{dx}\right) \text{ or } f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

At $x = a$

Viz the derivative of f at x i.e. $f'(x)$ or $D f(x)$.

Thus, the function is called the **derivative or differentiation of $f(x)$** with respect to x or

simply differentiation of $f(x)$ is denoted by $f'(x)$ or $D f(x)$ or $\frac{d}{dx}(f(x))$.

Thus
$$\frac{d}{dx}(f(x)) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \dots(1)$$

or
$$\frac{d}{dx}(f(x)) = \lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{-h} \quad \dots(2)$$

The differentiation or derivative of a function $f(x)$ is also called the differential coefficient of $f(x)$. But we shall be using the words differentiation or derivative only.

6.1.1 Geometrical Meaning of Derivative at a Point Consider the Curve

$$y = f(x).$$

Let $f(x)$ be differentiable at $x = c$. Let $P(c, f(c))$ be a point on the curve and let $Q(x, f(x))$ be a neighbouring point on the curve. Then slope of the chord

$$PQ = \frac{f(x) - f(c)}{x - c}$$

From the free body diagram (F. B.D), we have the limiting position of chord PQ .

Taking limit as $Q \rightarrow P$ i.e., $x \rightarrow c$ we get

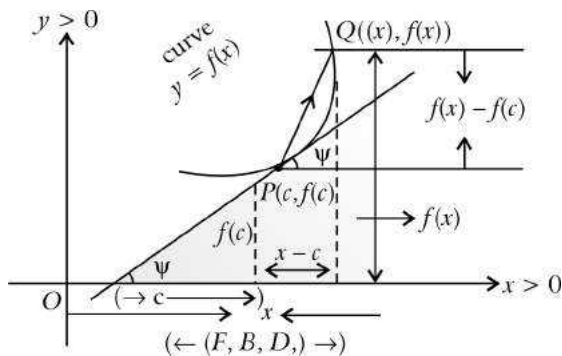


Fig. 6.1

Taking limit as $Q \rightarrow P$ i.e. $x \rightarrow c$ we get

$$\lim_{Q \rightarrow P} (\text{slope of the chord } PQ) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \quad \dots(1)$$

As $Q \rightarrow P$, Chord PQ becomes that tangent at P .

Hence from (1), we have

$$\text{slope of the tangent at } P = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \left[\frac{df(x)}{dx} \right]_{x=c} \text{ or } f'(c)$$

If free body diagram adopts the necessary and sufficient condition of $(Diff)^n$. Thus, the derivative of a function at a point $x = c$ is the slope of the tangent to the curve $y = f(x)$ at a point $(c, f(c))$. Also if $a(fu)^n$ is not differentiable at $x = c$ only if the point $(c, f(c))$ is a corner point of the curve $y = f(x)$.

Note: Differentiable = $(Diff)^n = Diff$

F.B.D = Free body diagram

Eqn = Equation

$(fu)^n$ = Function

D.C. = Differential coefficient

6.1.2 Geometrical Meaning of Derivative

Let us consider a function $y = f(x)$ in a rectangular coordinates system. We also consider a point $P(x, y)$ on the curve.

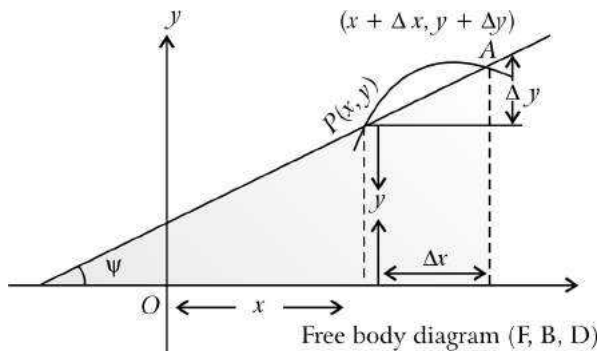


Fig. 6.2

If a point corresponding to an increased value of the argument $x + \Delta x$ is considered its ordinates value is given by.

$$y + \Delta y = f(x + \Delta x)$$

The point $(x + \Delta x, y + \Delta y)$ is represented by A. Hence PA is the secant to the curve.

Now; on $\Delta x, \Delta y \rightarrow 0$

$\Rightarrow PA \rightarrow 0$ (i.e. the distance PA tends to zero & to a single point P)

$\Rightarrow \lim_{\Delta x \rightarrow 0} (\text{slope of chord PA}) \rightarrow (\text{slope of the tangent at P})$

$$\text{or} \quad \tan \psi = \lim_{\Delta x \rightarrow 0} \tan \theta$$

$$\Rightarrow \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx} \text{ or } f'(x)$$

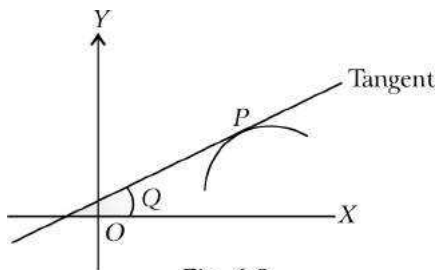


Fig. 6.3

means that the value of the derivative $f'(x)$ for a given value of x is equal to the tangent of the angle formed by the line tangent to the graph of function $y = f(x)$ at a point $P(x, y)$ with the positive.

Note: The meaning of the term rate of change of y with respect to x is that if x increased by additional unit change in y is given by $\left(\frac{dy}{dx}\right)$.

6.2 Differential Coefficients of Standard Function

- (i) $\frac{d}{dx}(\sin x) = \cos x$ (ii) $\frac{d}{dx}(\cos x) = -\sin x$
- (iii) $\frac{d}{dx}(\tan x) = \sec^2 x$ (iv) $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
- (v) $\frac{d}{dx}(\sec x) = \sec x \tan x$ (vi) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$
- (vii) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} \quad (-1 < x < 1)$
- (viii) $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}} \quad (-1 < x < 1) \text{ or } |x| < 1$
- (ix) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}, \quad -\infty < x < \infty \text{ or } x \in R$
- (x) $\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}, \quad -\infty < x < \infty$
- (xi) $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1 \text{ or } x \in R - [-1, 1]$
- (xii) $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}, \quad |x| > 1 \text{ or } x \in R - [-1, 1]$

Differential coefficients of standard function

6.2.1 Power Formulae in (Diff)ⁿ

(i) $\frac{d(x^n)}{dx} = nx^{n-1}$

Note: 1. $\frac{dy}{dx}$ is not a fraction i.e. $\frac{dy}{dx} \neq dy \div dx$

Since $\frac{dy}{dx} = \frac{d}{dx}(y)$ i.e. it is just an operation on real number.

2. For differential coefficients different notation or operation of (Diff)ⁿ can be written as;

$$\frac{dy}{dx} \text{ or } \frac{d(f(x))}{dx} \text{ or } f'(x) \text{ or } y_1 \text{ or } Dy$$

(ii) $\frac{d(c)}{dx} = 0$. Differentiation of a constant function is zero.

(iii) $\frac{d(\log_e x)}{dx} = \frac{1}{x}$ where $x > 0$

(iv) $\frac{d(e^x)}{dx} = e^x$

(v) $\frac{d(a^x)}{dx} = a^x \cdot \log_e a$ where $a > 0$

(vi) $\frac{d\{c \cdot f(x)\}}{dx} = c \cdot \frac{d}{dx}(f(x))$

i.e. the derivative of a constant times of function is the constant times the derivative of the function.

$$(vii) \quad \frac{d}{dx} [f(x) \pm g(x)] = \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x).$$

That is the derivative of the sum or diff. of two functions is the sum or difference of their derivatives.

Product Rule

$$\frac{d}{dx} [f(x) \cdot g(x)] = f(x) \cdot \frac{d}{dx} g(x) + \frac{d}{dx} f(x) \cdot g(x).$$

That is, derivative of the product of two functions = [first (fu)ⁿ × (derivative of second (fu)ⁿ + (second function) × (derivative of first function)]

6.2.2 Quotient Rule

$$\frac{d}{dx} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{g(x) \cdot \frac{d}{dx} \{f(x)\} - f(x) \frac{d}{dx} \{g(x)\}}{\{g(x)\}^2}$$

Remark: It is advisable to remember as follows

$$\frac{d}{dx} \left\{ \frac{N^r}{D^r} \right\} = \frac{D^r \cdot \frac{d}{dx} (N^r) - N^r \frac{d}{dx} (D^r)}{(D^r)^2}$$

Chain Rule

$$\frac{d}{dx} [u(v(x))] = \frac{du\{v(x)\}}{dv\{v(x)\}} \times \frac{dv(x)}{dx}$$

$$\frac{d}{dx} u[v\{w(x)\}] = \frac{du[v\{w(x)\}]}{dv\{w(x)\}} \times \frac{dv\{w(x)\}}{dw(x)} \times \frac{dw(x)}{dx}$$

$$\frac{d}{dx} \{f(x)\}^{\phi(x)} = \text{d. c. of } \{f(x)\}^{\phi(x)} \text{ w.r.t. } x \text{ taking } \phi(x) \text{ as a constant}$$

$$+ \text{d.c. of } \{f(x)\}^{\phi(x)} \text{ w.r.t. 'x' taking } f(x) \text{ as a constant.}$$

$$= \phi(x) \cdot \{f(x)\}^{\phi(x)-1} \frac{df(x)}{dx} + \{f(x)\}^{\phi(x)} \cdot \log f(x) \times \frac{d\phi(x)}{dx}$$

6.3 Differentiation of a Function Represented Parametrically

If y is a function of x such that

$x = \phi(t)$, $y = \psi(t)$ where 't' is parameter then

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \quad \text{i.e.} \quad \frac{\psi'(t)}{\phi'(t)}$$

6.3.1 Differentiation of One Function w.r.t. Another Function

If $y = \phi(x)$ and $z = \psi(x)$

Then $\frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}}$, i.e.

Relation between $\frac{dy}{dx}$ and $\frac{dx}{dy}$

$$\Rightarrow \frac{dy}{dx} \times \frac{dx}{dy} = 1$$

Higher derivative of a function if $y = f(x)$ then the derivative of $\frac{dy}{dx}$ w.r.t 'x' is called the second derivative of y w.r.t. x it is denoted by $\frac{d^2y}{dx^2}$.

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right); \frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right)$$

If y is the function of x given parametrically by $y = \phi(t)$, $x = \psi(t)$ then

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{\phi(t)}{\psi'(t)} \right) = \frac{d}{dt} \left(\frac{\phi(t)}{\psi'(t)} \right) \cdot \frac{dx}{dt}$$

6.3.2 Differentiation of a Function in a Form of Determinant

If $y = \begin{vmatrix} u(x) & v(x) & w(x) \\ p(x) & q(x) & r(x) \\ \lambda(x) & \mu(x) & \nu(x) \end{vmatrix}$ then

$$\frac{dy}{dx} = \begin{vmatrix} u'(x) & v'(x) & w'(x) \\ p(x) & q(x) & r(x) \\ \lambda(x) & \mu(x) & \nu(x) \end{vmatrix} + \begin{vmatrix} u(x) & v(x) & w(x) \\ p'(x) & q'(x) & r'(x) \\ \lambda(x) & \mu(x) & \nu(x) \end{vmatrix} + \begin{vmatrix} u(x) & v(x) & w(x) \\ p(x) & q(x) & r(x) \\ \lambda'(x) & \mu'(x) & \nu'(x) \end{vmatrix}$$

Note: 1. The $(Diff)^n$ can also be done column wise.

2. Relation between $\frac{dy}{dx}$ and $\frac{dx}{dy}$: Since $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ so $\frac{dy}{dx} = \frac{1}{dx/dy}$

6.3.3 Differential Operator $\left(\frac{dy}{dx} \right)$ of Parametric Functions

e.g. $x = \phi(t)$ and $y = \psi(t) \therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

6.3.4 Differentiation of a Function with Respect to Another Function

Let $u = f(x)$ and $v = g(x) \therefore \frac{du}{dv} = \frac{du/dx}{dv/dx}$

6.3.5 Logarithmic Differentiation

Let $y = [f(x)]^{g(x)}$

Taking logarithm of both the sides, we get

$$\log y = \log [f(x)]^{g(x)} = g(x) \cdot \log [f(x)]$$

(Diff)ⁿ. w.r.t. x , we get

$$\frac{1}{y} \frac{dy}{dx} = g(x) \cdot \frac{1}{f(x)} \frac{df(x)}{dx} + \log \{f(x)\} \cdot \frac{dg(x)}{dx}$$

$$\therefore \frac{dy}{dx} = y \left[\frac{g(x)}{f(x)} \cdot \frac{df(x)}{dx} + \log \{f(x)\} \cdot \frac{dg(x)}{dx} \right]$$

6.3.6 Higher Order Derivative

If given function $y = f(x)$

$\therefore$ The derivative of y w.r.t. ' x '; In general

$$\frac{dy}{dx} = \frac{d\{f(x)\}}{dx} \Rightarrow f'(x)$$

Hence to fix up the idea, we call

$$\frac{dy}{dx} = \text{First order derivative}$$

$$\frac{d^2 y}{dx^2} = \text{Second order derivative}$$

$$\frac{d^n y}{dx^n} = \text{nth order derivative}$$

Hence, If $y = f(x)$, then the other alternative notations for

	$\frac{dy}{dx},$	$\frac{d^2 y}{dx^2},$	$\frac{d^3 y}{dx^3},$	,	$\frac{d^n y}{dx^n}$
or	$y_1,$	$y_2,$	$y_3,$	,	y_n
or	$y',$	$y'',$	$y''',$	,	$y^{(n)}$
or	$Dy,$	$D^2 y,$	$D^3 y,$	,	$D^n y$
or	$f'(x),$	$f''(x),$	$f'''(x)$	,	$f^{(n)}$

The values of these derivatives at $x = a$ are denoted by

$$y_n(a), \quad y^n(a), \quad D^n y(a), \quad f^n(a), \quad \left(\frac{d^n y}{dx^n} \right)_{x=a}$$

Illustrative Examples

Example 1: Differentiate the following function w.r.t. 'x':

- (i) x^7 (ii) x^{-5} (iii) $\sqrt{x}$ (iv) $\frac{1}{x}$ (v) $\sqrt{x-3}$

Solution: Here, we have to use the power formula of derivative

i.e. $y = (x)^n$

$\therefore (Diff)^n$ on both the sides w.r.t. 'x' we get

$$\frac{dy}{dx} = \frac{d}{dx} (x)^n = n x^{n-1} \text{ where } n \neq -1$$

- (i) Let us consider,

$$y = x^7$$

So, $(Diff)^n$ on both the sides w.r.t. x , we get

$$\frac{dy}{dx} = \frac{d}{dx} (x^7) = 7 x^{7-1} = 7x^6$$

Similarly, we fix up the idea as done earlier;

(ii) $\frac{d}{dx} (x^{-5}) = (-5) x^{-5-1} = -5x^{-6} = \frac{-5}{x^6}.$

(iii) $\frac{d}{dx} \sqrt{x} = \frac{d}{dx} (x^{1/2}) = \frac{1}{2} x^{\left(\frac{1}{2}-1\right)} = \frac{1}{2} x^{-1/2}$

(iv) $\frac{d}{dx} \left(\frac{1}{x}\right) = \frac{d}{dx} (x^{-1}) = (-1) x^{-1-1} = -x^{-2} = -\frac{1}{x^2}.$

(v) $\frac{d}{dx} \sqrt{x-3} = \frac{d}{dx} (x^{-3/2}) = -\frac{3}{2} x^{\left(\frac{-3}{2}-1\right)} = -\frac{3}{2} x^{-5/2} = \frac{-3}{2x^{5/2}}.$

Example 2: Find the derivatives of the following functions.

- (i) $\frac{lx^2 + mx + n}{\sqrt{x}}$ (ii) $\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2$ (iii) $x \cdot (1+x)^2$

Solution:

- (i) Let us assume

$$y = \frac{lx^2 + mx + n}{\sqrt{x}} = \frac{lx^2}{\sqrt{x}} + \frac{mx}{\sqrt{x}} + \frac{n}{\sqrt{x}} = lx^{3/2} + mx^{1/2} + nx^{-1/2}$$

Now $(Diff)^n$ w.r.t x on both the sides, we get

$$\begin{aligned} \therefore Dy &= D(lx^{3/2}) + D(mx^{1/2}) + D(nx^{-1/2}) \\ &= lD(x^{3/2}) + mD(x^{1/2}) + nD(x^{-1/2}) \end{aligned}$$

$$\begin{aligned}
&= l \cdot \frac{3}{2} x^{\left(\frac{3}{2}-1\right)} + m \cdot \frac{1}{2} x^{\left(\frac{1}{2}-1\right)} + n \cdot \left(-\frac{1}{2}\right) x^{\left(\frac{1}{2}-1\right)} \\
&= \frac{3}{2} l x^{1/2} + \frac{m}{2} x^{-1/2} - \frac{n}{2} x^{-3/2} = \frac{3}{2} l \sqrt{x} + \frac{m}{2\sqrt{x}} - \frac{n}{2x^{3/2}}
\end{aligned}$$

(ii) Let us assume

$$y = \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 = x + \frac{1}{x} + 2 \cdot \sqrt{x} \cdot \frac{1}{\sqrt{x}} = x + x^{-1} + 2.$$

i.e. $(Diff)^n$ on both the sides w.r.t. 'x' we get;

$$Dy = D(x) + D(x^{-1}) + D(2) = 1 + (-1)x^{-1-1} + 0 = 1 - x^{-2} = 1 - \frac{1}{x^2}$$

(iii) Let us assume

$$y = x \cdot (1+x)^2$$

$(Diff)^n$. on both the sides w.r.t. x, we get

$$Dy = D(x + 2x^2 + x^3)$$

$$Dy = D(x) + 2D(x^2) + D(x^3) = 1 + 2 \cdot 2x + 3x^2 = 1 + 4x + 3x^2$$

Example 4: If $y = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$ To show that: $\frac{dy}{dx} - y + \frac{x^n}{n!} = 0$

Solution: We have

$$y = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

i.e. $(Diff)^n$ on both the sides w.r.t. x, we get

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx}(1) + \frac{1}{1!} \frac{d}{dx}(x) + \frac{1}{2!} \frac{d}{dx}x^2 + \frac{1}{3!} \frac{d}{dx}(x^3) \\
&\quad + \dots + \frac{1}{n!} \frac{d}{dx}(x^n). \\
&= 0 + \frac{1}{1!} \cdot 1 + \frac{1}{2!} \cdot 2x + \frac{1}{3!} \cdot 3x^2 + \dots + \frac{1}{n!} \cdot nx^{n-1} \\
&= 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^{n-1}}{(n-1)!}
\end{aligned}$$

Example 5: Differentiate the following w.r.t 'x':

- (i) $b \log_a x$ (ii) $ax + \frac{x^2}{b}$ (iii) $e^x + \log_e x + l^x$

Solution:

(i) We know that

$$b \log_a x = b \log_e x \log_a e$$

$\therefore (Diff)^x$ on both the sides w.r.t. 'x', we get

$$\frac{d}{dn} [b \cdot \log_a x] = \frac{d}{dx} [b \log_e x \cdot \log_a e] = b \log_a e \frac{d}{dx} [\log_e x] = b \log_a e \frac{1}{x} = \frac{b \log_a e}{x}.$$

$$(ii) \quad \frac{d}{dx} \left(ax + \frac{x^2}{b} \right) = a \frac{d}{dx} (x) + \frac{1}{b} \frac{d}{dx} (x^2) = a \cdot 1 + \frac{1}{b} \cdot 2x = a + \frac{2x}{b}$$

$$(iii) \quad \frac{d}{dx} (e^x + \log_e x + l^x) = \frac{d}{dx} (e^x) + \frac{d}{dx} (\log_e x) + \frac{d}{dx} (l^x) = e^x + \frac{1}{x} + l^x \cdot \log_e l$$

Example 6: If $y = ax^2$ then to prove that $\left(x \cdot \frac{dy}{dx} \right) = 2y$

Solution:

$$\therefore y = ax^2$$

(Diff)ⁿ on both the sides w.r.t. 'x', we get

$$\frac{dy}{dx} = \frac{d}{dx} (ax^2) = a \cdot \frac{d}{dx} x^2 = a \cdot 2x = 2ax$$

Now, $\frac{dy}{dx} = 2ax$ Multiply on both the sides with 'x', we get.

$$x \cdot \frac{dy}{dx} = 2ax^2 \Rightarrow x \cdot \frac{dy}{dx} = 2y \quad [\because ax^2 = y]$$

Example 7: Function $(2x^3 + 3^x - 4 \log_e x + 5)$ find out the differential coefficient with respect to x.

$$\begin{aligned} \text{Solution: } \frac{d}{dx} (2x^3 + 3^x - 4 \log_e x + 5) &= 2 \frac{d}{dx} (x^3) + \frac{d}{dx} (3^x) - 4 \frac{d}{dx} (\log_e x) + \frac{d}{dx} (5) \\ &= 2 \cdot 3x^2 + 3^x \cdot \log_e 3 - 4 \cdot \frac{1}{x} + 0 = (6x^2 + 3^x \cdot \log_e 3 - \frac{4}{x}) \end{aligned}$$

Example 8: Find the differential coefficient of the following function with respect to 'x'?

$$(i) \quad \sqrt{a + 2a^2 e^x + a^3 e^{2x}}$$

$$(ii) \quad \frac{(ax^2 + bx + c)}{\sqrt{x}}$$

$$(iii) \quad (a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n)$$

$$(iv) \quad \left(2 + \frac{3}{x} + \frac{4}{x^2} + \frac{6}{x^3} \right)$$

$$(v) \quad \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right)$$

$$(vi) \quad \frac{x^3 \log_e x + x - x^3 \cdot e^x}{x^3}$$

$$(vii) \quad 3\sqrt{[1 + 3 \log_a x + 3(\log_a x)^2 + (\log_a x)^3]}$$

$$(viii) \quad (3e^x - \frac{1}{5x} + 2^x - \sec x)$$

Solution:

$$\begin{aligned} (i) \quad \frac{d}{dx} [(a + 2a^2 e^x + a^3 e^{2x})]^{1/2} &= \frac{d}{dx} \sqrt{a(1 + 2ae^x + a^2 e^{2x})}^{1/2} = \frac{d}{dx} [a(1 + ae^x)^2]^{1/2} \\ &= \sqrt{a} \frac{d}{dx} [(1 + ae^x)] = \sqrt{a} \left[\frac{d}{dx} 1 + a \frac{d}{dx} e^x \right] = \sqrt{a} [0 + ae^x] = a^{3/2} e^x \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \frac{d}{dx} \cdot \frac{(ax^2 + bx + c)}{\sqrt{x}} &= \frac{d}{dx} \left[\frac{ax^2}{\sqrt{x}} + \frac{bx}{\sqrt{x}} + \frac{c}{\sqrt{x}} \right] \\
 &= \frac{d}{dx} [ax^{3/2} + bx^{1/2} + cx^{-1/2}] \\
 &= a \cdot \frac{d}{dx} (x)^{3/2} + b \cdot \frac{d}{dx} (x^{1/2}) + c \cdot \frac{d}{dx} (x)^{-1/2} \\
 &= a \cdot \frac{3}{2} \cdot x^{1/2} + b \cdot \frac{1}{2} x^{-1/2} + c \cdot \left(-\frac{1}{2}\right) x^{-3/2} = \left[\frac{3a\sqrt{x}}{2} + \frac{b}{2\sqrt{x}} - \frac{c}{2\sqrt{x^3}} \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \frac{d}{dx} (a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n) \\
 &= a_0 \frac{d}{dx} (x^n) + a_1 \frac{d}{dx} (x^{n-1}) + \dots + a_{n-1} \frac{d}{dx} (x) + \frac{d}{dx} (a_n) \\
 &= a_0 n x^{n-1} + a_1 (n-1) x^{n-2} + \dots + a_{n-1} \cdot 1 + 0 \\
 &= a_0 n x^{n-1} + a_1 (n-1) x^{n-2} + \dots + a_{n-1}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad \frac{d}{dx} \left[2 + \frac{3}{x} + \frac{4}{x^2} + \frac{6}{x^3} \right] &= \frac{d}{dx} (2) + 3 \frac{d}{dx} (x^{-1}) + 4 \frac{d}{dx} x^{-2} + 6 \frac{d}{dx} x^{-3} \\
 &= 0 + 3(-1)x^{-1-1} + 4(-2)x^{-2-1} + 6(-3)x^{-3-1} \\
 &= -3x^{-2} - 8x^{-3} - 18x^{-4} = -\left(\frac{3}{x^2} + \frac{8}{x^3} + \frac{18}{x^4} \right).
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad \frac{d}{dx} \left[\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right) \right] &= \frac{d}{dx} (x) - \frac{1}{2} \frac{d}{dx} (x^2) + \frac{1}{3} \frac{d}{dx} (x^3) - \frac{1}{4} \frac{d}{dx} (x^4) + \dots \\
 &= 1 - \frac{1}{2} \cdot 2x^{2-1} + \frac{1}{3} \cdot 3x^{3-1} - \frac{1}{4} \cdot 4x^{4-1} + \dots \\
 &= 1 - x + x^2 - x^3 + \dots
 \end{aligned}$$

$$\begin{aligned}
 \text{(vi)} \quad \frac{d}{dx} \left[\frac{x^3 \log_e x + x - x^3 \cdot e^x}{x^3} \right] &= \frac{d}{dx} (\log_e x + x^{-2} - e^x) \\
 &= \frac{d}{dx} (\log_e x) + \frac{d}{dx} (x^{-2}) - \frac{d}{dx} (e^x) \\
 &= \frac{1}{x} + (-2)x^{-2-1} - e^x = x^{-1} - 2x^{-3} - e^x = \frac{1}{x} - \frac{2}{x^3} - e^x.
 \end{aligned}$$

$$\text{(vii)} \quad \text{Let } \log_a x = t$$

$$\begin{aligned}
 \therefore \quad \frac{d}{dx} [3\sqrt[3]{1+3} \log_a x + 3(\log_a x)^2 + (\log_a x)^3] \\
 &= \frac{d}{dx} [(1+3t+3t^2+t^3)^{1/3}] = \frac{d}{dx} [\{(1+t)^3\}^{1/3}] \\
 &= \frac{d}{dx} (1+t) = \frac{d}{dx} (1 + \log_a x) = 0 + \frac{1}{x} \cdot \log_a e = \frac{1}{x} \cdot \log_a e.
 \end{aligned}$$

$$\begin{aligned}
 \text{(viii)} \quad \frac{d}{dx} \left(3e^x - \frac{1}{5x} + 2^x - \sec x \right) \\
 &= 3 \frac{d}{dx} (e^x) - \frac{1}{5} \frac{d}{dx} (x^{-1}) + \frac{d}{dx} (2^x) - \frac{d}{dx} (\sec x) \\
 &= 3 \cdot e^x - \frac{1}{5} (-1)x^{-2} + 2^x \cdot \log_e 2 - \sec x \cdot \tan x \\
 &= \left(3e^x + \frac{1}{5x^2} + 2^x \cdot \log_e 2 - \sec x \cdot \tan x \right)
 \end{aligned}$$

Example 9: If $y = \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right)$ Then to prove that $\frac{dy}{dx} = y$.

Solution: Given function is

$$y = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$y = e^x$$

$$\left[\because 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots = e^x \right]$$

Now taking $(\text{Diff})^n$ w.r.t. 'x', we get

$$\frac{dy}{dx} = \frac{d}{dx} (e^x) = e^x = y \Rightarrow \frac{dy}{dx} = y$$

Example 10: If $y = \sqrt{a/x} + \sqrt{x/a}$, then prove that

$$2xy \frac{dy}{dx} = \left(\frac{x}{a} - \frac{a}{x} \right)$$

[B.C.A. (Meerut) 2007]

Solution: Given function

$$y = \sqrt{x/a} + \sqrt{a/x} = \frac{1}{\sqrt{a}} x^{1/2} + \sqrt{a} x^{-1/2}$$

$\therefore (\text{Diff})^n$ w.r.t. 'x' on both the sides, we get

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} \left[\frac{1}{\sqrt{a}} x^{1/2} + \sqrt{a} x^{-1/2} \right] \\
 &= \frac{1}{\sqrt{a}} \frac{d}{dx} x^{1/2} + \sqrt{a} \frac{d}{dx} x^{-1/2} \\
 &= \frac{1}{\sqrt{a}} \cdot \frac{1}{2} x^{-1/2} + \sqrt{a} \left(-\frac{1}{2} \right) x \left(-\frac{1}{2} - 1 \right) = \frac{1}{2} \left(\frac{1}{\sqrt{ax}} - \frac{\sqrt{a}}{x\sqrt{x}} \right)
 \end{aligned}$$

Now multiply by $2xy$ on both the sides, we get

$$\Rightarrow 2xy \frac{dy}{dx} = xy \left(\frac{1}{\sqrt{ax}} - \frac{\sqrt{a}}{x\sqrt{x}} \right) = y \left(\sqrt{\frac{x}{a}} - \sqrt{\frac{a}{x}} \right)$$

Put the value of y , we get

$$\Rightarrow 2xy \frac{dy}{dx} = (\sqrt{x/a} - \sqrt{a/x}) \cdot (\sqrt{x/a} + \sqrt{a/x}) \text{ or } 2xy \frac{dy}{dx} = (x/a - a/x)$$

Example 11: If $y = \frac{(x-4)}{2\sqrt{x}}$, then find $\frac{dy}{dx}$ at $x = 4$.

Solution: Given function

$$y = \frac{(x-4)}{2\sqrt{x}} \quad [y = f(x)]$$

$$\therefore f(x) = \frac{1}{2} \times \frac{x-4}{x^{1/2}} = \frac{1}{2} \times (x^{1/2} - 4x^{-1/2})$$

$\therefore$ (Diff)ⁿ w.r.t 'x' on both the sides, we get

$$\begin{aligned} f'(x) &= \frac{1}{2} \frac{d}{dx} [x^{1/2} - 4x^{-1/2}] \\ &= \frac{1}{2} \left[\frac{d}{dx} x^{1/2} - 4 \frac{d}{dx} x^{-1/2} \right] \\ &= \frac{1}{2} \left[\frac{1}{2} x^{1/2-1} - 4 \cdot \left(-\frac{1}{2}\right) x^{\left(-\frac{1}{2}-1\right)} \right] = \frac{1}{4} \left[\frac{1}{\sqrt{x}} + \frac{4}{x\sqrt{x}} \right] \end{aligned}$$

$$\text{Now } f'(x) = \frac{x+4}{4x\sqrt{x}} \quad \dots(1)$$

Put $x = 4$ in (1) in both the sides we get

$$f'(4) = \frac{4+4}{4 \cdot 4\sqrt{4}} = \frac{8}{32} = \frac{1}{4}$$

Example 12: Find out the derivatives of the following functions with respect to x .

(i) e^x (ii) a^x (iii) $\log_e x$ (iv) $\log_a x$ [B.C.A. (Meerut) 2010]

Solution:

(i) Let us consider the function

$$f(x) = e^x$$

$$\text{Then } f(x+h) = e^{x+h}$$

$$\Rightarrow \frac{d}{dx} e^x = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} \quad \left[\because f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \right]$$

$$= \lim_{h \rightarrow 0} \frac{e^x(e^h - 1)}{h}$$

$$= \lim_{h \rightarrow 0} e^x \frac{\left[1 + \frac{h}{1!} + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots - 1 \right]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{e^x}{h} \left[h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots \right]$$

$$= \lim_{h \rightarrow 0} e^x \left[1 + \frac{h}{2!} + \frac{h^2}{3!} + \dots \right]$$

Now taking Lt. now Put $h=0$, we get

$$\frac{d}{dx}(e^x) = e^x \left[1 + \frac{0}{2!} + \frac{(0)^2}{3!} + \dots \right] = [e^x \times 1 + 0 + 0]$$

$$\Rightarrow \frac{d}{dx} e^x = e^x$$

(ii) Function $f(x) = a^x$

$$\Rightarrow f(x+h) = a^{x+h}$$

$$\begin{aligned} \therefore \frac{d}{dx}(a)^x &= \text{Lt}_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} \\ &= \text{Lt}_{h \rightarrow 0} \frac{a^x(a^h - 1)}{h} \\ &= \text{Lt}_{h \rightarrow 0} \frac{a^x \left(1 + h \frac{\log a}{1!} + \frac{(h \log a)^2}{2!} + \dots - 1 \right)}{h} \\ &= \text{Lt}_{h \rightarrow 0} \frac{a^x}{h} \left[h \log a + \frac{h^2 (\log a)^2}{2!} + \frac{h^3 (\log a)^3}{3!} + \dots \right] \\ &= \text{Lt}_{h \rightarrow 0} a^x \left[\log a + h \frac{(\log a)^2}{2!} + \text{higher power of } h \right] \end{aligned}$$

Now taking Lt. Put $h=0$, we get

$$\begin{aligned} \frac{d}{dx}(a)^x &= a^x \left[\log a + 0 \frac{(\log a)^2}{2!} + \dots \right] \\ &= [a^x \times \log_e a + 0 + \dots] \end{aligned}$$

$$\therefore \frac{d}{dx}(a)^x = a^x \log_e a$$

(iii) Function $f(x) = \log_e x$

Then $f(x+h) = \log_e(x+h)$

$$\begin{aligned} \therefore \frac{d}{dx}(\log_e x) &= \text{Lt}_{h \rightarrow 0} \frac{\log_e(x+h) - \log_e x}{h} \\ &= \text{Lt}_{h \rightarrow 0} \log \frac{\left(\frac{x+h}{x} \right)}{h} \\ &= \text{Lt}_{h \rightarrow 0} \frac{1}{h} \cdot \log \left(1 + \frac{h}{x} \right) \\ &= \text{Lt}_{h \rightarrow 0} \frac{1}{h} \left[\frac{h}{x} - \frac{1}{2} \left(\frac{h}{x} \right)^2 + \frac{1}{3} \left(\frac{h}{x} \right)^3 - \dots \right] \end{aligned}$$

$$= \lim_{h \rightarrow 0} \left[\left(\frac{1}{x} - \frac{1}{2} \frac{h}{x^2} + \frac{1}{3} \frac{h^2}{x^3} - \dots \right) \right]$$

Now taking $\lim$, Put $h=0$, we get

$$\begin{aligned} \frac{d}{dx} (\log_e x) &= \left[\frac{1}{x} - \frac{1}{2} \frac{0}{x^2} + \frac{1}{3} \frac{(0)^2}{x^3} - \dots \right] \\ &= \left[\frac{1}{x} - 0 + 0 - \dots \right] \end{aligned}$$

$$\therefore \frac{d}{dx} (\log_e x) = \frac{1}{x}$$

(iv) Function $f(x) = \log_a x$

$$\text{Then } f(x) = \log_e x \cdot \log_a e$$

(Changing the base from a to e)

$$= \log_a e \cdot \log_e x$$

Now $\log_a e$ is just a constant term.

Hence, we have now

$$\frac{d}{dx} f(x) = \frac{d}{dx} (\log_a e \cdot \log_e x)$$

$$\Rightarrow \frac{d}{dx} (\log_a x) = \log_a e \cdot \frac{d}{dx} (\log_e x) = \log_a e \cdot \frac{1}{x}$$

$$\therefore \frac{d}{dx} [\log_a x] = \frac{1}{x} \cdot \log_a e$$



Comprehensive Exercise 6.1

1. Find the derivatives or differential coefficients with respect to x of the following functions:

(i) (a) $\frac{2}{\sqrt{x}}$

(b) $\frac{1}{3\sqrt{x}}$

(c) $(x^4 + 5)$

(d) $(e^x - 5^x)$

(e) $4\sqrt{x}$

(f) 4^x

(g) $\log_e(\sqrt{x})$

(h) $5e^x$

(i) $x(1+x)^3$

(j) $\left(2 + \frac{3}{x} + \frac{4}{x^2} + \frac{6}{x^3}\right)$

(k) $\left(x + \frac{1}{x}\right)^2$

(l) $\frac{(x^2 + 1)(x + 3)}{x}$

(m) $ax + \frac{x^2}{b}$

(n) $ax^2 + bx + c$

(o) $\frac{4}{3}(x^{-5})$

(ii) If $y = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ show that $\frac{dy}{dx} - y + \frac{x^n}{n!} = 0$.

(iii) If $y = \frac{1}{3x^3}$, show that $3y + x \left(\frac{dy}{dx} \right) = 0$.

(iv) If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, prove it out $2x \cdot \left(\frac{dy}{dx} \right) + y - 2\sqrt{x} = 0$.

(v) If $y = \frac{1}{\sqrt[4]{x}}$, then prove out it $\left(\frac{dy}{dx} \right)_{x=16} = -\frac{1}{128}$

2. Differentiate the following:

(i) $x^n + 7e^x$

(ii) $b \log_e x + x^3 + \sqrt{x}$

(iii) $5 \log x + \sqrt{x} - 6$

(iv) $x(1 + x^2) + a^x$

(v) $3\sqrt{x} + 5 \log_e x - 11 \log_a x$

(vi) $\sqrt{a + 2a^2 e^x + a^3 e^{2x}}$

3. If $\sqrt{x} + \sqrt{y} = 5$, prove that at $(4, 9)$, $\frac{dy}{dx} = -3/2$.

[B.C.A. (Meerut) 2003]



Answers

1.	(a) $\frac{-1}{x^{3/2}}$	(b) $-\frac{1}{6} x^{-3/2}$	(c) $4x^3$
	(d) $(e^x - 5^x) \cdot \log_e 5$	(e) $2x^{-1/2}$	(f) $4^x \log_e 4$
	(g) $\frac{1}{2x}$	(h) $5e^x$	(i) $(1+x)^2 \cdot (1+4x)$
	(j) $-\left(\frac{3}{x^2} + \frac{8}{x^3} + \frac{18}{x^4}\right)$	(k) $2x - \frac{2}{x^3}$	(l) $2x + 3 - \frac{3}{x^2}$
	(m) $a + \frac{2x}{b}$	(n) $2ax + b$	(o) $-\frac{20}{3} x^{-6}$
2.	(i) $xn^{n-1} + 7e^x$	(ii) $\frac{b}{x} + 3x^2 + \frac{1}{2\sqrt{x}}$	
	(iii) $\frac{5}{x} + \frac{1}{2\sqrt{x}}$	(iv) $1 + 3x^2 + a^x \cdot \log_e a$	
	(v) $\frac{3}{2\sqrt{x}} + \frac{5}{x} - \frac{11}{x} \cdot \log_a e$	(vi) $a^{3/2} \cdot e^x$	

Illustrative Examples

Example 1: Find the differential coefficients of the following functions w.r.t. 'x'

(i) $5a^x + 2 \log x + 3 \cos x - 8 \sec x$ (ii) $\frac{7}{\sin x} + 4 \cot x + 9 \log_a x - 2/3 \tan x + 5c$

Solution:

(i) $D(5a^x + 2 \log x + 3 \cos x - 8 \sec x)$
 $= 5D(a^x) + 2D(\log x) + 3D(\cos x) - 8D(\sec x)$
 $= 5a^x \log_e a + 2 \cdot \frac{1}{x} + 3(-\sin x) - 8 \sec x \cdot \tan x$
 $= 5a^x \log_e a + \frac{2}{x} - 3 \sin x - 8 \sec x \cdot \tan x$

(ii) $D\left[\frac{7}{\sin x} + 4 \cot x + 9 \log_a x - \frac{2}{3} \tan x + 5c\right]$
 $= D(7 \cdot \operatorname{cosec} x) + 4D(\cot x) + 9 \cdot D(\log_a x) - \frac{2}{3} D(\tan x) + D(5c)$
 $= 7(-\operatorname{cosec} x \cot x) + 4D(\cot x) + 9 \cdot D(\log_a x) - \frac{2}{3} D(\tan x) + D(5c)$
 $= -7 \operatorname{cosec} x \cdot \cot x - 4 \operatorname{cosec}^2 x + \frac{9}{x} \log_a e - 2/3 \sec^2 x$

Example 2: Find the derivatives of the following functions w.r.t. 'x':

(i) $3 \sec x \cdot \sin x + 4 \operatorname{cosec} x \cdot \cos x - 5 \tan x \cdot \cot x$ (ii) $\sqrt{\frac{1 + \cos 2x}{1 - \cos 2x}}$

Solution: Hence, we have

(i) $y = 3 \sec x \cdot \sin x + 4 \operatorname{cosec} x \cdot \cos x - 5 \tan x \cdot \cot x$
 $= 3 \cdot \frac{1}{\cos x} \cdot \sin x + 4 \cdot \frac{1}{\sin x} \cdot \cos x - 5 \cdot \tan x \cdot \frac{1}{\tan x}$
 $\Rightarrow y = 3 \tan x + 4 \cot x - 5$
 $\therefore \frac{dy}{dx} = 3 \cdot \frac{d}{dx}(\tan x) + 4 \frac{d}{dx}(\cot x) - \frac{d}{dx}(5)$
 $= 3 \cdot \sec^2 x + 4(-\operatorname{cosec}^2 x) - 0 = 3 \cdot \sec^2 x - 4 \operatorname{cosec}^2 x.$

(ii) Let $y = \sqrt{\frac{1 + \cos 2x}{1 - \cos 2x}} = \sqrt{\frac{2 \cos^2 x}{2 \sin^2 x}} = \frac{\cos x}{\sin x} = \cot x$
 $\frac{dy}{dx} = -\operatorname{cosec}^2 x$

Example 3: Differentiate the following function w.r.t. 'x'.

(i) $e^x \cdot \log x$ (ii) $ax^2 \sin x$ (iii) $\sec x \cdot \tan x$

Solution:

(i) Let $y = e^x \cdot \log x$

$$\therefore \frac{dy}{dx} = \frac{d}{dx} \left(e^x \cdot \log x \right) = e^x \cdot \frac{d}{dx} \log x + \log x \cdot \frac{d}{dx} e^x = e^x \cdot \frac{1}{x} + (\log x) e^x$$

(ii) Let $y = ax^2 \cdot \sin x$

$$\Rightarrow \frac{dy}{dx} = a \frac{d}{dx} (x^2 \cdot \sin x) = a \left[x^2 \frac{d}{dx} (\sin x) + \sin x \frac{d}{dx} (x^2) \right]$$

$$= a [x^2 \cdot \cos x + (\sin x) \cdot 2x] = ax(x \cos x + 2 \sin x)$$

(iii) Let $y = \sec x \tan x$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} (\sec x \cdot \tan x) = \sec x \cdot \sec^2 x + \tan x \cdot \sec x \cdot \tan x$$

$$= \sec x (\sec^2 x + \tan^2 x) = \sec x (1 + \tan^2 x + \tan^2 x) = \sec x (1 + 2 \tan^2 x)$$

Example 4: Differentiate the following function w.r.t. 'x' :

(i) $7xa^x - 2 \sin x \cdot \cos x - 5 \tan x \cdot \cot x$ (ii) $\log_a \sqrt{x \sec x} - 11 \operatorname{cosec} x \cdot \cot x$

(iii) $e^x \log_e x \cdot \tan x$

Solution:

(i) Let $y = 7xa^x - 2 \sin x \cdot \cos x - 5 \tan x \cdot \cot x$

$$\therefore \frac{dy}{dx} = 7 \frac{d}{dx} (x a^x) - 2 \frac{d}{dx} (\sin x \cdot \cos x) - 5 \frac{d}{dx} (\tan x \cdot \cot x)$$

$$= 7 \left[x \cdot \frac{d}{dx} (a^x) + a^x \cdot \frac{d}{dx} (x) \right] - 2 \left[\sin x \cdot \frac{d}{dx} (\cos x) + \cos x \cdot \frac{d}{dx} (\sin x) \right] - 5 \frac{d}{dx} (1)$$

$$= 7 [xa^x \log_e a + a^x \cdot 1] - 2 [\sin x (-\sin x) + \cos x \cos x] - 5 \times 0$$

$$= 7a^x (1 + x \log_e a) - 2(\cos^2 x - \sin^2 x) = 7a^x (1 + x \log_e a) - 2 \cos 2x$$

(ii) Let $y = \log_a \sqrt{x \cdot \sec x} - 11 \operatorname{cosec} x \cdot \cot x$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} \frac{d}{dx} (\log_a x \cdot \sec x) - 11 \frac{d}{dx} (\operatorname{cosec} x \cdot \cot x)$$

$$\frac{dy}{dx} = \frac{1}{2} \left[\log_a x \cdot \frac{d}{dx} (\sec x) + \sec x \cdot \frac{d}{dx} (\log_a x) \right]$$

$$- 11 \left[\operatorname{cosec} x \cdot \frac{d}{dx} (\cot x) + \cot x \cdot \frac{d}{dx} (\operatorname{cosec} x) \right]$$

$$= \frac{1}{2} \left[\log_a x \cdot \sec x \cdot \tan x + (\sec x) \cdot \frac{1}{x} \log_a e \right]$$

$$- 11 \left[\operatorname{cosec} x (-\operatorname{cosec}^2 x) + \cot x (-\operatorname{cosec} x \cdot \cot x) \right]$$

$$= \frac{1}{2} \sec x \left[\tan x \log_a x + \frac{1}{x} \log_a e \right] + 11 \operatorname{cosec} x (\operatorname{cosec}^2 x + \cot^2 x)$$

(iii) Let $y = e^x \cdot \log_e x \cdot \tan x$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{d}{dx} [e^x (\log_e x) \cdot \tan x] = e^x \cdot \log_e x \cdot \frac{d}{dx} (\tan x) + \tan x \cdot \frac{d}{dx} (e^x \cdot \log_e x) \\ &= e^x \cdot \log_e x \cdot (\sec^2 x) + \tan x \left[e^x \cdot \frac{d}{dx} (\log_e x) + \log_e x \cdot \frac{d}{dx} (e^x) \right] \\ &= e^x \cdot \sec^2 x \cdot \log_e x + \tan x \left[e^x \cdot \frac{1}{x} + (\log_e x) \cdot e^x \right] \\ &= e^x \cdot \sec^2 x \cdot \log_e x + e^x \cdot \tan x \left(\frac{1}{x} + \log_e x \right)\end{aligned}$$

Example 5: Differentiate the following function w.r.t. 'x':

- (i) $\frac{e^x}{1+e^x}$ (ii) $\frac{\sin x + \cos x}{\log_e x}$ (iii) $\frac{e^x + \cot x}{\tan x - x^n}$
- (iv) $\frac{1 - \tan x}{\sec x}$ (v) $\frac{\tan x + \cot x}{\tan x - \cot x}$ (vi) $\frac{x \tan x}{\sec x + \tan x}$
- (vii) $\frac{\sin x - x \cdot \cos x}{x \sin x + \cos x}$

Solution:

(i) Let $y = \frac{e^x}{1+e^x}$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{(1+e^x) \frac{d}{dx} (e^x) - e^x \frac{d}{dx} (1+e^x)}{(1+e^x)^2} \\ &= \frac{(1+e^x) e^x - e^x (0+e^x)}{(1+e^x)^2} = \frac{(1+e^x - e^x) \cdot e^x}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2}\end{aligned}$$

(ii) Let $y = \frac{\sin x + \cos x}{\log_e x}$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{\sin x + \cos x}{\log_e x} \right) \\ &= \frac{(\log_e x) \cdot \frac{d}{dx} (\sin x + \cos x) - (\sin x + \cos x) \frac{d}{dx} (\log_e x)}{(\log_e x)^2} \\ &= \frac{\log_e x (\cos x - \sin x) - (\sin x + \cos x) \cdot 1/x}{(\log_e x)^2} \\ &= \frac{x(\cos x - \sin x) \log_e x - (\sin x + \cos x)}{x (\log_e x)^2}\end{aligned}$$

(iii) $y = \frac{e^x + \cot x}{\tan x - x^n}$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{(\tan x - x^n) \frac{d}{dx}(e^x + \cot x) - (e^x + \cot x) \frac{d}{dx}(\tan x - x^n)}{(\tan x - x^n)^2} \\ &= \frac{(\tan x - x^n)(e^x - \operatorname{cosec}^2 x) - (e^x + \cot x)(\sec^2 x - nx^{n-1})}{(\tan x - x^n)^2}\end{aligned}$$

$$(iv) \quad \text{Let } y = \frac{1 - \tan x}{\sec x}$$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{d}{dx} \cdot \left(\frac{1 - \tan x}{\sec x} \right) \\ &= \frac{\sec x \frac{d}{dx}(1 - \tan x) - (1 - \tan x) \frac{d}{dx} \sec x}{(\sec x)^2} \\ &= \frac{\sec x (-\sec^2 x) - (1 - \tan x) \sec x \tan x}{\sec^2 x} \\ &= \frac{\sec x [-\sec^2 x - \tan x + \tan^2 x]}{\sec^2 x} \\ &= -\frac{\sec^2 x - \tan^2 x + \tan x}{\sec x} = -\frac{1 + \tan^2 x - \tan^2 x + \tan x}{\sec x} \\ &= -\frac{1 + \tan x}{\sec x} = -\left(1 + \frac{\sin x}{\cos x}\right) \cdot \cos x = -(\cos x + \sin x)\end{aligned}$$

$$(v) \quad \text{Let } y = \frac{\tan x + \cot x}{\tan x - \cot x}$$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{(\tan x - \cot x) \cdot D(\tan x + \cot x) - (\tan x + \cot x) D(\tan x - \cot x)}{(\tan x - \cot x)^2} \\ &= \frac{(\tan x - \cot x) \cdot (\sec^2 x - \operatorname{cosec}^2 x) - (\tan x + \cot x) \cdot (\sec^2 x + \operatorname{cosec}^2 x)}{(\tan x - \cot x)^2} \\ &= \frac{-2 \tan x \cdot \operatorname{cosec}^2 x - 2 \cot x \cdot \sec^2 x}{(\tan x - \cot x)^2} \\ &= \frac{-2 \left[\frac{1}{\sin x \cdot \cos x} + \frac{1}{\sin x \cdot \cos x} \right]}{\left(\frac{\sin x}{\cos x} - \frac{\cos x}{\sin x} \right)^2} = \frac{-4}{\left(\frac{\sin^2 x - \cos^2 x}{\sin x \cdot \cos x} \right)^2} \\ &= \frac{-4 \sin x \cdot \cos x}{(\cos^2 x - \sin^2 x)^2} = \frac{-2 \sin 2x}{(\cos 2x)^2} = -2 \cdot \frac{1}{\cos 2x} \cdot \frac{\sin 2x}{\cos 2x} = -2 \sec 2x \cdot \tan 2x\end{aligned}$$

$$(vi) \quad \text{Let } y = \frac{x \tan x}{\sec x + \tan x}$$

$$\therefore \frac{dy}{dx} = \frac{(\sec x + \tan x) \frac{d}{dx}(x \cdot \tan x) - x \tan x \frac{d}{dx}(\sec x + \tan x)}{(\sec x + \tan x)^2}$$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{(\sec x + \tan x)(x \cdot \sec^2 x + \tan x \cdot 1) - x \tan x (\sec x \cdot \tan x + \sec^2 x)}{(\sec x + \tan x)^2} \\
 &= \frac{(\sec x + \tan x)(x \sec^2 x + \tan x) - x \sec x \cdot \tan x (\tan x + \sec x)}{(\sec x + \tan x)^2} \\
 &= \frac{(\sec x + \tan x)[x \sec^2 x + \tan x - x \sec x \cdot \tan x]}{(\sec x + \tan x)^2} \\
 &= \frac{x \sec^2 x + \tan x - x \sec x \tan x}{\sec x + \tan x} = \frac{x \sec x (\sec x - \tan x) + \tan x}{\sec x + \tan x}
 \end{aligned}$$

(vii) Let $y = \frac{\sin x - x \cos x}{x \sin x + \cos x}$

$$\begin{aligned}
 \therefore \frac{dy}{dx} &= \frac{(x \sin x + \cos x) \frac{d}{dx} (\sin x - x \cos x) - (\sin x - x \cos x) \frac{d}{dx} (x \sin x + \cos x)}{(x \sin x + \cos x)^2} \\
 &= \frac{(x \sin x + \cos x)(\cos x - (-x \sin x + \cos x)) - (\sin x - x \cos x)(x \cdot \cos x + \sin x - \sin x)}{(x \sin x + \cos x)^2} \\
 &= \frac{(x \sin x + \cos x)[x \cdot \sin x] - (\sin x - x \cos x)[x \cos x]}{(x \sin x + \cos x)^2} \\
 &= \frac{x^2 \sin^2 x + x \sin x \cdot \cos x - x \sin x \cdot \cos x + x^2 \cos^2 x}{(x \sin x + \cos x)^2} \\
 &= \frac{x^2 \sin^2 x + x^2 \cos^2 x}{(x \sin x + \cos x)^2} = \frac{x^2}{(x \sin x + \cos x)^2}
 \end{aligned}$$



Comprehensive Exercise 6.2

1. Find the differential coefficient of y , w.r.t. ' x '.

(i) $y = 3 \cos x + 5$

(ii) $y = \frac{9}{\sin x} + 3 \cot x + (a)^x$

(iii) $y = 3 \log x + \operatorname{cosec} x + (x)^7 - \frac{1}{x}$

(iv) $y = -3^x + \log_e x^2 + 5 \sec x$

(v) $y = ae^x + b \sin x + c \cdot \cos x$

(vi) $y = \sin x - \cos x + \tan x$

(vii) $y = \tan x + \frac{1}{\tan x}$

(viii) $y = 3 \operatorname{cosec} x \tan x - 7 \sec x \cdot \cot x$

2. Differentiate the following functions:

(i) $e^x \cdot \log_e x \cos x$

(ii) $7a^x \cdot x \sin x$

(iii) $\log x + \cos x$

(iv) $a^x \log_e x - \sin x \cdot \cos x$

(v) $x^5 \cdot \operatorname{cosec} x$

(vi) $2 \log_a x \cdot \sin x$

(vii) $x^2 \tan x - x \log x$

(viii) $e^x \cdot \sin x + x^3 \cdot \log x$

3. Differentiate the following functions w.r.t 'x' :

- (i) $\frac{e^x}{x}$ (ii) $\frac{e^x}{1+x}$ (iii) $\frac{e^x}{1+x^2}$ (iv) $\frac{e^x}{1+\sin x}$
 (v) $\frac{x^n}{\log_e x}$ (vi) $\frac{\sec x}{x+e^x}$ (vii) $\frac{\cot x}{\sqrt{x}}$ (viii) $\frac{2x+3}{3x+2}, x \neq -3/2$
 (ix) $\frac{\sqrt{a}-\sqrt{x}}{\sqrt{a}+\sqrt{x}}$ (x) $\frac{1+\sin x}{1-\sin x}$ (xi) $\frac{x \sin x}{1+\cos x}$

4. Find the differential coefficient at $x=0$ if $y = a^x + \sqrt{\frac{1+x}{1-x}}$.

[B.C.A. (Meerut) 2004]

5. If $y = \sqrt{\frac{x}{a}} + \sqrt{\frac{a}{x}}$ than $\frac{dy}{dn} = \frac{(n-a)}{2x\sqrt{ax}}$

[B.C.A. (Meerut) 2006]



Answers

1.	(i) $-3 \sin x$	(ii) $-9 \operatorname{cosec} x \cdot \cot x - 3 \operatorname{cosec}^2 x + a^x \cdot \log_e a$
	(iii) $\frac{3}{x} - \operatorname{cosec} x \cdot \cot x + 7x^6 + \frac{1}{x^2}$	(iv) $-3^x \cdot \log_e 3 + \frac{2}{x} + 5 \sec x \cdot \tan x$
	(v) $ae^x + b \cos x - c \sin x$	(vi) $\cos x + \sin x + \sec^2 x$
	(vii) $\sec^2 x - \operatorname{cosec}^2 x$	(viii) $3 \sec x \cdot \tan x + 7 \cdot \operatorname{cosec} x \cdot \cot x$
2.	(i) $-e^x \sin x \cdot \log_e x + \frac{1}{x} \cdot e^x \cos x + e^x \cdot \cos x \cdot \log x$	
	(ii) $7a^x(x \cdot \cos x + x \sin x \log_e a + \sin x)$	(iii) $\left[\frac{1}{x} \cdot \cos x - \sin x \log x \right]$
	(iv) $a^x \left[\frac{1}{x} \log_a e + \log_a x \cdot \log_e a \right] - \cos 2x$	(v) $x^4 \operatorname{cosec} x (5 - x \cot x)$
	(vi) $2 \left[\cos x \cdot \log_a x + \frac{\sin x}{x} \cdot \log_a e \right]$	(vii) $[x^2 \sec^2 x + 2x \cdot \tan x - 1 - \log x]$
	(viii) $e^x (\cos x + \sin x) + x^2 (1 + 3 \log x)$	
3.	(i) $\frac{e^x(x-1)}{x^2}$	(ii) $\frac{xe^x}{(1+x)^2}$
	(iii) $\frac{(1-x)^2 \cdot e^x}{(1+x^2)^2}$	
	(iv) $\frac{e^x(1+\sin x - \cos x)}{(1+\sin x)^2}$	(v) $\frac{x^{n-1}(n \cdot \log_e x - 1)}{(\log_e x)^2}$
	(vi) $\left\{ \frac{\sec x [e^x(\tan x - 1) + x \tan x - 1]}{(x+e^x)^2} \right\}$	
	(vii) $\frac{-2x \cdot \operatorname{cosec}^2 x + \cot x}{2x^{3/2}}$	(viii) $-\frac{5}{(3x+2)^2}$
	(ix) $-\frac{\sqrt{a}}{\sqrt{x}(\sqrt{a}+\sqrt{x})^2}$	
	(x) $\frac{2 \cos x}{(1-\sin x)^2}$	(xi) $\frac{x+\sin x}{1+\cos x}$
4.	$\log a + 1$	

6.4 Facts Towards Derivatives of Composite Functions

6.4.1 Meaning of Composite Functions

Let $f(x) = f_1(t)$ where $t = f_2(x)$

Then, eliminating (t) , we have $f(x) = f_1[f_2(x)]$

Hence, $f(x)$ is called a **composite function** or a **function of a function**.

Examples of composite function are : $\sin^2 x$, $\cos(x^2 + 5)$, $\tan \sqrt{x}$;

$$(2x^3 - 4x^2 + 5)^4, \log \sec x \text{ etc.}$$

6.4.2 Chain Rule of Differentiation

If y is a function of t , and t is a function of x ; then

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$



Illustrative Examples

Example 1: Find the derivative of the following function w.r.t. ' x ':

- | | | |
|--|--|--|
| (i) $(x^2 - 5)^2$ | (ii) $\sqrt{ax^2 + bx + c}$ | (iii) $\frac{1}{\sqrt{a^2 - x^2}}$ |
| (iv) $\sin x^3$ | (v) $(e^x)^3$ | (vi) $\log_e \sin x$ |
| (vii) $\log\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)$ | (viii) $\log \log x^2$ | (ix) $\frac{1 - \cos 2x}{1 + \cos 2x}$ |
| (x) $\cos(\cot x^2)$ | (xi) $\log \tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$ | (xii) $e^{ax} \cdot \sin(bx + c)$ |

Solution:

- (i) Let $y = (x^2 - 5)^2$ Putting $t = x^2 - 5$
- $\Rightarrow \frac{dt}{dx} = 2x$
- $\therefore y = t^2$ where $t = x^2 - 5$
- $\Rightarrow \frac{dy}{dt} = 2t$ and $\frac{dt}{dx} = 2x$.
- $\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = 2t \times 2x = 4x(x^2 - 5)$

(ii) Let $y = \sqrt{ax^2 + bx + c}$

Putting $ax^2 + bx + c = t$

$\therefore y = (t)^{1/2}$, where $t = ax^2 + bx + c$.

$\therefore \frac{dy}{dt} = \frac{1}{2} t^{-1/2} = \frac{1}{2\sqrt{t}}$ and $\frac{dt}{dx} = 2ax + b$.

$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{1}{2\sqrt{t}} \times (2ax + b)$

$\frac{dy}{dx} = \frac{2ax + b}{2\sqrt{ax^2 + bx + c}}$

(iii) Let $y = \frac{1}{\sqrt{a^2 - x^2}}$

Putting $a^2 - x^2 = t$

$\therefore y = \frac{1}{\sqrt{t}} = (t)^{-1/2}$ where $t = a^2 - x^2$

$\Rightarrow \frac{dy}{dt} = -\frac{1}{2} (t)^{-3/2}$ and $\frac{dt}{dx} = -2x$

$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = -1/2 t^{-3/2} (-2x) = xt^{-3/2} = x(a^2 - x^2)^{-3/2}$

$\frac{dy}{dx} = \frac{x}{(a^2 - x^2)^{3/2}}$

(iv) Let $y = \sin x^3$

Putting $t = x^3$

$y = \sin t$ where $t = x^3$

$\Rightarrow \frac{dy}{dt} = \cos t$ and $\frac{dt}{dx} = 3x^2$

$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = (\cos t) \times 3x^2 = 3x^2 \cdot \cos x^3$

(v) Let $y = (e^x)^3$

Putting $t = e^x$

$\therefore y = t^3$, where $t = e^x$

$\Rightarrow \frac{dy}{dt} = 3t^2$ and $\frac{dt}{dx} = e^x$

$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = 3t^2 \cdot e^x = 3(e^x)^2 \cdot e^x = 3(e^x)^3$

(vi) Let $y = \log_e \sin x$.

Putting $\sin x = t$, we have

$$y = \log_e t \text{ where } t = \sin x$$

$$\Rightarrow \frac{dy}{dt} = \frac{1}{t} \text{ and } \frac{dt}{dx} = \cos x$$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} \\ &= \frac{1}{t} \times \cos x = \frac{1}{\sin x} \times \cos x \end{aligned}$$

$$\Rightarrow \frac{dy}{dx} = \cot x$$

(vii) Let $y = \log \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)$

Putting $\sqrt{x} + \frac{1}{\sqrt{x}} = t$

i.e., $t = x^{1/2} + x^{-1/2}$

we have $\frac{dt}{dx} = \frac{1}{2} x^{-1/2} - \frac{1}{2} x^{-3/2} = \frac{1}{2} \left(\frac{1}{\sqrt{x}} - \frac{1}{x\sqrt{x}} \right) = \frac{1}{2\sqrt{x}} \left(1 - \frac{1}{x} \right) = \frac{x-1}{2x\sqrt{x}}$

Now $y = \log t$

$\therefore \frac{dy}{dt} = \frac{1}{t}$

Hence,
$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} = \frac{1}{t} \cdot \frac{x-1}{2x\sqrt{x}} \\ &= \frac{1}{\sqrt{x} + \frac{1}{\sqrt{x}}} \cdot \frac{x-1}{2x\sqrt{x}} = \frac{\sqrt{x}}{(x+1)} \cdot \frac{(x-1)}{2x\sqrt{x}} = \frac{x-1}{2x(x+1)} \end{aligned}$$

(viii) Let $y = \log \cdot \log x^2$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{d}{dx} (\log \cdot \log x^2) = \frac{1}{\log x^2} \cdot \frac{d}{dx} (\log x^2) = \frac{1}{2 \log x} \cdot \frac{1}{x^2} \frac{d}{dx} (x^2) \\ &= \frac{1}{2 \log x} \cdot \frac{1}{x^2} \cdot 2x = \frac{1}{x \cdot \log x} \end{aligned}$$

(ix) Let $y = \frac{1 - \cos 2x}{1 + \cos 2x} = \frac{2 \sin^2 x}{2 \cos^2 x} = \tan^2 x$

$$\therefore \frac{dy}{dx} = \frac{d}{dx} (\tan^2 x) = 2 \tan x \cdot \frac{d}{dx} (\tan x) = 2 \tan x \cdot \sec^2 x$$

(x) Let $y = \cos(\cot x^2)$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{d}{dx} [\cos(\cot x^2)] = -\sin(\cot x^2) \cdot \frac{d}{dx} (\cot x^2) = -\sin(\cot x^2) \cdot (-\operatorname{cosec}^2 x^2) \cdot \frac{d}{dx} (x^2) \\ &= \sin(\cot x^2) \operatorname{cosec}^2 x^2 \cdot 2x = 2x \cdot \sin(\cot x^2) \cdot \operatorname{cosec}^2 x^2 \end{aligned}$$

(xi) Let $y = \log \tan \left(\frac{\pi}{4} + \frac{x}{2} \right)$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{d}{dx} \left[\log \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right] \\ &= \frac{1}{\tan \left(\frac{\pi}{4} + \frac{x}{2} \right)} \cdot \frac{d}{dx} \left[\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right] \\ &= \frac{1}{\tan \left(\frac{\pi}{4} + \frac{x}{2} \right)} \cdot \sec^2 \left(\frac{\pi}{4} + \frac{x}{2} \right) \cdot \frac{d}{dx} \left(\frac{\pi}{4} + \frac{x}{2} \right) \\ &= \frac{\cos \left(\frac{\pi}{4} + \frac{x}{2} \right)}{\sin \left(\frac{\pi}{4} + \frac{x}{2} \right)} \cdot \frac{1}{\cos^2 \left(\frac{\pi}{4} + \frac{x}{2} \right)} \left[0 + \frac{1}{2} \right] \\ &= \frac{1}{2 \sin \left(\frac{\pi}{4} + \frac{x}{2} \right) \cos \left(\frac{\pi}{4} + \frac{x}{2} \right)} \\ &= \frac{1}{\sin 2 \left(\frac{\pi}{4} + \frac{x}{2} \right)} = \frac{1}{\sin \left(\frac{\pi}{2} + x \right)} = \frac{1}{\cos x} = \sec x \end{aligned}$$

(xii) Let $y = e^{ax} \cdot \sin(bx + c)$

$$\begin{aligned} \therefore \frac{dy}{dx} &= e^{ax} \cdot \frac{d}{dx} \sin(bx + c) + \sin(bx + c) \cdot \frac{d}{dx} e^{ax} \\ &= e^{ax} \cdot \cos(bx + c) \cdot \frac{d}{dx} (bx + c) + \sin(bx + c) \cdot e^{ax} \cdot \frac{d}{dx} (ax) \\ &= e^{ax} \cdot \cos(bx + c) \cdot b + \sin(bx + c) \cdot e^{ax} \cdot a \\ &= e^{ax} [b \cos(bx + c) + a \sin(bx + c)]. \end{aligned}$$



Comprehensive Exercise 6.3

1. Find the differential coefficients of the following functions w.r.t 'x':

- | | | |
|--|--|------------------------------------|
| (i) $\log \frac{x^2 + x + 1}{x^2 - x + 1}$ | (ii) $\frac{e^x + e^{-x}}{e^x - e^{-x}}$ | (iii) $\frac{x}{\sqrt{a^2 - x^2}}$ |
| (iv) $\sin^3 x$ | (v) $\tan^3 x$ | (vi) $(2x + 3)^{k-1}$ |
| (vii) $(\log x)^3$ | (viii) $\log(e^x + 1)$ | (ix) $\sqrt{\sin x}$ |
| (x) $\sqrt{(a + bx)^m}$ | (xi) $\log \tan(x/2)$ | (xii) $e^x \log \sin 2x$ |
| (xiii) $\frac{x}{\sqrt{1 - x^2}}$ | | |

2. If $y = \sin x \cdot \cos 2x$, prove that $\frac{dy}{dx} = y[\cot x - 2 \tan 2x]$.



Answers

1.	(i) $\frac{2(1-x^2)}{x^4+x^2+1}$	(ii) $-4(e^x - e^{-x})^{-2}$	(iii) $\frac{a^2}{(a^2 - x^2)^{3/2}}$
	(iv) $3 \sin^2 x \cos x$	(v) $3 \tan^2 x \sec^2 x$	(vi) $2(k-1)(2x+3)^{(k-2)}$
	(vii) $\frac{3(\log x)^2}{x}$	(viii) $\frac{e^x}{e^x+1}$	(ix) $\frac{\cos x}{2\sqrt{\sin x}}$
	(x) $\frac{1}{2} mb (a+bx)^{(m-2)/2}$	(xi) $\operatorname{cosec} x$	
	(xii) $e^x (2 \cot 2x + \log \sin 2x)$	(xiii) $(1-x^2)^{-3/2}$	

6.5 Derivatives of Inverse Functions

6.5.1 Derivative of $\sin^{-1} x$ with respect to 'x'

Let $y = \sin^{-1} x$

and $y + k = \sin^{-1}(x + h)$, where, $k \rightarrow 0$

when $k \rightarrow 0$

Then $x = \sin y$ and $x + h = \sin(y + k)$

$$\therefore \frac{d}{dx} \sin^{-1} x = \lim_{h \rightarrow 0} \frac{\sin^{-1}(x+h) - \sin^{-1} x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(x+h) - \sin^{-1} x}{(x+h) - x}$$

$$= \lim_{k \rightarrow 0} \frac{y+k-y}{\sin(y+k) - \sin y}$$

$$= \lim_{k \rightarrow 0} \frac{k}{2 \cos\left(y + \frac{k}{2}\right) \sin \frac{k}{2}}$$

$$= \lim_{k \rightarrow 0} \frac{\frac{k}{2}}{\cos\left(y + \frac{k}{2}\right) \cdot \sin\left(\frac{k}{2}\right)}$$

$$= \lim_{k \rightarrow 0} \frac{1}{\cos\left(y + \frac{k}{2}\right)} \cdot \lim_{h \rightarrow 0} \frac{k/2}{\sin k/2}$$

$$= \lim_{k \rightarrow 0} \frac{1}{\cos\left(y + \frac{k}{2}\right)} \times 1 = \frac{1}{\cos y} \cdot 1$$

$$\left[\therefore \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1 \right]$$

$$= \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

$$\therefore \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

6.5.2 Derivative of $\cos^{-1} x$ w.r.t. 'x'

Let $y = \cos^{-1} x$ and $y + k = \cos^{-1}(x + h)$

Where $k \rightarrow 0$. Where $h \rightarrow 0$.

Then, $x = \cos y$ and $x + h = \cos(y + k)$.

$$\begin{aligned} \therefore \frac{d}{dx} \cos^{-1} x &= \lim_{h \rightarrow 0} \frac{\cos^{-1}(x+h) - \cos^{-1} x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos^{-1}(x+h) - \cos^{-1} x}{(x+h) - x} \\ &= \lim_{k \rightarrow 0} \frac{y+k-y}{\cos(y+k) - \cos y} \\ &= \lim_{k \rightarrow 0} \frac{k}{2 \sin(y+k/2) \cdot \sin\left(-\frac{k}{2}\right)} \\ &= - \lim_{k \rightarrow 0} \frac{1}{\sin(y+k/2)} \cdot \frac{k/2}{(k/2)} \\ &= - \lim_{k \rightarrow 0} \frac{1}{\sin\left(y+\frac{k}{2}\right)} \cdot \lim_{k \rightarrow 0} \frac{k/2}{\sin k/2} \\ &= - \lim_{k \rightarrow 0} \frac{1}{\sin(y+k/2)} \times 1 \\ \frac{d}{dx} \cos^{-1} x &= - \frac{1}{\sin y} = \frac{-1}{\sqrt{1-\cos^2 y}} = \frac{-1}{\sqrt{1-x^2}} \end{aligned}$$

$$\therefore \frac{d}{dx} \cos^{-1} x = - \frac{1}{\sqrt{1-x^2}}$$

$$\left[\because \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1 \right]$$

6.5.3 Derivative of $\tan^{-1} x$ w.r.t. 'x'

Let $y = \tan^{-1} x$ and $y + k = \tan^{-1}(x + h)$

where $k \rightarrow 0$ where $h \rightarrow 0$

Then; $x = \tan y$ and $(x + h) = \tan(y + k)$

$$\begin{aligned} \therefore \frac{d}{dx} \tan^{-1} x &= \lim_{h \rightarrow 0} \frac{\tan^{-1}(x+h) - \tan^{-1} x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\tan^{-1}(x+h) - \tan^{-1} x}{(x+h) - x} \end{aligned}$$

$$\begin{aligned}
&= \lim_{k \rightarrow 0} \frac{y+k-y}{\tan(y+k) - \tan y} \\
&= \lim_{k \rightarrow 0} \frac{k}{\frac{\sin(y+k)}{\cos(y+k)} - \frac{\sin y}{\cos y}} \\
&= \lim_{k \rightarrow 0} \frac{k \cos(y+k) \cos y}{\sin(y+k) \cdot \cos y - \cos(y+k) \cdot \sin y} \\
&= \lim_{k \rightarrow 0} \frac{k \cos(y+k) \cdot \cos y}{\sin(y+k-y)} \\
&= \lim_{k \rightarrow 0} \frac{k \cos(y+k) \cdot \cos y}{\sin k} \\
&= \lim_{k \rightarrow 0} \frac{k}{\sin k} \cdot \lim_{k \rightarrow 0} [\cos(y+k) \cdot \cos y] \\
&= 1 \cdot \cos y \cdot \cos y = \cos^2 y \\
&= \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} \\
\Rightarrow \quad \frac{d}{dx} \tan^{-1} x &= \frac{1}{1+x^2}
\end{aligned}$$

6.5.4 Derivative of $\cot^{-1} x$ w.r.t. 'x'.

Let $y = \cot^{-1} x$ and $y+k = \cot^{-1}(x+h)$, where $k \rightarrow 0$ when $h \rightarrow 0$

Then $x = \cot y$ and $x+h = \cot(y+k)$

$$\begin{aligned}
\therefore \quad \frac{d}{dx} \cot^{-1} x &= \lim_{h \rightarrow 0} \frac{\cot^{-1}(x+h) - \cot^{-1} x}{h} \\
&= \lim_{h \rightarrow 0} \frac{\cot^{-1}(x+h) - \cot^{-1} x}{(x+h) - x} \\
&= \lim_{k \rightarrow 0} \frac{(y+k) - y}{\cot(y+k) - \cot y} \\
&= \lim_{k \rightarrow 0} \frac{k}{\frac{\cos(y+k)}{\sin(y+k)} - \frac{\cos y}{\sin y}} \\
&= \lim_{k \rightarrow 0} \frac{k \sin(y+k) \cdot \sin y}{\cos(y+k) \cdot \sin y - \sin(y+k) \cdot \cos y} \\
&= \lim_{k \rightarrow 0} \frac{k \sin(y+k) \cdot \sin y}{\sin[y - (y+k)]} \\
&= \lim_{k \rightarrow 0} \left[-\sin(y+k) \cdot \sin y \cdot \frac{k}{\sin k} \right] \\
&= - \lim_{k \rightarrow 0} [\sin(y+k) \cdot \sin y] \cdot \lim_{k \rightarrow 0} \frac{k}{\sin k} \\
&= -\sin y \cdot \sin y \cdot 1 = -\sin^2 y = -\frac{1}{\operatorname{cosec}^2 y} \\
&= -\frac{1}{1 + \cot^2 y} = -\frac{1}{1+x^2}
\end{aligned}$$

$\left[\therefore \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1 \right]$

$$\Rightarrow \frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

We can find out the derivatives of $\sec^{-1} x$ and $\operatorname{cosec}^{-1} x$, similarly as above

$$\Rightarrow \frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$$

$$\Rightarrow \frac{d}{dx} \operatorname{cosec}^{-1} x = \frac{-1}{x\sqrt{x^2-1}}$$

Key of tricks to find out the derivatives of inverse functions

- | | | |
|--------------------|--------------------|------------------------------------|
| (i) $\sin^{-1} x$ | (ii) $\cos^{-1} x$ | (iii) $\tan^{-1} x$ |
| (iv) $\sec^{-1} x$ | (v) $\cot^{-1} x$ | (vi) $\operatorname{cosec}^{-1} x$ |

Solution:

- (i) Consider $y = \sin^{-1} x \Rightarrow x = \sin y$

Now, $(\text{Diff})^n$ on both the sides, w.r.t. 'x', we get

$$\frac{d}{dx}(x) = \frac{d}{dx}(\sin y)$$

$$1 = \frac{d}{dy}(\sin y) \cdot \frac{dy}{dx}$$

$$\Rightarrow 1 = \cos y \cdot \frac{dy}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

$$\text{Therefore, } \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

- (ii) Consider the functions

$$y = \cos^{-1} x \Rightarrow x = \cos y$$

Now, $(\text{Diff})^n$ on both the sides w.r.t. 'x', we get

$$\frac{d}{dx}(x) = \frac{d}{dx}(\cos y)$$

$$1 = \frac{d}{dy}(\cos y) \cdot \frac{dy}{dx}$$

$$1 = -\sin y \cdot \frac{dy}{dx}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1-\cos^2 y}} = -\frac{1}{\sqrt{1-x^2}}$$

$$\therefore \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

Another trick you can apply here:

We know from Inverse circular functions

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \Rightarrow \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

$$\therefore \frac{d}{dx} \cos^{-1}(x) = \frac{d}{dx} \left(\frac{\pi}{2} - \sin^{-1} x \right) = 0 - \frac{d}{dx} \sin^{-1} x$$

$$\frac{d}{dx} \cos^{-1} x = - \frac{1}{\sqrt{1-x^2}}$$

(iii) Let $y = \tan^{-1} x \Rightarrow x = \tan y$

(Diff)ⁿ on both the sides w.r.t. 'x' we get;

$$\frac{d}{dx}(x) = \frac{d}{dx} \tan(y)$$

$$\Rightarrow 1 = \frac{d}{dy}(\tan y) \cdot \frac{dy}{dx}$$

$$1 = \sec^2 y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1 + x^2}$$

(iv) $\sec^{-1} x$

Let $y = \sec^{-1} x$, then $x = \sec y$

$$\Rightarrow \frac{d}{dx}(x) = \frac{d}{dx}(\sec y)$$

$$\Rightarrow 1 = \sec y \tan y \cdot \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sec y \tan y} = \frac{1}{\sec y \sqrt{\sec^2 y - 1}}$$

$$\therefore \frac{dy}{dx} = \frac{1}{x \sqrt{x^2 - 1}}$$

(v) $\cot^{-1}(x)$

Let $y = \cot^{-1} x$

$$\Rightarrow \cot y = x$$

$$\therefore \frac{d}{dx}(x) = \frac{d}{dx} \cot y$$

$$1 = \frac{d}{dy} \cot y \cdot \frac{dy}{dx}$$

$$= -\operatorname{cosec}^2 y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} = -\frac{1}{\operatorname{cosec}^2 y} = -\frac{1}{1 + \cot^2 y}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{1 + x^2}$$

(vi) Consider $y = \operatorname{cosec}^{-1} x$

$$\Rightarrow x = \operatorname{cosec} y$$

$$\therefore \frac{d}{dx}(x) = \frac{d}{dx}(\operatorname{cosec} y) = \frac{d}{dy}(\operatorname{cosec} y) \cdot \frac{dy}{dx}$$

$$1 = -\operatorname{cosec} y \cdot \cot y \cdot \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{\operatorname{cosec} y \cdot \cot y} = \frac{-1}{\operatorname{cosec} y \sqrt{\operatorname{cosec}^2 y - 1}}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{x\sqrt{x^2 - 1}}$$

Another Trick: From the Inverse circular functions

we know that $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$

$$\Rightarrow \operatorname{cosec}^{-1} x = \frac{\pi}{2} - \sec^{-1} x$$

$$\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{d}{dx}\left(\frac{\pi}{2} - \sec^{-1} x\right) = 0 - \frac{d}{dx} \sec^{-1} x$$

$$\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{x\sqrt{x^2 - 1}}.$$



Illustrative Examples

Example 1: Find $\frac{dy}{dx}$ if $y = \frac{\cos^{-1} x^2}{\cos(ax+b)}$

[B.C.A. (Rohilkhand) 2002]

Solution:

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{\cos(ax+b) \frac{d}{dx}(\cos^{-1} x^2) - \cos^{-1} x^2 \cdot \frac{d}{dx}[\cos(ax+b)]}{[\cos(ax+b)]^2} \\ &= \frac{\cos(ax+b) \left\{ \frac{-1}{\sqrt{1-x^4}} \right\} \cdot \frac{d}{dx}(x^2) - \cos^{-1} x^2 \cdot \{-\sin(ax+b)\} \frac{d}{dx}(ax+b)}{\cos^2(ax+b)} \\ &= \frac{\frac{-\cos(ax+b)}{\sqrt{1-x^4}} \cdot 2x + \sin(ax+b)(\cos^{-1} x^2) \cdot a}{\cos^2(ax+b)} \\ &= \frac{-2x}{(\sqrt{1-x^4}) \cos(ax+b)} + \frac{a \sin(ax+b) \cos^{-1} x^2}{\cos^2(ax+b)} \end{aligned}$$

Example 2: If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, show that:

$$(1-x^2) \frac{dy}{dx} = xy + 1$$

[B.C.A. (Agra) 2001, 2003, 2005]

Solution: We have $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{\sqrt{1-x^2} \frac{d}{dx} (\sin^{-1} x) - (\sin^{-1} x) \frac{d}{dx} (1-x^2)^{1/2}}{(\sqrt{1-x^2})^2} \\ &= \frac{\sqrt{1-x^2} \frac{d}{dx} (\sin^{-1} x) - (\sin^{-1} x) \frac{d}{dx} (1-x^2)^{1/2}}{(\sqrt{1-x^2})^2} \\ &= \frac{\sqrt{1-x^2} \frac{1}{\sqrt{1-x^2}} - (\sin^{-1} x) \cdot \left\{ \frac{1}{2} \cdot \frac{(-2x)}{\sqrt{1-x^2}} \right\}}{1-x^2} \end{aligned}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 + x \cdot \frac{\sin^{-1} x}{\sqrt{1-x^2}}}{1-x^2}$$

$$\Rightarrow (1-x^2) \frac{dy}{dx} = 1 + x \cdot \frac{\sin^{-1} x}{\sqrt{1-x^2}}$$

$$\Rightarrow (1-x^2) \frac{dy}{dx} = 1 + xy$$

Example 3: Find the differential coefficients w.r.t. 'x' :

(i) $\sin^{-1}(x/a)$

(ii) $\tan^{-1}(x/a)$

(iii) $\sec^{-1}(x/a)$

(iv) $\tan^{-1} \left(\frac{\sin x + \cos x}{\cos x - \sin x} \right)$

(v) $\sin^{-1}(2x)$

(vi) $\tan^{-1}(\log x)$

(vii) $\tan^{-1}(a^x \cdot x^2)$

(viii) $\cos \left(a \sin^{-1} \frac{1}{x} \right)$

(ix) $\sin(e^{\tan^{-1} x})$

Solution:

$$\begin{aligned} \text{(i)} \quad \frac{d}{dx} [\sin^{-1} x/a] &= \frac{1}{\sqrt{1-(x^2/a^2)}} \cdot \frac{d}{dx} (x/a) \\ &= \frac{a}{\sqrt{a^2-x^2}} \cdot \frac{1}{a} = \frac{1}{\sqrt{a^2-x^2}} \end{aligned}$$

$$\text{(ii)} \quad \frac{d}{dx} [\tan^{-1}(x/a)] = \frac{1}{1+(x/a)^2} \cdot \frac{d}{dx} (x/a) = \frac{a^2}{a^2+x^2} \cdot \frac{1}{a} \cdot 1 = \frac{a}{x^2+a^2}$$

$$\begin{aligned}
 \text{(iii)} \quad \frac{d}{dx} \sec^{-1}(x/a) &= \frac{1}{x/a \sqrt{(x/a)^2 - 1}} \times \frac{1}{a} \\
 &= \frac{1}{x/a \sqrt{\frac{x^2 - a^2}{a^2}}} \times \frac{1}{a} = \frac{a^2}{x \sqrt{x^2 - a^2}} \cdot \frac{1}{a} = \frac{a}{x \sqrt{x^2 - a^2}}
 \end{aligned}$$

$$\text{(iv)} \quad \text{Let } y = \tan^{-1} \left(\frac{\sin x + \cos x}{\cos x - \sin x} \right)$$

Now divide by $\cos x$ in N^r and D^r , we get

$$y = \tan^{-1} \left(\frac{\tan x + 1}{1 - \tan x} \right) = \tan^{-1} \left(\frac{1 + \tan x}{1 - \tan x} \right)$$

$$\therefore y = \tan^{-1} \left[\tan \left(\frac{\pi}{4} + x \right) \right] = \frac{\pi}{4} + x$$

$$\Rightarrow y = \left(\frac{\pi}{4} + x \right)$$

(Diff)ⁿ w.r.t x on both the sides, we get

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} \left(\frac{\pi}{4} + x \right)$$

$$\Rightarrow \frac{dy}{dx} = 0 + 1 \text{ or } \frac{dy}{dx} = 1$$

$$\text{(v)} \quad \frac{d}{dx} \sin^{-1}(2x) = \frac{1}{\sqrt{1 - (2x)^2}} \cdot \frac{d}{dx}(2x) = \frac{2}{\sqrt{1 - 4x^2}}$$

$$\text{(vi)} \quad \frac{d}{dx} \tan^{-1}(\log x) = \frac{1}{1 + (\log x)^2} \times \frac{d}{dx}(\log x) = \frac{1}{1 + (\log x)^2} \times \frac{1}{x} = \frac{1}{x[1 + (\log x)^2]}$$

$$\begin{aligned}
 \text{(vii)} \quad \frac{d}{dx} \tan^{-1}(a^{c \cdot x} \cdot x^2) &= \frac{1}{1 + (a^{c \cdot x} \cdot x^2)^2} \frac{d}{dx}(a^{c \cdot x} \cdot x^2) \\
 &= \frac{1}{1 + a^{2c \cdot x} \cdot x^4} \left[a^{c \cdot x} \cdot \frac{d}{dx} x^2 + x^2 \frac{d}{dx} a^{c \cdot x} \right] \\
 &= \frac{1}{1 + x^4 a^{2c \cdot x}} \left[a^{c \cdot x} \cdot 2x + x^2 a^{c \cdot x} \log a \cdot \frac{d}{dx}(c \cdot x) \right] = \frac{x a^{c \cdot x}}{1 + x^4 a^{2c \cdot x}} [2 + c \cdot x \log a]
 \end{aligned}$$

$$\begin{aligned}
 \text{(viii)} \quad \frac{d}{dx} \cos \left(a \sin^{-1} \frac{1}{x} \right) &= -\sin \left(a \sin^{-1} \frac{1}{x} \right) \frac{d}{dx} \left(a \sin^{-1} \left(\frac{1}{x} \right) \right) \\
 &= -\sin \left(a \sin^{-1} \left(\frac{1}{x} \right) \right) a \frac{1}{\sqrt{1 - \frac{1}{x^2}}} \frac{d}{dx} \left(\frac{1}{x} \right) \\
 &= -a \sin \left(a \sin^{-1} \left(\frac{1}{x} \right) \right) \frac{x}{\sqrt{x^2 - 1}} \frac{(-1)}{x^2} \\
 &= \frac{a}{x \sqrt{x^2 - 1}} \sin \left(a \sin^{-1} \left(\frac{1}{x} \right) \right)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ix)} \quad \frac{d}{dx} \sin(e^{\tan^{-1} x}) &= \cos(e^{\tan^{-1} x}) \cdot \frac{d}{dx} (e^{\tan^{-1} x}) \\
 &= \cos(e^{\tan^{-1} x}) \cdot e^{\tan^{-1} x} \cdot \frac{d}{dx} (\tan^{-1} x) \\
 &= \cos(e^{\tan^{-1} x}) \cdot e^{\tan^{-1} x} \cdot \frac{1}{1+x^2} = \frac{e^{\tan^{-1} x} \cos(e^{\tan^{-1} x})}{1+x^2}
 \end{aligned}$$

Example 4: If $y = e^{\tan^{-1} x} \log(\sec^2 x^3)$ then find the value of $\frac{dy}{dx}$:

Solution: Given function $y = e^{\tan^{-1} x} \cdot \log(\sec^2 x^3) = e^{\tan^{-1} x} \cdot \log(\sec x^3)^2$
 $= e^{\tan^{-1} x} \cdot 2 \log(\sec x^3)$

Now, differentiate on both the sides w. r. t. 'x', we get

$$\begin{aligned}
 \frac{dy}{dx} &= 2 \left[e^{\tan^{-1} x} \cdot \frac{d}{dx} [\log(\sec x^3)] + \log(\sec x^3) \cdot \frac{d}{dx} (e^{\tan^{-1} x}) \right] \\
 &= 2 \left[e^{\tan^{-1} x} \cdot \left\{ \frac{1}{\sec x^3} \cdot \frac{d}{dx} (\sec x^3) \right\} + \log(\sec x^3) \cdot \left\{ e^{\tan^{-1} x} \cdot \frac{d}{dx} (\tan^{-1} x) \right\} \right] \\
 &= 2 e^{\tan^{-1} x} \left[\frac{1}{\sec x^3} \cdot \sec x^3 \tan x^3 \cdot \frac{d}{dx} (x^3) + \log(\sec x^3) \cdot \frac{1}{1+x^2} \right] \\
 &= 2 e^{\tan^{-1} x} \left[(\tan x^3) \cdot 3x^2 + \frac{\log(\sec x^3)}{1+x^2} \right] \\
 &= e^{\tan^{-1} x} \left[6x^2 \tan x^3 + \frac{\log \sec^2 x^3}{1+x^2} \right]
 \end{aligned}$$

Example 5: To Prove that

$$\frac{d}{dx} \left[x\sqrt{a^2 - x^2} + a^2 \sin^{-1}(x/a) \right] = 2\sqrt{a^2 - x^2}$$

Solution: We have $\frac{d}{dx} \left[x\sqrt{a^2 - x^2} + a^2 \sin^{-1}(x/a) \right]$

$$\begin{aligned}
 &= \frac{d}{dx} \left[x(a^2 - x^2)^{1/2} \right] + a^2 \left[\frac{d}{dx} \sin^{-1}(x/a) \right] \\
 &= \left[x \cdot \frac{d}{dx} (a^2 - x^2)^{1/2} + (a^2 - x^2)^{1/2} \cdot \frac{d}{dx} (x) \right] + a^2 \cdot \frac{1}{\sqrt{1 - (x/a)^2}} \cdot \frac{d}{dx} (x/a)
 \end{aligned}$$

(By chain Rule)

$$\begin{aligned}
 &= x \cdot \frac{1}{2} (a^2 - x^2)^{-1/2} \cdot \frac{d}{dx} (a^2 - x^2) + (a^2 - x^2)^{1/2} + a^2 \cdot \frac{a}{\sqrt{(a^2 - x^2)}} \cdot \frac{1}{a} \\
 &= \left[\frac{x}{2\sqrt{a^2 - x^2}} (-2x) + \sqrt{a^2 - x^2} \right] + \frac{a^2}{\sqrt{(a^2 - x^2)}}
 \end{aligned}$$

$$= \left[\frac{-x^2 + (a^2 - x^2)}{\sqrt{a^2 - x^2}} \right] + \frac{a^2}{\sqrt{a^2 - x^2}} = \frac{2(a^2 - x^2)}{\sqrt{a^2 - x^2}} = 2\sqrt{a^2 - x^2}$$

Example 6: If $y = \tan^{-1} \sqrt{\frac{1 - \cos x}{1 + \cos x}}$ then find the value of $\frac{dy}{dx}$ w.r.t. 'x':

[B.C.A. (Meerut) 2009]

Solution:

$$\begin{aligned} y &= \tan^{-1} \sqrt{\frac{1 - \cos x}{1 + \cos x}} \\ &= \tan^{-1} \sqrt{\frac{1 - \left(1 - 2 \sin^2 \frac{x}{2}\right)}{1 + (2 \cos^2 \frac{x}{2} - 1)}} \\ y &= \tan^{-1}(\tan x/2) = \frac{1}{2}x \end{aligned}$$

Now, differentiate on both the sides, we get; $\frac{dy}{dx} = \frac{d}{dx}(x/2) = \frac{1}{2}$

Examples 7: Find the derivatives of the following functions w.r.t. 'x':

(i) $\cos^{-1} \sqrt{\cos x}$ (ii) $\tan^{-1}(\sin \sqrt{x})$ (iii) $\sqrt{\sec^{-1} x^2}$

Solution:

$$\begin{aligned} \text{(i)} \quad \frac{d}{dx} [\cos^{-1} \sqrt{\cos x}] &= -\frac{1}{\sqrt{1 - (\sqrt{\cos x})^2}} \cdot \frac{d}{dx} (\cos x)^{1/2} \\ &= -\frac{1}{\sqrt{1 - \cos x}} \cdot \frac{1}{2} (\cos x)^{-1/2} \cdot \frac{d}{dx} (\cos x) \\ &= -\frac{1}{\sqrt{1 - \cos x}} \cdot \frac{1}{2\sqrt{\cos x}} \cdot (-\sin x) = \frac{\sin x}{2\sqrt{\cos x} \sqrt{1 - \cos x}}. \end{aligned}$$

(ii) Let $y = \tan^{-1}(\sin \sqrt{x})$

$$\begin{aligned} \text{Then} \quad \frac{dy}{dx} &= \frac{1}{1 + (\sin \sqrt{x})^2} \cdot \frac{d}{dx} \sin(\sqrt{x}) \\ &= \frac{1}{1 + \sin^2 \sqrt{x}} \cdot \cos \sqrt{x} \cdot \frac{d}{dx} (x)^{1/2} \\ &= \frac{1}{1 + \sin^2 \sqrt{x}} \cdot \cos \sqrt{x} \cdot \frac{1}{2\sqrt{x}} = \frac{\cos \sqrt{x}}{2\sqrt{x}(1 + \sin^2 \sqrt{x})} \end{aligned}$$

(iii) Let $y = \sqrt{\sec^{-1} x^2} = (\sec^{-1} x^2)^{1/2}$

$$\begin{aligned} \therefore \quad \frac{dy}{dx} &= \frac{1}{2} (\sec^{-1} x^2)^{-1/2} \cdot \frac{d}{dx} (\sec^{-1} x^2) \\ &= \frac{1}{2\sqrt{\sec^{-1} x^2}} \cdot \frac{1}{x^2 \sqrt{x^4 - 1}} \cdot \frac{d}{dx} (x^2) \\ &= \frac{1}{2\sqrt{\sec^{-1} x^2}} \cdot \frac{1}{x^2 \sqrt{x^4 - 1}} \cdot 2x = \frac{1}{x\sqrt{x^4 - 1}\sqrt{\sec^{-1} x^2}} \end{aligned}$$

Example 8: Find $\frac{dy}{dx}$ if $y = e^{\tan^{-1} x} \cdot \log(\sec^2 x^3)$.

Solution: $y = e^{\tan^{-1} x} \cdot \log(\sec^2 x^3)$

(Diff)ⁿ on both the sides, with respect to x , we get

$$\begin{aligned}\frac{dy}{dx} &= e^{\tan^{-1} x} \frac{d}{dx} [\log(\sec^2 x^3)] + \log(\sec^2 x^3) \frac{d}{dx} (e^{\tan^{-1} x}) \\&= e^{\tan^{-1} x} \frac{1}{\sec^2 x^3} \frac{d}{dx} (\sec^2 x^3) + \log(\sec^2 x^3) e^{\tan^{-1} x} \frac{d}{dx} (\tan^{-1} x) \\&= e^{\tan^{-1} x} \cdot \frac{1}{\sec^2 x^3} \cdot 2 \cdot \sec x^3 \frac{d}{dx} (\sec x^3) + e^{\tan^{-1} x} \cdot \log(\sec^2 x^3) \cdot \frac{1}{1+x^2} \\&= 2e^{\tan^{-1} x} \cdot \frac{1}{\sec^2 x^3} \cdot (\sec^2 x^3 \tan x^3) \frac{d}{dx} (x^3) + \frac{e^{\tan^{-1} x} \cdot \log(\sec^2 x^3)}{1+x^2} \\&= 2e^{\tan^{-1} x} \cdot \tan x^3 \cdot 3x^2 + \frac{e^{\tan^{-1} x} \cdot \log(\sec^2 x^3)}{1+x^2} \\&= 2e^{\tan^{-1} x} \cdot \tan x^3 \cdot 3x^2 + \frac{e^{\tan^{-1} x} \cdot \log(\sec^2 x^3)}{1+x^2} \\&= e^{\tan^{-1} x} \left[6x^2 \tan x^3 + \frac{\log(\sec^2 x^3)}{1+x^2} \right]\end{aligned}$$

Examples 9: If $f(x) = \left(\frac{x^a}{x^b}\right)^{a+b} \cdot \left(\frac{x^b}{x^c}\right)^{b+c} \cdot \left(\frac{x^c}{x^a}\right)^{c+a}$; then find out $f'(x)$

Solution: We have $f(x) = (x^{a-b})^{a+b} \cdot (x^{b-c})^{b+c} \cdot (x^{c-a})^{c+a}$

$$\begin{aligned}&= x^{a^2-b^2} \cdot x^{b^2-c^2} \cdot x^{c^2-a^2} \\&= x^{a^2-b^2+b^2-c^2+c^2-a^2} = x^0 = 1\end{aligned}$$

$$\therefore f'(x) = 0 \quad \left\{ \frac{d}{dx} (x^0) \Leftrightarrow \frac{d}{dx} (1) = 0 \right\}$$

Example 10: If $y = (1+x^{1/4})(1+x^{1/2})(1-x^{1/4})$ then find $\frac{dy}{dx}$

Solution: We have $y = (1+x^{1/4})(1+x^{1/2})(1-x^{1/4}) = (1-x^{1/2})(1+x^{1/2}) = 1-x$

$$\therefore \frac{d}{dx} (y) = \frac{d}{dx} (1-x) = \frac{d}{dx} (1) - \frac{d}{dx} (x) = 0 - 1 = -1$$

Example 11: If S_n denotes the sum of n term of a G.P. whose common ratio is r , then

$$(r-1) \frac{dS_n}{dr} + S_n = nar^{n-1}$$

Solution: We have, $S_n = \frac{a(r^n-1)}{r-1}$

$$\Rightarrow (r-1)S_n = ar^n - a$$

On differentiating both the sides w.r.t. r , we get

$$(r-1) \frac{dS_n}{dr} + S_n = nar^{n-1} - 0$$

$$\Rightarrow (r-1) \frac{dS_n}{dr} + S_n = nar^{n-1}$$

Example 12: Differentiate $\frac{a^x}{x^n}$ with respect to x .

Solution: $(a)^x = 1 + x \log_e a + \frac{x^2 (\log_e a)^2}{2!} + \frac{x^3 (\log_e a)^3}{3!} + \dots$

$$\therefore \frac{(a)^x}{x^n} = \frac{1}{x^n} + \frac{1}{x^{n-1}} \log_e a + \frac{1}{x^{n-2}} \cdot \frac{(\log_e a)^2}{2!} + \frac{1}{x^{n-3}} \cdot \frac{(\log_e a)^3}{3!} + \dots$$

Differentiate on both the sides; we get

$$\begin{aligned} \frac{d}{dx} \left(\frac{a^x}{x^n} \right) &= \frac{d}{dx} \left[x^{-n} + x^{-n+1} \log_e a + x^{-n+2} \frac{(\log_e a)^2}{2!} + x^{-n+3} \frac{(\log_e a)^3}{3!} + \dots \right] \\ &= -nx^{-n-1} + (-n+1)x^{-n} \log_e a + (-n+2)x^{-n+1} \frac{(\log_e a)^2}{2!} \\ &\quad + (-n+3) \cdot x^{-n+2} \frac{(\log_e a)^3}{3!} + \dots \\ &= x^{-n} \left[-nx^{-1} + (-n+1) \log_e a + (-n+2)x \cdot \frac{(\log_e a)^2}{2!} \right. \\ &\quad \left. + (-n+3) \cdot x^2 \cdot \frac{(\log_e a)^3}{3!} + \dots \right] \\ &= x^{-n} \left[-n(x^{-1} + \log_e a + \frac{x}{2!} (\log_e a)^2 + \dots) \right. \\ &\quad \left. + \log_e a \{1 + x \log_e a + \frac{x^2}{2!} (\log_e a)^2 + \dots\} \right] \\ &= x^{-n} \left[-nx^{-1} \{1 + x \log_e a + \frac{x^2}{2!} (\log_e a)^2 + \dots\} + (\log_e a) \cdot a^x \right] \\ &= x^{-n} \left[-\frac{n}{x} \cdot a^x + (\log_e a) a^x \right] \\ &= \frac{a^x}{x^n} \left[-\frac{n}{x} + \log_e a \right] = \frac{a^x}{x^n} \left[\log_e a - \frac{n}{x} \right] \end{aligned}$$



Comprehensive Exercise 6.4

- Find the differential coefficient of the following functions w.r.t. 'x'.
 - $\cot^{-1}(\operatorname{cosec} x)$
 - $\tan^{-1}(\sin e^x)$
 - $\tan^{-1}(\log_e x)$
 - $(\sin^{-1} x)^2 - (\cos^{-1} x)^2$
 - $\sin(e^{\tan^{-1} x})$
 - $\log \cos^{-1} x$
 - $\log(\cos^{-1} x^2)$
 - $\cot(\cos^{-1} x)$
 - $\sec^2(\tan^{-1} x)$
 - $\cos \left\{ a \sin^{-1} \frac{1}{x} \right\}$
- If $y = x \sin^{-1} x + \sqrt{1-x^2}$. Then prove that $\frac{dy}{dx} = \sin^{-1} x$.
- If $y = \frac{x \sin^{-1} x}{\sqrt{1-x^2}}$. Prove that $(1-x^2) \frac{dy}{dx} = x + \frac{y}{x}$
- If $y = \tan^{-1} \sqrt{\frac{1-\cos x}{1+\cos x}}$. Prove that $\frac{dy}{dx} = \frac{1}{2}$
- If $y = \cos^{-1} \frac{(2 \cos x + 3 \sin x)}{\sqrt{13}}$. Prove that $\frac{dy}{dx} = 1$. [B.C.A. (2002, 2005, 2008)]



Answers

1.	(i) $\frac{\operatorname{cosec} x \cdot \cot x}{1 + \operatorname{cosec}^2 x}$	(ii) $\frac{e^x \cdot \cos e^x}{1 + \sin^2 e^x}$
	(iii) $\frac{1}{x[1 + (\log_e x)^2]}$	(iv) $\frac{2(\sin^{-1} x + \cos^{-1} x)}{\sqrt{1-x^2}}$
	(v) $\frac{1}{1+x^2} \cdot e^{\tan^{-1} x} \cdot \cos(e^{\tan^{-1} x})$	(vi) $-\frac{1}{(\sqrt{1-x^2}) \cos^{-1} x}$
	(vii) $\frac{-2x}{\sqrt{(1-x^4)} \cos^{-1} x^2}$	(viii) $\frac{1}{\sqrt{1-x^2}} \cdot \operatorname{cosec}^2(\cos^{-1} x)$
	(ix) $\frac{2x \cdot \sec^2(\tan^{-1} x)}{1+x^2}$	(x) $\frac{a \sin(a \operatorname{cosec}^{-1} x)}{x\sqrt{x^2-1}}$

Illustrative Examples

(Transformation)

Example 1: Differentiate the following function w.r.t. 'x'

(i) $\tan^{-1} \frac{2x}{1-x^2}$

(ii) $\sin^{-1}(3x-4x^3)$

(iii) $\sin^{-1}(2x\sqrt{1-x^2})$

(iv) $\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$

Solution:

(i) Let $y = \tan^{-1} \frac{2x}{1-x^2}$

Putting

$$x = \tan \theta \text{ so that}$$

$$\theta = \tan^{-1} x, \text{ we have}$$

$$y = \tan^{-1}\left(\frac{2 \tan \theta}{1 - \tan^2 \theta}\right) = \tan^{-1}(\tan 2\theta)$$

$$y = 2\theta$$

$\Rightarrow$

$$y = 2 \tan^{-1} x.$$

$\therefore$

$$\frac{dy}{dx} = 2 \times \frac{1}{1+x^2} = \frac{2}{1+x^2}$$

(ii) Let $y = \sin^{-1}(3x-4x^3).$

Putting

$$x = \sin \theta \Rightarrow \theta = \sin^{-1} x$$

we have

$$y = \sin^{-1}(3 \sin \theta - 4 \sin^3 \theta) = \sin^{-1}(\sin 3\theta) = 3\theta$$

$\Rightarrow$

$$y = 3 \sin^{-1} x$$

$\therefore$

$$\frac{dy}{dx} = 3 \times \frac{1}{\sqrt{1-x^2}} = \frac{3}{\sqrt{1-x^2}}$$

(iii) Let $y = \sin^{-1}(2x\sqrt{1-x^2})$

Putting

$$x = \sin \theta \Rightarrow \theta = \sin^{-1} x$$

we have

$$y = \sin^{-1}(2 \sin \theta \cdot \sqrt{1 - \sin^2 \theta})$$

$$= \sin^{-1}(2 \sin \theta \times \cos \theta) = \sin^{-1}(\sin 2\theta) = 2\theta$$

$\Rightarrow$

$$y = 2 \sin^{-1} x$$

$\therefore$

$$\frac{dy}{dx} = 2 \times \frac{1}{\sqrt{1-x^2}} = \frac{2}{\sqrt{1-x^2}}$$

(iv) Let $y = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$

Putting

$$x = \tan \theta \Rightarrow \theta = \tan^{-1} x, \text{ we have}$$

$$y = \tan^{-1} \left(\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \right) = \tan^{-1} (\tan 3\theta)$$

$$y = 3\theta = 3 \tan^{-1} x$$

$$\Rightarrow \frac{dy}{dx} = 3 \times \frac{1}{1+x^2} = \frac{3}{1+x^2}$$

Example 2: Differentiate the following functions w.r.t 'x' :

$$(i) \quad \sin^{-1} \left(\frac{1-x^2}{1+x^2} \right) \qquad (ii) \quad \sec^{-1} \left(\frac{x+x^{-1}}{x-x^{-1}} \right)$$

$$(iii) \quad \tan^{-1} \left(\frac{4\sqrt{x}}{(1-4x)} \right) \qquad (iv) \quad \cos^{-1} \left(\sqrt{\frac{1+x}{2}} \right)$$

Solution:

$$(i) \quad y = \sin^{-1} \left(\frac{1-x^2}{1+x^2} \right) \text{ Put } x = \tan \theta \Rightarrow \theta = \tan^{-1} x$$

$$\begin{aligned} \text{We have } y &= \sin^{-1} \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) \\ &= \sin^{-1} \left(\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} \right) \\ &= \sin^{-1} (\cos^2 \theta - \sin^2 \theta) \\ &= \sin^{-1} (\cos 2\theta) \\ &= \sin^{-1} [\sin(\pi/2 - 2\theta)] \\ &= \frac{\pi}{2} - 2\theta \end{aligned}$$

$$\Rightarrow y = \pi/2 - 2 \tan^{-1} x \qquad (\because x = \tan \theta)$$

$$\therefore \frac{dy}{dx} = 0 - 2 \times \frac{1}{1+x^2} = -\frac{2}{1+x^2}.$$

$$(ii) \quad \text{Let } y = \sec^{-1} \left(\frac{x+x^{-1}}{x-x^{-1}} \right) = \sec^{-1} \left[\frac{x + \frac{1}{x}}{x - \frac{1}{x}} \right] = \sec^{-1} \left(\frac{x^2 + 1}{x^2 - 1} \right)$$

$$\Rightarrow y = \cos^{-1} \left(\frac{x^2 - 1}{x^2 + 1} \right)$$

$$\text{Put } x = \cot \theta \Rightarrow \theta = \cot^{-1} x$$

$$\text{we have } y = \cos^{-1} \left(\frac{\cot^2 \theta - 1}{\cot^2 \theta + 1} \right) = \cos^{-1} \left[\frac{\frac{\cos^2 \theta}{\sin^2 \theta} - 1}{\frac{\cos^2 \theta}{\sin^2 \theta} + 1} \right]$$

$$= \cos^{-1} \left(\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} \right) = \cos^{-1}(\cos 2\theta) = 2\theta$$

$$\Rightarrow y = 2 \cot^{-1} x$$

$$\therefore \frac{dy}{dx} = 2 \times \left(-\frac{1}{1+x^2} \right) = \frac{-2}{1+x^2}$$

(iii) Let $y = \tan^{-1} \left(\frac{4\sqrt{x}}{1-4x} \right)$

Put $2\sqrt{x} = \tan \theta$ so that $\theta = \tan^{-1}(2\sqrt{x})$; we have

$$y = \tan^{-1} \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right) = \tan^{-1}(\tan 2\theta) = 2\theta$$

$$\Rightarrow y = 2 \tan^{-1}(2\sqrt{x})$$

$$\begin{aligned} \frac{dy}{dx} &= 2 \cdot \frac{d}{dx} (\tan^{-1} 2\sqrt{x}) = 2 \cdot \frac{1}{1+(2\sqrt{x})^2} \cdot \frac{d}{dx} (2x^{1/2}) \\ &= 2 \cdot \frac{1}{1+4x} \cdot 2 \cdot \frac{1}{2\sqrt{x}} = \frac{2}{(1+4x)\sqrt{x}} \end{aligned}$$

(iv) Let $y = \cot^{-1} \left(\sqrt{\frac{1+x}{2}} \right)$

Put $x = \cos \theta \Rightarrow \theta = \cos^{-1} x$; we have

$$\begin{aligned} y &= \cos^{-1} \sqrt{\frac{1+\cos \theta}{2}} \\ &= \cos^{-1} \sqrt{\frac{2 \cos^2 \theta/2}{2}} = \cos^{-1}(\cos \theta/2) = \theta/2 \end{aligned}$$

$$y = \frac{1}{2} \cos^{-1} x$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} \cdot \frac{d}{dx} (\cos^{-1} x)$$

$$\frac{dy}{dx} = \frac{1}{2} \cdot \left(\frac{-1}{\sqrt{1-x^2}} \right) = -\frac{1}{2\sqrt{1-x^2}}$$

Example 3: Differentiate the function in respect to 'x';

$$\tan^{-1} \left[\sqrt{\frac{1+\sin x}{1-\sin x}} \right]$$

Solution: Let $y = \tan^{-1} \left(\sqrt{\frac{1+\sin x}{1-\sin x}} \right)$

$$= \tan^{-1} \left[\sqrt{\frac{(\cos x/2 + \sin x/2)^2}{(\cos x/2 - \sin x/2)^2}} \right]$$

$$\begin{aligned}
 &= \tan^{-1} \left[\frac{\cos x/2 + \sin x/2}{\cos x/2 - \sin x/2} \right] \\
 &= \tan^{-1} \left[\frac{1 + \tan x/2}{1 - \tan x/2} \right] \\
 &= \tan^{-1} \left[\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right] = \frac{\pi}{4} + \frac{x}{2}
 \end{aligned}$$

$$\therefore \frac{dy}{dx} = 0 + \frac{1}{2} = \left(\frac{1}{2} \right)$$

Example 4: Differentiate the following functions w.r.t.'x';

$$(i) \tan^{-1}(\sqrt{1+x^2} + x) \quad (ii) \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$$

Solution:

$$(i) \text{ Let } y = \tan^{-1}(\sqrt{1+x^2} + x).$$

Put $x = \tan \theta \Rightarrow \theta = \tan^{-1} x$ we have

$$\begin{aligned}
 y &= \tan^{-1}(\sqrt{1+\tan^2 \theta} + \tan \theta) \\
 &= \tan^{-1}(\sec \theta + \tan \theta) \\
 &= \tan^{-1} \left(\frac{1 + \sin \theta}{\cos \theta} \right) \\
 &= \tan^{-1} \left[\frac{(\cos \theta/2 + \sin \theta/2)^2}{\cos^2 \theta/2 - \sin^2 \theta/2} \right] \\
 &= \tan^{-1} \left[\frac{\cos \theta/2 + \sin \theta/2}{\cos \theta/2 - \sin \theta/2} \right] \\
 &= \tan^{-1} \left[\frac{1 + \tan \theta/2}{1 - \tan \theta/2} \right] \\
 &= \tan^{-1} \left[\tan \left(\frac{\pi}{4} + \theta/2 \right) \right] \\
 y &= \frac{\pi}{4} + \frac{\theta}{2}
 \end{aligned}$$

$$\Rightarrow \frac{dy}{dx} = 0 + \frac{1}{2} \cdot \frac{1}{1+x^2} = \frac{1}{2(1+x^2)}$$

$$(ii) \text{ Let } y = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$$

Put $x = \tan \theta$ so that $\theta = \tan^{-1} x$.

$$\begin{aligned}
 \text{we have, } y &= \tan^{-1} \left(\frac{\sqrt{1+\tan^2 \theta}-1}{\tan \theta} \right) \\
 &= \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right)
 \end{aligned}$$

$$\begin{aligned}
 &= \tan^{-1} \left[\left(\frac{1}{\cos \theta} - 1 \right) \frac{\cos \theta}{\sin \theta} \right] \\
 &= \tan^{-1} \left[\frac{1 - \cos \theta}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} \right] \\
 &= \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) \\
 &= \tan^{-1} \left[\frac{2 \sin^2 \theta/2}{2 \sin \theta/2 \cdot \cos \theta/2} \right] \\
 &= \tan^{-1}(\tan \theta/2) \\
 \Rightarrow \quad y = \theta/2 &\Rightarrow y = \frac{1}{2} \tan^{-1} x \\
 \therefore \quad \frac{dy}{dx} &= \frac{1}{2} \cdot \frac{1}{1+x^2} = \frac{1}{2(1+x^2)}.
 \end{aligned}$$

Example 5: Find the derivative of the following function w.r.t. 'x';

(i) $\tan^{-1} \left(\frac{x}{\sqrt{a^2 - x^2}} \right)$ (ii) $\cos^{-1} \left(\frac{x}{\sqrt{x^2 + a^2}} \right)$

Solution:

(i) Let $y = \tan^{-1} \left(\frac{x}{\sqrt{a^2 - x^2}} \right)$

Put $x = a \sin \theta$, so that $\sin \theta = \left(\frac{x}{a} \right)$

$$\begin{aligned}
 \Rightarrow \quad \theta &= \sin^{-1} \left(\frac{x}{a} \right) \text{ we have} \\
 y &= \tan^{-1} \left(\frac{a \sin \theta}{\sqrt{a^2 - a^2 \sin^2 \theta}} \right) \\
 &= \tan^{-1} \left(\frac{a \sin \theta}{a \sqrt{1 - \sin^2 \theta}} \right) \\
 &= \tan^{-1} \left(\frac{\sin \theta}{\cos \theta} \right) \\
 &= \tan^{-1}(\tan \theta) \\
 \Rightarrow \quad y &= \theta \\
 y &= \sin^{-1}(x/a) \\
 \therefore \quad \frac{dy}{dx} &= \frac{1}{\sqrt{1 - \left(\frac{x^2}{a^2} \right)}} \cdot \frac{d}{dx} (x/a) = \frac{a}{\sqrt{a^2 - x^2}} \cdot \frac{1}{a} = \frac{1}{\sqrt{a^2 - x^2}}.
 \end{aligned}$$

(ii) Let $y = \cos^{-1} \left(\frac{x}{\sqrt{x^2 + a^2}} \right)$

Put $x = a \cot \theta \Rightarrow \theta = \cot^{-1}(x/a)$ we have

$$\begin{aligned} y &= \cos^{-1} \left(\frac{a \cot \theta}{\sqrt{a^2 \cot^2 \theta + a^2}} \right) \\ &= \cos^{-1} \left(\frac{a \cot \theta}{a \operatorname{cosec} \theta} \right) \\ &= \cos^{-1} \left(\frac{\cos \theta}{\sin \theta} \cdot \sin \theta \right) \\ &= \cos^{-1} (\cos \theta) = \theta \end{aligned}$$

$$\Rightarrow y = \theta \Rightarrow \cot^{-1}(x/a)$$

$$\frac{dy}{dx} = \frac{-1}{1+x^2/a^2} \cdot \frac{1}{a} = -\frac{a}{a^2+x^2}$$

Example 6: If $y = \frac{x \cos^{-1} x}{\sqrt{1-x^2}}$, find $\frac{dy}{dx}$.

Solution: Let $x = \cos \theta$ then $\theta = \cos^{-1} x$

and

$$\frac{d\theta}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

$$\therefore y = \frac{(\cos \theta) \cdot \theta}{\sqrt{1-\cos^2 \theta}} = \frac{\theta \cdot \cos \theta}{\sin \theta} = \theta \cot \theta$$

$$\therefore \frac{dy}{d\theta} = \theta \frac{d}{d\theta} (\cot \theta) + (\cot \theta) \cdot \frac{d}{d\theta} \theta.$$

$$= \theta (-\operatorname{cosec}^2 \theta) + \cot \theta \cdot 1$$

$$= \cot \theta - \theta \cdot \operatorname{cosec}^2 \theta.$$

$$= \frac{\cos \theta}{\sin \theta} - \theta \cdot \frac{1}{\sin^2 \theta}$$

$$= \frac{x}{\sqrt{1-x^2}} - (\cos^{-1} x) \cdot \frac{1}{1-x^2} = \frac{x}{\sqrt{1-x^2}} - \frac{\cos^{-1} x}{1-x^2}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= \left(\frac{x}{\sqrt{1-x^2}} - \frac{\cos^{-1} x}{1-x^2} \right) \left(-\frac{1}{\sqrt{1-x^2}} \right) = \frac{\cos^{-1} x}{(1-x^2)^{3/2}} - \frac{x}{1-x^2}.$$

Example 7: Find $\frac{d}{dx}$ if (i) $y = \sec^{-1} \left(\frac{\sqrt{x}+1}{\sqrt{x}-1} \right) + \sin^{-1} \left(\frac{\sqrt{x}-1}{\sqrt{x}+1} \right)$

(ii) $y = \sin^{-1} [x \cdot \sqrt{1-x} - \sqrt{x} \sqrt{1-x^2}]$.

Solution: (i) Given function

$$y = \sec^{-1} \left(\frac{\sqrt{x}+1}{\sqrt{x}-1} \right) + \sin^{-1} \left(\frac{\sqrt{x}-1}{\sqrt{x}+1} \right)$$

$$\begin{aligned}
 &= \cos^{-1} \left(\frac{\sqrt{x}-1}{\sqrt{x}+1} \right) + \sin^{-1} \left(\frac{\sqrt{x}-1}{\sqrt{x}+1} \right) \quad \left(\because \sec^{-1} x = \cos^{-1} \frac{1}{x} \right) \\
 &= \frac{\pi}{2} \quad \left(\because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \right)
 \end{aligned}$$

$$\therefore \frac{dy}{dx} = 0$$

(ii) Given function $y = \sin^{-1} [x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}]$

Put $x = \sin \theta \Rightarrow \theta = \sin^{-1} x$.

and $\sqrt{x} = \sin \phi \Rightarrow \phi = \sin^{-1} \sqrt{x}$

we have now;

$$\begin{aligned}
 y &= \sin^{-1} [\sin \theta \cdot \sqrt{1-\sin^2 \phi} - \sin \phi \cdot \sqrt{1-\sin^2 \theta}] \\
 &= \sin^{-1} [\sin \theta \cdot \cos \phi - \sin \phi \cos \theta] \\
 &= \sin^{-1} [\sin(\theta - \phi)] \\
 &= \theta - \phi
 \end{aligned}$$

$$\Rightarrow y = \sin^{-1} x - \sin^{-1} \sqrt{x}$$

$$\begin{aligned}
 \therefore \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{d}{dx} (x)^{1/2} \\
 &= \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2} (x)^{-1/2} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x}\sqrt{1-x}}.
 \end{aligned}$$

Example 8: Find $\frac{dy}{dx}$ if

(i) $y = \sin \left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right)$ (ii) $y = \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right)$

Solution:

(i) Given function

$$y = \sin \left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right)$$

Put $x = \cos 2\theta$, we get

$$\begin{aligned}
 y &= \sin \left[2 \tan^{-1} \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} \right] \\
 &= \sin \left[2 \tan^{-1} \sqrt{\frac{2 \sin^2 \theta}{2 \cos^2 \theta}} \right] \\
 &= \sin [2 \tan^{-1} (\tan \theta)] \\
 &= \sin 2\theta \\
 &= \sqrt{1-\cos^2 2\theta}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \quad y &= \sqrt{1-x^2} = (1-x^2)^{1/2} \\
 \therefore \quad \frac{dy}{dx} &= \frac{1}{2} (1-x^2)^{-1/2} \cdot \frac{d}{dx} (1-x^2) \\
 &= \frac{1}{2\sqrt{1-x^2}} \cdot (0-2x) \\
 &= -\frac{x}{\sqrt{1-x^2}}
 \end{aligned}$$

(ii) Given function

$$y = \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right)$$

Put $x^2 = \cos 2\theta$, we have

$$\begin{aligned}
 y &= \tan^{-1} \left[\frac{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}} \right] \\
 &= \tan^{-1} \left[\frac{\sqrt{2\cos^2 \theta} + \sqrt{2\sin^2 \theta}}{\sqrt{2\cos^2 \theta} - \sqrt{2\sin^2 \theta}} \right] \\
 &= \tan^{-1} \left[\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} \right] \\
 &= \tan^{-1} \left[\frac{1 + \frac{\sin \theta}{\cos \theta}}{1 - \frac{\sin \theta}{\cos \theta}} \right] \\
 &= \tan^{-1} \left(\frac{1 + \tan \theta}{1 - \tan \theta} \right) \\
 &= \tan^{-1} \left[\tan \left(\frac{\pi}{4} + \theta \right) \right] \\
 &= \frac{\pi}{4} + \theta \\
 &= \frac{\pi}{4} + \frac{1}{2} \cos^{-1} x^2 \quad (\because \cos 2\theta = x^2)
 \end{aligned}$$

$$\begin{aligned}
 \therefore \quad \frac{dy}{dx} &= 0 + \frac{1}{2} \frac{d}{dx} (\cos^{-1} x^2) \\
 &= 0 + \frac{1}{2} \left(-\frac{1}{\sqrt{1-x^4}} \right) \cdot \frac{d}{dx} (x^2) \\
 &= -\frac{1}{2} \frac{1}{\sqrt{1-x^4}} \cdot 2x = -\frac{x}{\sqrt{1-x^4}}.
 \end{aligned}$$



Comprehensive Exercise 6.5

1. Find the derivative of the following trigonometrical functions using suitable transformations:

(i) $\sin^{-1}(\sqrt{1-x^2})$

(ii) $\cos^{-1}(\sqrt{1-x^2})$

(iii) $\sin^{-1}(1-2x^2)$

(iv) $\cos^{-1}(2x^2-1)$

(v) $\cot^{-1}\left(\frac{1-x^2}{2x}\right)$

(vi) $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$

(vii) $\cos^{-1}(4x^3-3x)$

(viii) $\tan^{-1}\left[\frac{3a^2x-x^3}{a(a^2-3x^2)}\right]$

(ix) $\tan^{-1}\left(\frac{a+x}{1-ax}\right)$

(x) $\cot^{-1}\left(\frac{1+x}{1-x}\right)$

(xi) $\tan^{-1}\left(\frac{1-\cos x}{\sin x}\right)$

(xii) $\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$

(xiii) $\sin^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{2}\right)$

(xiv) $\tan^{-1}\left(\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}}\right)$

(xv) $\sin^2\left(\cot^{-1}\sqrt{\frac{1+x}{1-x}}\right)$

2. If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right) + \sec^{-1}\left(\frac{1+x^2}{1-x^2}\right)$; then prove that $\frac{dy}{dx} = \frac{4}{1+x^2}$.



Answers

1.	(i) $-\frac{1}{\sqrt{1-x^2}}$	(ii) $\frac{1}{\sqrt{1-x^2}}$	(iii) $\frac{-2}{\sqrt{1-x^2}}$
	(iv) $\frac{-2}{\sqrt{1-x^2}}$	(v) $\frac{2}{1+x^2}$	(vi) $\frac{-2}{\sqrt{1-x^2}}$
	(vii) $\frac{-3}{\sqrt{1-x^2}}$	(viii) $\frac{3a}{a^2+x^2}$	(ix) $\frac{1}{1+x^2}$
	(x) $-\frac{1}{1+x^2}$	(xi) $\frac{1}{2}$	(xii) -1
	(xiii) $\frac{1}{2\sqrt{1-x^2}}$	(xiv) $-\frac{1}{2\sqrt{1-x^2}}$	(xv) $-\frac{1}{2}$

6.6 Logarithmic Differentiation

When a given function contains a variable as its exponent or the function is the product or quotient of a number of function: the derivative cannot be obtained easily by using standard formula. In such cases;

To find the derivative, we Proceed as follows

- (i) Given function as y
- (ii) Find $\log y$
- (iii) After $(Diff)^n$; calculate $\left(\frac{dy}{dx}\right)$



Illustrative Examples

Example 1: Find the derivatives of the following functions:

- (i) x^x
- (ii) $x^{\sqrt{x}}$
- (iii) $x^{\cos^{-1}x}$
- (iv) $(\log x)^{\sin x}$

Solution:

- (i) $y = (x)^x$

Taking log on both the sides, we get

$$\log y = \log(x)^x = x \log x$$

$(Diff)^n$; both sides w.r.t. 'x', we get

$$\frac{1}{y} \frac{dy}{dx} = x \cdot \frac{1}{x} + (\log x) \cdot 1$$

$$\Rightarrow \frac{dy}{dx} = y [1 + \log x]$$

$$i.e. \frac{dy}{dx} = (x)^x [1 + \log x]$$

- (ii) Let $y = x^{\sqrt{x}}$.

Taking log on both the sides, we get

$$\log y = \log(x)^{\sqrt{x}}$$

$$\Rightarrow \log y = \sqrt{x} \cdot \log x.$$

$(Diff)^n$ on both the sides w.r.t. 'x', we get

$$\frac{1}{y} \cdot \frac{dy}{dx} = \sqrt{x} \cdot \frac{d}{dx} (\log x) + (\log x) \frac{d}{dx} (x)^{1/2} = \sqrt{x} \cdot \frac{1}{x} + (\log x) \cdot \frac{1}{2} \cdot (x)^{-1/2}$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{1}{\sqrt{x}} + \frac{1}{2\sqrt{x}} \log x$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{\sqrt{x}} \cdot \left(1 + \frac{1}{2} \log x\right)$$

$$\frac{dy}{dx} = \frac{(x)^{\sqrt{x}}}{\sqrt{x}} \left(1 + \frac{1}{2} \log x\right).$$

(iii) Let $y = x^{\cos^{-1} x}$. then;

Taking log on both the sides;

$$\log y = \cos^{-1} x \cdot \log(x)$$

(Diff)ⁿ on both the sides; w.r.t. 'x', we get;

$$\frac{1}{y} \cdot \frac{dy}{dx} = (\cos^{-1} x) \cdot \frac{1}{x} + (\log x) \left(-\frac{1}{\sqrt{1-x^2}}\right)$$

$$\Rightarrow \frac{dy}{dx} = y \left[\frac{1}{x} \cos^{-1} x - \frac{1}{\sqrt{1-x^2}} \log x \right]$$

$$\frac{dy}{dx} = x^{\cos^{-1} x} \left[\frac{1}{x} \cos^{-1} x - \frac{1}{\sqrt{1-x^2}} \log x \right].$$

(iv) Let $y = (\log x)^{\sin x}$

Taking log on both the sides, we get;

$$\log y = \sin x \log(\log x).$$

(Diff)ⁿ on both the sides, w.r.t. 'x', we get

$$\frac{1}{y} \cdot \frac{dy}{dx} = \sin x \cdot \frac{d}{dx} \log[\log x] + \log(\log x) \frac{d}{dx} (\sin x)$$

$$= \sin x \cdot \frac{1}{\log x} \cdot \frac{1}{x} + \log(\log x) \cdot \cos x$$

$$\Rightarrow \frac{dy}{dx} = y \left[\frac{\sin x}{x \log x} + \cos x \cdot \log(\log x) \right]$$

$$\Rightarrow \frac{dy}{dx} = (\log x)^{\sin x} \left[\frac{\sin x}{x \cdot \log x} + \cos x \log(\log x) \right]$$

Example 2: Find the differential coefficients of the following functions w.r.t. 'x'

$$(i) \frac{1}{(x+p)(x+q)(x+r)} \quad (ii) \frac{(x-\sin x)^{3/2}}{\sqrt{x}} \quad (iii) \frac{(x+1)^2 \sqrt{(x-1)}}{(x+4)^3 \cdot e^x}$$

Solution:

$$(i) \quad \text{Let} \quad y = \frac{1}{(x+p)(x+q)(x+r)}$$

Taking log on both the sides, we get

$$\log y = \log \frac{1}{(x+p)(x+q)(x+r)}$$

$$\Rightarrow \log y = \log 1 - \log(x+p) - \log(x+q) - \log(x+r)$$

(Diff)ⁿ on both the sides w.r.t. 'x', we get

$$\begin{aligned}\frac{1}{y} \cdot \frac{dy}{dx} &= 0 - \frac{1}{x+p} - \frac{1}{x+q} - \frac{1}{x+r} \\ \Rightarrow \frac{dy}{dx} &= -y \left[\frac{1}{x+p} + \frac{1}{x+q} + \frac{1}{x+r} \right] \\ \Rightarrow \frac{dy}{dx} &= -\frac{1}{(x+p)(x+q)(x+r)} \left[\frac{1}{x+p} + \frac{1}{x+q} + \frac{1}{x+r} \right]\end{aligned}$$

(ii) Let $y = \frac{(x - \sin x)^{3/2}}{\sqrt{x}}$ then

Taking log on both the sides, we get

$$\begin{aligned}\log y &= \log \frac{(x - \sin x)^{3/2}}{(x)^{1/2}} \\ \Rightarrow \log y &= \frac{3}{2} \log(x - \sin x) - \frac{1}{2} \log x.\end{aligned}$$

(Diff)ⁿ w.r.t. 'x' on both the sides, we get

$$\begin{aligned}\frac{1}{y} \cdot \frac{dy}{dx} &= \frac{3}{2} \cdot \frac{1}{x - \sin x} \cdot (1 - \cos x) - \frac{1}{2} \cdot \frac{1}{x} \\ \Rightarrow \frac{dy}{dx} &= y \left[\frac{3}{2} \cdot \frac{1 - \cos x}{x - \sin x} - \frac{1}{2x} \right] \\ \Rightarrow \frac{dy}{dx} &= \frac{(x - \sin x)^{3/2}}{2\sqrt{x}} \left[\frac{3(1 - \cos x)}{(x - \sin x)} - \frac{1}{x} \right]\end{aligned}$$

(iii) Let $y = \frac{(x+1)^2 \sqrt{x-1}}{(x+4)^3 \cdot e^x}$ then

Taking log on both the sides, we get

$$\begin{aligned}\log y &= \log \frac{(x+1)^2 \cdot (\sqrt{x-1})}{(x+4)^3 \cdot e^x} \\ \Rightarrow \{\log y &= \log(x+1)^2 \sqrt{x-1}\} - \log\{(x+4)^3 e^x\} \\ &= 2 \log(x+1) + \frac{1}{2} \log(x-1) - \{3 \log(x+4) + x \log e\} \\ \log y &= 2 \log(x+1) + \frac{1}{2} \log(x-1) - 3 \log(x+4) - x \quad (\because \log e = 1)\end{aligned}$$

(Diff)ⁿ on both the sides, we get

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= 2 \cdot \frac{1}{x+1} + \frac{1}{2} \cdot \frac{1}{x-1} - 3 \cdot \frac{1}{x+4} - 1 \\ \Rightarrow \frac{dy}{dx} &= y \left[\frac{2}{x+1} + \frac{1}{2(x-1)} - \frac{3}{x+4} - 1 \right] \\ \Rightarrow \frac{dy}{dx} &= \frac{(x+1)^2 \sqrt{x-1}}{(x+4)^3 \cdot e^x} \left[\frac{2}{x+1} + \frac{1}{2(x-1)} - \frac{3}{x+4} - 1 \right]\end{aligned}$$

Example 3: Find the differential coefficients of

$$\left(1 + \frac{1}{x}\right)^x + (x)^{1+1/x} \text{ w.r.t. 'x'.$$

Solution: Let the given function is

$$y = \left(1 + \frac{1}{x}\right)^x + (x)^{1+1/x}$$

Let $u = \left(1 + \frac{1}{x}\right)^x$ and $v = (x)^{1+1/x}$

$$\Rightarrow y = u + v$$

(Diff)ⁿ on both the sides, we get ; w.r.t. 'x';

$$\frac{dy}{dx} = \left(\frac{du}{dx} + \frac{dv}{dx}\right) \quad \dots(1)$$

Now $u = \left(1 + \frac{1}{x}\right)^x$

$$\log u = x \cdot \log \left(1 + \frac{1}{x}\right)$$

(Diff)ⁿ on both the sides. w.r.t. 'x', we get

$$\begin{aligned} \frac{1}{u} \cdot \frac{du}{dx} &= x \cdot \frac{d}{dx} \log \left(1 + \frac{1}{x}\right) + \log \left(1 + \frac{1}{x}\right) \frac{d}{dx} (x) \\ &= x \cdot \frac{1}{1 + \frac{1}{x}} \frac{d}{dx} \left(1 + \frac{1}{x}\right) + \log \left(1 + \frac{1}{x}\right) \cdot 1 \end{aligned}$$

$$\Rightarrow \frac{du}{dx} = u \left[\frac{x^2}{x+1} \cdot \left(0 - \frac{1}{x^2}\right) + \log \left(1 + \frac{1}{x}\right) \right]$$

$$\Rightarrow \frac{du}{dx} = \left(1 + \frac{1}{x}\right)^x \left[\log \left(1 + \frac{1}{x}\right) - \frac{1}{x+1} \right] \quad \dots(2)$$

Again $v = x \left(1 + \frac{1}{x}\right)$

$$\therefore \log v = \left(1 + \frac{1}{x}\right) \log x.$$

$$\frac{1}{v} \cdot \frac{dv}{dx} = \left(1 + \frac{1}{x}\right) \cdot \frac{1}{x} + \log x \left(0 - \frac{1}{x^2}\right)$$

$$\Rightarrow \frac{dv}{dx} = v \cdot \left[\frac{1}{x} + \frac{1}{x^2} - \frac{1}{x^2} \log x \right]$$

$$\Rightarrow \frac{dv}{dx} = x \left(1 + \frac{1}{x}\right) \left[\frac{1}{x} + \frac{1}{x^2} (1 - \log x) \right] \quad \dots(3)$$

$\therefore$ From (1), (2) and (3) we get;

$$\frac{dy}{dx} = \left(1 + \frac{1}{x}\right)^x \left[\log \left(1 + \frac{1}{x}\right) - \frac{1}{1+x} \right] + x \left(1 + \frac{1}{x}\right) \left[\frac{1}{x} + \frac{1}{x^2} (1 - \log x) \right]$$

Example 4: Find the differential coefficients of $(\sin x)^{\cos x} + (\cos x)^{\sin x}$ w.r.t. 'x'.

Solution: Let $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$.

Also, $(\sin x)^{\cos x} = u$ and $(\cos x)^{\sin x} = v$.

Then, we have

$$\begin{aligned} y &= u + v \\ \frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \end{aligned} \quad \dots(1)$$

Now,

$$u = (\sin x)^{\cos x}$$

$$\Rightarrow \log u = \cos x \log \sin x$$

$(\text{Diff})^n$ w.r.t. 'x' we get;

$$\begin{aligned} \frac{1}{u} \cdot \frac{du}{dx} &= \cos x \cdot \frac{d}{dx} (\log \sin x) + \log \sin x \cdot \frac{d}{dx} (\cos x) \\ \Rightarrow \frac{du}{dx} &= u \left[\cos x \cdot \frac{1}{\sin x} \cdot \cos x + (\log \sin x) \cdot (-\sin x) \right] \\ \frac{du}{dx} &= (\sin x)^{\cos x} \cdot (\cos x \cdot \cot x - \sin x \cdot \log \sin x) \end{aligned} \quad \dots(2)$$

Again

$$v = (\cos x)^{\sin x}$$

$$\Rightarrow \log v = \sin x \cdot \log \cos x$$

$$\begin{aligned} \frac{1}{v} \cdot \frac{dv}{dx} &= \sin x \cdot \frac{1}{\cos x} (-\sin x) + (\log \cos x) \cdot \cos x \\ \Rightarrow \frac{dv}{dx} &= v [-\sin x \cdot \tan x + (\log \cos x) \cdot \cos x] \\ \Rightarrow \frac{dv}{dx} &= (\cos x)^{\sin x} [\cos x \cdot \log \cos x - \sin x \cdot \tan x] \end{aligned} \quad \dots(3)$$

From (1), (2), and (3), we get

$$\begin{aligned} \frac{dy}{dx} &= (\sin x)^{\cos x} (\cos x \cdot \cot x - \sin x \cdot \log \sin x) \\ &\quad + (\cos x)^{\sin x} (\cos x \cdot \log \cos x - \sin x \cdot \tan x). \end{aligned}$$

Example 5: If $y = \sqrt{\frac{1-x}{1+x}}$, Prove that $(1-x^2) \frac{dy}{dx} + y = 0$

Solution: Let $y = \left(\frac{1-x}{1+x} \right)^{1/2}$.

Taking log on both the sides. w.r.t. 'x', we get

$$\begin{aligned} \log y &= \frac{1}{2} \log \frac{1-x}{1+x} \\ \Rightarrow \log y &= \frac{1}{2} [\log(1-x) - \log(1+x)] \end{aligned}$$

$(\text{Diff})^n$ on both the sides, w.r.t. 'x', we get

$$\begin{aligned} \frac{1}{y} \cdot \frac{dy}{dx} &= \frac{1}{2} \left[\frac{1}{1-x} \cdot \frac{d}{dx} (1-x) - \frac{1}{1+x} \cdot \frac{d}{dx} (1+x) \right] \\ &= \frac{1}{2} \left[\frac{1}{1-x} \cdot (-1) - \frac{1}{1+x} \cdot (1) \right] \end{aligned}$$

$$\begin{aligned}
 &= -\frac{1}{2} \left[\frac{1}{1-x} + \frac{1}{1+x} \right] \\
 &= -\frac{1}{2} \left[\frac{1+x+1-x}{(1-x)(1+x)} \right] = -\frac{1}{2} \cdot \frac{2}{1-x^2} \\
 \Rightarrow \quad &\frac{1}{y} \cdot \frac{dy}{dx} = -\frac{1}{1-x^2} \\
 \Rightarrow \quad &\frac{1}{y} \cdot \frac{dy}{dx} + \frac{1}{1-x^2} = 0 \\
 \Rightarrow \quad &(1-x^2) \cdot \frac{dy}{dx} + y = 0
 \end{aligned}$$

Example 6: If $y = e^{x+e^{x+e^{x \dots \infty}}}$; Prove that

$$\frac{dy}{dx} = \frac{y}{1-y}$$

[B.C.A. (Meerut) 2006, 2007]

Solution: Let $y = e^{x+e^{x+e^{x \dots \infty}}}$

$$\Rightarrow y = e^{x+y}$$

Taking log on both the sides, we get

$$\log_e y = \log_e e^{(x+y)}$$

$$\log_e y = (x+y) \log_e e \quad (\because \log_e e = 1)$$

$$\log_e y = (x+y).$$

Now, (diff) on both the sides w.r.t. 'x', we get

$$\frac{1}{y} \cdot \frac{dy}{dx} = 1 + \frac{dy}{dx}$$

$$\Rightarrow \left(\frac{1}{y} - 1 \right) \frac{dy}{dx} = 1$$

$$\Rightarrow \left(\frac{1-y}{y} \right) \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{1-y}$$

Example 7: If $y = \left[x + \sqrt{x^2 - 1} \right]^m$, prove that

$$(x^2 - 1) \left(\frac{dy}{dx} \right)^2 = m^2 y^2$$

[B.C.A. (Meerut) 2009]

Solution: We have

$$y = \left[x + \sqrt{x^2 - 1} \right]^m \quad \dots(1)$$

$$\frac{dy}{dx} = m \left[x + \sqrt{x^2 - 1} \right]^{m-1} \left[1 + \frac{1}{2} (x^2 - 1)^{-1/2} \times 2x \right]$$

$$\begin{aligned}\frac{dy}{dx} &= m \left[x + \sqrt{x^2 - 1} \right]^{m-1} \left[1 + \frac{x}{\sqrt{x^2 - 1}} \right] \\ \frac{dy}{dx} &= m \left[x + \sqrt{x^2 - 1} \right]^{m-1} \left[\frac{\sqrt{x^2 - 1} - 1 + x}{\sqrt{x^2 - 1}} \right] \\ \frac{dy}{dx} &= \frac{m \left[x + \sqrt{x^2 - 1} \right]^m}{\sqrt{x^2 - 1}} \quad \dots(2)\end{aligned}$$

From (1) $\frac{dy}{dx} = \frac{m y}{\sqrt{x^2 - 1}}$

Taking square on both sides

$$\begin{aligned}\left(\frac{dy}{dx} \right)^2 &= \frac{m^2 y^2}{(x^2 - 1)} \\ \Rightarrow (x^2 - 1) \left(\frac{dy}{dx} \right)^2 &= m^2 y^2\end{aligned}$$

Example 8: Find $\frac{dy}{dx}$, if $\sin(x + y) = \log(x + y)$.

[B.C.A. (Meerut) 2009]

Solution: Differential w.r.t. (x)

$$\begin{aligned}\cos(x + y) \left[1 + \frac{dy}{dx} \right] &= \frac{1}{(x + y)} \left[1 + \frac{dy}{dx} \right] \\ (x + y) \cos(x + y) + (x + y) \cos(x + y) \frac{dy}{dx} &= 1 + \frac{dy}{dx} \\ [(x + y) \cos(x + y) - 1] \frac{dy}{dx} &= 1 - (x + y) \cos(x + y) \\ \Rightarrow \frac{dy}{dx} &= \frac{1 - (x + y) \cos(x + y)}{(x + y) \cos(x + y) - 1} = -1\end{aligned}$$



Comprehensive Exercise 6.6

1. Differentiates the following function w.r.t. 'x';

- | | | |
|---|---|---------------------------------------|
| (i) $(1 + x)^x$ | (ii) $\frac{1}{x^x}$ | (iii) $(x)^{\sin x}$ |
| (iv) $(x)^{\tan x}$ | (v) $x^{\sin^{-1} x}$ | (vi) $(\cos x)^{\log x}$ |
| (vii) $(\log x)^{\cos x}$ | (viii) $(\tan x)^{\cot x}$ | (ix) $(\sin x)^{\cos^{-1} x}$ |
| (x) $e^x \cdot \cos^3 x \cdot \sin^2 x$ | (xi) $\frac{(5)^x}{(x)^5}$ | (xii) $\frac{(x-a)(x-b)}{\sqrt{x-c}}$ |
| (xiii) $(x)^{1/x} + \left(\frac{1}{x} \right)^{1/x}$ | (xiv) $x^{\log x} + (\log x)^x$ | |
| (xv) $(\sin x)^x + (x)^{\sin x}$ | (xvi) $(\cot x)^{\tan x} + (\tan x)^{\cot x}$ | |

[B.C.A. (Meerut) 2009]

2. If $y = x^{x^{x^{\dots\infty}}}$, prove that $x \cdot \frac{dy}{dx} = \frac{y^2}{1 - y \log x}$. [B.C.A. (Meerut) 2004]
3. If $y = (a)^{x^a^{x^{\dots\infty}}}$, then prove that $\frac{y^2 \log y}{x(1 - y \log x \cdot \log y)}$.
4. If $y = (\sin x)^{(\sin x)^{(\sin x)^{\dots\infty}}}$ prove that $\frac{dy}{dx} = \frac{y^2 \cot x}{1 - y \cdot \log \sin x}$.
5. Find $\frac{dy}{dx}$ if $x^m y^n = (x + y)^{m+n}$. [B.C.A. (Meerut) 2003]



Answers

1.	(i) $(1+x)^x \left[\frac{x}{1+x} + \log(1+x) \right]$ (ii) $(x)^{1/x} \left(\frac{1 - \log x}{x^2} \right)$ (iii) $(x)^{\sin x} \left(\frac{\sin x}{x} + \cos x \cdot \log x \right)$ (iv) $(x)^{\tan x} \left(\frac{\tan x}{x} + \sec^2 \log x \right)$ (v) $x^{\sin^{-1} x} \left[\frac{\sin^{-1} x}{x} + \frac{\log x}{\sqrt{1-x^2}} \right]$ (vi) $(\cos x)^{\log x} \left[\frac{1}{x} \log \cos x - \tan x \cdot \log x \right]$ (vii) $(\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log(\log x) \right]$ (viii) $(\tan x)^{\cot x} \operatorname{cosec}^2 x [1 - \log(\tan x)]$ (ix) $(\sin x)^{\cos^{-1} x} \left[\cot x \cdot \cos^{-1} x - \frac{\log \sin x}{\sqrt{1-x^2}} \right]$ (x) $e^x \cdot \cos^3 x \cdot \sin^2 x [1 - 3 \tan x + 2 \cot x]$ (xi) $\frac{(5)^x}{x^5} \cdot \left(\log 5 - \frac{5}{x} \right)$ (xii) $\frac{(x-a)(x-b)}{\sqrt{x-c}} \left[\frac{1}{x-a} + \frac{1}{x-b} - \frac{1}{2(x-c)} \right]$ (xiii) $(x)^{1/x} \left(\frac{1 - \log x}{x^2} \right) - \left(\frac{1}{x} \right)^x (1 + \log x)$ (xiv) $(x)^{\log x} \cdot \frac{2 \log x}{x} + (\log x)^x \left[\log(\log x) + \frac{1}{(\log x)} \right]$ (xv) $(\sin x)^x (x \cot x + \log \sin x) + (x)^{\sin x} \left(\frac{\sin x}{x} + \cos x \log x \right)$ (xvi) $(\cot x)^{\tan x} \sec^2 x (\log \cot x - 1) + (\tan x)^{\cot x} \cdot \operatorname{cosec}^2 x (1 - \log \tan x)$
5.	$\left(\frac{y}{x} \right)$.

6.7 Derivative of an Implicit Function

When independent variable x and dependent variable y occur together in an equation and it is not convenient or possible to express y in terms of x . Then we say that y is an implicit function of x . In such cases, the variables x and y are connected by a relation in the form $f(x, y) = 0$.

In order to find $\frac{dy}{dx}$ from $f(x, y) = 0$, adopt the following procedure :

- (i) Differentiate the term w.r.t. 'x', noting that y is a function of x .
- (ii) Solve the equation, thus obtained for $\frac{dy}{dx}$ as shown in the examples.



Illustrative Examples

(Implicit Function)

Example 1: If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$; find $\frac{dy}{dx}$

Solution: The given equation is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

(Diff)ⁿ w.r.t. 'x' each term of the given equation, we get

$$2ax + 2h\left(x \cdot \frac{dy}{dx} + y \cdot 1\right) + 2by \cdot \frac{dy}{dx} + 2g + 2f \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow ax + h\left(x \cdot \frac{dy}{dx} + y\right) + by \cdot \frac{dy}{dx} + g + f \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow (hx + by + f) \frac{dy}{dx} = -(ax + hy + g)$$

$$\therefore \left(\frac{dy}{dx}\right) = -\frac{(ax + hy + g)}{(hx + by + f)}$$

Example 2: If $y^x = x^{\sin y}$, find $\frac{dy}{dx}$.

[B.C.A. (Meerut) 2002, 2005]

Solution: We have $(y)^x = (x)^{\sin y}$

$$\therefore \log(y)^x = \log(x)^{\sin y}$$

$$\Rightarrow x \log y = \sin y \cdot \log(x)$$

(Diff)ⁿ on both the sides w.r.t. 'x', we get

$$x \cdot \frac{1}{y} \cdot \frac{dy}{dx} + (\log y) \cdot 1 = (\sin y) \cdot \frac{1}{x} + (\log x) \cos y \cdot \frac{dy}{dx}$$

$$\Rightarrow \left(\frac{x}{y} - \cos y \cdot \log x \right) \frac{dy}{dx} = \frac{\sin y}{x} - \log y$$

$$\Rightarrow \left(\frac{x - y \cos y \cdot \log x}{y} \right) \frac{dy}{dx} = \frac{\sin y - x \log y}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y (\sin y - x \log y)}{x (x - y \cos y \cdot \log x)}$$

Example 3: If $(x)^y = (e)^{x-y}$, prove $\frac{dy}{dx} = \frac{\log_e x}{(1 + \log_e x)^2}$ [B.C.A. (Garhwal) 2005, 2008]

Solution: Given function $(x)^y = (e)^{x-y}$

Taking log on both the sides, we get

$$\log_e (x)^y = \log_e (e)^{x-y}$$

$$\Rightarrow y \log_e x = (x - y) \log_e e \quad (\because \log_e e = 1)$$

$$\Rightarrow y \cdot \log_e x = x - y$$

$$\Rightarrow y(1 + \log_e x) = x \Rightarrow y = \frac{x}{1 + \log_e x}$$

$$(Diff)^n \text{ we get, } \frac{dy}{dx} = \frac{(1 + \log_e x) \frac{d}{dx} x - x \frac{d}{dx} (1 + \log_e x)}{(1 + \log_e x)^2}$$

$$= \frac{(1 + \log_e x) \cdot 1 - x \left(0 + \frac{1}{x} \right)}{(1 + \log_e x)^2}$$

$$= \frac{1 + \log_e x - 1}{(1 + \log_e x)^2} = \frac{\log_e x}{(1 + \log_e x)^2}$$

Example 4: If $\sin y = x \sin (a + y)$; prove that $\frac{dy}{dx} = \frac{\sin^2 (a + y)}{\sin a}$ [B.C.A. (Kanpur) 2004]

Solution: We have $\sin y = x \sin (a + y)$... (1)

(Diff)ⁿ it w.r.t. 'x', on both the sides, we get

$$\cos y \frac{dy}{dx} = x \cos (a + y) \frac{dy}{dx} + \sin (a + y) \cdot 1$$

$$\Rightarrow [\cos y - x \cos (a + y)] \frac{dy}{dx} = \sin (a + y)$$

$$\Rightarrow \left[\cos y - \frac{\sin y}{\sin (a + y)} \cdot \cos (a + y) \right] \frac{dy}{dx} = \sin (a + y); \text{ From (1)}$$

$$\Rightarrow [\sin (a + y) \cos y - \cos (a + y) \cdot \sin y] \frac{dy}{dx} = \sin^2 (a + y)$$

$$\Rightarrow \sin (a + y - y) \frac{dy}{dx} = \sin^2 (a + y) \Rightarrow \sin a \cdot \frac{dy}{dx} = \sin^2 (a + y)$$

$$\frac{dy}{dx} = \frac{\sin^2 (a + y)}{\sin a}$$

Example 5: If $y = \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}$; Prove that $\frac{dy}{dx} = \frac{\cos x}{2y-1}$

[B.C.A. (Rohilkhand) 2001, 2005]

Solution: We have; $y = \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}$

$$\Rightarrow y = \sqrt{\sin x + y}$$

$$\Rightarrow y^2 = \sin x + y$$

$(Diff)^n$ on both the sides w.r.t. 'x', we get

$$2y \cdot \frac{dy}{dx} = \cos x + \frac{dy}{dx}$$

$$\Rightarrow (2y-1) \frac{dy}{dx} = \cos x$$

$$\frac{dy}{dx} = \frac{\cos x}{2y-1}$$

Example 6: If $x\sqrt{1+y} + y\sqrt{1+x} = 0$; prove that $\frac{dy}{dx} = -(1+x)^{-2}$

Solution: We have

$$x\sqrt{1+y} + y\sqrt{1+x} = 0$$

$$\Rightarrow x\sqrt{1+y} = -y\sqrt{1+x}$$

$$\Rightarrow x^2(1+y) = y^2(1+x)$$

$$\Rightarrow x^2 - y^2 = xy^2 - x^2y$$

$$\Rightarrow (x-y)(x+y) = -xy(x-y)$$

$$\Rightarrow (x-y)(x+y+xy) = 0$$

$$\Rightarrow x = y \quad \text{or} \quad x + y + xy = 0$$

Since $x = y$ does not satisfy the given equation therefore, we get

$$x + y + xy = 0$$

$$y(1+x) = -x$$

$$\Rightarrow y = -\frac{x}{1+x}$$

$\therefore (Diff)^n$ w.r.t. 'x', we get

$$\frac{dy}{dx} = -\left[\frac{(1+x) \cdot 1 - x \cdot 1}{(1+x)^2} \right]$$

$$\frac{dy}{dx} = -\frac{1}{(1+x)^2}$$

$$\Rightarrow \frac{dy}{dx} = -(1+x)^{-2}$$

Example 7: Find $\frac{dy}{dx}$ when $(x)^y + (y)^x = a$.

Solution: The given equation is $(x)^y + (y)^x = a$

$(Diff)^n$ w.r.t. 'x', we get

$$\frac{d}{dx} (x)^y + \frac{d}{dx} (y)^x = 0 \quad \dots(1)$$

Now put $(x)^y = u$, Then;

$$\begin{aligned} \log u &= y \log x \\ \Rightarrow \frac{1}{u} \cdot \frac{du}{dx} &= y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx} \\ \frac{du}{dx} &= u \left(\frac{y}{x} + \frac{dy}{dx} \log x \right) \\ \frac{d}{dx} (x)^y &= (x)^y \left(\frac{y}{x} + \frac{dy}{dx} \log x \right) \end{aligned}$$

Again, put $(y)^x = v$, Then;

$$\begin{aligned} \log v &= x \cdot \log y \\ \frac{1}{v} \cdot \frac{dv}{dx} &= x \cdot \frac{d}{dx} \log y + (\log y) \cdot 1 \\ \Rightarrow \frac{dv}{dx} &= v \left(x \cdot \frac{1}{y} \cdot \frac{dy}{dx} + \log y \right) \\ \Rightarrow \frac{d}{dx} (y)^x &= (y)^x \left(\frac{x}{y} \cdot \frac{dy}{dx} + \log y \right) \end{aligned}$$

Hence, from (1), we get

$$\begin{aligned} (x)^y (y/x + \frac{dy}{dx} \log x) + (y)^x \left(\frac{x}{y} \cdot \frac{dy}{dx} + \log y \right) &= 0 \\ \Rightarrow \frac{dy}{dx} \left(x^y \cdot \log x + y^x \cdot \frac{x}{y} \right) &= - \left(x^y \cdot \frac{y}{x} + y^x \log y \right) \\ \Rightarrow \frac{dy}{dx} (x^y \log x + xy^{x-1}) &= - (y \cdot x^{y-1} + y^x \log y) \\ \Rightarrow \frac{dy}{dx} &= \left(- \frac{yx^{y-1} + y^x \cdot \log y}{xy^{x-1} + x^y \log x} \right) \end{aligned}$$



Comprehensive Exercise 6.7

1. Find $\frac{dy}{dx}$ when;

(i) $x^2 + y^2 = a^2$ (ii) $xy - \log(xy) = \log 2$

(iii) if $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$. Prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$

[B.C.A. (Meerut) 2003]

2. (i) $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}$; Prove that $\frac{dy}{dx} = \frac{1}{2y-1}$

(ii) if $y = \sqrt{e^x + \sqrt{e^x + \sqrt{e^x + \dots \infty}}}$; Prove that $\frac{dy}{dx} = \frac{e^x}{2y-1}$

(iii) if $x = y + \frac{1}{y + \frac{1}{y + \dots \infty}}$; Then prove that $\frac{dy}{dx} = 2x^2 + y^2 - 3xy$

[Hint: Take $x = \left(y + \frac{1}{x}\right)$; Then $(\text{Diff})^n$ and take $1 = (x^2 - xy)$]

3. If $\cos y = x \cos(b+y)$; prove that $\frac{dy}{dx} = \frac{\cos^2(b+y)}{\sin b}$

[B.C.A. (Kanpur) 2003]



Answers

1.	(i) $\frac{-x}{y}$	(ii) $\frac{-y}{x}$
----	--------------------	---------------------

6.7.1 Derivative of Function in the Forms of Parametric

Let y be a function of x . Some times x and y both are expressed as a function of another variable. If $x = \Phi(t)$ and $y = \Psi(t)$ are functions of a variable t , then the functions :

$$x = \phi(t) \quad \text{and} \quad y = \psi(t)$$

are said to represent the function 'y' of 'x' in **Parametric form** and the variable t is called the **parameter** of the function 'y' of 'x'.

Note: We always find $\frac{dy}{dx}$ in such cases without first eliminating the parameters by the

formula $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

Illustrative Examples

(Parametric form)

Example 1: If $\begin{cases} x = a \sec \theta \\ y = b \tan \theta \end{cases}$ find $\frac{dy}{dx}$

Solution: We have $x = a \sec \theta$ (Diff)ⁿ w.r.t. 'θ', we get

$$\frac{dx}{d\theta} = a \sec \theta \tan \theta$$

Also,

$$y = b \tan \theta \text{ (Diff)}^n \text{ w.r.t. } \theta, \text{ we get}$$

$$\frac{dy}{d\theta} = b \sec^2 \theta$$

∴

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} = \frac{b \sec^2 \theta}{a \sec \theta \cdot \tan \theta} \\ &= \frac{b \sec \theta}{a \tan \theta} = \frac{b}{a} \cdot \frac{1}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} \\ &= \frac{b}{a} \cdot \frac{1}{\sin \theta} = \frac{b}{a} \operatorname{cosec} \theta \end{aligned}$$

Example 2: Find $\frac{dy}{dx}$ from the following equations:

$$x = a(t + \sin t)$$

$$y = a(1 + \cos t)$$

Solution:

$$x = a(t + \sin t)$$

⇒

$$\frac{dx}{dt} = a(1 + \cos t)$$

Again

$$y = a(1 + \cos t)$$

$$\frac{dy}{dt} = a(0 - \sin t)$$

⇒

$$\frac{dy}{dx} = -a \sin t$$

Hence

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} = \frac{-a \sin t}{a(1 + \cos t)} \\ &= -\frac{2 \sin(t/2) \cdot \cos(t/2)}{1 + 2 \cos^2(t/2) - 1} \\ &= -\frac{2 \sin(t/2) \cdot \cos(t/2)}{2 \cos^2(t/2)} = -\frac{\sin(t/2)}{\cos(t/2)} = -\tan t/2 \end{aligned}$$

Example 3: Find $\frac{dy}{dx}$ when; $x = \frac{3at}{1+t^3}$, $y = \frac{3at^2}{1+t^3}$

Solution:

$$x = 3a \cdot \frac{t}{1+t^3}$$

$$\Rightarrow \frac{dx}{dt} = 3a \left[\frac{(1+t^3) \frac{d}{dt}(t) - t \cdot \frac{d}{dt}(1+t^3)}{(1+t^3)^2} \right]$$

$$= 3a \left[\frac{(1+t^3) \cdot 1 - t \cdot 3t^2}{(1+t^3)^2} \right] = 3a \cdot \frac{(1-2t^3)}{(1+t^3)^2}$$

Again $y = 3a \cdot \frac{t^2}{1+t^3}$

$$\therefore \frac{dy}{dt} = 3a \cdot \left[\frac{(1+t^3) \frac{d}{dt} t^2 - t^2 \cdot \frac{d}{dt}(1+t^3)}{(1+t^3)^2} \right] = 3a \cdot \left[\frac{(1+t^3) \cdot 2t - t^2 \cdot 3t^2}{(1+t^3)^2} \right]$$

$$= 3a \cdot \frac{t(2+2t^3-3t^3)}{(1+t^3)^2} = 3a \cdot \frac{t(2-t^3)}{(1+t^3)^2}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{3at(2-t^3)}{(1+t^3)^2} \times \frac{(1+t^3)^2}{3a(1-2t^3)} = \frac{t(2-t^3)}{1-2t^3}$$

Example 4: Find $\frac{dy}{dx}$, when: $\left. \begin{array}{l} x = a(\cos t + \log \tan(t/2)) \\ y = a \sin t \end{array} \right\}$

[B.C.A. (Meerut) 2002, 2005, 2006, 2007]

Solution: We have $x = a(\cos t + \log \tan(t/2))$

$$\frac{dx}{dt} = a \left[-\sin t + \frac{1}{\tan(t/2)} \cdot \frac{d}{dt}(\tan t/2) \right]$$

$$= a \left[-\sin t + \cot(t/2) \cdot \sec^2(t/2) \cdot \frac{1}{2} \right]$$

$$= a \left[-\sin t + \frac{\cos(t/2)}{2 \sin(t/2) \cdot \cos^2(t/2)} \right]$$

$$= a \left[-\sin t + \frac{1}{2 \sin(t/2) \cdot \cos(t/2)} \right]$$

$$= a \left[-\sin t + \frac{1}{\sin t} \right] = a \left[\frac{1 - \sin^2 t}{\sin t} \right] = \frac{a \cos^2 t}{\sin t}$$

Again $y = a \sin t$

$$\therefore \frac{dy}{dt} = a \cos t$$

Hence $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{a \cos t}{\frac{a \cos^2 t}{\sin t}} = \frac{\sin t}{\cos t}$

$$\frac{dy}{dx} = \tan t$$

Example 5: If $y = x + \tan x$, show that $\cos^2 x \frac{d^2 y}{dx^2} - 2y + 2x = 0$

[B.C.A. (Meerut) 2008]

Solution: We have $y = x + \tan x$

...(1)

(Diff)ⁿ (1) w.r.t. (x), we get

$$\frac{dy}{dx} = 1 + \sec^2 x$$

Again (Diff)ⁿ w.r.t 'x', we get

$$\frac{d^2 y}{dx^2} = 2 \sec x \cdot \sec x \tan x$$

$$\frac{d^2 y}{dx^2} = \frac{2 \sin x}{\cos^3 x}$$

$$\cos^2 x \frac{d^2 y}{dx^2} = \frac{2 \sin x}{\cos x}$$

$$\cos^2 x \frac{d^2 y}{dx^2} = 2 \tan x$$

[from (1)]

$$\cos^2 x \frac{d^2 y}{dx^2} = 2 [y - x]$$

$$\Rightarrow \cos^2 x \frac{d^2 y}{dx^2} - 2y + 2x = 0$$

Example 6: If $x = a (\cos t + t \sin t)$, $y = a (\sin t - t \cos t)$ Find $\frac{d^2 y}{dx^2}$

[B.C.A. (Meerut) 2008]

Solution: We have $x = a (\cos t + t \sin t)$ and $y = a (\sin t - t \cos t)$

$$\Rightarrow \frac{dx}{dt} = a (-\sin t + \sin t + t \cos t) = at \cos t$$

$$\text{and } \frac{dy}{dt} = a (\cos t - \cos t + t \sin t) = at \sin t$$

$$\therefore \frac{dy}{dx} = \frac{(dy/dt)}{(dx/dt)} = \frac{a t \sin t}{a t \cos t} = \tan t$$

$$\Rightarrow \frac{dy}{dx} = \tan t \quad \dots(1)$$

Again (Diff)ⁿ w.r.t. 'x'

$$\frac{d^2 y}{dx^2} = \sec^2 t \frac{dt}{dx} = \sec^2 t \times \frac{1}{a t \cos t}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{\sec^3 t}{at}$$



Comprehensive Exercise 6.8

- Find $\frac{dy}{dx}$, when
 - $x = at^2$, $y = 2at$
 - $x = a \cos \theta$, $y = a \sin \theta$
 - $x = a \sec \theta$, $y = a \tan \theta$
 - $x = a \cos^3 \theta$, $y = a \sin^3 \theta$
 - $x = a(t - \sin t)$, $y = a(1 - \cos t)$
 - $x = \log t + \sin t$, $y = e^t + \cos t$
 - $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$
- If $x = \sin^{-1} \left(\frac{2t}{1+t^2} \right)$, $y = \tan^{-1} \left(\frac{2t}{1-t^2} \right)$; prove that $\frac{dy}{dx} = 1$ [B.C.A. (Agra) 2006]
 [Hint.: $t = \tan \theta$, $x = 2\theta$ and $y = 2\theta$]
- If $y = (ax + b)^{p/q}$, find $\frac{d^2y}{dx^2}$. [B.C.A. (Kanpur) 2007, 2009]
- If $y = \tan^{-1} x^3$, then prove $\frac{d^2y}{dx^2} = \frac{6x(1-2x^6)}{(1+x^6)^2}$
- If $y = \tan x + \sec x$, prove that $\frac{d^2y}{dx^2} = \frac{\cos x}{(1 - \sin x)^2}$
- If $y = \left[\log \left\{ x + \sqrt{x^2 + 1} \right\} \right]^2$; prove that $(x^2 + 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} = 2$



Answers

1.	(i) $\frac{1}{t}$	(ii) $-\cot \theta$	(iii) $\operatorname{cosec} \theta$
	(iv) $-\tan \theta$	(v) $\cot \frac{t}{2}$	(vi) $\frac{t(e^t - \sin t)}{1 + t \cos t}$
	(vii) $\tan t$		
3.	$\frac{P}{q} \left(\frac{P}{q} - 1 \right) a^2 (ax + b)^{P/q - 2}$		

6.7.2 Derivative of a Function with Respect to Another Function

let u and v be two functions of x such that

$$u = f_1(x), \quad v = f_2(x)$$

i.e. $(\text{Diff})^n$ the function w.r.t. 'x', we get

$$\frac{du}{dx} = f_1'(x), \quad \frac{dv}{dx} = f_2'(x)$$

∴ Derivative of u w.r.t. v is

$$\frac{du}{dv} = \frac{du}{dx} \times \frac{dx}{dv}$$

$$\Rightarrow \frac{du}{dv} = \frac{f_1'(x)}{f_2'(x)}$$

$$i.e. \quad \frac{du}{dv} = \frac{\text{Derivative of } u \text{ w.r.t. } x}{\text{Derivative of } v \text{ w.r.t. } x}$$

Illustrative Examples

Example 1: Differentiate $\sin^{-1} \left(\frac{1-x}{1+x} \right)$ w.r.t. $\sqrt{x}$.

[B.C.A. (Meerut) 2005]

Solution: Suppose $u = \sin^{-1} \left(\frac{1-x}{1+x} \right)$ and $v = \sqrt{x}$

Then we have $u = \sin^{-1} \left(\frac{1-v^2}{1+v^2} \right)$ We have to find $\frac{du}{dv}$

$$\begin{aligned} \text{Putting } v = \tan \theta, \text{ we get } u &= \sin^{-1} \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) \\ &= \sin^{-1} \left(\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} \right) \\ &= \sin^{-1} (\cos^2 \theta - \sin^2 \theta) \\ &= \sin^{-1} (\cos 2\theta) = \sin^{-1} (\sin (\pi/2 - 2\theta)) \end{aligned}$$

$$\Rightarrow u = \frac{\pi}{2} - 2\theta$$

$$\Rightarrow u = \frac{\pi}{2} - 2 \tan^{-1} v; \quad (\because \tan \theta = v)$$

$$\begin{aligned} \frac{du}{dv} &= 0 - 2 \cdot \frac{1}{1+v^2} \\ &= -\frac{2}{1+x}; \quad (\because v = \sqrt{x}) \end{aligned}$$

Hence, the required derivative is $-\frac{2}{1+x}$

Example 2: Find the differential coefficient of $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ w.r.t. $\tan^{-1} x$.

Solution: Let $u = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ and $v = \tan^{-1} x$

Now putting $x = \tan \theta$, we have;

$$\begin{aligned}
 u &= \tan^{-1} \left[\frac{\sqrt{1 + \tan^2 \theta} - 1}{\tan \theta} \right] \\
 &= \tan^{-1} \left[\frac{\sec \theta - 1}{\tan \theta} \right] \\
 &= \tan^{-1} \left[\left(\frac{1}{\cos \theta} - 1 \right) \cdot \frac{\cos \theta}{\sin \theta} \right] \\
 &= \tan^{-1} \left[\frac{1 - \cos \theta}{\sin \theta} \right] \\
 &= \tan^{-1} \left[\frac{1 - (1 - 2 \sin^2 \theta / 2)}{2 \sin \theta / 2 \cdot \cos \theta / 2} \right] \\
 &= \tan^{-1} (\tan \theta / 2)
 \end{aligned}$$

$$\Rightarrow u = \theta / 2$$

$$\text{Also, } v = \tan^{-1} (\tan \theta) \quad (\because x = \tan \theta)$$

$$\Rightarrow v = \theta$$

$$u = \frac{v}{2}$$

Hence, the required differential coefficient

$$\frac{du}{dv} = \frac{1}{2}$$

Example 3: Find the differential coefficient of $\tan^{-1} \left(\frac{2x}{1-x^2} \right)$ with respect to

$$\sin^{-1} \left(\frac{2x}{1+x^2} \right).$$

[B.C.A. (Meerut) 2001]

Solution: Let $u = \tan^{-1} \frac{2x}{1-x^2}$ and $v = \sin^{-1} \frac{2x}{1+x^2}$

Now putting $x = \tan \theta$, we have;

$$u = \tan^{-1} \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right) = \tan^{-1} (\tan 2\theta) = 2\theta$$

$$\Rightarrow u = 2 \tan^{-1} x \quad \frac{du}{dx} = \frac{2}{1+x^2}$$

$$\begin{aligned}
 \text{Also, } v &= \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) \\
 &= \sin^{-1} \left(\frac{2 \tan \theta}{\sec^2 \theta} \right) \\
 &= \sin^{-1} \left(\frac{2 \sin \theta \cdot \cos^2 \theta}{\cos \theta \times 1} \right) \\
 &= \sin^{-1} (2 \sin \theta \cos \theta)
 \end{aligned}$$

$$\begin{aligned}
 &= \sin^{-1} (\sin 2\theta) \\
 &= 2\theta \\
 \Rightarrow \quad &v = 2 \tan^{-1} x \\
 \therefore \quad &\frac{dv}{dx} = \frac{2}{1+x^2} \\
 \therefore \quad &\frac{du}{dv} = \frac{du}{dx} \times \frac{dx}{dv} = \frac{2/(1+x^2)}{2/(1+x^2)} = 1
 \end{aligned}$$

Example 4: Find the differential coefficient of $(\log x)^{\tan x}$ w.r.t. $\sin (m \cdot \cos^{-1} x)$.

Solution: Let $u = (\log x)^{\tan x}$

Then $\log u = \tan x \cdot \log (\log x)$

$(\text{Diff})^n$ w.r.t. 'x' on both the sides, we get

$$\begin{aligned}
 \frac{1}{u} \cdot \frac{du}{dx} &= \tan x \cdot \frac{1}{\log x} \cdot \frac{1}{x} + [\log (\log x)] \sec^2 x \\
 \Rightarrow \quad \frac{du}{dx} &= u \left[\frac{\tan x}{x \cdot \log x} + \sec^2 x \log (\log x) \right] \\
 \frac{du}{dx} &= (\log x)^{\tan x} \left[\frac{\tan x}{x \cdot \log x} + \sec^2 x \log (\log x) \right]
 \end{aligned}$$

Again, let

$$\begin{aligned}
 v &= \sin (m \cos^{-1} x) \\
 \frac{dv}{dx} &= \cos (m \cos^{-1} x) \cdot m \left(-\frac{1}{\sqrt{1-x^2}} \right) = -\frac{m}{\sqrt{1-x^2}} \cos (m \cdot \cos^{-1} x)
 \end{aligned}$$

Hence, the required differential coefficient is

$$\begin{aligned}
 \frac{du}{dv} &= \frac{du}{dx} \times \frac{dx}{dv} \\
 &= \frac{(\log x)^{\tan x} \left[\frac{\tan x}{x \cdot \log x} + \sec^2 x \cdot \log (\log x) \right]}{-\frac{m}{\sqrt{1-x^2}} \cos (m \cos^{-1} x)} \\
 &= \frac{-\sqrt{1-x^2} (\log x)^{\tan x} \left[\frac{\tan x}{x \log x} + \sec^2 x \log (\log x) \right]}{m \cos (m \cos^{-1} x)}
 \end{aligned}$$

Example 5: If $y = \sin \left[\sqrt{\sin x \sqrt{x}} \right]$, find $\frac{dy}{dx}$.

[B.C.A (Meerut) 2003]

Solution: Putting $\sqrt{x} = t$, $\sin \sqrt{x} = \sin t = u$ and $\sqrt{\sin \sqrt{x}} = \sqrt{u} = v$, we get

$$\therefore \quad \frac{dy}{dv} = \cos v, \quad \frac{dv}{du} = \frac{1}{2} u^{-1/2} = \frac{1}{2\sqrt{u}}$$

$$\frac{du}{dt} = \cos t \text{ and } \frac{dt}{dx} = \frac{1}{2\sqrt{x}}$$

So,

$$\begin{aligned} \frac{dy}{dx} &= \left(\frac{dy}{dv} \times \frac{dv}{du} \times \frac{du}{dt} \times \frac{dt}{dx} \right) \\ &= \left(\cos v \cdot \frac{1}{2\sqrt{u}} \cdot \cos t \cdot \frac{1}{2\sqrt{x}} \right) \\ &= \left(\cos \sqrt{u} \cdot \frac{1}{2\sqrt{u}} \cos t \cdot \frac{1}{2\sqrt{x}} \right) \quad (v = \sqrt{u}) \\ &= \frac{1}{4} \cos \left(\sqrt{\sin t} \right) \cdot \frac{1}{\sqrt{\sin \sqrt{x}}} \cdot \cos \sqrt{x} \cdot \frac{1}{\sqrt{x}} \quad [\because t = \sqrt{x}] \\ &= \frac{\cos \left(\sqrt{\sin \sqrt{x}} \right)}{4\sqrt{x}\sqrt{\sin \sqrt{x}}} \cos \sqrt{x} \end{aligned}$$



Comprehensive Exercise 6.9

1. Find the differential coefficient of the following functions:

(i) e^t w.r.t. $\sqrt{t}$

(ii) $\log x$ w.r.t. $\tan x$

(iii) $\log(x^2 + 2x + 1)$ w.r.t. $(x^2 + 2x)$

(iv) $e^{\sin^{-1} x}$ w.r.t. $\sin^{-1} x$

(v) $e^{\sin x}$ w.r.t. $\cos x$

(vi) $\sec^{-1} \left(\frac{1}{2x^2 - 1} \right)$ w.r.t. $\sqrt{1 - x^2}$

(vii) $\sin^{-1} (2x\sqrt{1 - x^2})$ w.r.t. $\sin^{-1} (3x - 4x^3)$

(viii) $x^{\sin x}$ w.r.t. $(\sin x)^x$

2. If $y = \sqrt{\frac{1 + e^x}{1 - e^x}}$; show that $\frac{dy}{dx} = \frac{e^x}{(1 - e^x)\sqrt{1 - e^{2x}}}$

3. If $f(x) = \sin(\log x)$ and $y = f\left[\left(\frac{2x + 3}{3 - 2x}\right)\right]$

Prove that $\frac{dy}{dx} = \left(\frac{12}{9 - 4x^2} \right) \cdot \cos \log \left(\frac{2x + 3}{3 - 2x} \right)$

4. Find the derivatives of the following functions w.r.t. 'x':

(i) $\tan^{-1} \left(\frac{a \cos x - b \sin x}{a \sin x + b \cos x} \right)$

(ii) $\tan^{-1} \left(\frac{2 + \tan x}{2 - \tan x} \right)$

5. If $\sin y = x \cos(a + y)$ then show that $\frac{dy}{dx} = \frac{\cos^2(a + y)}{\cos a}$ and find the value of $\frac{dy}{dx}$ at $x = 0$.

[B.C.A. (Meerut) 2003, 2004, 2005]

6. Find the derivative of $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$, w.r.t. $\tan^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2} \right)$ at $x=0$.
7. Find the differential coefficient of the function
 $f(x) = \log_x \sin x^2 + (\sin x^2) \log_e x$ with respect to $\sqrt{x+1}$.
8. If $y = t^2 + t^3$ and $x = t - t^4$ find $\frac{d^2 y}{dx^2}$.



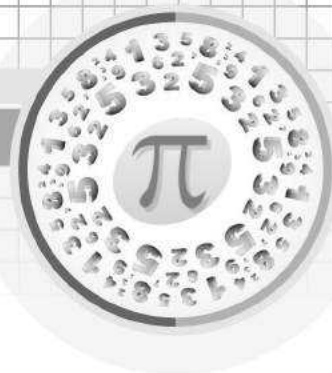
Answers

1.	(i) $2e^t \sqrt{t}$	(ii) $\frac{1}{x} \cos^2 x$
	(iii) $\frac{1}{(1+x)^2}$	(iv) $e^{\sin^{-1} x}$
	(v) $-e^{\sin x} \cdot \cot x$	(vi) $\frac{2}{x}$
	(vii) $\frac{2}{3}$	(viii) $\frac{(x)^{\sin x} \left[\frac{\sin x}{x} + \cos x \cdot \log x \right]}{(\sin x)^x [x \cot x + \log \sin x]}$
4.	(i) -1	(ii) $\frac{2 \sec^2 x}{4 + \tan^2 x}$
5.	$\cos a$	
6.	$\frac{1}{4}$	
7.	$2\sqrt{x+1} \left[\frac{1}{\log_e x} \left\{ 2x \cdot \cot x^2 - \frac{1}{x} \log_x \sin x^2 \right\} \right.$ $\left. + (\sin x^2)^{\log_e x} \left\{ \frac{1}{x} \cdot \log_e x^2 + 2x \log_e x \cot x^2 \right\} \right]$	
8.	$\frac{2(1+3t+8t^3+6t^4)}{(1-4t^3)^3}$	





Unit-3



Chapter

7

Rolles and Mean Value Theorems

7.1 Rolle's Theorem

If a function $f(x)$ such that

- (i) $f(x)$ is continuous in the closed interval $[a, b]$
- (ii) $f'(x)$ exists for every point in the open interval $]a, b[$
- (iii) $f(a) = f(b)$, then there exists at least one value of x , say c where $a < c < b$, such that
$$f'(c) = 0$$

Proof: Since $f(x)$ is continuous in closed interval $[a, b]$, therefore it is bounded and attains its bound in $[a, b]$. If

$$m = \inf\{f(x) : x \in [a, b]\}$$

$$M = \sup\{f(x) : x \in [a, b]\}$$

Then there exists two numbers c and d in the interval $[a, b]$ such that

$$f(c) = M \text{ and } f(d) = m$$

Now, there are two possibilities, either $m = M$ or $m \neq M$

- (i) If $M = m$, then $f(x) = M = m$
$$\Rightarrow f(x) \text{ is a constant function } \forall x \in [a, b]$$
$$\Rightarrow f'(x) = 0 \quad \forall x \in [a, b]$$

Hence, the theorem is proved in that case.

- (ii) If $M \neq m$, then at least one of the number M and m must be different from the equal values $f(a)$ and $f(b)$ [$\because f(a) = f(b)$] i.e., either M or m is not equal to $f(a)$ and $f(b)$. Let $M \neq f(a)$ then $M \neq f(b)$. Thus $f(c) \neq f(a)$ and $f(c) \neq f(b)$. Therefore, $c \neq a$ and $c \neq b$ or $a < c < b$.

We have $M = f(c) = \sup\{f(x): x \in [a, b]\}$

Now $f(c)$ is supremum of f on $[a, b]$, therefore

$$f(x) \leq f(c) \quad \forall x \in [a, b] \quad \dots(1)$$

or
$$f(c-h) \leq f(c) \quad \forall (c-h) \in [a, b]$$

or
$$\frac{f(c-h) - f(c)}{-h} \geq 0 \quad \dots(2)$$

or
$$\lim_{h \rightarrow 0} \frac{f(c-h) - f(c)}{-h} \geq \lim_{h \rightarrow 0} 0$$

or
$$Lf'(c) \geq 0 \quad \dots(3)$$

By the same argument as above, we get

$$Rf'(c) \leq 0 \quad \dots(4)$$

Since $f'(c)$ exists, hence $Lf'(c) = Rf'(c) = f'(c)$ $\dots(5)$

From (3), (4) and (5) we conclude that

$$f'(c) = 0$$

Similarly when $m \neq f(d) = f(a)$ or $f(b)$ we can prove that $f'(d) = 0$.

This prove the theorem.

Note:

1. There may be more than one point like c at which $f'(x)$ vanishes.
2. The converse of the Rolle's theorem is not true i.e., $f'(x)$ may vanish at a point $c \in]a, b[$ without satisfying all the three conditions of Rolle's theorem. In other words, the three conditions of Rolle's theorem are sufficient but not necessary conditions for $f'(x) = 0$ for some $x \in]a, b[$
3. Rolle's theorem will not hold good
 - (i) If $f(x)$ is discontinuous at some point in the interval $a \leq x \leq b$.
 - (ii) If $f'(x)$ does not exists at some point in the interval $a < x < b$.
 - (iii) If $f(a) \neq f(b)$.

7.1.1 Geometrical Interpretation of Rolle's Theorem

If a function is continuous on a closed and bounded interval i.e., its graph has no break and is derivable i.e., has no sharp bend, then there exists at least one point on the graph, the tangent at which is parallel to the x - axis.

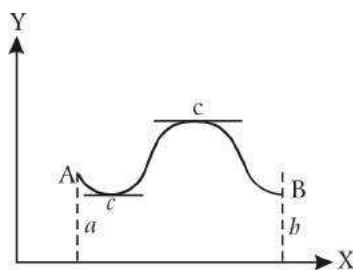


Fig. 7.1

7.1.2 Algebraic Interpretation of Rolle's Theorem

Between two zeros a and b of $f(x)$ i.e. between two roots a and b of $f(x) = 0$ there exists at least one zero of $f'(x)$.



Illustrative Examples

(Rolle's Theorem)

Example 1: Verify Rolle's theorem for

$$f(x) = x^4 - 1 \text{ in } [-1, 1]$$

Can we apply the theorem for $f(x) = |x|$ in $[-2, 2]$

[B.C.A. (Meerut) 2007]

Solution: **Ist Part:** The function $f(x) = x^4 - 1$ is a polynomial in x of degree 4 so it is continuous in the closed interval $[-1, 1]$ and is differentiable in the open interval $] -1, 1[$.

Also

$$f(1) = 1 - 1 = 0 \quad \text{and} \quad f(-1) = (-1)^4 - 1 = 1 - 1 = 0$$

So that

$$f(1) = f(-1)$$

Hence, $f(x)$ satisfying all conditions of Rolle's theorem in $[-1, 1]$, then there exists a point c such that

$$f'(c) = 0, \text{ differentiate } f(x) = x^4 - 1 \text{ w.r.t. } x \text{ we get,}$$

$$f'(x) = 4x^3 \quad \text{put } x = c$$

$$\Rightarrow f'(c) = 4c^3$$

$$\Rightarrow 0 = 4c^3$$

$$\Rightarrow c = 0$$

$$\therefore c \in] -1, 1[$$

Such that $-1 < 0 < 1$

Hence, the Rolle's theorem is verified.

IInd Part: We cannot apply Rolle's theorem for $f(x) = |x|$ in $[-2, 2]$ because $f(x) = |x|$ is not differentiable at $x = 0$ which is a point of the open interval $] -2, 2 [$. Thus $f(x)$ does not satisfy a condition of Rolle's theorem. Hence Rolle's theorem is not applicable here.

Example 2: Verify Rolle's theorem for the following:

$$f(x) = x^2 \text{ in } [-1, 1]$$

Solution: We have $f(x) = x^2$ taking $a = -1$, $b = 1$ then,

- (i) $f(x) = x^2 + 0x + 0$ is a polynomial and we know that every polynomial in x is continuous for every finite value of x , $f(x)$ is continuous in $[-1, 1]$.
- (ii) $f'(x) = 2x$ which exists for every value of x in $]-1, 1[$.
- (iii) $f(a) = f(-1) = (-1)^2 = 1$

$$\text{and} \quad f(b) = f(1) = (1)^2 = 1$$

$$\therefore \quad f(-1) = f(1)$$

Hence, $f(x)$ satisfies all the three conditions of Rolle's theorem. Therefore, there must exist at least one point $c \in]-1, 1[$ such that $f'(c) = 0$

$$\text{i.e.,} \quad 2c = 0 \Rightarrow c = 0$$

This is a point in the interval $]-1, 1[$ and hence the theorem is verified.

Example 3: Examine Rolle's theorem for the function

$$f(x) = |x| \text{ in the interval }]-1, 1[$$

Solution: We have
$$f(x) = \begin{cases} x & \text{if } 0 \leq x \leq 1 \\ -x & \text{if } -1 \leq x < 0 \end{cases}$$

The point $x = 0 \in]-1, 1[$. Let us check the differentiability of the function at $x = 0$.

$$\begin{aligned} Rf'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}, h > 0 \\ &= \lim_{h \rightarrow 0} \frac{|h| - 0}{h} = 1 \\ Lf'(0) &= \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{|-h| - 0}{-h} = \lim_{h \rightarrow 0} \left(\frac{h}{-h} \right) = -1 \end{aligned}$$

Thus, $Rf'(0) \neq Lf'(0)$ and therefore function is not differentiable at $x = 0$. This shows that function is not differentiable in $]-1, 1[$.

Hence, Rolle's theorem cannot be applied for the function $f(x) = |x|$ in the interval $]-1, 1[$.

Example 4: Verify Rolle's theorem when

$$f(x) = e^x \sin x, \quad a = 0, \quad b = \pi$$

Solution: The function $f(x) = e^x \sin x$ is continuous as well as differentiable on R . So $f(x)$ is continuous in $[0, \pi]$ and differentiable in $]0, \pi[$.

We have $f(0) = e^0 \sin 0 = 0$

and $f(\pi) = e^\pi \sin \pi = 0$

$\therefore f(0) = f(\pi)$

Thus, $f(x)$ satisfies all the three conditions of Rolle's theorem in $[0, \pi]$. Therefore there must exist at least one number say c in the open interval $]0, \pi[$ for which

$$f'(c) = 0$$

Now, $f'(x) = e^x \sin x + e^x \cos x = e^x (\sin x + \cos x)$

Put $x = c \Rightarrow f'(c) = e^c (\sin c + \cos c)$

$$\Rightarrow 0 = e^c (\sin c + \cos c)$$

But $e^c \neq 0 \Rightarrow \sin c + \cos c = 0$

$$\Rightarrow \tan c = -1$$

$$\Rightarrow c = \frac{3\pi}{4} \in]0, \pi[$$

Thus, the Rolle's theorem is verified.

Example 5: Discuss the applicability of Rolle's theorem to

$$f(x) = \log \left\{ \frac{x^2 + ab}{(a+b)x} \right\} \text{ in the interval } [a, b], 0 < a < b$$

Solution: We have $f(a) = \log \left[\frac{a^2 + ab}{(a+b)a} \right] = \log 1 = 0$

$$f(b) = \log \left[\frac{b^2 + ab}{(a+b)b} \right] = \log 1 = 0$$

Thus, $f(a) = f(b) = 0$

Also $Rf'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[\log \left\{ \frac{(x+h)^2 + ab}{(a+b)(x+h)} \right\} - \log \left\{ \frac{x^2 + ab}{(a+b)x} \right\} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[\log \frac{(x^2 + 2hx + h^2 + ab)(a+b)x}{(a+b)(x+h)(x^2 + ab)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[\log \frac{(x^2 + 2hx + h^2 + ab)}{x^2 + ab} \times \frac{x}{x+h} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[\log \left\{ 1 + \frac{2hx + h^2}{x^2 + ab} \right\} - \log \left\{ 1 + \frac{h}{x} \right\} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{2hx + h^2}{x^2 + ab} - \frac{h}{x} + \dots \right]$$

$$= \frac{2x}{x^2 + ab} - \frac{1}{x} \quad \left[\because \log(1+x) = x - \frac{x^2}{2} + \dots \right]$$

Again

$$\begin{aligned} Lf'(x) &= \lim_{h \rightarrow 0} \left\{ \frac{f(x-h) - f(x)}{-h} \right\} \\ &= \lim_{h \rightarrow 0} \frac{1}{(-h)} \left\{ \frac{-2hx + h^2}{x^2 + ab} - \frac{(-h)}{x} \right\} \\ &= \frac{2x}{x^2 + ab} - \frac{1}{x} \end{aligned}$$

Since $Rf'(x) = Lf'(x)$, $f(x)$ is differentiable for all values of x in $[a, b]$. We know continuity is the necessary condition for differentiability, so $f(x)$ is continuous for all value of x . Thus all the conditions of Rolle's theorem satisfying then there exist a point c such that $f'(c) = 0$

i.e.,
$$f'(x) = \frac{2x}{x^2 + ab} - \frac{1}{x}$$

Put $x = c$

$$\Rightarrow f'(c) = 0$$

$$\Rightarrow \frac{2c}{c^2 + ab} - \frac{1}{c} = 0$$

$$\Rightarrow 2c^2 - (c^2 + ab) = 0$$

$$\Rightarrow c = \sqrt{ab}$$

This is the geometrical mean of a and b and lies between a and b i.e., $c \in]a, b[$. Hence Rolle's theorem is verified.

Example 6: Verify Rolle's theorem for

$$f(x) = x(x+3)e^{-x/2} \text{ in } [-3, 0]$$

Solution: (i) We have $f(x) = (x^2 + 3x)e^{-x/2}$. The function $(x^2 + 3x)$ and $e^{-x/2}$ are polynomial and exponential functions, therefore continuous. Also their product is continuous i.e., $f(x) = x(x+3)e^{-x/2}$ is continuous in $[-3, 0]$.

$$(ii) \quad f(-3) = f(0) = 0$$

$$\begin{aligned} (iii) \quad f'(x) &= (2x+3)e^{-x/2} + (x^2+3x)e^{-x/2} \left(-\frac{1}{2} \right) = e^{-x/2} \left[2x+3 - \frac{x^2}{2} - \frac{3}{2}x \right] \\ &= -\frac{1}{2}e^{-x/2} [x^2 - x - 6] \end{aligned}$$

which exists in the interval $]-3, 0[$

Thus, the function satisfies all the three conditions of Rolle's theorem, therefore there exists a point c .

Such that $f'(c) = 0$

$$\Rightarrow f'(c) = -\frac{1}{2}e^{-c/2}(c^2 - c - 6) = 0$$

$$\Rightarrow c^2 - c - 6 = 0$$

$$\Rightarrow c = 3, -2$$

But, $c = 3 \notin [-3, 0]$ while $c = -2 \in]-3, 0[$

Hence, Rolle's theorem is verified.

Example 7: Verify Rolle's theorem for the function

[B.C.A. (Meerut) 2011]

$$f(x) = x^3 - 6x^2 + 11x - 6$$

Solution: We have $f(x) = x^3 - 6x^2 + 11x - 6$

Put $f(x) = 0$, we get

$$x^3 - 6x^2 + 11x - 6 = 0$$

$$\Rightarrow x^2(x-1) - 5x(x-1) + 6(x-1) = 0$$

$$\Rightarrow (x-1)(x^2 - 5x + 6) = 0$$

$$\Rightarrow (x-1)(x-2)(x-3) = 0$$

$$\Rightarrow x = 1, 2, 3$$

Let us consider the closed interval $[1, 3]$

$$(i) \quad f(1) = f(3) = 0$$

(ii) $f(x)$ is a polynomial in x , hence it is continuous in closed interval $[1, 3]$

(iii) $f'(x) = 3x^2 - 12x + 11$ which exists in an open interval $]1, 3[$,

All the conditions of Rolle's theorem are satisfying in $[1, 3]$, therefore there exists a point c such that

$$f'(c) = 0$$

$$\Rightarrow f'(c) = 3c^2 - 12c + 11 = 0$$

$$\Rightarrow c = \frac{12 \pm \sqrt{144 - 132}}{6} = 2 \pm \frac{1}{\sqrt{3}} = 2 \pm \frac{1}{1.732}$$

$$\Rightarrow c = 2 \pm 0.58 = 2.58, 1.42$$

Both these points lie in the interval $]1, 3[$ and therefore, the theorem is verified.

Example 8: Verify Rolle's theorem for the following functions.

(i) $f(x) = (x-4)^5(x-3)^4$ in the interval $[3, 4]$

(ii) $f(x) = e^x(\sin x - \cos x)$ in $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$

Solution: (i) We have $f(x) = (x-4)^5(x-3)^4$ which is a polynomial in x of degree 9 and a polynomial function is continuous as well as differentiable on $\mathbb{R}$. So $f(x)$ is continuous in $[3, 4]$ and differentiable in $]3, 4[$.

Also $f(3) = 0$ and $f(4) = 0$ so that $f(3) = f(4)$

Thus all the three conditions of Rolle's theorem are satisfied so that there is at least one value of x in the open interval $]3, 4[$ where $f'(x) = 0$

$$\begin{aligned}
 \text{Now,} \quad f'(x) &= 5(x-4)^4(x-3)^4 + 4(x-3)^3(x-4)^5 \\
 &= (x-3)^3(x-4)^4[5(x-3) + 4(x-4)] \\
 &= (x-3)^3(x-4)^4[9x-31]
 \end{aligned}$$

$$\text{Put} \quad f'(x) = 0 \Rightarrow (x-3)^4(x-4)^4(9x-31) = 0$$

$$\Rightarrow \quad x = 3, 4, \frac{31}{9}$$

out of these values the value $\frac{31}{9} = 3\frac{4}{9}$ is a point which lies in the open interval $]3, 4[$. Since $3 < 3\frac{4}{9} < 4$. Hence the Rolle's theorem is verified.

(ii) Here $f(x) = e^x (\sin x - \cos x)$

$$\text{we have} \quad f\left(\frac{\pi}{4}\right) = e^{\pi/4} \left(\sin \frac{\pi}{4} - \cos \frac{\pi}{4}\right) = 0$$

$$\begin{aligned}
 f\left(\frac{5\pi}{4}\right) &= e^{5\pi/4} \left(\sin \frac{5\pi}{4} - \cos \frac{5\pi}{4}\right) \\
 &= e^{5\pi/4} \left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) = 0
 \end{aligned}$$

$$\therefore \quad f\left(\frac{\pi}{4}\right) = f\left(\frac{5\pi}{4}\right)$$

The function $f(x)$ is continuous as well as differentiable for all real values of x and so $f(x)$ is continuous in $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$ and differentiable $\left]\frac{\pi}{4}, \frac{5\pi}{4}\right[$. Thus $f(x)$ satisfies all the three conditions of Rolle's theorem in $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$.

$\therefore$ There must exist at least one real number x in the open interval $\left]\frac{\pi}{4}, \frac{5\pi}{4}\right[$

at which $f'(x) = 0$.

$$\begin{aligned}
 \text{Now,} \quad f'(x) &= e^x(\cos x + \sin x) + e^x(\sin x - \cos x) \\
 &= 2e^x \sin x
 \end{aligned}$$

$$\text{From} \quad f'(x) = 0 \Rightarrow 2e^x \sin x = 0 \quad \text{or} \quad \sin x = 0 \quad [\because e^x \neq 0 \quad \forall x \in \mathbb{R}]$$

$$\text{or} \quad \sin x = 0 \quad \text{or} \quad x = 0, \pm \pi, \pm 2\pi, \pm 3\pi, \dots$$

Out of these values $x = \pi$ lies in the open interval $\left]\frac{\pi}{4}, \frac{5\pi}{4}\right[$.

Thus the Rolle's theorem is verified.

Example 9: Verify Rolle's theorem for

$$f(x) = (x-a)^m(x-b)^n$$

Solution: We have $f(x) = (x-a)^m(x-b)^n$

As m and n are positive integers, $(x-a)^m$ and $(x-b)^n$ are polynomials in x on being expanded by binomial theorem. Hence $f(x)$ is also a polynomial in x . Consequently $f(x)$ is continuous and differentiable in the closed interval $[a, b]$. Also

$$f(a) = f(b) = 0$$

Thus all the three conditions of Rolle's theorem are satisfied so that there is at least one value of x in the open interval $]a, b[$ where $f'(x) = 0$.

Now
$$f'(x) = (x-a)^m \cdot n(x-a)^{n-1} + m(x-a)^{m-1}(x-b)^n$$

Solving the equation $f'(x) = 0$, we get $x = a, b, (na + mb)/(m + n)$

Out of these values the value $(na + mb)/(m + n)$ is a point which lies in the open interval $]a, b[$, since it divides the interval $]a, b[$ internally in the ratio $m:n$. Hence the Rolle's theorem is verified.

Example 10: Show that between any two roots of $e^x \cos x = 1$ there exists at least one root of $e^x \sin x - 1 = 0$.

Solution: If $x = a$ and $x = b$ are two distinct roots of $e^x \cos x = 1$ then

$$e^a \cos a = 1 \text{ and } e^b \cos b = 1 \quad \dots(1)$$

Let f be the function defined as follows:

$$f(x) = e^{-x} - \cos x$$

We observe that

(i) f is continuous in $[a, b]$ as both e^{-x} and $\cos x$ are continuous.

(ii) $f'(x) = -e^{-x} + \sin x$, which exists $\forall x \in]a, b[$; so f is differentiable in $]a, b[$.

$$(iii) \quad f(a) = e^{-a} - \cos a = (e^{-a} - e^{-a}) = 0 \quad [\text{By (1)}]$$

$$f(b) = e^{-b} - \cos b = (e^{-b} - e^{-b}) = 0 \quad [\text{By (1)}]$$

i.e.,
$$f(a) = f(b) = 0.$$

Thus, f satisfies all the three conditions of Rolle's theorem in $[a, b]$. Hence there is at least one value of x in the open interval $]a, b[$, say c , such that $f'(c) = 0$.

Now
$$f'(c) = 0 \Rightarrow -e^{-c} + \sin c = 0 \Rightarrow e^c \sin c - 1 = 0$$

$$\Rightarrow c \text{ is a root of the equation } e^x \sin x - 1 = 0.$$

Hence, between any two roots of the equation $e^x \cos x = 1$ there is at least one root of the equation $e^x \sin x - 1 = 0$.

Example 11: The function f is defined in $[0,1]$ as follows:

$$f(x) = 1 \text{ for } 0 \leq x < \frac{1}{2}$$

$$f(x) = 2 \text{ for } \frac{1}{2} \leq x \leq 1$$

Show that $f(x)$ satisfies none of the conditions of Rolle's theorem, yet $f'(x) = 0$ for many point in $[0,1]$.

Solution: Here $f\left(\frac{1}{2}+0\right) = 2, f\left(\frac{1}{2}-0\right) = 1$.

Since $f\left(\frac{1}{2}+0\right) \neq f\left(\frac{1}{2}-0\right)$, f is discontinuous at $x = \frac{1}{2}$ and so it is not differentiable at $x = \frac{1}{2}$. Also $f(0) = 1, f(1) = 2$ so that $f(0) \neq f(1)$.

Thus all the three conditions of Rolle's theorem are not satisfied by f in $[0,1]$. But f is a constant function in $\left[0, \frac{1}{2}\right]$ and in $\left[\frac{1}{2}, 1\right]$.

Hence $f'(x) = 0$ for many points in $[0,1]$.

Example 12: Let $\frac{a_0}{n+1} + \frac{a_1}{n} + \frac{a_2}{n-1} + \dots + \frac{a_{n-1}}{2} + a_n = 0$. Show that there exists at least one real x between 0 and 1 such that

$$a_0 x^n + a_1 x^{n-1} + \dots + a_n = 0$$

Solution: Consider the function, f , defined by

$$f(x) = a_0 \frac{x^{n+1}}{n+1} + a_1 \frac{x^n}{n} + \dots + a_{n-1} \frac{x^2}{2} + a_n x$$

Since $f(x)$ is a polynomial, it is continuous and differentiable for all x . Consequently $f(x)$ is continuous in the closed interval $[0,1]$ and differentiable in the open interval $]0,1[$.

Also $f(0) = 0$ and $f(1) = \frac{a_0}{n+1} + \frac{a_1}{n} + \dots + \frac{a_{n-1}}{2} + a_n = 0$ (given)

i.e., $f(0) = f(1)$

Thus all the three conditions of Rolle's theorem are satisfied. Hence there is at least one value of x in the open interval $]0,1[$ where $f'(x) = 0$

i.e., $a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n = 0$.

Example 13: Show that there is no real number k for which the equation $x^3 - 3x + k = 0$, has two distinct roots in $]0,1[$.

Solution: Suppose, if possible, there are two distinct roots a, b of the given equation in $]0,1[$ such that $0 < a < b < 1$. Let

$$f(x) = x^3 - 3x + k.$$

Since $f(x)$ is polynomial, so it is continuous and differentiable for all values of x i.e., $f(x)$ is continuous in $[a, b]$ and differentiable in $]a, b[$. Also we have $f(a) = f(b) = 0$.

Thus f satisfies all the three conditions of Rolle's theorem in $[a, b]$. Hence there is a value c of x in $]a, b[$ such that $f'(c) = 0$.

Now $f'(x) = 0 \Rightarrow 3x^2 - 3 = 0 \Rightarrow x = \pm 1$, which contradicts the fact that $a < c < b$, as $0 < a < b < 1$.

Hence our assumption is wrong. So there cannot be two distinct roots of $f(x) = 0$ in $]0, 1[$ for any value of k .

Example 14: If the functions f, g, h are continuous on $[a, b]$ and twice differentiable on $]a, b[$, prove that there exist $\xi, \eta \in]a, b[$ such that

$$\begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix} = \frac{1}{2}(b-c)(c-a)(a-b) \begin{vmatrix} f(a) & f'(\xi) & f''(\eta) \\ g(a) & g'(\xi) & g''(\eta) \\ h(a) & h'(\xi) & h''(\eta) \end{vmatrix}$$

where $a < c < b$.

Solution: Consider the function F defined by

$$F(x) = \begin{vmatrix} f(a) & f(b) & f(x) \\ g(a) & g(b) & g(x) \\ h(a) & h(b) & h(x) \end{vmatrix} - \frac{(x-a)(x-b)}{(c-a)(c-b)} \begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix}$$

We have $F(a) = F(b) = F(c) = 0$. The function F satisfies the conditions of Rolle's theorem on $[a, c]$ and $[c, b]$. Hence, we get

$$F'(x_1) = 0 = F'(x_2) \text{ where } a < x_1 < c \text{ and } c < x_2 < b.$$

Again, applying Rolle's theorem for $F'(x)$ on $[x_1, x_2]$, we get

$$F''(\eta) = 0, \text{ where } x_1 < \eta < x_2$$

Now

$$F''(\eta) = 0$$

$$\Rightarrow \begin{vmatrix} f(a) & f(b) & f''(\eta) \\ g(a) & g(b) & g''(\eta) \\ h(a) & h(b) & h''(\eta) \end{vmatrix} = \frac{2}{(c-a)(c-b)} \begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix} \quad \dots(i)$$

Again, let ϕ be the function defined by

$$\phi(x) = \begin{vmatrix} f(a) & f(x) & f''(\eta) \\ g(a) & g(x) & g''(\eta) \\ h(a) & h(x) & h''(\eta) \end{vmatrix} - \frac{2(x-a)}{(c-a)(c-b)(b-a)} \begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix}$$

Then $\phi(a) = 0$ and using (1), we have $\phi(b) = 0$.

Therefore applying Rolle's theorem for ϕ on $[a, b]$, we have $\phi'(\xi) = 0$ where $a < \xi < b$

$$\text{i.e.,} \quad \begin{vmatrix} f(a) & f'(\xi) & f''(\eta) \\ g(a) & g'(\xi) & g''(\eta) \\ h(a) & h'(\xi) & h''(\eta) \end{vmatrix} = \frac{2}{(c-a)(c-b)(b-a)} \begin{vmatrix} f(a) & f(b) & f(c) \\ g(a) & g(b) & g(c) \\ h(a) & h(b) & h(c) \end{vmatrix}$$



Comprehensive Exercise 7.1

(Rolle's Theorem)

1. Show that Rolle's theorem is not applicable for the function

$$f(x) = \begin{cases} x^2 + 1 & \text{if } 0 \leq x \leq 1 \\ 3 - x & \text{if } 1 < x \leq 2 \end{cases}$$

2. If $f(x) = \begin{vmatrix} \sin x & \sin \alpha & \sin \beta \\ \cos x & \cos \alpha & \cos \beta \\ \tan x & \tan \alpha & \tan \beta \end{vmatrix}$ Where, $0 < \alpha < \beta < \frac{\pi}{2}$

Show that $f'(\xi) = 0$ where $\alpha < \xi < \beta$

3. Verify Rolle's theorem for

$$f(x) = 3x^4 - 4x^2 + 5 \text{ in } [-1, 1]$$

4. Discuss the application of Rolles theorem for the functions

(i) $f(x) = \tan x$ in the interval $[0, \pi]$

(ii) $f(x) = \cos\left(\frac{1}{x}\right)$ in the interval $[-1, 1]$

(iii) $f(x) = \sec x$ in the interval $[-\pi, \pi]$

(iv) $f(x) = \sin x$ in the interval $[-\pi, \pi]$

5. Discuss the Rolle's theorem for the functions

(i) $f(x) = x^2 - 3x + 2$ in the interval $[1, 2]$

(ii) $f(x) = |x - 2|$ in the interval $[1, 3]$



Answers

4. (i) Rolle's theorems is not applicable.
 (ii) Rolle's theorem is not applicable.
 (iii) Rolle's theorem is applicable and verified also
 (iv) Rolle's theorem is applicable.
5. (i) Rolle's theorem is applicable.
 (ii) Rolle's Theorem is not applicable.

7.2 Lagrange's Mean Value Theorem

Theorem 1: If a function $f(x)$ is

- (i) continuous in a closed interval $[a, b]$
 - (ii) differentiable in the open interval $]a, b[$ i.e., $a < x < b$,
- then there exists at least one value c of x lying in the open interval $]a, b[$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c) \quad [\text{B.C.A. (Meerut) 2011}]$$

Proof: Consider the function

$$\phi(x) = f(x) + Ax \quad \dots(1)$$

where A is a constant to be so chosen that

$$\phi(a) = \phi(b)$$

$$\Rightarrow f(a) + Aa = f(b) + Ab$$

$$\Rightarrow A = -\frac{f(b) - f(a)}{(b - a)} \quad \dots(2)$$

- (i) The function $f(x)$ is continuous in $[a, b]$ and Ax is a polynomial, so continuous in $[a, b]$, we know that sum of two continuous function is continuous, therefore.

$$f(x) + Ax \text{ is continuous}$$

- (ii) The function $f(x)$ is differentiable in $]a, b[$ and Ax is polynomial, so differentiable then $\phi(x) = f(x) + Ax$ is differentiable in $]a, b[$

$$\phi(a) = \phi(b)$$

- (iii) Thus, $\phi(x)$ satisfies all the three conditions of Rolle's theorem. Therefore, there exists a point $c \in]a, b[$ such that

$$\phi'(c) = 0$$

But $\phi'(x) = f'(x) + A$

Put $x = c \Rightarrow \phi'(c) = f'(c) + A$

$$\Rightarrow 0 = f'(c) + A$$

$$\Rightarrow -A = f'(c)$$

[From (2)]

$$\Rightarrow \frac{f(b) - f(a)}{(b - a)} = f'(c)$$

which proves the theorem.

7.2.1 Another form of Lagrange's Mean Value Theorem

If a function $f(x)$ is such that it is

- (i) Continuous on the closed interval $[a, a + h]$
- (ii) Differentiable on the open interval $]a, a + h[$

Then there exist at least a number θ such that

$$\frac{f(a+h) - f(a)}{h} = f'(a + \theta h) \quad \text{where } 0 < \theta < 1$$

Proof: Let $a + h = b$, then by above theorem, we have

$$\frac{f(a+h) - f(a)}{h} = f'(c) \quad \dots(1)$$

where $h = b - a$. Since the number c lies between a and $a + h$ must be greater than a by a function h . Therefore it may be written as $c = a + \theta h$, where θ is some positive fraction lying between 0 and 1 i.e., $0 < \theta < 1$

Hence from (1), we have

$$\frac{f(a+h) - f(a)}{h} = f'(a + \theta h)$$

7.2.2 Geometrical Interpretation of the Mean Value Theorem

Let $y = f(x)$ and let ACB be the given graph of $y = f(x)$ in $[a, b]$. The coordinates of the point A are $(a, f(a))$ and those of B are $(b, f(b))$. If the chord AB makes an angle α with the x -axis, then

$$\tan \alpha = \frac{f(b) - f(a)}{b - a} = f'(c)$$

by Lagrange's is mean value theorem where $a < c < b$. Thus, Lagrange's mean value theorem says that there is some point c in $]a, b[$ such that the tangent to the curve at this point is parallel to the chord joining the points on the graph with abscissae a and b .

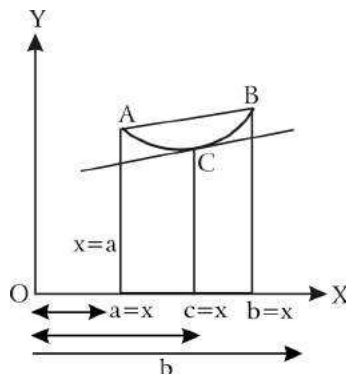


Fig. 7.2

7.2.3 Some Important Deductions from the Mean Value Theorem

Theorem 2: If a function f is continuous on $[a, b]$, differentiable on $]a, b[$ and if $f'(x) = 0$ for all x in $[a, b]$, then $f(x)$ has constant value throughout $[a, b]$.

Proof: Let c be any point of $[a, b]$. Then the function $f(x)$ is continuous $[a, c]$ and differentiable on $]a, c[$. Thus $f(x)$ satisfies all the conditions of Lagrange's mean theorem on $[a, c]$. Consequently there exists a real number d between a and c i.e., $a < d < c$ such that

$$f(c) - f(a) = (c - a)f'(d)$$

But by hypothesis $f'(x) = 0$ throughout the interval $]a, b[$, therefore, in particular $f'(d) = 0$ and hence

$$f(c) - f(a) = 0 \quad \text{or} \quad f(c) = f(a)$$

Since c is any point of $]a, b[$, therefore it gives that

$$f(x) = f(a) \quad \forall \quad x \in]a, b[$$

Thus $f(x)$ has a constant value throughout $[a, b]$.

Theorem 3: Let $f(x)$ be a real valued function, continuous on $[a, b]$ and differentiable in $]a, b[$, then

- (i) $f(x)$ is constant if $f'(x) = 0$ on $]a, b[$
- (ii) $f(x)$ is strictly increasing on $[a, b]$ if $f'(x) > 0$ on $]a, b[$
- (iii) $f(x)$ is strictly decreasing on $]a, b[$ if $f'(x) < 0$ on $]a, b[$

[B.C.A (Rohilkhand) 2003, 2007]

Proof: Let x_1, x_2 where $(x_2 > x_1)$ be any two distinct points on $[a, b]$ then by Lagrange's mean value theorem.

$$\frac{f(x_2) - f(x_1)}{(x_2 - x_1)} = f'(c), \text{ where } x_1 < c < x_2 \quad \dots(1)$$

- (i) Since $f'(x) = 0$ in $]a, b[$, therefore $f'(c) = 0$ then from (1), we have

$$\frac{f(x_2) - f(x_1)}{(x_2 - x_1)} = 0$$

$$\Rightarrow f(x_2) - f(x_1) = 0$$

$$\Rightarrow f(x_2) = f(x_1)$$

as $x_2 - x_1 > 0$. Since x_1 and x_2 are any two arbitrary point of $[a, b]$ such that $f(x_1) = f(x_2)$. Thus $f(x)$ has a constant value throughout $[a, b]$. Hence, $f(x)$ is a constant function.

- (ii) Since $f'(x) > 0$ in $]a, b[$. Therefore, $f'(c) > 0$, then from (1), we have

$$\frac{f(x_2) - f(x_1)}{(x_2 - x_1)} > 0$$

$$\Rightarrow f(x_2) - f(x_1) > 0$$

$$\Rightarrow f(x_2) > f(x_1)$$

Thus, $f(x_2) > f(x_1)$, where $x_2 > x_1$ and $x_1, x_2 \in]a, b[$

Therefore, f is strictly increasing on $[a, b]$

- (iii) Since $f'(x) < 0$ in $x \in]a, b[$, there fore $f'(c) < 0$ and hence from (1) we have

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} < 0$$

$$\Rightarrow f(x_2) - f(x_1) < 0$$

$$\Rightarrow f(x_2) < f(x_1)$$

Thus, $f(x_2) < f(x_1)$, when $x_2 > x_1$ and $x_1, x_2 \in]a, b[$. Therefore, f is strictly decreasing on $[a, b]$.

Note:

1. If $f(x)$ is continuous on $[a, b]$ then $f(x)$ strictly increasing or strictly decreasing on $[a, b]$, as $f'(x) > 0$ or < 0 in $]a, b[$.
2. For a strictly increasing function $f(x)$, the derivative $f'(x)$ need not be strictly positive
For example, consider $f(x) = x^3, x \in [-5, 5]$, Here $f(x)$ is strictly increasing but $f'(x) = 3x^2$, which is zero at $x = 0 \in]-1, 1[$.

Illustrative Examples

(Lagrange's Theorem)

Example 15: If $f(x) = (x-1)(x-3)(x-5)$, $a=0$, $b=4$. find the value of c such that $f'(c)$ has the same value as the slope of the chord joining the points for which $x=0$ and $x=4$.

[B.C.A. (Meerut) 2006]

Solution: We know that a polynomial function is continuous as well as differentiable for all x . Since

$$f(x) = (x-1)(x-3)(x-5) = x^3 - 9x^2 + 23x - 15$$

is a polynomial in x , therefore, it is continuous in $[0,4]$ and differentiable in $]0,4[$.

Hence, $f(x)$ satisfies all the conditions of Lagrange's mean value theorem, therefore, there exists some point $c \in]0,4[$ such that

$$\begin{aligned} \frac{f(4) - f(0)}{4 - 0} &= f'(c) \\ \Rightarrow \frac{(-3) - (-15)}{4 - 0} &= f'(c) \\ \Rightarrow 3 &= f'(c) \end{aligned} \quad \dots(1)$$

Since $f(x) = x^3 - 9x^2 + 23x - 15$

$$\begin{aligned} \Rightarrow f'(x) &= 3x^2 - 18x + 23 \\ \Rightarrow f'(c) &= 3c^2 - 18c + 23 \\ \Rightarrow 3 &= 3c^2 - 18c + 23 \\ \Rightarrow 0 &= (3c - 10)(c - 2) \\ \Rightarrow c &= 2, 10/3 \in]0,4[\end{aligned}$$

Such that the slope to the curve at these points is same as the slope of chord joining the points for which $x=0$ and $x=4$.

Example 16: If $f(x) = (x-1)(x-2)(x-3)$ and $a=0, b=4$, find c using Lagrange's mean value theorem.

[B.C.A. (Meerut) 2005; B.C.A. (Rohilkhand) 2004, 2007]

Solution: We have

$$f(x) = (x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6$$

$$\therefore f(a) = f(0) = -6 \text{ and } f(b) = f(4) = 6$$

$$\therefore \frac{f(b) - f(a)}{(b-a)} = \frac{6 - (-6)}{4 - 0} = \frac{12}{4} = 3$$

Also

$$f'(x) = 3x^2 - 12x + 11 \Rightarrow f'(c) = 3c^2 - 12c + 11$$

Putting these value in Lagrange's mean value theorem

$$\frac{f(b) - f(a)}{(b-a)} = f'(c), \quad (a < c < b), \text{ we get}$$

$$3 = 3c^2 - 12c + 11 \text{ or } 3c^2 - 12c + 8 = 0$$

$$\text{or } c = \frac{12 \pm \sqrt{144 - 96}}{6} = 2 \pm \frac{2\sqrt{3}}{3}$$

As both of these value of c lie in the open interval $]0,4[$, hence both of these are the required value of c .

Example 17: Discuss the applicability of Lagrange's mean value theorem to $f(x) = x^3$ in $[-1,1]$ [B.C.A. (Meerut) 2006]

Solution: The function $f(x) = x^3$ is a polynomial and so it is continuous and differentiable at all $x \in R$. In particular it is continuous in the closed interval $[-1,1]$ and differentiable in the open interval $] -1,1[$

Since, $f(x)$ satisfies all the conditions of Lagrange's mean value theorem, therefore, there must exists, some $c \in] -1,1[$ such that

$$\frac{f(1) - f(-1)}{1 - (-1)} = f'(c)$$

$$\text{or } \frac{1 - (-1)}{2} = 3c^2$$

$$[f'(x) = 3x^2]$$

$$\text{or } c = \pm \frac{1}{\sqrt{3}} = \pm \frac{1.73}{3} = \pm 0.57$$

As both of these value of c lie in the open interval $] -1,1[$, hence both of these are the required value of c and this verifies Lagrange's mean value theorem.

Example 18: Find c of the mean value theorem, if

$$f(x) = x(x-1)(x-2), \quad a=0, b=1/2 \quad [\text{B.C.A. (Meerut) 2003, 2004}]$$

Solution: We have $f(a) = f(0) = 0$

$$f(b) = f\left(\frac{1}{2}\right) = \frac{1}{2}\left(\frac{1}{2} - 1\right)\left(\frac{1}{2} - 2\right) = \frac{3}{8}$$

$$\therefore \frac{f(b) - f(a)}{b - a} = \frac{\frac{3}{8} - 0}{1/2 - 0} = \frac{3}{4}$$

$$\text{Now } f(x) = x^3 - 3x^2 + 2x$$

$$\Rightarrow f'(x) = 3x^2 - 6x + 2$$

$$\Rightarrow f'(c) = 3c^2 - 6c + 2$$

Put these value in Lagrange's mean value theorem

$$\frac{f(b) - f(a)}{(b - a)} = f'(c), \quad (a < c < b), \text{ we get}$$

$$\frac{3}{4} = 3c^2 - 6c + 2 \quad \text{or } 12c^2 - 24c + 5 = 0$$

$$c = \frac{24 \pm \sqrt{24 \times 24 - 4 \times 12 \times 5}}{24} = \frac{24 \pm 4\sqrt{36 - 15}}{24}$$

$$= 1 \pm \frac{\sqrt{21}}{6}$$

Out of these values of c only $1 - \frac{\sqrt{21}}{6}$ lie in the open interval $]0, \frac{1}{2}[$ which is therefore the required value of c .

Example 19: Find c of mean value theorem when

$$f(x) = x^3 - 3x - 2 \text{ in } [-2, 3] \quad [\text{B.C.A. (Meerut) 2007; (Rothak) 2011}]$$

Solution: We have

$$f(x) = x^3 - 3x - 2, a = -2, b = 3$$

$$\therefore f(a) = f(-2) = (-2)^3 - 3(-2) - 2 = -8 + 6 - 2 = -4$$

$$f(b) = f(3) = (3)^3 - 3 \cdot 3 - 2 = 16$$

$$\therefore \frac{f(b) - f(a)}{(b - a)} = \frac{16 - (-4)}{3 - (-2)} = \frac{20}{5} = 4$$

$$\text{Also } f'(x) = 3x^2 - 3 \text{ gives } f'(c) = 3c^2 - 3$$

Put these values in Lagrange's mean value theorem

$$\frac{f(b) - f(a)}{(b - a)} = f'(c), (a < c < b)$$

$$4 = 3c^2 - 3 \text{ or } c = \pm\sqrt{7/3}$$

As both these values of c lie in the open interval $] -2, 3[$, hence both of these are the required values of c .

Example 20: Verify L.M.V theorem for $f(x) = 2x^2 - 7x + 10$ in $[2, 5]$.

Solution: Here $f(x)$ is a polynomial and hence it is continuous and differentiable in $[2, 5]$. Also $a = 2, b = 5$.

$$\therefore f(a) = f(2) = 2(2)^2 - 7(2) + 10 = 4$$

$$\text{and } f(b) = f(5) = 2(5)^2 - 7(5) + 10 = 25$$

Also $f'(x) = 4x - 7$ then by L.M.V theorem, we find

$$f'(c) = \frac{f(b) - f(a)}{(b - a)}$$

$$\Rightarrow 4c - 7 = \frac{25 - 4}{5 - 2} \Rightarrow 4c - 7 = 7$$

$$\Rightarrow 4c = 14 \text{ or } c = 7/2 = 3.5 \in]3, 5[$$

$\Rightarrow$ Lagrange's mean value's theorem is verified.

Example 21: Find c of the mean value theorem when

$$f(x) = \cosh x \text{ in } [-1, 3]$$

[B.C.A. (Rohilkhand) 2008; (Delhi) 2011]

Solution: Here $f(x) = \cosh x = \frac{1}{2} (e^x + e^{-x})$

$$\Rightarrow f'(x) = \frac{e^x - e^{-x}}{2} = \sinh x$$

$$f(a) = f(-1) = \frac{e^{-1} + e^1}{2} = \cosh(1)$$

$$f(b) = f(3) = \left(\frac{e^3 + e^{-3}}{2} \right) = \cosh 3$$

$$\therefore \frac{f(b) - f(a)}{(b-a)} = f'(c)$$

$$\Rightarrow \frac{\cosh 3 - \cosh 1}{3-1} = \sinh c$$

$$\Rightarrow \sinh c = \frac{10.068 - 1.5431}{4}$$

$$\Rightarrow \sinh c = 2.1312$$

$$\Rightarrow c = 1.5 \text{ (Approximately)}$$

Example 22: If $f(x) = \log x$, find all numbers strictly between e^2 and e^3 such that

$$f'(x) = \frac{f(e^3) - f(e^2)}{(e^3 - e^2)}$$

[B.C.A. (Meerut) 2008; (Agra) 2006]

Solution: We have $f(x) = \log x$

$$\Rightarrow f'(x) = \frac{1}{x}$$

$$\Rightarrow f(e^3) = \log e^3 = 3 \log e = 3$$

$$\text{And } f(e^2) = \log e^2 = 2 \log e = 2$$

Put these values in

$$f'(x) = \frac{f(e^3) - f(e^2)}{(e^3 - e^2)}$$

$$\Rightarrow \frac{1}{x} = \frac{3-2}{e^3 - e^2}$$

$$\Rightarrow \frac{1}{x} = \frac{1}{e^3 - e^2}$$

$$\Rightarrow x = e^3 - e^2 \quad \forall x$$

This is required value of x such that

$$e^2 < x < e^3$$

Example 23: Test if Lagrange's mean value theorem holds for the function $f(x) = |x|$ in the interval $[-1, 1]$.

Solution: The function $f(x)$ is continuous throughout the closed interval $[-1, 1]$ but it is not differentiable in $] -1, 1[$. Hence Lagrange's mean value theorem does not hold for the function $f(x) = |x|$ in the interval $[-1, 1]$.

Example 24: Verify the mean value theorem for

$$f(x) = \log_e x \text{ in } [1, e.]$$

[B.C.A. (Meerut) 2002, 2003]

Solution: We know that every logarithmic function is continuous and differentiable in its domain, therefore $f(x) = \log x$ is continuous in closed interval $[1, e.]$ and differentiable in the open interval $]1, e.[$

Therefore, all conditions of Lagrange's mean value theorem are satisfied. Now, by Lagrange's mean value theorem there exists $c \in]1, e.[$ such that

$$\begin{aligned} f'(c) &= \frac{f(e) - f(1)}{e - 1}, f(x) = \log x \Rightarrow f'(x) = \frac{1}{x} \\ \Rightarrow \frac{1}{c} &= \frac{f(e) - f(1)}{e - 1} \\ \frac{1}{c} &= \frac{\log_e e - \log 1}{e - 1} \\ \Rightarrow \frac{1}{c} &= \frac{1}{e - 1} \\ \Rightarrow c &= e - 1 = 2.71 - 1 = 1.71 \in]1, e.[\end{aligned}$$

Hence Lagrange's mean value theorem is verifying in $]1, e[$

Example 25: If $f(x+h) = f(x) + hf'(x+\theta h)$, $0 < \theta < 1$, find the value of θ when

$$f(x) = x^2$$

[B.C.A. (Rohilkhand) 2005, 2008]

Solution: We have $f(x) = x^2 \Rightarrow f(x+h) = (x+h)^2$

And

$$f'(x) = 2x$$

Put

$$x = x + \theta h \Rightarrow f'(x + \theta h) = 2(x + \theta h)$$

Put these value in

$$f(x+h) = f(x) + hf'(x+\theta h)$$

$\therefore$

$$(x+h)^2 = x^2 + h2(x+\theta h)$$

$\Rightarrow$

$$x^2 + h^2 + 2xh = x^2 + 2hx + 2\theta h^2$$

$\Rightarrow$

$$h^2 - 2\theta h^2 = 0$$

$\Rightarrow$

$$h^2(1-2\theta) = 0, \text{ but } h^2 \neq 0$$

$\Rightarrow$

$$1 - 2\theta = 0$$

$\Rightarrow$

$$\theta = \frac{1}{2}$$

Example 26: A function $f(x)$ is continuous in the closed interval $[0,1]$ and differentiable in the open interval $]0,1[$, prove that

$$f'(x_1) = f(1) - f(0), \text{ where } 0 < x_1 < 1$$

Solution: Here $a=0, b=1$ so that

$$\frac{f(b) - f(a)}{b - a} = \frac{f(1) - f(0)}{1 - 0} = f(1) - f(0)$$

If we take $c = x_1$, and substitute these values in the result of Lagrange's mean value theorem, we get

$$f(1) - f(0) = f'(x_1) \text{ where } 0 < x_1 < 1$$

This is a particular case of Lagrange's mean value theorem. Students can give an independent proof of this.

Example 27: Separate the intervals in which the polynomial

$$2x^3 - 15x^2 + 36x + 1 \text{ is increasing or decreasing.}$$

Solution: We have $f(x) = 2x^3 - 15x^2 + 36x + 1$

$$\therefore f'(x) = 6x^2 - 30x + 36 = 6(x-2)(x-3)$$

$$\text{Now, } f'(x) > 0 \text{ for } x < 2 \text{ or for } x > 3$$

$$f'(x) < 0 \text{ for } 2 < x < 3, \text{ and } f'(x) = 0 \text{ for } x = 2, 3$$

Thus $f'(x)$ is +ive in the intervals $]-\infty, 2[$ and $]3, \infty[$ and negative in the interval $]2, 3[$.

Hence, f is monotonically increasing in the intervals $]-\infty, 2[$, $]3, \infty[$ and monotonically decreasing in the interval $]2, 3[$.

Example 28: Use the function $f(x) = x^{1/x}, x > 0$ to determine the bigger of the two number e^π and π^e .

Solution: Let $f(x) = x^{1/x}$. Then $\log f(x) = \frac{1}{x} \log_e x$.

Differentiating w.r.t. x , we get

$$\frac{1}{f(x)} f'(x) = \frac{1}{x} \cdot \frac{1}{x} - \frac{1}{x^2} \log_e x \quad \text{or} \quad f'(x) = \frac{x^{1/x}}{x^2} [1 - \log_e x]$$

$$\text{For } x > e, f'(x) < 0 \quad [\because \log_e x > 1 \text{ for } x > e]$$

$\therefore f(x)$ is a decreasing function of x for $x > e$.

$$\text{Hence, } \pi > e \Rightarrow f(\pi) < f(e) \Rightarrow \pi^{1/\pi} < e^{1/e}$$

$$\Rightarrow (\pi^{1/\pi})^{e\pi} < (e^{1/e})^{e\pi} \Rightarrow \pi^e < e^\pi \Rightarrow e^\pi \text{ is bigger than } \pi^e.$$

Example 29: Show that the set of all x for which $\log(1+x) \leq x$ is equal to $[0, \infty[$.

Solution: Let $f(x) = \log(1+x) - x$. Then $f(0) = 0$.

$$\text{Now } f'(x) = \frac{1}{1+x} - 1 = -\frac{x}{1+x} \text{ which is } \leq 0 \text{ for } x \geq 0.$$

$\therefore f(x)$ is a decreasing function in $[0, \infty[$

$$\Rightarrow f(x) \leq f(0) \text{ for } x \geq 0$$

$$\Rightarrow \log(1+x) - x \leq 0 \quad \text{or} \quad \log(1+x) \leq x \text{ for } x \in [0, \infty[.$$

Example 30: Show that

$$\frac{x}{1+x} < \log(1+x) < x \text{ for } x > 0. \quad [\text{B.C.A. (Meerut) 2008, 2011}]$$

Solution: Let $f(x) = \log(1+x) - \frac{x}{1+x}$

$$\therefore f(0) = 0$$

$$\text{Then } f'(x) = \frac{1}{1+x} - \frac{1 \cdot (1+x) - x \cdot 1}{(1+x)^2} = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2}$$

We observe that $f'(x) > 0$ for $x > 0$. Hence $f(x)$ is monotonically increasing in the interval $[0, \infty[$. Therefore

$$f(x) > f(0) \text{ for } x > 0 \text{ i.e., } \left[\log(1+x) - \frac{x}{1+x} \right] > 0 \text{ for } x > 0$$

$$\text{i.e., } \log(1+x) > \frac{x}{1+x} \text{ for } x > 0 \quad \dots(1)$$

$$\text{Again, let } \phi(x) = x - \log(1+x)$$

$$\therefore \phi(0) = 0$$

$$\text{Then } \phi'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x}$$

We observe that $\phi'(x) > 0$ for $x > 0$. Hence $\phi(x)$ is monotonically increasing in the interval $[0, \infty[$. Therefore

$$\phi(x) > \phi(0) \text{ for } x > 0 \text{ i.e., } [x - \log(1+x)] > 0 \text{ for } x > 0$$

$$\text{i.e., } x > \log(1+x) \text{ for } x > 0 \quad \dots(2)$$

From (1) and (2), we get

$$\frac{x}{1+x} < \log(1+x) < x \text{ for } x > 0$$

Example 31: Use Lagrange's mean value theorem to prove that

$$1+x < e^x < 1+xe^x \quad \forall x > 0 \quad [\text{B.C.A. (Meerut) 2005}]$$

Solution: Consider the function $f(x) = e^x$ in $[0, x]$.

Then f is continuous in $[0, x]$ and differentiable in $]0, x[$. Consequently by Lagrange's mean value theorem there exists $c \in]0, x[$ such that

$$f'(c) = \frac{f(x) - f(0)}{x - 0} \text{ or } e^c = \frac{e^x - 1}{x} \quad \dots(1)$$

Now $0 < c < x \Rightarrow e^0 < e^c < e^x \quad \dots(2)$

From (1) and (2), $e^0 < \frac{e^x - 1}{x} < e^x \quad \forall x > 0$

or $1 < \frac{e^x - 1}{x} < e^x$ or $x < e^x - 1 < xe^x$

or $1 + x < e^x < 1 + xe^x$, which proves the required result.

Example 32: Let $f: [0,1] \rightarrow \mathbb{R}$ be defined by

$$f(x) = (x-1)^2 + 2 \quad \forall x \in [0,1]$$

Find the equation of the tangent to the graph of this curve which is parallel to the chord joining the points (0,3) and (1,2) of the curve.

Solution: Since $f(x)$ is a polynomial function, therefore it is continuous on $[0,1]$ and differentiable in $]0,1[$. Hence, by Lagrange's mean value theorem, there is some $c \in]0,1[$ such that

$$\frac{f(1) - f(0)}{1 - 0} = f'(c) \quad \text{or} \quad \frac{2 - 3}{1} = f'(c) \quad \text{or} \quad -1 = f'(c)$$

Now $f'(x) = 2(x-1)$ gives $f'(c) = 2(c-1)$

Thus $2(c-1) = -1$ i. e., $c = \frac{1}{2}$

$\therefore f(c) = \frac{9}{4}$, so that the point of contact of the tangent is $\left(\frac{1}{2}, \frac{9}{4}\right)$ and its slope is $f'(c) = -1$.

Hence the equation of the required tangent is

$$y - \frac{9}{4} = -1 \left(x - \frac{1}{2} \right) \quad \text{or} \quad 4x + 4y = 11$$

Example 33: Show that any chord of the parabola $y = Ax^2 + Bx + C$ is parallel to the tangent at the point whose abscissa is same as that of the middle point of the chord.

Solution: Let a and b ($a < b$) be the abscissae of the ends of the chord and let $f(x) = Ax^2 + Bx + C$. Since f is a polynomial function, f is continuous on $[a,b]$ and differentiable in $]a,b[$. Consequently by Lagrange's mean value theorem there exists $c \in]a,b[$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

$$\text{i.e., } Ab^2 + Bb + C - Aa^2 - Ba - C = (b-a)(2Ac + B) \quad [\because f'(c) = 2Ac + B]$$

On simplification it gives $c = \frac{1}{2}(a+b)$ i.e., the abscissa of the point at which the tangent is parallel to the chord is the same as that of the middle point of the chord.

Example 34: Compute the value of θ in the first mean value theorem

$$f(x+h) = f(x) + hf'(x+\theta h), \text{ if } f(x) = ax^2 + bx + c$$

Solution: Here

$$f(x) = ax^2 + bx + c$$

$$\therefore f(x+h) = a(x+h)^2 + b(x+h) + c$$

$$f'(x) = 2ax + b, f'(x+\theta h) = 2a(x+\theta h) + b$$

Substituting all these values in the Lagrange's mean value theorem, we get

$$a(x+h)^2 + b(x+h) + c = ax^2 + bx + c + h[2a(x+\theta h) + b] \quad \dots(1)$$

The relation (1) being identically true for all values of x , hence when $x \rightarrow 0$, we have

$$ah^2 + bh + c = c + h[2a\theta h + b]$$

$$\text{or} \quad ah^2 = 2a\theta h^2 \text{ or } \theta = \frac{1}{2}$$

Example 35: If $x > 0$, then show that

$$x - \frac{x^2}{2} < \log(1+x) < x - \frac{x^2}{2(1+x)}$$

[B.C.A (Meerut) 2004]

Solution: Let $f(x) = \log(1+x) - \left(x - \frac{x^2}{2}\right)$, then $f(0) = 0$.

$$\therefore f'(x) = \frac{1}{1+x} - 1 + x = \frac{x^2}{1+x} > 0 \quad [\because x > 0]$$

$\Rightarrow f(x)$ is an increasing function of x , for $x > 0$

$$\Rightarrow f(x) > f(0), \text{ for } x > 0 \Rightarrow f(x) > 0 \quad [\because f(0) = 0]$$

$$\Rightarrow \log(1+x) - \left(x - \frac{x^2}{2}\right) > 0, \text{ for } x > 0$$

$$\therefore x - \frac{x^2}{2} < \log(1+x), \text{ for } x > 0$$

$$\text{Let } g(x) = x - \frac{x^2}{2(1+x)} - \log(1+x), \text{ then } g(0) = 0$$

$$\therefore g'(x) = 1 - \frac{1}{2} \left\{ \frac{2x+x^2}{(1+x)^2} \right\} - \frac{1}{(1+x)} = \frac{x^2}{2(1+x)^2} > 0, \quad (\because x > 0)$$

$\Rightarrow g(x)$ is an increasing function for $x > 0$.

$$\Rightarrow g(x) > g(0), \text{ for } x > 0 \Rightarrow g(x) > 0 \quad [\because g(0) = 0]$$

$$\Rightarrow x - \frac{x^2}{2(1+x)} - \log(1+x) > 0, \text{ for } x > 0$$

$$\therefore \log(1+x) < x - \frac{x^2}{2(1+x)}, \text{ for } x > 0$$

$$\text{Hence } x - \frac{x^2}{2} < \log(1+x) < x - \frac{x^2}{2(1+x)}, \text{ for } x > 0$$

Example 36: If $x > 0$, then show that

$$x - \frac{x^2}{2} + \frac{x^3}{3(1+x)} < \log(1+x) < x - \frac{x^2}{2} + \frac{x^3}{3} \quad [\text{B.C.A. (Meerut) 2005}]$$

Solution: Let $f(x) = \log(1+x) - \left[x - \frac{x^2}{2} + \frac{x^3}{3(1+x)} \right]$. Then $f(0) = 0$

$$\therefore f'(x) = \frac{1}{1+x} - 1 + x - \frac{3x^2 + 2x^3}{3(1+x)^2} = \frac{x^3}{3(1+x)^2} > 0, \text{ for } x > 0$$

$\Rightarrow f$ is increasing, for $x > 0 \Rightarrow f(x) > f(0) = 0$ for $x > 0$

$$\Rightarrow x - \frac{x^2}{2} + \frac{x^3}{3(1+x)} < \log(1+x), \text{ if } x > 0$$

Again let $g(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \log(1+x)$, then $g(0) = 0$

$$\therefore g'(x) = 1 - x + x^2 - \frac{1}{1+x} = \frac{x^3}{1+x} > 0, \text{ for } x > 0$$

$\Rightarrow g$ is increasing, for $x > 0 \Rightarrow g(x) > g(0) = 0$ for $x > 0$

$$\Rightarrow \log(1+x) < x - \frac{x^2}{2} + \frac{x^3}{3} \text{ if } x > 0$$

Example 37: If $0 < x < 1$, then show that $x < \sin^{-1} x < \frac{x}{\sqrt{1-x^2}}$.

Solution: Let $f(x) = \sin^{-1} x - x$, so that $f(0) = 0$.

$$\therefore f'(x) = \frac{1}{\sqrt{1-x^2}} - 1 = \frac{1 - \sqrt{1-x^2}}{\sqrt{1-x^2}} > 0, \text{ for } 0 < x < 1$$

$\Rightarrow f$ is strictly increasing in $0 < x < 1$

$$\Rightarrow f(x) > f(0) = 0 \text{ for } 0 < x < 1 \quad (\because x > 0)$$

$$\Rightarrow \sin^{-1} x - x > 0, \text{ for } 0 < x < 1$$

$$\Rightarrow \sin^{-1} x > x, \text{ for } 0 < x < 1$$

$$\Rightarrow x < \sin^{-1} x, \text{ for } 0 < x < 1$$

Again let $g(x) = \frac{x}{\sqrt{1-x^2}} - \sin^{-1} x$, then $g(0) = 0$

$$\begin{aligned} \therefore g'(x) &= \frac{1 \cdot \sqrt{1-x^2} - \frac{x(-2x)}{2\sqrt{1-x^2}}}{(1-x^2)} - \frac{1}{\sqrt{1-x^2}} \\ &= \frac{1}{(1-x^2)\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = \frac{x^2}{(1-x^2)^{3/2}} > 0 \end{aligned}$$

$\Rightarrow g(x)$ is strictly increasing, for $0 < x < 1$

$$\Rightarrow g(x) > g(0) = 0 \text{ for } 0 < x < 1 \quad (\because x > 0)$$

$$\Rightarrow \frac{x}{\sqrt{1-x^2}} > \sin^{-1} x \text{ and hence } \sin^{-1} x < \frac{x}{\sqrt{1-x^2}} \text{ for } 0 < x < 1$$

Example 38: Using Lagrange's means value theorem, show that

$$\frac{x}{1+x} < \log(1+x) < x, \text{ if } x > 0 \quad [\text{B.C.A. (Rohilkhand) 2003, 2008}]$$

Solution: Let $f(x) = \log(1+x)$ in $[0, x] \Rightarrow f'(x) = 1/(1+x)$

We know that every logarithmic function is continuous and differentiable in its domain. Therefore, $f(x) = \log(1+x)$ is continuous in closed interval $[0, x]$ and differentiable in the open interval $]0, x[$.

So by Lagrange's mean value theorem, there exists some $\theta, 0 < \theta < 1$ such that

$$\frac{f(x) - f(0)}{x - 0} = f'(\theta x) \quad \text{or} \quad \log(1+x) = \frac{x}{1+\theta x}$$

$$\therefore \quad 0 < \theta < 1 \Rightarrow 0x < \theta x < x \text{ and } f(0) = 0 \quad \dots(1)$$

Now $0 < \theta < 1$ and $x > 0 \Rightarrow \theta x < x \Rightarrow 1 + \theta x < 1 + x$

$$\Rightarrow \quad \frac{1}{1+\theta x} > \frac{1}{1+x} \Rightarrow \frac{x}{1+\theta x} > \frac{x}{1+x} \Rightarrow \frac{x}{1+x} < \frac{x}{1+\theta x} \quad \dots(2)$$

Again $0 < \theta < 1$ and $x > 0 \Rightarrow 0 < \theta x \Rightarrow 1 < 1 + \theta x$

$$\Rightarrow \quad \frac{1}{1+\theta x} < 1 \Rightarrow \frac{x}{1+\theta x} < x \quad \dots(3)$$

From (1), (2) and (3), we at once conclude that

$$\frac{x}{1+x} < \log(1+x) < x, \text{ for } x > 0$$

Example 39: Applying Lagrange's mean value theorem to the function defined by $f(x) = \log(1+x)$ for all $x > 0$, show that

$$0 < [\log(1+x)]^{-1} - x^{-1} < 1, \text{ for } x > 0$$

Solution: First proceed like example 38, to obtain

$$\frac{x}{1+x} < \log(1+x) < x \Rightarrow \frac{1+x}{x} > \frac{1}{\log(1+x)} > \frac{1}{x}$$

$$\Rightarrow \quad 1 + \frac{1}{x} > \frac{1}{\log(1+x)} > \frac{1}{x} \Rightarrow 1 > \frac{1}{\log(1+x)} - \frac{1}{x} > 0$$

Hence, $0 < [\log(1+x)]^{-1} - x^{-1} < 1$

Example 40: Prove that $\tan x > x$, whenever $0 < x < \pi/2$.

Solution: Let c be any real number such that $0 < c < \pi/2$.

Let $f(x) = \tan x - x \quad \forall x \in [0, c]$

Clearly, $f(x)$ is continuous as well as differentiable on $[0, c]$.

Now $f'(x) = \sec^2 x - 1 = \tan^2 x > 0$, for $0 < x < c$

Thus, $f(x)$ is strictly increasing in $[0, c] \Rightarrow f(x) > f(0)$, for $c > 0$

But $f(0) = 0$. Therefore, $f(c) > 0 \Rightarrow \tan c - c > 0$

Since c is any real number such that $0 < c < \pi/2$, therefore, $\tan x > x$, whenever $0 < x < \pi/2$.

Example 41: Let $f(x)$ be defined for all $x \in R$ and let $|f(x) - f(y)| \leq (x - y)^2, \forall x, y \in R$. Show that $f(x)$ is a constant function.

Solution: We have $|f(x) - f(y)| \leq (x - y)^2 \quad \forall x, y \in R$... (1)
 $[\because |x|^2 = x^2, \quad \forall x \in R]$

Let x be any real number and let y be chosen arbitrarily close to x but not equal to x . Then (1) gives

$$\left| \frac{f(x) - f(y)}{x - y} \right| \leq |x - y|$$

Taking limit when $y \rightarrow x$, we get

$$\Rightarrow \lim_{y \rightarrow x} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \lim_{y \rightarrow x} |x - y|$$

$$\Rightarrow |f'(x)| \leq 0 \quad \left[\because \lim_{y \rightarrow x} \frac{f(x) - f(y)}{x - y} = f'(x) \right] \quad \dots (2)$$

But $|f'(x)| \geq 0$, therefore, $|f'(x)| = 0 \Rightarrow f'(x) = 0 \Rightarrow f(x)$ is constant.

Example 42: Use the mean value theorem to show that

$$1 - \frac{a}{b} < \log \frac{b}{a} < \frac{b}{a} - 1, \text{ where } b > a$$

Hence, obtain $(1/2) < \log 2 < 1$.

[B.C.A. (Meerut) 2002]

Solution: Let $f(x) = \log x - \left(1 - \frac{1}{x}\right)$, then $f(1) = 0$

$$\therefore f'(x) = \frac{1}{x} - \frac{1}{x^2} \Rightarrow f'(x) = \frac{x-1}{x^2} > 0, \quad \text{if } x > 1$$

$\Rightarrow f(x)$ is an increasing function of x , for $x > 1$.

$$\Rightarrow f(x) > f(1), \text{ for } x > 1 \Rightarrow f(x) > 0 \quad [\because f(1) = 0]$$

$$\Rightarrow \log x - \left(1 - \frac{1}{x}\right) > 0, \text{ for } x > 1 \Rightarrow 1 - \frac{1}{x} < \log x, x > 1 \quad \dots (1)$$

Again let $g(x) = (x-1) - \log x$, then $g(1) = 0$

$$\therefore g'(x) = 1 - \frac{1}{x} \Rightarrow g'(x) = \frac{x-1}{x} > 0, \text{ if } x > 1$$

$\Rightarrow g(x)$ is an increasing function of x , for $x > 1$.

$$\Rightarrow g(x) > g(1), \text{ for } x > 1 \Rightarrow g(x) > 0 \quad [\because g(1) = 0]$$

$$\Rightarrow (x-1) - \log x > 0 \Rightarrow \log x < x-1 \quad \dots (2)$$

From (1) and (2), we get

$$1 - \frac{1}{x} < \log x < x-1 \text{ for } x > 1 \quad \dots (3)$$

In particular if $x = \frac{b}{a} > 1$, then equation (3) gives

$$1 - \frac{a}{b} < \log \frac{b}{a} < \frac{b}{a} - 1 \quad \text{for } \frac{b}{a} > 1$$

$$\text{or} \quad 1 - \frac{a}{b} < \log \frac{b}{a} < \frac{b}{a} - 1 \quad \text{for } b > a \quad \dots(4)$$

which proves the required first result.

If we take $b = 2$ and $a = 1$; $b > a$, then (4) gives

$$1 - \frac{1}{2} < \log 2 < 2 - 1 \text{ or } \frac{1}{2} < \log 2 < 1$$

which proves the required second result.

Proved

Example 43: Given that $f''(x)$ exists for x in $[a, b]$ and $\frac{f(b) - f(c)}{b - c} = \frac{f(c) - f(a)}{c - a}$, $a < c < b$
 $\dots(1)$

prove that there exists a point $t \in (a, b)$ s.t. $f''(t) = 0$

Solution: Since $f''(x)$ exists in $[a, b]$, the function f and f' are both continuous and differentiable in $[a, b]$ and hence also in the intervals $[a, c]$ and $[c, b]$. Applying LMV theorem in the intervals $[a, c]$ and $[c, b]$, we shall get two points $x_1 \in (a, c)$ and $x_2 \in (c, b)$ such that

$$f'(x_1) = \frac{f(c) - f(a)}{c - a} \quad \text{and} \quad f'(x_2) = \frac{f(b) - f(c)}{b - c}$$

Using the given result (1), we get

$$f'(x_1) = f'(x_2)$$

Now $[x_1, x_2] \subset [a, b]$, therefore, $f'(x)$ satisfies all the conditions of Rolle's Theorem in the interval $[x_1, x_2]$. Therefore, we shall get a point $t \in (x_1, x_2)$ s.t.

$$\left[\frac{d}{dx} f'(x) \right]_{x=t} = 0 \quad \text{or} \quad f''(t) = 0$$

Since $t \in (x_1, x_2) \Rightarrow t \in (a, b)$ also.

Example 44: Applying the Lagrange's mean value theorem prove that: If $f'(x)$ is positive for all points in the open interval (a, b) then $f(x)$ is strictly increasing throughout the closed interval $[a, b]$.
[B.C.A. (Meerut) 2003]

Solution: Since $f'(x) > 0$, it means $f(x)$ is differentiable and hence it is continuous also. Suppose x_1, x_2 are any two distinct point of $[a, b]$ s.t. $x_1 < x_2$. Then $[x_1, x_2] \subset [a, b]$. Now all the conditions of L.M.V. Theorem are satisfied by the function f in $[x_1, x_2]$. Applying mean value theorem, we get

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c) \quad \text{where } x_1 < c < x_2$$

But given that $f'(c) > 0$ and $x_2 - x_1$ is also positive as $x_1 < x_2$. So necessarily $f(x_2) - f(x_1) > 0$, i.e., $f(x_2) > f(x_1)$ necessarily.

$\Rightarrow f(x)$ is strictly increasing.



Comprehensive Exercise 7.2

(Lagrange's Mean Value Theorem)

- Determine whether $f(x) = \frac{1}{x}$, $-1 < x < 0$ is strictly increasing, decreasing or neither of these. [B.C.A. (Meerut) 2008]
- Explain Lagrange's mean value theorem for the function $f(x) = \frac{1}{x}$ in the interval $] -1, 1[$
- Verify Lagrange's mean value theorem for the function
 - $f(x) = x(x-1)$ in $[1, 2]$
 - $f(x) = (x-1)^2 + 2$ in $x \in [0, 1]$
 - $f(x) = 2x^2 + 3x + 4$ in $[1, 2]$
 - $f(x) = x^3 - 3x - 2$ in $[-2, 3]$
 - $f(x) = x^2 - 3x - 1$ in $\left[\frac{-11}{7}, \frac{13}{7}\right]$
 - $f(x) = (x-a)^m(x-b)^n$ in $[a, b]$, where $m, n \in \mathbb{N}$
- Find c of mean value theorem if $f(x) = x^3 - 5x^2 - 3x$, $a = 1, b = 3$
- Find the interval in which the function $f(x) = (x^4 + 6x^3 + 17x^2 + 32x + 32)e^{-x}$ is increasing or decreasing.
- Prove that $\cos x + x \sin x > 1$ in the interval $]0, \frac{\pi}{2}[$
- Find the interval in which the polynomial $2x^3 - 15x^2 + 36x + 1$ is increasing or decreasing.
- Verify Lagrange's mean value theorem for
 - $f(x) = lx^2 + mx + n \quad \forall x \in [a, b]$
 - $f(x) = \sqrt{x^2 - 4}$ in interval $[2, 4]$
 - $f(x) = \sin x$ in interval $\left[0, \frac{\pi}{2}\right]$
- If $f(x) = \sin^2 x \quad \forall x \in [0, \pi]$ on what subinterval of $[0, \pi]$ is f increasing and decreasing.



Answers

1.	Decreasing
2.	The function is not continuous and differentiable at $x = 0$, so mean value theorem does not hold.
4.	$c = \frac{7}{3}$
5.	Monotonically increasing in the intervals $[-2, -1]$ and $[0, 1]$ and monotonically decreasing in the interval $]-\infty, -2[$, $[-1, 0]$ and $[1, \infty]$
7.	Monotonically increasing in the intervals $]-\infty, 2[$, $[3, \infty]$ and monotonically decreasing in the interval $[2, 3]$, because $f'(x) = 0$ when $x = 2$ and $x = 3$
9.	$f(x)$ is increasing in the interval $\left[0, \frac{\pi}{2}\right]$ and decreasing in the interval $\left[\frac{\pi}{2}, \pi\right]$.

7.3 Cauchy's Mean Value Theorem

If two functions $f(x)$ and $g(x)$ are

- (i) Continuous in a closed interval $[a, b]$
- (ii) Differentiable in the open interval $]a, b[$,
- (iii) $g'(x) \neq 0$ for any point of the open interval $]a, b[$, then there exists atleast one value c of x in the open interval $]a, b[$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}, \quad a < c < b$$

Proof: First of all we shall prove $g(b) - g(a) \neq 0$ i.e., $g(b) \neq g(a)$. If possible let $g(b) = g(a)$ or $g(b) - g(a) = 0$, then the function $g(x)$ satisfies all the three conditions of Rolle's theorem and therefore, there exists a point $x \in]a, b[$ such that $g'(x) = 0$ which contradicts the hypothesis (iii) that $g'(x) \neq 0 \quad \forall x \in]a, b[$.

Hence, $g(b) - g(a) \neq 0$

Now, let us consider function $\phi(x)$ defined on $[a, b]$,

$$\phi(x) = f(x) + Ag(x) \quad \dots(1)$$

Where A is a constant to be chosen such that

$$\phi(a) = \phi(b)$$

$$\Rightarrow f(a) + Ag(a) = f(b) + Ag(b)$$

$$\Rightarrow -A = \frac{f(b) - f(a)}{g(b) - g(a)} \quad \dots(2)$$

- (i) Now f and g are continuous on $[a, b]$, therefore, $\phi(x)$ is also continuous on $[a, b]$
 (ii) Again, f and g are differentiable on $]a, b[$ therefore ϕ is also differentiable on $]a, b[$

$$\phi(a) = \phi(b)$$

- (iii) Thus the function $\phi(x)$ satisfies the conditions of Rolle's theorem in the interval $[a, b]$. Consequently there exists, at least one value of c in $]a, b[$ such that

$$\phi'(c) = 0$$

But

$$\phi'(x) = f'(x) + Ag'(x)$$

$\Rightarrow$

$$\phi'(c) = f'(c) + Ag'(c)$$

$\Rightarrow$

$$0 = f'(c) + Ag'(c)$$

$\Rightarrow$

$$-A = \frac{f'(c)}{g'(c)} \quad \dots(3)$$

From (2) and (3), we get

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

which proves the theorem.

7.3.1 Alternative form of Cauchy's Mean Value Theorem

If two functions $f(x)$ and $g(x)$ are such that both are

- (i) Continuous on $[a, a+h]$
 (ii) Differentiable on $]a, a+h[$
 (iii) $g'(x) \neq 0 \quad \forall x \in]a, a+h[$

Then there exists at least one real value number θ between 0 and 1 such that

$$\frac{f(a+h) - f(a)}{g(a+h) - g(a)} = \frac{f'(a+\theta h)}{g'(a+\theta h)}, \quad 0 < \theta < 1$$



Illustrative Examples

(Cauchy Mean Value Theorem)

Example 45: Verify Cauchy's mean value theorem for the functions x^2 and x^3 in the interval $[1, 2]$

Solution: Let $f(x) = x^2$ and $g(x) = x^3$

The function $f(x)$ and $g(x)$ both are continuous and differentiable for all values of $x \in \mathbb{R}$. Therefore, $f(x)$ and $g(x)$ both are continuous in $[1, 2]$ and differentiable in $]1, 2[$ such that

$$\frac{f(2) - f(1)}{g(2) - g(1)} = \frac{f'(c)}{g'(c)} \quad \dots(1)$$

We have

$$f(2) = 4, f(1) = 1, g(2) = 8, g(1) = 1$$

Also

$$f'(x) = 2x, \quad g'(x) = 3x^2$$

$\Rightarrow$

$$f'(c) = 2c, \quad g'(c) = 3c^2$$

Then from (1) we get

$$\frac{4-1}{8-1} = \frac{2c}{3c^2}$$

$\Rightarrow$

$$9c^2 - 14c = 0$$

$\Rightarrow$

$$c(9c - 14) = 0 \text{ or } c = 0 \text{ or } c = 14/9$$

$\therefore$

$$c = 0 \notin]1, 2[\text{ whereas } c = \frac{14}{9} \in]1, 2[$$

Hence Cauchy's mean value theorem is verified.

Example 46: Verify Cauchy's mean value theorem for

$$f(x) = \sin x, \quad g(x) = \cos x \text{ in } \left[-\frac{\pi}{2}, 0\right]$$

Solution: Here both the functions $f(x)$ and $g(x)$ are continuous in the closed interval $\left[-\frac{\pi}{2}, 0\right]$ and differentiable in the open interval $\left]-\frac{\pi}{2}, 0\right[$. Also $g'(x) = -\sin x \neq 0$ for any point in the open interval $\left]-\frac{\pi}{2}, 0\right[$. Hence by Cauchy's mean value theorem there exists at least one real number in the open interval $\left]-\frac{\pi}{2}, 0\right[$ such that

$$\frac{f(0) - f\left(-\frac{\pi}{2}\right)}{g(0) - g\left(-\frac{\pi}{2}\right)} = \frac{f'(c)}{g'(c)} \quad \dots(1)$$

$$\Rightarrow \frac{\sin 0 - \sin\left(-\frac{\pi}{2}\right)}{\cos 0 - \cos\left(-\frac{\pi}{2}\right)} = \frac{1}{1} = 1$$

Also

$$f'(x) = \cos x, \quad g'(x) = -\sin x$$

$\therefore$

$$\frac{f'(c)}{g'(c)} = \frac{\cos c}{-\sin c} = -\cot c$$

Put these values in (1), we get $-\cot c = 1$ or $\cot c = -1$ whose solution $c = \frac{-\pi}{4}$ lies in the open interval $\left]-\frac{\pi}{2}, 0\right[$. Hence Cauchy's mean value theorem is verified.

Example 47: Show that $\frac{\sin \alpha - \sin \beta}{\cos \beta - \cos \alpha} = \cot \theta$

Where

$$0 < \alpha < \beta < \frac{\pi}{2}$$

Solution: Let $f(x) = \sin x$ and $g(x) = \cos x$

$$\text{for } x \in [\alpha, \beta], \text{ where } 0 < \alpha < \beta < \frac{\pi}{2}$$

$$\therefore f'(x) = \cos x \quad \text{and} \quad g'(x) = -\sin x$$

Here both the function $f(x)$ and $g(x)$ are continuous in the $[\alpha, \beta]$ and differentiable $]\alpha, \beta[$. Also $g'(x) = -\sin x \neq 0$ for any point in the open interval $]\alpha, \beta[$. Hence by Cauchy's mean value theorem there exists at least one real number say θ , in the open interval $]\alpha, \beta[$ such that

$$\begin{aligned} \frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} &= \frac{f'(\theta)}{g'(\theta)} \\ \Rightarrow \frac{\sin \beta - \sin \alpha}{\cos \beta - \cos \alpha} &= \frac{\cos(\theta)}{-\sin(\theta)} = -\cot \theta \\ \Rightarrow \frac{\sin \alpha - \sin \beta}{\cos \beta - \cos \alpha} &= \cot \theta \quad \text{where} \quad 0 < \alpha < \theta < \beta < \frac{\pi}{2} \end{aligned}$$

Example 48: Find c of Cauchy mean value theorem for the following pair of functions

(i) $f(x) = e^x, g(x) = e^{-x}$ in $[a, b]$

(ii) $f(x) = \sqrt{x}, g(x) = \frac{1}{\sqrt{x}}$ in $[a, b]$

Solution: (i) If $f(x) = e^x$ and $f(a) = e^a$ and $f(b) = e^b$

If $g(x) = e^{-x}$, then $g(a) = e^{-a}$ and $g(b) = e^{-b}$

We have $f'(x) = e^x, g'(x) = -e^{-x}$, therefore

$$f'(c) = e^c \quad \text{and} \quad g'(c) = -e^{-c}$$

By Cauchy's mean value theorem, we have

$$\begin{aligned} \frac{f(b) - f(a)}{g(b) - g(a)} &= \frac{f'(c)}{g'(c)} \\ \Rightarrow \frac{e^b - e^a}{e^{-b} - e^{-a}} &= \frac{e^c}{-e^{-c}} \\ \Rightarrow \frac{e^b - e^a}{\frac{1}{e^b} - \frac{1}{e^a}} &= -e^{2c} \\ \Rightarrow \frac{(e^b - e^a)e^b e^a}{(e^a - e^b)} &= -e^{2c} \\ \Rightarrow -e^a e^b &= -e^{2c} \\ \Rightarrow e^{a+b} &= e^{2c} \\ \Rightarrow a + b &= 2c \\ \Rightarrow c &= \frac{1}{2}(a + b) \end{aligned}$$

(ii) If $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{\sqrt{x}}$, then

$$f(a) = \sqrt{a}, f(b) = \sqrt{b}, g(a) = \frac{1}{\sqrt{a}} \quad \text{and} \quad g(b) = \frac{1}{\sqrt{b}}$$

We have $f'(x) = \frac{1}{2\sqrt{x}}$ and $g'(x) = \frac{-1}{2x\sqrt{x}}$

Therefore, $f'(c) = \frac{1}{2\sqrt{c}}$ and $g'(c) = \frac{-1}{2c\sqrt{c}}$

By Cauchy's mean value theorem

$$\begin{aligned} \frac{f(b) - f(a)}{g(b) - g(a)} &= \frac{f'(c)}{g'(c)} \\ \Rightarrow \frac{(\sqrt{b} - \sqrt{a})}{\left(\frac{1}{\sqrt{b}} - \frac{1}{\sqrt{a}}\right)} &= \frac{\frac{1}{2\sqrt{c}}}{\frac{-1}{2c\sqrt{c}}} \\ \Rightarrow (\sqrt{b} - \sqrt{a}) \times \frac{\sqrt{a}\sqrt{b}}{(\sqrt{a} - \sqrt{b})} &= -c \\ \Rightarrow -\sqrt{a}\sqrt{b} &= -c \text{ or } c = \sqrt{ab} \end{aligned}$$

Example 49: If in a Cauchy's mean value theorem, we write $f(x) = \frac{1}{x^2}$ and $g(x) = \frac{1}{x}$ then show that c is Harmonic mean of a and b .

Solution: If $f(x) = \frac{1}{x^2}$ and $g(x) = \frac{1}{x}$, then we have

$$f(a) = \frac{1}{a^2}, f(b) = \frac{1}{b^2}, g(a) = \frac{1}{a}, g(b) = \frac{1}{b}$$

We have $f'(x) = -2x^{-3}$ and $g'(x) = -x^{-2}$

Then $f'(c) = -2c^{-3}$ and $g'(c) = -c^{-2}$

By Cauchy's mean value theorem, we have

$$\begin{aligned} \frac{f(b) - f(a)}{g(b) - g(a)} &= \frac{f'(c)}{g'(c)} \\ \frac{\frac{1}{b^2} - \frac{1}{a^2}}{\frac{1}{b} - \frac{1}{a}} &= \frac{-2c^{-3}}{-c^{-2}} \\ \Rightarrow \frac{(a^2 - b^2)}{(a-b)(ab)} &= \frac{2}{c} \text{ or } \frac{a+b}{ab} = \frac{2}{c} \text{ or } c = \frac{2ab}{a+b} \end{aligned}$$

Hence, c is a Harmonic mean of a and b .

Example 50: Find the value of $\lim_{x \rightarrow 1} \left\{ \frac{\cos \frac{\pi x}{2}}{\log \left(\frac{1}{x} \right)} \right\}$, by using Cauchy's mean value theorem.

Solution: Let $f(x) = \cos \left(\frac{\pi x}{2} \right)$, $g(x) = \log x$, $a = x$, $b = 1$. Putting these values in Cauchy's mean value theorem

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}, \quad a < c < b$$

$$\Rightarrow \frac{\cos\left(\frac{\pi}{2}\right) - \cos\left(\frac{\pi x}{2}\right)}{\log 1 - \log x} = \frac{-\frac{\pi}{2} \sin\left(\frac{\pi c}{2}\right)}{\frac{1}{c}}, \quad x < c < 1$$

Taking limits as $x \rightarrow 1$ which implies that $c \rightarrow 1$, we get

$$\lim_{x \rightarrow 1} \left\{ \frac{0 - \cos\left(\frac{\pi x}{2}\right)}{\log\left(\frac{1}{x}\right)} \right\} = \lim_{c \rightarrow 1} \left\{ \frac{-\frac{\pi}{2} \sin\left(\frac{\pi c}{2}\right)}{\frac{1}{c}} \right\}$$

or
$$\lim_{x \rightarrow 1} \left\{ \frac{-\cos\left(\frac{\pi x}{2}\right)}{\log\left(\frac{1}{x}\right)} \right\} = -\frac{\pi}{2} \quad \text{as } \sin\left(\frac{\pi c}{2}\right) \rightarrow 1 \text{ as } c \rightarrow 1$$

or
$$\lim_{x \rightarrow 1} \left\{ \frac{\cos\left(\frac{\pi x}{2}\right)}{\log\left(\frac{1}{x}\right)} \right\} = \frac{\pi}{2}$$

Example 51: Deduce from Cauchy's mean value theorem

$$f(b) - f(a) = cf'(c) \log\left(\frac{b}{a}\right)$$

where $f(x)$ is continuous and differentiable in $[a, b]$ and $a < c < b$.

Solution: Cauchy's mean value theorem is

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}, \quad a < c < b$$

Now let $g(x) = \log x$, then $g'(x) = \frac{1}{x}$ and $g(b) - g(a) = \log b - \log a = \log\left(\frac{b}{a}\right)$

From (1), we have
$$\frac{f(b) - f(a)}{\log\left(\frac{b}{a}\right)} = \frac{f'(c)}{\frac{1}{c}}$$

$$\Rightarrow f(b) - f(a) = cf'(c) \log\left(\frac{b}{a}\right)$$

Example 52: If $f(x)$, $g(x)$ and $h(x)$ are function such that

- (i) all the three are continuous on $[a, b]$
- (ii) all the three are differentiable in $]a, b[$, then

$$\begin{vmatrix} f'(c) & g'(c) & h'(c) \\ f(b) & g(b) & h(b) \\ f(a) & g(a) & h(a) \end{vmatrix} = 0 \quad \text{where } a < c < b$$

Solution: Consider the following function

$$\phi(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f(b) & g(b) & h(b) \\ f(a) & g(a) & h(a) \end{vmatrix} \quad \dots(1)$$

On expanding the determinant, we see that the function $\phi(x)$ is of the form $Af(x) + Bg(x) + Ch(x)$, where A, B, C are some real numbers.

In view of the given condition (i) and (ii), $\phi(x)$ is continuous on $[a, b]$ and differentiable in $]a, b[$. Also $\phi(a) = \phi(b) = 0$ because then the two rows of the determinant become identical. Thus $\phi(x)$ satisfies all the conditions of Rolle's theorem. Hence there exists a value c of x lying between a and b such that $\phi'(c) = 0$

$$i.e., \quad \begin{vmatrix} f'(c) & g'(c) & h'(c) \\ f(b) & g(b) & h(b) \\ f(a) & g(a) & h(a) \end{vmatrix} = 0 \quad \dots(2)$$



Comprehensive Exercise 7.3

(Cauchy Mean Value Theorem)

- Find the value of $c \in \left]0, \frac{\pi}{2}\right[$ for the pair of function $f(x) = x^2$, $g(x) = \cos x$, which give the result of Cauchy's mean value theorem in interval $\left[0, \frac{\pi}{2}\right]$ for the function $f(x)$ and $g(x)$.
- Verify Cauchy's mean value theorem for the following functions .
 - $f(x) = x^2, g(x) = x^4$ in $[2, 5]$
 - $f(x) = \log x, g(x) = x$ in $[1, e]$
- Find c of Cauchy's mean value theorem for the pair of functions.

$$f(x) = x(x-1)(x-2), \quad g(x) = (x+1)(x+2) \text{ in } \left[0, \frac{1}{2}\right]$$



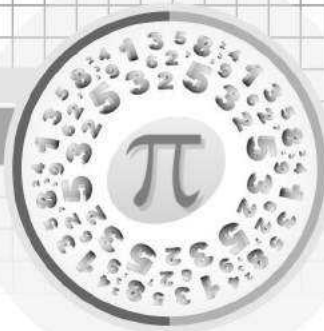
Answers

1.	The root of the equation $\sin c - \frac{8c}{\pi^2} = 0$ is the required value of c .
3.	$c = 0.24$ (Approximately)





Unit-3



Chapter

8

Expansions of Functions

8.1 Maclaurin's Theorem

Let $f(x)$ be a function of x which possesses continuous derivatives of all orders in the interval $[0, x]$. Assuming that $f(x)$ can be expanded as a infinite power series in x , we have

$$f(x) = f(0) + \frac{x}{(1)!} f'(0) + \frac{x^2}{(2)!} f''(0) + \frac{x^3}{(3)!} f'''(0) + \dots + \frac{x^n}{(n)!} f^n(0) + \dots$$

Proof: Suppose $f(x) = A_0 + A_1x + A_2x^2 + A_3x^3 + \dots$... (1)

Differentiate successively, we get

$$f'(x) = A_1 + 2A_2x + 3A_3x^2 + 4A_4x^3 + \dots$$

$$f''(x) = 2.1A_2 + 3.2A_3x + 4.3A_4x^2 + \dots$$

$$f'''(x) = 3.2.1A_3 + 4.3.2A_4x + \dots \text{ and so on}$$

Putting $x = 0$ in each of these relations, we get

$$f(0) = A_0, f'(0) = A_1, f''(0) = (2)!A_2, f'''(0) = (3)!A_3$$

Substituting these values of $A_0, A_1, A_2, \dots$ in (1) we get

$$f(x) = f(0) + x f'(0) + \frac{x^2}{(2)!} f''(0) + \frac{x^3}{(3)!} f'''(0) + \dots + \frac{x^n}{(n)!} f^n(0) + \dots$$

This is Maclaurian's theorem. If we denote $f(x)$ by y then Maclaurian's theorem can be written as

$$y = (y)_0 + \frac{x}{(1)!} (y_1)_0 + \frac{x^2}{(2)!} (y_2)_0 + \frac{x^3}{(3)!} (y_3)_0 + \dots + \frac{x^n}{(n)!} (y_n)_0 + \dots$$

or

$$y = y(0) + \frac{x}{(1)!} y_1(0) + \frac{x^2}{(2)!} y_2(0) + \dots + \frac{x^n}{(n)!} y_n(0) + \dots$$

8.2 Taylor's Theorem

Let $f(x)$ be a function of x which possesses continuous derivatives of all orders in the interval $[a, a+h]$. Assuming that $f(a+h)$ can be expanded as an infinite power series in h , we have

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{(2)!} f''(a) + \dots + \frac{h^n}{(n)!} f^n(a) + \dots$$

Proof: Suppose $f(a+h) = A_0 + A_1 h + A_2 h^2 + A_3 h^3 + \dots$... (1)

Now by successive differentiation, we get

$$f'(a+h) = A_1 + 2A_2 h + 3A_3 h^2 + \dots$$

$$f''(a+h) = 2 \cdot 1 A_2 + 3 \cdot 2 A_3 h + \dots$$

$$f'''(a+h) = 3 \cdot 2 \cdot 1 A_3 + \dots \text{ and so on}$$

Putting $h=0$ in each of the above relations, we get

$$f(a) = A_0, f'(a) = A_1, f''(a) = (2)! A_2$$

$$f'''(a) = (3)! A_3 \text{ and so on.}$$

$$\therefore A_0 = f(a), A_1 = f'(a), A_2 = \frac{1}{(2)!} f''(a), A_3 = \frac{1}{(3)!} f'''(a) \dots \text{ and so on}$$

Put these values in (1) we get

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{(2)!} f''(a) + \frac{h^3}{(3)!} f'''(a) + \dots + \frac{h^n}{(n)!} f^n(a) + \dots$$

Note:

1. The Taylor's theorem in the power of $(x-a)$

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{(2)!} f''(a) + \dots + \frac{(x-a)^n}{(n)!} f^n(a) + \dots$$

2. If we expand $f(x+h)$, by Taylor's theorem as a power series in h , then

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{(2)!} f''(x) + \dots + \frac{h^n}{(n)!} f^n(x) + \dots$$

8.3 Some Important Expansions

1. Expansion of e^x :

Let $f(x) = e^x$. Then $f(0) = e^0 = 1$

Differentiate w.r.t (x) successively we get

$$f'(x) = e^x, f''(x) = e^x, f'''(x) = e^x \dots\dots f^n(x) = e^x$$

Put $x = 0$ in above relation we find

$$f'(0) = 1, f''(0) = 1, f'''(0) = 1 \dots\dots f^n(0) = 1$$

Put these values in Maclaurian's series, we get.

$$f(x) = f(0) + xf'(0) + \frac{x^2}{(2)!} f''(0) + \frac{x^3}{(3)!} f'''(0) + \dots\dots$$

$$\Rightarrow e^x = 1 + x + \frac{x^2}{(2)!} + \frac{x^2}{(3)!} + \dots\dots + \frac{x^n}{(n)!}$$

2. Expansion of $\sin x$:

[B.C.A. (Rohilkhand) 2003, 2005, 2008]

Let $f(x) = \sin x, x \in R$

$$\Rightarrow f'(x) = \cos x, f''(x) = -\sin x, f'''(x) = -\cos x$$

$$\text{In general } f^n(x) = \sin\left(x + \frac{n\pi}{2}\right)$$

Put $x = 0$ in above relation

$$\Rightarrow f(0) = 0, f'(0) = 1, f''(0) = 0, f'''(0) = -1, f^n(0) = \sin\left(\frac{n\pi}{2}\right)$$

$$\text{When } n = 2m, f^n(0) = \sin m\pi = 0$$

$$\begin{aligned} \text{When } n = 2m + 1, f^n(0) &= \sin(2m + 1) \frac{\pi}{2} = \sin\left(m\pi + \frac{\pi}{2}\right) \\ &= (-1)^m \sin\left(\frac{\pi}{2}\right) = (-1)^m \end{aligned}$$

Substituting these values in Maclaurin's series, we get

$$\sin x = 0 + x.1 + 0 + \frac{x^3}{(3)!}(-1) + 0 + \dots\dots + 0 + (-1)^m \frac{x^{2m+1}}{(2m+1)!} + \dots\dots$$

$$\text{or } \sin x = x - \frac{x^3}{(3)!} + \frac{x^5}{(5)!} - \dots\dots + (-1)^m \frac{x^{2m+1}}{(2m+1)!}$$

3. Expansion of $\cos x$:

Let $f(x) = \cos x \quad \forall x \in R$

Then, $f'(x) = -\sin x, f''(x) = -\cos x, f'''(x) = \sin x$

$$f^n(x) = \cos\left(x + \frac{n\pi}{2}\right) \quad \forall n \in N$$

Put $x = 0$ in above relation we get

$$f(0) = 1, f'(0) = 0, f''(0) = -1, f'''(0) = 0, f^{iv}(0) = 1$$

$$f^{2n}(0) = (-1)^n, \quad f^{2n+1}(0) = 0 \text{ and so on}$$

Put these value in Maclaurian series

$$f(x) = f(0) + x f'(0) + \frac{x^2}{(2)!} f''(0) + \frac{x^3}{(3)!} f'''(0) + \dots$$

$$\Rightarrow \cos x = 1 - \frac{x^2}{(2)!} + \frac{x^4}{(4)!} - \frac{x^6}{(6)!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!}$$

4. Expansion of $\log(1+x)$:

[B.C.A. (Meerut) 2011]

Let $f(x) = \log(1+x) \Rightarrow f(0) = \log 1 = 0$

$$f'(x) = \frac{1}{1+x}, \quad f''(x) = -\frac{1}{(1+x)^2}, \quad f'''(x) = \frac{(-1)^2(2)!}{(1+x)^3}$$

In general $f^n(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}$

Thus $f(0) = 0, \quad f'(0) = 1, \quad f''(0) = (-1)^1(1)!$

$$f'''(0) = (-1)^2(2)!, \quad f^{iv}(0) = (-1)^3(3)!$$

$$f^n(0) = (-1)^{n-1}(n-1)! \text{ and so on}$$

Put these value in Maclaurin's series we get

$$f(x) = f(0) + x f'(0) + \frac{x^2}{(2)!} f''(0) + \frac{x^3}{(3)!} f'''(0) + \dots$$

or $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n-1} \frac{x^n}{n} + \dots$

5. Expansion of $(1+x)^m$:

Let $f(x) = (1+x)^m$

$$\Rightarrow f'(x) = m(1+x)^{m-1}, \quad f''(x) = m(m-1)(1+x)^{m-2}$$

In general $f^n(x) = m(m-1)(m-2) \dots (m-(n-1))(1+x)^{m-n}$

$$\Rightarrow f(0) = 1, \quad f'(0) = m, \quad f''(0) = m(m-1), \quad f'''(0) = m(m-1)(m-2)$$

$$f^n(0) = m(m-1)(m-2) \dots m-(n-1),$$

Put these value in Maclaurian's series

$$f(x) = f(0) + x f'(0) + \frac{x^2}{(2)!} f''(0) + \frac{x^3}{(3)!} f'''(0) + \dots$$

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{(2)!} x^2 + \dots + \frac{m(m-1)(m-2) \dots m-(n-1)}{(n)!} x^n$$

Illustrative Examples

(Maclaurion's Series)

Example 1: Obtain by Maclaurin's theorem the first five terms in the expansion of $\log(1 + \sin x)$ [B.C.A. (Meerut) 2002, 2004, 2008]

Solution: Let $y = \log(1 + \sin x) \Rightarrow (y)_0 = 0$

$$y_1 = \frac{\cos x}{1 + \sin x} \Rightarrow (y_1)_0 = 1$$

$$y_2 = \frac{-\sin x(1 + \sin x) - \cos^2 x}{(1 + \sin x)^2} = \frac{-(1 + \sin x)}{(1 + \sin x)^2} = -\frac{1}{1 + \sin x} \Rightarrow (y_2)_0 = -1$$

$$y_3 = \frac{\cos x}{(1 + \sin x)^2} = \frac{\cos x}{1 + \sin x} \cdot \frac{1}{1 + \sin x} = -y_1 y_2$$

$$\Rightarrow (y_3)_0 = -(y_1)_0 (y_2)_0 = (-1)(-1) = 1$$

$$(y_4)_0 = -y_1 y_3 - y_2^2$$

$$\Rightarrow (y_4)_0 = -(y_1)_0 (y_3)_0 - (y_2)_0^2 = -1 - 1 = -2$$

$$\text{and } (y_5)_0 = -y_1 y_4 - y_2 y_3 - 2 y_2 y_3 = -y_1 y_4 - 3 y_2 y_3$$

$$\Rightarrow (y_5)_0 = (-1)(-2) - 3(-1)(1) = 2 + 3 = 5$$

Put these values in Maclaurin's series, we get

$$y = (y)_0 + x(y_1)_0 + \frac{x^2}{(2)!} (y_2)_0 + \frac{x^3}{(3)!} (y_3)_0 + \frac{x^4}{(4)!} (y_4)_0 + \frac{x^5}{(5)!} (y_5)_0 + \dots$$

$$\therefore \log(1 + \sin x) = 0 + x + \frac{x^2}{(2)!} (-1) + \frac{x^3}{(3)!} (1) + \frac{x^4}{(4)!} (-2) + \frac{x^5}{(5)!} (5) + \dots$$

$$\log(1 + \sin x) = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \frac{x^5}{24} - \dots$$

Example 2: Find the first four terms in the expansion of $\log(1 + \tan x)$ in the power of x .

[B.C.A. (Meerut) 2005]

Solution: Let $y = \log(1 + \tan x) \Rightarrow (y)_0 = 0$

$$\text{Now } e^y = 1 + \tan x$$

$$\text{Differentiating both sides w.r.t. } x, \text{ we get } e^y y_1 = \sec^2 x \quad \dots(1)$$

$$\text{Putting } x = 0 \Rightarrow e^{(y)_0} (y_1)_0 = \sec^2 0 \Rightarrow (y_1)_0 = 1$$

Again differentiating (1), we get

$$e^y y_1^2 + e^y y_2 = 2 \sec^2 x \tan x$$

$$\text{or } e^y (y_1^2 + y_2) = 2 \sec^2 x \tan x \quad \dots(2)$$

Putting $x = 0$ in (2), we get

$$1 + (y_2)_0 = 0 \text{ or } (y_2)_0 = -1$$

Again differentiating (2) w.r.t. (x) we get

$$e^y y_1 (y_1^2 + y_2) + e^y (2 y_1 y_2 + y_3) = 2 \sec^4 x + 4 \sec^2 x \tan^2 x$$

$$\text{or } e^y (y_1^3 + 3 y_1 y_2 + y_3) = 2 \sec^4 x + 4 \sec^2 x \tan^2 x \quad \dots(3)$$

Putting $x = 0$ in (3), we get

$$1 + 3 \cdot 1(-1) + (y_3)_0 = 2 \text{ or } (y_3)_0 = 4$$

Differentiating (3), we get

$$\begin{aligned} e^y y_1 (y_1^3 + 3 y_1 y_2 + y_3) + e^y (3 y_1^2 y_2 + 3 y_2^2 + 3 y_1 y_3 + y_4) \\ = 8 \sec^4 x \tan x + 8 \sec^2 x \tan^3 x + 8 \sec^4 x \tan x \end{aligned}$$

$$\text{or } e^y (y_1^4 + 6 y_1^2 y_2 + 4 y_1 y_3 + 3 y_2^2 + y_4) = 16 \sec^4 x \tan x + 8 \sec^2 x \tan^3 x \quad \dots(4)$$

Putting $x = 0$ in (4), we get

$$1 + 6 \cdot 1(-1) + 4 \cdot 1 \cdot 4 + 3(-1)^2 + (y_4)_0 = 0$$

$$\text{or } (y_4)_0 + 14 = 0 \text{ or } (y_4)_0 = -14$$

Now putting these values in Maclaurin's theorem, we get

$$\log(1 + \tan x) = 0 + x \cdot 1 + \frac{x^2}{(2)!}(-1) + \frac{x^3}{(3)!} \cdot 4 + \frac{x^4}{(4)!}(-14) + \dots$$

$$\text{or } \log(1 + \tan x) = x - \frac{x^2}{2} + \frac{2}{3}x^3 - \frac{7}{12}x^4 + \dots$$

Example 3: Apply Maclaurin's theorem to find the expansion in ascending powers of x of $\log_e(1 + e^x)$ to terms containing x^4 .

Solution: Let $y = \log_e(1 + e^x)$. Then $(y)_0 = \log_e(1 + e^0) = \log_e 2$.

$$\text{Now, } y_1 = \frac{e^x}{1 + e^x} = \frac{(1 + e^x) - 1}{1 + e^x} = 1 - \frac{1}{1 + e^x} \text{ so that } (y_1)_0 = 1 - \frac{1}{2} = \frac{1}{2},$$

$$y_2 = 0 + \frac{e^x}{(1 + e^x)^2} = \frac{e^x}{1 + e^x} \cdot \frac{1}{1 + e^x} = y_1(1 - y_1) = y_1 - y_1^2,$$

$$\text{So that } (y_2)_0 = (y_1)_0 - [(y_1)_0]^2 = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4},$$

$$y_3 = y_2 - 2 y_1 y_2 \text{ so that } (y_3)_0 = \frac{1}{4} - 2 \cdot \frac{1}{2} \cdot \frac{1}{4} = 0,$$

$$y_4 = y_3 - 2 y_2^2 - 2 y_1 y_3 \text{ so that } (y_4)_0 = 0 - 2 \cdot \left(\frac{1}{4}\right)^2 - 0 = -\frac{1}{8}, \text{ and so on.}$$

Now by Maclaurin's theorem, we have

$$y = (y)_0 + x(y_1)_0 + \frac{x^2}{2!}(y_2)_0 + \frac{x^3}{3!}(y_3)_0 + \frac{x^4}{4!}(y_4)_0 + \dots$$

$$\therefore \log(1 + e^x) = \log 2 + x \cdot \frac{1}{2} + \frac{x^2}{2!} \cdot \frac{1}{4} + \frac{x^3}{3!} \cdot 0 + \frac{x^4}{4!} \cdot \left(-\frac{1}{8}\right) + \dots$$

$$= \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \dots$$

Example 4: Assuming the validity of expansion, show that

$$\log \sec x = \frac{x^2}{2} + \frac{x^4}{12} + \dots$$

[B.C.A. (Rohilkhand) 2002, 2006, 2008]

Solution: Let $f(x) = \log \sec x \Rightarrow f(0) = \log(\sec 0) = \log(1) = 0$

$$f'(x) = \tan x \Rightarrow f'(0) = \tan 0 = 0$$

$$f''(x) = \sec^2 x = 1 + \tan^2 x \Rightarrow f''(0) = 1 + 0 = 1$$

We can also write $f''(x) = 1 + \{f'(x)\}^2$

$$\therefore f'''(x) = 2f'(x)f''(x) \Rightarrow f'''(0) = 2 \cdot f'(0)f''(0) = 2 \cdot 0 \cdot 1 = 0$$

$$\text{and } f^{(iv)}(x) = 2f'(x)f'''(x) + 2\{f''(x)\}^2 \Rightarrow f^{(iv)}(0) = 0 + 2 \cdot 1 = 2$$

Since Maclaurin's expansion is valid for the function $f(x) = \log \sec x$, therefore,

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots \quad \dots(1)$$

Putting the values of $f(0), f'(0), f''(0), f'''(0), f^{(iv)}(0)$ in (1), we get

$$\log \sec x = \frac{x^2}{2!} + \frac{x^4}{4!} + \dots = \frac{x^2}{2} + \frac{x^4}{12} + \dots$$

Example 5: Assuming the validity of expansion, show that

$$e^x \cos x = 1 + x - \frac{2x^3}{3!} - \frac{2^2 x^4}{4!} - \frac{2^2 x^5}{5!} + \dots$$

[B.C.A. (Meerut) 2003]

Solution: Let $f(x) = e^x \cos x \Rightarrow f(0) = e^0 \cdot \cos 0 = 1 \cdot 1 = 1$

Thus f has derivatives of all orders, and we have $f(0) = 1, f'(0) = 1,$

$$f''(0) = 0, f'''(0) = -2, f^{(iv)}(0) = -4, f^{(v)}(0) = -4 \text{ and so on.}$$

Since Maclaurin's expansion is valid for the function $f(x) = e^x \cos x$, therefore

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots \quad \dots(1)$$

Putting the values of $f(0), f'(0), f''(0), f'''(0), f^{(iv)}(0)$ in (1), we get

$$e^x \cos x = 1 + x - \frac{2x^3}{3!} - \frac{2^2 x^4}{4!} - \frac{2^2 x^5}{5!} + \dots$$

Example 6: Expand by Maclaurin's theorem $f(x) = \frac{e^x}{1+e^x}$ as far as the terms of x^3 .

[B.C.A. (Rohilkhand) 2005]

Solution:
$$f(x) = \frac{1+e^x-1}{1+e^x} = 1 - \frac{1}{1+e^x} \Rightarrow f(0) = 1 - \frac{1}{1+e^0} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow f'(x) = 0 + \frac{e^x}{(1+e^x)^2} = \frac{e^x}{1+e^x} \cdot \frac{1}{1+e^x}$$

$$\Rightarrow f'(x) = f(x)(1 - f(x)) = f(x) - [f(x)]^2 \Rightarrow f'(0) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$f''(x) = f'(x) - 2f(x)f'(x) \Rightarrow f''(0) = \frac{1}{4} - 2 \cdot \frac{1}{2} \cdot \frac{1}{4} = 0$$

$$f'''(x) = f''(x) - 2[f'(x)]^2 - 2f(x)f''(x)$$

$$\Rightarrow f'''(0) = 0 - 2\left(\frac{1}{4}\right)^2 - 0 = -\frac{1}{8} \text{ and so on.}$$

Since, function is differentiable any number of times, Maclaurin's expansion is valid for the function $f(x)$,

$$\Rightarrow f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots \quad \dots(1)$$

Putting the values of $f(0)$, $f'(0)$, $f''(0)$, $f'''(0)$, $f^{(iv)}(0)$ in (1), we get

$$f(x) = \frac{e^x}{1+e^x} = \frac{1}{2} + x \cdot \frac{1}{4} + \frac{x^2}{2!} \cdot 0 + \frac{x^3}{3!} \left(-\frac{1}{8}\right) + \dots$$

or
$$\frac{e^x}{1+e^x} = \frac{1}{2} + \frac{x}{4} - \frac{1}{48}x^3 + \dots$$

Example 7: Apply Maclaurin's theorem to find the expansion in ascending powers of x of $\log_e(1+e^x)$ upto the term containing x^4 . [B.C.A. (Delhi) 2004, 2004]

Solution: Let $f(x) = \log(1+e^x) \Rightarrow f(0) = \log_e(1+e^0) = \log_e 2$

$$f'(x) = \frac{e^x}{1+e^x} = \frac{(1+e^x)-1}{1+e^x} = 1 - \frac{1}{1+e^x} \Rightarrow f'(0) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$f''(x) = 0 + \frac{e^x}{(1+e^x)^2} = \frac{e^x}{(1+e^x)} \cdot \frac{1}{(1+e^x)} = f'(x)[1 - f'(x)]$$

$$\Rightarrow f''(x) = f'(x) - [f'(x)]^2 \Rightarrow f''(0) = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$f'''(x) = f''(x) - 2f'(x)f''(x) \Rightarrow f'''(0) = \frac{1}{4} - 2 \cdot \frac{1}{2} \cdot \frac{1}{4} = 0$$

$$f^{(iv)}(x) = f'''(x) - 2[f''(x)]^2 - 2f'(x)f'''(x)$$

$$\Rightarrow f^{(iv)}(0) = 0 - 2 \cdot \left(\frac{1}{4}\right)^2 - 0 = -\frac{1}{8} \text{ and so on.}$$

Since, Maclaurin's expansion is valid for the function $f(x) = \log(1+e^x)$. Therefore

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots \quad \dots(1)$$

Putting the values of $f(0)$, $f'(0)$, $f''(0)$, $f'''(0)$, $f^{(iv)}(0)$ in (1), we get

$$\begin{aligned} \log_e(1+e^x) &= \log 2 + x \cdot \frac{1}{2} + \frac{x^2}{2!} \cdot \frac{1}{4} + \frac{x^3}{3!} \cdot 0 + \frac{x^4}{4!} \left(-\frac{1}{8}\right) + \dots \\ &= \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \dots \end{aligned}$$

Example 8: Expand $\log\{x + \sqrt{1+x^2}\}$ in ascending powers of x and find the general term.

Solution: Let $y = \log\{x + \sqrt{1+x^2}\}$

$$\text{Then } y_1 = \frac{1}{x + \sqrt{1+x^2}} \cdot \left\{ 1 + \frac{2x}{2\sqrt{1+x^2}} \right\} = \frac{1}{\sqrt{1+x^2}} \quad \dots(2)$$

$$\therefore y_1^2(1+x^2) - 1 = 0$$

Differentiating again, we get $(1+x^2)2y_1y_2 + 2xy_1^2 = 0$

$$\text{or } 2y_1[(1+x^2)y_2 + xy_1] = 0$$

$$\text{or } (1+x^2)y_2 + xy_1 = 0 \text{ since } 2y_1 \neq 0 \quad \dots(3)$$

Now differentiating (3) n times by Leibnitz's theorem, we get

$$(1+x^2)y_{n+2} + n.y_{n+1}.2x + \frac{n(n-1)}{1.2}y_n.2 + y_{n+1}.x + n.y_n.1 = 0$$

$$\text{or } (1+x^2)y_{n+2} + (2n+1)xy_{n+1} + n^2y_n = 0 \quad \dots(4)$$

Putting $x=0$ in (1), (2), (3) and (4), we get

$$(y)_0 = 0, (y_1)_0 = 1, (y_2)_0 = 0$$

$$\text{And } (y_{n+2})_0 = -n^2(y_n)_0 \quad \dots(5)$$

Now Putting $n=1, 3, 5, \dots$ in (5), we get

$$(y_3)_0 = -1^2(y_1)_0 = -1^2$$

$$(y_5)_0 = (-3^2)(y_3)_0 = (-3^2)(-1^2) = 3^2.1^2$$

$$(y_7)_0 = (-5^2)(y_5)_0 = (-5^2)(-3^2)(-1^2) = -5^2.3^2.1^2, \text{ and so on}$$

Putting $n-2$ in place of n in (5), we get

$$(y_n)_0 = \{-(n-2)^2\}(y_{n-2})_0 \quad \dots(6)$$

$$= \{-(n-2)^2\}\{-(n-4)^2\}(y_{n-4})_0$$

$$[\because \text{replacing } n \text{ by } n-2 \text{ in (6), we have } (y_{n-2})_0 = -(n-4)^2(y_{n-4})_0]$$

Thus if n is odd, we have

$$(y_n)_0 = \{-(n-2)^2\}\{-(n-4)^2\} \dots (-5^2)(-3^2)(-1^2).1$$

$$= (-1)^{(n-1)/2} (n-2)^2 (n-4)^2 \dots 5^2.3^2.1^2 \quad \dots(7)$$

Again, putting $n=2, 4, 6, \dots$ in (5) we get

$$(y_4)_0 = -2^2.(y_2)_0 = 0, (y_6)_0 = -4^2.(y_4)_0 = 0 \text{ and so on.}$$

Thus, if n is even, we have $(y_n)_0 = 0$.

Now by Maclaurin's theorem, we have

$$y = (y)_0 + \frac{x}{1!}(y_1)_0 + \frac{x^2}{2!}(y_2)_0 + \frac{x^3}{3!}(y_3)_0 + \dots$$

$$\begin{aligned}
 \therefore \log\{x + \sqrt{1+x^2}\} &= 0 + x.1 + \frac{x^2}{2!}.0 + \frac{x^3}{3!}.(-1^2) + \frac{x^4}{4!}.0 \\
 &\quad + \frac{x^5}{5!}(3^2.1^2) + \frac{x^6}{6!}.0 + \frac{x^7}{7!}(-5^2.3^2.1^2) + \dots \\
 &= x - \frac{x^3}{3!}.1^2 + \frac{x^5}{5!}(3^2.1^2) - \frac{x^7}{7!}(5^2.3^2.1^2) + \dots
 \end{aligned}$$

The general term = $\frac{x^n}{n!}(y_n)_0$, where $(y_n)_0$ is given by (7) when n is odd and $(y_n)_0 = 0$, when n is even.

Put $(2n-1)$ in place of n in (2), we find that

$$((y_{2n-1}))_0 = (-1)^{n-1}(2n-3)^2 \dots 5^2 3^2 1^2$$

$$\begin{aligned}
 \text{Thus } \log\{x + \sqrt{1+x^2}\} &= x - 1^2 \cdot \frac{x^3}{(3)!} + 1^2 3^2 \cdot \frac{x^5}{(5)!} - 1^2 3^2 5^2 \cdot \frac{x^7}{7} + \dots \\
 &\quad + (-1)^{n-1} 1^2 3^2 5^2 \dots (2n-3)^2 \frac{x^{2n-1}}{(2n-1)!}
 \end{aligned}$$



Illustrative Examples

(Taylor's Theorem)

Example 9: Expand $\sin x$ in powers of $\left(x - \frac{\pi}{2}\right)$ by using Taylor's series.

[B.C.A. (Meerut) 2003, 2005]

Solution: Let $f(x) = \sin x$, we want to expand $f(x)$ in the powers of $\left(x - \frac{\pi}{2}\right)$

$$\begin{aligned}
 f(x) &= f\left[\frac{\pi}{2} + \left(x - \frac{\pi}{2}\right)\right] \text{ (Expand by Taylor's theorem)} \\
 &= f\left(\frac{\pi}{2}\right) + \left(x - \frac{\pi}{2}\right) f'\left(\frac{\pi}{2}\right) + \frac{\left(x - \frac{\pi}{2}\right)^2}{(2)!} f''\left(\frac{\pi}{2}\right) + \frac{\left(x - \frac{\pi}{2}\right)^3}{(3)!} f'''\left(\frac{\pi}{2}\right) + \dots \quad \dots (1)
 \end{aligned}$$

Now, $f(x) = \sin x$, therefore $f\left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} = 1$

$$f'(x) = \cos x, \text{ gives } f'\left(\frac{\pi}{2}\right) = \cos \frac{\pi}{2} = 0$$

$$f''(x) = -\sin x, \text{ gives } f''\left(\frac{\pi}{2}\right) = -\sin\left(\frac{\pi}{2}\right) = -1$$

$$f'''(x) = -\cos x \text{ gives } f'''\left(\frac{\pi}{2}\right) = -\cos \frac{\pi}{2} = 0$$

$$f^{(iv)}(x) = \sin x \text{ gives } f^{(iv)}\left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} = 1$$

Substituting these values in (1) we get

$$\begin{aligned}\sin x &= 1 + \left(x - \frac{\pi}{2}\right) \cdot 0 + \frac{1}{(2)!} \left(x - \frac{\pi}{2}\right)^2 (-1) + \frac{1}{(3)!} \left(x - \frac{\pi}{2}\right)^3 \cdot 0 + \frac{1}{(4)!} \left(x - \frac{\pi}{2}\right)^4 \cdot 1 + \dots \\ &= 1 - \frac{1}{(2)!} \left(x - \frac{\pi}{2}\right)^2 + \frac{1}{4!} \left(x - \frac{\pi}{2}\right)^4 + \dots\end{aligned}$$

Example 10: Expand $\log \sin(x+h)$ in powers of h by Taylor's theorem.

Solution: Let $f(x+h) = \log \sin(x+h)$

Then by Taylor's theorem, we have

$$\begin{aligned}\log \sin(x+h) &= f(x+h) \\ &= f(x) + \frac{h}{(1)!} f'(x) + \frac{h^2}{(2)!} f''(x) + \frac{h^3}{(3)!} f'''(x) + \dots\end{aligned}$$

Now $f(x+h) = \log \sin(x+h)$

$$\therefore f(x) = \log \sin x \quad f'(x) = \frac{1}{\sin x} \cos x = \cot x$$

$$f''(x) = -\operatorname{cosec}^2 x$$

$$f'''(x) = 2 \operatorname{cosec} x \cdot \operatorname{cosec} x \cot x = 2 \operatorname{cosec}^2 x \cot x$$

$$\text{Hence, } \log \sin(x+h) = \log \sin x + h \cot x - \frac{h^2}{(2)!} \operatorname{cosec}^2 x + \frac{2}{3!} h^3 \operatorname{cosec}^2 x \cot x + \dots$$



Comprehensive Exercise

(Maclaurian's and Taylor's Theorem)

Expand the following functions by Maclaurin's theorem.

1. a^x 2. $\tan x$ 3. $e^{x \cos x}$ 4. $\sec x$ 5. $e^{\sin x}$

6. Show that

$$(i) e^x \cos x = 1 + x - \frac{2x^3}{(3)!} - \frac{2^2 x^4}{(4)!} - \frac{2^2 x^5}{(5)!} + \dots + \cos\left(\frac{n\pi}{4}\right) \frac{2^{n/2} x^n}{(n)!} + \dots$$

$$(ii) e^x \sin x = x + x^2 - \frac{2}{(3)!} x^3 - \frac{2^2}{(5)!} x^5 + \dots + \sin\left(\frac{n\pi}{4}\right) \frac{2^{n/2} x^n}{(n)!}$$

7. (iii) $e^{ax} \sin bx = bx + abx^2 + \frac{3a^2 b - b^3}{(3)!} x^3 + \dots$

$$+ \frac{(a^2 + b^2)^{n/2}}{(n)!} x^n \sin\left(n \tan^{-1}\left(\frac{b}{a}\right)\right)$$

$$(iv) e^{ax} \cos bx = 1 + ax + \frac{a^2 - b^2}{2} x^2 + \frac{a(a^2 - 3b^2)}{(3)!} x^3 + \dots$$

$$+ \frac{(a^2 + b^2)^{n/2}}{(n)!} x^n \cos\left(n \tan^{-1}\left(\frac{b}{a}\right)\right) + \dots$$

8. (i) Expand $\sin^{-1}(x+h)$ in powers of x as far as the term x^3 .
 (ii) Prove that $\log(x+h) = \log h + \frac{x}{h} - \frac{x^2}{2h^2} + \frac{x^3}{3h^3} + \dots$
9. Expand $\tan^{-1} x$ in power of $\left(x - \frac{\pi}{4}\right)$
10. (i) Expand $2x^3 + 7x^2 + x - 1$ in the power of $x - 2$
 (ii) Write the value of α , if the Taylor's theorem
 $2x^3 + 7x^2 + x - 1 = \alpha + 53(x-2) + 19(x-2)^2 + 2(x-2)^3$
11. Expand $\log \sin x$ in the power of $(x-a)$
12. $f(mx) = f(x) + (m-1)xf'(x) + \frac{(m-1)^2}{(2)!} x^2 f''(x) + \frac{(m-1)^3}{(3)!} x^3 f'''(x) + \dots$



Answers

1.	$1 + x \log a + \frac{(x \log a)^2}{(2)!} + \dots + \frac{(x \log a)^n}{(n)!} + \dots$
2.	$x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots$
3.	$1 + x + \frac{x^2}{2} - \frac{x^3}{3} - \frac{11}{24}x^4 - \frac{x^5}{5} + \dots$
4.	$1 + \frac{x^2}{(2)!} + \frac{5x^4}{(4)!} + \frac{61}{(6)!}x^6 + \dots$
5.	$1 + x + \frac{x^2}{2} - \frac{x^4}{8}$
8.	(i) $\sin^{-1} h + x(1-h^2)^{-1/2} + \frac{x^2}{(2)!} h(1-h^2)^{-3/2} + \frac{x^3}{(3)!} (1-h^2)^{-5/2} (1+2h^2) + \dots$
9.	$\tan^{-1}\left(\frac{\pi}{4}\right) + \left(x - \frac{\pi}{4}\right) / \left(1 + \frac{\pi^2}{16}\right) - \pi \left(x - \frac{\pi}{4}\right)^2 / \left\{4 \left(1 + \frac{\pi^2}{16}\right)^2\right\} + \dots$
10.	(i) $45 + 53(x-2) + 19(x-2)^2 + 2(x-2)^3$ (ii) 45
11.	$\log \sin a + (x-a) \cot a - \frac{(x-a)^2}{(2)!} \operatorname{cosec}^2 a + \frac{(x-a)^3}{(3)!} 2 \operatorname{cosec}^2 x \cot a + \dots$



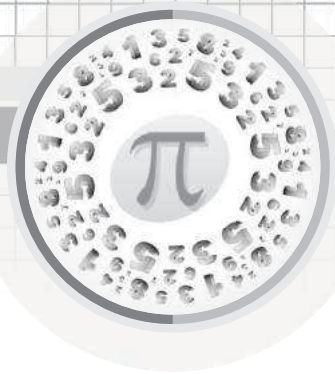


Unit-3

Chapter

9

Indeterminate Forms



9.1 Indeterminate Forms

If $f(x)$ and $g(x)$ be any two functions of x , then we know that

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{f(a)}{g(a)}$$

But if both $f(a)$ and $g(a)$ are zero

i.e. $\lim_{x \rightarrow a} f(x) \rightarrow 0$ and $\lim_{x \rightarrow a} g(x) \rightarrow 0$

Then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \left(\frac{0}{0} \right)$

This is called **indeterminate form**. This does not mean that the limit of $\frac{f(x)}{g(x)}$ is meaningless or that it does not exist.

The other indeterminate forms are:

$$\frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 1^\infty, 0^\circ \text{ and } \infty^\circ$$

Note: $\log 1 = 0, \log e = 1, \log \infty = \infty, \log 0 = -\infty, \frac{1}{0} = \infty, \frac{1}{\infty} = 0, e^\infty = \infty, e^{-\infty} = 0$

9.1.1 The form $\left(\frac{0}{0}\right)$, L' Hospital's Rule

If $f(x)$ and $g(x)$ be two functions of x which can be expanded by Taylor's theorem in the neighborhood of $x = a$ and if $f(a) = g(a) = 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided, the latter limit exists, finite or infinite.

Proof: We have by Taylor's theorem

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \left\{ \frac{f(a) + (x-a)f'(a) + \frac{(x-a)^2}{(2)!} f''(a) + \dots + R_1}{g(a) + (x-a)g'(a) + \frac{(x-a)^2}{(2)!} g''(a) + \dots + R_2} \right\}$$

Where $R_1 = \frac{(x-a)^n}{(n)!} f^n\{a + \theta_1(x-a)\}, \quad 0 < \theta_1 < 1$

And $R_2 = \frac{(x-a)^n}{(n)!} g^n\{a + \theta_2(x-a)\}, \quad 0 < \theta_2 < 1$

Now, since $f(a) = 0$ and $g(a) = 0$ we get, after dividing both numerator and denominator by $(x-a)$

$$\begin{aligned} \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \left\{ \frac{f'(a) + \frac{(x-a)}{(2)!} f''(a) + \dots}{g'(a) + \frac{(x-a)}{(2)!} g''(a) + \dots} \right\} \\ &= \frac{f'(a)}{g'(a)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \end{aligned}$$

This is known as **L'Hospital's rule**

Note:

If $f'(a) = f''(a) = \dots = f^{n-1}(a) = 0$

and $g'(a) = g''(a) = \dots = g^{n-1}(a) = 0$

But $f^n(a)$ and $g^n(a)$ are not both zero, then by repeated application of L. Hospital's rule.

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f^n(x)}{g^n(x)}$$

Rule: If the limit of $\frac{f(x)}{g(x)}$ as $x \rightarrow a$ takes the form $\left(\frac{0}{0}\right)$, differentiate the numerator and

denominator separately w.r.t. x and obtain a new function $\frac{f'(x)}{g'(x)}$. Now as $x \rightarrow a$, if it again

takes the form $\left(\frac{0}{0}\right)$, differentiate the numerator and denominator again w.r.t. x and repeat the above process, till indeterminate form persists.

9.1.2 Method of Expansion (Algebraic Method)

The following expansions should be remembered.

$$(i) \quad (1+x)^n = 1 + nx + \frac{n(n-1)}{(2)!} x^2 + \frac{n(n-1)(n-2)}{(3)!} x^3 + \dots$$

$$(ii) \quad (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots, |x| < 1$$

$$(iii) \quad a^x = 1 + x \log a + \frac{x^2}{(2)!} (\log a)^2 + \frac{x^3}{(3)!} (\log a)^3 + \dots$$

$$(iv) \quad e^x = 1 + x + \frac{x^2}{(2)!} + \frac{x^3}{(3)!} + \dots$$

$$(v) \quad \sin x = x - \frac{x^3}{(3)!} + \frac{x^5}{(5)!} - \dots$$

$$(vi) \quad \cos x = 1 - \frac{x^2}{(2)!} + \frac{x^4}{(4)!} - \dots$$

$$(vii) \quad \tan x = x + \frac{x^3}{3} + \frac{2}{15} x^5 + \dots$$

$$(viii) \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, |x| < 1$$

$$(ix) \quad \log(1-x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots\right), |x| < 1$$

$$(x) \quad \sin^{-1} x = x + \frac{x^3}{6} + \frac{3}{40} x^5 + \dots$$

$$(xi) \quad \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$(xii) \quad \sinh x = x + \frac{x^3}{(3)!} + \frac{x^5}{(5)!} + \dots$$

$$(xiii) \quad \cosh x = 1 + \frac{x^2}{(2)!} + \frac{x^4}{(4)!} + \dots$$

Illustrative Examples

Example 1: Evaluate

$$(i) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$(ii) \quad \lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$$

[B.C.A. (Garhwal) 2011]

$$(iii) \quad \lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x}$$

$$(iv) \quad \lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2}$$

[B.C.A. (Meerut) 2007]

$$(v) \quad \lim_{x \rightarrow 1} \frac{x^5 - 2x^3 - 4x^2 + 9x - 4}{x^4 - 2x^3 + 2x - 1}$$

[B.C.A. (Rohilkhand) 2002]

Solution: (i) We have $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \left(\frac{\cos x}{1} \right) = \frac{1}{1} = 1$$

(ii) $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$ $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{\cos ax \cdot a}{\cos bx \cdot b} = \frac{a}{b}$$

(iii) $\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x}$ $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{n(1+x)^{n-1}}{1} = n$$

(iv) We have $\lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2}$ $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{xe^x + e^x - \frac{1}{1+x}}{2x}$$
 $\left(\frac{0}{0}\right)$ form

$$= \lim_{x \rightarrow 0} \frac{xe^x + e^x + e^x + \frac{1}{(1+x)^2}}{2}$$
 $\left(\frac{0}{0}\right)$ form

$$= \frac{0+1+1+1}{2} = \frac{3}{2}$$

(v) $\lim_{x \rightarrow 1} \frac{x^5 - 2x^3 - 4x^2 + 9x - 4}{x^4 - 2x^3 + 2x - 1}$ $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 1} \frac{5x^4 - 6x^2 - 8x + 9}{4x^3 - 6x^2 + 2}$$
 $\left(\frac{0}{0}\right)$ form

Again by L' Hospital's rule

$$= \lim_{x \rightarrow 1} \frac{20x^3 - 12x - 8}{12x^2 - 12x}$$
 $\left(\frac{0}{0}\right)$ form

Again by L' Hospital's rule

$$= \lim_{x \rightarrow 1} \frac{60x^2 - 12}{24x - 12} = \frac{60 - 12}{24 - 12} = \frac{48}{12} = 4$$

Example 2: Evaluate

(i) $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$

[B.C.A. (Meerut) 2011]

(ii) $\lim_{x \rightarrow 0} \frac{x - \tan x}{x^3}$

(iii) $\lim_{x \rightarrow 0} \frac{\log x}{x-1}$

(iv) $\lim_{x \rightarrow 0} \frac{e^{ax} - e^{-ax}}{\log(1+bx)}$

(v) $\lim_{x \rightarrow 0} \frac{\log(1+kx^2)}{1-\cos x}$

(vi) $\lim_{x \rightarrow b} \frac{x^b - b^x}{x^x - b^b}$

Solution: (i) We have $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$ $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \left\{ \frac{1 - \cos x}{3x^2} \right\}$$

 $\left(\frac{0}{0}\right)$ form

$$= \lim_{x \rightarrow 0} \left\{ \frac{\sin x}{6x} \right\}$$

 $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule $= \lim_{x \rightarrow 0} \left\{ \frac{\cos x}{6} \right\} = \frac{1}{6}$

(ii) We have $\lim_{x \rightarrow 0} \left\{ \frac{x - \tan x}{x^3} \right\}$

 $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule $= \lim_{x \rightarrow 0} \frac{1 - \sec^2 x}{3x^2}$

 $\left(\frac{0}{0}\right)$ form

Again applying L' Hospital's rule $\lim_{x \rightarrow 0} \left(\frac{-2 \sec^2 x \tan x}{6x} \right)$

$$= \lim_{x \rightarrow 0} \left(\frac{-4 \sec^2 x \tan^2 x - 2 \sec^4 x}{6} \right) = \frac{-2}{6} = \frac{-1}{3}$$

(iii) $\lim_{x \rightarrow 1} \frac{\log x}{x-1}$

 $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \left(\frac{1/x}{1} \right) = 1$$

(iv) We have $\lim_{x \rightarrow 0} \frac{e^{ax} - e^{-ax}}{\log(1+bx)}$

 $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{ae^{ax} + e^{-ax}a}{b \left(\frac{1}{1+bx} \right)} = \frac{2a}{b}$$

(v) We have $\lim_{x \rightarrow 0} \frac{\log(1+kx^2)}{(1-\cos x)}$

 $\left(\frac{0}{0}\right)$ form

Applying L' Hospital's rule

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{\frac{1}{1+kx^2} \times 2kx}{\sin x} \left(\frac{0}{0} \right) \text{ form} \\
 &= \lim_{x \rightarrow 0} \frac{2kx}{(1+kx^2) \sin x} \left(\frac{0}{0} \right) \text{ form}
 \end{aligned}$$

Again applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \left\{ \frac{2k}{(2kx) \sin x + (1+kx^2) \cos x} \right\} = 2k$$

(vi) We have $\lim_{x \rightarrow b} \frac{x^b - b^x}{x^x - b^b} \left(\frac{0}{0} \right) \text{ form}$

Let $y = x^x \Rightarrow \log y = x \log x$

Differentiate w.r.t. 'x'

$$\frac{1}{y} \frac{dy}{dx} = x \cdot \frac{1}{x} + \log x$$

$$\Rightarrow \frac{dy}{dx} = y(1 + \log x) = x^x(1 + \log x)$$

$$\Rightarrow \frac{d}{dx}(x^x) = x^x(1 + \log x)$$

$$\begin{aligned}
 \therefore \lim_{x \rightarrow b} \frac{x^b - b^x}{x^x - b^b} &= \lim_{x \rightarrow b} \frac{bx^{b-1} - b^x \log b}{x^x(1 + \log x) - 0} \quad (\text{By L. Hospital's rule}) \\
 &= \frac{b \cdot b^{b-1} - b^b \log b}{b^b(1 + \log b)} = \frac{b^b(1 - \log b)}{b^b(1 + \log b)} = \frac{1 - \log b}{1 + \log b}.
 \end{aligned}$$

Example 3: Evaluate

(i) $\lim_{x \rightarrow 0} \frac{e^x + \log\left(\frac{1-x}{e}\right)}{\tan x - x}$ [B.C.A. (Rohilkhand) 2000, 2003]

(ii) $\lim_{x \rightarrow 0} \frac{x \cos x - \log(1+x)}{x^2}$ [B.C.A. (Meerut) 2004]

(iii) $\lim_{x \rightarrow 0} \frac{e^x - 2 \cos x + e^{-x}}{x \sin x}$ [B.C.A. (Rohilkhand) 2000]

(iv) $\lim_{x \rightarrow 0} \frac{e^x \sin x - x - x^2}{x^2 + x \log(1-x)}$ [B.C.A. (Kanpur) 2011]

Solution: (i) We have $\lim_{x \rightarrow 0} \frac{e^x + \log\left(\frac{1-x}{e}\right)}{\tan x - x}$

$$= \lim_{x \rightarrow 0} \frac{e^x + \log(1-x) - \log e}{\tan x - x} \left(\frac{0}{0} \right) \text{ form}$$

By L' Hospital's rule $\lim_{x \rightarrow 0} \frac{e^x - \frac{1}{1-x}}{\sec^2 x - 1} \quad \left(\frac{0}{0}\right) \text{ form}$

Again by L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{e^x - \frac{1}{(1-x)^2}}{2 \sec^2 x \tan x} \quad \left(\frac{0}{0}\right) \text{ form}$$

Again by L' Hospital's rule

$$= \lim_{x \rightarrow 0} \left\{ \frac{e^x - \frac{2}{(1-x)^3}}{2 \sec^4 x + 4 \sec^2 x \tan^2 x} \right\} = \frac{-1}{2}$$

(ii) $\lim_{x \rightarrow 0} \frac{x \cos x - \log(1+x)}{x^2} \quad \left(\frac{0}{0}\right) \text{ form}$

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{-x \sin x + \cos x - \frac{1}{1+x}}{2x} \quad \left(\frac{0}{0}\right) \text{ form}$$

Again by L' Hospital rule

$$= \lim_{x \rightarrow 0} \frac{-\sin x - \cos x - x \cos x + \frac{1}{(1+x)^2}}{2} = \frac{1}{2}$$

(iii) $\lim_{x \rightarrow 0} \frac{e^x - 2 \cos x + e^{-x}}{x \sin x} \quad \left(\frac{0}{0}\right) \text{ form}$

Applying L' Hospital rule

$$= \lim_{x \rightarrow 0} \frac{e^x + 2 \sin x - e^{-x}}{\sin x + x \cos x} \quad \left(\frac{0}{0}\right) \text{ form}$$

Again by L' Hospital

$$= \lim_{x \rightarrow 0} \left\{ \frac{e^x + 2 \cos x + e^{-x}}{\cos x + \cos x - x \sin x} \right\} = \frac{4}{2} = 2.$$

(iv) $\lim_{x \rightarrow 0} \frac{e^x \sin x - x - x^2}{x^2 + x \log(1-x)} \quad \left(\frac{0}{0}\right) \text{ form}$

$$= \lim_{x \rightarrow 0} \frac{\left(1+x+\frac{x^2}{(2)!}+\frac{x^3}{(3)!}+..\right)\left(x-\frac{x^3}{(3)!}+\frac{x^5}{(5)!}+...\right)-x-x^2}{x^2+x\left(-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}+.....\right)}$$

$$= \lim_{x \rightarrow 0} \frac{x+x^2-\frac{x^3}{6}+\frac{x^3}{2}+.....-x-x^2}{\frac{-1}{2}x^3+.....}$$

$$= \lim_{x \rightarrow 0} \frac{x^3 \left(\frac{1}{3} + \dots \right)}{x^3 \left(-\frac{1}{2} + \dots \right)} = \frac{1/3}{-1/2} = \frac{-2}{3}$$

Example 4: Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$

Solution: We have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5} & \quad \left(\frac{0}{0} \right) \text{ form} \\ &= \lim_{x \rightarrow 0} \frac{\left(x - \frac{x^3}{(3)!} + \frac{x^5}{(5)!} + \dots \right) - x + \frac{x^3}{6}}{x^5} \\ &= \lim_{x \rightarrow 0} \frac{\frac{x^5}{(5)!} - \frac{x^7}{(7)!}}{x^5} = \lim_{x \rightarrow 0} \left(\frac{1}{(5)!} - \frac{x^2}{(7)!} + \dots \right) = \frac{1}{(5)!} = \frac{1}{120} \end{aligned}$$

Example 5: Evaluate

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x}$$

[B.C.A (Meerut) 2003]

Solution: We have

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x} \quad \left(\frac{0}{0} \right) \text{ form}$$

As

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

Let

$$y = (1+x)^{1/x}, \text{ then}$$

$$\log y = \frac{1}{x} \log(1+x) = \frac{1}{x} \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \right)$$

$$= \left(1 - \frac{x}{2} + \frac{x^2}{3} - \dots \right)$$

$$= 1 + z \quad \left(z = -\frac{x}{2} + \frac{x^2}{3} - \dots \right)$$

$$y = e^{1+z} = e e^z = e \left[1 + \frac{z}{(1)!} + \frac{z^2}{(2)!} + \dots \right]$$

$$= e \left[1 + \left(-\frac{x}{2} + \frac{x^2}{3} - \dots \right) + \frac{1}{2} \left(-\frac{x}{2} + \frac{x^2}{3} - \dots \right)^2 + \dots \right]$$

$$= e \left[1 - \frac{x}{2} + \frac{x^2}{3} + \frac{x^2}{8} + \dots \right]$$

$$= e \left[1 - \frac{x}{2} + \frac{11}{24} x^2 + \dots \right]$$

Now,
$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x} = \lim_{x \rightarrow 0} \frac{e \left[1 - \frac{x}{2} + \frac{11}{24} x^2 + \dots \right] - e}{x}$$

$$= \lim_{x \rightarrow 0} \frac{e \left[-\frac{x}{2} + \frac{11}{24} x^2 + \dots \right]}{x}$$

$$= \lim_{x \rightarrow 0} e \left[-\frac{1}{2} + \frac{11}{24} x + \dots \right] = \frac{-e}{2}$$

Example 6: Evaluate $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2 \log(1+x)}{x \sin x}$

[B.C.A. (Meerut) 2006]

Solution: We have

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2 \log(1+x)}{x \sin x} \quad \left(\frac{0}{0} \right) \text{ form}$$

By L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - \frac{2}{1+x}}{\sin x + x \cos x} \quad \left(\frac{0}{0} \right) \text{ form}$$

$$= \lim_{x \rightarrow 0} \frac{e^x - e^{-x} + \frac{2}{(1+x)^2}}{\cos x + \cos x - x \sin x} = \frac{1-1+2}{1+1-0} = 1$$

Example 7: Evaluate $\lim_{x \rightarrow 0} \frac{x^{1/2} \tan x}{(e^x - 1)^{3/2}}$

Solution: We have $\lim_{x \rightarrow 0} \frac{x^{1/2} \tan x}{(e^x - 1)^{3/2}}$

$$= \lim_{x \rightarrow 0} \frac{x^{1/2} \tan x}{\left[1 + x + \frac{x^2}{(2)!} + \dots - 1 \right]^{3/2}} = \lim_{x \rightarrow 0} \frac{x^{1/2} \tan x}{\left(x + \frac{x^2}{2} + \dots \right)^{3/2}}$$

$$= \lim_{x \rightarrow 0} \frac{x^{1/2} \tan x}{(x)^{3/2} \left[1 + \frac{x}{2} + \dots \right]^{3/2}} = \lim_{x \rightarrow 0} \frac{\tan x}{x \left[1 + \frac{x}{2} + \dots \right]^{3/2}}$$

$$= \lim_{x \rightarrow 0} \left[\frac{1}{\left(1 + \frac{x}{2} + \dots\right)^{3/2}} \cdot \frac{\tan x}{x} \right] = 1 \cdot 1 = 1$$

Example 8: Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^2 \tan x}$

[B.C.A. (Meerut) 2003, 2007]

Solution: We have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan x - x}{x^2 \tan x} & \quad \left(\frac{0}{0}\right) \text{ form} \\ &= \lim_{x \rightarrow 0} \frac{x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots - x}{x^2 \left[x + \frac{x^3}{3} + \dots \right]} \\ &= \lim_{x \rightarrow 0} \frac{x^3 \left[\frac{1}{3} + \frac{2}{15}x^2 + \dots \right]}{x^3 \left[1 + \frac{x^2}{3} + \dots \right]} = \frac{1}{3} \end{aligned}$$

Example 9: $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$

[B.C.A. (Rohilkhand) 2005]

Solution: We have

$$\lim_{x \rightarrow 0} \left(\frac{a^x - b^x}{x} \right) \quad \left(\frac{0}{0}\right) \text{ form}$$

Applying L. Hospital's rule

$$= \lim_{x \rightarrow 0} \left(\frac{a^x \log a - b^x \log b}{1} \right) = \log a - \log b = \log \left(\frac{a}{b} \right)$$

Example 10: Find the value of a and b in order that

$$\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} \text{ may be equal to } 1.$$

Solution: We have $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} \quad \left(\frac{0}{0}\right) \text{ form}$

$$= \lim_{x \rightarrow 0} \frac{x \left(1 + a \left(1 - \frac{x^2}{(2)!} + \frac{x^4}{(4)!} \right) \dots \right) - b \left(x - \frac{x^3}{(3)!} + \frac{x^5}{(5)!} + \dots \right)}{x^3} = 1$$

$$= \lim_{x \rightarrow 0} \left\{ \frac{1+a-b}{x^2} + x^3 \frac{\left(-\frac{a}{2} + \frac{b}{6}\right) + \dots}{x^3} \right\} = 1$$

$$\Rightarrow 1+a-b=0 \text{ and } -\frac{a}{2} + \frac{b}{6} = 1$$

Solve these equations, we get $a = -\frac{5}{2}, b = -\frac{3}{2}$

Example 11: Find the value of a, b, c so that

$$\lim_{x \rightarrow 0} \frac{ae^x - b \cos x + ce^{-x}}{x \sin x} = 2$$

Solution: We have

$$\lim_{x \rightarrow 0} \frac{ae^x - b \cos x + ce^{-x}}{x \sin x} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{a \left(1 + x + \frac{x^2}{(2)!} + \frac{x^3}{(3)!} + \dots \right) - b \left(1 - \frac{x^2}{(2)!} + \frac{x^4}{(4)!} + \dots \right) + c \left(1 - x + \frac{x^2}{(2)!} - \frac{x^3}{(3)!} + \dots \right)}{x \left[x - \frac{x^3}{(3)!} + \frac{x^5}{(5)!} + \dots \right]} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{(a-b+c) + x(a-c) + x^2 \left(\frac{a}{2} + \frac{b}{2} + \frac{c}{2} \right) + \dots}{x^2 \left(1 - \frac{x^2}{(3)!} + \frac{x^4}{(5)!} + \dots \right)} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\frac{a-b+c}{x^2} + \frac{a-c}{x} + \frac{a}{2} + \frac{b}{2} + \frac{c}{2}}{1 - \frac{x^2}{(3)!} + \frac{x^4}{(5)!}} = 2$$

$$\Rightarrow a-b+c=0, a-c=0 \text{ and } a+b+c=4$$

Solve these equations, we get

$$a=1, b=2 \text{ and } c=1$$



Comprehensive Exercise 9.1

Evaluate

$$1. \quad \lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2}}{x^2}$$

$$2. \quad \lim_{x \rightarrow 0} \frac{\log(1 - x^2)}{\log \cos x}$$

$$3. \quad \lim_{x \rightarrow 0} \frac{\tan x}{x}$$

$$4. \quad \lim_{x \rightarrow 0} \frac{a^x - 1}{b^x - 1}$$

$$5. \quad \lim_{x \rightarrow 0} \frac{x - \sin x}{\tan^3 x}$$

$$6. \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2}$$

$$7. \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \log(1 + x)}$$

$$8. \quad \lim_{x \rightarrow 0} \frac{\log(1 + x^3)}{\sin^3 x}$$

$$9. \quad \lim_{x \rightarrow 0} \frac{\sin x \sin^{-1} x}{x^2}$$

$$10. \quad \lim_{x \rightarrow 0} \frac{e^x - e^{x \cos x}}{x - \sin x}$$

$$11. \quad \lim_{x \rightarrow b} \frac{a^b - b^a}{a^a - b^b}$$

$$12. \quad \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x^2 \sin x}$$

$$13. \quad \lim_{x \rightarrow 0} \frac{\cosh x - \cos x}{x \sin x}$$

$$14. \quad \lim_{x \rightarrow 0} \frac{e^x - \log(e + ex)}{x^2}$$

$$15. \quad \lim_{x \rightarrow 1/2} \frac{\cos^2 \pi x}{e^{2x} - 2ex}$$

$$16. \quad \lim_{x \rightarrow 0} \frac{a^x - 1 - x \log a}{x^2}$$

$$17. \quad \lim_{x \rightarrow 0} \frac{\sin 2x + 2 \sin^2 x - 2 \sin x}{\cos x - \cos^2 x}$$

$$18. \quad \lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{ex}{2}}{x^2}$$

$$19. \quad \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

$$20. \quad \text{Find the values of } a, b, c \text{ so that } \lim_{x \rightarrow 0} \frac{x(a + b \cos x) - c \sin x}{x^5} = 1.$$



Answers

1.	$\frac{1}{2}$	2.	2	3.	1
4.	$\log_b a$	5.	$\frac{1}{6}$	6.	$\frac{1}{6}$
7.	$\frac{1}{2}$	8.	1	9.	1
10.	3	11.	$\frac{1 - \log b}{1 + \log b}$	12.	$\frac{1}{3}$
13.	1	14.	1	15.	$\frac{\pi^2}{2e}$
16.	$\frac{1}{2}(\log a)^2$	17.	4	18.	$\frac{11e}{24}$
19.	$\frac{1}{2}$	20.	$a = 120, \quad b = 60, \quad c = 180$		

9.1.3 When form $\left(\frac{\infty}{\infty}\right)$, then Applying L' Hospital's Rule

Note: $\log 0 = -\infty$, $\log \infty = \infty$

Example 12: Evaluate

(i) $\lim_{x \rightarrow \infty} \frac{x^3}{e^x}$

[B.C.A. (Meerut) 2008]

(ii) $\lim_{x \rightarrow 0} \frac{\log x}{\cot x}$

[B.C.A. (Meerut) 2000]

(iii) $\lim_{x \rightarrow \infty} \frac{\log x}{a^x}$

(iv) $\lim_{x \rightarrow \pi/2} \frac{\log\left(x - \frac{\pi}{2}\right)}{\tan x}$

(v) $\lim_{x \rightarrow 0} \frac{\log x^2}{\cot x^2}$

(vi) $\lim_{x \rightarrow \infty} x^m e^{-x}$

Solution: (i) We have $\lim_{x \rightarrow \infty} \left(\frac{x^3}{e^x}\right)$

$\left(\frac{\infty}{\infty} \text{ form}\right)$

Applying L' Hospital's rule = $\lim_{x \rightarrow \infty} \left(\frac{3x^2}{e^x}\right)$

$\left(\frac{\infty}{\infty} \text{ form}\right)$

$$\begin{aligned}
 &= \lim_{x \rightarrow \infty} \left(\frac{6x}{e^x} \right) \quad \left(\frac{\infty}{\infty} \text{ form} \right) \\
 &= \lim_{x \rightarrow \infty} \left(\frac{6}{e^x} \right) = \frac{6}{e^\infty} = \frac{6}{\infty} = 0
 \end{aligned}$$

(ii) We have $\lim_{x \rightarrow 0} \frac{\log x}{\cot x} \quad \left(\frac{-\infty}{\infty} \text{ form} \right)$

Applying L' Hospital rule

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left(\frac{1/x}{-\operatorname{cosec}^2 x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{-\sin^2 x}{x} \right) \quad \left(\frac{0}{0} \text{ form} \right)
 \end{aligned}$$

Again by L' Hospital's rule

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left(\frac{-2 \sin x \cos x}{1} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{-\sin^2 x}{1} \right) = \frac{-0}{1} = 0
 \end{aligned}$$

(iii) $\lim_{x \rightarrow \infty} \left(\frac{\log x}{a^x} \right) \quad \left(\frac{\infty}{\infty} \text{ form} \right)$

By L' Hospital's rule

$$= \lim_{x \rightarrow \infty} \left(\frac{1/x}{a^x \log a} \right) = \frac{0}{\infty} = 0 \times 0 = 0$$

(iv) $\lim_{x \rightarrow \pi/2} \left\{ \frac{\log \left(x - \frac{\pi}{2} \right)}{\tan x} \right\} \quad \left(\frac{\infty}{\infty} \text{ form} \right)$

By L' Hospital's rule

$$\begin{aligned}
 &= \lim_{x \rightarrow \pi/2} \left\{ \frac{1}{\sec^2 x} \right\} = \lim_{x \rightarrow \pi/2} \left\{ \frac{\cos^2 x}{x - \pi/2} \right\} \\
 &= \lim_{x \rightarrow \pi/2} \frac{-2 \cos x \sin x}{1} = \frac{0}{1} = 0 \text{ [By L' Hospital's rule]}
 \end{aligned}$$

(v) $\lim_{x \rightarrow 0} \frac{\log x^2}{\cot x^2} \quad \left(\frac{\infty}{\infty} \text{ form} \right)$

By L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{1/x^2 \times 2x}{-\operatorname{cosec}^2 x^2 \times 2x}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left(\frac{-\sin^2 x^2}{x^2} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sin x^2}{x^2} \right) (-\sin x^2) = 1 \times 0 = 0
 \end{aligned}$$

$$\begin{aligned}
 \text{(vi)} \quad \lim_{x \rightarrow \infty} x^m e^{-x} &= \lim_{x \rightarrow \infty} \frac{x^m}{e^x} && \left(\frac{\infty}{\infty} \text{ form} \right) \\
 &= \lim_{x \rightarrow \infty} \frac{mx^{m-1}}{e^x} && \left(\frac{\infty}{\infty} \text{ form} \right)
 \end{aligned}$$

Again by L' Hospital's rule

$$= \lim_{x \rightarrow \infty} \frac{m(m-1)x^{m-2}}{e^x} \quad \left(\frac{\infty}{\infty} \text{ form} \right)$$

Continuous upto n time

$$= \lim_{x \rightarrow \infty} \frac{m(m-1)(m-2)\dots 3.2.1}{e^x} = \lim_{x \rightarrow \infty} \frac{(m)!}{e^\infty} = 0$$

Example 13: Evaluate $\lim_{x \rightarrow 0} \frac{\log \log (1-x^2)}{\log \log \cos x}$ $\left(\frac{\infty}{\infty} \text{ form} \right)$

Applying L' Hospital's rule

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{\frac{1}{\log(1-x^2)} \cdot \frac{1}{1-x^2} (-2x)}{\frac{1}{\log \cos x} \cdot \frac{1}{\cos x} \cdot (-\sin x)} \\
 &= 2 \lim_{x \rightarrow 0} \frac{x \cos x \log \cos x}{\sin x (1-x^2) \log(1-x^2)} \\
 &= 2 \lim_{x \rightarrow 0} \frac{x}{\sin x} \cdot \lim_{x \rightarrow 0} \frac{\cos x}{1-x^2} \cdot \lim_{x \rightarrow 0} \frac{\log \cos x}{\log(1-x^2)} \\
 &= 2 \times 1 \times 1 \times \lim_{x \rightarrow 0} \frac{\log \cos x}{\log(1-x^2)} \\
 &= 2 \lim_{x \rightarrow 0} \frac{\frac{1}{\cos x} (-\sin x)}{\frac{1}{1-x^2} (-2x)} \\
 &= 2 \times \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \cdot \frac{1-x^2}{\cos x} \right) = 1.
 \end{aligned}$$

9.1.4 Form $(0 \times \infty)$. This form can be written in the form $\left(\frac{0}{\infty}\right)$ or $\left(\frac{\infty}{0}\right)$ i.e.,

$$0 \times \infty = \frac{0}{1/\infty} = \frac{0}{0}$$

And

$$0 \times \infty = \frac{\infty}{1/0} = \frac{\infty}{\infty}$$

Then used L' Hospital rule or otherwise.

Illustrative Examples

Example 14: Evaluate

(i) $\lim_{x \rightarrow 1} \left(\sec \frac{\pi}{2x} \right) \log x$

[B.C.A. (Meerut) 2002]

(ii) $\lim_{x \rightarrow 0} x \log x$

(iii) $\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2}$

[B.C.A. (kanpur) 2006]

Solution:

(i) $\lim_{x \rightarrow 1} \sec \left(\frac{\pi}{2x} \right) \log x$

[form $\infty \times 0$]

$$= \lim_{x \rightarrow 1} \frac{\log x}{\cos \left(\frac{\pi}{2x} \right)}$$

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 1} \frac{1/x}{-\sin \left(\frac{\pi}{2x} \right) \left(-\frac{\pi}{2x^2} \right)} = \lim_{x \rightarrow 1} \frac{2x}{\pi} \operatorname{cosec} \left(\frac{\pi}{2x} \right) = \frac{2}{\pi}$$

(ii) $\lim_{x \rightarrow 0} x \log x$

[$0 \times \infty$ form]

$$= \lim_{x \rightarrow 0} \frac{\log x}{1/x} \quad \left(\frac{\infty}{\infty} \text{ form} \right)$$

By L' Hospital's rule

$$= \lim_{x \rightarrow 0} \left(\frac{1/x}{-1/x^2} \right) = \lim_{x \rightarrow 0} (-x) = 0$$

(iii) Similar to (ii)

9.1.5 Form $[\infty - \infty]$. If $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$

Then $\lim_{x \rightarrow a} [f(x) - g(x)]$ becomes of the form $(\infty - \infty)$ we write it as

$$\lim_{x \rightarrow a} \left[\frac{\frac{1}{g(x)} - \frac{1}{f(x)}}{\frac{1}{f(x) \cdot g(x)}} \right] \quad \left[\text{form } \frac{0}{0} \right]$$

Then applying L' Hospital's rule

Illustrative Examples

Example 15: Evaluate

(i) $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$

[B.C.A. (Meerut) 2000, 2006]

(ii) $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \cot^2 x \right)$

[B.C.A. (Meerut) 2006]

Solution: (i) We have $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$

$(\infty - \infty \text{ form})$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin^2 x - x^2}{x^2 \sin^2 x} \right) \quad \left(\frac{0}{0} \text{ form} \right)$$

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \left(\frac{2 \sin x \cos x - 2x}{2x \cdot \sin^2 x + x^2 (2 \sin x \cos x)} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin 2x - 2x}{2x \cdot \sin^2 x + x^2 \sin 2x} \quad \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{x \rightarrow 0} \left\{ \frac{2 \cos 2x - 2}{2 \sin^2 x + 4x \sin x \cos x + 2x \sin 2x + x^2 2 \cos 2x} \right\}$$

$$= \lim_{x \rightarrow 0} \frac{2 \cos 2x - 2}{2 \sin^2 x + 4x \sin 2x + 4x^2 \cos 2x} \quad \left(\frac{0}{0} \text{ form} \right)$$

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{-4 \sin 2x}{4 \sin x \cos x + 4 \sin 2x + 8x \cos 2x} + 4x \cos 2x + 2x^2 (-2 \sin 2x)$$

$$= \lim_{x \rightarrow 0} \frac{-4 \sin 2x}{2 \sin 2x + 4 \sin 2x + 8x \cos 2x} + 4x \cos 2x - 4x^2 \sin 2x$$

$$= \lim_{x \rightarrow 0} \frac{-4 \sin 2x}{2 \sin 2x + 4 \sin 2x + 12x \cos 2x - 4x^2 \sin 2x} \quad \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{x \rightarrow 0} \frac{-8 \cos 2x}{4 \cos 2x + 8 \cos 2x + 12 \cos 2x - 24x \sin 2x} - 8x \sin 2x - 8x^2 \cos 2x$$

$$= \frac{-8}{24} = \frac{-1}{3}$$

$$(ii) \quad \lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \cot^2 x \right) \quad [(\infty - \infty) \text{ form}]$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{\cos^2 x}{\sin^2 x} \right) = \lim_{x \rightarrow 0} \frac{\sin^2 x - x^2 \cos^2 x}{x^2 \sin^2 x} \\ &= \lim_{x \rightarrow 0} \left[\frac{\sin^2 x - x^2 \cos^2 x}{x^4} \cdot \frac{x^2}{\sin^2 x} \right] \\ &= \lim_{x \rightarrow 0} \left[\frac{\sin^2 x - x^2 \cos^2 x}{x^4} \right] \\ &= \lim_{x \rightarrow 0} \frac{\left[x^2 - \frac{2x^4}{(3)!} + \dots \right] - x^2 [1 - x^2 + \dots]}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{x^2 - \frac{x^4}{3} + \dots - x^2 + x^4 + \dots}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{\frac{2}{3}x^4 + \dots}{x^4} = \frac{2}{3} \end{aligned}$$

Example 16: (i) $\lim_{x \rightarrow 0} \left[\frac{1}{x} - \frac{1}{x^2} \log(1+x) \right]$

[B.C.A. (Kanpur) 2004, 2007]

$$(ii) \quad \lim_{x \rightarrow 0} \left\{ \frac{1}{x(1+x)} - \frac{\log(1+x)}{x^2} \right\}$$

[B.C.A. (Agra) 2003]

$$(iii) \quad \lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$$

[B.C.A. (Garhwal) 2010]

$$(iv) \quad \lim_{x \rightarrow 1} \left[\frac{2}{x^2 - 1} - \frac{1}{x - 1} \right]$$

[B.C.A. (Bundelkhand) 2000]

Solution: (i) $\lim_{x \rightarrow 0} \left[\frac{1}{x} - \frac{1}{x^2} \log(1+x) \right] = \lim_{x \rightarrow 0} \left[\frac{x - \log(1+x)}{x^2} \right] \quad \left(\frac{0}{0} \text{ form} \right)$

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \left\{ \frac{1 - \frac{1}{1+x}}{2x} \right\} = \lim_{x \rightarrow 0} \frac{1+x-1}{2x(1+x)} = \lim_{x \rightarrow 0} \frac{1}{2(1+x)} = \frac{1}{2}.$$

$$(ii) \quad \text{We have } \lim_{x \rightarrow 0} \left[\frac{1}{x(1+x)} - \frac{\log(1+x)}{x^2} \right]$$

$$= \lim_{x \rightarrow 0} \left[\frac{x - (1+x) \log(1+x)}{x^2(1+x)} \right] \quad \left(\frac{0}{0} \text{ form} \right)$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{1 - (1+x) \cdot \frac{1}{(1+x)} - \log(1+x)}{2x + 3x^2} \\
 &= \lim_{x \rightarrow 0} \frac{-\log(1+x)}{2x + 3x^2} \quad \left(\frac{0}{0} \text{ form} \right)
 \end{aligned}$$

Applying L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{1+x}}{2+6x} = -\frac{1}{2}$$

$$(iii) \quad \lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right) = \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{\cos x}{\sin x} \right) = \lim_{x \rightarrow 0} \left(\frac{\sin x - x \cos x}{x \sin x} \right) \left(\frac{0}{0} \text{ form} \right)$$

Applying L' Hospital's rule

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left(\frac{\cos x - \cos x + x \sin x}{\sin x + x \cos x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{x \sin x}{\sin x + x \cos x} \right) \quad \left(\frac{0}{0} \text{ form} \right)
 \end{aligned}$$

Applying L' Hospital's rule

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left(\frac{x \cos x + \sin x}{\cos x + \cos x - x \sin x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{x \cos x + \sin x}{2 \cos x - x \sin x} \right) = \frac{0}{2} = 0
 \end{aligned}$$

$$\begin{aligned}
 (iv) \quad \lim_{x \rightarrow 1} \left[\frac{2}{x^2 - 1} - \frac{1}{x - 1} \right] &\quad (\infty - \infty \text{ form}) \\
 &= \lim_{x \rightarrow 1} \left\{ \frac{2 - (x+1)}{x^2 - 1} \right\} = \lim_{x \rightarrow 1} \left\{ \frac{1-x}{x^2 - 1} \right\} = \lim_{x \rightarrow 1} \left\{ \frac{-1}{2x} \right\} = -\frac{1}{2}.
 \end{aligned}$$

9.1.6 When the forms $0^0, 1^\infty, \infty^0$

Suppose $\lim_{x \rightarrow a} [f(x)]^{\phi(x)}$ take any one of these forms

Then let $y = \lim_{x \rightarrow a} [f(x)]^{\phi(x)}$

Taking logarithm of both sides, we get

$$\log y = \lim_{x \rightarrow a} \phi(x) \cdot \log f(x)$$

Now in any of the above form $\log y$ takes the form $0 \times \infty$ which is changed to form $\left(\frac{0}{0} \right)$ or $\left(\frac{\infty}{\infty} \right)$. Then applying L' Hospital's rule.

Illustrative Examples

Example 17: Evaluate

(i) $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x$

[B.C.A. (Garhwal) 2009]

(ii) $\lim_{x \rightarrow 0} (1+x)^{1/x}$

[B.C.A. (Meerut) 2005]

(iii) $\lim_{x \rightarrow 0} (\cos x)^{\cot^2 x}$

[B.C.A. (Rohilkhand) 2007; (Delhi) 2011]

(iv) $\lim_{x \rightarrow 0} (\cos x)^{1/x}$

[B.C.A. (Meerut) 2001, 2002, 2006]

Solution: (i) Let $y = \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x$ [form 1^∞]

$$\begin{aligned} \therefore \log y &= \lim_{x \rightarrow \infty} \left\{ x \log \left(1 + \frac{a}{x}\right) \right\} \quad [0 \times \infty \text{ form}] \\ &= \lim_{x \rightarrow \infty} \left\{ \frac{\log \left(1 + \frac{a}{x}\right)}{1/x} \right\} \quad \left(\frac{0}{0} \text{ form} \right) \end{aligned}$$

Applying L' Hospital's rule

$$= \lim_{x \rightarrow \infty} \left\{ \frac{\frac{1}{1 + \frac{a}{x}} \left(-\frac{a}{x^2}\right)}{\left(-\frac{1}{x^2}\right)} \right\} = \lim_{x \rightarrow \infty} \left(\frac{a}{1 + \frac{a}{x}} \right) = a \Rightarrow y = e^a$$

(ii) We have $\lim_{x \rightarrow 0} (1+x)^{1/x}$

Let $y = \lim_{x \rightarrow 0} (1+x)^{1/x}$ [1^∞ form]

$$\begin{aligned} \Rightarrow \log y &= \lim_{x \rightarrow 0} \left\{ \frac{\log(1+x)}{x} \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \frac{\frac{1}{1+x}}{\frac{1}{1}} \right\} = \lim_{x \rightarrow 0} \left\{ \frac{1}{1+x} \right\} = 1 \end{aligned}$$

$\Rightarrow y = e^1 = e$

or $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$

$$\begin{aligned}
 \text{(iii) We have } y &= \lim_{x \rightarrow 0} (\cos x)^{\cot^2 x} && [1^\infty \text{ form}] \\
 \therefore \log y &= \lim_{x \rightarrow 0} \cot^2 x \cdot \log(\cos x) && (0 \times \infty \text{ form}) \\
 &= \lim_{x \rightarrow 0} \left\{ \frac{\log \cos x}{\tan^2 x} \right\} && \left(\frac{0}{0} \text{ form} \right)
 \end{aligned}$$

Applying L' Hospital's rule

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left\{ \frac{\frac{1}{\cos x} \times -\sin x}{2 \tan x \cdot \sec^2 x} \right\} \\
 &= \lim_{x \rightarrow 0} \left\{ \frac{-\tan x}{2 \tan x \cdot \sec^2 x} \right\} \\
 &= \lim_{x \rightarrow 0} \left\{ \frac{-1}{2 \sec^2 x} \right\} = \frac{-1}{2} \\
 \therefore y &= e^{-1/2} \\
 \text{or } \lim_{x \rightarrow 0} (\cos x)^{\cot^2 x} &= \frac{-1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) We have } y &= \lim_{x \rightarrow 0} (\cos x)^{1/x} && [1^\infty \text{ form}] \\
 \therefore \log y &= \lim_{x \rightarrow 0} \left(\frac{1}{x} \right) \log \cos x && [\infty \times 0 \text{ form}] \\
 &= \lim_{x \rightarrow 0} \left(\frac{\log \cos x}{x} \right) && \left[\frac{0}{0} \text{ form} \right]
 \end{aligned}$$

$$\text{Applying L. Hospital's rule} = \lim_{x \rightarrow 0} \frac{\frac{1}{\cos x} (-\sin x)}{1} = 0$$

$$\therefore y = e^0 = 1$$

Example 18: Evaluate

$$\text{(i) } \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x} \quad [\text{B.C.A. (Rohtak) 2009}]$$

$$\text{(ii) } \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2} \quad [\text{B.C.A. (Rohilkhand) 2005}]$$

$$\text{(iii) } \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^3} \quad [\text{B.C.A. (Rohilkhand) 2000, 2003}]$$

$$\text{(iv) } \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x} \quad [\text{B.C.A. (Meerut) 2003}]$$

$$\text{(v) } \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2} \quad [\text{B.C.A. (Meerut) 2005, 2008}]$$

$$(vi) \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^3}$$

[B.C.A. (Meerut) 2006]

Solution: (i) We have $y = \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x}$ [1^∞ form]

$$\begin{aligned} \therefore \log y &= \lim_{x \rightarrow 0} \frac{1}{x} \log \left(\frac{\tan x}{x} \right) \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \log \frac{\left\{ x + \frac{x^3}{3} + \frac{2}{15} x^5 + \dots \right\}}{x} \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \log \left\{ 1 + \frac{x^2}{3} + \frac{2}{15} x^4 + \dots \right\} \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \log (1+z) \text{ where } z = \frac{x^2}{3} + \frac{2}{15} x^4 + \dots \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \left\{ z - \frac{z^2}{2} + \dots \right\} \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \left\{ \left(\frac{x^2}{3} + \frac{2}{15} x^4 + \dots \right) - \frac{1}{2} \left(\frac{x^2}{3} + \frac{2}{15} x^4 + \dots \right)^2 \right\} \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \left\{ \frac{x^2}{3} + \left(\frac{2}{15} - \frac{1}{18} \right) x^4 + \dots \right\} = \lim_{x \rightarrow 0} \frac{1}{x} \left\{ \frac{x^2}{3} + \frac{7}{90} x^4 + \dots \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \frac{x}{3} + \frac{7}{90} x^3 + \dots \right\} = 0 \\ \therefore y &= e^0 = 1. \end{aligned}$$

(ii) We have $y = \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2}$ [1^∞ form]

$$\begin{aligned} \therefore \log y &= \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left(\frac{\tan x}{x} \right) \\ &= \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left[\frac{\left(x + \frac{x^3}{3} + \frac{2}{15} x^5 + \dots \right)}{x} \right] \\ &= \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left(1 + \frac{x^2}{3} + \frac{2}{15} x^4 + \dots \right) \\ &= \lim_{x \rightarrow 0} \frac{1}{x^2} \log(1+z), \text{ where } z = \frac{x^2}{3} + \frac{2}{15} x^4 + \dots \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ z - \frac{z^2}{2} + \dots \right\} \\
&= \lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ \frac{x^2}{3} + \frac{2}{15} x^4 + \dots - \frac{1}{2} \left(\frac{x^2}{3} + \frac{2}{15} x^4 + \dots \right)^2 + \dots \right\} \\
&= \lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ \frac{x^2}{3} + \frac{7}{90} x^4 + \dots \right\} \\
&= \lim_{x \rightarrow 0} \left\{ \frac{1}{3} + \frac{7}{90} x^2 + \dots \right\} = \frac{1}{3}
\end{aligned}$$

$$\therefore y = e^{1/3}$$

(iii) Similar to (ii) $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^3} = \infty$.

(iv) We have $y = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x}$ [1^∞ form]

$$\begin{aligned}
\therefore \log y &= \lim_{x \rightarrow 0} \frac{1}{x} \left\{ \log \left(\frac{\sin x}{x} \right) \right\} \\
&= \lim_{x \rightarrow 0} \frac{1}{x} \log \left(\frac{x - \frac{x^3}{(3)!} + \frac{x^5}{(5)!} + \dots}{x} \right) \\
&= \lim_{x \rightarrow 0} \frac{1}{x} \log \left\{ 1 - \left(\frac{x^2}{6} - \frac{x^4}{120} + \dots \right) \right\} \\
&= \lim_{x \rightarrow 0} \frac{1}{x} \log (1 - z) \qquad \left[z = \frac{x^2}{6} - \frac{x^4}{120} \right] \\
&= \lim_{x \rightarrow 0} \frac{1}{x} \left(-z - \frac{z^2}{2} - \dots \right) \\
&= \lim_{x \rightarrow 0} \frac{1}{x} \left(- \left(\frac{x^2}{6} - \frac{x^4}{120} \right) - \frac{1}{2} \left(\frac{x^2}{6} - \frac{x^4}{120} \right)^2 - \dots \right) \\
&= \lim_{x \rightarrow 0} \frac{1}{x} \left[-\frac{x^2}{6} + \left(\frac{x^4}{120} - \frac{x^4}{72} \right) + \dots \right] \\
&= \lim_{x \rightarrow 0} \frac{1}{x} \left[-\frac{x^2}{6} - \frac{x^4}{180} + \dots \right] \\
&= \lim_{x \rightarrow 0} \left[-\frac{x}{6} - \frac{x^3}{180} + \dots \right] = 0
\end{aligned}$$

$$\therefore y = e^0 = 1$$

or $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x} = 1$

(v) We have

$$y = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}$$

$$\begin{aligned} \therefore \log y &= \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left\{ \frac{x - \frac{x^3}{(3)!} + \frac{x^5}{(5)!}}{x} \right\} \\ &= \lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ \log \left[1 - \frac{x^2}{(3)!} + \frac{x^4}{(5)!} + \dots \right] \right\} \\ &= \lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ -\frac{x^2}{6} - \frac{x^4}{180} + \dots \right\} \\ &= \lim_{x \rightarrow 0} \left\{ -\frac{1}{6} - \frac{x^2}{180} + \dots \right\} = -\frac{1}{6} \\ \Rightarrow y &= e^{-1/6} \end{aligned}$$

(vi) We have

$$y = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^3}$$

$$\begin{aligned} \therefore \log y &= \lim_{x \rightarrow 0} \frac{1}{x^3} \log \left(\frac{\sin x}{x} \right) \\ \log y &= \lim_{x \rightarrow 0} \frac{1}{x^3} \log \left(\frac{x - \frac{x^3}{3!} + \frac{x^5}{(5)!} + \dots}{x} \right) \\ &= \lim_{x \rightarrow 0} \frac{1}{x^3} \log \left\{ 1 - \frac{x^2}{6} + \frac{x^4}{(5)!} + \dots \right\} \\ &= \lim_{x \rightarrow 0} \frac{1}{x^3} \left\{ -\frac{x^2}{6} - \frac{x^4}{180} + \dots \right\} = -\infty \\ y &= e^{-\infty} = 0 \end{aligned}$$

$$\therefore \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^3} = 0$$

Example 19: $\lim_{x \rightarrow 0} \left(\frac{\sinh x}{x} \right)^{1/x^2}$

Solution: Let

$$y = \lim_{x \rightarrow 0} \left(\frac{\sinh x}{x} \right)^{1/x^2}$$

$$\begin{aligned} \log y &= \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left(\frac{\sinh x}{x} \right) \\ &= \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left\{ \frac{x + \frac{x^3}{(3)!} + \frac{x^5}{(5)!} + \dots}{x} \right\} \end{aligned}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{1}{x^2} \log \left\{ 1 + \frac{x^2}{6} + \frac{x^4}{120} + \dots \right\} \\
 &= \lim_{x \rightarrow 0} \frac{1}{x^2} \{ \log(1+z) \}, \text{ where } z = \frac{x^2}{6} + \frac{x^4}{120} \\
 &= \lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ z - \frac{z^2}{2} + \dots \right\} = \lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ \frac{x^2}{6} - \frac{x^4}{180} + \dots \right\} = \frac{1}{6}
 \end{aligned}$$

$$\therefore y = e^{1/6}$$

Example 20: Evaluate, $\lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)}$

Solution: We have

$$y = \lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)}$$

$$\therefore \log y = \lim_{x \rightarrow a} \tan\left(\frac{\pi x}{2a}\right) \log\left(2 - \frac{x}{a}\right) \quad [\infty \times 0 \text{ form}]$$

$$= \lim_{x \rightarrow a} \frac{\log\left(2 - \frac{x}{a}\right)}{\cot\left(\frac{\pi x}{2a}\right)} \quad \left[\frac{0}{0} \text{ form} \right]$$

$$\begin{aligned}
 \text{Applying L. Hospital's rule} &= \lim_{x \rightarrow 0} \frac{\left(2 - \frac{x}{a}\right)^{-1} \left(-\frac{1}{a}\right)}{-\operatorname{cosec}^2\left(\frac{\pi x}{2a}\right) \times \frac{\pi}{2a}} \\
 &= \lim_{x \rightarrow a} \frac{1}{a} \frac{2a}{\pi} \frac{1}{\left(2 - \frac{x}{a}\right)} \sin^2\left(\frac{\pi x}{2a}\right) = \frac{2}{\pi}
 \end{aligned}$$

$$\therefore y = e^{2/\pi}$$

Example 21: Evaluate $\lim_{x \rightarrow 0} x^x$

Solution: Let $y = \lim_{x \rightarrow 0} x^x$

$$\begin{aligned}
 \therefore \log y &= \lim_{x \rightarrow 0} x \log x \quad [0 \times \infty \text{ form}] \\
 &= \lim_{x \rightarrow 0} \left(\frac{\log x}{1/x} \right) \quad \left[\frac{\infty}{\infty} \text{ form} \right]
 \end{aligned}$$

Applying L. Hospital's rule

$$= \lim_{x \rightarrow 0} \left(\frac{1/x}{-1/x^2} \right)$$

$$= \lim_{x \rightarrow 0} (-x) = 0$$

$$\therefore y = e^0 = 1$$

Example 22: Evaluate $\lim_{x \rightarrow 0} (\sin x)^{\tan x}$

Solution: We have

$$y = \lim_{x \rightarrow 0} (\sin x)^{\tan x}$$

$$\therefore \log y = \lim_{x \rightarrow 0} \tan x \log (\sin x) \quad [0 \times \infty \text{ form}]$$

$$= \lim_{x \rightarrow 0} \frac{\log (\sin x)}{\cot x} \quad \left[\frac{0}{0} \text{ form} \right]$$

Applying L. Hospital's rule = $\lim_{x \rightarrow 0} \left\{ \frac{\frac{1}{\sin x} \times \cos x}{-\operatorname{cosec}^2 x} \right\} \quad \left(\frac{\infty}{\infty} \text{ form} \right)$

$$= \lim_{x \rightarrow 0} \left\{ -\frac{\sin^2 x}{\tan x} \right\}$$

$$= \lim_{x \rightarrow 0} \left\{ \frac{-2 \sin x \cos x}{\sec^2 x} \right\} = 0$$

$$\therefore y = e^0 = 1$$

Example 23: Evaluate $\lim_{x \rightarrow \infty} \left(\frac{\pi}{2} - \tan^{-1} x \right)^{1/x}$

Solution: Let

$$y = \lim_{x \rightarrow \infty} \left(\frac{\pi}{2} - \tan^{-1} x \right)^{1/x}$$

$$\therefore \log y = \lim_{x \rightarrow \infty} \frac{1}{x} \log \left(\frac{\pi}{2} - \tan^{-1} x \right)$$

$$= \lim_{x \rightarrow \infty} \left\{ \frac{\log \left(\frac{\pi}{2} - \tan^{-1} x \right)}{x} \right\}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{\pi/2 - \tan^{-1} x} \times -\frac{1}{1+x^2}}{1}$$

$$= \lim_{x \rightarrow \infty} \left\{ \frac{-\frac{1}{1+x^2}}{\pi/2 - \tan^{-1} x} \right\}$$

$$= \lim_{x \rightarrow \infty} \left\{ \frac{1}{(1+x^2)^2} 2x \middle/ -\frac{1}{1+x^2} \right\} = \lim_{x \rightarrow \infty} \left\{ \frac{-2x}{1+x^2} \right\}$$

$$= \lim_{x \rightarrow \infty} \left\{ \frac{-2}{2x} \right\} = 0$$

$$\therefore y = e^0 = 1.$$

Example 24: Evaluate $\lim_{x \rightarrow \pi/2} (\sec x)^{\cot x}$

Solution: Let $y = \lim_{x \rightarrow \pi/2} (\sec x)^{\cot x}$

$$\begin{aligned} \therefore \log y &= \lim_{x \rightarrow \pi/2} \cot x \log (\sec x), & [0 \times \infty] \\ &= \lim_{x \rightarrow \pi/2} \left\{ \frac{\log(\sec x)}{\tan x} \right\}, & \left(\frac{\infty}{\infty} \text{ form} \right) \end{aligned}$$

Applying L. Hospital's rule

$$\begin{aligned} \log y &= \lim_{x \rightarrow \pi/2} \left\{ \frac{\frac{1}{\sec x} \times \sec x \tan x}{\sec^2 x} \right\} \\ &= \lim_{x \rightarrow \pi/2} \left\{ \frac{\tan x}{\sec^2 x} \right\} \\ &= \lim_{x \rightarrow \pi/2} \{ \sin x \cos x \} = 0 \end{aligned}$$

$$\therefore y = e^0 = 1$$



Comprehensive Exercise 9.2

Evaluate

- $\lim_{x \rightarrow \pi/2} (\sec x - \tan x)$
- $\lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x)$
- $\lim_{x \rightarrow 0} \frac{\cot x - \frac{1}{x}}{x}$
- $\lim_{x \rightarrow 1} \left[\frac{x}{x-1} - \frac{1}{\log x} \right]$
- $\lim_{x \rightarrow \pi/2} (\sin x)^{\tan x}$
- $\lim_{x \rightarrow \infty} (1 + x^{-2})^x$
- $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$
- $\lim_{x \rightarrow 0} \left(\frac{\sinh x}{x} \right)^{1/x}$
- $\lim_{x \rightarrow 0} \left\{ \frac{2(\cosh x - 1)}{x^2} \right\}^{1/x^2}$
- $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{1/x}$
- $\lim_{x \rightarrow \pi/4} (\tan x)^{\tan 2x}$
- $\lim_{x \rightarrow 0} \frac{x^x - 1}{x}$

13. $\lim_{x \rightarrow \infty} \left(\frac{1}{x} \right)^{1/x}$
14. $\lim_{x \rightarrow 0} (\operatorname{cosec} x)^{1/\log x}$
15. $\lim_{x \rightarrow 0} (\cot x)^{\sin x}$
16. $\lim_{x \rightarrow 0} (a^x + x)^{1/x}$
17. $\lim_{x \rightarrow \pi/2} \left(\sec x - \frac{1}{1 - \sin x} \right)$
18. $\lim_{x \rightarrow 0} \left(\operatorname{cosec}^3 x - \frac{1}{x^3} \right)$
19. $\lim_{x \rightarrow 0} \left[\frac{1}{e^x - 1} - \frac{1}{x} \right]$
20. $\lim_{x \rightarrow \infty} a^x \sin \left(\frac{b}{a^x} \right), a > 1$
21. $\lim_{x \rightarrow \infty} 2^x \sin \left(\frac{a}{2^x} \right)$
22. $\lim_{x \rightarrow 0} x^m (\log x)^n$
23. $\lim_{x \rightarrow \infty} (a^{1/x} - 1)x$
24. $\lim_{x \rightarrow \infty} \left\{ x \tan \left(\frac{1}{x} \right) \right\}$
25. $\lim_{x \rightarrow 0} \frac{\log(\tan^2 2x)}{\log(\tan^2 x)}$



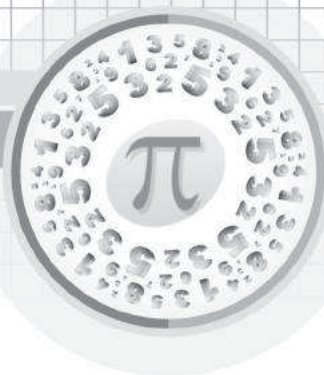
Answers

1.	0	2.	0	3.	$-\frac{1}{3}$	4.	$\frac{1}{2}$	5.	1	6.	1
7.	$e^{-1/2}$	8.	1	9.	$e^{(1/12)}$	10.	$\sqrt{ab}$	11.	$\frac{1}{e}$	12.	$-\infty$
13.	1	14.	$\frac{1}{e}$	15.	1	16.	ae	17.	∞	18.	∞
19.	$-\frac{1}{2}$	20.	b	21.	a	22.	0	23.	$\log a$	24.	1
25.	1										





Unit-3



Chapter

10

Maxima and Minima

10.1 Maximum and Minimum Values of a Function

10.1.1 Maximum Value

A function $f(x)$ is said to attain the maximum value in an interval I at a point $a \in I$, if

$$f(a) \geq f(x) \quad \forall x \in I$$

i.e., $f(a)$ is called the **maximum value** or **greatest value** or **absolute maximum** value of $f(x)$ in I , and a is called the point of maximum of $f(x)$ in I .

10.1.2 Minimum Value

A function is said to attain a minimum value in an interval I at a point $a \in I$

if
$$f(a) \leq f(x) \quad \forall x \in I$$

i.e., $f(a)$ is called the **minimum value** or **least value** or **absolute minimum value**.

10.2 Local Maxima and Local Minima

In domain of a function, there may be points where the function does not attain the maximum or minimum value but the values at these points are greater or less than the values of the function in the small neighbourhood of the points. Such points are known as the points of **local maximum** or **local minimum**.

10.3 Extreme Values

The points at which a function $f(x)$ has either local maximum values or local minimum values are called the extreme points or turning points and both the local maximum and local minimum values are called extreme values.

It is obvious that a function $f(x)$ has an extreme value at $x = a$ if $f(x)$ is either a local maximum or a local minimum value. Therefore, $f(x) - f(a)$ has the same sign at $x = a$ for all values of x in a deleted neighbourhood of the point $x = a$.

All extreme values of $f(x)$ are also critical values for $f(x)$. In other words, any non-critical value for $f(x)$ cannot be an extremum for $f(x)$.

10.4 Tests for Finding Maxima or Minima

I. First Derivative Test: In order to find the extremum values of a given function $y = f(x)$ by using the first derivative test, we proceed according to the following steps.

- (i) Find $\frac{dy}{dx}$ and equate it to zero.
- (ii) Solve this equation for real values of x , let these values be $a, b, c, \dots$
- (iii) Consider the value $x = a$, study the sign of $\frac{dy}{dx}$ for values of x slightly less than and slightly greater than a .
- (iv) If $\frac{dy}{dx}$ changes sign from positive to negative, then $f(x)$ has **maximum** at $x = a$.
- (v) If $\frac{dy}{dx}$ changes sign from negative to positive, then $f(x)$ is **minimum** at $x = a$.
- (vi) If $\frac{dy}{dx}$ does not change sign, then $x = a$ is a point **neither of maximum nor of minimum**.

II. Higher Order Derivative Test:

- (i) Find $f'(x)$ by differentiating the given function $f(x)$ with respect to x .
- (ii) Solve the equation $f'(x) = 0$ for real values of x . Let these values be $a_1, a_2, a_3, \dots$
- (iii) Find $f''(x)$ and substitute in it by turns $a_1, a_2, a_3, \dots$
- (iv) If $f''(a_1)$ is **negative**, we have a maximum at $x = a_1$.
- (v) If $f''(a_1)$ is **positive**, we have minimum at $x = a_1$.
- (vi) If $f''(a_1) = 0$, find $f'''(x)$ and substitute in it a_1 . If $f'''(a_1) \neq 0$, there is **neither** a maximum nor a minimum at $x = a_1$.
- (vii) If $f'''(a_1) = 0$ find $f^{(iv)}(x)$ and substitute in it a_1 . If $f^{(iv)}(a_1)$ is **negative**, we have a maximum at $x = a_1$, and if $f^{(iv)}$ is positive, a minimum. If it is zero, find $f^{(v)}$, and so on.

Illustrative Examples

(Maximum and Minimum)

Example 1: Find the points of local maxima or local minima of the function,

$$f(x) = x^3 - 6x^2 + 9x + 15$$

using the first derivative test. Also find the local maximum or local minimum values, as the case may be.

[B.C.A (Allahabad) 2009]

Solution: We have,

$$f(x) = x^3 - 6x^2 + 9x + 15$$

$$\Rightarrow f'(x) = 3x^2 - 12x + 9$$

For local maximum or local minimum, we have

$$f'(x) = 0$$

$$\Rightarrow 3x^2 - 12x + 9 = 0$$

$$\Rightarrow x^2 - 4x + 3 = 0$$

$$\Rightarrow (x - 1)(x - 3) = 0$$

$$\Rightarrow x = 1, 3$$

Now, to test the local maximum or local minimum or neither of them at $x = 1$ and $x = 3$.

When $x = 1$, in this case, if x is slightly less than 1, it is clear that $(x - 1)$ is negative and $(x - 3)$ is also negative and hence,

$$f''(x) = 3(x - 1)(x - 3) \text{ is positive.}$$

When x is slightly greater than 1, $(x - 1)$ is positive and $(x - 3)$ is negative. Therefore, $f'(x)$ is negative.

Thus, $f'(x)$ changes sign from positive to negative, as x increase through $x = 1$. Therefore, $x = 1$ is a point of local maximum.

$$\begin{aligned}\therefore \text{The local maximum value} &= f(1) \\ &= 1^3 - 6 \cdot 1^2 + 9 \cdot 1 + 15 = 19.\end{aligned}$$

When $x = 3$, in this case, if x is slightly less than 3, then $(x - 1)$ is positive and $(x - 3)$ is negative.

$$\therefore f'(x) = 3(x - 1)(x - 3) \text{ is negative.}$$

Again, when x is slightly greater than 3, then $(x - 1)$ as well as $(x - 3)$ are positive.

Thus, $f'(x)$ changes sign from negative to positive as x increases through $x = 3$. Therefore, $x = 3$ is a point of local minimum.

$$\therefore \text{The local minimum value} = f(3) = 15.$$

Hence, the local maximum value is 19 at $x = 1$, and the local minimum value is 15 at $x = 3$.

Example 2: Find the local maxima or local minima, if any of

(i) $f(x) = \frac{1}{x^2 + 2}$

(ii) $f(x) = x^3 - 3x$

In each case, find the local maximum or the local minimum values, as the case may be.

Solution: (i) We have, $f(x) = \frac{1}{x^2 + 2} \Rightarrow f'(x) = \frac{-2x}{(x^2 + 2)^2}$

For a local maxima or minima, we have $f'(x) = 0$

$$\therefore f'(x) = 0 \Rightarrow \frac{-2x}{(x^2 + 2)^2} = 0 \Rightarrow x = 0$$

Now, when x is slightly less than 0 i.e., when x is negative, then $f'(x) = \frac{-2x}{(x^2 + 2)^2}$

is positive. When x is slightly greater than 0 then $f'(x) = \frac{-2x}{(x^2 + 2)^2} < 0$.

Thus, $f'(x)$ change sign from positive to negative. So, $x = 0$ is a point of local maximum.

And, local maximum value is $f(0) = \frac{1}{(0^2 + 2)} = \frac{1}{2}$.

(ii) We have, $f(x) = x^3 - 3x \Rightarrow f'(x) = 3x^2 - 3$

Now, for a maximum or minimum, we have

$$f'(x) = 0 \Rightarrow 3x^2 - 3 = 0$$

$$\Rightarrow x = \pm 1$$

Thus, $x = 1$ and $x = -1$, are the conditions for local maxima or local minima.

When $x = 1$, we know that $f'(x) = 3(x^2 - 1)$.

When x is slightly less than 1 then x^2 is slightly less than 1 and therefore, $3(x^2 - 1)$ is negative. Again, when x is slightly more than 1 then x^2 is slightly more than 1 and therefore, $3(x^2 - 1)$ is positive.

Thus, $f'(x)$ changes values from negative to positive as x increases through 1. So, $x = 1$ is a point of local minimum.

Local minimum value $= f(1) = (1^3 - 3 \times 1) = -2$

When $x = -1$, when x is slightly less than -1 then x^2 is slightly more than 1. So, $3(x^2 - 1)$ is positive. Again, when x is slightly more than -1 then x^2 is slightly less than 1, so, $3(x^2 - 1)$ is negative. Thus, $f'(x)$ changes sign from positive to negative as x increase through -1 . So, $x = -1$ is a point of local maximum.

Local maximum value $= f(-1) = (-1)^3 - 3 \times (-1) = 2$.

Example 3: Find the local maxima or local minima of the function

$$f(x) = \sin 2x - x, -\frac{\pi}{2} < x < \frac{\pi}{2}$$

using the first derivative test.

Solution: We have,

$$\begin{aligned} f(x) &= \sin 2x - x \\ \Rightarrow f'(x) &= 2 \cos 2x - 1 \end{aligned}$$

For local maximum or local minimum, we have

$$\begin{aligned} f'(x) &= 0 \Rightarrow 2 \cos 2x - 1 = 0 \\ \Rightarrow \cos 2x &= \frac{1}{2} = \cos \left(-\frac{\pi}{3} \right) \text{ or } \cos \left(\frac{\pi}{3} \right) \\ \therefore 2x &= -\frac{\pi}{3}, \frac{\pi}{3} \text{ or } x = -\frac{\pi}{6}, \frac{\pi}{6} \end{aligned}$$

when $x = -\frac{\pi}{6}$. In this case, if x is slightly less than $-\frac{\pi}{6}$, let us take for simplicity, $x = -\frac{\pi}{4}$.

Then,

$$f' \left(-\frac{\pi}{4} \right) = 2 \cos \left(\frac{-2\pi}{4} \right) - 1 = 2 \cos \left(\frac{\pi}{2} \right) - 1 = 0 - 1 = -1$$

and if x is slightly greater than $-\frac{\pi}{6}$ let $x = 0$. Then,

$$f'(0) = 2 \cos 0 - 1 = 1$$

Thus, $f'(x)$ changes sign from negative to positive as x increases through $x = -\frac{\pi}{6}$.

Therefore, x is a point of local minimum.

$$\therefore \text{local minimum value} = f \left(-\frac{\pi}{6} \right) = \sin \left(\frac{-2\pi}{6} \right) - \left(-\frac{\pi}{6} \right) = \frac{-\sqrt{3}}{2} + \frac{\pi}{6}$$

when $x = \frac{\pi}{6}$, in this case, when x is slightly less than $\frac{\pi}{6}$, let us consider $x = 0$. Then,

$$f'(0) = 2 \cos 0 - 1 = 1$$

and when x is slightly greater than $x = \frac{\pi}{6}$, let us take $x = \frac{\pi}{4}$. Then,

$$f' \left(\frac{\pi}{4} \right) = 2 \cos \frac{2\pi}{4} - 1 = -1$$

Thus, $f'(x)$ changes sign from positive to negative as x increases through $x = \frac{\pi}{6}$. Therefore,

$x = \frac{\pi}{6}$ is a point of local maximum.

$$\therefore \text{local maximum value} = f \left(\frac{\pi}{6} \right) = \sin \left(\frac{2\pi}{6} \right) - \frac{\pi}{6} = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

Example 4: Show that $(x^5 - 5x^4 + 5x^3 - 1)$ has a local maximum when $x = 1$, a local minimum when $x = 3$ and neither when $x = 0$.

Solution: We have,

$$f(x) = x^5 - 5x^4 + 5x^3 - 1$$

$$\Rightarrow f'(x) = 5x^4 - 20x^3 + 15x^2 = 5x^2(x-3)(x-1)$$

For a local maxima or minima, we have $f'(x) = 0$

$$\therefore 15x^2(x-3)(x-1) = 0 \Rightarrow x = 0 \text{ or } x = 3 \text{ or } x = 1$$

when $x = 1$, in this case, when x is slightly less than 1 then, $f'(x) = 5x^2(x-3)(x-1)$ is positive and when x is slightly more than 1 then $f'(x)$ is negative. Thus, $f'(x)$ change sign from positive to negative. So, $x = 1$ is a point of local maximum.

When $x = 3$, in that case, when x is slightly less than 3 then, $f'(x) = 5x^2(x-3)(x-1)$ is negative and when x is slightly greater than 3 then $f'(x)$ is positive. So, $x = 3$ is a point of local minimum.

When $x = 0$, in that case, when x is slightly less than 0 then $f'(x) = 5x^2(x-3)(x-1)$ is positive and also when x is slightly more than 0 then $f'(x)$ is positive. Thus $f'(x)$ changes sign from negative to positive.

Thus, $f'(x)$ does not change sign as it passes through 0.

So, it is neither a point of local maximum nor a point of local minimum i.e., $x = 0$ is a point of inflexion.

Example 5: Find the maximum and minimum values of the function.

$$x^3 - 2x^2 + x + 6.$$

Solution: We have,

$$f(x) = x^3 - 2x^2 + x + 6$$

$$\Rightarrow f'(x) = 3x^2 - 4x + 1$$

$$\Rightarrow f''(x) = 6x - 4$$

For maximum or minimum value of $f(x)$, we have

$$f'(x) = 0 \Rightarrow 3x^2 - 4x + 1 = 0$$

$$\Rightarrow (x-1)(3x-1) = 0$$

$$\Rightarrow x = 1, \frac{1}{3}$$

At $x = 1$, $f''(x) = 6 \times 1 - 4 = 2$ which is positive.

$\therefore f(x)$ is minimum at $x = 1$.

The minimum value of $f(x) = x^3 - 2x^2 + x + 6$

$$f(1) = (1)^3 - 2(1)^2 + 1 + 6 = 6$$

when $x = 1/3$ then, $f''(1/3) = -2$ which is negative

$\therefore f(x)$ is maximum at $x = 1/3$

The maximum value of $f(x) = x^3 - 2x^2 + x + 6$ at $x = 1/3$

$$= \left(\frac{1}{3}\right)^3 - 2\left(\frac{1}{3}\right)^2 + \frac{1}{3} + 6 = 6\frac{4}{27}$$

Hence, maximum value is $6\frac{4}{27}$ at $x = \frac{1}{3}$ and minimum value is 6 at $x = 1$.

Example 6: Find the values of x for which the function

$$f(x) = x^5 - 5x^4 + 5x^3 - 1$$

is maximum or minimum. Show that $x = 0$, the function is neither maximum nor minimum. [B.C.A. (Meerut) 2007]

Solution: We have, $f(x) = x^5 - 5x^4 + 5x^3 - 1$

$$f'(x) = 5x^4 - 20x^3 + 15x^2$$

$$f''(x) = 20x^3 - 60x^2 + 30x = 10x(2x^2 - 6x + 3)$$

For maximum or minimum value of $f(x)$, we have

$$f'(x) = 0 \Rightarrow 5x^2(x^2 - 4x + 3) = 0$$

$$\Rightarrow 5x^2(x - 1)(x - 3) = 0$$

$$\Rightarrow x = 0, 1, 3$$

$$\text{At } x = 1, f''(1) = -10, \text{ which is negative}$$

$\therefore f(x)$ is maximum at $x = 1$.

$$\text{At } x = 3, f''(3) = 5 \times (3)^2(13)^2 - 4 \times 3 + 3 = 90, \text{ which is positive.}$$

$\therefore f(x)$ is minimum at $x = 3$

Hence, for $x = 1$, $f(x)$ is maximum and for $x = 3$, $f(x)$ is minimum.

At $x = 0$, we have

$$f''(x) = 20x^3 - 60x^2 + 30x$$

$$\Rightarrow f''(0) = 0$$

$$\text{Then, } f'''(x) = 60x^2 - 120x + 30$$

$$\text{At } x = 0, \text{ we have } f'''(0) = 30 \neq 0$$

Hence, the given function has neither maximum nor minimum at $x = 0$.

Example 7: Find the maximum value of the function xy^2 if, $x + y = a$.

Solution: Let

$$u = xy^2$$

$$\Rightarrow u = (a - y)y^2 \quad [\because x + y = a]$$

$$\Rightarrow u = ay^2 - y^3$$

$$\Rightarrow \frac{du}{dy} = 2ay - 3y^2$$

$$\Rightarrow \frac{d^2 u}{dy^2} = 2a - 6y$$

For maximum or minimum value of x , we have

$$\frac{du}{dy} = 0 \Rightarrow 2ay - 3y^2 = 0$$

$$\Rightarrow y(2a - 3y) = 0$$

$$\Rightarrow y = 0, \frac{2a}{3}$$

Since, the value of y cannot be zero, therefore $y = \frac{2a}{3}$

$$\therefore \text{The value of } \frac{d^2 u}{dy^2} \text{ at } \left(y = \frac{2a}{3}\right) = 2a - 6y$$

$$= 2a - 6 \times \frac{2a}{3} = -2a, \text{ which is negative.}$$

$$\therefore u \text{ is maximum when } y = \frac{2a}{3}$$

Then maximum

$$u = (a - y) y^2$$

$$= \left(a - \frac{2a}{3}\right) \left(\frac{2a}{3}\right)^2 = \frac{4a^3}{27}$$

Example 8: Find the maximum value of $\frac{\log x}{x}$, when $0 < x < \infty$.

Solution: Let $y = \frac{\log x}{x}$

$$\Rightarrow \frac{dy}{dx} = \frac{x \cdot \frac{1}{x} - (\log x) \cdot 1}{x^2} = \frac{1 - \log x}{x^2}$$

For the maximum or minimum value of y , we have

$$\frac{dy}{dx} = 0 \Rightarrow \frac{1 - \log x}{x^2} = 0$$

$$\Rightarrow \log x = 1 \text{ or } \log x = \log e$$

$$\Rightarrow x = e$$

$$\text{Now, } \frac{d^2 y}{dx^2} = \frac{\left(0 - \frac{1}{x}\right) x^2 - (1 - \log x) 2x}{x^4}$$

$$= \frac{2 \log x - 3}{x^3}$$

$$\text{At } x = e, \frac{d^2 y}{dx^2} = \frac{2 \log e - 3}{e^3} = \frac{2 - 3}{e^3} = \frac{-1}{e^3} \text{ which is negative.}$$

$\therefore$ At $x = e$, the function is maximum.

$$\text{The maximum value of the function} = \frac{\log e}{e} = \frac{1}{e} = e^{-1}.$$

Example 9: Prove that $\sin^p \theta \cos^q \theta$ is maximum when

$$\theta = \tan^{-1} \sqrt{\frac{p}{q}}$$

[B.C.A. (Meerut) 2006; B.C.A. (Rohilkhand) 2003, 2006]

Solution: Let $y = \sin^p \theta \cos^q \theta$

$$\begin{aligned} \Rightarrow \frac{dy}{d\theta} &= p \sin^{p-1} \theta \cos^q \theta + \sin^p \theta q \cos^{q-1} \theta (-\sin \theta) \\ &= p \sin^{p-1} \theta \cos^{q-1} \theta - q \sin^{p+1} \theta \cos^{q-1} \theta \\ &= \sin^{p-1} \theta \cos^{q-1} \theta [p \cos^2 \theta - q \sin^2 \theta] \end{aligned}$$

For maximum or minimum value of y

$$\begin{aligned} \Rightarrow \frac{dy}{d\theta} &= 0 \\ \Rightarrow \sin^{p-1} \theta \cos^{q-1} \theta (p \cos^2 \theta - q \sin^2 \theta) &= 0 \\ \Rightarrow p \cos^2 \theta - q \sin^2 \theta &= 0 \\ \Rightarrow \tan^2 \theta &= \frac{p}{q} \\ \Rightarrow \tan \theta &= \pm \sqrt{\frac{p}{q}} \end{aligned}$$

$$\begin{aligned} \text{Now, } \frac{dy}{d\theta} &= \sin^{p-1} \theta \cos^{q+1} \theta (p - q \tan^2 \theta) \\ &= \sin^p \theta \cos^q \theta \left(p \frac{\cos \theta}{\sin \theta} - q \tan^2 \theta \frac{\cos \theta}{\sin \theta} \right) \\ &= \sin^p \theta \cos^q \theta (p \cot \theta - q \tan \theta) \\ &= y (p \cot \theta - q \tan \theta) \\ \therefore \frac{d^2 y}{d\theta^2} &= \frac{dy}{d\theta} (p \cot \theta - q \tan \theta) + y (-p \operatorname{cosec}^2 \theta - q \sec^2 \theta) \\ &= -y (p \operatorname{cosec}^2 \theta + q \sec^2 \theta) \quad \left[\because \frac{dy}{d\theta} = 0 \right] \\ &= -y (p + q + p \cot^2 \theta + q \tan^2 \theta) \end{aligned}$$

$$\therefore \text{ The value of } \frac{d^2 y}{d\theta^2} \text{ at } \left(\theta = \tan^{-1} \sqrt{\frac{p}{q}} \right) = -y \left[p + q + p \cdot \frac{q}{p} + q \cdot \frac{p}{q} \right]$$

Which is negative.

$$\therefore y \text{ is maximum when } \theta = \tan^{-1} \sqrt{\frac{p}{q}}$$

Example 10: Find the maximum and minimum values of the function $x + \sin 2x$ if $0 < x < 2\pi$.

Solution: Let

$$y = x + \sin 2x$$

$$\Rightarrow \frac{dy}{dx} = 1 + 2 \cos 2x$$

For maximum or minimum of y put $\frac{dy}{dx} = 0$

$$\therefore 1 + 2 \cos 2x = 0$$

$$\Rightarrow \cos 2x = \frac{-1}{2} = \cos \frac{2\pi}{3}$$

$$\Rightarrow 2x = 2n\pi \pm \frac{2\pi}{3}$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{3}$$

But $0 < x < 2\pi$, therefore,

$$x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

Now,
$$\frac{d^2y}{dx^2} = -4 \sin 2x$$

$$\therefore \text{At } x = \frac{\pi}{3}, \text{ we have } \frac{d^2y}{dx^2} = -4 \sin \frac{2\pi}{3} = \frac{-4\sqrt{3}}{2}, \text{ which is negative}$$

$$\therefore y \text{ is maximum at } x = \frac{\pi}{3} \text{ and the maximum value is}$$

$$= \frac{\pi}{3} + \sin 2 \cdot \frac{\pi}{3} = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$$

At $x = \frac{2\pi}{3}, \frac{d^2y}{dx^2} = -4 \sin \frac{4\pi}{3} = -4 \left(-\frac{\sqrt{3}}{2} \right)$

Which is positive.

$$\therefore y \text{ is minimum at } x = \frac{2\pi}{3} \text{ and the minimum value is}$$

$$= \frac{2\pi}{3} + \sin \frac{4\pi}{3} = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

At $x = \frac{4\pi}{3}, \frac{d^2y}{dx^2} = -4 \sin \frac{8\pi}{3} = -4 \left(\frac{\sqrt{3}}{2} \right)$ which is negative

$$\therefore y \text{ is maximum at } x = \frac{4\pi}{3} \text{ and maximum value}$$

$$= \frac{4\pi}{3} + \sin \frac{8\pi}{3} = \frac{4\pi}{3} + \frac{\sqrt{3}}{2}$$

Again at $x = \frac{5\pi}{3}$

$$\frac{d^2y}{dx^2} = -4 \sin \frac{10\pi}{3} = -4 \sin \left(4\pi - \frac{2\pi}{3} \right)$$

$$= 4 \sin \frac{2\pi}{3}, \text{ which is positive.}$$

$$\therefore y \text{ is minimum at } x = \frac{5\pi}{3} \text{ and the minimum value} = \frac{5\pi}{3} + \sin \frac{10\pi}{3} = \frac{5\pi}{3} - \frac{\sqrt{3}}{2}$$

Example 11: Find the maximum and minimum values of the function

$$y = \frac{40}{3x^4 + 8x^3 - 18x^2 + 60}$$

Solution: Let

$$y = \frac{40}{3x^4 + 8x^3 - 18x^2 + 60}$$

and

$$u = 3x^4 + 8x^3 - 18x^2 + 60$$

Then, y is maximum or minimum accordingly as u is minimum or maximum.

Now,
$$\frac{du}{dx} = 12x^3 + 24x^2 - 36x$$

$$\Rightarrow \frac{d^2u}{dx^2} = 36x^2 + 48x - 36$$

For maximum or minimum value of u , we have,

$$\frac{du}{dx} = 0$$

$$\Rightarrow 12x^3 + 24x^2 - 36x = 0$$

$$\Rightarrow 12x(x^2 + 2x - 3) = 0$$

$$\Rightarrow x(x-1)(x+3) = 0$$

$$\Rightarrow x = 0, 1, -3$$

At $x = 0$, $\frac{d^2u}{dx^2} = 36 \times 0 + 48 \times 0 - 36 = -36$ which is negative

$\therefore u$ is maximum at $x = 0$ i.e., y is minimum at $x = 0$

$\therefore$ The minimum value of y is at $x = 0$ and is

$$= \frac{40}{3 \times 0 + 8 \times 0 - 18 \times 0 + 60} = \frac{40}{60} = \frac{2}{3}$$

At $x = 1$, $\frac{d^2u}{dx^2} = 36 \times 1^2 + 48 \times 1 - 36 = 48$, which is positive.

$\therefore u$ is minimum at $x = 1$ i.e., y is maximum at $x = 1$

$\therefore$ The maximum value of y is at $x = 1$

$$= \frac{40}{3 \times (1)^4 + 8 \times (1)^3 - 18 \times (1)^2 + 60} = \frac{40}{3 + 8 - 18 + 60} = \frac{40}{53}$$

At $x = -3$, $\frac{d^2u}{dx^2} = 36 \times (-3)^2 + 48(-3) - 36 = 144$,

which is positive.

$\therefore u$ is minimum and so y is maximum at $x = -3$

$\therefore$ The maximum value of y is

$$= \frac{40}{3 \times (-3)^4 + 8 \times (-3)^3 - 18 \times (-3)^2 + 60}$$

$$= \frac{40}{243 - 216 - 162 + 60} = -\frac{40}{75} = -\frac{8}{15}$$

Example 12: Prove that the maximum value of $\sin x + \cos x$ is $\sqrt{2}$.

Solution: Let $y = \sin x + \cos x$

$$\Rightarrow \frac{dy}{dx} = \cos x - \sin x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\sin x - \cos x$$

For maximum or minimum value of y , we have,

$$\frac{dy}{dx} = 0 \Rightarrow \cos x - \sin x = 0$$

$$\tan x = 1 \text{ or } x = \frac{\pi}{4}, \frac{5\pi}{4} \quad \text{if } 0 < x < 2\pi$$

$$\begin{aligned} \text{At } x = \frac{\pi}{4}, \text{ the value of } \frac{d^2y}{dx^2} &= -\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \\ &= -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{2}{\sqrt{2}} \text{ which is negative.} \end{aligned}$$

$$\therefore y \text{ is maximum at } x = \frac{\pi}{4}$$

$$\text{The maximum value} = \sin \frac{\pi}{4} + \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$\text{At } x = \frac{5\pi}{4}, \text{ we have,}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= -\cos \frac{5\pi}{4} - \sin \frac{5\pi}{4} = \cos \frac{\pi}{4} + \sin \frac{\pi}{4} \\ &= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} \text{ which is positive.} \end{aligned}$$

$$\therefore y \text{ is minimum at } x = \frac{5\pi}{4}$$

Hence, the maximum value of y is $\sqrt{2}$.

Example 13: Find the points of local maxima or local minima of the function

$$f(x) = (\sin^4 x + \cos^4 x) \text{ in } 0 < x < \frac{\pi}{2}$$

Solution: $f(x) = \sin^4 x + \cos^4 x$

$$\begin{aligned} \Rightarrow f'(x) &= 4 \sin^3 x \cos x + 4 \cos^3 x (-\sin x) \\ &= -4 \sin x \cos x (\cos^2 x - \sin^2 x) \\ &= -2 \sin 2x \cos 2x = -\sin 4x \end{aligned}$$

And, $f''(x) = -4 \cos 4x$

Now, $f'(x) = 0 \Rightarrow -\sin 4x = 0 \Rightarrow 4x = \pi \text{ or } x = \frac{\pi}{4}$

$\therefore x = \frac{\pi}{4}$ is a point of local maximum or local minimum

Now, $f''\left(\frac{\pi}{4}\right) = -4 \cos \pi = 4 > 0$

$\therefore x = \frac{\pi}{4}$ is a point of local minimum

$$\text{Local minimum value} = f\left(\frac{\pi}{4}\right) = \frac{1}{2}$$

Example 14: Prove that the maximum value of $\left(\frac{1}{x}\right)^x$ is $e^{1/e}$.

[B.C.A. (Rohilkhand) 2005, 2008]

Solution: Let $y = \left(\frac{1}{x}\right)^x = x^{-x}$

Then, $\log y = -x \log x$

$$\therefore \frac{1}{y} \frac{dy}{dx} = -x \times \frac{1}{x} - \log x = -(1 + \log x)$$

$$\text{or } \frac{dy}{dx} = -y(1 + \log x)$$

$$\text{And, } \frac{d^2 y}{dx^2} = -y \cdot \frac{1}{x} - (1 + \log x) \frac{dy}{dx} = -\left(\frac{1}{x}\right)^{x+1} - (1 + \log x) \frac{dy}{dx}$$

$$\text{Now, } \frac{dy}{dx} = 0 \Rightarrow 1 + \log x = 0$$

$$\Rightarrow \log x = -1$$

$$\Rightarrow \log x = -\log e$$

$$\Rightarrow \log x = \log \left(\frac{1}{e}\right)$$

$$\Rightarrow x = \frac{1}{e}$$

$$\text{Also, } \frac{d^2 y}{dx^2} \text{ at } x = \left(\frac{1}{e}\right) \text{ is } -e^{\left(1 + \frac{1}{e}\right)} < 0 \quad \left[\because \frac{dy}{dx} = 0 \right]$$

So, $x = \frac{1}{e}$ is a point of local maximum.

$$\text{Local maximum value} = \left[\text{value of } y \text{ when } x = \frac{1}{e} \right] = e^{1/e}$$

Example 15: Show that $\sin x (1 + \cos x)$, $x \in [0, \pi]$ is maximum at $x = (\pi/3)$.

Solution: Let $f(x) = \sin x (1 + \cos x)$

$$\begin{aligned} \Rightarrow f'(x) &= -\sin^2 x + \cos x (1 + \cos x) \\ &= 2 \cos^2 x + \cos x - 1 \\ &= (2 \cos x - 1)(\cos x + 1) \end{aligned}$$

$$\text{And, } f''(x) = (-4 \cos x \sin x - \sin x) = -\sin x (1 + 4 \cos x)$$

$$\text{Now, } f'(x) = 0 \Rightarrow (2 \cos x - 1)(\cos x + 1) = 0$$

$$\Rightarrow \cos x = \frac{1}{2} \text{ or } \cos x = -1 \Rightarrow x = \frac{\pi}{3} \text{ or } x = \pi$$

But,
$$f''\left(\frac{\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right)\left(1 + 4\cos\frac{\pi}{3}\right) = \frac{-3\sqrt{3}}{2} < 0$$

$\therefore x = \left(\frac{\pi}{3}\right)$ is a point of local maximum

Now,
$$f(0) = 0, f(\pi) = 0 \text{ and } f\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{4}$$

Example 16: Investigate for maximum and minimum values, the function

$$(\sin x + \cos 2x) \quad [\text{B.C.A. (Meerut) 2005, 2008; B.C.A. (Delhi) 2007}]$$

Solution: Let $f(x) = \sin x + \cos 2x$ (1)

Differentiating w.r.t. x , we get

$$f'(x) = \cos x - 2\sin 2x$$

and

$$f''(x) = -\sin x - 4\cos 2x$$

For maximum or minimum put $f'(x) = 0$, we get

$$\cos x - 2\sin 2x = 0$$

or

$$\cos x - 4\sin x \cos x = 0$$

or

$$\cos x(1 - 4\sin x) = 0$$

$\Rightarrow$

$$\cos x = 0 \text{ or } 1 - 4\sin x = 0$$

$\Rightarrow$

$$x = \frac{\pi}{2} \text{ or } x = \sin^{-1}\left(\frac{1}{4}\right)$$

When

$$x = \frac{\pi}{2} \text{ then } f''\left(\frac{\pi}{2}\right) = -\sin\frac{\pi}{2} - 4\cos\pi$$

$$= -1 + 4 = 3 = \text{positive}$$

$\therefore f(x)$ is minimum when $x = \frac{\pi}{2}$ and minimum value of $f(x)$ at $x = \frac{\pi}{2}$ is

$$f\left(\frac{\pi}{2}\right) = \sin\frac{\pi}{2} + \cos\pi = 1 - 1 = 0$$

When $x = \sin^{-1}\left(\frac{1}{4}\right)$ then $f''\left(\sin^{-1}\left(\frac{1}{4}\right)\right) = -\frac{1}{4} - 4\left(1 - 2\left(\frac{1}{16}\right)\right)$

There is maximum at $x = \sin^{-1}\left(\frac{1}{4}\right) = \frac{-15}{4}$ (negative) and maximum value of

$$f(x) = \sin x + (1 - 2\sin^2 x) = \frac{1}{4} + \left(1 - 2 \times \frac{1}{16}\right) = \frac{9}{8}$$

Example 17: Show that $f(x) = x^x$ is minimum at $x = \frac{1}{e}$

[B.C.A. (Meerut) 2001]

Solution: Let $y = f(x) = x^x$

Taking log on both sides, we get

$$\log y = \log x^x$$

or

$$\log y = x \log x \quad \dots(1)$$

Differentiate w.r.t. (x) , we get

$$\frac{1}{y} \cdot \frac{dy}{dx} = x \cdot \frac{1}{x} + \log x$$

$$\Rightarrow \frac{dy}{dx} = x^x (1 + \log x) \quad \dots(2)$$

Again differentiate w.r.t (x), we get

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} (x^x) \cdot (1 + \log x) + x^x \frac{d}{dx} (1 + \log x) \\ &= x^x (1 + \log x) (1 + \log x) + x^x \left(\frac{1}{x} \right) \end{aligned} \quad [\text{from 2}]$$

$$\frac{d^2 y}{dx^2} = x^x (1 + \log x)^2 + x^{x-1}$$

For maximum or minimum put $\frac{dy}{dx} = 0$, we get

$$x^x (1 + \log x) = 0 \text{ but } x^x \neq 0 \text{ then}$$

$$1 + \log x = 0 \text{ or } \log x = -1 \text{ or } x = e^{-1} = \frac{1}{e}$$

Then $\left(\frac{d^2 y}{dx^2} \right)_{\left(x = \frac{1}{e} \right)} = \left(\frac{1}{e} \right)^{\frac{1}{e}} \times 0 + \left(\frac{1}{e} \right)^{\left(\frac{1}{e} - 1 \right)} = \frac{1}{e^{\left(\frac{1}{e} - 1 \right)}} = e^{\left(1 - \frac{1}{e} \right)} = \text{positive}$

Hence, the function $f(x) = x^x$ is minimum at $x = \frac{1}{e}$

Example 18: Investigate the maxima and minima of

$$f(x) = 2x^3 - 21x^2 + 36x - 20$$

[B.C.A (Lucknow) 2010;
B.C.A. (Meerut) 2003]

Solution: Let $f(x) = 2x^3 - 21x^2 + 36x - 20$...(1)

Differentiate w.r.t. x we get

$$f'(x) = 6x^2 - 42x + 36$$

and

$$f''(x) = 12x - 42 \quad \dots(2)$$

For maximum and minimum put $f'(x) = 0$, we get

$$6x^2 - 42x + 36 = 0$$

or

$$x^2 - 7x + 6 = 0$$

or

$$(x - 6)(x - 1) = 0$$

or

$$x = 1, x = 6$$

when $x = 1$, then $f''(1) = 12 - 42 = -30 < 0$

i.e., negative, then $f(x)$ is maximum at $x = 1$

The maximum value = $f(1) = 2 \times 1^3 - 21 \times 1^2 + 36 \times 1 - 20 = -3$

When $x = 6$, then $f''(6) = 12 \times 6 - 42 = 72 - 42 = 30 > 0$

i.e., positive, then $f(x)$ is minimum at $x = 6$

The minimum value = $f(6) = 2 \times (6)^3 - 21 \times (6)^2 + 36(6) - 20 = -128$

Example 19: Prove that $f(x) = x^2 \log\left(\frac{1}{x}\right)$ is maximum when $x = \frac{1}{\sqrt{e}}$.

[B.C.A. (Meerut) 2002, 2004]

Solution: We have $f(x) = x^2 \log\left(\frac{1}{x}\right)$... (1)

Differentiating w.r.t. x , we get

$$f'(x) = 2x \times \log\left(\frac{1}{x}\right) + x^2 \times \frac{1}{\left(\frac{1}{x}\right)} \left(-\frac{1}{x^2}\right) = 2x \log\left(\frac{1}{x}\right) - x$$

$$f'(x) = x \left[2 \log\left(\frac{1}{x}\right) - 1 \right] \quad \dots (2)$$

Again Differentiating w.r.t. x , we get

$$f''(x) = 2 \log\left(\frac{1}{x}\right) - 1 + 2x \times \frac{1}{\left(\frac{1}{x}\right)} \left(-\frac{1}{x^2}\right)$$

$$\begin{aligned} \Rightarrow f''(x) &= 2 \left\{ \log\left(\frac{1}{x}\right) \right\} - 2 - 1 \\ &= 2 \log \frac{1}{x} - 3 \quad \dots (3) \end{aligned}$$

For maximum or minimum put $f'(x) = 0$, we get

$$0 = x \left[2 \log\left(\frac{1}{x}\right) - 1 \right]$$

$$\Rightarrow \text{either } \begin{cases} 2 \log\left(\frac{1}{x}\right) - 1 = 0 \text{ or } 2 \log\left(\frac{1}{x}\right) = 1 \\ \text{or } \log\left(\frac{1}{x}\right) = \frac{1}{2} \text{ or } \frac{1}{x} = e^{\frac{1}{2}} \\ \text{or } x = \frac{1}{\sqrt{e}} \end{cases}$$

or $x = 0$

$$\text{When } x = \frac{1}{\sqrt{e}} \text{ then } f''\left(\frac{1}{\sqrt{e}}\right) = 2 \log\left(\frac{1}{\frac{1}{\sqrt{e}}}\right) - 3$$

$$= 2 \log(e)^{1/2} - 3 = 2 \times \frac{1}{2} \log e - 3 = 1 \times 1 - 3 = -2 < 0$$

$$\Rightarrow f(x) \text{ is maximum at } x = \frac{1}{\sqrt{e}}$$



Comprehensive Exercise 10.1

1. Prove that the following functions do not have maxima or minima.
(i) e^x (ii) $x^3 - 6x^2 + 24x + 4$ (iii) $\log x$
[Hint: $f'(x) \neq 0$]
 2. Find the points of local maxima or local minima, if any, of the following functions, using the first derivative test. Also, find the local maximum or local minimum value, as the case may be.
(i) $x^3 - 3x$ (ii) $\sin x - \cos x, 0 < x < 2\pi$ (iii) $(x-1)(x+2)^2$
 3. Find the maximum value of the function $(x-1)(x-2)(x-3)$.
[B.C.A. (Rohilkhand) 2005]
 4. Find the maximum or minimum values of the following functions:
(i) $x^3 - 2x^2 + x + 6$ (ii) $ax + \frac{b}{x}$ (iii) $3x^4 - 10x^3 + 6x^2 + 5$
 5. Show that the function $x^3 - 3x^2 + 3x + 7$ has neither maximum nor minimum at $x = 1$.
 6. Show that the function $x^5 - 5x^4 + 5x^3 - 10$ is neither maximum nor minimum at $x = 0$.
 7. Find the maximum and minimum values of $a \sin^2 \theta + b \cos^2 \theta$ when $a > b$ and $0 < \theta < 2\pi$.
 8. Show that $\frac{x^2 + x + 1}{x^2 - x + 1}$ has 3 for its maximum value and $\frac{1}{3}$ for its minimum value.
 9. Show that the value of x^x is minimum when $x = \frac{1}{e}$.
 10. Show that the function $\sin x (1 + \cos x)$ has maximum at $x = \frac{\pi}{3}$.
 11. Prove that $\frac{x}{1 + x \tan x}$ is maximum when $x = \cos x$.
 12. Find the maximum and minimum values of the function $\frac{\sin x}{1 + \sin x}$ when $0 < x < 2\pi$.
-



Answers

2. (i) $x = -1$, point of local max.; local max. value is 2
 $x = 1$, point of local min.; local min. value is -2
 (ii) $x = \frac{3\pi}{4}$ is the point of local max; and the value is $\sqrt{2}$
 $x = \frac{7\pi}{4}$ is the point of local min; and the value is $-\sqrt{2}$
 (iii) $x = 0$, point of local minimum; local minimum value is -4 .
 $x = -2$, point of local max; local max value is 0.
3. $\frac{2}{3\sqrt{3}}$
4. (i) Max. value $6\frac{4}{27}$ at $x = \frac{1}{3}$; Min. value 6 at $x = 1$
 (ii) Max. value $-2a\sqrt{\frac{a}{b}}$ at $x = -\sqrt{\frac{b}{a}}$; and Min. value $= 2a\sqrt{\frac{a}{b}}$ at $x = \sqrt{\frac{b}{a}}$
 (iii) Max. value $5\frac{7}{16}$ at $x = \frac{1}{2}$; Min. value are 5 at $x = 0$, and -3 at $x = 2$
7. Maximum value is a and Minimum value is b .
12. Maximum value $= \frac{1}{2\sqrt{2}}$ at $x = \frac{\pi}{4}$
 and Minimum value $= -\frac{1}{2\sqrt{2}}$ at $x = \frac{5\pi}{4}$

10.5 Application of Maxima and Minima to Practical Problems

Example 20: Amongst all pairs of positive numbers with product 256, find those whose sum is the least.

Solution: Let the required numbers be x and $\frac{256}{x}$

Let,
$$S = \left(x + \frac{256}{x} \right)$$

Then,
$$\frac{dS}{dx} = \left(1 - \frac{256}{x^2} \right) \text{ and } \frac{d^2S}{dx^2} = \frac{512}{x^3}$$

For a maxima or minima, we have $\frac{dS}{dx} = 0$

Now,
$$\frac{dS}{dx} = 0 \Rightarrow \left(1 - \frac{256}{x^2} \right) = 0 \Rightarrow x^2 = 256 \text{ or } x = 16$$

Also,
$$\left[\frac{d^2 S}{dx^2} \right]_{x=16} = \left(\frac{512}{16 \times 16 \times 16} \right) = \frac{1}{8} > 0$$

So, $x = 16$ is a point of minimum.

Hence, the required numbers are 16 and 16.

Example 21: Find two positive numbers x and y such that $(x + y) = 60$ and xy^3 is maximum.

Solution: Let $(x + y) = 60$ and let $P = xy^3$

Then,
$$P = (60 - y) y^3 \quad [\because x = (60 - y)]$$

$$\therefore \frac{dP}{dy} = 3y^2 (60 - y) + y^3 (-1) = (180y^2 - 4y^3) = 4y^2 (45 - y)$$

and,
$$\frac{d^2 P}{dy^2} = (360y - 12y^2) = 12y (30 - y)$$

Now,
$$\frac{dP}{dy} = 0 \Rightarrow 4y^2 (45 - y) = 0 \Rightarrow y = 0 \text{ or } y = 45$$

Neglecting $y = 0$, we are left with $y = 45$

Now,
$$\left[\frac{d^2 P}{dy^2} \right]_{y=45} = (12 \times 45) (30 - 45) = -8100 < 0$$

So, $y = 45$ is a point of maximum.

Hence, the numbers are 45 and 15.

Example 22: Find two numbers whose sum is 16 and the sum of whose cubes is minimum.

Solution: Let the numbers be x and $(16 - x)$

Let
$$S = x^3 + (16 - x)^3$$

Then,
$$\frac{dS}{dx} = 3x^2 - 3(16 - x)^2 \text{ and } \frac{d^2 S}{dx^2} = 6x + 6(16 - x) = 96$$

Now,
$$\frac{dS}{dx} = 0 \Rightarrow 3x^2 - 3(16 - x)^2 = 0 \Rightarrow x = 8$$

And,
$$\left[\frac{d^2 S}{dx^2} \right]_{x=8} = 96 > 0$$

$\therefore x = 8$ is a point of minimum.

Hence, the required numbers are 8 and 8.

Example 23: Show that maximum rectangle that can be inscribed in a circle is a square.

[B.C.A (Meerut) 2005]

Solution: Let PQRS be the rectangle inscribed in the circle with centre O and radius a . Let $PQ = 2x$.

From O draw OL perpendicular to PQ, then

$$OL = \sqrt{(a^2 - x^2)}$$

$$\therefore QR = 2\sqrt{(a^2 - x^2)}$$

$\therefore$ If A be the area of the rectangle, then

$$A = PQ \times QR = 2x \cdot 2\sqrt{(a^2 - x^2)}$$

$$= 4x\sqrt{(a^2 - x^2)}$$

$$\frac{dA}{dx} = 4 \left[x \left(\frac{-2x}{2\sqrt{(a^2 - x^2)}} \right) + \sqrt{(a^2 - x^2)} \right] = \frac{4(a^2 - 2x^2)}{\sqrt{(a^2 - x^2)}}$$

$$\frac{d^2 A}{dx^2} = 4 \left[\frac{\sqrt{a^2 - x^2}(-4x) - (a^2 - 2x^2)(1/3)(a^2 - x^2)^{-1/2}(-2x)}{(a^2 - x^2)} \right]$$

$$= -4 \left[\frac{4x(a^2 - x^2) - x(a^2 - 2x^2)}{(a^2 - x^2)^{3/2}} \right]$$

Equating dA/dx to zero, we get

$$a^2 - 2x^2 = 0 \quad \text{or} \quad x = a/\sqrt{2}.$$

$$\text{At } x = a/\sqrt{2}, \quad \frac{d^2 A}{dx^2} = \frac{-16(a/\sqrt{2})\{a^2 - (a^2/2)\}}{\{a^2 - (a^2/2)\}^{1/2}} = \text{negative}$$

$$\therefore A \text{ is max when } x = a/\sqrt{2} \Rightarrow PQ = 2x = \sqrt{2}a \text{ and } QR = \sqrt{2}a$$

$$\Rightarrow PQ = QR \text{ when } x = a/\sqrt{2} \Rightarrow \text{Maximum rectangle is a square.}$$

Hence the max rectangle inscribed in a circle is a square.

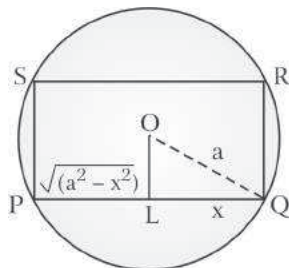


Fig. 10.1

Example 24: Show that the height of the closed cylinder of given volume and least surface is equal to its diameter.

[B.C.A. (Meerut) 2005]

Solution: Let h be the height and r the radius of the base of the cylinder.

$$\therefore V = \pi r^2 h, \text{ where } V \text{ is given} \quad \dots(1)$$

$$\text{and} \quad S = 2\pi rh + 2\pi r^2 \quad \dots(2)$$

$$\text{From (1), we get } h = V/\pi r^2 \quad \dots(3)$$

Substituting this value of h in (2), we get

$$S = 2\pi r(V/\pi r^2) + 2\pi r^2 = (2V/r) + 2\pi r^2$$

$$\therefore \frac{dS}{dr} = -\frac{2V}{r^2} + 4\pi r \quad \text{and} \quad \frac{d^2S}{dr^2} = \frac{4V}{r^3} + 4\pi$$

Equating dS/dr to zero, we get

$$\frac{2V}{r^2} = 4\pi r \quad \text{or} \quad V = 2\pi r^3$$

Using this value of V in d^2S/dr^2 , we get

$$\frac{d^2S}{dr^2} = 8\pi + 4\pi = 12\pi = \text{positive.}$$

Hence, S is least for $V = 2\pi r^3$. For least S , from (3), we get

$$h = \frac{V}{\pi r^2} = \frac{2\pi r^3}{\pi r^2} = 2r = \text{diameter of the base of the cylinder.}$$

Example 25: Prove that a conical tent of a given capacity will require the least amount of canvas when the height is $\sqrt{2}$ times the radius of the base. [B.C.A. (Meerut) 2006]

Solution: Let h be the height, r the radius of the base and l the slant height of the conical tent. Let V be the given capacity (volume) and S the area of the curved surface of the conical tent. then we have

$$V = \frac{1}{3}\pi r^2 h \quad \text{or} \quad h = \frac{3V}{\pi r^2} \quad \dots(1)$$

and

$$S = \pi l r = \pi \sqrt{(h^2 + r^2)}.r$$

or

$$S^2 = \pi^2 r^2 (h^2 + r^2) = u \text{ (say)} \quad \dots(2)$$

Substituting the value of h in (2), we get

$$u = \pi^2 r^2 \left[\frac{9V^2}{\pi^2 r^4} + r^2 \right] = \frac{9V^2}{r^2} + \pi^2 r^4$$

$$\frac{du}{dr} = -\frac{18V^2}{r^3} + 4\pi^2 r^3 \quad \text{and} \quad \frac{d^2u}{dr^2} = \frac{54V^2}{r^4} + 12\pi^2 r^2$$

Equating du/dr to zero, we get

$$18V^2 = 4\pi^2 r^6 \quad \text{or} \quad V = \frac{2}{3\sqrt{2}}\pi r^3$$

And for this value of V we find that d^2u/dr^2 is positive, i.e., u [or S^2]

from (2) is least. Hence S is least when $V = \frac{2}{3\sqrt{2}}\pi r^3$

$$\text{i.e., when} \quad \frac{2}{3}\pi r^3 / \sqrt{2} = \frac{1}{2}\pi r^2 h,$$

Using (1)

$$\text{i.e., when} \quad h = r\sqrt{2}$$

Example 26: Show that the radius of the right circular cylinder of greatest curved surface which can be inscribed in a given cone is half that of the cone.

[B.C.A. (Meerut) 2008, 2001]

Solution: Suppose r is the radius and h is the height of the given cone.

$\Rightarrow OB = r, OA = h$ where O is the centre of the base circle.

Suppose x is the radius and H is the height of the cylinder inscribed in the given cone.

Now triangles AOB and ADE are similar, therefore

$$\frac{AD}{AO} = \frac{DE}{OB} \text{ or } \frac{h-H}{h} = \frac{x}{r}$$

$$\Rightarrow 1 - \frac{H}{h} = \frac{x}{r} \Rightarrow \frac{H}{h} = 1 - \frac{x}{r}$$

$$\Rightarrow H = h \left(1 - \frac{x}{r} \right)$$

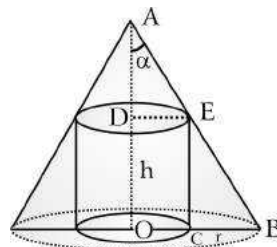


Fig. 10.2

Now the curved surface of the cylinder

$$= S = 2\pi \cdot x \cdot H = 2\pi \cdot x \cdot h \left(1 - \frac{x}{r} \right) = \frac{2\pi h}{r} (rx - x^2)$$

$$\Rightarrow \frac{dS}{dx} = \frac{2\pi h}{r} (r - 2x). \text{ Solving } \frac{dS}{dx} = 0, \text{ we get } r - 2x = 0 \text{ or } x = r/2$$

$$\text{Also } \frac{d^2S}{dx^2} = \frac{2\pi h}{r} \cdot (-2) < 0$$

$\Rightarrow S$ is greatest when $x = r/2$, i.e., when radius of the cylinder is half of that of cone.

Example 27: Show that the volume of the greatest cylinder, which can be inscribed in a cone of height h and semi-vertical angle α , is $(4/27) \pi h^3 \tan^3 \alpha$. [B.C.A. (Meerut) 2001]

Solution: Let r be the radius of the given cone whose height is h and semi vertical angle is α (both given). Then $OB = r \Rightarrow$ By triangle AOB , we get $\tan \alpha = r/h \Rightarrow r = h \tan \alpha$.

Suppose x is the radius and H is the height of the inscribed cylinder. Then by triangle AMN .

$$\tan \alpha = \frac{x}{h-H} \Rightarrow h-H = \frac{x}{\tan \alpha} = x \cot \alpha \Rightarrow H = h - x \cot \alpha$$

Now volume of the cylinder $= V = \pi x^2 H$

$$\Rightarrow V = \pi x^2 (h - x \cot \alpha) = \pi (hx^2 - x^3 \cot \alpha)$$

We are to maximize this V as a function of x . For, differentiating it w.r.t. x , we get

$$\frac{dV}{dx} = \pi (h \cdot 2x - 3x^2 \cot \alpha) = \pi x (2h - 3x \cot \alpha)$$

Solving $dV/dx = 0$, we get $x = 0, 2h \tan \alpha / 3$

$$\text{Now } \frac{d^2V}{dx^2} = \pi (2h - 6x \cot \alpha) = 2\pi (h - 3x \cot \alpha)$$

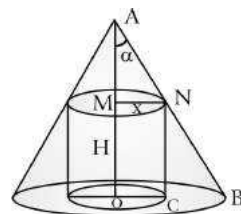


Fig. 10.3

Putting $x = 0$, we get $\left(\frac{d^2V}{dx^2}\right) = 2\pi h > 0 \Rightarrow x = 0$ is point of minima.

Putting $x = \frac{2h \tan \alpha}{3}$, we get $\frac{d^2V}{dx^2} = 2\pi \left(h - \frac{3}{\tan \alpha} \cdot \frac{2h}{3} \tan \alpha\right) < 0$

$\Rightarrow V$ is maximum when $x = 2h \tan \alpha / 3$ (point of maxima)

$\Rightarrow$ maximum volume $= V = \pi (h - x \cot \alpha) \cdot x^2$

$$\Rightarrow V_{\max} = \pi \left(h - \frac{2}{3} h \tan \alpha \cdot \frac{1}{\tan \alpha}\right) \cdot \left(\frac{2h \tan \alpha}{3}\right)^2 = \frac{\pi h}{3} \cdot \frac{4h^2 \tan^2 \alpha}{9}$$

$$\Rightarrow \text{Max volume} = \frac{4}{27} \pi h^3 \tan^2 \alpha$$

Example 28: Show that the volume of the largest cone that can be inscribed in a sphere of radius 'a', is $(8/27)$ of the volume of the sphere.

Solution: Suppose r is the radius of the cone and $OM = x$. Then

$$r^2 + x^2 = a^2 \quad \dots(1)$$

$\Rightarrow$ height of the cone $= HC = x + a$

Now volume of the cone $= V = \frac{1}{3} \pi r^2 \cdot (x + a)$

$$(1) \Rightarrow V = \frac{1}{3} \pi (a^2 - x^2)(x + a) \quad \dots(2)$$

We are to maximise this V . Diff. it w.r.t. x ,

$$\frac{dV}{dx} = \frac{\pi}{3} [-2x(x + a) + (a^2 + x^2) \cdot 1] = \frac{\pi}{3} [a^2 - 2ax - 3x^2] \quad \dots(3)$$

Now solving $dV/dx = 0$, we get $a^2 - 2ax - 3x^2 = 0$

$$\text{or } (a^2 - x^2) - 2ax - 2x^2 = 0 \quad \text{or } (a + x)(a - x) - 2x(a + x) = 0$$

$$\Rightarrow (x + a)(a - x - 2x) = 0 \Rightarrow x = -a \quad \text{or } a/3.$$

But x can not be $-a$ (negative). Therefore only $x = a/3$.

Now differentiating (3) again w.r.t. x , we get

$$\frac{d^2V}{dx^2} = \frac{\pi}{3} (-2a - 6x) < 0, \text{ when } x = \frac{a}{3}$$

$$\Rightarrow V \text{ is maximum when } x = a/3. \text{ Then maxi valume (V)} = \frac{\pi}{3} \left(a^2 - \frac{a^2}{9}\right) \left(\frac{a}{3} + a\right)$$

$$= \frac{\pi}{3} \cdot \frac{8a^2}{9} \cdot \frac{4a}{3} = \frac{32\pi a^3}{81} \quad \dots(4)$$

$$\text{Also valume of the sphere } V_1 = \frac{4}{3} \pi a^3 \quad \dots(5)$$

$$\therefore \frac{V}{V_1} = \frac{8}{27} \Rightarrow V = \frac{8}{27} V_1$$

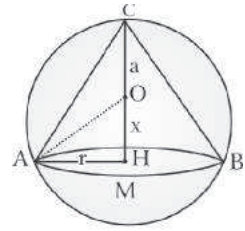


Fig. 10.4

Example 29: Assuming the petrol burnt (per hour) in driving a motor boat varies as the cube of its velocity, show that most economical speed when going against a current of c kms per hour is $(3/2)c$ kms per hour. [B.C.A. (Rohilkhand) 2000; B.C.A. (Meerut) 2004]

Solution: Let the velocity of the motor boat be v km/ hour. Then velocity of the boat relative to the current $(v - c)$ km per hour. Let the total distance to be covered be a km.

Then time taken $= a / (v - c)$ hours

Also we are given that the petrol burnt in one hour $= kv^3$ where k is constant of proportionality.

$\therefore$ If P be the total amount of petrol burnt in covering a kms , then

$$P = kv^3 \frac{a}{(v - c)} \quad \dots(1)$$

$$\therefore \frac{dP}{dv} = ak \left[\frac{(v - c)3v^2 - v^3(1)}{(v - c)^2} \right] = \frac{ak(2v^3 - 3cv^2)}{(v - c)^2}$$

$$\begin{aligned} \text{And } \frac{d^2P}{dv^2} &= \frac{ak}{(v - c)^4} [(v - c)^2(6v^2 - 6vc) - (2v^3 - 3cv^2)2(v - c)] \\ &= \frac{ak}{(v - c)^3} [6v(v - c)^2 - 2v^2(2v - 3c)] \end{aligned}$$

Equating dP / dv to zero, we get

$$2v^3 - 3cv^2 = 0, \text{ which gives either } v = 0 \text{ or } \frac{3}{2}c.$$

If $v = 0, P = 0$, i.e., no petrol is burnt.

If $v = 3c/2, d^2P / dv^2 = \text{Positive}$, i.e., P is minimum,

$\therefore$ Petrol burnt is minimum when $v = 3c/2$, i.e., the most economical speed is $(3c/2)$ km per hour.

Example 30: The fuel charges for running a train are proportional to the square of the speed generated in kilometres/hour and costs ₹. 48 per hour at 16 km per hour. What is the most economic speed if the fixed charges, i.e., salaries *etc.*, amount to ₹. 300 per hour? [B.C.A. (Meerut) 2006]

Solution: Let the speed be v km per hour and the cost of fuel per hour in rupees $= kv^2$ where k is constant.

Therefore, $48 = k(16)^2$ ($\because$ at 16 km per hour speed, charges are ₹. 48) or $k = 3/16$.

Again the time in covering the distance of 80 km $= 80 / v$ hrs.

$$\text{Therefore, the fuel charges for this time} = kv^2 \frac{80}{v} = \frac{3}{16} \times (80)v = 15v$$

$$\text{and the fixe charges} = 300 \times \frac{80}{v} = \frac{24,000}{v}$$

$$\text{Let } z \text{ denote the total charges; then } z = 15v + \frac{24,000}{v}$$

$$\therefore \frac{dz}{dv} = 15 - \frac{24,000}{v^2} = 0 \text{ for maxima or minima of } z$$

$$\text{i.e., } 15v^2 = 24,000 \text{ or } v^2 = 1,600 \text{ or } v = 40 \text{ km per hour}$$

$$\text{Again } \frac{d^2z}{dv^2} = \frac{48,000}{v^3}, \text{ which is a positive quantity at } v = 40$$

$\therefore z$ is minimum when $v = 40$, i.e., the most economical speed is 40 km per hour.

Example 31: Prove that the area of a right-angled triangle of a given hypotenuse is maximum when the triangle is isosceles.

Solution: Let h be the hypotenuse of the triangle and x be its altitude.

Then, the base of the triangle $= \sqrt{h^2 - x^2}$

$$\text{Now, } A = \frac{1}{2} x \sqrt{h^2 - x^2}$$

$$\therefore \frac{dA}{dx} = \frac{1}{2} \left\{ x \cdot \frac{1}{2} (h^2 - x^2)^{-1/2} (-2x) + \sqrt{h^2 - x^2} \right\} = \frac{1}{2} \left(\frac{h^2 - 2x^2}{\sqrt{h^2 - x^2}} \right)$$

$$\text{and, } \frac{d^2A}{dx^2} = \frac{(\sqrt{h^2 - x^2})(-4x) - (h^2 - 2x^2) \cdot \frac{1}{2} (h^2 - x^2)^{-1/2} (-2x)}{2(h^2 - x^2)} \\ = \frac{(2x^3 - 3xh^2)}{2(h^2 - x^2)^{3/2}}$$

$$\text{Now, } \frac{dA}{dx} = 0 \Rightarrow (h^2 - 2x^2) = 0 \Rightarrow x = \frac{h}{\sqrt{2}}$$

$$\therefore \left[\frac{d^2A}{dx^2} \right]_{x=\frac{h}{\sqrt{2}}} = -2 < 0$$

Thus, A is maximum at $x = \left(\frac{h}{\sqrt{2}} \right)$

$$\therefore \text{base} = \sqrt{h^2 - x^2} = \sqrt{\left(h^2 - \frac{h^2}{2} \right)} = \frac{h}{\sqrt{2}} = \text{altitude.}$$

Hence, the triangle is isosceles.

Example 32: If the sum of the lengths of the hypotenuse and a side of a right-angled triangle is given, show that the area of the triangle is maximum when the angle between them is $(\pi/3)$.

Solution: Let us consider a right-angled triangle with base $= x$ and hypotenuse $= y$. Let $x + y = k$, where k is a constant. Let θ be the angle between the base and the hypotenuse. Let A be the area of the triangle. Then,

$$A = \frac{1}{2} \times BC \times AC = \frac{1}{2} x \sqrt{y^2 - x^2}$$

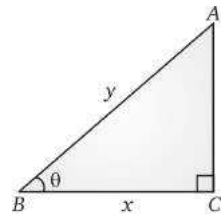


Fig. 10.5

$$\therefore A^2 = \frac{x^2 (y^2 - x^2)}{4} = \frac{x^2 [(k-x)^2 - x^2]}{4} \quad [\because y = (k-x)]$$

$$\text{or} \quad A^2 = \frac{k^2 x^2 - 2kx^3}{4} \quad \dots(1)$$

$$\text{On differentiating (1), we get } 2A \cdot \frac{dA}{dx} = \frac{2k^2 x - 6kx^2}{4} \quad \dots(2)$$

$$\text{or} \quad \frac{dA}{dx} = \frac{k^2 x - 3kx^2}{4A}$$

$$\text{Now,} \quad \frac{dA}{dx} = 0 \Rightarrow (k^2 x - 3kx^2) = 0 \Rightarrow x = \frac{k}{3} \quad [\text{neglecting } x = 0]$$

Now, differentiating (2), we get

$$2 \left(\frac{dA}{dx} \right)^2 + 2A \cdot \frac{d^2 A}{dx^2} = \frac{2k^2 - 12kx}{4} \quad \dots(3)$$

$$\text{Putting } \frac{dA}{dx} = 0 \text{ and } x = \frac{k}{3} \text{ in (3), we get } \frac{d^2 A}{dx^2} = \frac{-k^2}{4A} < 0$$

Thus, A is maximum when $x = (k/3)$.

$$\text{Now,} \quad x = \frac{k}{3} \Rightarrow y = \left(k - \frac{k}{3} \right) = \frac{2k}{3}$$

$$\therefore \quad \frac{x}{y} = \cos \theta \Rightarrow \cos \theta = \frac{(k/3)}{(2k/3)} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

Example 33: Show that the triangle of maximum area that can be inscribed in a given circle is an equilateral triangle.

Solution: Let ABC be a triangle inscribed in a given circle with centre O and radius r . For maximum area, the vertex A should be at a maximum distance from the base BC .

Therefore, A must lie on the diameter, perpendicular to BC . Thus, $AD \perp BC$.

So, triangle ABC must be isosceles.

$$\text{Let} \quad \angle CAD = \theta,$$

$$\text{Now,} \quad BC = 2 CD = 2 OC \sin 2\theta = 2 r \sin 2\theta$$

$$\text{and,} \quad AD = (OA + OD) = (r + r \cos 2\theta)$$

Let A be the area of the triangle.

$$\text{Then,} \quad A = \frac{1}{2} BC \times AD = r^2 \sin 2\theta (1 + \cos 2\theta)$$

$$\begin{aligned} \therefore \quad \frac{dA}{d\theta} &= r^2 [\sin 2\theta (-2\sin 2\theta) + (1 + \cos 2\theta) \cdot 2 \cos 2\theta] \\ &= r^2 [2 (\cos^2 2\theta - \sin^2 2\theta) + 2 \cos 2\theta] \\ &= 2r^2 [\cos 4\theta + \cos 2\theta] \end{aligned}$$

$$\text{And,} \quad \frac{d^2 A}{d\theta^2} = 2r^2 [-4 \sin 4\theta - 2 \sin 2\theta] = -4r^2 (2 \sin 4\theta + \sin 2\theta)$$

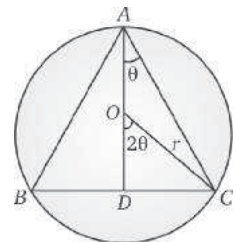


Fig. 10.6

Now, $\frac{dA}{d\theta} = 0 \Rightarrow \cos 4\theta + \cos 2\theta = 0$

$\Rightarrow \cos 4\theta = -\cos 2\theta = \cos (\pi - 2\theta)$

$\Rightarrow 4\theta = \pi - 2\theta \Rightarrow \theta = \frac{\pi}{6}$

And, $\left[\frac{d^2 A}{d\theta^2} \right]_{\theta=(\pi/6)} = -4r^2 \left(2 \sin \frac{2\pi}{3} + \sin \frac{\pi}{3} \right) = -6r^2 \sqrt{3} < 0$

$\therefore \theta = \frac{\pi}{6}$ is a point of maximum.

So, in this case, each angle of the triangle is $(\pi/3)$

Hence, ABC is an equilateral triangle.

Example 34: Prove that the least perimeter of an isosceles triangle in which a circle of radius r can be inscribed, is $6r\sqrt{3}$.

Solution: Let ABC be an isosceles triangle circumscribing the circle of radius r (centre at O). Suppose $\angle MAC = \theta$. Then in the triangle ONA , we have

$$\frac{AN}{r} = \cot \theta \quad \text{and} \quad \frac{OA}{r} = \operatorname{cosec} \theta$$

$\Rightarrow AN = r \cot \theta \quad \text{and} \quad OA = r \operatorname{cosec} \theta$

$\Rightarrow AM = r + r \operatorname{cosec} \theta = r(1 + \operatorname{cosec} \theta)$

Further in the triangle AMC , $\frac{MC}{MA} = \tan \theta$

$\Rightarrow MC = r(1 + \operatorname{cosec} \theta) \tan \theta$. Also $CN = MC$

Now perimeter of the $\triangle ABC = P = 2 [MC + CN + NA]$

$\Rightarrow P = 2 [2MC + NA] = 2 [2r(1 + \operatorname{cosec} \theta) \tan \theta + r \cot \theta]$

$\Rightarrow P = 2r[2(1 + \operatorname{cosec} \theta) \tan \theta + \cot \theta] = 2r[2 \tan \theta + 2 \sec \theta + \cot \theta] \quad \dots(1)$

$\Rightarrow \frac{dP}{d\theta} = 2r[2 \sec^2 \theta + 2 \sec \theta \tan \theta - \operatorname{cosec}^2 \theta] \quad \dots(2)$

Solving $dP / d\theta = 0$, we get $2 \sec^2 \theta + 2 \sec \theta \tan \theta - \operatorname{cosec}^2 \theta = 0$

or $\frac{2}{\cos^2 \theta} + \frac{2 \sin \theta}{\cos^2 \theta} - \frac{1}{\sin^2 \theta} = 0$ or $2 \sin^2 \theta + 3 \sin^3 \theta - \cos^2 \theta = 0$

or $2 \sin^2 \theta + 2 \sin^3 \theta - (1 - \sin^2 \theta) = 0$ or $2 \sin^3 \theta + 3 \sin^2 \theta - 1 = 0$

or $2 \sin^3 \theta + 2 \sin^2 \theta + \sin^2 \theta - 1 = 0$

or $2 \sin^2 \theta (\sin \theta + 1) + (\sin \theta + 1)(\sin \theta - 1) = 0$

or $(\sin \theta + 1)(2 \sin^2 \theta + \sin \theta - 1) = 0$

or $(\sin \theta + 1)(\sin \theta + 1)(2 \sin \theta - 1) = 0$

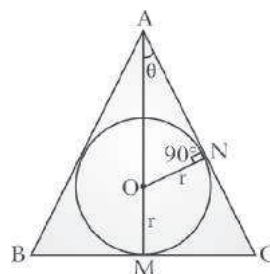


Fig. 10.7

$$\Rightarrow \sin \theta = -1 \text{ or } \sin \theta = 1/2 \Rightarrow \theta = -\pi/2 \text{ or } \pi/6$$

But θ is an acute angle, so it can not be $-\pi/2$

$$\Rightarrow \text{only } \theta = \pi/6$$

$$\text{Further } \frac{d^2 P}{d\theta^2} = 2r' [2.2 \sec^2 \theta \tan \theta + 2(\sec^2 \theta \tan \theta + \sec^3 \theta) + 2 \operatorname{cosec}^2 \theta \cot \theta]$$

$$\Rightarrow \left(\frac{d^2 P}{d\theta^2} \right)_{(\theta=\pi/6)} = 2r \left[4 \sec^2 \frac{\pi}{6} \tan \frac{\pi}{6} + 2 \left(\sec^2 \frac{\pi}{6} \tan \frac{\pi}{6} + \sec^3 \frac{\pi}{6} \right) \right] + 2 \operatorname{cosec}^2 \frac{\pi}{6} \cot \frac{\pi}{6} > 0$$

$\Rightarrow P$ will be least when $\theta = \pi/6$. Then

$$\begin{aligned} \text{Min value of } P &= 2r \left[2 \tan \frac{\pi}{6} + 2 \sec \frac{\pi}{6} + \cot \frac{\pi}{6} \right] = 2r \left[2 \cdot \frac{1}{\sqrt{3}} + \frac{2.2}{\sqrt{3}} + \sqrt{3} \right] \\ &= 2r \cdot \frac{9}{\sqrt{3}} = \frac{2r9\sqrt{3}}{\sqrt{3}\sqrt{3}} = \frac{2.9\sqrt{3}}{3} r = 6\sqrt{3} r \quad \text{Proved.} \end{aligned}$$

Example 35: Show that of all the rectangles with a given perimeter, the square has the largest area.

Solution: Let a be the fixed perimeter.

Consider a rectangle with sides x and y and perimeter a .

Let A be the area of the rectangle

$$\text{Then, } 2x + 2y = a \quad \dots(1)$$

$$\text{And, } A = xy = x \left(\frac{a-2x}{2} \right) \quad [\text{using (1)}]$$

$$\therefore \frac{dA}{dx} = \left(\frac{1}{2} a - 2x \right) = \frac{1}{2} (a - 4x) \text{ and } \frac{d^2 A}{dx^2} = -2$$

$$\text{Now, } \frac{dA}{dx} = 0 \Rightarrow \frac{1}{2} (a - 4x) = 0 \Rightarrow x = \frac{a}{4}$$

$$\text{Also, } \left[\frac{d^2 A}{dx^2} \right]_{x=(a/4)} = -2 < 0$$

$\therefore x = (a/4)$ is a point of maximum.

$$\text{Also, from (i), we have } y = \frac{1}{2} (a - 2x) = \frac{1}{2} \left(a - \frac{a}{2} \right) = \frac{1}{4} a, \text{ when } x = \frac{a}{4}$$

$$\text{Thus, } x = y \quad [\text{each} = (a/4)]$$

Hence, the rectangle is a square.

Example 36: Show that of all the rectangles of a given area, the square has the smallest perimeter. [B.C.A. (Meerut) 2003]

Solution: Let A be the fixed area.

Consider a rectangle with sides x and y and area A .

Let P be the perimeter of this rectangle.

$$\text{Then, } A = xy \Rightarrow y = \frac{A}{x} \quad \dots(1)$$

$$\text{And, } P = 2x + 2y = 2x + \frac{2A}{x} \quad [\text{using (1)}]$$

$$\text{Now, } P = \left(2x + \frac{2A}{x}\right) \Rightarrow \frac{dP}{dx} = \left(2 - \frac{2A}{x^2}\right) \text{ and } \frac{d^2P}{dx^2} = \frac{4A}{x^3}$$

$$\text{Now, } \frac{dP}{dx} = 0 \Rightarrow 2 - \frac{2A}{x^2} = 0 \Rightarrow x = \sqrt{A}$$

$$\text{And, } \left[\frac{d^2P}{dx^2}\right]_{x=\sqrt{A}} = \frac{4A}{A^{3/2}} = \frac{4}{\sqrt{A}} > 0$$

So, $x = \sqrt{A}$ is a point of minimum.

$$\text{Moreover, } x = \sqrt{A} \Rightarrow y = \frac{A}{x} = \frac{A}{\sqrt{A}} = \sqrt{A} = x$$

So, when the perimeter is smallest, the rectangle is a square.

Example 37: The combined resistance R of two resistors R_1 and R_2 where $R_1, R_2 > 0$ is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}.$$

If $R_1 + R_2 = C$ (constant), show that the maximum resistance R is obtained by choosing $R_1 = R_2$. B.C.A. (Delhi) 2011

Solution: We have $R = \frac{R_1 R_2}{(R_1 + R_2)}$ and $R_1 + R_2 = C$

$$\therefore R = \frac{R_1 (C - R_1)}{C} \quad [\because R_2 = (C - R_1)]$$

$$\text{So, } \frac{dR}{dR_1} = \frac{C - 2R_1}{C} \text{ and } \frac{d^2R}{dR_1^2} = -\frac{2}{C}$$

For a maxima or minima, we have $\frac{dR}{dR_1} = 0$

$$\text{Now, } \frac{dR}{dR_1} = 0 \Rightarrow \frac{C - 2R_1}{C} = 0 \text{ or } R_1 = \frac{C}{2}$$

$$\text{And, } \left[\frac{d^2R}{dR_1^2}\right]_{R_1=(C/2)} = \frac{-2}{C} < 0$$

Thus, $R_1 = \frac{C}{2}$ is a point of maximum.

When, $R_1 = \frac{C}{2}$, we have $R_2 = \left(C - \frac{C}{2}\right) = \frac{C}{2}$. Hence, $R_1 = R_2$

Example 38: A beam of length l is supported at one end. If W is the uniform load per unit length, the bending moment M at a distance x from the end is given by $M = \left(\frac{lx}{2} - \frac{Wx^2}{2} \right)$. Find the point on the beam at which the bending moment has the maximum value.

Solution: $M = \left(\frac{lx}{2} - \frac{Wx^2}{2} \right) \Rightarrow \frac{dM}{dx} = \left(\frac{l}{2} - Wx \right)$ and $\frac{d^2M}{dx^2} = -W$

For a maxima or minima, we have $\frac{dM}{dx} = 0$

Now, $\frac{dM}{dx} = 0 \Rightarrow \frac{l}{2} - Wx = 0 \Rightarrow x = \frac{l}{2W}$

Also, $\frac{d^2M}{dx^2} = -W < 0$ for all values of x

$\therefore x = \frac{l}{2W}$ is a point of maxima.

So, the required point is at a distance of $(l/2W)$ from the supporting end.

Example 39: An open box is to be made out of a piece of cardboard measuring $(24 \text{ cm} \times 24 \text{ cm})$ by cutting off equal squares from the corners and turning up the sides. Find the height of the box when it has maximum volume.

Solution: Let the length of the side of each square cut off from the corners be x cm.

Then, height of the box $= x$ cm

$\therefore V = (24 - 2x)^2 \times x = 4x^3 - 96x^2 + 576x$

$\Rightarrow \frac{dV}{dx} = 12(x^2 - 16x + 48)$

and, $\frac{d^2V}{dx^2} = 24(x - 8)$

Now, $\frac{dV}{dx} = 0 \Rightarrow x^2 - 16x + 48 = 0$ i.e., $(x - 12)(x - 4) = 0$

$\Rightarrow x = 4$ [$\because x \neq 12$]

$\left[\frac{d^2V}{dx^2} \right]_{x=4} = -96 < 0. \therefore V$ is maximum at $x = 4$

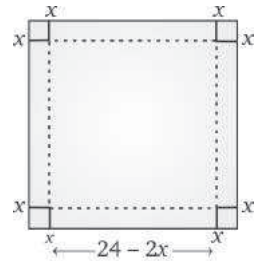


Fig. 10.8

Hence, the volume of the box is maximum when its height is 4 cm.

Example 40: Show that the height of a cylinder which is open at the top, having a given surface and the greatest volume, is equal to the radius of its base.

Solution: Let r be the radius, h the height and S is given surface area of an open cylinder.

Then, $S = (\pi r^2 + 2\pi rh)$ or $h = \left(\frac{S - \pi r^2}{2\pi r} \right)$... (1)

Let V be the volume of the cylinder

Then, $V = \pi r^2 h = \pi r^2 \cdot \left(\frac{S - \pi r^2}{2\pi r} \right) = \left(\frac{rS - \pi r^3}{2} \right)$ [using (1)]

$\therefore \frac{dV}{dr} = \left(\frac{S - 3\pi r^2}{2} \right)$ and $\frac{d^2V}{dr^2} = -3\pi r$

Now, $\frac{dV}{dr} = 0 \Rightarrow S - 3\pi r^2 = 0 \Rightarrow \pi r^2 + 2\pi rh - 3\pi r^2 = 0 \Rightarrow h = r$

And, $\left(\frac{d^2V}{dr^2} \right)_{r=h} = -3\pi h < 0$. Thus, $r = h$ is a point of maximum.

So, V is maximum when $r = h$

Example 41: Two sides of a triangle are given. Find the angle between them such that the area is maximum.

Solution: Let a and b be the lengths of the given sides and let θ be the angle between them. Let A be the area of the triangle.

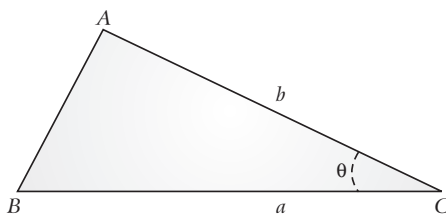


Fig. 10.9

Then, $A = \frac{1}{2} ab \sin \theta$

$\therefore \frac{dA}{d\theta} = \frac{1}{2} ab \cos \theta$ and $\frac{d^2A}{d\theta^2} = -\frac{1}{2} ab \sin \theta$

Now, $\frac{dA}{d\theta} = 0 \Rightarrow \frac{1}{2} ab \cos \theta = 0 \Rightarrow \cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$

And, $\left[\frac{d^2A}{d\theta^2} \right]_{\theta=(\pi/2)} = -\frac{1}{2} ab \sin \frac{\pi}{2} = -\frac{1}{2} ab < 0$

$\therefore \theta = \frac{\pi}{2}$ is a point of maximum.

Hence, the area of the triangle is maximum when the angle between the given sides is $(\pi/2)$.

Example 42: Show that of all the rectangles inscribed in a given fixed circle, the square has the maximum area. [B.C.A. (Meerut) 2005]

Solution: Let $ABCD$ be a rectangle inscribed in a given circle with centre O and radius r .

Let

$$\angle CAB = \theta$$

Then,

$$AC = 2r, AB = 2r \cos \theta \text{ and } BC = 2r \sin \theta$$

Let A be the area of rectangle $ABCD$

Then,

$$A = AB \times BC = 4r^2 \sin \theta \cos \theta = 2r^2 \sin 2\theta$$

Thus,

$$A = 2r^2 \sin 2\theta, \text{ where } r \text{ is constant.}$$

$$\therefore \frac{dA}{d\theta} = 4r^2 \cos 2\theta \text{ and } \frac{d^2 A}{d\theta^2} = -8r^2 \sin 2\theta$$

$$\text{Now, } \frac{dA}{d\theta} = 0 \Rightarrow 4r^2 \cos 2\theta = 0 \Rightarrow \cos 2\theta = 0 \Rightarrow 2\theta = \frac{\pi}{2}, \text{ i.e., } \theta = \frac{\pi}{4}$$

$$\text{And, } \left[\frac{d^2 A}{d\theta^2} \right]_{\theta = (\pi/4)} = -8r^2 \sin \frac{\pi}{2} = -8r^2 < 0$$

$\therefore \theta = (\pi/4)$ is a point of maximum.

Thus, area is maximum when $\theta = (\pi/4)$

$$\text{In this case, } AB = 2r \cos \frac{\pi}{4} = r\sqrt{2}$$

and,

$$BC = 2r \sin \frac{\pi}{4} = r\sqrt{2}$$

Thus,

$$AB = BC \text{ and therefore, } ABCD \text{ is a square.}$$

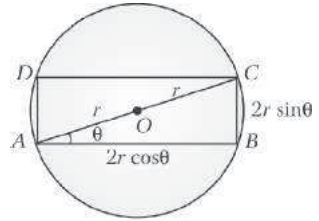


Fig. 10.10

Example 43: Show that the semi-vertical angle of a right circular cone of given surface area and maximum volume is $\sin^{-1}(1/3)$.

[B.C.A. (Meerut) 2002, 2003]

[B.C.A. (Rohilkhand) 2000, 2007]

Solution: Let r be the radius, l the slant height and h the height of the cone.

Let S denote the surface and V the volume of the cone.

Then,

$$S = (\pi r^2 + \pi r l) = \text{constant.}$$

$$\therefore l = \left(\frac{S}{\pi r} - r \right) \quad \dots(1)$$

$$\text{Now, } V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi r^2 \sqrt{l^2 - r^2}$$

$$\therefore V^2 = \frac{1}{9} \pi^2 r^4 (l^2 - r^2) = \frac{1}{9} \pi^2 r^4 \cdot \left\{ \left(\frac{S}{\pi r} - r \right)^2 - r^2 \right\} \quad [\text{using (1)}]$$

$$= \frac{1}{9} S (S r^2 - 2\pi r^4)$$

$$\text{Thus, } V^2 = \left(\frac{1}{9} S^2 r^2 - \frac{2\pi}{9} S r^4 \right)$$

$$\therefore 2V \cdot \frac{dV}{dr} = \left(\frac{2}{9} S^2 r - \frac{8\pi S}{9} r^3 \right) = \frac{2rS}{9} (S - 4\pi r^2) \quad \dots(2)$$

$$\therefore \frac{dV}{dr} = 0 \Rightarrow r = 0 \text{ or } (S - 4\pi r^2) = 0 \Rightarrow r^2 = \frac{S}{4\pi} \quad (\text{neglecting } r = 0)$$

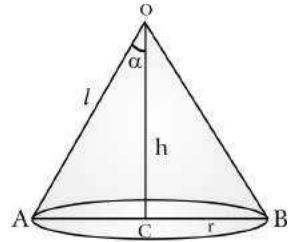


Fig. 10.11

On differentiating (2), we get $2 \left(\frac{dV}{dr} \right)^2 + 2V \cdot \frac{d^2V}{dr^2} = \frac{1}{9} S (2S - 24\pi r^2)$

Putting $\frac{dV}{dr} = 0$ and $r^2 = \frac{S}{4\pi}$, we get

$$2V \cdot \frac{d^2V}{dr^2} = \frac{1}{9} S (2S - 6S) = -\frac{4}{9} S^2 < 0$$

$\therefore$ When the volume is maximum, we have

$$r^2 = \frac{S}{4\pi} = \frac{(\pi r^2 + \pi r l)}{4\pi} \Rightarrow l = 3r$$

Now, if α is the semi-vertical angle of the cone then

$$\frac{r}{l} = \sin \alpha \Rightarrow \frac{r}{3r} = \sin \alpha \Rightarrow \sin \alpha = \frac{1}{3} \Rightarrow \alpha = \sin^{-1}\left(\frac{1}{3}\right)$$

Hence, the semivertical angle of a right cone of a given surface and maximum volume is $\sin^{-1}(1/3)$.

Example 44: Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$. Find the volume of the largest cylinder inscribed in a sphere of radius R .

Solution: Let r be the radius and h the height of the inscribed cylinder $ABCD$. Let V be its volume.

$$\text{Then, } V = \pi r^2 h \quad \dots(1)$$

$$\text{Clearly, } AC = 2R$$

$$\text{Also, } AC^2 = AB^2 + BC^2$$

$$\Rightarrow (2R)^2 = (2r)^2 + h^2$$

$$\Rightarrow r^2 = \frac{1}{4} (4R^2 - h^2) \quad \dots(2)$$

Using (2) in (1), we get

$$V = \frac{\pi h}{4} (4R^2 - h^2) \Rightarrow \frac{dV}{dh} = \left(\pi R^2 - \frac{3}{4} \pi h^2 \right) \text{ and } \frac{d^2V}{dh^2} = -\frac{3}{2} \pi h$$

For a maxima or minima, we have $(dV/dh) = 0$.

$$\text{Now, } \frac{dV}{dh} = 0 \Rightarrow \pi R^2 - \frac{3}{4} \pi h^2 = 0 \Rightarrow h = \frac{2R}{\sqrt{3}}$$

$$\therefore \left[\frac{d^2V}{dh^2} \right]_{h=(2R/\sqrt{3})} = -\frac{3}{2} \pi \times \frac{2R}{\sqrt{3}} = -\pi R \sqrt{3} < 0$$

So, V is maximum when $h = \frac{2R}{\sqrt{3}}$

Hence, the height of the cylinder of maximum volume is $\frac{2R}{\sqrt{3}}$.

$$\text{Largest volume of the cylinder} = \pi \times \frac{1}{4} \left[4R^2 - \frac{4R^2}{3} \right] \times \frac{2R}{\sqrt{3}} = \frac{4\pi R^3}{3\sqrt{3}}$$

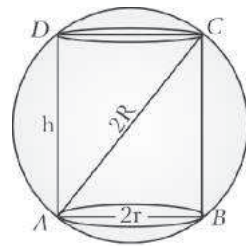


Fig. 10.12

Example 45: Show that the cone of greatest volume which can be inscribed in a given sphere is such that three times its altitude is twice the diameter of the sphere. Find the volume of the largest cone inscribed in a sphere of radius R .

Solution: Let R be the radius of the given sphere with centre O , and let V be the volume of the inscribed cone, h be its height and r be the radius of the base.

In the given figure, we have $OD = AD - AO = (h - R)$

$$\therefore R^2 = (h - R)^2 + r^2 \text{ or } r^2 = h(2R - h) \quad \dots(1)$$

Now, $V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi h^2 (2R - h)$ [using (1)]

$$\therefore \frac{dV}{dh} = \frac{1}{3} \pi h (4R - 3h), \text{ and } \frac{d^2V}{dh^2} = \left(\frac{4}{3} \pi R - 2\pi h \right)$$

For a maxima or minima, we have $\frac{dV}{dh} = 0$

$$\text{Now, } \frac{dV}{dh} = 0 \Rightarrow \frac{1}{3} \pi h (4R - 3h) = 0$$

$$\Rightarrow h = 0 \text{ or } (4R - 3h) = 0 \Rightarrow h = \frac{4}{3} R \quad [\because h \neq 0]$$

$$\text{And, } \left[\frac{d^2V}{dh^2} \right]_{h=(4/3)R} = -\frac{4\pi R}{3} < 0$$

So, V is maximum when $h = \frac{4}{3} R$, i.e., when $3h = 2(2R)$

i.e., 3 times the height = 2 times the diameter

$$\text{Volume of the largest cone} = \frac{1}{3} \pi \times \frac{16R^2}{9} \times \left(2R - \frac{4R}{3} \right) = \frac{32\pi R^3}{81}$$

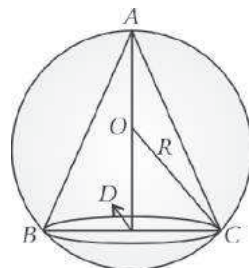


Fig. 10.13

Example 46: A wire of length 25 m is to be cut into two pieces. One of the wires is to be made into a square and the other into a circle. What should be the length of the two pieces so that the combined area of the square and the circle is minimum?

Solution: Let x metres and $(25 - x)$ metres be the required lengths.

Let a be the side of the square formed and r be the radius of the circle formed. Then,

$$4a = x \text{ and } 2\pi r = 25 - x \text{ or } a = \frac{x}{4} \text{ and } r = \frac{25 - x}{2\pi}$$

$$\therefore \text{Area of the square} = \left(\frac{x^2}{16} \right) \text{ sq. metres.}$$

$$\text{Area of the circle} = \pi \left(\frac{25 - x}{2\pi} \right)^2 = \frac{(25 - x)^2}{4\pi}$$

$$\text{Now, the combined area } A = \frac{x^2}{16} + \frac{(25 - x)^2}{4\pi}$$

$$\therefore \frac{dA}{dx} = \frac{x}{8} - \frac{(25 - x)}{2\pi} = \frac{(\pi + 4)x - 100}{8\pi}, \text{ and } \frac{d^2A}{dx^2} = \frac{(\pi + 4)}{8\pi}$$

For a maxima or minima, we have $\frac{dA}{dx} = 0$

$$\text{Now, } \frac{dA}{dx} = 0 \Rightarrow (\pi + 4)x - 100 = 0 \Rightarrow x = \frac{100}{(\pi + 4)}$$

$$\text{And, } \frac{d^2A}{dx^2} = \frac{(\pi + 4)}{8\pi} > 0 \text{ for all values of } x$$

$$x = \frac{100}{(\pi + 4)} \text{ is a point of minimum}$$

$\therefore$ Length of the pieces are $\left(\frac{100}{\pi + 4}\right)$ metres and $\left(\frac{25\pi}{\pi + 4}\right)$ metres.

Example 47: Show that a cylinder of a given volume which is open at the top has minimum total surface area, provided its height is equal to the radius of its base.

[B.C.A. (Meerut) 2005]

Solution: Let r be the radius and h the height of the cylinder of given volume V .

$$\text{Then, } V = \pi r^2 h \Rightarrow h = \frac{V}{\pi r^2} \quad \dots(1)$$

Let the total surface area be S . Then,

$$S = \pi r^2 + 2\pi r h = \left(\pi r^2 + 2\pi r \cdot \frac{V}{\pi r^2} \right) \quad [\text{using (1)}]$$

$$\therefore S = \left(\pi r^2 + \frac{2V}{r} \right)$$

$$\text{So, } \frac{dS}{dr} = \left(2\pi r - \frac{2V}{r^2} \right) \text{ and } \frac{d^2S}{dr^2} = \left(2\pi + \frac{4V}{r^3} \right)$$

For a maxima or minima, we have $\frac{dS}{dr} = 0$

$$\text{Now, } \frac{dS}{dr} = 0 \Rightarrow 2\pi r - \frac{2V}{r^2} = 0 \Rightarrow 2\pi r^3 = 2V = 2\pi r^2 h \Rightarrow r = h$$

$$\left(\frac{d^2S}{dr^2} \right)_{r=h} = \left(2\pi + \frac{4V}{h^3} \right) > 0$$

Thus, $r = h$ is a point of minimum.

This shows that the total surface area of the cylinder is minimum when $h = r$, i.e., when the height of the cylinder is equal to the radius of the base.

Example 48: Prove that the perimeter of a right-angled triangle of given hypotenuse is maximum when the triangle is isosceles.

Solution: Let h be the hypotenuse and θ be the angle between the hypotenuse and the base.

Then, base = $h \cos \theta$ and altitude = $h \sin \theta$

Let P be the perimeter.

$$\text{Then, } P = h + h \cos \theta + h \sin \theta$$

$$\therefore \frac{dP}{d\theta} = -h \sin \theta + h \cos \theta = h (\cos \theta - \sin \theta)$$

$$\text{And, } \frac{d^2P}{d\theta^2} = h (-\sin \theta - \cos \theta) = -h (\sin \theta + \cos \theta)$$

$$\text{Now, } \frac{dP}{d\theta} = 0 \Rightarrow h (\cos \theta - \sin \theta) = 0$$

$$\Rightarrow \cos \theta = \sin \theta, \text{ i.e., } \tan \theta = 1 \Rightarrow \theta = (\pi/4)$$

$$\text{Also, } \left[\frac{d^2 P}{d\theta^2} \right]_{\theta = (\pi/4)} = -h \left[\sin \frac{\pi}{4} + \cos \frac{\pi}{4} \right] = -h\sqrt{2} < 0$$

$\therefore P$ is maximum when $\theta = (\pi/4)$

$$\text{In this case, base} = h \cos \frac{\pi}{4} = \frac{h}{\sqrt{2}} \text{ and altitude} = h \sin \frac{\pi}{4} = \frac{h}{\sqrt{2}}$$

$\therefore$ base = altitude, and hence the triangle is isosceles.

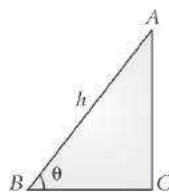


Fig. 10.14

Example 49: Show that the semi-vertical angle of a cone of maximum volume and of given slant height is $\tan^{-1} \sqrt{2}$.

Solution: Let α be the semi-vertical angle of a cone of given slant height l . Then, height of the cone = $l \cos \alpha$, radius of the base = $l \sin \alpha$.

Let V be the volume of the cone. Then,

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi (l^2 \sin^2 \alpha) (l \cos \alpha)$$

$$\text{or } V = \frac{1}{3} \pi l^3 \sin^2 \alpha \cos \alpha$$

$$\therefore \frac{dV}{d\alpha} = \frac{1}{3} \pi l^3 (-\sin^3 \alpha + 2 \sin \alpha \cos^2 \alpha) = \frac{1}{3} \pi l^3 \sin \alpha (2 \cos^2 \alpha - \sin^2 \alpha)$$

$$\begin{aligned} \text{And, } \frac{d^2 V}{d\alpha^2} &= \frac{1}{3} \pi l^3 \cos \alpha (2 \cos^2 \alpha - \sin^2 \alpha) \\ &+ \frac{1}{3} \pi l^3 \sin \alpha (-4 \sin \alpha \cos \alpha - 2 \sin \alpha \cos \alpha) \\ &= \frac{1}{3} \pi l^3 \cos^3 \alpha (2 - 7 \tan^2 \alpha) \end{aligned}$$

Now, for a maxima or minima, we have $\frac{dV}{d\alpha} = 0$

$$\text{Now, } \frac{dV}{d\alpha} = 0 \Rightarrow \frac{1}{3} \pi l^3 \sin \alpha (2 \cos^2 \alpha - \sin^2 \alpha) = 0$$

$$\Rightarrow \sin \alpha = 0 \text{ or } 2 \cos^2 \alpha - \sin^2 \alpha = 0$$

$$\Rightarrow \alpha = 0 \text{ or } \alpha = \tan^{-1} \sqrt{2}$$

Neglecting $\alpha = 0$, we have $\alpha = \tan^{-1} \sqrt{2}$, i.e., $\tan \alpha = \sqrt{2}$

$$\text{Now, } \tan \alpha = \sqrt{2} \Rightarrow \sec^2 \alpha = 3 \Rightarrow \cos \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \left[\frac{d^2 V}{d\alpha^2} \right]_{\alpha = \tan^{-1} \sqrt{2}} = \frac{1}{3} \pi l^3 \cdot \left(\frac{1}{\sqrt{3}} \right)^3 [2 - 14] = -\frac{4\pi l^3}{3\sqrt{3}} < 0$$

Thus, V is maximum when $\alpha = \tan^{-1} \sqrt{2}$, i.e., when the semi-vertical angle of the cone is $\tan^{-1} \sqrt{2}$.

Example 50: Show that the surface area of a closed cuboid with square base and given volume is minimum when it is a cube.

Solution: Let V be the fixed volume of a closed cuboid with length a , breadth a and height h .

Let S be its surface area.

$$\text{Then, } V = (a \times a \times h) \text{ or } h = \frac{V}{a^2} \quad \dots(1)$$

Now, $S = 2(a^2 + ah + ah) = 2(a^2 + 2ah) = 2\left(a^2 + \frac{2V}{a}\right)$ [using (1)]

i.e., $S = 2\left(a^2 + \frac{2V}{a}\right)$

$\therefore \frac{dS}{da} = 2\left(2a - \frac{2V}{a^2}\right)$ and $\frac{d^2S}{da^2} = \left(4 + \frac{8V}{a^3}\right)$

Now, $\frac{dS}{da} = 0 \Rightarrow V = a^3 \Rightarrow a \times a \times h = a^3 \Rightarrow h = a.$

Now, when $h = a$, we have $V = a^3$

$\therefore \left[\frac{d^2S}{da^2}\right]_{h=a} = \left(4 + \frac{8a^3}{a^3}\right) = 12 > 0$

So, S is minimum when length = a , breadth = a and height = a , i.e., when it is a cube.

Example 51: Show that the height of a closed cylinder of given surface and maximum volume is equal to the diameter of its base. [B.C.A. (Meerut) 2004; (Rohilkhand) 2008]

Solution: Let S be the fixed surface area of a closed cylinder of radius r and height h . Let V be its volume.

Then, $S = (2\pi r^2 + 2\pi rh) \Rightarrow h = \left(\frac{S - 2\pi r^2}{2\pi r}\right)$... (1)

And, $V = \pi r^2 h = \pi r^2 \left(\frac{S - 2\pi r^2}{2\pi r}\right) = \left(\frac{rS - 2\pi r^3}{2}\right)$ [using (1)]

$\therefore \frac{dV}{dr} = \left(\frac{S - 6\pi r^2}{2}\right)$ and $\frac{d^2V}{dr^2} = -6\pi r$

Now, $\frac{dV}{dr} = 0 \Rightarrow (S - 6\pi r^2) = 0 \Rightarrow S = 6\pi r^2$

$\Rightarrow (2\pi r^2 + 2\pi rh) = 6\pi r^2 \Rightarrow h = 2r$, i.e., $r = \frac{1}{2}h$

And, $\left[\frac{d^2V}{dr^2}\right]_{r=(h/2)} = -6\pi \times \frac{1}{2}h = -3\pi h < 0$

So, V is maximum when $h = 2r$

Example 52: A closed right circular cylinder has a volume of 2156 cubic units. What should be the radius of its base so that its total surface area may be maximum?

Solution: Let r be the radius and h the height of the cylinder.

Then, $\pi r^2 h = 2156 \Rightarrow h = \left(\frac{2156}{\pi r^2}\right)$. Let S be its total surface area.

Then, $S = (2\pi r^2 + 2\pi rh) = \left(2\pi r^2 + 2\pi r \times \frac{2156}{\pi r^2}\right) = \left(2\pi r^2 + \frac{4312}{r}\right)$

$\therefore \frac{dS}{dr} = \left(4\pi r - \frac{4312}{r^2}\right)$ and $\frac{d^2S}{dr^2} = \left(4\pi + \frac{8624}{r^3}\right)$

Now, $\frac{dS}{dr} = 0 \Rightarrow \left(4\pi r - \frac{4312}{r^2}\right) = 0$

$$\Rightarrow 4\pi r^3 = 4312 \Rightarrow r = \left(\frac{4312}{4\pi}\right)^{1/3} = \left(\frac{4312 \times 7}{4 \times 22}\right)^{1/3} = 7$$

And $\left[\frac{d^2S}{dr^2}\right]_{r=7} = \left(4\pi + \frac{8624}{343}\right) > 0$

$\therefore S$ is maximum when $r = 7$



Comprehensive Exercise 10.2

- Find two numbers whose sum is 6 and the sum of whose cubes is minimum.
- Prove that a conical tent of given capacity will require the minimum canvas when the ratio between the height of the cone and radius of base is $\sqrt{2} : 1$.
- Inside a cone of height h and semi-vertical angle α , a cylinder is inscribed. Show that its maximum volume is $\frac{4}{27}\pi h^3 \tan^2 \alpha$.
[B.C.A. (Meerut) 2001]
- If the sum of the lengths of the hypotenuse and a side of a right-angle triangle is given, prove that the area of the triangle is maximum when the angle between the given side and the hypotenuse is $\frac{\pi}{3}$.
- The three sides of a trapezium are equal. If each of them is 6 cm long, find the area of the trapezium, when it is maximum.
- The sum of two numbers is constant. Prove that their product is maximum, if each number is half of their sum.
- Divide 15 into two parts such that the product of the square of one part and the cube of the other is maximum.
- Prove that the semi-vertical angle of a cone of given slant height and maximum volume is $\tan^{-1}(\sqrt{2})$.
- Show that the semi-vertical angle of a right cone of given total surface and maximum volume is $\sin^{-1}\left(\frac{1}{3}\right)$.
- Show that the height of the cylinder of maximum volume that can be made inside a given sphere is $\frac{2a}{\sqrt{3}}$, where a is radius of cylinder
- The velocity of waves of wavelength x on deep water is proportional to $\sqrt{\frac{x}{2} + \frac{2}{x}}$. Find the value of x for which the velocity is minimum.
- Show that the surface area of a closed cuboid with square base and given volume is minimum, if it is a cube.



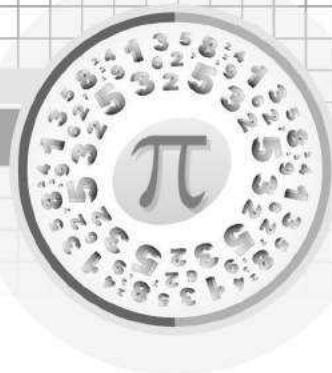
Answers

- | | | | |
|------------|-------------------------|---------|-------|
| 1. 3 and 3 | 5. $27\sqrt{2}$ sq. cm. | 7. 6, 9 | 11. 2 |
|------------|-------------------------|---------|-------|





Unit-3



Chapter

11

Curve Tracing

11.1 Tracing of Curves

In this chapter we shall deal with the tracing of Cartesian curves only whose equations are given in the forms

i.e. $x^2 + y^2 = a^2$, $x^2 = 4ay$, $x^4 + y^4 = 4x^3y^3$

11.1.1 Procedure for Tracing Curves given in Cartesian form

The following points should be remembered for the tracing of Cartesian curves.

1. Symmetry

- (i) If all powers of y in the given equation of curve is even, then there is symmetry about x -axis *i.e.*, the shape of the curve above and below the axis of x is symmetrical.

For example: The parabola $y^2 = 4ax$

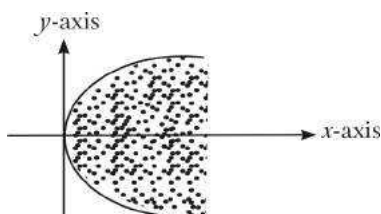


Fig. 11.1

- (ii) If all powers of x in the given equation of curve is even, then there is symmetry about y -axis *i.e.*, the shape of the curve in left and right of y -axis is same or symmetrical.

For example: The parabola $x^2 = 4ay$.

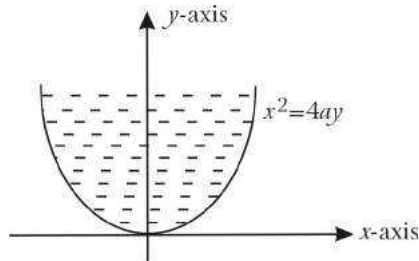


Fig. 11.2

- (iii) If all powers of x and y in given equation of curve is even, then there is symmetric about both axes *i.e.*, the shape of curve above and below to x and y axis is same or symmetrical.

For example: The circle $x^2 + y^2 = a^2$

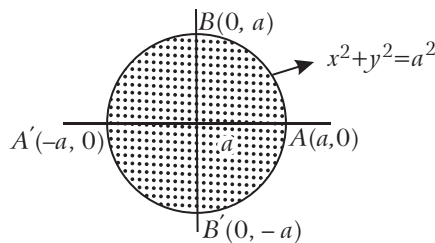


Fig. 11.3

- (iv) Put $x = y$ and $y = x$ in given equation of curve and equation of curve remain unchanged, then there is symmetry about the line $y = x$ *i.e.*, the line which passed through origin and make an angle 45° with the positive direction of x -axis.

For example: The rectangular hyperbola $xy = c^2$

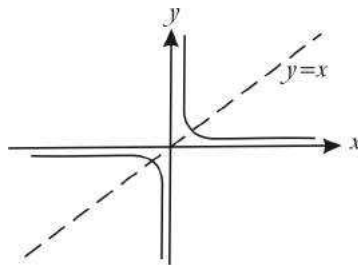


Fig. 11.4

- (v) If we put $x = -x$ and $y = -y$ in given equation of curve and equation of curve remain unchanged then there is symmetry in the opposite quadrants.

For example: The Folium of descartes $x^3 + y^3 = 3axy$

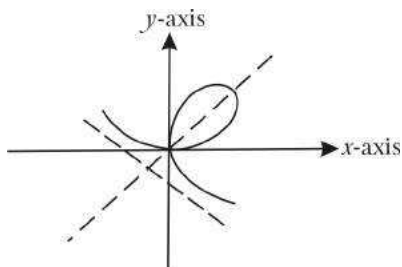


Fig. 11.5

2. Curve passed through origin or not

- (i) If there is separate constant term in the given equation of curve then curve does not passed through origin *i.e.*, $x^2 + y^2 = a^2$ does not passed through origin.
- (ii) If there are no separate constant term in the given equation of curve then curve passed through origin *i.e.*, $y = x^3$ passed through origin.

3. Tangent at origin:

The tangent at origin is obtain by equating to zero the lowest degree term in the given equation of curve.

For example: The curve $x^3 + y^3 = 3axy$ passes through origin. The lowest degree term in this equation is $3ax$. Equating this term to zero we get the equation of the tangent at origin as $xy = 0$ or $x = 0$ and $y = 0$.

4. Points of intersection with axes

Find the points where the curve intersects the co-ordinate axes, putting $y = 0$ in the given equation of curve we find x . Again put $x = 0$ in the given equation of curve we find y .

For Example: $y^2(a+x) = x^2(3a-x)$

Putting $y = 0$ we get $x^2(3a-x) = 0$ or $x = 0$ and $x = 3a$. That is $(0, 0)$ and $(3a, 0)$ are points of intersection similarly put $x = 0$ in given equation of curve, we get $y = 0$ *i.e.*, $(0, 0)$ is point of intersection.

5. Tangent at any point (h, k)

To find tangents at any point (h, k) , shift the origin to (h, k) . To do it, put $x = x + h$ and $y = y + k$ in given equation of curve. Then the tangents at this new origin are obtained by equating to zero the lowest degree terms.

For example: In the curve $9ay^2 = x(x-3a)^2$ we find putting $y = 0$ that the curve crosses x -axis at $(0, 0)$ and $(3a, 0)$. For shifting the origin to $(3a, 0)$, we should put $x = x + 3a$ and $y = y + 0$ in given equation of curve to obtain the new equation of the curve referred to $(3a, 0)$ as the new origin.

Thus, the new equation of the curve becomes

$$9a(y+0)^2 = (x+3a)(x+3a-3a)^2 \text{ or } 9ay^2 = (x+3a)x^2$$

Equating the lowest degree term to zero the tangent at the new origin is

$$9ay^2 - 3ax^2 = 0$$

$$\text{i.e., } 3y^2 - x^2 = 0 \text{ or } y^2 = \frac{1}{3}x^2 \text{ or } y = \pm \frac{1}{\sqrt{3}}x = \pm (\tan 30^\circ)x.$$

i.e., tangents at $(3a, 0)$ are inclined to x -axis at $\pm 30^\circ$.

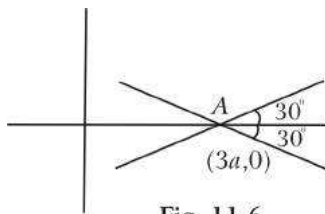


Fig. 11.6

6. Asymptotes

- (i) The asymptote parallel to x -axis is obtained by equating to zero the coefficient of the highest degree term of x provided it is not constant if it is constant then no asymptote parallel to x -axis.
- (ii) The asymptote parallel to y -axis is obtained by equating to zero the coefficient of the highest degree term of y provided it is not constant if it is constant then no asymptote parallel to y -axis.

7. Regions where the curve does not exist

Solve for y or x as case may be and find out those values of x lying in the region, say values of x between a and b for which y is imaginary, then the curve does not exist in the region bounded by lines $x = a$ and $x = b$.

For example: In curve $a^2y^2 = x^2(x-a)(2a-x)$ we find that all values of x less than a , y^2 is negative i.e., y is imaginary.

$\therefore$ The curve does not exist for values of $x > a$. Similarly for the values of $x > 2a$, the curve does not exist. Hence the curve exists for values of x lying between a and $2a$.

8. Observe how y varies as x increase from zero to infinity and as x decrease from zero to $-\infty$.
9. **Point of inflexion**

The point of inflexion is obtained by equating to zero to $\frac{d^2y}{dx^2}$ or $\frac{d^2x}{dy^2}$

Example 1: Trace the curve $y^2 = x^3$ (i)

Solution:

- Symmetry:** Since all power of y is even, then symmetry about x -axis.
- Curve passed through origin.
- Tangent at origin:** The tangent at origin are $y^2 = 0$ or $y = 0$ i.e., x -axis is tangent at the origin.
- Points of intersection:** Put $y = 0$ in (i) we get $x = 0$. Again put $x = 0$ in (i) we get $y = 0$. Hence we do not get any new point but the origin.
- No asymptotes
- For negative values of x , y^2 is negative i.e., y is imaginary or the curve does not exist for negative value of x .
- As x increase from 0 to ∞ , y also increase from 0 to ∞

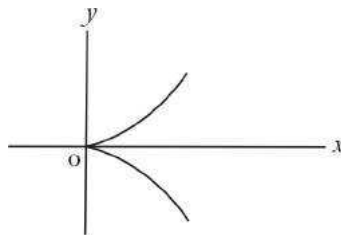


Fig. 11.7

With above data the shape of the curve is shown in Fig.11.7

Example 2: Trace the curve $y = x(x^2 - 1)$ (i)

Solution:

- Symmetry:** Symmetry in opposite quadrants.
- Curve passed through origin.
- Tangent at origin:** Equating the lowest degree terms to zero, the equation of the tangent at origin is

$$x + y = 0 \quad \text{or} \quad y = -x$$
 i.e., the line which passed through $(0,0)$ and make an angle 135° with positive x -axis.
- Points of intersection:** Put $y = 0$ in (i) we get $x = 0, x = \pm 1$ i.e., $(0,0), (-1,0), (1,0)$
Again put $x = 0$ in (ii) we get $y = 0$ Hence, $(1,0), (-1,0)$
- (i) **Tangent at $(1, 0)$:** Put $x = x + 1, y = y + 0$ in (i), we get

$$y = (x + 1)((x + 1)^2 - 1)$$

$$\text{or} \quad y = (x + 1)(x^2 + 2x)$$

$$\text{or} \quad y = x^3 + 3x^2 + 2x$$

$\therefore$ The tangent at the new origin is $y = 2x$

- (ii) Tangent at $(-1, 0)$: Put $x = x-1$, $y = y+0$
in (i), we get

$$y = (x-1)((x-1)^2 - 1)$$

or

$$y = (x-1)(x^2 - 2x)$$

$$y = x^3 - 3x^2 + 2x$$

$\therefore$ The Tangent at the new origin is
 $y = 2x$

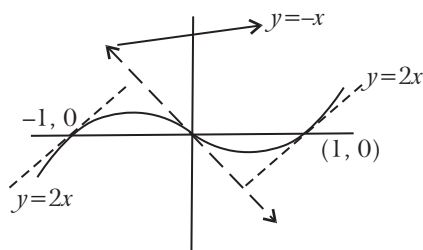


Fig. 11.8

6. No asymptotes
7. When x increase from 1 to ∞ then y increase from 0 to ∞ .

With the above data the shape of the curve is shown in figure 11.8.

Example 3: Trace the curve $x^2 y^2 = a^2 (x^2 + y^2)$

...(i)

Solution:

1. **Symmetry:** The Symmetry about both axes.
2. Curve passed through origin.
3. **Tangent at origin:** Tangent at origin are $x^2 + y^2 = 0$ which gives imaginary tangents. So $(0,0)$ is a conjugate point.
4. **Points of intersection:** Put $x=0$ in (i) we get $y=0$. Again put $y=0$ in (i) we get $x=0$. Hence we do not get any new points but the origin.
5. **Asymptotes:** The asymptote parallel to x -axis is

$$y^2 - a^2 = 0 \Rightarrow y = \pm a$$

The asymptote $||$ to y -axis is $x^2 - a^2 = 0 \Rightarrow x = \pm a$

6. **Region:** Solve for y , we get $y^2 = \frac{a^2 x^2}{x^2 - a^2}$

if $x^2 < a^2$, then y^2 is negative or y is imaginary,
then curve does not exist between the lines
 $x = -a$ and $x = a$

Similarly if solve for y , we get $x^2 = \frac{a^2 y^2}{y^2 - a^2}$

if $y^2 < a^2$, then x^2 is negative or x is imaginary
then curve does not exist between the lines
 $y = -a$ and $y = a$

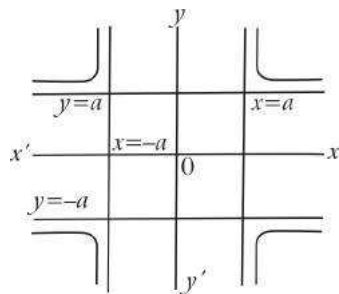


Fig. 11.9

$\therefore$ The shape of the curve is shown in the figure 11.9.

Example 4: Trace the curve $y = \frac{x^2 + 1}{x^2 - 1}$... (i)

Solution:

1. **Symmetry:** The symmetry about y -axis.
2. Curve does not pass through the origin.
3. **Points of intersection:** Putting $x = 0$, in (i) we find $y = -1$ i.e., the curve cuts the y -axis at $(0, -1)$
4. **Tangent at $(0, -1)$:** Put $x = x + 0, y = y - 1$ in (i)

we get
$$y - 1 = \frac{x^2 + 1}{x^2 - 1}$$

or
$$(y - 1)(x^2 - 1) = x^2 + 1 \quad \text{or} \quad xy^2 - y - 2x^2 = 0$$

$\therefore$ The tangent at the new origin is $y = 0$ i.e., the new x -axis.

5. **Asymptotes:** The given equation of curve can be written as $x^2(y - 1) - y - 1 = 0$

- (i) The asymptote parallel to x -axis is $y - 1 = 0$ or $y = 1$
- (ii) The asymptote parallel to y -axis is $x^2 - 1 = 0$ or $x = \pm 1$

7. **Region:** As x increase from 1 to ∞ , y decrease from ∞ to 1 and as x increase from 0 to 1, y decrease from -1 to $-\infty$.

$\therefore$ For values of $x > 1$, y is positive and for values of x lying between 0 and 1, y is negative.

The shape of the curve is as shown in figure 11.10

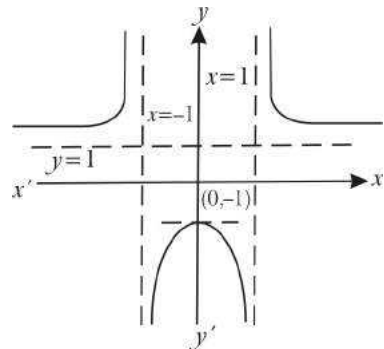


Fig. 11.10

Example 5: Trace the curve $x^2 = \frac{y - 1}{y + 1}$

Solution: We observe from the equation of the given curve that:

1. The equation is of even degree in x , and therefore, the curve is symmetric about y -axis.
2. The curve does not pass through the origin.
3. The curve meets the y -axis at $(0, 1)$ and this point $\frac{dy}{dx} = 0$, therefore the tangent is parallel to x -axis.
4. The asymptote parallel to x -axis is $y = -1$ and asymptotes parallel to y axis is $x = \pm 1$ which are obtained respectively by equating to zero the coefficients of highest degree in x and y . Also there is no oblique asymptote.

5. When $|x| < 1$, then $y > 1$ and when $x \rightarrow 1^-$, then $y \rightarrow +\infty$,

When $|x| > 1$, then $y < -1$ and when $x \rightarrow 1^+$ then $y \rightarrow -\infty$.

From the equation of the curve $x^2 = \frac{y-1}{y+1}$ it is

clear that when

$-1 < y < 1$, then the value of x^2 is negative; i.e., x is imaginary.

It follows that there is no branch of the curve in between $y = -1$ and $y = 1$.

Hence, keeping in view the symmetry, of the graph of the curve will be shown in the figure 11.11

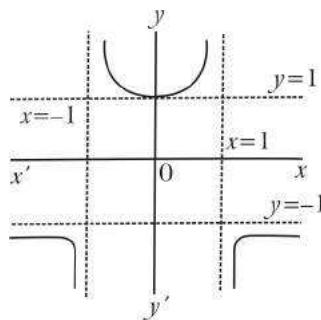


Fig. 11.11

Example 6: Trace the curve $y^2 = \frac{(1+x^2)x^2}{1-x^2}$.

Solution: We observe from the equation of the curve that

1. There is no change in the equation of the curve by replacing x with $-x$ and y with $-y$, hence the curve is symmetric in opposite quadrants.
2. The asymptotes parallel to y -axis are $x = \pm 1$.
3. The curve passes through origin. Also the equations of tangents at origin to the given curve, by equating to zero the lowest degree terms, are given by

$$y^2 - x^2 = 0; \text{ i.e., } y = \pm x.$$

Hence the tangents at the origin are $y = x$ and $y = -x$.

4. If $1 - x^2 < 0$; i.e., $x \geq \pm 1$, then y is imaginary, hence there is no branch of the curve on the left-side of the line $x = -1$ and the right-side of the line $x = 1$.

Also when $x \rightarrow -1^+$, then $y \rightarrow \pm \infty$ and when $x \rightarrow 1^-$, then also $y \rightarrow \pm \infty$.

On the basis of the above facts the graph of the given curve will be as shown in the above figure 11.12.

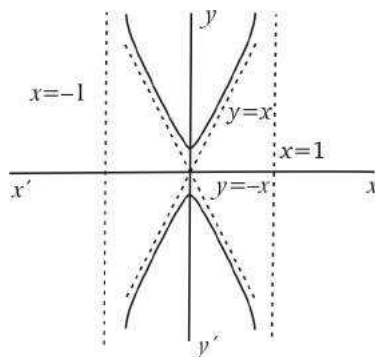


Fig. 11.12

Example 7: Trace the curve $x = (y-1)(y-2)(y-3)$.

Solution: (1) No symmetry.

(2) The curve does not through $(0, 0)$

(3) When $y = 0, x = -6$, and when $x = 0, y = 1, 2, 3$.

$\therefore$ The curve cuts the axes at the points $(-6, 0), (0, 1), (0, 2), (0, 3)$.

The given curve is

$$x = y^3 - 6y^2 + 11y - 6 \quad \dots(i)$$

$$\frac{dx}{dy} = 3y^2 - 12y + 11, \quad \frac{dy}{dx} = \frac{1}{(3y^2 - 12y + 11)}$$

$$\left(\frac{dy}{dx}\right)_{(-6,0)} = \frac{1}{11}, \quad \left(\frac{dy}{dx}\right)_{(0,1)} = \frac{1}{2}, \quad \left(\frac{dy}{dx}\right)_{(0,2)} = -\frac{1}{2}$$

$$\left(\frac{dy}{dx}\right)_{(0,3)} = \frac{1}{2}.$$

Hence the tangents at $(-6, 0), (0, 1), (0, 2)$ and $(0, 3)$ are inclined at angles $\tan^{-1}(1/11), \tan^{-1}(1/2), (3\pi/4)$ and $\tan^{-1}(1/2)$ with the x -axis.

(4) No asymptote.

(5) Solving the given equation for x

$$x = (y-1)(y-2)(y-3)$$

(I) When y is positive.

(i) When $0 < y < 1, x = -ve$ and $x = 0$ at $y = 1$.

(ii) When $1 < y < 2, x = +ve$ and $x = 0$ at $y = 2$.

(iii) When $2 < y < 3, x = -ve$ and $x = 0$ at $y = 3$.

(iv) When $y > 3, x = +ve$ and $y \rightarrow \infty, x \rightarrow \infty$.

(II) When $y < 0, x = -ve$ and $y \rightarrow -\infty, x \rightarrow -\infty$.

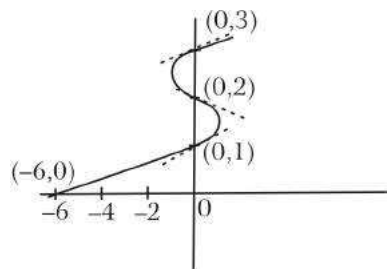


Fig. 11.13

(6) From the equation (i), we have

$$\frac{d^2x}{dy^2} = 6y - 12; \quad \frac{d^3x}{dy^3} = 6.$$

Clearly $d^2x/dy^2 = 0$ and $d^3x/dy^3 \neq 0$ at $(0, 2)$. Hence there is a point of inflexion at $(0, 2)$ only.

$\therefore$ The curve is as shown in the figure 11.13.

Example 8: Trace the curve $x^2y^2 = (1+y^2)(4-y^2)$.

[B.C.A. (Agra) 2004]

Solution: (1) The curve is symmetrical about x -axis.

(2) The curve does not pass through $(0, 0)$.

(3) When $y = 0$, we do not get any value of x .

When $x = 0$, $y = -1, 2, -2$. Thus the curve cuts the y -axis at $(0, 2)$, $(0, -1)$ and $(0, -2)$.

To find tangent at $(0, 2)$. Shifting the origin to $(0, 2)$, we get

$$x^2(y+2)^2 = \{1+(2+y)\}^2 \{4-(2+y)^2\}$$

$$\text{or} \quad x^2(y+2)^2 = (3+y)^2(-4y-y^2).$$

$\therefore$ The tangent at new origin is $y = 0$, i.e., the tangent at $(0, 2)$ is parallel to original x -axis. **To find tangent at $(0, -1)$.** Shifting the origin to $(0, -1)$, we get

$$x^2(y-1)^2 = \{1+(y-1)\}^2 \{4-(y-1)^2\}$$

$$\text{or} \quad x^2(y-1)^2 = (3+2y-y^2).$$

$\therefore$ The tangent at new origin are $x^2 = 3y^2$, i.e., $y = \pm (1/\sqrt{3})x$. Thus $(0, -1)$ is **node** and the tangents at this point the inclined at angles $\pm 30^\circ$ with the x -axis.

Similarly the tangents at $(0, -2)$ is parallel to x -axis.

(4) **Asymptote:** The equation of the curve may be written as

$$y^2(x^2 + y^2) + 2y^3 - 3y^2 - 8y - 4 = 0.$$

The linear factors in the highest power terms are y , which is constant then no asymptote parallel to y axis. Hence the asymptote may be only parallel to x -axis. Equating to zero the coefficient of highest power of x (i.e., of x^2), the required asymptote is $y^2 = 0$, i.e., $y = 0$ (x -axis).

(5) Solving the given equation for x , we get

$$x^2 = (1+y)^2(4-y^2)/y^2.$$

(I) **When y is positive.**

(i) When $0 < y < 2$, $x^2 = +ve \Rightarrow y$ is real.

(ii) When $y > 2$, $x^2 = -ve \Rightarrow x$ is imaginary.

Hence when $y > 0$, then the curve exists in the region $0 < y < 2$. Again when $y \rightarrow 0$, $x^2 \rightarrow \infty$.

(II) **When y is negative.**

(i) When $-1 < y < 0$, $x^2 = +ve \Rightarrow x$ is real.

(ii) When $-2 < y < -1$, $x^2 = +ve \Rightarrow x$ is real.

(iii) When $y < -2$, $x^2 = -ve \Rightarrow x$ is imaginary.

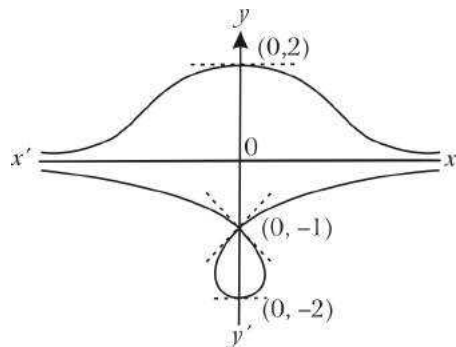


Fig. 11.14

Hence when $y < 0$, then the curve does not exist in the region $y > -2$.

$\therefore$ The curve is as shown in the figure 11.14.

Example 9: Trace the curve $x^3 y = x + 1$.

Solution: (1) No symmetry.

(2) The curve does not pass through $(0, 0)$.

(3) The curve cuts the axes at $(-1, 0)$.

To find tangent at $(-1, 0)$. The equation of the curve may be written as

$$y = x^{-2} + x^{-3}$$

$$dy/dx = -2x^{-3} - 3x^{-4}, (dy/dx) \text{ at } (-1, 0) = -1.$$

$\therefore$ The tangent at $(-1, 0)$ is inclined at an angle $(3\pi/4)$ with the axis of x .

Also $d^2y/dx^2 = -6x^{-4} + 12x^{-5} \neq 0$ at $(-1, 0)$.

Then $(-1, 0)$ is not a point of inflexion.

(4) **Asymptote:** Clearly the asymptotes are $x=0$, $y=0$, i.e., both axes are the asymptotes.

(5) Solving the given equation for y , we get

$$y = (x+1)/x^3.$$

(I) If x is positive, then $y = +ve$.

$\therefore$ The curve exists in the first quadrant. When $x \rightarrow \infty$, $y \rightarrow 0$ and when $x \rightarrow 0$ (from the right) then $y \rightarrow \infty$.

(II) If x is negative.

(i) When $-1 < x < 0$, $y = -ve$.

$\therefore$ the curve exists in the third quadrant and when $x \rightarrow 0$ (from the left) then $y \rightarrow -\infty$.

(ii) What $x < -1$, $y = +ve$.

$\therefore$ The curve exists in the second quadrant.

$\therefore$ The curves is as shown in the figure 11.15.

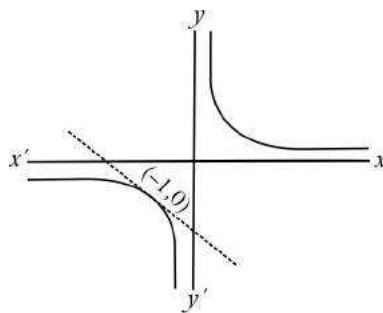


Fig. 11.15

Example 10: Trace the curve $y^2 x = a^2(x-a)$

...(i)

Solution:

1. **Symmetry:** The symmetry about x -axis.
2. Curve does not pass through the origin.

3. **Points of intersection:** Putting $y = 0$ in (i) we get $x = a$ i.e. curve crosses x -axis at $(a, 0)$.
4. **Tangent at $(a, 0)$:** Shift the origin to $(a, 0)$, the equation of the curve becomes

$$y'^2 (x + a) = a^2 x$$

Then tangent at new origin is $x = 0$ i.e. new y -axis i.e., a line through $(a, 0)$ parallel to old y -axis.

5. **Asymptotes**

(i) The asymptote parallel to x -axis is $y^2 - a^2 = 0$ or $y = \pm a$

(ii) The asymptote parallel to y -axis is $x = 0$

6. **Region:** Solve for $y^2 = \frac{a^2 (x - a)}{x}$

When x lying between 0 and a , y^2 is negative i.e., y is imaginary i.e., the curve does not exist between the lines $x = 0$ and $x = a$

$\therefore$ The shape of the curve is shown in figure 11.16.

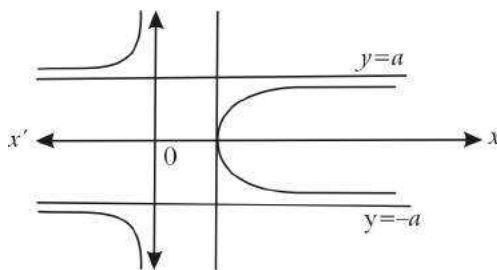


Fig. 11.16

Example 11: Trace the curve $y^2 (2a - x) = x^3$

[B.C.A. (Agra) 2005, 2007]

Solution:

1. **Symmetry:** The symmetry about x -axis.
2. Curve passed through origin.
3. **Tangent at origin:** The tangent at origin are $y^2 = 0$ or $y = 0$ i.e., there is cusp at origin.
4. **Points of intersection:** The curve crosses the axes only at the origin.
5. **Asymptotes:**
 - (i) No asymptote parallel to x -axis.
 - (ii) Asymptote parallel to y -axis is $x - 2a = 0$ or $x = 2a$

6. **Region:** Solve for $y^2 = \frac{x^3}{(2a - x)}$

When $x > 2a$ then y^2 is negative or y is imaginary then curve does not lie in that region.
When $x < 0$ then y^2 is negative or y is imaginary then curve does not lie in that region.

The shape of curve is shown in figure 11.17.

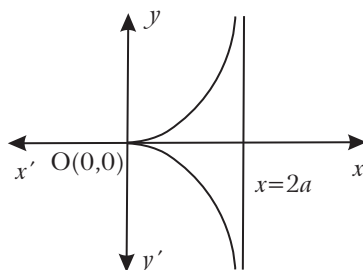


Fig. 11.17

Example 12: Trace the curve $y^2 (x^2 - 1) = x$

Solution:

- Symmetry:** The symmetry about x -axis.
- Curve passes through origin.
- Tangent at origin:** The tangent at origin is $x = 0$ i.e., y -axis.
- Points of intersection:** The points of intersection is only at $(0, 0)$.
- Asymptotes:**
 - The asymptote parallel to x -axis is $y = 0$.
 - The asymptote parallel to y -axis is $x^2 - 1 = 0$ or $x = \pm 1$
- Region:** Solve for $y^2 = x(x^2 - 1)$
 When $x < -1$ then y^2 is negative i.e., y is imaginary.
 $\therefore$ The curve does not exist for $x < -1$.
 Similarly for values of x lying between 0 and 1, y^2 is negative i.e., curve does not exist.
 $\therefore$ The shape of the curve shown in figure 11.18.

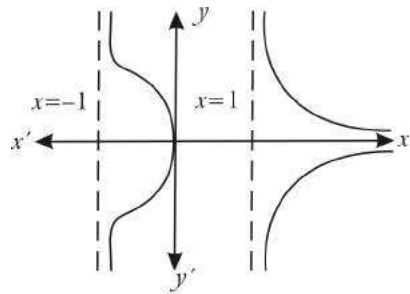


Fig. 11.18

Example 13: Trace the curve $x^2 (x^2 - 4a^2) = y^2 (x^2 - a^2)$.

Solution: (1) Symmetry about both the axes.

- The curve passes through the origin and the tangents at $(0, 0)$ are $4a^2 x^2 = a^2 y^2$, i.e., $y = \pm 2x$ and so $(0, 0)$ is a node.
- The curve crosses the axis of x at $(0, 0)$, $(2a, 0)$ and $(-2a, 0)$.

Shifting the origin to $(2a, 0)$, the new equation of the curve is

$$(x + 2a)^2 (x^2 + 4ax) = y^2 (x^2 + 4ax + 3a^2).$$

The tangent at new origin is $16a^3 x = 0$ i.e., $x = 0$ and so the tangent at $(2a, 0)$ is parallel to the original y -axis.

- Asymptotes:** The equation of the curve may be written as

$$x^2 (y - x)(y + x) - a^2 y^2 + 4a^2 x^2 = 0. \quad \dots(i)$$

Equating to zero the coefficient of y^2 (i.e., highest power of y) the asymptotes parallel to y -axis are given by

$$x^2 - a^2 = 0 \quad \text{or} \quad x = \pm a.$$

Now let $y - x = k$, be the asymptote corresponding to the factor $y - x$.

$\therefore$ Putting $y = x + k_1$ in (1), we get

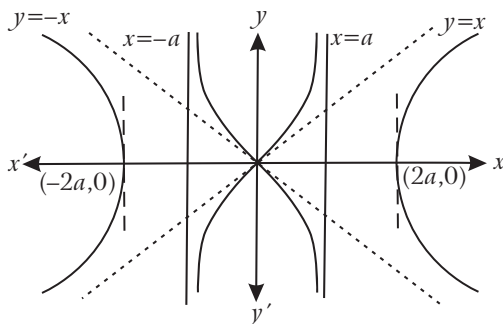


Fig. 11.19

$$x^2 k_1 (kx + k_1) - a^2 (x + k_1)^2 + 4a^2 x^2 = 0$$

Equating to zero the coefficient of x^2 , we get $k_1 = 0$

$\therefore y - x = 0$ is an asymptote.

Similarly $y + x = 0$ is also asymptote.

(5) Solve for y ,

$$y^2 = x^2 (x^2 - 4a^2) / (x^2 - a^2) \quad \dots(ii)$$

If x is positive:

- (i) When $0 < x < a$, $y^2 = +ve \Rightarrow y$ is real and so the curve exists in this region.
- (ii) When $a < x < 2a$, y is imaginary and so the curve does not exist in this region.
- (iii) When $x > 2a$, $y^2 = +ve \Rightarrow y$ is real and so the curve exists in this region. Also when $x \rightarrow \infty$, $y \rightarrow \infty$.

We need not consider the negative value of x as the curve is symmetrical about y -axis.

(6) In the positive quadrant, we have from (ii),

$$y = 2x \frac{\{1 - x^2 / (4a^2)\}^{1/2}}{(1 - x^2 / a^2)^{1/2}}$$

$\therefore$ When $0 < x < a$, $y > 2x$. So the curve touches the line $y = 2x$ above it at the origin.

$\therefore$ The shape of the curve is as shown in the figure 11.19.

Example 14: Trace the curve $x(x - 2a)y^2 = a^2(x - a)(x - 3a)$.

Solution: (1) Symmetry about x -axis.

(2) The curve does not pass through the origin.

- (3) The curve crosses the x -axis at $(a, 0)$, $(3a, 0)$.

The tangents are parallel to y -axis at these points.

- (4) **Asymptote:** Equating to zero the coefficient of highest power of x , the asymptotes parallel to x -axis are $x = 0$, $x = 2a$. Similarly asymptotes parallel to y -axis are $y = \pm a$.

- (5) Solving for y , we have

$$y^2 = \frac{a^2 (x-a)(x-3a)}{x(x-2a)}$$

- (I) If x is positive,
 (i) when $0 < x < a$, $y^2 = -ve \Rightarrow y$ is imaginary,
 (ii) when $a < x < 2a$, $y^2 = +ve \Rightarrow y$ is real.

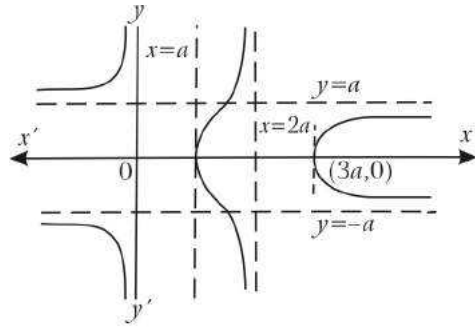


Fig. 11.20

- (iii) when $2a < x < 3a$, $y^2 = -ve \Rightarrow y$ is imaginary,
 (iv) when $x > 3a$, $y^2 = +ve \Rightarrow y$ is real.
 (II) If x is negative, $y^2 = +ve \Rightarrow y$ is real.

From above discussion the curve exists in the regions given by $a < x < 2a$, $x > 3a$ and $x < 0$. Combining above facts, the shape of the curve is as shown in the figure 11.20.

Example 15: Trace the curve $4ay^2 = x(x-2a)^2$... (i)

Solution:

- Symmetry:** The symmetry about x -axis.
- Curve passes through origin.
- Tangent at origin:** The tangent at origin is $x = 0$ i.e., y -axis.
- Points of intersection:** Put $y = 0$ in (i), we get $x = 0$ and $x = 2a$ i.e., the curve crosses the x -axis at $(0, 0)$, $(2a, 0)$.
- Tangent at $(2a, 0)$:** Shift the origin to $(2a, 0)$ i.e., put $x = x + 2a$, $y = y + 0$ in (i)

we get $4ay^2 = (x+2a)(x+2a-2a)^2$

or $4ay^2 = x^2(x+2a)$

or $4ay^2 = x^3 + 2ax^2$

$\therefore$ The tangent at new origin are given by

$$4ay^2 - 2ax^2 = 0$$

or $2y^2 - x^2 = 0$ or $y = \pm \frac{x}{\sqrt{2}}$

whose inclinations to the x -axis are $\pm \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ i.e., $\pm \frac{\pi}{4}$

6. **Asymptote:** No asymptotes parallel to x and y axis.

7. **Region:** Solve for $y^2 = \frac{x(x-2a)^2}{4a}$

When $x < a$ then y^2 is negative or y is imaginary. Hence curve does not lie in that region.

- $\therefore$ The shape of the curve is shown in figure 11.21.

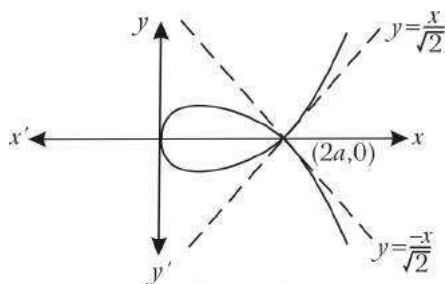


Fig. 11.21

Example 16: Trace the curve $y^2(a+x) = x^2(a-x)$, $a > 0$

...(i)

[B.C.A. (Agra) 2004]

Solution:

- Symmetry:** The symmetry about x -axis.
- Curve passed through origin.
- Tangent at origin:** Tangent at origin is $ay^2 - ax^2 = 0$ or $y^2 = x^2$ or $y = \pm x$
- Points of intersection:** Put $y = 0$ in (i) we get $x = 0$ and $x = a$ i.e., the curve crosses x -axis at $(0, 0)$ $(a, 0)$.
- Tangent at $(a, 0)$:** Put $x = x + a$ and $y = y + 0$ in (i), we get

$$y^2(a+x+a) = (x+a)^2(a-x-a) \text{ or } y^2(2a+x) = -x(x+a)^2$$

Hence, the tangent at new origin is $x = 0$ i.e., new y -axis i.e., a straight line through $(a, 0)$ parallel to y -axis.

6. **Asymptotes:**
- Asymptote parallel to y -axis is $a+x=0$ or $x=-a$
 - No asymptote parallel to x -axis.

7. **Region:** Solve for $y^2 = \frac{x^2(a-x)}{(a+x)}$

When $x > a$ then y^2 is negative or y is imaginary then curve does not lie in that region. Similarly when $x < -a$, the curve does not exist.

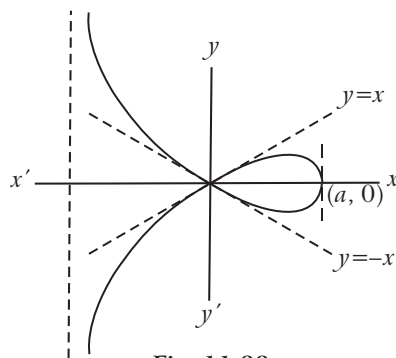


Fig. 11.22

The shape of curve as shown in figure 11.22.

Example 17: Trace the curve $x^3 + y^3 = 3axy$... (i)

Solution:

1. **Symmetry:** Put $x = y$ and $y = x$ in (i) the equation of curve remain unchanged then symmetry about the line $y = x$.
2. Curve passed through origin.
3. **Tangent at origin:** The tangent at origin are
 $xy = 0 \Rightarrow x = 0$ or $y = 0$ i.e., y and x -axis are tangent.

4. **Points of intersection:** Put $y = x$ in (i), we find

$$2x^3 = 3ax^2 \text{ or } x^2(2x - 3a) = 0$$

$$\Rightarrow x = 0 \text{ or } x = \frac{3a}{2}$$

i.e., $(0, 0), \left(\frac{3a}{2}, \frac{3a}{2}\right)$, are the points of intersection.

5. Tangent at $\left(\frac{3a}{2}, \frac{3a}{2}\right)$: Differentiations (i) w.r.t. (x) we find

$$\frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$$

$$\therefore \left(\frac{dy}{dx}\right) \text{ at } \left(\frac{3a}{2}, \frac{3a}{2}\right) = \frac{a\left(\frac{3a}{2}\right) - \left(\frac{3a}{2}\right)^2}{\left(\frac{3a}{2}\right)^2 - a\left(\frac{3a}{2}\right)} = -1$$

$\therefore$ The slope of the tangent of the curve at $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ is -1

i.e., it inclined to x -axis at an angle 135°

6. **Asymptotes:** No asymptotes parallel to x and y -axis.
- $\therefore$ The shape of the curve is shown in Figure 11.23.

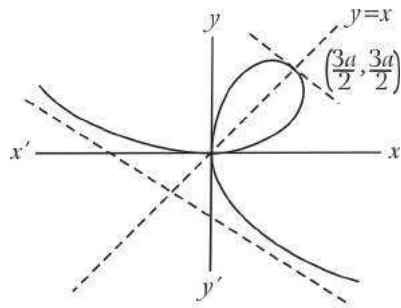


Fig. 11.23

Example 18: Trace the curve $a^3 y^2 = (x-a)^4 (x-b)$, $a > b$.

Solution:

- (1) Symmetry about x -axis.
- (2) The curve does not pass through the origin.
- (3) The curve cuts the axes at $(b, 0)$ and $(a, 0)$.

To find tangent at $(b, 0)$. Shifting the origin to $(b, 0)$, the new equation of the curve is

$$a^3 y^2 = (x + b - a)^4 x.$$

$\therefore$ The tangent at new origin is $x = 0$, i.e., the tangent at $(b, 0)$ is parallel to original y -axis.

To find tangent at $(a, 0)$. Shifting the origin to $(a, 0)$, the equation of the curve becomes

$$a^3 y^2 = x^4 (x + a - b).$$

The tangent at new origin is $y^2 = 0$ or $y = 0$, i.e., the tangent at $(a, 0)$ is parallel to original x -axis.

- (4) No asymptotes.
- (5) Solving for y , $y^2 = (x-a)^4 (x-b)/a^3$
- (I) If x is positive.
 - (i) When $0 \leq x < b$, y is imaginary and so the curve does not exist in the region.
 - (ii) When $b < x < a$, y is real and $y = 0$ at $x = a, b$. The curve exists in this region.
 - (iii) When $x > a$, y is real, the curve lies in this region. Also $y^2 \rightarrow \infty$ as $x \rightarrow \infty$.

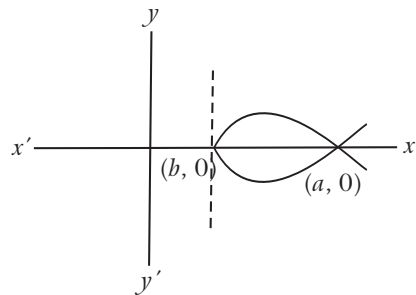


Fig. 11.24

- (II) If $x < 0$, y is imaginary. So the curve does not exist in this region.

Hence the shape of the curve is as shown by the figure 11.24.

Example 19: Trace the curve $y^2 = (x-a)(x-b)(x-c)$, $a > b > c$.

Solution: (1) Symmetry about x -axis.

- (2) Does not pass through $(0, 0)$.
- (3) The curve crosses the axes at $(c, 0)$, $(b, 0)$ and $(a, 0)$ and the tangents at these points are parallel to y -axis.
- (4) No asymptotes.
- (5) From the curve $y^2 = (x-a)(x-b)(x-c)$.
 - (I) If x is positive.
 - (i) When $0 < x < c$, $y^2 = -ve \Rightarrow y$ is imaginary, so the curve does not exist in this region.
 - (ii) When $c < x < b$, $y^2 = +ve \Rightarrow y$ is real and so the curve exists in this region.
 - (iii) When $b < x < a$, $y^2 = -ve \Rightarrow y$ is imaginary and so the curve does not exist in this region.

(iv) When $x > a$, $y^2 = +ve \Rightarrow y$ is real and so the curve exists in this region. Also $y^2 \rightarrow \infty$ as $x \rightarrow \infty$.

(II) When $x < 0$, y is imaginary and so the curve does not exist in this region.

(6) Clearly $dy/dx \rightarrow \infty$ as $x \rightarrow \infty$ i.e., the curve becomes parallel to y -axis as $x \rightarrow \infty$.

$\therefore$ The shape of the curve is as shown in the figure.

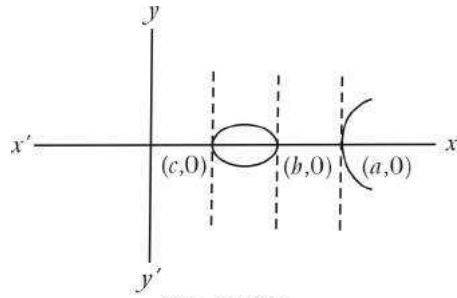


Fig. 11.25

Example 20: Trace the curve $y^2(x-a) = x^2(x+a)$.

Solution: (1) Symmetry about x -axis.

- (2) $(0, 0)$ satisfies the equation of the curve. The tangents at $(0, 0)$ are given by $y^2 = -x^2$, i.e., $y = \pm ix$ which are imaginary. Hence $(0, 0)$ is a **conjugate point** and so the curve will not pass through the origin.
- (3) The curve crosses the axes at $(-a, 0)$. Shifting the origin to $(-a, 0)$, the equation becomes

$$y^2(x-2a) = (x-a)^2 x.$$

The tangent at new origin is $x = 0$, i.e., the tangent at $(-a, 0)$ is parallel to the y -axis.

- (4) **Asymptotes:** The equation of the curve may be written as

$$x(y-x)(y+x) - ay^2 - ax^2 = 0 \quad \dots(1)$$

The asymptote parallel to y -axis is $x-a=0$. Now let $y-x=k$ or $y=x+k$ be the asymptote corresponding to the linear factor $y-x$. Putting $y = k_1 + x$ in (1), we get

$$xk_1(k_2 + 2x) - a(k_1 + x)^2 - ax^2 = 0.$$

Equating to zero the coefficient of x^2 , we get

$$2k_1 - 2a = 0 \quad \text{or} \quad k_1 = a.$$

$\therefore y-x=a$ is an asymptote.

Again let $y+x=k_2$, $\therefore$ putting $y = k_2 - x$ in (i), we get

$$x(k_2 - 2x)k_2 - a(k_2 - x)^2 - ax^2 = 0$$

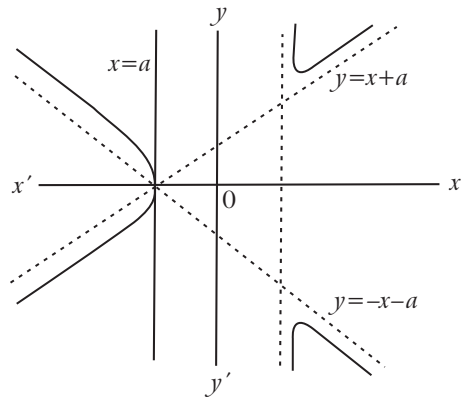


Fig. 11.26

Equating to zero the coefficient of x^2 , we get $k_2 = -a$.

$\therefore y + x = -a$ is the asymptote corresponding to $y + x$.

So the all asymptotes of the curve are

$$x = a, y = x + a, y = -x - a.$$

(5) Solving for y

$$y^2 = x^2(x + a) / (x - a). \quad \dots(ii)$$

(I) If x is +ve.

(i) When $0 < x < a$, $y^2 = -ve \Rightarrow y$ is imaginary.

(ii) When $x > a$, $y^2 = +ve \Rightarrow y$ is real. Also $y^2 \rightarrow \infty$ as $x \rightarrow \infty$.

$\therefore$ The curve exists in the region $x > a$.

(II) If x is negative.

(i) When $-a, x < 0$, y is imaginary and (ii) when $x < -a$, y is real. So the curve lies in the region $x < -a$. Also $y^2 \rightarrow \infty$ as $x \rightarrow -\infty$.

$\therefore$ The shape of the curve is as shown in the figure 11.26.

Example 21: Trace the curve $y(a^2 + x^2) = a^2x$.

Solution: (1) Symmetry in opposite quadrants since the equation remains unchanged by substituting $-x$ for x and $-y$ for y .

(2) The curve passes through the origin and the tangent at this point is $y = x$.

(3) The curve cuts the axes at $(0, 0)$ only.

(4) The only asymptote is $y = 0$, i.e., x -axis.

(5) Solving for y , we have $y = a^2x / (a^2 + x^2)$.

If x is positive, y is positive and when x is negative, y is negative and so the curve exists in the first and third quadrants. Also $y \rightarrow 0$ as $x \rightarrow \infty$ and it shows that the x -axis is the asymptote in this region.

(6) Differentiating the given equation of the curve, we get

$$\frac{dy}{dx} = \frac{a^2(a^2 - x^2)}{(a^2 + x^2)^2}, \quad \frac{d^2y}{dx^2} = \frac{-2a^2x(3a^2 - x^2)}{(a^2 + x^2)^3}$$

$$\text{Now, } (a^2 + x^2)^3 \frac{d^2y}{dx^2} = -6a^4x + 2a^2x^3.$$

Differentiating again w.r.t. x , we get

$$(a^2 + x^2)^3 \frac{d^3y}{dx^3} + 6x(a^2 + x^2)^2 \frac{d^2y}{dx^2} = -6a^4 + 6a^2x^2. \quad \dots(i)$$

Putting $a^2y/dx^2 = 0$, we get $x = 0, \pm a\sqrt{3}$ and $a^3y/dx^3 \neq 0$ at

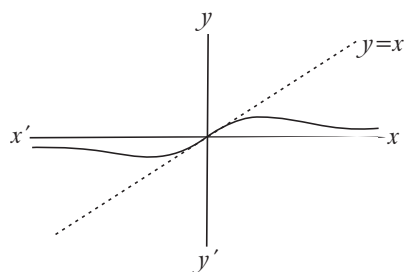


Fig. 11.27

these points. Hence there are points of inflexion at $x = 0, \pm a\sqrt{3}$.

- (7) $dy/x = 0 \Rightarrow x = \pm a$ and so the tangent is parallel to x -axis at $x = \pm a$. Combining the above facts the curve is as shown in the figure 11.27.

Example 22: Trace the curve $y^2 (x^2 - 1) = x$.

Solution: (I) Symmetry about x -axis.

- (2) Passes through the origin and the tangent at origin is $x = 0$, i.e., y -axis.
 (3) The curve cuts the axes at $(0, 0)$ only.
 (4) **Asymptotes:** The asymptotes are $y = 0, x = \pm 1$.
 (5) Solving for

$$y^2 = x/(x^2 - 1)$$

(I) If x is positive.

- (i) When $0 < x < 1$, y is imaginary and so the curve does not exist in this region.

- (ii) When $x > 1$, y is real. Also $y^2 \rightarrow 0$ as $x \rightarrow \infty$ and it shows that $y = 0$ is the asymptote of the curve in this region. Also $y^2 \rightarrow \infty$ as $x \rightarrow 1$ (from the right).

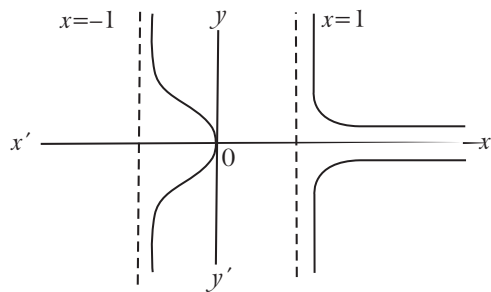


Fig. 11.28

(II) If x is negative.

- (i) When $-1 < x < 0$, y is real, and $y^2 \rightarrow \infty$ as $x \rightarrow -1$ (from the right). So the curve exists in this region and $x = -1$ is the asymptote.
 (ii) When $x < -1$, y is imaginary and so the curve does not exist in this region.

The shape of the curve is as shown in the figure.

Example 23: Trace the curve $x^2 (x^2 + y^2) = a^2 (x^2 - y^2)$

or $y^2 (a^2 + x^2) = x^2 (a^2 - x^2)$... (i)

[B.C.A. (Agra) 2000, 2001]

Solution:

- Symmetry:** The symmetry about both axis.
- Curve passed through origin.
- Tangent at origin:** The tangent at origin are $y^2 = x^2$ or $y = \pm x$. These tangents being real and distinct, node is expected at the origin.
- Points of intersection:** Putting $y = 0$ in (i) we get $x = 0, \pm a$ i.e., the curve crosses x -axis at $(0, 0)$, $(a, 0)$ and $(-a, 0)$.

5. **Tangent at $(a, 0)$:** Shift the origin to $(a, 0)$ put $x = x + a$ and $y = y + 0$ in (i), we find

$$y^2 \{a^2 + (x+a)^2\} = (x+a)^2 \{(a^2 - (x+a)^2)\}$$

or $y^2 \{x^2 + 2ax + 2a^2\} = (x+a)^2 \{-x^2 - 2ax\}$

Equating the lowest degree term to zero, we get the equation of the tangent at the new origin as $x = 0$ i.e., new y -axis similarly tangent at $(-a, 0)$ is y -axis.

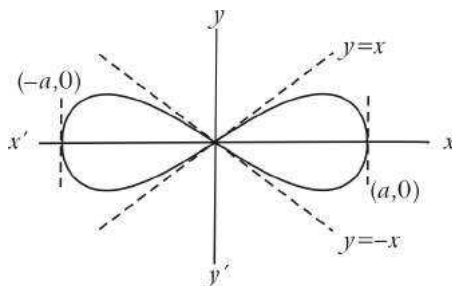


Fig. 11.29

6. **Asymptotes:** No asymptotes parallel to x and y axis.
7. **Region:** Solve for y i.e.,
 $y^2 = x^2(a^2 - x^2)/(a^2 + x^2)$ when
 $x^2 > a^2$ then y^2 is negative or y is imaginary then curve does not lie in that region.
 $\therefore$ The shape of the curve is shown in figure.

Example 24: Trace the curve $xy^2 + (x+a)^2(x+2a) = 0$

...(i)

[B.C.A. (Agra) 2008]

Solution:

- Symmetry:** The symmetry about x -axis.
- Curve does not pass through origin.
- Points of intersection:** Put $x = 0$ in (i) we get $y = -\infty$. Again put $y = 0$ in (i), we get $x = -a$, $x = -2a$

Hence $(-a, 0)$, $(-2a, 0)$ are the points of intersection.

- Tangent at $(-a, 0)$:** Put $x = x - a$, $y = y + 0$ in (i), we get

$$(x-a)y^2 + (x-a+a)^2(x-a+2a) = 0$$

or $(x-a)y^2 + x^2(x+a) = 0$

Equate to zero the lowest degree term i.e.,

$$-ay^2 + ax^2 = 0$$

or $y^2 = x^2$

or $y = \pm x$

Now to find tangent at $(-2a, 0)$: Put $x = x - 2a$, $y = y + 0$ in (i), we get

$$(x-2a)y^2 + (x-2a+a)^2(x+2a-2a) = 0$$

or $(x-2a)y^2 + (x-a)^2x = 0$

Equate to zero the lowest degree term i.e., $a^2x = 0$ or $x = 0$

i.e., y -axis is tangent at $(-2a, 0)$.

5. **Asymptote:** The asymptote parallel to y -axis is $x=0$ and no asymptote parallel to x -axis.

6. **Region:** Solve for

$$y^2 = \frac{-(x+a)^2 (x-2a)}{x}$$

when $x < a$ then y^2 is positive or y

is real then curve lies in that

region, when $x > 2a$ then y^2 is negative or y is imaginary then curve does not lie in that region.

The shape of the curve is shown in figure 11.30.

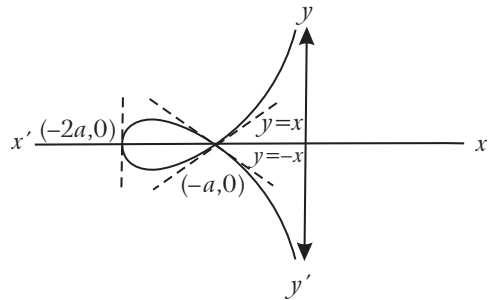


Fig. 11.30

Example 25: Trace the curve $y^2 (x^2 + y^2) + a^2 (x^2 - y^2) = 0$

or $y^2 (a^2 + x^2) = x^2 (a^2 - x^2)$... (i)

[B.C.A. (Agra) 2000, 2001]

Solution:

- Symmetry:** The symmetry about both axes.
- Curve passed through origin.
- Tangent at origin:** The tangent at origin are $y^2 = x^2$ or $y = \pm x$
- Points of intersection:** Put $y=0$ in (i), we get

$$0 = x^2 (a^2 - x^2) \quad \text{or} \quad x=0, \quad x=\pm a$$

i.e., $(0, 0)$, $(a, 0)$, $(-a, 0)$ are the points of intersection.

5. **Tangent at $(a, 0)$:** Put $x = x+a$, $y = y+0$ in (i), we get

$$y^2 (a^2 + (x+a)^2) = (x+a)^2 (a^2 - (x+a)^2)$$

$$\text{or } y^2 (x^2 + 2a^2 + 2ax) = (x+a)^2 (-x^2 - 2ax)$$

Equate to zero the lowest degree we get the tangent at new origin i.e., $x=0$ or y -axis is tangent. Similarly tangent at $(-a, 0)$ is y -axis.

6. **Asymptotes:** No asymptote parallel to x and y -axis.

7. **Region:** Solve for y , we get

$$y^2 = \frac{x^2 (a^2 - x^2)}{(a^2 + x^2)}$$

When $x > a$ then y^2 is negative or y

is negative then curve does not lie in that region.

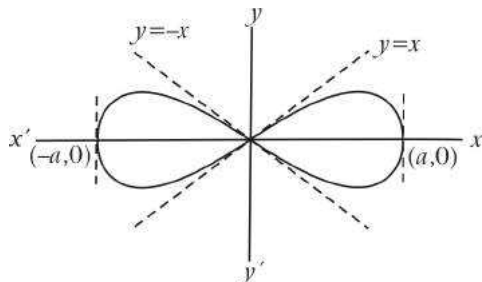


Fig. 11.31

When $x < -a$ then y^2 is negative or y is imaginary then the curve does not lie in that region.

Hence, the shape of the curve is shown in figure 11.31.

Example 26: Trace the curve $y^2(x^2 + y^2) + a^2(x^2 - y^2) = 0$... (i)

[B.C.A. (Agra) 2000, 2002]

Solution:

1. **Symmetry:** Symmetry about both axis.
2. The curve passes through origin.
3. **Tangent at origin:** The tangents at origin are $x^2 - y^2 = 0$ or $y = \pm x$
These two tangents being real and distinct, hence node is at origin.

4. **Point of intersection:** Put $x = 0$ in (i), we get

$$y^4 - a^2 y^2 = 0 \quad \text{or} \quad y^2(y^2 - a^2) = 0$$

$$\text{or} \quad y = 0, \quad y = \pm a$$

$\therefore$ The curve crosses the y -axis at $(0, 0)$, $(0, a)$ and $(0, -a)$

5. **Tangent at $(0, a)$:** Shifting the origin to $(0, a)$ by putting $x = x + 0$, $y = y + a$ in (i), we get

$$(y+a)^2(x^2 + (y+a)^2) + a^2(x^2 - (y+a)^2) = 0$$

$\therefore$ The tangent at new origin is $y = 0$ i.e., x -axis
similarly tangent at $(0, -a)$ is also x -axis.

6. **Asymptotes:** No asymptotes parallel to x and y -axis.

7. **Region:** Solve for x i.e., $x^2 = \frac{y^2(a^2 - y^2)}{(a^2 + y^2)}$

when $y > a$ then x^2 is negative or x is imaginary
then curve does not lie in that region.

The shape of the curve is shown in figure 11.32.

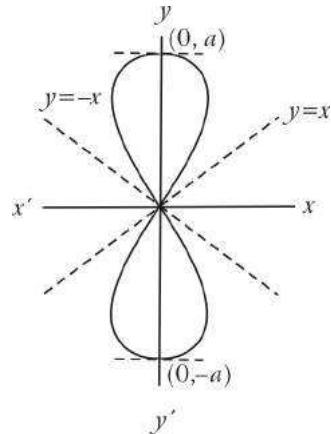


Fig. 11.32

Example 27: Trace the curve $y^3 = a^2 x - x^3$... (i)

[B.C.A. (Agra) 2003]

Solution:

1. **Symmetry:** Put $x = -x$ and $y = -y$ in given equation of the curve and equation of curve remains unchanged, so there is symmetry in opposite quadrants.
2. Curve passes through origin.

3. **Tangent at origin:** The tangent at origin is $x=0$ i.e., y -axis.
4. **Points of intersection:** Putting $y=0$ in (i) we, get

$$x(a^2 - x^2) = 0 \quad \text{i.e., } x=0, \quad x = \pm a$$

so curve crosses x -axis, at $(0,0)$, $(a,0)$ and $(-a,0)$.

5. **Tangent at $(a,0)$:** Shifting the origin to $(a,0)$, the equation of the curve becomes

$$y^3 = a^2(x+a) - (x+a)^3$$

$$\text{or } y^3 = -x^3 - 3x^2a - 2a^2x$$

Equate to zero the lowest degree term i.e., $x=0$ or y -axis is tangent at $(a,0)$

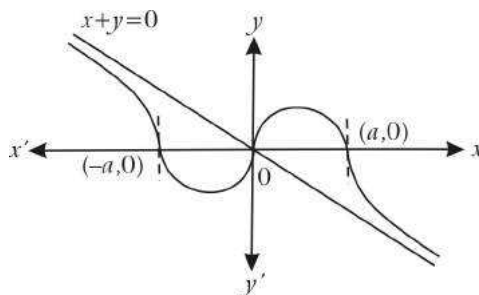


Fig. 11.33

6. **Asymptotes:** $x+y=0$ is an asymptote to the curve.
 7. As $x \rightarrow \infty$, $y \rightarrow -\infty$ and as $x \rightarrow -\infty$, $y \rightarrow \infty$
- $\therefore$ The shape of the curve is shown in the figure 11.33.

Example 28: Trace the curve $x^5 + y^5 = 5a^2x^2y$

...(i)

[B.C.A. (Lucknow) 2008]

Solution:

1. **Symmetry:** Symmetry in opposite quadrants.
2. Curve passed through origin.
3. **Tangent at origin:** The tangent at origin is $x^2y=0$ or $x=0$ and $y=0$ i.e., y -axis and x -axis is tangent.
4. **Points of intersection:** Put $y=0$ in (i), we get $x=0$
Again put $x=0$ in (i) we get $y=0$.
Hence there is no point of intersection except $(0,0)$.
5. **Asymptote:** No asymptote parallel to x and y -axis but $x+y=0$ is oblique asymptote.
6. **Region:** In order to discuss the form of the curve near the origin neglect x^5 and consider $y^5 = 5a^2x^2y$ or $y^4 = 5a^2x^2$
 $\therefore x=0$ is a tangent to this branch at origin.
This branch may be written as $y^2 = \pm \sqrt{5}ax$
Again neglect y^5 the equation of the curve becomes

$$x^5 - 5a^2x^2y = 0 \quad \text{or} \quad x^3 = 5a^2y$$

$\therefore y=0$ is a tangent to this branch. Also for this branch i.e., $y = \frac{x^3}{5a^2}$ we find that

$\frac{d^2y}{dx^2} = 0$ gives $x=0$ i.e., $(0, 0)$ is point of inflexion of this branch.

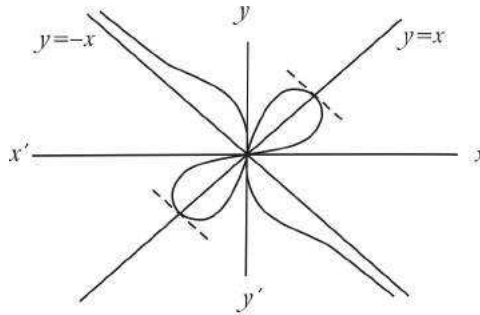


Fig. 11.34

Example 29: Trace the curve $x^{2/3} + y^{2/3} = a^{2/3}$... (i)

Solution:

1. **Symmetry:** We can write (i) as $\left(\frac{x^2}{a^2}\right)^{1/3} + \left(\frac{y^2}{a^2}\right)^{1/3} = 1$

This shows that symmetry about both axes.

2. Curve does not pass through origin.
3. **Points of intersection.** Put $y=0$ in (i), we get $(x)^{2/3} = a^{2/3}$
 $\Rightarrow x^2 = a^2 \quad \text{or} \quad x = \pm a$

i.e., the curve crosses the x-axis at $(a, 0)$ and $(-a, 0)$ similarly the curve crosses the y-axis at $(0, a)$ and $(0, -a)$.

4. **Tangent at $(a, 0)$:** Differentiating (i) w.r.t. (x) , we get $\frac{dy}{dx} = -\left(\frac{y}{x}\right)^{1/3}$

Then $\left(\frac{dy}{dx}\right)_{(a,0)} = 0$ i.e., tangent at $(a,0)$ is parallel to x-axis.

Again $\left(\frac{dy}{dx}\right)_{(0,a)} = -\infty$ i.e., tangent at $(0, a)$ is parallel to y-axis.

5. **Asymptotes:** No asymptote parallel to x and y-axis.

6. **Region:** Solve for x and y i.e., $\left(\frac{y^2}{a^2}\right)^{1/3} = 1 - \left(\frac{x^2}{a^2}\right)^{1/3}$

When $\left(\frac{x}{a}\right) > 1$ or $x > a$ then $\frac{y^2}{a^2} < 0$ or $y^2 < 0$ or y is imaginary then curve does not lie in the region $y > a$. Due to symmetry the curve does not exist for $x < -a$ and for $y < -a$.

The shape of the curve with the help of above data is

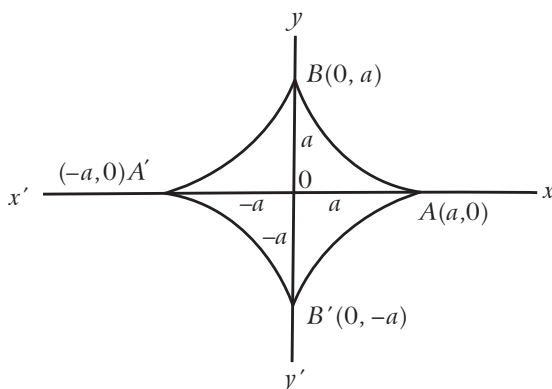


Fig. 11.35

Example 30: Trace the curve $x^2 y^2 = a^2 (y^2 - x^2)$

Solution:

1. **Symmetry:** The symmetry about both axes.
2. The curve passes through the origin.
3. **Tangent at origin:** The tangent at origin are $y^2 - x^2 = 0$ or $y = \pm x$.
4. **Points of intersection:** There is no point of intersection except origin.
5. **Asymptote:** Asymptote parallel to y -axis is $x^2 = a^2$ or $x = \pm a$ and no asymptote parallel to x -axis.
6. **Region:** Solve for y , we get

$$y^2 = -a^2 x^2 / (x^2 - a^2)$$

or

$$y^2 = a^2 x^2 / (a^2 - x^2)$$

When $x^2 > a^2$ then y^2 is negative or y is imaginary then curve does not exist in that region.

- $\therefore$ The shape of the curve is shown in figure 11.36

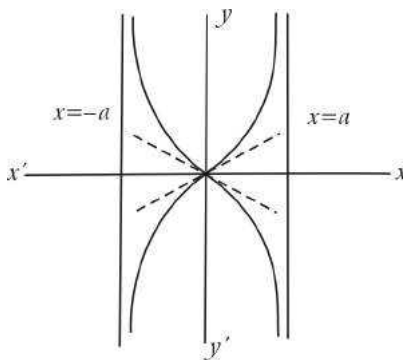


Fig. 11.36



Comprehensive Exercise

1. $ay^2 = x^3$

2. $y(x^2 + 4a^2) = 8a^3$

3. $9ay^2 = x(x - 3a)^2$

4. $xy^2 = 4a^2(2a - x)$

5. $a^2 y^2 = x^3(2a - x)$

6. $x^2 y^2 = x^2 - a^2$

7. $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$

8. $x^6 + y^6 = a^2 x^2 y^2$

9. $a^4 y^2 = a^2 x^4 - x^6$

10. $y^2(a - x) = x^2(a + x)$

[B.C.A. (Kanpur) 2001, 2006]

11. $x^2 = y^2(x + a)^3$



Answers

1.

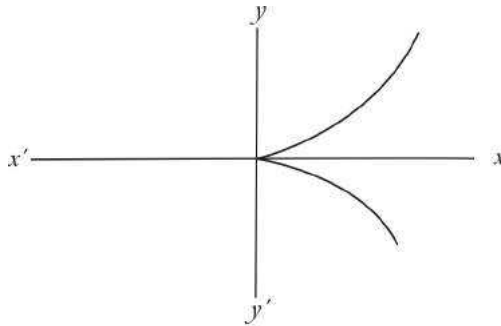


Fig. 11.37

2.

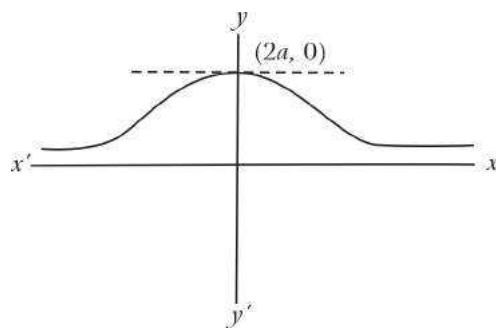


Fig. 11.38

3.

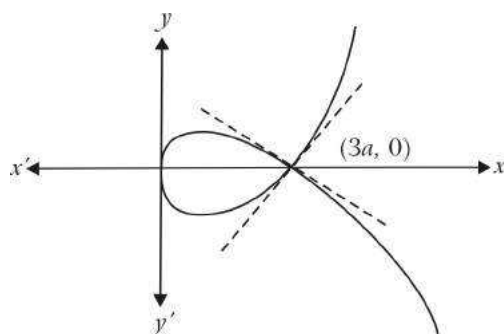


Fig. 11.39

4.

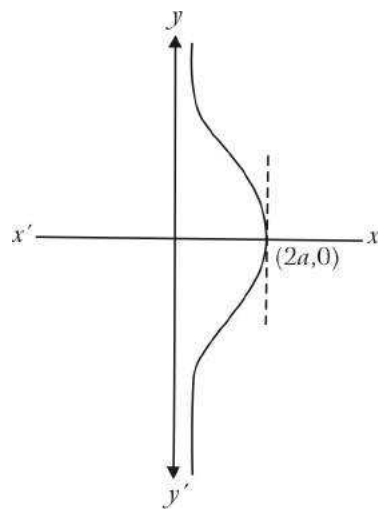


Fig. 11.41

5.

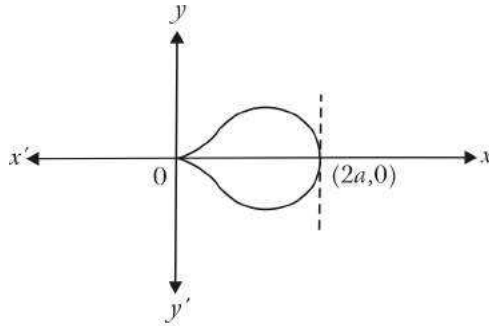


Fig. 11.41

6.

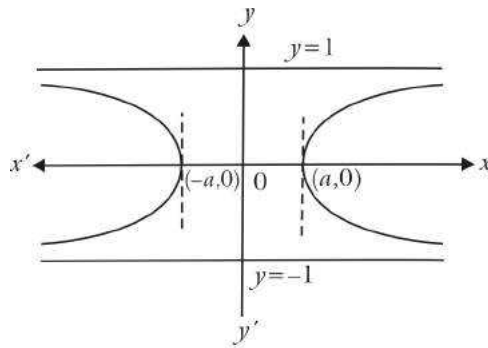


Fig. 11.42

7.

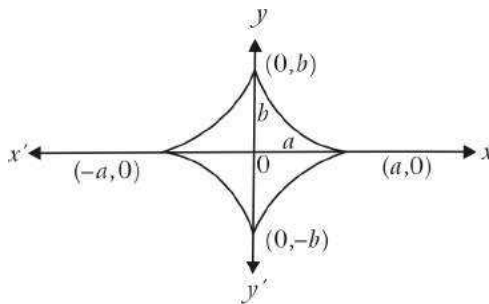


Fig. 11.43

8.

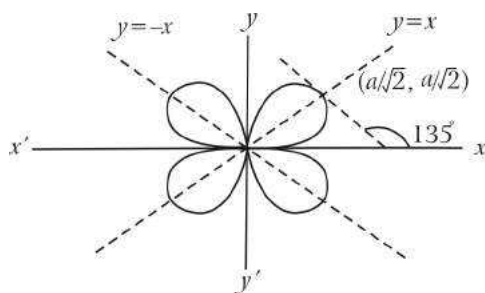


Fig. 11.44

9.

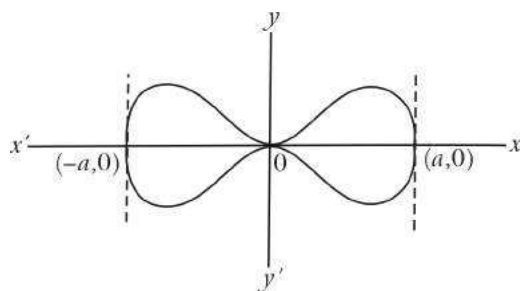


Fig. 11.45

10.

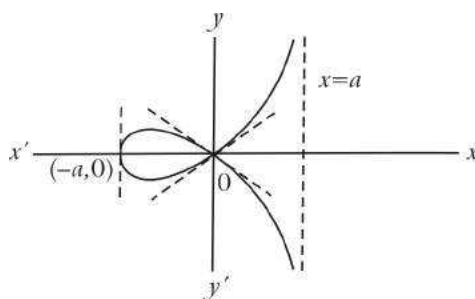


Fig. 11.46

11.

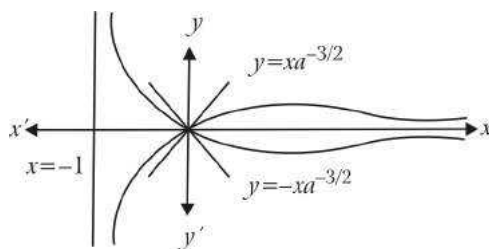
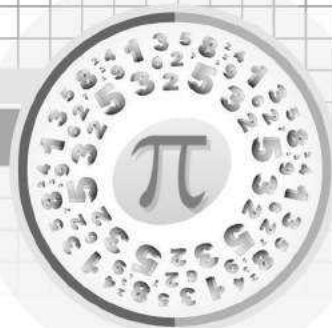


Fig. 11.47





Unit-3



Chapter

12

Successive Differentiation and Leibnitz Theorem

12.1 Successive Differentiation

If y be any function of x then its derivative $\frac{dy}{dx}$ will be a function of x whose differentiation can be done. Hence we can find again differential coefficient of $\frac{dy}{dx}$. The derivative of $\frac{dy}{dx}$; i.e., $\frac{d}{dx} \left(\frac{dy}{dx} \right)$ is called **second derivative** of y . Similarly derivative of second derivative is called third derivative. Similarly, fourth, fifth etc. The successive derivatives of y with respect to x are denoted by

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \dots$$

The n^{th} derivative or differential coefficient of y w.r.t. to x is denoted by $\frac{d^n y}{dx^n}$.

If $y = f(x)$ then the successive differential coefficients are also denoted by the following notations:

$y_1,$	$y_2,$	$y_3,$	$\dots,$	$y_n,$
$y',$	$y'',$	y'''	$\dots,$	$y^{(n)},$
$f'(x),$	$f''(x),$	$f'''(x),$	$\dots,$	$f^{(n)}(x),$
$Dy,$	$D^2y,$	$D^3y,$	$\dots,$	$D^n y,$
$Df(x),$	$D^2 f(x),$	$D^3 f(x),$	$\dots,$	$D^n f(x),$

Note: The value of differential coefficient at $x = a$ is usually denoted by $(y_n)_{x=a}$ or $(y_n)_a$.
If $y = f(x)$ then it is denoted by $f^{(n)}(a)$.

Example 1: If $y = A \sin mx + B \cos mx$, then prove that

$$\frac{d^2 y}{dx^2} + m^2 y = 0 \quad \text{or} \quad y_2 + m^2 y = 0.$$

[B.C.A. (Agra) 2000, B.C.A. (Kanpur) 2005]

Solution: We have

$$y = A \sin mx + B \cos mx \quad \dots(1)$$

Differentiating both side of (1) with respect to x , we find

$$\frac{dy}{dx} = Am \cos mx - Bm \sin mx \quad \dots(2)$$

Differentiating (2) with respect to x , we find

$$\frac{d^2 y}{dx^2} = -Am^2 \sin mx - Bm^2 \cos mx = -m^2 (A \sin mx + B \cos mx)$$

$$\text{or} \quad \frac{d^2 y}{dx^2} = -m^2 y \quad [\text{From (1)}]$$

$$\text{or} \quad \frac{d^2 y}{dx^2} + m^2 y = 0.$$

Example 2: If $y = e^{ax} \sin bx$, then prove that $y_2 - 2ay_1 + (a^2 + b^2)y = 0$.

[B.C.A. (Rohilkhand) 2006, 2008]

Solution: We have $y = e^{ax} \sin bx \quad \dots(1)$

Differentiating both sides of (1) with respect to x , we find

$$y_1 = ae^{ax} \sin bx + be^{ax} \cos bx = ay + be^{ax} \cos bx \quad [\text{From (1)}]$$

$$\text{or} \quad y_1 - ay = be^{ax} \cos bx \quad \dots(2)$$

Differentiating both sides of (2) w.r. to x , we find

$$y_2 - ay_1 = abe^{ax} \cos bx - b^2 e^{ax} \sin bx$$

$$\text{or} \quad y_2 - ay_1 = a(y_1 - ay) - b^2 y \quad [\text{From (1) and (2)}]$$

$$\text{or} \quad y_2 - 2ay_1 + (a^2 + b^2)y = 0.$$

Example 3: If $y = \sin (m \sin^{-1} x)$, then prove that

$$(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + m^2 y = 0.$$

[B.C.A. (Delhi) 2004, 2008, 2011]

Solution: We have $y = \sin (m \sin^{-1} x) \quad \dots(1)$

Differentiating both sides of (1) w.r. to x , we get

$$\frac{dy}{dx} = \cos (m \sin^{-1} x) \cdot \frac{m}{\sqrt{1 - x^2}}$$

$$\Rightarrow \sqrt{1-x^2} \frac{dy}{dx} = m \cos (m \sin^{-1} x)$$

Squaring both sides, we get

$$(1-x^2) \left(\frac{dy}{dx} \right)^2 = m^2 \cos^2 (m \sin^{-1} x)$$

$$\Rightarrow (1-x^2) \left(\frac{dy}{dx} \right)^2 = m^2 [1 - \sin^2 (m \sin^{-1} x)]$$

$$\Rightarrow (1-x^2) \left(\frac{dy}{dx} \right)^2 = m^2 (1-y^2)$$

Again, differentiating w.r. to x , we get

$$2(1-x^2) \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} - 2x \left(\frac{dy}{dx} \right)^2 = -2m^2 y \frac{dy}{dx}$$

Dividing by $2 \frac{dy}{dx}$, we find

$$(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = -m^2 y \quad \text{or} \quad (1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + m^2 y = 0.$$

Example 4: If $y = Ae^{-kt} \cos (pt + c)$, then prove that

$$\frac{d^2y}{dt^2} + 2k \frac{dy}{dt} + n^2 y = 0, \text{ where } n^2 = p^2 + k^2.$$

[B.C.A. (Rohtak) 2007, 2011]

Solution: We have $y = Ae^{-kt} \cos (pt + c)$

...(1)

Differentiating (1) w.r. to t , we get

$$\frac{dy}{dt} = -kAe^{-kt} \cos (pt + c) - pAe^{-kt} \sin (pt + c)$$

$$= -ky - pAe^{-kt} \sin (pt + c) \quad \dots(2)$$

Again, differentiating w.r. to t , we get

$$\frac{d^2y}{dt^2} = -k \frac{dy}{dt} + pAe^{-kt} \sin (pt + c) - p^2 Ae^{-kt} \cos (pt + c)$$

$$= -k \frac{dy}{dt} - k \left(\frac{dy}{dt} + ky \right) - p^2 y \quad [\text{From (2)}]$$

$$\text{or} \quad \frac{d^2y}{dt^2} + 2k \frac{dy}{dt} + (k^2 + p^2) y = 0$$

$$\text{or} \quad \frac{d^2y}{dt^2} + 2k \frac{dy}{dt} + n^2 y = 0 \quad [\because n^2 = p^2 + k^2]$$

Example 5: If $x = f(t)$, $y = g(t)$, then prove that

$$\frac{d^2 y}{dx^2} = \frac{x' y'' - y' x''}{x'^3}, \text{ where } x' = \frac{dx}{dt}, y' = \frac{dy}{dt} \text{ etc.} \quad [\text{B.C.A. (Agra) 2004}]$$

Solution: Here $\frac{dx}{dt} = x'$ and $\frac{dy}{dt} = y'$, so

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{y'}{x'} \quad \dots(1)$$

Hence

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{y'}{x'} \right) = \frac{d}{dt} \left(\frac{y'}{x'} \right) \cdot \frac{dt}{dx} \\ &= \frac{x' y'' - y' x''}{x'^2} \cdot \frac{1}{x'} = \frac{x' y'' - y' x''}{x'^3}. \end{aligned}$$

Example 6: If $y = x \log [x / (a + bx)]$, then prove that $x^3 \frac{d^2 y}{dx^2} = \left[x \frac{dy}{dx} - y \right]^2$.
[B.C.A. (Agra) 2002]

Solution: We have, $\frac{y}{x} = \log x - \log (a + bx)$... (1)

Differentiating (1) w.r. to x , we get

$$\frac{x \frac{dy}{dx} - y}{x^2} = \frac{1}{x} - \frac{b}{a + bx} \quad \text{or} \quad x \frac{dy}{dx} - y = \frac{ax}{a + bx} \quad \dots(2)$$

Differentiating (2) w.r. to x , we get

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} - \frac{dy}{dx} = \frac{a(a + bx) - ax \cdot b}{(a + bx)^2}$$

or
$$x \frac{d^2 y}{dx^2} = \frac{a^2}{(a + bx)^2} = \left(\frac{a}{a + bx} \right)^2$$

or
$$x^3 \frac{d^2 y}{dx^2} = \left(\frac{ax}{a + bx} \right)^2 \quad [\text{On multiplying by } x^2]$$

or
$$x^3 \frac{d^2 y}{dx^2} = \left(x \frac{dy}{dx} - y \right)^2 \quad [\text{From (2)}]$$

Example 7: If $p^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta$, then prove that

$$p + \frac{d^2 p}{d\theta^2} = \frac{a^2 b^2}{p^3}. \quad [\text{B.C.A. (Agra) 2002, B.C.A. (Awadh) 2008, B.C.A. (Meerut) 2011}]$$

Solution: We have

$$p^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta \quad \dots(1)$$

Differentiating (1) w.r. to θ , we get

$$2p \frac{dp}{d\theta} = a^2 (-2 \cos \theta \sin \theta) + b^2 (2 \sin \theta \cos \theta)$$

$$\therefore 2p \frac{dp}{d\theta} = (b^2 - a^2) \sin 2\theta. \quad \dots(2)$$

Again, differentiating (2) w.r. to θ , we get

$$2p \frac{d^2p}{d\theta^2} + 2 \frac{dp}{d\theta} \cdot \frac{dp}{d\theta} = 2(b^2 - a^2) \cos 2\theta$$

Thus
$$p \frac{d^2p}{d\theta^2} + \left(\frac{dp}{d\theta}\right)^2 = (b^2 - a^2) \cos 2\theta$$

$$\therefore p^3 \frac{d^2p}{d\theta^2} = p^2 (b^2 - a^2) \cos 2\theta - p^2 \left(\frac{dp}{d\theta}\right)^2$$

Adding p^4 on both sides, we get

$$\begin{aligned} p^4 + p^3 \frac{d^2p}{d\theta^2} &= p^4 + p^2 (b^2 - a^2) \cos 2\theta - p^2 \left(\frac{dp}{d\theta}\right)^2 \\ &= p^2 [p^2 + (b^2 - a^2) \cos 2\theta] - \left(p \frac{dp}{d\theta}\right)^2 \\ &= p^2 [a^2 \cos^2 \theta + b^2 \sin^2 \theta + (b^2 - a^2) (\cos^2 \theta - \sin^2 \theta)] \\ &\quad - (b^2 - a^2)^2 \sin^2 \theta \cos^2 \theta \\ &\quad \text{[From equations (1) and (2)]} \\ &= p^2 [b^2 \cos^2 \theta + a^2 \sin^2 \theta] - (b^2 - a^2)^2 \sin^2 \theta \cos^2 \theta \\ &= (a^2 \cos^2 \theta + b^2 \sin^2 \theta) (b^2 \cos^2 \theta + a^2 \sin^2 \theta) \\ &\quad - (b^2 - a^2)^2 \sin^2 \theta \cos^2 \theta \\ &= a^2 b^2 (\cos^4 \theta + \sin^4 \theta + 2 \sin^2 \theta \cos^2 \theta) \\ &= a^2 b^2 (\cos^2 \theta + \sin^2 \theta)^2 = a^2 b^2. \end{aligned}$$

Dividing both sides by p^3 , we get

$$p + \frac{d^2p}{d\theta^2} = \frac{a^2 b^2}{p^3}.$$

12.2 Nth Derivative of Some Standard Functions

(I) If $y = e^{ax+b}$, then $y_1 = ae^{ax+b}$, $y_2 = a^2 e^{ax+b}$... etc.

In general $y_n = a^n e^{ax+b}$

$$D^n (e^{ax+b}) = a^n e^{ax+b}$$

Particular Case 1: If $a=1$, $b=0$ then $D^n e^x = e^x$

Particular Case 2: If $y = a^x = e^{x \log a}$ then $y_n = (\log a)^n e^{x \log a} = (\log a)^n a^x$.

$$D^n a^x = (\log a)^n a^x$$

(II) $y = (ax + b)^m$, then

$$y_1 = m (ax + b)^{m-1} a$$

$$y_2 = m (m-1) (ax + b)^{m-2} a^2$$

$$y_3 = m (m-1) (m-2) (ax + b)^{m-3} a^3 \text{ etc.}$$

$$y_n = m (m-1) (m-2) \dots (m-n+1) (ax + b)^{m-n} a^n$$

$$D^n (ax + b)^m = m(m-1)(m-2)\dots(m-n+1) a^n (ax + b)^{m-n} \text{ if } n < m$$

Particular Cases 1: If m is a positive integer, then

$$\begin{aligned} D^n (ax + b)^m &= \frac{m(m-1)(m-2)\dots(m-n+1)(m-n)\dots 2.1}{(m-n)\dots 2.1} \cdot a^n (ax + b)^{m-n} \\ &= \frac{m!}{(m-n)!} a^n (ax + b)^{m-n} \end{aligned}$$

If $m < n$, then $D^n (ax + b)^m = 0$

2. If m is a positive integer and $m = n$, then

$$D^n (ax + b)^n = \frac{n!}{0!} a^n (ax + b)^0 = n! a^n$$

If $a = 1, b = 0$, then $D^n x^n = n!$

3. If m is a negative integer then let $m = -p$ where p is a +ve integer, then

$$\begin{aligned} D^n (ax + b)^{-p} &= (-p)(-p-1)\dots\{-p-(n-1)\} a^n (ax + b)^{-p-n} \\ &= (-1)^n p(p+1)\dots(p+n-1) a^n (ax + b)^{-p-n} \\ &= \frac{(-1)^n (p+n-1)!}{(p-1)!} \cdot a^n (ax + b)^{-p-n} \end{aligned}$$

4. If $m = -1$ i.e., $p = 1$ then

$$D^n (ax + b)^{-1} = (-1)^n n! a^n (ax + b)^{-n-1}$$

(III) If $y = \log (ax + b)$, then

[B.C.A. (Agra) 2003]

$$y_1 = \frac{a}{ax + b} = a (ax + b)^{-1}$$

$$y_2 = (-1) a^2 (ax + b)^{-2}$$

$$y_3 = (-1)(-2) a^3 (ax + b)^{-3} \text{ etc.}$$

$$y_n = (-1)(-2)(-3)\dots\{-(n-1)\} a^n (ax + b)^{-n}$$

$$= (-1)^{n-1} (n-1)! a^n (ax + b)^{-n}$$

$$\therefore D^n \log (ax + b) = \frac{(-1)^{n-1} (n-1)! a^n}{(ax + b)^n}.$$

(IV) If $y = \sin (ax + b)$, then

[B.C.A. (Agra) 2009]

$$y_1 = a \cos (ax + b) = a \sin \left(ax + b + \frac{1}{2} \pi\right)$$

$$y_2 = a^2 \cos \left(ax + b + \frac{1}{2} \pi\right) = a^2 \sin \left(ax + b + \frac{1}{2} \cdot 2\pi\right)$$

$$\dots \dots \dots \dots \dots \dots \dots \dots \dots$$

$$y_n = a^n \sin \left(ax + b + \frac{1}{2} \cdot n\pi\right)$$

$$D^n \sin (ax + b) = a^n \sin \left(ax + b + \frac{1}{2} n\pi\right)$$

$$\text{Similarly, } D^n \cos (ax + b) = a^n \cos \left(ax + b + \frac{1}{2} n\pi\right)$$

(V) If $y = e^{ax} \cos (bx + c)$, then

$$y_1 = e^{ax} \cdot a \cos (bx + c) - e^{ax} \cdot b \sin (bx + c)$$

$$= e^{ax} [a \cos (bx + c) - b \sin (bx + c)]$$

Let

$$a = r \cos \phi, b = r \sin \phi$$

Thus

$$r^2 = a^2 + b^2, \phi = \tan^{-1} (b / a) \quad \dots(1)$$

$\therefore$

$$y_1 = e^{ax} [r \cos \phi \cos (bx + c) - r \sin \phi \sin (bx + c)]$$

$$= r \cdot e^{ax} \cos (bx + c + \phi)$$

Similarly,

$$y_2 = r^2 e^{ax} \cos (bx + c + 2\phi), \text{ etc.}$$

In general

$$y_n = r^n e^{ax} \cos (bx + c + n\phi)$$

$$\therefore D^n \{e^{ax} \cos (bx + c)\} = r^n e^{ax} \cos (bx + c + n\phi)$$

where

$$r^2 = a^2 + b^2, \phi = \tan^{-1} (b / a)$$

Similarly, proceeding as above, we have

$$D^n \{e^{ax} \sin (bx + c)\} = r^n e^{ax} \sin (bx + c + n\phi)$$

$$\text{where } r^2 = a^2 + b^2, \phi = \tan^{-1} (b / a)$$

Example 8: Find the n^{th} differential coefficients of:

(a) $\{\log (ax + b) / (cx + d)\}$

[B.C.A. (Bundelkhand) 2008]

(b) $\sin ax \sin bx$

[B.C.A. (Avadh) 2003]

(c) $\cos^4 x$

[B.C.A. (Agra) 2006]

(d) $\cos x \cos 2x \cos 3x$

[B.C.A. (Agra) 2001]

(e) $\cos^2 x \sin^3 x$

[B.C.A. (Aligarh) 2011]

(f) $e^{ax} \cos^2 x \sin x$

[B.C.A. (Rohtak) 2008]

(g) $e^{ax} \sin bx \cos cx$

[B.C.A. (Kurukshetra) 2011]

(h) $\sin 3x \cos x$

[B.C.A. (Agra) 2002]

Solution: (a) $y = \log \{(ax + b) / (cx + d)\} = \log (ax + b) - \log (cx + d)$

Differentiating n times on both sides, we find

$$\begin{aligned} \therefore y_n &= \frac{(-1)^{n-1} (n-1)! a^n}{(ax + b)^n} - \frac{(-1)^{n-1} (n-1)! c^n}{(cx + d)^n} \\ &= (-1)^{n-1} (n-1)! \left[\frac{a^n}{(ax + b)^n} - \frac{c^n}{(cx + d)^n} \right]. \end{aligned}$$

(b) $y = \sin ax \cos bx = \frac{1}{2} [2 \sin ax \cos bx]$

$$= \frac{1}{2} [\sin (a + b)x + \sin (a - b)x]$$

Differentiating n times on both sides, we find

$$y_n = \frac{1}{2} [(a + b)^n \sin \{(a + b)x + \frac{1}{2} n\pi\} + (a - b)^n \sin \{(a - b)x + \frac{1}{2} n\pi\}].$$

(c) $y = \cos^4 x = (\cos^2 x)^2 = \left[\frac{1}{2} (1 + \cos 2x)\right]^2 = \frac{1}{4} (1 + 2 \cos 2x + \cos^2 2x)$

$$= \frac{1}{4} [1 + 2 \cos 2x + \frac{1}{2} (1 + \cos 4x)] = (3/8) + (1/2) \cos 2x + (1/8) \cos 4x$$

Differentiating n times on both sides, we find

$$\begin{aligned} \therefore y_n &= 0 + \frac{1}{2} \cdot 2^n \cos (2x + \frac{1}{2} n\pi) + (1/8) 4^n \cos (4x + \frac{1}{2} n\pi) \\ &= 2^{n-1} \cos (2x + \frac{1}{2} n\pi) + 2^{2n-3} \cos (4x + \frac{1}{2} n\pi). \end{aligned}$$

(d) $y = \cos x \cos 2x \cos 3x = \frac{1}{2} (2 \cos x \cos 2x) \cos 3x = \frac{1}{2} (\cos 3x + \cos x) \cos 3x$

$$= \frac{1}{4} [2 \cos^2 3x + 2 \cos 3x \cos x] = \frac{1}{4} [1 + \cos 6x + \cos 4x + \cos 2x]$$

Differentiating n times on both sides, we find

$$\therefore y_n = \frac{1}{4} [(6^n \cos (6x + \frac{1}{2} n\pi) + 4^n \cos (4x + \frac{1}{2} n\pi) + 2^n \cos (2x + \frac{1}{2} n\pi)].$$

(e) $y = \cos^2 x \sin^3 x = \frac{1}{2} (1 + \cos 2x) \cdot \frac{1}{4} (3 \sin x - \sin 3x)$

$$= \frac{1}{8} (3 \sin x - \sin 3x + 3 \cos 2x \sin x - \sin 3x \cos 2x)$$

$$= \frac{1}{8} \left[3 \sin x - \sin 3x + \frac{3}{2} (\sin 3x - \sin x) - \frac{1}{2} (\sin 5x + \sin x) \right]$$

$$= \frac{1}{16} [2 \sin x + \sin 3x - \sin 5x]$$

Differentiating n times on both sides, we find

$$\therefore y_n = \frac{1}{16} \left[2^n \sin \left(x + \frac{1}{2} n\pi \right) + 3^n \sin \left(3x + \frac{1}{2} n\pi \right) - 5^n \sin \left(5x + \frac{1}{2} n\pi \right) \right].$$

$$\begin{aligned}
 \text{(f)} \quad y &= e^{ax} \cos^2 x \sin x = e^{ax} \cdot \frac{1}{2} (1 + \cos 2x) \sin x \\
 &= \frac{1}{2} e^{ax} \sin x + \frac{1}{4} (2 \cos 2x \sin x) e^{ax} \\
 &= \frac{1}{2} e^{ax} \sin x + \frac{1}{4} (\sin 3x - \sin x) e^{ax} = \frac{1}{4} [e^{ax} \sin x + e^{ax} \sin 3x]
 \end{aligned}$$

Differentiating n times on both sides, we find

$$y_n = \frac{1}{4} [r^n e^{ax} \sin(x + n\phi) + R^n e^{ax} \sin(3x + n\theta)]$$

$$\begin{aligned}
 \text{where} \quad r &= \sqrt{a^2 + 1}, \quad R = \sqrt{a^2 + 9} \\
 \phi &= \tan^{-1}(1/a), \quad \theta = \tan^{-1}(3/a).
 \end{aligned}$$

$$\begin{aligned}
 \text{(g)} \quad y &= e^{ax} \sin bx \cos cx = e^{ax} \cdot \frac{1}{2} (\sin bx \cos cx) \\
 &= \frac{1}{2} e^{ax} [\sin(b+c)x + \sin(b-c)x] \\
 &= \frac{1}{2} [e^{ax} \sin(b+c)x + e^{ax} \sin(b-c)x]
 \end{aligned}$$

Differentiating n times on both sides, we find

$$\therefore y_n = \frac{1}{2} [r^n e^{ax} \sin\{(b+c)x + n\phi\} + R^n e^{ax} \sin\{(b-c)x + n\theta\}]$$

$$\begin{aligned}
 \text{where} \quad r &= \{a^2 + (b+c)^2\}^{1/2}, \quad R = \{a^2 + (b-c)^2\}^{1/2} \\
 \phi &= \tan^{-1}\{(b+c)/a\}, \quad \theta = \tan^{-1}\{(b-c)/a\}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(h)} \quad y &= \sin 3x \cos x = \frac{1}{2} (2 \sin 3x \cos x) = \frac{1}{2} [\sin 4x + \sin 2x] \\
 \therefore y_n &= \frac{1}{2} [4^n \sin(4x + \frac{1}{2} n\pi) + 2^n \sin(2x + \frac{1}{2} n\pi)].
 \end{aligned}$$

Example 9: Find the n^{th} differential coefficients of:

$$\begin{aligned}
 \text{(a)} \quad & \frac{x}{(x-a)(x-b)(x-c)} & \text{(b)} \quad & \frac{1}{1-5x+6x^2} \\
 \text{(c)} \quad & \frac{x}{2x^2+3x+1} & \text{(d)} \quad & \frac{x^4}{(x-1)(x-2)}, n \geq 3 \\
 \text{(e)} \quad & \frac{1}{x^2-a^2} & \text{(f)} \quad & \frac{1}{(x-1)^3(x-2)}.
 \end{aligned}$$

Solution: (a) Let $y = \frac{x}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$

$$\begin{aligned}
 \text{Now} \quad A &= \lim_{x \rightarrow a} (x-a) \cdot \frac{x}{(x-a)(x-b)(x-c)} \\
 &= \lim_{x \rightarrow a} \frac{x}{(x-b)(x-c)} = \frac{a}{(a-b)(a-c)}
 \end{aligned}$$

Similarly, $B = \frac{b}{(b-a)(b-c)}, C = \frac{c}{(c-a)(c-b)}$

$$\therefore y = \frac{a(x-a)^{-1}}{(a-b)(a-c)} + \frac{b(x-b)^{-1}}{(b-a)(b-c)} + \frac{c(x-c)^{-1}}{(c-a)(c-b)}$$

Differentiating n times on both sides, we find

$$\begin{aligned} \therefore y_n &= \frac{a}{(a-b)(a-c)} \cdot \frac{(-1)^n n!}{(x-a)^{n+1}} + \frac{b}{(b-a)(b-c)} \cdot \frac{(-1)^n n!}{(x-b)^{n+1}} \\ &\quad + \frac{c}{(c-a)(c-b)} \cdot \frac{(-1)^n n!}{(x-c)^{n+1}} \\ &= (-1)^n n! \left\{ \frac{a}{(a-b)(a-c)(x-a)^{n+1}} + \frac{b}{(b-a)(b-c)(x-b)^{n+1}} \right. \\ &\quad \left. + \frac{c}{(c-a)(c-b)(x-c)^{n+1}} \right\} \end{aligned}$$

(b) Let $y = \frac{1}{6x^2 - 5x + 1} = \frac{1}{(2x-1)(3x-1)}$

or $y = \frac{2}{2x-1} - \frac{3}{3x-1}$ (Resolving into partial fractions)

$$= 2(2x-1)^{-1} - 3(3x-1)^{-1}$$

Differentiating n time on both sides and used, we get

$$\begin{aligned} D^n (ax+b)^{-1} &= (-1)^n n! a^n (ax+b)^{-n-1} \\ y_n &= 2(-1)^n n! 2^n (2x-1)^{-n-1} - 3 \cdot 3^n (-1)^n n! (3x-1)^{-n-1} \\ &= (-1)^n n! [2^{n+1} (2x-1)^{-n-1} - 3^{n+1} (3x-1)^{-n-1}]. \end{aligned}$$

(c) $y = \frac{x}{(2x^2 + 3x + 1)} = \frac{x}{(2x+1)(x+1)} = \frac{1}{x+1} - \frac{1}{2x+1}$

$$\therefore y = (x+1)^{-1} - (2x+1)^{-1}$$

Differentiating n times on both sides, we find

$$\begin{aligned} \therefore y_n &= (-1)^n n! (x+1)^{-n-1} - (-1)^n n! 2^n (2x+1)^{-n-1} \\ &= (-1)^n n! [(x+1)^{-n-1} - 2^n (2x+1)^{-n-1}]. \end{aligned}$$

(d) $y = \frac{x^4}{(x-1)(x-2)} = x^2 + 3x + 7 + \frac{15x-14}{(x-1)(x-2)}$

$$= x^2 + 3x + 7 + \frac{16}{x-2} - \frac{1}{x-1}$$

Differentiating n times on both sides, we find

$$\begin{aligned} \therefore y_n &= 0 + 0 + 0 + 16(-1)^n n! (x-2)^{-n-1} - (-1)^n n! (x-1)^{-n-1} \\ &= (-1)^n n! [16(x-2)^{-n-1} - (x-1)^{-n-1}] \text{ where } n \geq 3. \end{aligned}$$

$$(e) \quad y = \frac{1}{x^2 - a^2} = \frac{1}{(x-a)(x+a)}$$

$$\text{or} \quad y = \frac{1}{2a} \left\{ \frac{1}{x-a} - \frac{1}{x+a} \right\} = \frac{1}{2a} \{ (x-a)^{-1} - (x+a)^{-1} \}$$

Differentiating n times on both sides, we find

$$\therefore y_n = \frac{1}{2a} (-1)^n n! \{ (x-a)^{-n-1} - (x+a)^{-n-1} \}.$$

$$(f) \quad y = \frac{1}{(x-1)^3 (x-2)}$$

To resolve into partial fractions, let $x-1=t$,

$$\text{Then we get } y = \frac{1}{t^3 (t-1)} = \frac{1}{t^3} \cdot \frac{1}{(-1+t)}$$

$$\begin{aligned} \therefore y &= \frac{1}{t^3} \left[-1 - t - t^2 + \frac{t^3}{t-1} \right] = \frac{-1}{t^3} - \frac{1}{t^2} - \frac{1}{t} + \frac{1}{t-1} \\ &= -\frac{1}{(x-1)^3} - \frac{1}{(x-1)^2} - \frac{1}{(x-1)} + \frac{1}{x-2} \end{aligned} \quad \dots(1)$$

Now, we know

$$D^n (ax+b)^{-p} = \frac{(-1)^n (p+n-1)! \cdot a^n}{(p-1)! (ax+b)^{p+n}}$$

Hence differentiating (1) n times, we get

$$\begin{aligned} \therefore y_n &= -\frac{(-1)^n (3+n-1)!}{(3-1)! (x-1)^{n+3}} - \frac{(-1)^n (2+n-1)!}{(2-1)! (x-1)^{n+2}} - \frac{(-1)^n n!}{(x-1)^{n+1}} + \frac{(-1)^n n!}{(x-2)^{n+1}} \\ &= (-1)^n \left[\frac{-(n+2)!}{2! (x-1)^{n+3}} - \frac{(n+1)!}{(x-1)^{n+2}} - \frac{n!}{(x-1)^{n+1}} + \frac{n!}{(x-2)^{n+1}} \right] \\ &= (-1)^{n+1} n! \left[\frac{(n+2)(n+1)}{2(x-1)^{n+3}} + \frac{n+1}{(x-1)^{n+2}} + \frac{1}{(x-1)^{n+1}} - \frac{1}{(x-2)^{n+1}} \right]. \end{aligned}$$

Example 10: Prove that the value of the n^{th} differential coefficient of $\frac{x^3}{x^2-1}$ for $x=0$ is

zero if n is even and is $-n!$ if n is odd.

Solution: $y = \frac{x^3}{x^2-1} = x + \frac{x}{(x-1)(x+1)} = x + \frac{1}{(1+1)(x-1)} + \frac{-1}{(-1-1)(x+1)}$

$$= x + \frac{1}{2(x-1)} + \frac{1}{2(x+1)} = x + \frac{1}{2} [(x-1)^{-1} + (x+1)^{-1}]$$

$$\therefore y_n = \frac{(-1)^n n!}{2} \left[\frac{1}{(x-1)^{n+1}} + \frac{1}{(x+1)^{n+1}} \right] (n > 1)$$

Putting $x = 0$, we have

$$(y_n)_0 = \frac{(-1)^n n!}{2} \left[\frac{1}{(-1)^{n+1}} + \frac{1}{(1)^{n+1}} \right] = \frac{(-1)^n n!}{2} \left[\frac{1}{(-1)^{n+1}} + 1 \right] \quad \dots(1)$$

Case I: If n is even then $(n+1)$ is odd

$$\therefore (-1)^{n+1} = -1$$

$$\therefore \text{From (1)} \quad (y_n)_0 = 0$$

Case II: If n is odd, then $(n+1)$ is even

$$\therefore (-1)^{n+1} = 1$$

$$\therefore \text{From (1)} \quad (y_n)_0 = \frac{1}{2} (-1)^n (n)! [1 + 1] = -(n)!$$

Example 11: Find the n^{th} differential coefficient of the following:

(a) $\tan^{-1} (x/a)$

(b) $\tan^{-1} x$

[B.C.A. (Agra) 2006]

(c) $\tan^{-1} \left\{ \frac{1+x}{1-x} \right\}$

(d) $\tan^{-1} \left\{ \frac{2x}{1-x^2} \right\}$

(e) $\tan^{-1} \left\{ \frac{\sqrt{1+x^2} - 1}{x} \right\}$

(f) $\frac{x}{x^2 + a^2}$

Solution: (a) we have $y = \tan^{-1} (x/a)$, then

Differentiating w.r.t. x on both sides, we get

$$y_1 = \frac{1}{1 + (x^2/a^2)} \cdot \frac{1}{a} = \frac{a}{x^2 + a^2}$$

$$\begin{aligned} y_1 &= \frac{a}{(x + ai)(x - ai)} = \frac{a}{(-ia - ia)(x + ai)} + \frac{a}{(ai + ai)(x - ai)} \\ &= \frac{1}{2i} \left[\frac{1}{(x - ai)} - \frac{1}{(x + ai)} \right] = \frac{1}{2i} [(x - ai)^{-1} - (x + ai)^{-1}] \end{aligned}$$

Now differentiating both sides $(n-1)$ times, we have

$$y_n = \frac{(-1)^{n-1} (n-1)!}{2i} [(x - ai)^{-n} - (x + ai)^{-n}] \quad \dots(1)$$

Put $x = r \cos \phi$ and $a = r \sin \phi$

$$(x - ai)^{-n} = r^{-n} (\cos \phi - i \sin \phi)^{-n} = r^{-n} (\cos n\phi + i \sin n\phi)$$

Similarly, $(x + ai)^{-n} = r^{-n} (\cos n\phi - i \sin n\phi)$

$$\therefore (x - ai)^{-n} - (x + ai)^{-n} = r^{-n} (2i \sin n\phi)$$

Substituting values in equation (1), we get

$$\begin{aligned} y_n &= (-1)^{n-1} (n-1)! r^{-n} \sin n\phi \\ &= (-1)^{n-1} (n-1)! (a/\sin \phi)^{-n} \sin n\phi \quad [\because r = a/\sin \phi] \\ &= (-1)^{n-1} (n-1)! a^{-n} \sin^n \phi \sin n\phi, \quad \text{where } \phi = \tan^{-1} (a/x). \end{aligned}$$

(b) $y = \tan^{-1} x$

Proceeding as (a) above or put $a=1$ in above result, we have

$$y_n = (-1)^{n-1} (n-1)! \sin^n \phi \sin n\phi$$

where

$$\phi = \tan^{-1} (1/x).$$

(c) $y = \tan^{-1} \left\{ \frac{1+x}{1-x} \right\} = \tan^{-1} \left\{ \frac{1+x}{1-1/x} \right\} = \tan^{-1} 1 + \tan^{-1} x$

Differentiating w.r.t. x on both sides, we get

$$\therefore y_1 = \frac{1}{1+x^2}$$

Now proceeding as (a) above when $a=1$, we get

$$y_n = (-1)^{n-1} \sin^n \phi \sin n\phi, \quad \text{where } \phi = \tan^{-1} (1/x).$$

(d) $y = \tan^{-1} \left\{ \frac{2x}{1-x^2} \right\} = 2 \tan^{-1} x$, therefore from (a)

Differentiating n times on both sides, we find

$$y_n = 2 (-1)^{n-1} (n-1)! \sin^n \phi \sin n\phi, \quad \text{where } \phi = \tan^{-1} (1/x).$$

(e) $y = \tan^{-1} \left\{ \frac{\sqrt{1+x^2} - 1}{x} \right\}$

Let $x = \tan \theta$, then $1+x^2 = 1+\tan^2 \theta = \sec^2 \theta$

$$\begin{aligned} \therefore y &= \tan^{-1} \left\{ \frac{\sec \theta - 1}{\tan \theta} \right\} = \tan^{-1} \left\{ \frac{1 - \cos \theta}{\sin \theta} \right\} \\ &= \tan^{-1} \left\{ \frac{2 \sin^2 \frac{1}{2} \theta}{2 \sin \frac{1}{2} \theta \cos \frac{1}{2} \theta} \right\} = \tan^{-1} \left(\tan \frac{1}{2} \theta \right) = \frac{1}{2} \theta = \frac{1}{2} \tan^{-1} x \end{aligned}$$

therefore from (a) when $a=1$, we have

$$y = \frac{1}{2} (-1)^{n-1} (n-1)! \sin^n \phi \sin n\phi, \quad \text{where } \phi = \tan^{-1} (1/x).$$

(f) $y = \frac{x}{x^2 + a^2} = \frac{x}{(x+ai)(x-ai)} = \frac{1}{2} \left[\frac{1}{(x+ai)} + \frac{1}{x-ai} \right]$

Differentiating n times w.r.t. x , we get

$$y_n = \frac{1}{2} (-1)^n n! [(x+ai)^{-n-1} + (x-ai)^{-n-1}]$$

Put $x = r \cos \phi$, $a = r \sin \phi$

$$\begin{aligned} y_n &= \frac{1}{2} (-1)^n n! r^{-n-1} [(\cos \phi + i \sin \phi)^{-n-1} + (\cos \phi - i \sin \phi)^{-n-1}] \\ &= \frac{1}{2} (-1)^n n! r^{-n-1} [\cos(n+1)\phi - i \sin(n+1)\phi + \cos(n+1)\phi + i \sin(n+1)\phi] \\ &= \frac{1}{2} (-1)^n n! r^{-n-1} [2 \cos(n+1)\phi] \\ &= (-1)^n n! (a/\sin \phi)^{-n-1} \cos(n+1)\phi \quad [\therefore r = a/\sin \phi] \end{aligned}$$

$$= (-1)^n n! a^{-n-1} \sin^{n+1} \phi \cos (n+1) \phi, \text{ where } \phi = \tan^{-1} (a/x).$$

Example 12: If $u = \sin nx + \cos nx$, then prove that

$$u_r = n^r \{1 + (-1)^r \sin 2nx\}^{1/2},$$

where u_r is the r^{th} differential coefficient of u w.r.t. x .

[B.C.A. (Bhopal) 2004, 2006, 2009, 2010]

Solution: We have $u_r = \sin nx + \cos nx$ differentiating n times on both sides, we find

$$\begin{aligned} u_r &= n^r \left[\sin \left(nx + \frac{r\pi}{2} \right) + \cos \left(nx + \frac{r\pi}{2} \right) \right] \\ &= n^r \left\{ \sin^2 \left(nx + \frac{r\pi}{2} \right) + \cos^2 \left(nx + \frac{r\pi}{2} \right) + 2 \sin \left(nx + \frac{r\pi}{2} \right) \cos \left(nx + \frac{r\pi}{2} \right) \right\}^{1/2} \\ &= n^r [1 + \sin (2nx + r\pi)]^{1/2} \end{aligned}$$

[Squaring and then taking square root]

Now if r is even then $\sin (2nx + r\pi) = \sin 2nx$ and if r is odd then $\sin (2nx + r\pi) = -\sin 2nx$.

Therefore for all values (even or odd) of r , we get

$$\sin (2nx + r\pi) = (-1)^r \sin 2nx$$

$$\text{Hence } u_r = n^r \{1 + (-1)^r \sin 2nx\}^{1/2}.$$



Comprehensive Exercise 12.1

(On Successive Differentiation)

1. If $x = a(t - \sin t)$, $y = a(1 + \cos t)$ prove that $\frac{d^2 y}{dx^2} = \frac{1}{4a} \operatorname{cosec}^4 (t/2)$.

[B.C.A. (Kanpur) 2003, 2005]

2. If $y = \frac{a + bx}{c + dx}$, then show that $y_n = \frac{(-1)^n n! d^{n-1} (ad - bc)}{(c + dx)^{n+1}}$.

[B.C.A (Lucknow) 2008]

Find y_n for the following:

3. If $y = \left(\frac{1}{x} \right)^x$ then prove that $y_2(1) = 0$

[B.C.A. (Agra) 2005]

4. If $y^3 - 3ax^2 + x^3 = 0$, then prove that

$$\frac{d^2 y}{dx^2} + \frac{2a^2 x^2}{y^5} = 0.$$

[B.C.A. (Agra) 2008]

5. $\sin^2 x$.

6. $\sin^3 x$.

[B.C.A. (Kurukshetra) 2010]

Find y_n for the following:

7. $y = e^x \sin^2 x.$ 8. $y = e^{ax} \cos^3 bx.$ [B.C.A. (Agra) 2006]

9. $y = e^{ax} \sin x \cos x$ 10. $y = e^{ax} \sin^3 bx.$

11. $y = e^x \sin^2 x \sin 2x.$ 12. $y = \sin x \sin 2x \cos 3x.$

13. $y = (a^2 - x^2)^{-1}.$

14. (a) $y = \sin^{-1} \frac{2x}{1+x^2}$ (b) $y = \cos^{-1} \frac{2x}{1+x^2}$ [B.C.A. (Agra) 2000]

(c) $y = \tan^{-1} \left\{ \frac{\sqrt{1-x^2} - 1}{x} \right\}.$ [B.C.A. (Agra) 2001]



Answers

5.	$-2^{n-1} \cos (2x + n\pi / 2)$
6.	$\frac{1}{4} [3 \sin (x + n\pi / 2) - 3^n \cos (3x + n\pi / 2)]$
7.	$\frac{1}{2} [1 - 5^{n-1} \cos (2x + n \tan^{-1} 2)]$
8.	$\frac{1}{4} (a^2 + 9b^2)^{a/2} e^{ax} \cos \left[3bx + \tan^{-1} \left(\frac{3b}{a} \right) + \frac{3}{4} (a^2 + b^2)^{n/2} e^{ax} \cos \left\{ bx + n \tan^{-1} \left(\frac{b}{a} \right) \right\} \right]$
9.	$[(a^2 + 4)^{n/2}] e^{ax} \sin (2x + n \tan^{-1} 2 / a)$
10.	$(3/4) (a^2 + b^2)^{n/2} e^{ax} \sin \{bx + n \tan^{-1} (b/a)\} - \frac{1}{4} (a^2 + 9b^2)^{n/2} e^{ax} \sin [3bx + n \tan^{-1} (3b/a)]$
11.	$(e^x / 4) [2 \cdot 5^{n/2} \sin (2x + n \tan^{-1} 2) - (17)^{n/2} \sin (4x + n \tan^{-1} 4)]$
12.	$\frac{1}{4} [4^n \cos (4x + n\pi / 2) - 6^n \cos (6x + n\pi / 2) + 2^n \cos (2x + n\pi / 2)]$
13.	$\frac{1}{2a} (-1)^n n! \{ (x+a)^{-n-1} - (x-a)^{-n-1} \}$
14.	(a) $y_n = 2 (-1)^{n-1} ! \sin^n \theta \sin n\theta$, where $\theta = \cot^{-1} x$ (b) $y_n = 2 (-1)^n (n-1)! \sin^n \theta \sin n\theta$, where $\theta = \cot^{-1} x$ (c) $y_n = \frac{1}{2} (-1)^{n-1} ! \sin^n \theta \sin n\theta$, where $\theta = \cot^{-1} x$

12.3 Leibnitz's Theorem

The n^{th} differential coefficient of the product of two functions is conveniently evaluated by the use of this theorem.

Theorem 1: If u and v are two functions of x , then

[B.C.A. (Agra) 2003, 2006, 2007, 2008, 2009]

$$D^n(uv) = D^n u \cdot v + {}^nC_1 D^{n-1} u Dv + {}^nC_2 D^{n-2} u D^2 v + \dots + {}^nC_r D^{n-r} u D^r v + \dots + u D^n v.$$

Proof: We shall prove this theorem by mathematical induction. By actual differentiation, we have

$$D(uv) = Du \cdot v + u \cdot Dv$$

$$D^2(uv) = D^2 u \cdot v + 2Du \cdot Dv + u \cdot D^2 v$$

or

$$D^2(uv) = (D^2 u) v + {}^2C_1 Du \cdot Dv + {}^2C_2 u \cdot D^2 v$$

Thus the statement is true for $n=1, 2$.

Now let the theorem be true for some particular value m of n i.e.,

$$D^m(uv) = D^m u \cdot v + {}^mC_1 D^{m-1} u \cdot Dv + {}^mC_2 D^{m-2} u \cdot D^2 v + \dots + {}^mC_r D^{m-r} u \cdot D^r v + \dots + u \cdot D^m v \text{ is true.}$$

Now differentiating w.r.t. x again, we have

$$\begin{aligned} D^{m+1}(uv) &= \{D^{m+1} u \cdot v + D^m u \cdot Dv\} + {}^mC_1 \{D^m u \cdot Dv + D^{m-1} u \cdot D^2 v\} + \dots \\ &\quad + {}^mC_r \{D^{m-r+1} u \cdot D^r v + D^{m-r} u \cdot D^{r+1} v\} + \dots \\ &\quad + \{Du \cdot D^m v + u \cdot D^{m+1} v\} \end{aligned}$$

Rearranging the terms as follows:

$$\begin{aligned} D^{m+1}(uv) &= D^{m+1} u \cdot v + (1 + {}^mC_1) D^m u \cdot Dv + \dots \\ &\quad + ({}^mC_r + {}^mC_{r+1}) D^{m-r} u \cdot D^{r+1} v + \dots + u D^{m+1} v \end{aligned}$$

$$\text{But } {}^mC_r + {}^mC_{r+1} = {}^{m+1}C_{r+1} \quad [\text{From Algebra}]$$

$$\begin{aligned} \text{Hence } D^{m+1} u \cdot v &= D^{m+1} u \cdot v + {}^{m+1}C_1 D^m u \cdot Dv + \dots \\ &\quad + {}^{m+1}C_{r+1} D^{m-r} u \cdot D^{r+1} v + \dots + u \cdot D^{m+1} v \quad \dots(1) \end{aligned}$$

The result (1) shows that the theorem is true for $n=m+1$ if it is true for $n=m$. We have already proved that the theorem is true for $n=1, 2$. Hence by mathematical induction it follows that the theorem is true for all positive integral values of n .

Note 1: While applying Leibnitz's theorem if one of the two functions is such that its higher differential coefficients become zero then this function should be taken as v and the remaining function u .

Note 2: The formula of Leibnitz's theorem can be rewritten in the following form, [by taking successive integration of $D^n(u)$ and successive differentiation of v]

$$D^n(u \cdot v) = D^n u \cdot v + {}^nC_1 \left\{ \int (D^n u) dx \right\} Dv + {}^nC_2 \left[\int \left\{ \int (D^n u) dx \right\} dx \right] D^2 v + \dots + u \cdot D^n v.$$

Example 13: If $y = x^2 e^x$ then prove that

$$y_n = \frac{1}{2} n(n-1) \frac{d^2 y}{dx^2} - n(n-2) \frac{dy}{dx} + \frac{1}{2} (n-1)(n-2) y.$$

Solution:

$$y = x^2 e^x \quad \dots(1)$$

$$\therefore \frac{dy}{dx} = x^2 e^x + 2xe^x = y + 2xe^x \quad \dots(2)$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{dy}{dx} + 2[xe^x + e^x] = \frac{dy}{dx} + \left(\frac{dy}{dx} - y \right) + 2e^x \quad [\text{From (1)}] \\ &= 2 \frac{dy}{dx} - y + 2e^x \quad \dots(3) \end{aligned}$$

Differentiating n times on both sides, we find

$$\begin{aligned} \text{Now } D^n (x^2 e^x) &= y_n = D^n (e^x) \cdot x^2 + {}^n C_1 D^{n-1} (e^x) \cdot 2x + {}^n C_2 D^{n-2} (e^x) \cdot 2 \\ &= e^x \cdot x^2 + 2nx e^x + n(n-1) e^x \\ &= y + n \left(\frac{dy}{dx} - y \right) + n(n-1) \times \frac{1}{2} \left(\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y \right) \\ &\quad [\text{From (1), (2) and (3)}] \\ &= \frac{1}{2} n(n-1) \frac{d^2 y}{dx^2} - n(n-2) \frac{dy}{dx} + \frac{1}{2} (n-1)(n-2) y. \end{aligned}$$

Example 14: Find the n^{th} differential coefficients of the following:

- (a) $x^3 \cos x$ [B.C.A. (Agra) 2001] (b) $x^2 \tan^{-1} x$
 (c) $e^x \log x$ (d) $x^3 \log x$.

Solution: (a) Clearly the fourth and higher differential coefficients of x^3 are all zero, therefore, for the sake of convenience we shall take x^3 as v and $\cos x$ as u . Applying Leibnitz's theorem, we get,

$$\begin{aligned} D^n [(\cos x) \cdot x^3] &= D^n (\cos x) \cdot x^3 + {}^n C_1 D^{n-1} (\cos x) D(x^3) + {}^n C_2 D^{n-2} (\cos x) D^2 (x^3) \\ &\quad + {}^n C_3 D^{n-3} (\cos x) D^3 (x^3) \\ &= x^3 \cos \left(x + \frac{1}{2} n\pi \right) + n [\cos \{x + \frac{1}{2} (n-1) \pi\}] 3x^2 + \frac{n(n-1)}{1 \cdot 2} [\cos \{x + \frac{1}{2} (n-2) \pi\}] 3 \cdot 2 \cdot x \\ &\quad + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} [\cos \{x + \frac{1}{2} (n-3) \pi\}] \cdot 3 \cdot 2 \cdot 1 \\ &= x^3 \cos \left(x + \frac{1}{2} n\pi \right) + 3nx^2 \sin \left(x + \frac{1}{2} n\pi \right) - 3n(n-1)x \cos \left(x + \frac{1}{2} n\pi \right) \\ &\quad - n(n-1)(n-2) \sin \left(x + \frac{1}{2} n\pi \right) \\ &= x \{x^2 - 3n(n-1)\} \cos \left(x + \frac{1}{2} n\pi \right) + n \{3x^2 - (n-1)(n-2)\} \sin \left(x + \frac{1}{2} n\pi \right). \end{aligned}$$

(b) Taking $u = \tan^{-1} x$, $v = x^2$

$$D^n [(\tan^{-1} x) x^2] = D^n (\tan^{-1} x) \cdot x^2 + {}^nC_1 D^{n-1} (\tan^{-1} x) \cdot D(x^2) + {}^nC_2 D^{n-2} (\tan^{-1} x) D^2(x^2) \quad [\because D^3(x^2) = 0 \text{ etc.}]$$

but $D^n (\tan^{-1} x) = (-1)^{n-1} (n-1)! \sin^n \phi \sin n\phi$ where $\phi = \tan^{-1} (1/x)$.

$$\begin{aligned} \therefore D^n [(\tan^{-1} x) x^2] &= \{(-1)^{n-1} (n-1)! \sin^n \phi \sin n\phi\} \cdot x^2 \\ &\quad + \{n \cdot (-1)^{n-2} (n-2)! \sin^{n-1} \phi \sin (n-1)\phi\} 2x \\ &\quad + \left[\frac{n(n-1)}{1 \cdot 2} (-1)^{n-3} (n-3)! \sin^{n-2} \phi \sin (n-2)\phi \right] \cdot 2 \\ &= (-1)^{n-1} (n-3)! [(n-1)(n-2)x^2 \sin^n \phi \sin n\phi \\ &\quad - 2nx(n-2) \sin^{n-1} \phi \sin (n-1)\phi + n(n-1) \sin^{n-2} \phi \sin (n-2)\phi]. \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad D^n (e^x \log x) &= D^n (e^x) \cdot \log x + {}^nC_1 D^{n-1} (e^x) \cdot D(\log x) \\ &\quad + {}^nC_2 D^{n-2} (e^x) \cdot D^2(\log x) + \dots + e^x D^n (\log x) \\ &= e^x \log x + {}^nC_1 e^x \cdot \frac{1}{x} + {}^nC_2 e^x \left(-\frac{1}{x^2} \right) + \dots + \frac{e^x \{(-1)^{n-1} (n-1)!\}}{x^n} \\ &= e^x [\log x + {}^nC_1 x^{-1} - {}^nC_2 x^{-2} + \dots + (-1)^{n-1} (n-1)! \cdot x^{-n}]. \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad D^n [\log x \cdot x^3] &= D^n (\log x) \cdot x^3 + {}^nC_1 D^{n-1} (\log x) D(x^3) + {}^nC_2 D^{n-2} (\log x) D^2(x^3) \\ &\quad + {}^nC_3 D^{n-3} (\log x) D^3(x^3) \\ &= \frac{(-1)^{n-1} (n-1)!}{x^n} x^3 + n \frac{(-1)^{n-2} (n-2)!}{x^{n-1}} \cdot 3x^2 \\ &\quad + \frac{n(n-1)}{1 \cdot 2} \cdot \frac{(-1)^{n-3} (n-3)!}{x^{n-2}} \cdot 6x + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} \cdot \frac{(-1)^{n-4} (n-4)!}{x^{n-3}} \cdot 6 \\ &= \frac{(-1)^{n-1} (n-4)!}{x^{n-3}} [(n-1)(n-2)(n-3) - 3n(n-2)(n-3) \\ &\quad + 3n(n-1)(n-3) - n(n-1)(n-2)] \\ &= (-1)^{n-1} (n-4)! x^{3-n} [(n-1)(n-2)(n-3-n) \\ &\quad + 3n(n-3)(n-1-n+2)] \\ &= (-1)^{n-1} (n-4)! x^{3-n} [3(n^2 - 3n + 2) + 3(n^2 - 3n)] \\ &= 6(-1)^n (n-4)! x^{3-n}. \end{aligned}$$

Example 15: Prove that

$$\frac{d^n}{dx^n} \left(\frac{\log x}{x} \right) = \frac{(-1)^n n!}{x^{n+1}} \left[\log x - 1 - \frac{1}{2} - \frac{1}{3} - \dots - \frac{1}{n} \right].$$

Solution: $y = \log x \cdot x^{-1}$.

Differentiating n times by Leibnitz's theorem taking $\log x$ at first function u and $\frac{1}{x}$ as v , we get

$$\begin{aligned} y_n &= \frac{(-1)^{n-1} (n-1)!}{x^n} \cdot \frac{1}{x} + n \frac{(-1)^{n-2} (n-2)!}{x^{n-1}} \left(-\frac{1}{x^2} \right) + \frac{n(n-1) (-1)^{n-3} (n-3)!}{1 \cdot 2 x^{n-2}} \cdot \frac{(-1) (-2)}{x^3} \\ &\quad + \frac{n(n-1) (n-2)}{1 \cdot 2 \cdot 3} \cdot \frac{(-1)^{n-4} (n-4)!}{x^{n-3}} \cdot \frac{(-1) (-2) (-3)}{x^4} + \dots + (\log x) \cdot \frac{(-1)^n n!}{x^{n+1}} \\ &= \frac{(-1)^n n!}{x^{n+1}} \left[-\frac{1}{n} - \frac{1}{n-1} - \frac{1}{n-2} - \frac{1}{n-3} - \dots + \log x \right] \\ &= \frac{(-1)^n n!}{x^{n+1}} \left[\log x - 1 - \frac{1}{2} - \frac{1}{3} - \dots - \frac{1}{n} \right]. \end{aligned}$$

Example 16: If $y = x^{n-1} \log x$, then prove that $y_n = \frac{(n-1)!}{x}$.

Solution: Let $y = x^{n-1} \log x$

Differentiating n times on both sides we find

$$\begin{aligned} y_n &= D^n (x^{n-1} \log x) = D \{D^{n-1} (x^{n-1} \log x)\} \\ &= D \left\{ (n-1)! \cdot \log x + {}^nC_1 (n-1)! \cdot \frac{x}{1} \cdot \frac{1}{x} + {}^nC_2 (n-2)! \left(\frac{x^2}{1 \cdot 2} \right) \cdot \left(-\frac{1}{x^2} \right) + \dots \right\} \\ &= D \{ (n-1)! \log x + \text{terms not containing } x \} \\ &= (n-1)! \cdot \frac{1}{x} + 0 = \frac{(n-1)!}{x}. \end{aligned}$$

Example 17: If $I_n = \frac{d^n}{dx^n} (x^n \log x)$, prove that $I_n = nI_{n-1} + (n-1)!$

Hence show that $I_n = n! (\log x + 1 + \frac{1}{2} + \frac{1}{3} + \dots + 1/n)$.

Solution: $I_n = D^n (x^n \log x) = D^{n-1} [D (x^n \log x)]$

$$\begin{aligned} &= D^{n-1} [x^n (1/x) + nx^{n-1} \log x] \\ &= n \cdot D^{n-1} (x^{n-1} \log x) + D^{n-1} (x^{n-1}) = nI_{n-1} + (n-1)! \end{aligned} \quad \dots(1)$$

Now replacing n by $n-1$ in the above relation (1), we have

$$\begin{aligned} I_{n-1} &= (n-1) I_{n-2} + (n-2)! \\ \therefore I_n &= n [(n-1) I_{n-2} + (n-2)!] + (n-1)! \\ &= n(n-1) I_{n-2} + n(n-2)! + (n-1)! \\ &= \frac{n!}{(n-2)!} I_{n-2} + \frac{n!}{(n-1)} + \frac{n!}{n} \end{aligned}$$

Thus applying the reduction formula (1) $(n-2)$ times again, we get

$$I_n = \frac{n!}{0!} I_0 + \frac{n!}{1!} + \frac{n!}{2!} + \frac{n!}{3!} + \dots + \frac{n!}{n-1} + \frac{n!}{n}$$

$$= n! [\log x + 1 + \frac{1}{2} + \frac{1}{3} + \dots + \{1/(n-1)\} + (1/n)]$$

$$[\therefore I_0 = D^0 (x^0 \log x) = \log x]$$

Example 18: If $y = \sin (m \sin^{-1} x)$, then show that

$$(1-x^2) y_2 - xy_1 + m^2 y = 0$$

[B.C.A. (Bilaspur) 2007]

and deduce from it that

$$(1-x^2) y_{n+2} - (2n+1) xy_{n+1} - (n^2 - m^2) y_n = 0.$$

Solution: We have $y = \sin (m \sin^{-1} x)$

...(1)

Differentiating (1) w.r. to x , we get

$$y_1 = \cos (m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\text{or } y_1^2 (1-x^2) = m^2 \{1 - \sin^2 (m \sin^{-1} x)\}$$

$$\text{or } (1-x^2) y_1^2 = m^2 (1-y^2) \quad \dots(2)$$

Again, differentiating w.r. to x , we get

$$(1-x^2) 2 y_1 y_2 - 2xy_1^2 = -2m^2 y y_1$$

Dividing by $2 y_1$, we get

$$(1-x^2) y_2 - xy_1 + m^2 y = 0 \quad \dots(3)$$

Differentiating (3) n times by Leibnitz's theorem, we get

$$D^n \{(1-x^2) y_2\} + D^n \{-xy_1\} + D^n \{m^2 y\} = D^n(0) = 0$$

Now, by Leibnitz's theorem, we get

$$D^n \{(1-x^2) y_n\} = y_{n+1} \cdot (1-x^2) + ny_{n+1} \cdot (-2x) + \frac{n(n-1)}{2!} y_n \cdot (-2)$$

$$D^n \{-xy_1\} = y_{n+1} \cdot (-x) + ny_n \cdot (-1)$$

$$D^n \{m^2 y\} = m^2 y_n$$

$$\text{Adding, } (1-x^2) y_{n+2} - (2n+1) xy_{n+1} - (n^2 - m^2) y_n = 0.$$

Example 19: If $y = a \cos (\log x) + b \sin (\log x)$, then prove that

$$x^2 y_2 + xy_1 + x = 0$$

and

$$x^2 y_{n+2} + (2n+1) xy_{n+1} + (n^2 + 1) y_n = 0.$$

Solution: $y = a \cos (\log x) + b \sin (\log x)$

...(1)

Differentiating w.r.t. (x) on both sides, we get,

$$\therefore y_1 = -\frac{a \sin (\log x)}{x} + \frac{b \cos (\log x)}{x}$$

or $xy_1 = -a \sin(\log x) + b \cos(\log x)$

Differentiating again w.r.t. x ,

$$xy_2 + y_1 = -\frac{a \cos(\log x)}{x} - \frac{b \sin(\log x)}{x}$$

or $x^2 y_2 + xy_1 = -[a \cos(\log x) + b \sin(\log x)] = -y$ [use (1)]

or $x^2 y_2 + xy_1 + y = 0$

Now differentiating each term n times by Leibnitz's theorem, we get

$$[y_{n+2} \cdot x^2 + {}^nC_1 \cdot y_{n+1} \cdot 2x + {}^nC_2 \cdot y_n \cdot 2] + [y_{n+1} \cdot x + {}^nC_1 \cdot y_n \cdot 1] + y_n = 0$$

or $x^2 y_{n+2} + (2n+1) xy_{n+1} + (n^2 + 1) y_n = 0.$

Example 20: If $x + y = 1$, then prove that

$$\frac{d^n}{dx^n} (x^n y^n) = n! \{y^n - ({}^nC_1)^2 y^{n-1} x + ({}^nC_2)^2 y^{n-2} x^2 + \dots + (-1)^n x^n\}.$$

[B.C.A. (Rohilkhand) 2007]

Solution: We have $x + y = 1 \Rightarrow y = 1 - x$

Differentiating w.r.t. (x)

$$\begin{aligned} \therefore \frac{d^n}{dx^n} (x^n y^n) &= D^n \{x^n (1-x)^n\} \\ &= (D^n x^n) (1-x)^n + {}^nC_1 (D^{n-1} x^n) \cdot D(1-x)^n \\ &\quad + {}^nC_2 (D^{n-2} x^n) \cdot D^2 (1-x)^n + \dots + x^n D^n (1-x)^n \\ &= n! (1-x) + {}^nC_1 \frac{n!}{2!} x \cdot (-1)^n (1-x)^{n-1} \\ &\quad + {}^nC_1 \frac{n!}{2!} x^2 (-1)^2 n(n-1) (1-x)^{n-2} + \dots + x^n (-1)^n n! \\ &= n! \left[(1-x)^n - n \cdot {}^nC_1 (1-x)^{n-1} x + \frac{(n-1)}{2!} C_2 (1-x)^{n-2} \cdot x^2 + \dots + (-1)^n x^n \right] \\ &= n! [y^n - ({}^nC_1)^2 y^{n-1} x + ({}^nC_2)^2 y^{n-2} x^2 + \dots + (-1)^n x^n] \end{aligned}$$

Since $n = {}^nC_1 \cdot \frac{n(n-1)}{2!} = nC_2$ etc.

Example 21: If $y^{1/m} + y^{-1/m} = 2x$, then prove that

$$(x^2 - 1) y_{n+2} + (2n+1) x \cdot y_{n+1} + (n^2 - m^2) y_n = 0.$$

[B.C.A. (Agra) 2002]

Solution: $y^{1/m} + y^{-1/m} = 2x$

$$y^{1/m} + 1/(y^{1/m}) = 2x \quad \text{or} \quad (y^{1/m})^2 - 2xy^{1/m} + 1 = 0$$

$\therefore$ This is a quadratic equation in $y^{1/m}$.

$$\therefore y^{1/m} = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1}$$

$$\text{or} \quad y = [x \pm \sqrt{x^2 - 1}]^m \quad \dots(1)$$

Differentiating w.r.t. (x) on both sides, we get,

$$\therefore \quad y_1 = m [x \pm \sqrt{x^2 - 1}]^{m-1} \left\{ 1 \pm \frac{x}{\sqrt{x^2 - 1}} \right\}$$

$$\sqrt{x^2 - 1} \cdot y_1 = m [x \pm \sqrt{x^2 - 1}]^m = \pm my \quad [\text{from (1)}]$$

Squaring both sides, we have

$$y_1^2 (x^2 - 1) = m^2 y^2$$

Differentiating again, we get,

$$2 y_1 y_2 (x^2 - 1) + y_1^2 \cdot 2x = 2m^2 y y_1$$

Dividing by $2 y_1$, we get $y_2 (x^2 - 1) + x y_1 - m^2 y = 0$

Differentiating each term n times by Leibnitz's theorem, we get,

$$[(x^2 - 1) y_{n+2} + {}^nC_1 y_{n+1} \cdot (2x) + {}^nC_2 y_n \cdot 2] + [x y_{n+1} + {}^nC_1 y_n \cdot 1] - m^2 y_n = 0$$

$$\text{or} \quad (x^2 - 1) y_{n+2} + (2n + 1) x y_{n+1} + (n^2 - m^2) y_n = 0.$$

Example 22: If $\cos^{-1} \left(\frac{y}{b} \right) = \log \left(\frac{x}{n} \right)^n$, then prove that $x^2 y_2 + x y_1 + n^2 y = 0$

$$\text{and} \quad x^2 y_{n+2} + (2n + 1) x y_{n+1} + 2n^2 y_n = 0. \quad [\text{B.C.A. (Agra) 2002, 2004, 2008}]$$

Solution: We have $y = b \cos \log (x/n)^n$

Differentiating w.r.t. (x) times on both sides, we get, ...(1)

$$y_1 = -b \sin \log (x/n)^n \cdot n \left(\frac{1}{x/n} \right) \cdot 1/n$$

$$\text{or} \quad x y_1 = -bn \sin \log (x/n)^n.$$

Differentiating again, we get

$$x y_2 + 1 \cdot y_1 = -bn \cos \log (x/n)^n \cdot n \left(\frac{1}{x/n} \right) n$$

$$\text{or} \quad x^2 y_2 + x y_1 = -n^2 b \cos \log (x/n)^n = -n^2 y$$

$$\text{or} \quad x^2 y_2 + x y_1 + n^2 y = 0$$

Differentiating each term n times by Leibnitz's theorem, we get

$$y_{n+2} x^2 + {}^nC_1 \cdot y_{n+1} \cdot 2x + {}^nC_2 y_n \cdot 2 + y_{n+1} \cdot x + {}^nC_1 y_n + n^2 y_n = 0$$

$$\text{or} \quad x^2 y_{n+2} + (2n + 1) x y_{n+1} + 2n^2 y_n = 0.$$

Example 23: If $y = e^{\tan^{-1} x}$, then prove that [B.C.A. (Delhi) 2005, 2009]

$$(1 + x^2) y_2 + (2x - 1) y_1 = 0$$

$$\text{and} \quad (1 + x^2) y_{n+2} + [2(n + 1)x - 1] y_{n+1} + n(n + 1) y_n = 0.$$

Solution: We have $y = e^{\tan^{-1} x}$

Differentiating w.r.t. (x) on both sides, we get,

$$\therefore y_1 = e^{\tan^{-1} x} \cdot \frac{1}{1+x^2} \quad \text{or} \quad y_1 = \frac{y}{1+x^2} \quad \text{or} \quad (1+x^2) y_1 = y$$

Differentiating again each term w.r.t. x , we get

$$(1+x^2) y_2 + 2xy_1 = y_1$$

$$\text{or} \quad (1+x^2) y_2 + (2x-1) y_1 = 0$$

Now differentiating each term n times by Leibnitz's theorem, we have

$$y_{n+2} (1+x^2) + {}^nC_1 y_{n+1} \cdot 2x + {}^nC_2 y_n \cdot 2 + y_{n+1} (2x-1) + {}^nC_1 y_n \cdot 2 = 0$$

$$\text{or} \quad (1+x^2) y_{n+2} + [2(n+1)x-1] y_{n+1} + n(n+1) y_n = 0.$$



Comprehensive Exercise 12.2

(On Leibnitz Theorem)

- Find the fourth differential-coefficients of $x^4 e^{ax}$, $x^3 \log x$, $x^2 \sin 3x$.
[B.C.A. (Bundelkhand) 2007]
- Find n^{th} differential-coefficient of the following
 - $x^2 e^{ax}$ [B.C.A. (Bhopal) 2011]
 - $x^2 (ax+b)^m$
 - $x^2 \sin x$ [B.C.A. (Kanpur) 2005]
 - $x^2 \log_e x$ [B.C.A. (Rohtak) 2011]
 - $x^3 \cos x$
 - $\sin^2 x \sin 2x$ [B.C.A. (Kurukshetra) 2009]
 - $\sin^4 x$ [B.C.A. (Agra) 2006]
- Differentiate n times the equation $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + a^2 y = 0$.
[B.C.A. (Avadh) 2004]
- If $I_n = \frac{d^n}{dx^n} \{x^n \log_e x\}$, then prove that $\left[\frac{1}{n} + \frac{1}{n-1} + \frac{1}{n-2} + \dots + \log x \right] \cdot (n!)$.
- If $y = \sin^{-1} x$, then prove that $(1-x^2) y_2 - xy_1 = 0$.
Hence prove that: $(1-x^2) y_{n+2} - (2n+1) xy_{n+1} - n^2 y_n = 0$.
[B.C.A. (Agra) 2005]

6. If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, then prove that $(1-x^2) y_{n+1} - (2n+1) xy_n - n^2 y_{n-1} = 0$.
[B.C.A. (Delhi) 2009]
7. If $y = (\sin^{-1} x)^2$, then prove that $(1-x^2) y_{n+2} - (2n+1) xy_{n+1} - n^2 y_n = 0$.
[B.C.A. (Agra) 2000]
8. If $y = [\log\{x + \sqrt{1+x^2}\}]^2$, then prove that $(1+x^2) y_{n+2} + (2n+1) xy_{n+1} + n^2 y_n = 0$.
[B.C.A. (Rohilkhand) 2003]
9. If $\cos^{-1}(y/b) = \log(x/m)^m$, then prove that $x^2 y_{n+2} + (2n+1) xy_{n+1} + (n^2 + m^2) y_n = 0$.
[B.C.A. (Delhi) 2011]
10. If $y = x(a^2 + x^2)^{-1}$, then prove that

$$y_n = (-1)^n n! a^{-n-1} \sin^{n+1} \phi \cos(n+1)\phi,$$
where $\phi = \tan^{-1}(a/x)$.
11. Prove that n^{th} differential-coefficient of $x^n(1-x)^n$ is equal to:

$$n!(1-x)^n \left(1 - \frac{n^2}{1^2} \cdot \frac{x}{1-x} + \frac{n^2(n-1)^2}{1^2 \cdot 2^2} \cdot \frac{x^2}{(1-x)^2} - \dots + (-1)^n \cdot \frac{x^n}{(1-x)^n} \right).$$
12. State and prove that Leibnitz's theorem.
[B.C.A. (Agra) 2003, 2006, 2007, 2008, 2009]
13. $y = e^{a \sin^{-1} x}$ or $x = \sin \log(y/a)$, show that

$$(1-x^2) y_{n+2} - (2n+1) xy_{n+1} - (x^2 + a^2) y_n = 0.$$
[B.C.A. (Agra) 2003]
14. $y = (x + \sqrt{1+x^2})^m$, prove that $(1+x^2) y_{n+2} + (2n+1) xy_{n+1} + (n^2 - m^2) y_n = 0$.
15. $y = (x^2 - 1)^n$, prove that $(x^2 - 1) y_{n+2} + 2xy_{n+1} - n(n+1) y_n = 0$, hence if

$$P_n = \frac{d^n}{dx^n} (x^2 - 1)^n$$
, show that $\frac{d}{dx} \{(1-x^2) \frac{dP_n}{dx}\} + n(n+1) P_n = 0$.



Answers

1.	$e^{ax} (a^4 x^4 + 12a^3 x^3 + 54a^2 x^2 + 72ax + 18); 6/x;$ $3^3 (3x^2 - 4) \sin 3x - 6^3 x \cos 3x - 64 e^{2x} \sin 2x,$ $(a^2 + b^2)^2 x e^{ax} \sin \{bx + 4 \tan^{-1} (b/a)\} + 4(a^2 + b^2)^{3/2} a^{ax} \sin \{bx + 3 \tan^{-1} (b/a)\}$
2.	(i) $a^{n-2} e^{ax} \{a^2 x^2 + 2nax + n(n-1)\}$ (ii) $m(m-1)(m-2) \dots (m-n+2) a^{n-2} (ax+b)^{m-n} \{(m-n+2) \cdot (m-n+1) a^2 x^2$ $+ 2n(m-n+2) a(ax+b)x + n(n-1)(ax+b)^2\}$ (iii) $x^2 \sin [x + n\pi/2] + 2nx \sin [x + (n-1)\pi/2] + n(n+1) \sin [x + (n-2)\pi/2]$ (iv) $2(-1)^{n-1} (n-3)!/x^{n-2}$ (v) $x \{x^2 - 3n(n-1)\} \cos (x + n\pi/2) + n \{3x^2 - (n-1)(n-2)\} \sin (x + n\pi/2)$ (vi) $2^{n-1} \sin \left(2x + \frac{n\pi}{2}\right) - 2^{n-2} \sin \left(4x + \frac{n\pi}{2}\right)$ (vii) $-2^{n-1} \cos \left\{2x + \frac{n\pi}{2}\right\} + 2^{2n-3} \cos \left(4x + \frac{n\pi}{2}\right)$
3.	$(1-x^2)y_{n+2} + (2n+1)xy_{n+1} - (n^2 - a^2)y_n = 0$

12.4 n^{th} Differential Coefficient for $(x=0)$

This method will be clear from the following examples.

Example 24: If $y = [\log \{x + \sqrt{(1+x^2)}\}]^2$, then prove that

$$(y_{n+2})_0 = -n^2 (y_n)_0$$

hence find $(y_n)_0$.

[B.C.A. (Lucknow) 2003, 2005]

Solution: $y = [\log \{x + \sqrt{(1+x^2)}\}]^2 \quad \dots (1)$

$$\therefore y_1 = 2 \log [x + \sqrt{(1+x^2)}] \times \frac{1}{[x + \sqrt{(1+x^2)}]} \left[1 + \frac{1}{2\sqrt{(1-x^2)}} \cdot 2x \right]$$

$$\text{or } y_1 = \frac{2 \log [x + \sqrt{(1+x^2)}]}{\sqrt{(1+x^2)}} \quad \dots(2)$$

Squaring both sides, we get,

$$\text{or } (1+x^2) y_1^2 = 4 [\log x + (1+x^2)]^2 = 4y \quad \text{or } (1+x^2) y_1^2 = 4y$$

Differentiating again, we get,

$$(1+x^2) 2 y_1 y_2 + y_1^2 \cdot 2x = 4 y_1$$

Dividing by $2 y_1$, we get

$$y_2 (1+x^2) + x y_1 = 2 \quad \dots(3)$$

Differentiating each term n times, we get,

$$[y_{n+2} (1+x^2) + {}^nC_1 \cdot y_{n+1} \cdot 2x + {}^nC_2 \cdot y_n \cdot 2] + [y_{n+1} \cdot x + {}^nC_1 \cdot y_n \cdot 1] = 0$$

$$\text{or } (1+x^2) y_{n+2} + (2n+1) x y_{n+1} + (n^2 - n + n) y_n = 0$$

$$\text{or } (1+x^2) y_{n+2} + (2n+1) x y_{n+1} + n^2 y_n = 0$$

Putting $x=0$, we get

$$(y_{n+2})_0 = -n^2 (y_n)_0 \quad \dots(4)$$

Again when $x=0$, then from (1), (2) and (3), we have,

$$(y_0)_0 = 0, \quad y_1 = 0$$

$$(y_2)(1+0) + 0 \cdot 0 = 2 \quad \text{or } (y_2)_0 = 2$$

$$\text{From (4), } y_3 = -1^2 \cdot y_1 = 0 \quad (\text{Taking } n=1)$$

$$y_5 = -3^2 \cdot y_3 = 0, \text{ etc.}$$

$\therefore$ **When n is odd integer** $(y_n)_0 = 0$

Again

$$y_2 = 2$$

$$y_4 = -2^2 \cdot y_2 = -2^2 \cdot 2$$

$$y_6 = -4^2 y_4 = -4^2 (-2^2) \cdot 2 = (-1)^2 4^2 \cdot 2^2 \cdot 2$$

$$y_8 = -6^2 y_6 = -6^2 \cdot (-4^2) (-2^2) \cdot 2 = (-1)^3 6^2 \cdot 4^2 \cdot 1^2 \cdot 2$$

$\therefore$ **When n is even integer, then**

$$(y_n)_0 = (-1)^{n/2-1} (n-2)^2 (n-4)^2 \dots 6^2 \cdot 4^2 \cdot 2^2 \cdot 2$$

Example 25: If $y = (\sinh^{-1} x)^2$, then prove that

$$(1+x^2) y_{n+2} + (2n+1) x y_{n+1} + n^2 y_n = 0$$

hence find $(y_n)_0$.

Solution: $y = (\sinh^{-1} x)^2 = [\log \{x + \sqrt{(1+x^2)}\}]^2$

Now see Ex. 25.

Example 26: If $y = \tan^{-1} x$, then prove that

$$(1 + x^2) y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$$

hence find $(y_n)_0$.

[B.C.A. (Kanpur) 2005]

Solution:

$$y = \tan^{-1} x \quad \dots(1)$$

$$\therefore y_1 = \frac{1}{1+x^2} \quad \dots(2)$$

$$\text{or} \quad (1+x^2)y_1 = 1.$$

Differentiating again

$$(1+x^2)y_2 + y_1 \cdot 2x = 0 \quad \text{or} \quad y_2(1+x^2) + 2xy_1 = 0 \quad \dots(3)$$

Differentiating each term n times by Leibnitz's theorem, we get,

$$[y_{n+2}(1+x^2) + {}^nC_1 \cdot y_{n+1} \cdot 2x + {}^nC_2 \cdot y_n \cdot 2] + 2[y_{n+1} \cdot x + {}^nC_1 y_n \cdot 1] = 0$$

$$\text{or} \quad (1+x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$$

Putting $x=0$, we have

$$(y_{n+2})_0 = -n(n+1)(y_n)_0 \quad \dots(4)$$

But from (1), (2) and (3), we get

$$(y)_0 = \tan^{-1} 0 = 0, \quad (y_1)_0 = \frac{1}{1+0} = 1, \quad (y_2)_0 = 0$$

Putting $n=2, 4, 6, \dots$ in equation (4), we get,

$$(y_4)_0 = (y_6)_0 = (y_8)_0 = \dots = 0$$

$$\therefore \text{When } n \text{ is even} \quad (y_n)_0 = 0$$

$$\text{Again,} \quad (y_1)_0 = 1$$

$$\therefore \text{From (4),} \quad (y_3)_0 = -1 \cdot 2 (y_1)_0 = -1 \cdot 2 \cdot 1 = (-1) \cdot 2!$$

$$(y_5)_0 = -3 \cdot 4 (y_3)_0 = -3 \cdot 4 (-1) 2! = (-1)^2 4!$$

$$\therefore \text{When } n \text{ is odd, then we have}$$

$$(y_n)_0 = (-1)^{(n-1)/2} (n-1)!$$

Example 27: If $y = e^{a \cos^{-1} x}$, prove that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0 \quad [\text{B.C.A. (Indore) 2004, 2008}]$$

Also find the n^{th} differential coefficient of $e^{a \cos^{-1} x}$ at $x=0$.

Solution:

$$y = e^{a \cos^{-1} x} \quad \dots(1)$$

$$\therefore y_1 = \frac{-a}{\sqrt{1-x^2}} e^{a \cos^{-1} x}$$

$$\text{or} \quad \sqrt{1-x^2} y_1 = -ay \quad \dots(2)$$

$$\therefore (1-x^2) y_1^2 = a^2 y^2 \quad [\text{Squaring}]$$

Differentiating again, we have

$$(1-x^2) 2 y_1 y_2 + y_1^2 (-2x) = 2a^2 y y_1 \quad \text{or} \quad y_2 (1-x^2) - x y_1 = a^2 y \quad \dots(3)$$

Differentiating each term n times, we get,

$$y_{n+2} (1-x^2) + {}^nC_1 \cdot y_{n+1} (-2x) + {}^nC_2 \cdot y_n (-2) - [y_{n+1} \cdot x + {}^nC_1 y_n \cdot 1] = a^2 y_n$$

$$(1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 + a^2) y_n = 0$$

Putting $x=0$, we get,

$$(y_{n+2})_0 = (a^2 + n^2) (y_n)_0 \quad \dots(4)$$

(1), (2), (3), we get,

$$(y_0) = e^{a \cos^{-1} 0} = e^{a\pi/2}, (y_1)_0 = -ae^{a\pi/2}, (y_2)_0 = -a^2 e^{a\pi/2}$$

$$\therefore \text{From (4)} \quad (y_4)_0 = (a^2 + 2^2) (y_2)_0 = (a^2 + 2^2) a^2 e^{a\pi/2}$$

$$\text{Similarly,} \quad (y_6)_0 = (a^2 + 4^2) (a^2 + 2^2) a^2 e^{n\pi/2}$$

$\therefore$ When n is even, then

$$(y_n)_0 = [a^2 + (n-2)^2] [a^2 + (n-4)^2] \dots (a^2 + 4^2) (a^2 + 2^2) a^2 e^{a\pi/2}$$

$$\text{Again} \quad (y_1)_0 = -ae^{a\pi/2}$$

$$\text{Now from (4)} \quad (y_3)_0 = -(a^2 + 1^2) (y_1)_0 = -(a^2 + 1^2) ae^{a\pi/2}$$

$$\text{Similarly,} \quad (y_5)_0 = -(a^2 + 3^2) (a^2 + 1^2) ae^{a\pi/2}$$

$\therefore$ When n is odd, then

$$(y_n)_0 = -[a^2 + (n-2)^2] [a^2 + (n-4)^2] \dots (a^2 + 3^2) (a^2 + 1^2) ae^{a\pi/2}.$$

Example 28: If $y = [x + \sqrt{(1+x^2)}]^m$, then find $(y_n)_0$.

[B.C.A. (Agra) 2004]

$$\text{Solution:} \quad y = [x + \sqrt{(1+x^2)}]^m \quad \dots(1)$$

$$y_1 = m [x + \sqrt{(1+x^2)}]^{m-1} \left(1 + \frac{1}{2} \cdot \frac{2x}{\sqrt{(1+x^2)}} \right)$$

$$\therefore = m [x + \sqrt{(1+x^2)}]^{m-1} \left(\frac{x + \sqrt{(1+x^2)}}{\sqrt{(1+x^2)}} \right)$$

$$\therefore \sqrt{(1+x^2)} y_1 = m y \quad \dots(2)$$

$$\text{or} \quad (1+x^2) y_1^2 = m^2 y^2$$

Differentiating again,

$$(1+x^2) 2 y_1 y_2 + y_1^2 \cdot 2x = 2m^2 y y_1$$

$$\text{or} \quad (1+x^2) y_2 + x y_1 - m^2 y = 0 \quad \dots(3)$$

Now differentiating n times, we have

$$[y_{n+2} (1+x^2) + {}^nC_1 \cdot y_{n+1} \cdot 2x + {}^nC_2 \cdot y_n \cdot 2] + [y_{n+1} \cdot x + {}^nC_1 \cdot y_n \cdot 1] - m^2 y_n = 0$$

$$\text{or} \quad (1+x^2) y_{n+2} + (2n+1) x y_{n+1} + (n^2 - m^2) y_n = 0$$

Putting $x=0$, we get

$$(y_{n+2})_0 = (m^2 - n^2) (y_n)_0 \quad \dots(4)$$

Again put $x=0$ in (1), (2) and (3), we have

$$(y)_0 = 1, (y_1)_0 = m, 1 = m, (y_2)_0 = m^2 \cdot 1 = m^2$$

$$\text{From (4),} \quad (y_4)_0 = (m^2 - 2^2) (y_2)_0 = (m^2 - 2^2) \cdot m^2$$

$$\text{Similarly,} \quad (y_6)_0 = (m^2 - 4^2) (m^2 - 2^2) \cdot m^2$$

$\therefore$ When n is even, then

$$(y_n)_0 = [m^2 - (n-2)^2] [m^2 - (n-4)^2] \dots (m^2 - 4^2) (m^2 - 2^2) m^2$$

$$\text{Also} \quad (y_1)_0 = m$$

$$\text{Now from (4),} \quad (y_3)_0 = (m^2 - 1^2) (y_1)_0 = (m^2 - 1^2) m$$

$$\text{Similarly,} \quad (y_5)_0 = (m^2 - 3^2) (m^2 - 1^2) m$$

$\therefore$ When n is odd, then

$$(y_n)_0 = [m^2 - (n-2)^2] [m^2 - (n-4)^2] \dots (m^2 - 3^2) (m^2 - 1^2) m.$$

Example 29: If $y = \sin^{-1} x$, then find $(y_n)_0$.

$$\text{Solution:} \quad y = \sin^{-1} x \quad \dots(1)$$

$$\therefore \quad y_1 = \frac{1}{\sqrt{1-x^2}} \quad \dots(2)$$

$$\Rightarrow \quad (1-x^2) y_1^2 = 1 \quad [\text{Squaring}]$$

Differentiating again, we have

$$(1-x^2) 2 y_1 y_2 + y_1^2 (-2x) = 0 \quad \text{or} \quad y_2 (1-x^2) - x y_1 = 0 \quad \dots(3)$$

$\therefore$ Differentiating n times, we get,

$$y_{n+2} (1 - x^2) + {}^nC_1 y_{n+1} (-2x) + {}^nC_2 y_n (-2) - [y_{n+1} \cdot x + {}^nC_1 \cdot y_n \cdot 1] = 0$$

Putting $x = 0$, we have

$$(y_{n+2})_0 - \left[\frac{n(n-1)}{2} \cdot 2 + n \right] (y_n)_0 = 0$$

$$(y_{n+2})_0 = n^2 (y_n)_0 \quad \dots(4)$$

Again put $x = 0$ in (1), (2) and (3), we have

$$(y)_0 = 0, (y_1)_0 = 1, (y_2)_0 = 0$$

$$\therefore (y_4)_0 = 0 = (y_6)_0 = \dots \quad [\text{from (4)}]$$

$$\therefore \text{When } n \text{ is even, then } (y_n)_0 = 0$$

$$\text{Again } (y_1)_0 = 1$$

$$\text{Again } (y_3)_0 = 1^2 (y_1)_0 = 1^2 \cdot 1 \quad [\text{from (4)}]$$

$$\text{Similarly, } (y_5)_0 = 3^2 \cdot 1^2 \cdot 1$$

$$\therefore \text{When } n \text{ is odd, then}$$

$$(y_n)_0 = (n-2)^2 (n-4)^2 \dots 3^2 \cdot 1.$$

Example 30: If $y = \sin (a \sin^{-1} x)$, then find $(y_n)_0$.

[B.C.A. (Agra) 2001]

$$\text{Solution: } y = \sin (a \sin^{-1} x). \quad \dots(1)$$

$$\therefore y_1 = \cos (a \sin^{-1} x) \frac{a}{\sqrt{(1-x^2)}}$$

$$\text{or } \frac{y_1 \sqrt{(1-x^2)}}{a} = \cos (a \sin^{-1} x) \quad \dots(2)$$

Squaring (1) and (2) and adding, we get

$$y^2 + \frac{(1-x^2) y_1^2}{a^2} = 1 \quad \text{or} \quad a^2 y^2 + (1-x^2) y_1^2 = a^2.$$

Differentiating again

$$a^2 2 y y_1 + (1-x^2) 2 y_1 y_2 + y_1^2 (-2x) = 0$$

$$\text{or } y_2 (1-x^2) - x y_1 + a^2 y = 0. \quad \dots(3)$$

Now differentiating each term n times, we have

$$y_{n+2} (1-x^2) + {}^nC_1 y_{n+1} (-2x) + {}^nC_2 y_n (-2) - [y_{n+1} x + {}^nC_1 y_n \cdot 1] + a^2 y_n = 0$$

$$\text{or } (1-x^2) y_{n+2} - (2n+1) x \cdot y_{n+1} - (n^2 - a^2) y_n = 0.$$

Putting $x = 0$, we get

$$(y_{n+2})_0 = (n^2 - a^2) (y_n)_0. \quad \dots(4)$$

But from (1) and (3), we have

$$(y_1)_0 = a \quad \text{and} \quad (y_2)_0 = 0.$$

$\therefore$ From (4), $(y_2)_0 = 0 = (y_4)_0 = (y_6)_0 = \dots$

$\therefore$ **When n is even integer** $(y_n)_0 = 0$.

Again

$$(y_1)_0 = a$$

$$(y_3)_0 = (1^2 - a^2) (y_1)_0 = (1^2 - a^2) a.$$

Similarly

$$(y_5)_0 = (3^2 - a^2) (1^2 - a^2) a.$$

$\therefore$ **When n is odd integer**

$$(y_n)_0 = [(n-2)^2 - a^2] [(n-4)^2 - a^2] \dots (5^2 - a^2) (3^2 - a^2) (1^2 - a^2) a.$$



Comprehensive Exercise 12.3

1. If $y = (\sin^{-1} x)$, then find $(y_n)_0$.

2. If $x = \sin [(1/\theta) \log_e y]$, then prove that

$$(1 - x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 + a^2) y_n = 0$$

and find also the value of $(y_n)_0$.

or

If $y = e^{a \sin^{-1} x}$ find y_n at $x = 0$.

3. If $y = \tan^{-1} x$, then prove that

$$(1 + x^2) y_2 + 2xy_1 = 0 \text{ and find } (y_x)_0$$

4. Show that $y = \cos (m \sin^{-1} x)$ satisfies the following equations:

$$(1 - x^2) y'' - xy' + m^2 y = 0.$$

Using Leibnitz's theorem differentiate this equation n times and find the value of $(y_n)_0$.

5. If $y = (\sin^{-1} x)^2$ then prove that $(1 - x^2) y_2 - xy_1 = 2$ and

$$(1 - x^2) y_{n+2} - x(2n+1) y_{n+1} - x^2 y_n = 0.$$



Answers

1.	0 when n is even and $(n-2)^2 (n-4)^2 \dots 3^2 \cdot 1^2$, when n is odd.
2.	$a(a^2 + 1^2)(a^2 + 3^2) \dots \{a^2 + (n-2)^2\}$, when n is odd. $a^2(a^2 + 2^2)(a^2 + 4^2) \dots \{a^2 + (n-2)^2\}$, when n is even.
3.	$\frac{n-1}{2} (-1)^{\frac{n-1}{2}} (n-1)!$, when n is odd.
4.	0 when n is odd and $-\{(n-2)^2 - m^2\} \{(n-4)^2 - m^2\} \dots (2^2 - m^2)(m^2)$, when n is even.



UNIT

4

13. Indefinite Integral

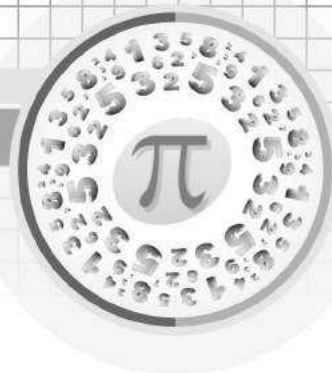
14. Definite Integral

15. Reduction Formulae

16. Gamma and Beta Functions



Unit-4



Chapter 13

Indefinite Integral

13.1 Integration

13.1.1 Elementary Integration

If $F(x)$ and $f(x)$ are two functions such that

$$\frac{d}{dx} F(x) = f(x);$$

i.e. $f(x)$ is the differential coefficient of $F(x)$ then $F(x)$ is the integral or anti-derivative of $f(x)$. We express this symbolically as:

$$\int f(x) dx = F(x) + c$$

For example:

$$(i) \quad \frac{d}{dx} x^2 = 2x \Rightarrow \int 2x dx = x^2 + c$$

$$(ii) \quad \frac{d}{dx} e^x = e^x \Rightarrow \int e^x dx = e^x + c$$

$$(iii) \quad \frac{d}{dx} \sin x = \cos x \Rightarrow \int \cos x . dx = \sin x + c$$

$$(iv) \quad \frac{d}{dx} \log x = \frac{1}{x} \Rightarrow \int \frac{1}{x} dx = \log x + c \quad \text{or} \quad \int f(x) dx = F(x) + c$$

$$\text{Sign of Integration} \int f(x) = F(x) + c.$$

Variable of Integration

Constant of Integration

or $\int dx \rightarrow$ Integral w.r.t. 'x'.

Note:

1. $F(x) + c$ known as general integral of function $f(x)$.
2. Generally c is not written with indefinite integrals but its presence is always kept in mind.

13.1.2 The Integral of x^n

By differentiation of $\frac{x^{n+1}}{n+1}$, we have

$$\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = \frac{1}{n+1} \cdot \frac{d}{dx} (x^{n+1}) = \frac{1}{n+1} \cdot (n+1) x^{n+1-1} = x^n$$

$\therefore$ By the definition of Integral, we have

$$\int x^n \cdot dx = \frac{x^{n+1}}{n+1} + c \text{ where } n \neq -1$$

Note:

1. To find the integral of x^n , where n is any constant not equal to -1 , increase the index of x by 1 and divide by the new index.
2.
$$\int dx = \int 1 \cdot dx = \int x^0 \cdot dx = \frac{x^{0+1}}{0+1} + c$$

i.e.,
$$\int dx = x + c$$

For example:

$$(i) \quad \int x^4 \cdot dx = \frac{x^{4+1}}{4+1} + c = \frac{1}{5} x^5 + c$$

$$(ii) \quad \int \sqrt{x} \cdot dx = \int x^{1/2} \cdot dx = \frac{x^{\left(\frac{1}{2}+1\right)}}{\left(\frac{1}{2}+1\right)} + c = \frac{2}{3} x^{3/2} + c$$

$$(iii) \quad \int a^{2 \log_a x} dx = \int a^{\log_a x^2} dx \quad (\because a^{\log_a x} = x)$$

$$= \int x^2 \cdot dx = \frac{x^{2+1}}{2+1} + c = \frac{1}{3} x^3 + c$$

13.1.3 Fundamental Integration Formulae

Since, $\frac{d}{dx} \{g(x)\} = f(x) \Leftrightarrow \int f(x) \cdot dx = g(x) + c$,

Hence, based upon this definition and various standard differentiation formulae; we obtain the following integration formula:

$$(i) \quad \frac{d}{dx} \left(\frac{x^{n+1}}{x+1} \right) = x^n, n \neq -1 \Rightarrow \int x^n dx = \frac{x^{n+1}}{x+1} + c, n \neq -1$$

(Known as power formula in integration)

Similarly,

Derivatives

$$(ii) \quad \frac{d}{dx} (\log |x|) = \frac{1}{x}$$

$$(iii) \quad \frac{d}{dx} (e^x) = e^x$$

$$(iv) \quad \frac{d}{dx} \left(\frac{a^x}{\log_e a} \right) = a^x, a > 0, a \neq 1$$

$$(v) \quad \frac{d}{dx} (-\cos x) = \sin x$$

$$(vi) \quad \frac{d}{dx} (\sin x) = \cos x$$

$$(vii) \quad \frac{d}{dx} (\tan x) = \sec^2 x$$

$$(viii) \quad \frac{d}{dx} (-\cot x) = \operatorname{cosec}^2 x$$

$$(ix) \quad \frac{d}{dx} (\sec x) = \sec x \cdot \tan x$$

$$(x) \quad \frac{d}{dx} (-\operatorname{cosec} x) = \operatorname{cosec} x \cdot \cot x$$

$$(xi) \quad \frac{d}{dx} (\log |\sin x|) = \cot x$$

$$(xii) \quad \frac{d}{dx} (-\log |\cos x|) = \tan x$$

$$(xiii) \quad \frac{d}{dx} (\log |\sec x + \tan x|) = \sec x$$

$$(xiv) \quad \frac{d}{dx} (\log |\operatorname{cosec} x - \cot x|) = \operatorname{cosec} x$$

$$(xv) \quad \frac{d}{dx} (\sin^{-1} x/a) = \frac{1}{\sqrt{a^2 - x^2}}$$

$$(xvi) \quad \frac{d}{dx} \cos^{-1} (x/a) = -\frac{1}{\sqrt{a^2 - x^2}}$$

$$(xvii) \quad \frac{d}{dx} \left(\frac{1}{a} \tan^{-1} (x/a) \right) = \frac{1}{a^2 + x^2}$$

Anti-derivatives

$$(ii) \quad \int \frac{1}{x} dx = \log |x| + c, \text{ where } x \neq 0$$

$$(iii) \quad \int e^x \cdot dx = e^x + c$$

$$(iv) \quad \int a^x \cdot dx = \frac{a^x}{\log_e a} + c$$

$$(v) \quad \int \sin x dx = -\cos x + c$$

$$(vi) \quad \int \cos x \cdot dx = \sin x + c$$

$$(vii) \quad \int \sec^2 x \cdot dx = \tan x + c$$

$$(viii) \quad \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$(ix) \quad \int \sec x \cdot \tan x dx = \sec x + c$$

$$(x) \quad \int \operatorname{cosec} x \cdot \cot x \cdot dx = -\operatorname{cosec} x + c$$

$$(xi) \quad \int \cot x \cdot dx = \log |\sin x| + c$$

$$(xii) \quad \int \tan x \cdot dx = -\log |\cos x| + c$$

$$(xiii) \quad \int \sec x \cdot dx = \log |\sec x + \tan x| + c$$

$$(xiv) \quad \int \operatorname{cosec} x \cdot dx = \log |\operatorname{cosec} x - \cot x| + c$$

$$(xv) \quad \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} (x/a) + c$$

$$(xvi) \quad \int \frac{-1}{\sqrt{a^2 - x^2}} dx = \cos^{-1} (x/a) + c$$

$$(xvii) \quad \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} (x/a) + c$$

$$(xviii) \quad \frac{d}{dx} \left(\frac{1}{a} \cot^{-1} x/a \right) = -\frac{1}{a^2 + x^2} \quad (xviii) \quad \int \frac{-1}{a^2 + x^2} dx = \frac{1}{a} \cot^{-1} (x/a) + c$$

$$(xix) \quad \frac{d}{dx} \left(\frac{1}{a} \sec^{-1} x/a \right) = \frac{1}{x \sqrt{x^2 - a^2}} \quad (xix) \quad \int \frac{dx}{x \sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} (x/a) + c$$

$$(xx) \quad \frac{d}{dx} \left(\frac{1}{a} \operatorname{cosec}^{-1} x/a \right) = -\frac{1}{x \sqrt{x^2 - a^2}} \quad (xx) \quad \int -\frac{dx}{x \sqrt{x^2 - a^2}} = \frac{1}{a} \operatorname{cosec}^{-1} \left(\frac{x}{a} \right) + c$$

Illustrative Examples

Example 1: Evaluate the following:

$$(i) \int \left(\sqrt{\frac{a}{x}} + \sqrt{\frac{x}{a}} \right) dx \quad (ii) \int \frac{(1+x)^3}{x^2} dx \quad [\text{B.C.A. (Kanpur) 2002}]$$

$$(iii) \int (10 + 10^x) \cdot dx \quad (iv) \int \frac{x^4}{x^2 + 1} dx \quad [\text{B.C.A. (Meerut) 2006, 2007}]$$

Solution:

(i) Let

$$I = \int (\sqrt{a/x} + \sqrt{x/a}) dx$$

$$I = \int \frac{\sqrt{a}}{\sqrt{x}} dx + \int \frac{\sqrt{x}}{\sqrt{a}} dx$$

$$I = \sqrt{a} \int x^{-1/2} dx + \frac{1}{\sqrt{a}} \int x^{1/2} dx$$

$$I = \sqrt{a} \cdot \frac{x^{-1/2+1}}{-(1/2)+1} + \frac{1}{\sqrt{a}} \cdot \frac{x^{1/2+1}}{1/2+1} + c$$

$$I = \sqrt{a} \cdot 2 x^{1/2} + \frac{1}{\sqrt{a}} \cdot \frac{2}{3} x^{3/2} + c$$

(ii) Let

$$I = \int \frac{(1+x)^3}{x^2} dx = \int \frac{1+3x+3x^2+x^3}{x^2} \cdot dx$$

$$I = \int \frac{1}{x^2} dx + \int \frac{3}{x} dx + \int 3 \cdot dx + \int x \cdot dx$$

$$I = \int x^{-2} dx + 3 \int \frac{1}{x} \cdot dx + 3 \int 1 \cdot dx + \int x \cdot dx$$

$$I = \frac{x^{-1}}{-1} + 3 \log x + 3 \cdot x + \frac{x^2}{2} + c$$

(iii) Let

$$I = \int (10 + 10^x) dx = \int 10 dx + \int 10^x dx = 10 \int 1 \cdot dx + \int 10^x dx$$

$$I = 10x + \frac{10^x}{\log_e 10} + c$$

$$\begin{aligned}
 \text{(iv) Let } I &= \int \frac{x^4}{x^2 + 1} dx = \int \frac{(x^4 - 1) + 1}{x^2 + 1} dx \\
 I &= \int \frac{(x^2 + 1)(x^2 - 1) + 1}{(x^2 + 1)} dx \\
 I &= \int \left(x^2 - 1 + \frac{1}{x^2 + 1} \right) dx \\
 I &= \int x^2 dx - 1 \int dx + \int \frac{1}{x^2 + 1} dx \\
 I &= \frac{x^3}{3} - 1 \cdot x + \tan^{-1} x + c
 \end{aligned}$$

Example 2: Evaluate:

$$\begin{aligned}
 \text{(i) } \int \tan^2 x \cdot dx & \qquad \text{(ii) } \int (\tan x + \cot x)^2 dx \\
 \text{(iii) } \int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cdot \cos^2 x} dx & \qquad \text{(iv) } \int \frac{\cos 2x}{\sin^2 x - \cos^2 x} dx
 \end{aligned}$$

Solution:

$$\begin{aligned}
 \text{(i) Let } I &= \int \tan^2 x \cdot dx = \int (\sec^2 x - 1) \cdot dx = \int \sec^2 x \cdot dx - \int 1 \cdot dx \\
 I &= \tan x - x + c \\
 \text{(ii) Let } I &= \int (\tan x + \cot x)^2 dx \\
 I &= \int (\tan^2 x + \cot^2 x + 2 \tan x \cdot \cot x) dx \\
 I &= \int (\tan^2 x + \cot^2 x + 2) dx \\
 I &= \int (1 + \tan^2 x) + (1 + \cot^2 x) dx \\
 I &= \int \sec^2 x dx + \int \operatorname{cosec}^2 x \cdot dx \\
 I &= \tan x - \cot x + c \\
 \text{(iii) Let } I &= \int \frac{(\sin^2 x - \cos^2 x)}{\sin^2 x \cdot \cos^2 x} dx \\
 I &= \int \frac{\sin^2 x}{\sin^2 x \cdot \cos^2 x} \cdot dx - \int \frac{\cos^2 x}{\sin^2 x \cdot \cos^2 x} dx \\
 I &= \int \frac{1}{\cos^2 x} dx - \int \frac{1}{\sin^2 x} \cdot dx \\
 I &= \int \sec^2 x dx - \int \operatorname{cosec}^2 x \cdot dx \\
 I &= \tan x + \cot x + c \\
 \text{(iv) Let } I &= \int \frac{\cos 2x}{\sin^2 x - \cos^2 x} \cdot dx = \int \frac{\cos^2 x - \sin^2 x}{\sin^2 x - \cos^2 x} \cdot dx \\
 &= \int (-1) dx = - \int 1 \cdot dx \\
 I &= -x + c
 \end{aligned}$$

Example 3: Evaluate these:

- (i) $\int \sqrt{1 + \sin 2x} \, dx$ (ii) $\int \sqrt{1 - \cos 2x} \, dx$
 (iii) $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} \, dx$ (iv) $\int \frac{1}{1 + \sin x} \, dx$
 (v) $\int \frac{dx}{1 + \cos x}$ (vi) $\int \frac{\sec x}{\sec x + \tan x} \, dx$
 (vii) $\int \frac{\sec x + \tan x}{\sec x - \tan x} \, dx$

Solution:

- (i) Let $I = \int \sqrt{1 + \sin 2x} \, dx$

$$= \int \sqrt{(\sin^2 x + \cos^2 x + 2 \sin x \cdot \cos x)} \, dx$$

$$= \int \sqrt{(\sin x + \cos x)^2} \, dx$$

$$= \int (\sin x + \cos x) \, dx$$

$$= \int \sin x \, dx + \int \cos x \, dx$$

$$\Rightarrow I = -\cos x + \sin x + c$$
- (ii) Let $I = \int \sqrt{1 - \cos 2x} \, dx$

$$= \int \sqrt{2 \sin^2 x} \, dx$$

$$= \sqrt{2} \int \sin x \, dx$$

$$I = \sqrt{2} (-\cos x) + c = -\sqrt{2} \cos x + c$$
- (iii) Let $I = \int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} \, dx = \int \frac{\sin x + \cos x}{\sqrt{(\sin^2 x + \cos^2 x) + 2 \sin x \cdot \cos x}} \, dx$

$$I = \int \frac{\sin x + \cos x}{\sqrt{(\sin x + \cos x)^2}} \, dx$$

$$I = \int \frac{\sin x + \cos x}{\sin x + \cos x} \, dx$$

$$I = \int dx = x + c$$
- (iv) Let $I = \int \frac{1}{1 + \sin x} \, dx = \int \left(\frac{1}{1 + \sin x} \times \frac{1 - \sin x}{1 - \sin x} \right) dx$

$$= \int \frac{1 - \sin x}{1 - \sin^2 x} \, dx$$

$$= \int \frac{1 - \sin x}{\cos^2 x} \, dx$$
-

$$= \int \left(\frac{1}{\cos^2 x} - \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} \right) \cdot dx$$

$$= \int (\sec^2 x - \sec x \cdot \tan x) dx$$

$$= \int \sec^2 x \cdot dx - \int \sec x \cdot \tan x \cdot dx$$

$$I = \tan x - \sec x + c$$

$$(v) \quad \text{Let} \quad I = \int \frac{1}{1 + \cos x} dx = \int \frac{1}{1 + \cos x} \cdot \frac{1 - \cos x}{1 - \cos x} dx$$

$$= \int \frac{1 - \cos x}{1 - \cos^2 x} dx$$

$$= \int \frac{1 - \cos x}{\sin^2 x} \cdot dx = \int \left(\frac{1}{\sin^2 x} - \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} \right) dx$$

$$= \int (\operatorname{cosec}^2 x - \operatorname{cosec} x \cdot \cot x) dx$$

$$= \int \operatorname{cosec}^2 x dx - \int \operatorname{cosec} x \cdot \cot x \cdot dx$$

$$I = -\cot x + \operatorname{cosec} x + c$$

$$(vi) \quad \text{Let} \quad I = \int \frac{\sec x}{\sec x + \tan x} dx = \int \frac{\sec x (\sec x - \tan x)}{(\sec x + \tan x)(\sec x - \tan x)} dx$$

$$= \int \frac{\sec^2 x - \sec x \cdot \tan x}{\sec^2 x - \tan^2 x} dx = \int (\sec^2 x - \sec x \cdot \tan x) dx$$

$$= \int \sec^2 x \cdot dx - \int \sec x \cdot \tan x \cdot dx$$

$$I = \tan x - \sec x + c$$

$$(vii) \quad \text{Let} \quad I = \int \frac{\sec x + \tan x}{\sec x - \tan x} dx = \int \frac{\sec x + \tan x}{\sec x - \tan x} \cdot \frac{\sec x + \tan x}{\sec x + \tan x} dx$$

$$= \int \frac{(\sec x + \tan x)^2}{\sec^2 x - \tan^2 x} dx$$

$$= \int (\sec x + \tan x)^2 \cdot dx$$

$$= \int (\sec^2 x + \tan^2 x + 2 \sec x \cdot \tan x) dx$$

$$= \int (\sec^2 x + (\sec^2 x - 1) + 2 \sec x \cdot \tan x) dx$$

$$= \int (2 \sec^2 x - 1 + 2 \sec x \cdot \tan x) dx$$

$$= 2 \int \sec^2 x \cdot dx + 2 \int \sec x \cdot \tan x \cdot dx - \int dx$$

$$I = 2 \tan x + 2 \sec x - x + c$$



Comprehensive Exercise 13.1

1. Evaluate the following:

(i) $\int x^8 dx$

(ii) $\int x^{5/4} dx$

(iii) $\int \frac{1}{x^{3/2}} dx$

(iv) $\int \frac{5}{x} \cdot dx$

(v) $\int 2a^x dx$

(vi) $\int \log_x x dx$

(vii) $\int 3^{2 \log_3 x} dx$

(viii) $\int (1 - x^2)^{-1/2} dx$

2. (i) $\int (x^4 + \frac{1}{x} - e^x + 1) dx$

(ii) $\int (7x^6 - 8x^3 + 4) dx$

(iii) $\int (\sin x - 2/x + e^x) dx$

(iv) $\int (x+1)(x+2)^2 \cdot dx$

(v) $\int \frac{ax^3 + bx + d}{x^2} dx$

(vi) $\int \left(x - \frac{1}{x}\right)^2 dx$

(vii) $\int \frac{3x^2 + 2x}{x^3 + x^2 + 1} dx$

[B.C.A. (Meerut) 2007]

3. (i) $\int (\sec^2 x - \operatorname{cosec}^2 x) dx$

(ii) $\int \left(\frac{4 - 7 \sin x}{\cos^2 x}\right) dx$

(iii) $\int \frac{\cos x}{\sin^2 x} dx$

(iv) $\int \left(\frac{2 \cos x}{3 \sin^2 x} - 1\right) dx$

(v) $\int \frac{\cos 2x}{\sin^2 x \cdot \cos^2 x} dx$

(vi) $\int \frac{1}{1 - \cos 2x} dx$

(vii) $\int \frac{\sec \theta}{\tan^2 \theta} \cdot d\theta$

4. (i) $\int \cot^2 x \cdot dx$

(ii) $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} \cdot dx$

(iii) $\int \frac{1}{\sin^2 x \cos^2 x} dx$

(iv) $\int \frac{\sin x + \operatorname{cosec} x}{\tan x} dx$

(v) $\int \frac{\cos 2x + \sin^2 x}{\cos^2 x} dx$

(vi) $\int \frac{\sin x}{1 + \sin x} dx$

5. (i) $\int \sin^{-1} (\cos x) dx$

(ii) $\int \frac{e^{\log x}}{x} \cdot dx$

(iii) $\int b^{x+a} dx$

(iv) $\int \frac{\sin x}{\sqrt{1 + \sin x}} \cdot dx$

[B.C.A. (Meerut) 2002]



Answers

1.	(i) $\frac{x^9}{9} + c$	(ii) $\frac{4}{9} x^{9/4} + c$
	(iii) $-\frac{2}{\sqrt{x}} + c$	(iv) $5 \log_e x + c$
	(v) $\frac{2a^x}{\log_e a} + c$	(vi) $x + c$
	(viii) $\frac{1}{3} x^3 + c$	(ix) $\sin^{-1} x + c$
2.	(i) $\frac{1}{5} x^5 + \log x - e^x + x + c$	(ii) $x^7 - 2x^4 + 4x + c$
	(iii) $-\cos x - 2 \log x + e^x + c$	(iv) $\frac{1}{4} x^4 + \frac{5}{3} x^3 + 4x^2 + 4x + c$
	(v) $\frac{1}{2} ax^2 + b \log x - \frac{d}{x} + c$	(vi) $\frac{1}{3} x^3 - \frac{1}{x} - 2x = c$
	(vii) $\log (x^3 + x^2 + 1)$	
3.	(i) $\tan x + \cot x + c$	(ii) $4 \tan x - 7 \sec x + c$
	(iii) $-\operatorname{cosec} x + c$	(iv) $(-2/3) \operatorname{cosec} x - x + c$
	(v) $-\cot x - \tan x + c$	(vi) $-\frac{1}{2} \cot x + c$
	(vii) $-\operatorname{cosec} \theta + c$	
4.	(i) $-\cot x - x + c$	(ii) $2 \sin x + 2 x \cos \alpha + c$
	(iii) $\tan x - \cot x + c$	(iv) $\sin x - \operatorname{cosec} x + c$
	(v) $(x + c)$	(vi) $x - \tan x + \sec x + c$
5.	(i) $(\pi/2)x - \frac{x^2}{2} + c$	(ii) $(x + c)$
	(iii) $\frac{b^{x+a}}{\log_e b} + c$	(iv) $-\sqrt{2} \cos (x/2 + \pi/8) + \sqrt{2} \log \tan (x/2 + \pi/8)$

13.2 Methods of Integration

- (i) Integration by substitution
- (ii) Integration by parts

13.2.1 Integration by Substitution

If $\phi(x)$ is continuously differentiable function, then to evaluate

$$\int f\{\phi(x)\} \phi'(x) dx$$

we substitute $\phi(x) = t$, so that $\phi'(x) \cdot dx = dt$

These substitutions reduce the above integral to the form $\int f(t) \cdot dt$ after evaluating this integral; we replace the value of t .

13.2.2 Integral of the form $\int f(ax + b) dx$

Let
$$I = \int f(ax + b) \cdot dx$$

Putting $ax + b = t$ i.e., $a dx = dt \Rightarrow dx = \frac{1}{a} dt$

$\therefore I = \int f(t) \frac{1}{a} \cdot dt, I = \frac{1}{a} \int f(t) \cdot dt = \frac{1}{a} F(t) + c$

Where $\frac{d}{dt} F(t) = f(t)$

$$I = \frac{1}{a} F(ax + b) + c$$

13.2.3 Evaluation of $\int (ax + b)^n dx$

Let
$$I = \int (ax + b)^n \cdot dx$$

Put $ax + b = t \Rightarrow a dx = dt \Rightarrow dx = \frac{1}{a} dt$

$\therefore I = \frac{1}{a} \int (t)^n \cdot dt = \frac{1}{a} \cdot \frac{t^{n+1}}{n+1} + c, n \neq -1$

Then, $\int (ax + b)^n \cdot dx = \frac{1}{a} \cdot \frac{(ax + b)^{n+1}}{n+1} + c, (n \neq -1)$

13.2.4 Evaluation of $\int \frac{1}{ax + b} dx$

Let
$$\begin{aligned} I &= \int \frac{1}{ax + b} \cdot dx \\ &= \int \frac{1}{t} \cdot \frac{dt}{a} \text{ where } ax + b = t \\ &= \frac{1}{a} \int \frac{1}{t} \cdot dt = \frac{1}{a} \log t + c \end{aligned}$$

Thus,
$$I = \int \frac{1}{ax + b} dx = \frac{1}{a} \log (ax + b) + c$$

13.2.5 Evaluation of $\int \sin(ax + b) \cdot dx$

Let
$$\begin{aligned} I &= \int \sin(ax + b) \cdot dx \\ I &= \int \sin t \cdot \frac{dt}{a}, \text{ where } ax + b = t \\ &= \frac{1}{a} \int \sin t \cdot dt = \frac{1}{a} (-\cos t) = -\frac{1}{a} \cos t + c \end{aligned}$$

Thus,
$$\int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + c$$

Note:

If in any of the standard integration formulae, in place of x we have $(ax + b)$ then the same formula is applicable but we must divide it by the derivative of $(ax + b)$; i.e. by a

13.2.6 Evaluation of Few Special Integrals

$$(i) \int \frac{1}{a^2 + x^2} dx \quad (ii) \int \frac{1}{\sqrt{a^2 - x^2}} dx \quad (iii) \int \frac{1}{x \sqrt{x^2 - a^2}} dx$$

Solution:

$$(i) \quad \text{Let} \quad I = \int \frac{1}{a^2 + x^2} dx \quad \text{Put } x = at \Rightarrow dx = a dt$$

$$i.e. \quad \int \frac{dx}{a^2 + x^2} = \int \frac{a dt}{a^2 + a^2 t^2} = \frac{a}{a^2} \int \frac{dt}{1 + t^2} = \frac{1}{a} \tan^{-1} t + c$$

$$\Rightarrow \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + c$$

$$(ii) \quad \text{Let } I = \int \frac{dx}{\sqrt{a^2 - x^2}} = \int \frac{a dt}{\sqrt{a^2 - t^2 a^2}} = \frac{a}{a} \int \frac{dt}{\sqrt{1 - t^2}} \quad (\because x = at \Rightarrow dx = a dt)$$

$$= \int \frac{dt}{\sqrt{1 - t^2}} = \sin^{-1} t + c = \sin^{-1} (x/a) + c$$

$$(iii) \quad \text{Let} \quad I = \int \frac{dx}{x \sqrt{x^2 - a^2}} = \int \frac{a \cdot dt}{at \sqrt{a^2 t^2 - a^2}}$$

$$= \int \frac{dt}{t \sqrt{a^2 t^2 - a^2}} = \frac{1}{a} \int \frac{dt}{t \sqrt{t^2 - 1}}$$

$$I = \frac{1}{a} \sec^{-1} (t) + c = \frac{1}{a} \sec^{-1} (x/a) + c$$

**Illustrative Examples**

Example 1: Integrate the following:

$$(i) \int (5x - 2)^3 \cdot dx$$

$$(ii) \int \frac{dx}{(a + bx)^4}$$

$$(iii) \int \sqrt{4 - 3x} \cdot dx$$

$$(iv) \int e^{4 - 3x} \cdot dx$$

$$(v) \int a^{3x+2} \cdot dx$$

Solution:

$$(i) \quad \text{Let } I = \int (5x - 2)^3 \cdot dx, \quad \text{Put } 5x - 2 = t \Rightarrow 5 dx = dt \Rightarrow dx = \frac{1}{5} \cdot dt$$

$$\therefore I = \int (5x - 2)^3 \cdot dx = \int t^3 \cdot \frac{1}{5} \cdot dt$$

$$= \frac{1}{5} \int t^3 \cdot dt = \frac{1}{5} \cdot \frac{t^4}{4} + c = \frac{1}{20} t^4 + c$$

$$I = \frac{1}{20} (5x - 2)^4 + c$$

$$(ii) \quad \text{Let } I = \int \frac{dx}{(a + bx)^4} = \int \frac{1}{t^4} \cdot \frac{1}{b} \cdot dt$$

$$\text{Put } a + bx = t \Rightarrow b dx = dt \Rightarrow dx = \frac{1}{b} dt$$

$$= \frac{1}{b} \int t^{-4} \cdot dt = \frac{1}{b} \cdot \frac{t^{-3}}{-3} + c$$

$$I = -\frac{1}{3b} (a + bx)^{-3} + c$$

$$I = -\frac{1}{3b (a + bx)^3} + c$$

$$(iii) \quad \text{Let } I = \int \sqrt{4 - 3x} \cdot dx = \int (4 - 3x)^{1/2} \cdot dx = \frac{(4 - 3x)^{1/2 + 1}}{\frac{1}{2} + 1} \cdot \left(-\frac{1}{3}\right) + c$$

$$= \frac{2}{3} \cdot \left(-\frac{1}{3}\right) (4 - 3x)^{3/2} + c$$

$$I = -\frac{2}{9} (4 - 3x)^{3/2} + c$$

$$(iv) \quad \text{Let } I = \int e^{4 - 3x} \cdot dx = e^{4 - 3x} \cdot \left(-\frac{1}{3}\right) + c = -\frac{1}{3} e^{4 - 3x} + c$$

$$(v) \quad \text{Let } I = \int a^{3x+2} \cdot dx = \frac{a^{3x+2}}{\log a} \cdot \frac{1}{3} + c = \frac{1}{3 \log a} \cdot a^{3x+2} + c$$

Example 2: Integrate these:

$$(i) \quad \int \sin (2x + \pi / 2) dx$$

$$(ii) \quad \int \sec^2 5x \cdot dx$$

$$(iii) \quad \int \frac{dx}{1 + \cos x}$$

$$(iv) \quad \int (\cos^4 x - \sin^4 x) dx$$

Solution:

$$(i) \quad \text{Let } I = \int \sin (2x + \pi / 2) \cdot dx = -\cos \left(2x + \frac{\pi}{2}\right) \cdot \frac{1}{2} + c = -\frac{1}{2} \cos \left(2x + \frac{\pi}{2}\right) + c$$

$$(ii) \quad \text{Let } I = \int \sec^2 5x \cdot dx = \tan (5x) \cdot \frac{1}{5} + c = \frac{1}{5} \tan 5x + c$$

$$(iii) \quad \text{Let } I = \int \frac{1}{1 + \cos x} \cdot dx = \int \frac{1}{1 + 2 \cos^2 x / 2 - 1} dx$$

$$= \frac{1}{2} \int \sec^2 x / 2 \cdot dx = \frac{1}{2} \left(\tan \frac{x}{2} \right) \cdot \frac{1}{1/2} + c = \tan (x / 2) + c$$

$$\begin{aligned}
 \text{(iv) Let } I &= \int (\cos^4 x - \sin^4 x) \cdot dx \\
 &= \int (\cos^2 x + \sin^2 x) (\cos^2 x - \sin^2 x) \cdot dx \\
 &= \int 1 \cdot (\cos^2 x - \sin^2 x) \cdot dx \\
 &= \int \cos 2x \cdot dx = (\sin 2x) \cdot \frac{1}{2} + c \\
 I &= \frac{1}{2} \sin 2x + c
 \end{aligned}$$

Example 3: Integrate

$$\text{(i) } \int \sin^2 x \cdot dx \qquad \text{(ii) } \int \sqrt{1 - \sin x} \cdot dx$$

Solution:

$$\begin{aligned}
 \text{(i) Let } I &= \int \sin^2 x \cdot dx = \frac{1}{2} \int (1 - \cos 2x) \cdot dx \\
 &= \frac{1}{2} \int dx - \frac{1}{2} \int \cos 2x \cdot dx = \frac{1}{2} x - \frac{1}{2} \cdot \frac{\sin 2x}{2} + c \\
 I &= \frac{1}{2} x - \frac{1}{4} \sin 2x + c \\
 \text{(ii) Let } I &= \int \sqrt{1 - \sin x} \cdot dx \\
 I &= \int \sqrt{\cos^2 (x/2) + \sin^2 (x/2) - 2 \sin (x/2) \cdot \cos (x/2)} \cdot dx \\
 &= \int \sqrt{(\cos (x/2) - \sin (x/2))^2} \cdot dx \\
 &= \int (\cos (x/2) - \sin (x/2)) \cdot dx \\
 &= \int \cos (x/2) \cdot dx - \int \sin (x/2) \cdot dx \\
 &= \frac{\sin (x/2)}{1/2} + \frac{\cos (x/2)}{1/2} + c \\
 I &= 2 (\sin (x/2) + \cos (x/2)) + c
 \end{aligned}$$

Example 4: Integrate these

$$\text{(i) } \int \sin 4x \cdot \cos 6x \cdot dx \qquad \text{(ii) } \int \cos x \cdot \cos 2x \cdot \cos 3x \cdot dx$$

Solution:

$$\begin{aligned}
 \text{(i) Let } I &= \int \sin 4x \cdot \cos 6x \cdot dx \\
 &= \frac{1}{2} \int 2 \cos 6x \cdot \sin 4x \cdot dx \\
 &= \frac{1}{2} \int [\sin 10x - \sin 2x] \cdot dx \\
 &= \frac{1}{2} \left[\int \sin 10x \cdot dx - \int \sin 2x \cdot dx \right] \\
 &= \frac{1}{2} \left[-\frac{\cos (10x)}{10} - \left(-\frac{\cos 2x}{2} \right) \right] + c
 \end{aligned}$$

$$I = \frac{1}{2} \left[\frac{1}{2} \cos 2x - \frac{1}{10} \cos 10x \right] + c$$

$$\Rightarrow I = \frac{1}{4} \cdot \cos 2x - \frac{1}{20} \cos 10x + c$$

(ii) Let $I = \int \cos x \cdot \cos 2x \cdot \cos 3x \cdot dx$

$$= \frac{1}{2} \int (2 \cos 3x \cdot \cos x) \cos 2x \cdot dx$$

$$= \frac{1}{2} \int (\cos 4x + \cos 2x) \cdot \cos 2x \cdot dx$$

$$= \frac{1}{2} \int (\cos 4x \cdot \cos 2x + \cos^2 2x) \cdot dx$$

$$= \frac{1}{2} \cdot \frac{1}{2} \int [2 \cos 4x \cdot \cos 2x + 2 \cos^2 2x] \cdot dx$$

$$= \frac{1}{4} \left[\frac{\sin 6x}{6} + \frac{\sin 2x}{2} + x + \frac{\sin 4x}{4} \right] + c$$

$$I = \frac{1}{4} \left[x + \frac{1}{2} \sin 2x + \frac{1}{4} \sin 4x + \frac{1}{6} \sin 6x \right] + c$$

Example 5: Integrate these

(i) $\int \frac{1}{b^2 + a^2 x^2} dx$ (ii) $\int \frac{1}{\sqrt{1 - (3x - 2)^2}} dx$ (iii) $\int \frac{dx}{(x - 2) \sqrt{x^2 - 4x + 3}}$

Solution:

(i) Let
$$I = \int \frac{dx}{b^2 + a^2 x^2} = \int \frac{dx}{b^2 + (ax)^2}$$

$$= \int \frac{1}{b^2 + t^2} \cdot \frac{1}{a} \cdot dt \quad \left(\because ax = t \Rightarrow dx = \frac{dt}{a} \right)$$

$$= \frac{1}{a} \int \frac{1}{t^2 + b^2} \cdot dt$$

$$= \frac{1}{a} \cdot \frac{1}{b} \cdot \tan^{-1} \frac{t}{b} + c$$

$$I = \frac{1}{ab} \tan^{-1} \left(\frac{ax}{b} \right) + c$$

(ii) Let
$$I = \int \frac{dx}{\sqrt{1 - (3x - 2)^2}} = \int \frac{1}{\sqrt{1 - t^2}} \cdot \frac{1}{3} \cdot dt$$

Let $3x - 2 = t \Rightarrow 3 \cdot dx = dt \Rightarrow dx = \frac{1}{3} \cdot dt$

$$I = \frac{1}{3} \sin^{-1} t + c = \frac{1}{3} \sin^{-1} (3x - 2) + c$$

$$\begin{aligned}
 \text{(iii) Let } I &= \int \frac{dx}{(x-2)\sqrt{x^2-4x+3}} = \int \frac{dx}{(x-2)\sqrt{(x^2-4x+4)-1}} \\
 &= \int \frac{dx}{(x-2)\sqrt{(x-2)^2-1}} \quad [\text{Put } x-2=t] \\
 &= \int \frac{dt}{t\sqrt{t^2-1}} = \sec^{-1} t + c = \sec^{-1} (x-2) + c
 \end{aligned}$$

13.2.7 Integral of the form $\int f(x^n) x^{n-1} dx$

$$\text{Put } x^n = t \Rightarrow nx^{n-1} dx = dt$$

$$\Rightarrow x^{n-1} dx = \frac{1}{n} dt$$

$$\therefore \int f(x^n) x^{n-1} dx = \int f(t) \cdot \frac{1}{n} \cdot dt = \frac{1}{n} F(t) + c$$

$$\text{Where } \frac{d}{dt} F(t) = f(t)$$

$$\text{Thus } \int f(x^n) x^{n-1} dx = \frac{1}{n} F(x^n) + c$$

13.2.8 Integral of the form $\int [f(x)]^n \cdot f'(x) \cdot dx$

$$\text{Put } f(x) = t \Rightarrow f'(x) \cdot dx = dt$$

$$= \int [f(x)]^n \cdot f'(x) \cdot dx = \int t^n \cdot dt$$

$$= \frac{t^{n+1}}{n+1} + c$$

$$I = \frac{1}{n+1} [f(x)]^{n+1} + c$$



Illustrative Examples

Example 1: Evaluate the following integral functions:

$$\int 4x^3 \cdot \sin x^4 \cdot dx$$

Solution: Let $I = \int 4x^3 \cdot \sin x^4 \cdot dx$

$$\text{Put } x^4 = t \Rightarrow 4x^3 \cdot dx = dt$$

$$I = \int \sin x^4 \cdot 4x^3 \cdot dx = \int \sin t \cdot dt = -\cos t + c$$

$$I = -\cos(x^4) + c$$

Example 2: Evaluate $\int \sin^3 x \cdot \cos x \cdot dx$

Solution: Let $I = \int \sin^3 x \cdot \cos x \cdot dx$

Let : $\sin x = t$ Then, $\cos x \cdot dx = dt$

$$\Rightarrow I = \int t^3 \cdot dt = \frac{t^4}{4} + c = \frac{(\sin x)^4}{4} + c$$

$$I = \frac{1}{4} \sin^4 x + c$$

Example 3: Evaluate $\int \frac{\sin x}{1 + \cos^2 x} dx$

Solution: $I = \int \frac{\sin x}{1 + \cos^2 x} dx$

Put $\cos x = t \Rightarrow -\sin x \cdot dx = dt$

$$I = - \int \frac{dt}{1 + t^2} = -\tan^{-1} t + c$$

$$I = -\tan^{-1} (\cos x) + c$$

Example 4: Evaluate $\int x^2 \tan^4 x^3 \sec^2 x^3 \cdot dx$

Solution: Let $I = \int x^2 \tan^4 x^3 \cdot \sec^2 x^3 \cdot dx$

Put $\tan x^3 = t$

$$\Rightarrow (\sec^2 x^3) \cdot 3x^2 \cdot dx = dt$$

$$\Rightarrow x^2 \cdot \sec^2 x^3 \cdot dx = \frac{1}{3} dt$$

$$\therefore \int \tan^4 x^3 \cdot (x^2 \cdot \sec^2 x^3) \cdot dx = \frac{1}{3} \int t^4 \cdot dt = \frac{1}{3} \cdot \frac{t^5}{5} + c$$

$$I = \frac{1}{15} \tan^5 x^3 + c$$

Example 5: Evaluate the following integrals:

(i) $\int \frac{\sin x}{\sqrt{9 - \cos^2 x}} dx$

(ii) $\int \frac{\tan^{-1} x}{1 + x^2} dx$

(iii) $\int \sec x \cdot \log (\sec x + \tan x) dx$

(iv) $\int \frac{2x^3}{4 + x^8} dx$

(v) $\int \frac{dx}{2\sqrt{x(1-x)}}$

(vi) $\int \frac{dx}{x(1 + \log x)^n}$

Solution:

(i) $I = \int \frac{\sin x}{\sqrt{9 - \cos^2 x}} dx$

Put $\cos x = t \Rightarrow -\sin x \cdot dx = dt$

$$\therefore I = \int \frac{-dt}{\sqrt{9 - t^2}} = -\sin^{-1} (t/3) + c = -\sin^{-1} \left(\frac{\cos x}{3} \right) + c$$

$$(ii) \quad I = \int \frac{\tan^{-1} x}{1+x^2} dx$$

$$\text{Put } \tan^{-1} x = t \Rightarrow \frac{1}{1+x^2} dx = dt$$

$$\therefore I = \int t \cdot dt = \frac{t^2}{2} + c = \frac{1}{2} (\tan^{-1} x)^2 + c$$

$$(iii) \quad I = \int \sec x \cdot \log (\sec x + \tan x) \cdot dx$$

$$\text{Put } \log (\sec x + \tan x) = t$$

$$\Rightarrow \frac{1}{\sec x + \tan x} \cdot (\sec x \tan x + \sec^2 x) \cdot dx = dt$$

$$\Rightarrow \frac{\sec x (\tan x + \sec x)}{(\sec x + \tan x)} dx = dt$$

$$\sec x \cdot dx = dt$$

$$\therefore I = \int t \cdot dt$$

$$= \frac{t^2}{2} + c$$

$$I = \frac{1}{2} [\log (\sec x + \tan x)]^2 + c$$

$$(iv) \quad \int \frac{2x^3}{4+x^8} dx$$

$$\text{Put } x^4 = t \Rightarrow 4x^3 dx = dt \Rightarrow 2x^3 dx = \frac{1}{2} dt$$

$$\therefore I = \frac{1}{2} \int \frac{dt}{4+t^2} = \frac{1}{2} \cdot \frac{1}{2} \tan^{-1} (t/2) + c$$

$$I = \frac{1}{4} \cdot \tan^{-1} \left(\frac{x^4}{2} \right) + c$$

$$(v) \quad \int \frac{dx}{2\sqrt{x}(1-x)}$$

$$I = \int \frac{dx}{2\sqrt{x}\sqrt{1-x}} \quad \text{Put } \sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}} dx = dt$$

$$I = \int \frac{dt}{\sqrt{1-t^2}} = \sin^{-1} t + c$$

$$I = \sin^{-1} (\sqrt{x}) + c$$

$$(vi) \quad \int \frac{dx}{x(1+\log x)^n} \quad \text{Put } (1+\log x) = t \Rightarrow \frac{1}{x} \cdot dx = dt$$

$$\Rightarrow I = \int \frac{1}{t^n} dt = \int t^{-n} dt = \frac{t^{-n+1}}{-n+1} + c$$

$$I = \frac{1}{1-n} (1+\log x)^{-(n-1)} + c$$

Example 6: Evaluate $\int \frac{e^x (1+x)}{\cos^2 (xe^x)} dx$

Let $xe^x = t \Rightarrow (xe^x + e^x \cdot 1) dx = dt$
 $\Rightarrow e^x (x+1) dx = dt$

$$I = \int \frac{e^x (1+x)}{\cos^2 (xe^x)} dx$$

$$I = \int \frac{dt}{\cos^2 t} = \int \sec^2 t \cdot dt = \tan t + c$$

$$I = \tan (xe^x) + c.$$

13.2.9 Integration of the form $\int \frac{f'(x)}{f(x)} dx$

where $f'(x)$ is the differential coefficient of $f(x)$ w.r.t. 'x', then

$$f(x) = t \Rightarrow f'(x) dx = dt.$$

$$\therefore \int \frac{dt}{t} = \log t = \log f(x) + c$$

Hence, $\int \frac{f'(x)}{f(x)} = \log f(x) + c$

13.3 Some Standard Results

(i) **Integration of $\tan x$ w.r.t. 'x' :**

We have, $I = \int \tan x dx = \int \frac{\sin x}{\cos x} dx$

Putting $\cos x = t \Rightarrow -\sin x dx = dt$

or $\sin x dx = -dt$

$$I = - \int \frac{dt}{t} = -\log t = -\log (\cos x) = \log (\cos x)^{-1} + c$$

$$= \log \sec x + c$$

$$\Rightarrow I = \int \tan x dx = \log \sec x + c$$

(ii) **Integration of $\cot x$ w.r.t. 'x' :**

$$I = \int \cot x \cdot dx = \int \frac{\cos x}{\sin x} dx$$

Putting $\sin x = t \Rightarrow \cos x \cdot dx = dt$

$$\therefore I = \int \frac{dt}{t} = \log t + c = \log \sin x + c$$

$$\therefore \int \cot x dx = \log \sin x + c$$

(iii) **Integration of cosec x w.r.t. ' x '. i.e., $\int \text{cosec } x \cdot dx$**

We have, $I = \int \text{cosec } x \cdot dx$

We have, $I = \int \text{cosec } x \cdot dx = \int \frac{1}{\sin x} dx$

$$= \int \frac{1}{2 \sin(x/2) \cdot \cos(x/2)} \cdot dx = \int \frac{\frac{1}{2} \sec^2(x/2)}{\tan(x/2)} dx$$

(on dividing the Nr and Dr. by $\cos^2 x/2$)

Now putting $\tan x/2 = t$, we have $\frac{1}{2} \sec^2 x/2 \cdot dx = dt$

$$\int \text{cosec } x \cdot dx = \int \frac{dt}{t} = \log t = \log \tan x/2$$

Hence, $I = \int \text{cosec } x \cdot dx = \log \tan x/2 + c$

Similarly, we can do as;

$$\int \sec x \cdot dx = \log \tan \left(\frac{\pi}{4} + \frac{x}{2} \right)$$

or

$$\int \sec x \cdot dx = \log (\sec x + \tan x)$$

Illustrative Examples

Evaluate the following integrals:

Example 1: $I = \int \frac{e^x}{1 + e^x} dx$

Solution: Put $1 + e^x = t \Rightarrow e^x \cdot dx = dt$

$$I = \int \frac{dt}{t} = \log t + c = \log(1 + e^x) + c$$

Example 2: $I = \int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx.$

[B.C.A. (Meerut) 2008]

Solution: Let $e^x + e^{-x} = t \Rightarrow (e^x - e^{-x}) dx = dt$

$\therefore I = \int \frac{dt}{t} = \log t + c = \log (e^x - e^{-x}) + c$

Example 3: $I = \int \frac{1 - \tan x}{1 + \tan x} dx$

Solution: $I = \int \frac{1 - \sin x / \cos x}{1 + \sin x / \cos x} dx$

$$= \int \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right) dx$$

Now put $\cos x + \sin x = t$

$$\Rightarrow (-\sin x + \cos x) dx = dt$$

$$\Rightarrow (\cos x - \sin x) dx = dt$$

$$\begin{aligned} \therefore \int \frac{1 - \tan x}{1 + \tan x} dx &= \int \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right) dx = \int \frac{1}{t} dt = \log t + c \\ &= \log (\cos x + \sin x) + c \end{aligned}$$

Example 4: $I = \int \frac{\sin x + x \cos x}{x \sin x} dx$

Solution: Put $x \sin x = t \Rightarrow (x \cos x + \sin x \cdot 1) dx = dt \Rightarrow (\sin x + x \cos x) dx = dt$

$$\therefore I = \int \frac{\sin x + x \cos x}{x \sin x} dx = \int \frac{dt}{t} = \log t + c = \log (x \sin x) + c$$

Example 5: $I = \int \frac{1}{x(1 + \log x)} dx$

Solution: Put $1 + \log x = t \Rightarrow \frac{1}{x} \cdot dx = dt$

$$\begin{aligned} \therefore I &= \int \frac{1}{x(1 + \log x)} dx = \int \frac{\frac{1}{x}}{1 + \log x} dx \\ &= \int \frac{dt}{t} = \log t + c = \log (1 + \log x) + c \end{aligned}$$

Example 6: $I = \int \frac{\sin 2x \cdot dx}{a \cos^2 x + b \sin^2 x}$

Solution: Putting $a \cos^2 x + b \sin^2 x = t$

$$\Rightarrow (a \cdot 2 \cos x (-\sin x) + b \cdot 2 \sin x \cdot \cos x) dx = dt$$

$$\Rightarrow (b - a) \cdot 2 \sin x \cdot \cos x \cdot dx = dt$$

$$(b - a) \sin 2x dx = dt$$

$$\Rightarrow \sin 2x dx = \frac{1}{b - a} dt$$

$$\begin{aligned} \therefore \int \frac{\sin 2x \cdot dx}{a \cos^2 x + b \sin^2 x} &= \frac{1}{b - a} \int \frac{dt}{t} \\ &= \frac{1}{b - a} \log t + c = \frac{1}{b - a} \log (a \cos^2 x + b \sin^2 x) + c \end{aligned}$$

Example 7: $I = \int \frac{\sin x}{\sin(x - \alpha)} dx$

Solution: Putting $x - \alpha = t \Rightarrow dx = dt$

we have;

$$\begin{aligned} I &= \int \frac{\sin(t + \alpha)}{\sin t} \cdot dt \\ &= \int \frac{\sin \alpha \cdot \cos t + \cos \alpha \cdot \sin t}{\sin t} \cdot dt \\ &= \int (\sin \alpha \cdot \cot t + \cos \alpha) dt \\ &= \sin \alpha \int \cot t dt + \cos \alpha \int dt \\ &= \sin \alpha \cdot \log \sin t + (\cos \alpha) t + c \\ &= \sin \alpha \cdot \log \sin(x - \alpha) + (x - \alpha) \cos \alpha + c \end{aligned}$$

Example 8: $I = \int \frac{\cos(x - \alpha)}{\cos x} \cdot dx$

Solution: We have $I = \int \frac{\cos x \cos \alpha + \sin x \cdot \sin \alpha}{\cos x} dx$

$$\begin{aligned} I &= \int [\cos \alpha + \sin \alpha \tan x] dx \\ &= \cos \alpha \int dx + \sin \alpha \int \tan x \cdot dx \\ &= (\cos \alpha) \cdot x + \sin \alpha \cdot \log \sec x + c \\ I &= x \cos \alpha + \sin \alpha \cdot \log \sec x + c \end{aligned}$$

Example 9: $I = \int \frac{dx}{\sin(x - a) \cdot \sin(x - b)}$

[B.C.A. (Meerut) 2005, 2009]

Solution: We have

$$\begin{aligned} I &= \frac{1}{\sin(a - b)} \int \frac{\sin(a - b) dx}{\sin(x - a) \sin(x - b)} \\ &= \frac{1}{\sin(a - b)} \int \frac{\sin[(x - b) - (x - a)]}{\sin(x - a) \sin(x - b)} dx \\ &= \frac{1}{\sin(a - b)} \int \frac{\sin(x - b) \cdot \cos(x - a) - \cos(x - b) \sin(x - a)}{\sin(x - a) \cdot \sin(x - b)} dx \\ &= \frac{1}{\sin(a - b)} \int \cot(x - a) - \cot(x - b) \cdot dx \\ &= \frac{1}{\sin(a - b)} [\log \sin(x - a) - \log \sin(x - b)] + c \\ I &= \operatorname{cosec}(a - b) \times \log \frac{\sin(x - a)}{\sin(x - b)} + c \end{aligned}$$

Example 10: $I = \int \frac{1}{\sqrt{1 + \sin x}} dx$

Solution: We have,
$$\begin{aligned} I &= \int \frac{dx}{\sqrt{1 - \cos(\pi/2 + x)}} \\ &= \int \frac{dx}{\sqrt{2 \sin^2(\pi/4 + x/2)}} \\ &= \frac{1}{\sqrt{2}} \int \operatorname{cosec}\left(\frac{\pi}{4} + \frac{x}{2}\right) dx \end{aligned}$$

Put $\frac{x}{2} + \frac{\pi}{4} = t \Rightarrow dx = 2 dt$

$$I = \frac{2}{\sqrt{2}} \int \operatorname{cosec} t \cdot dt = \sqrt{2} \log \tan \frac{t}{2} + c = \sqrt{2} \log \tan \left(\frac{\pi}{8} + \frac{x}{4} \right) + c$$

Example 11: $I = \int \frac{dx}{1 + \cot x}$

Solution: We have,
$$\begin{aligned} I &= \int \frac{\sin x}{\sin x + \cos x} dx \\ &= \int \frac{\sin x (\cos x - \sin x)}{(\sin x + \cos x)(\cos x - \sin x)} dx \\ &= \int \frac{\sin x \cdot \cos x - \sin^2 x}{\cos^2 x - \sin^2 x} dx \\ &= \frac{1}{2} \int \frac{2 \sin x \cdot \cos x - 2 \sin^2 x}{\cos 2x} dx \\ &= \frac{1}{2} \int \frac{\sin 2x - (1 - \cos 2x)}{\cos 2x} dx \\ &= \frac{1}{2} \int (\tan 2x - \sec 2x + 1) dx \\ &= \frac{1}{2} \left[\frac{1}{2} \log \sec 2x - \frac{1}{2} \log (\sec 2x + \tan 2x) + x \right] + c \\ &= \frac{1}{2} x + \frac{1}{4} \log \sec 2x - \frac{1}{4} \log (\sec 2x + \tan 2x) + c \end{aligned}$$

Example 12: $I = \int \frac{\cot x}{\log \sin x} dx$

Solution: We have

$$\log \sin x = t, \quad \frac{1}{\sin x} \cos x \cdot dx = dt \Rightarrow \cot x \cdot dx = dt$$

$$\therefore I = \int \frac{\cot x}{\log \sin x} dx = \int \frac{dt}{t} = \log t + c = \log (\log \sin x) + c$$



Comprehensive Exercise 13.2

1. (i) $(x+2)^3$ (ii) $(1-2x)^4$
 (iii) $(4+5x)^3$ (iv) $\frac{3x^2+2x}{x^3+x^2}$ [B.C.A. (Meerut) 2006, 2007]
2. (i) $\frac{1}{(1+x)^4}$ (ii) $\frac{1}{(3x+4)^2}$
3. (i) $\sin 3x$ (ii) $\sin x/2$
 (iii) $\sin (2x+3)$ (iv) $\cos (4x+5)$
4. (i) $\frac{x}{a+x}$ (ii) $\frac{a+x}{a-x}$
5. (i) $\sqrt{1-5x} + 9e^{3x}$ (ii) $\sin \frac{1}{3}(b+x) + \cos \frac{1}{2}(b-x)$
 (iii) $5a^x + 6a \cos (5x+2)$ (iv) $\int \cot x e^{n \log (\sin x)} dx$ [B.C.A. (Meerut) 2003, 2004]



Answers

1.	(i) $\frac{1}{4}(x+2)^4 + c$ (iii) $\frac{1}{20}(4+5x)^4 + c$	(ii) $-\frac{1}{10}(1-2x)^5 + c$ (iv) $\log(x^3 + x^2)$
2.	(i) $-\frac{1}{3(1+x)^3} + c$	(ii) $-\frac{1}{3(3x+4)} + c$
3.	(i) $-\frac{1}{3}\cos 3x + c$ (iii) $-\frac{1}{2}\cos (2x+3) + c$	(ii) $-2\cos x/2 + c$ (iv) $\frac{1}{4}\sin (4x+5) + c$
4.	(i) $x - a \log (a+x) + c$	(ii) $-x - 2a \log (a-x) + c$
5.	(i) $-\frac{2}{15}(1-5x)^{3/2} + 3e^{3x} + c$ (iii) $\frac{a^x}{\log a} + \frac{6}{5}a \sin (5x+2) + c$	(ii) $-3\cos \frac{1}{3}(b+x) - 2\sin \frac{1}{2}(b-x) + c$ (iv) $\frac{1}{n}e^{n \log (\sin x)} + c$

13.4 Some Standard Result

$$(i) \quad \int f(x) \cdot dx = g(x) + c, \text{ then } \int f(ax + b) dx = \frac{1}{a} g(ax + b) + c ;$$

$$(ii) \quad \int \frac{1}{x} dx = \log |x| + c, \text{ then } \int \frac{1}{ax + b} dx = \frac{1}{a} \log |ax + b| + c$$

Thus, in any fundamental integral formula given as above. If in place of x we have $(ax + b)$, then same formula is applicable but we must divide it by coefficient of x or derivative of $(ax + b)$ i.e. a

We gave the list of some frequently used formula :

$$(i) \quad \int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{a(n+1)} + c \quad (n \neq -1)$$

$$(ii) \quad \int \frac{1}{ax + b} dx = \frac{1}{a} \log |ax + b| + c$$

$$(iii) \quad \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + c$$

$$(iv) \quad \int a^{bx+d} dx = \frac{1}{b} \cdot \frac{a^{bx+d}}{\log a} + c$$

$$(v) \quad \int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + c$$

$$(vi) \quad \int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + c$$

$$(vii) \quad \int \sec^2(ax + b) dx = \frac{1}{a} \tan(ax + b) + c$$

$$(viii) \quad \int \operatorname{cosec}^2(ax + b) dx = -\frac{1}{a} \cot(ax + b) + c$$

$$(ix) \quad \int \sec(ax + b) \tan(ax + b) dx = \frac{1}{a} \sec(ax + b) + c$$

$$(x) \quad \int \operatorname{cosec}(ax + b) \cot(ax + b) dx = -\frac{1}{a} \operatorname{cosec}(ax + b) + c$$

$$(xi) \quad \int \tan(ax + b) dx = -\frac{1}{a} \log |\cos(ax + b)| + c$$

$$(xii) \quad \int \cot(ax + b) dx = \frac{1}{a} \log |\sin(ax + b)| + c$$

$$(xiii) \quad \int \sec(ax + b) dx = \frac{1}{a} \log |\sec(ax + b) + \tan(ax + b)| + c$$

$$(xiv) \quad \int \operatorname{cosec}(ax + b) dx = \frac{1}{a} \log |\operatorname{cosec}(ax + b) - \cot(ax + b)| + c$$

Illustrative Examples

Evaluate the following integrals:

Example 1: $I = \int \frac{2^x + 3^x}{5^x} dx.$

[B.C.A. (Kanpur) 2003]

Solution: We have,

$$I = \int \left(\frac{2^x}{5^x} + \frac{3^x}{5^x} \right) dx = \int \left[\left(\frac{2}{5} \right)^x + \left(\frac{3}{5} \right)^x \right] dx = \frac{(2/5)^x}{\log_e (2/5)} + \frac{(3/5)^x}{\log_e (3/5)} + c$$

Example 2: $\int \frac{(a^x + b^x)^2}{a^x \cdot b^x} dx.$

[B.C.A. (Garhwal) 2005]

Solution: We have,

$$\begin{aligned} I &= \int \frac{a^{2x} + b^{2x} + 2a^x b^x}{a^x \cdot b^x} \cdot dx = \int \left(\frac{a^x}{b^x} + \frac{b^x}{a^x} + 2 \right) \cdot dx = \int \left\{ \left(\frac{a}{b} \right)^x + \left(\frac{b}{a} \right)^x + 2 \right\} dx \\ &= \frac{(a/b)^x}{\log_e (a/b)} + \frac{(b/a)^x}{\log_e (b/a)} + 2x + c \end{aligned}$$

Example 3: $\int \frac{e^{5 \log_e x} - e^{4 \log_e x}}{e^{3 \log_e x} - e^{2 \log_e x}} dx$

Solution: We have,

$$I = \int \frac{x^5 - x^4}{x^3 - x^2} dx [\because e^{a \log_e x} = x^a] = \int \frac{x^4 (x-1)}{x^2 (x-1)} dx = \int x^2 \cdot dx = \frac{x^3}{3} + c = e^{3 \log_e x} + c$$

Example 4: $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x} dx$

[B.C.A. (Bhopal) 2008]

Solution: We have,

$$I = \int \frac{(\sin^2 x + \cos^2 x)^3 - 3 \sin^2 x \cdot \cos^2 x (\sin^2 x + \cos^2 x)}{\sin^2 x \cdot \cos^2 x} dx$$

$$I = \int \frac{1 - 3 \sin^2 x \cdot \cos^2 x}{\sin^2 x \cdot \cos^2 x} \cdot dx$$

$$I = \int \left(\frac{1}{\sin^2 x \cdot \cos^2 x} - 3 \right) dx$$

$$I = \int (\sec^2 x + \operatorname{cosec}^2 x - 3) dx$$

$$I = \tan x - \cot x - 3x + c$$

Example 5: $\int \sec^n x \cdot \tan x \cdot dx$

We have,

$$\begin{aligned} I &= \int (\sec x)^{n-1} \sec x \cdot \tan x \cdot dx \\ &= \int \sec^{n-1} x \cdot d(\sec x) \\ &= \int (\sec x)^{n-1} d(\sec x) = \frac{(\sec x)^n}{n} + c \end{aligned}$$

Example 6: $\int \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} dx$

[B.C.A. (Bundelkhand) 2004]

Solution: We have,

$$\begin{aligned} I &= \int \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} \cdot dx \\ &= \int (\sin^{-1} x)^3 d(\sin^{-1} x) = \frac{(\sin^{-1} x)^4}{4} + c \end{aligned}$$

Example 7: $\int \frac{1}{x^2 (x^4 + 1)^{3/4}} dx$

Solution: We have,

$$\begin{aligned} I &= \int \frac{1}{x^2 \left\{ x^4 \left(1 + \frac{1}{x^4} \right) \right\}^{3/4}} dx \\ &= \int \frac{1}{x^5 \left(1 + \frac{1}{x^4} \right)^{3/4}} dx \\ &= -\frac{1}{4} \int \frac{1}{\left(1 + \frac{1}{x^4} \right)^{3/4}} \times \left(-\frac{4}{x^5} \right) dx \\ &= -\frac{1}{4} \int \left(1 + \frac{1}{x^4} \right)^{-3/4} \cdot d \left(1 + \frac{1}{x^4} \right) \\ &= -\frac{1}{4} \frac{\left(1 + \frac{1}{x^4} \right)^{-\frac{3}{4}+1}}{\left(-\frac{3}{4} + 1 \right)} + c = -\left(1 + \frac{1}{x^4} \right)^{1/4} + c \end{aligned}$$

Example 8: $\int \frac{\sqrt{5+x^{10}}}{x^{16}} dx$

Solution: We have, $I = \int \frac{\sqrt{5+x^{10}}}{x^{16}} dx$

$$= \int \sqrt{\frac{5+x^{10}}{x^{10}}} \cdot \frac{1}{x^{11}} dx$$

$$\begin{aligned}
&= \int \sqrt{\frac{5}{x^{10}} + 1} \cdot \frac{1}{x^{11}} \cdot dx \\
&= -\frac{1}{50} \int \sqrt{1 + \frac{5}{x^{10}}} \cdot \left(-\frac{50}{x^{11}}\right) dx \\
&= -\frac{1}{50} \int \sqrt{1 + \frac{5}{x^{10}}} \cdot d\left(1 + \frac{5}{x^{10}}\right) \\
&= -\frac{1}{50} \left[\frac{\left(1 + \frac{5}{x^{10}}\right)^{3/2}}{3/2} \right] + c = -\frac{1}{75} \left(1 + \frac{5}{x^{10}}\right)^{3/2} + c
\end{aligned}$$

Example 9: $\int \sqrt[3]{\frac{\sin^n x}{\cos^{n+6} x}} \cdot dx$

Solution: We have, $I = \int \sqrt[3]{\frac{\sin^n x}{\cos^n x \cdot \cos^6 x}} \cdot dx$

$$\begin{aligned}
&= \int \sqrt[3]{\tan^n x \cdot \sec^2 x} \cdot dx \\
&= \int (\tan x)^{n/3} \cdot d(\tan x) \\
&= \frac{(\tan x)^{\frac{n}{3}+1}}{\frac{n}{3}+1} + c = \frac{3}{3+n} (\tan x)^{\frac{n}{3}+1} + c
\end{aligned}$$

Example 10: $\int \frac{ax+b}{cx+d} dx$

Solution: We have, $I = (a/c) \int \frac{(cx+d) - \left(d - \frac{bc}{a}\right)}{cx+d} dx$

$$= (a/c) \int dx - \frac{ad-bc}{c} \int \frac{1}{cx+d} dx = (a/c)x - \left(\frac{ad-bc}{c^2}\right) \log |cx+d| + c_1$$

Example 11: $\int \tan 2x \cdot \tan 3x \cdot \tan 5x \cdot dx$

Solution: We have $\tan 5x = \tan (2x + 3x)$

$$= \frac{\tan 2x + \tan 3x}{1 - \tan 2x \cdot \tan 3x}$$

$$\Rightarrow \tan 5x - \tan 2x \cdot \tan 3x \cdot \tan 5x = \tan 2x + \tan 3x$$

$$\Rightarrow \tan 2x \cdot \tan 3x \cdot \tan 5x = \tan 5x - \tan 2x - \tan 3x$$

$$\begin{aligned}
\therefore \int \tan 2x \cdot \tan 3x \cdot \tan 5x \cdot dx &= \int (\tan 5x - \tan 2x - \tan 3x) dx \\
&= \frac{1}{5} \log |\sec 5x| - \frac{1}{2} \log |\sec 2x| - \frac{1}{3} \log |\sec 3x| + c
\end{aligned}$$

Example 12: $\int \frac{1}{\cos(x-a) \cdot \cos(x-b)} dx$

We have,

$$\begin{aligned} I &= \int \frac{1}{\cos(x-a) \cdot \cos(x-b)} dx \\ I &= \frac{1}{\sin(a-b)} \int \frac{\sin(a-b)}{\cos(x-a) \cdot \cos(x-b)} dx \\ I &= \frac{1}{\sin(a-b)} \int \frac{\sin\{(x-b)-(x-a)\}}{\cos(x-a) \cos(x-b)} \\ I &= \frac{1}{\sin(a-b)} \int [\tan(x-b) - \tan(x-a)] dx \\ &= \frac{1}{\sin(a-b)} [-\log_e |\cos(x-b)| + \log_e |\cos(x-a)| + c \end{aligned}$$

Example 13: $\int \frac{\cos 5x + \cos 4x}{1 - 2 \cos 3x} dx$

Solution: We have,

$$\begin{aligned} I &= \int \frac{\sin 3x (\cos 5x + \cos 4x)}{\sin 3x - 2 \sin 3x \cdot \cos 3x} \cdot dx \\ &= \int \frac{\left(2 \sin \frac{3x}{2} \cdot \cos \frac{3x}{2}\right) \left(2 \cos \frac{9x}{2} \cdot \cos \frac{x}{2}\right)}{\sin 3x - \sin 6x} dx \\ &= 4 \int \frac{\sin \frac{3x}{2} \cdot \cos \frac{3x}{2} \cdot \cos \frac{9x}{2} \cos \frac{x}{2}}{-2 \cos \frac{9x}{2} \cdot \sin \frac{3x}{2}} \\ &= - \int 2 \cos \frac{3x}{2} \cdot \cos \frac{x}{2} \cdot dx \\ &= - \int (\cos 2x + \cos x) dx \\ &= - \left(\frac{\sin 2x}{2} + \sin x \right) + c \end{aligned}$$

Example 14: $\int \frac{\tan x}{a + b \tan^2 x} dx$

Solution: We have, $I = \int \frac{\tan x}{a + b \tan^2 x} dx = \int \frac{\sin x \cdot \cos x}{a \cos^2 x + b \sin^2 x} dx$

We observe that $\sin x \cdot \cos x$ appears in the differentiation of $a \cos^2 x + b \sin^2 x$, so we substitute $a \cos^2 x + b \sin^2 x = t$

$$\Rightarrow d(a \cos^2 x + b \sin^2 x) = dt$$

$$\Rightarrow \frac{d}{dx} (a \cos^2 x + b \sin^2 x) \cdot dx = dt$$

$$\Rightarrow (-2a \cos x \cdot \sin x + 2b \sin x \cdot \cos x) dx = dt \Rightarrow dx = \frac{dt}{2(b-a) \sin x \cdot \cos x}$$

$$\begin{aligned}
 \therefore \int \frac{\tan x}{a + b \tan^2 x} dx &= \int \frac{\sin x \cdot \cos x}{a \cos^2 x + b \sin^2 x} dx \\
 &= \int \frac{\sin x \cdot \cos x}{t} \cdot \frac{dt}{2(b-a) \cdot \sin x \cdot \cos x} \\
 &= \frac{1}{2(b-a)} \int \frac{1}{t} dt \\
 &= \frac{1}{2(b-a)} \log |t| + c = \frac{1}{2(b-a)} \log |a \cos^2 x + b \sin^2 x| + c
 \end{aligned}$$

Example 15: $\int \frac{1}{\sqrt{\sin^3 x \cdot \sin(x+a)}} dx$

Solution: We have,

$$\begin{aligned}
 I &= \int \frac{1}{\sqrt{\sin^3 x \sin(x+a)}} dx \\
 &= \int \frac{1}{\sqrt{\sin^3 x (\sin x \cdot \cos a + \cos x \cdot \sin a)}} dx \\
 &= \int \frac{1}{\sqrt{\sin^4 x (\cos a + \cot x \cdot \sin a)}} dx \\
 &= \int \frac{\operatorname{cosec}^2 x}{\sqrt{\cos a + \cot x \cdot \sin a}} dx \\
 &= -\frac{1}{\sin a} \int \frac{1}{\sqrt{\cos a + \cot x \cdot \sin a}} d(\cos a + \cot x \sin a) \\
 &= -\frac{1}{\sin a} \int \frac{1}{\sqrt{t}} dt \quad \text{where, } t = \cos a + \cot x \cdot \sin a \\
 &= -\frac{2}{\sin a} \sqrt{\cos a + \cot x \cdot \sin a} + c
 \end{aligned}$$

Example 16: $\int \frac{1}{x^6 + x^4} \cdot dx$

Solution: We have

Let $I = \int \frac{1}{x^6 + x^4} dx = \int \frac{1}{x^4(x^2 + 1)} dx$

On putting $x = \tan \theta$ and $dx = \sec^2 \theta \cdot d\theta$

we get;

$$\begin{aligned}
 I &= \int \frac{1}{\tan^4 \theta (1 + \tan^2 \theta)} \sec^2 \theta \cdot d\theta \\
 I &= \int \cot^4 \theta \cdot d\theta \\
 &= \int \cot^2 \theta (\operatorname{cosec}^2 \theta - 1) \cdot d\theta \\
 &= \int \cot^2 \theta (\operatorname{cosec}^2 \theta) d\theta - \int \cot^2 \theta d\theta
 \end{aligned}$$

$$\begin{aligned}
 &= - \int \cot^2 \theta \cdot d(\cot \theta) - \int (\operatorname{cosec}^2 \theta - 1) d\theta \\
 &= -\frac{\cot^3 \theta}{3} + \cot \theta + \theta + c = -\frac{1}{3x^3} + \frac{1}{x} + \tan^{-1} x + c
 \end{aligned}$$

Example 17: $\int \frac{\sin^4 x}{\cos^8 x} \cdot dx$

Let
$$I = \int \frac{\sin^4 x}{\cos^8 x} \cdot dx$$

$$I = \int \frac{\sin^4 x}{\frac{\cos^8 x}{\cos^4 x}} dx \quad [\text{Dividing } N^r, D^r \text{ by } \cos^4 x]$$

$$I = \int \tan^4 x \cdot \sec^4 x \cdot dx$$

$$= \int \tan^4 x (1 + \tan^2 x) \cdot \sec^2 x \cdot dx$$

$$= \int \tan^4 x (1 + \tan^2 x) d(\tan x)$$

$$= \int t^4 (1 + t^2) dt, \quad [\text{Put } t = \tan x]$$

$$= \frac{t^5}{5} + \frac{t^7}{7} + c = \frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} + c$$

13.5 Some Special Integral

- (i) $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} (x/a) + c$
- (ii) $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c$
- (iii) $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + c$
- (iv) $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} (x/a) + c$
- (v) $\int \frac{dx}{\sqrt{a^2 + x^2}} = \log |x + \sqrt{x^2 + a^2}| + c$
- (vi) $\int \frac{dx}{\sqrt{x^2 - a^2}} = \log |x + \sqrt{x^2 - a^2}| + c$
- (vii) $\int \sqrt{a^2 - x^2} dx = \frac{1}{2} x \sqrt{a^2 - x^2} + \frac{1}{2} a^2 \sin^{-1} (x/a) + c$
- (viii) $\int \sqrt{a^2 + x^2} dx = \frac{1}{2} x \sqrt{a^2 + x^2} + \frac{1}{2} a^2 \log |x + \sqrt{a^2 + x^2}| + c$
- (ix) $\int \sqrt{x^2 - a^2} dx = \frac{1}{2} x \sqrt{x^2 - a^2} - \frac{1}{2} a^2 \log |x + \sqrt{x^2 - a^2}| + c$

13.6 Some Important Substitution

	Expression	Substitution
(i)	$a^2 + x^2$	$x = a \tan \theta$ or $a \cot \theta$
(ii)	$a^2 - x^2$	$x = a \sin \theta$ or $a \cos \theta$
(iii)	$x^2 - a^2$	$x = a \sec \theta$ or $a \operatorname{cosec} \theta$
(iv)	$\sqrt{\frac{a-x}{a+x}}$ or $\sqrt{\frac{a+x}{a-x}}$	$x = a \cos 2\theta$
(v)	$\sqrt{\frac{x-\alpha}{\beta-x}}$ or $\sqrt{(x-\alpha)(x-\beta)}$	$x = \alpha \cos^2 \theta + \beta \sin^2 \theta$

13.6.1 Application of these Formulae

Type (A) The standard integrals such as:

$$\begin{aligned}
 \text{(i)} \quad & \int \frac{dx}{ax^2 + bx + c} & \text{(ii)} \quad & \int \frac{dx}{\sqrt{ax^2 + bx + c}} \\
 \text{(iii)} \quad & \int \sqrt{ax^2 + bx + c} \, dx
 \end{aligned}$$

Here, $ax^2 + bx + c$ can be factorized;

$$ax^2 + bx + c = a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right) = a \left\{ \left(x + \frac{b}{2a} \right)^2 + \left(\frac{c}{a} \right) - \frac{b^2}{4a^2} \right\}$$

Make the substitution $x + \frac{b}{2a} = t$

Type (B)

$$\begin{aligned}
 \text{(i)} \quad & \int \frac{(px + q) \, dx}{ax^2 + bx + c} & \text{(ii)} \quad & \int \frac{(px + q)}{\sqrt{ax^2 + bx + c}} \, dx \\
 \text{(iii)} \quad & \int (px + q) \sqrt{ax^2 + bx + c} \, dx
 \end{aligned}$$

The linear factor $(px + q)$ is expressed in terms of the derivative of the quadratic factor $ax^2 + bx + c$ together with a constant as:

$$px + q = \lambda \frac{d}{dx} (ax^2 + bx + c) + \mu$$

$$\Rightarrow px + q = \lambda (2x + b) + \mu$$

Here, we have to find λ and μ and replace $(px + q)$ by $\{\lambda (2ax + b) + \mu\}$ in (i), (ii) and (iii).

Illustrative Examples

Example 1: Evaluate $\int \frac{dx}{\sqrt{x^2 + a^2}}$

Solution: Put $x = a \tan \theta \Rightarrow dx = a \sec^2 \theta \cdot d\theta$

$$\begin{aligned} \therefore \int \frac{dx}{\sqrt{x^2 + a^2}} &= \int \frac{a \sec^2 \theta \cdot d\theta}{\sqrt{a^2 \tan^2 \theta + a^2}} \\ &= \int \frac{a \sec^2 \theta}{a \sec \theta} \cdot d\theta \\ &= \int \sec \theta \cdot d\theta \\ &= \log (\sec \theta + \tan \theta) + c' \\ &= \log (\tan \theta + \sqrt{(\tan^2 \theta + 1)}) + c' \\ &= \log (x/a + \sqrt{x^2/a^2 + 1}) + c' \\ &= \log \left(\frac{x + \sqrt{x^2 + a^2}}{a} \right) + c' \\ &= \log \left(x + \sqrt{x^2 + a^2} \right) - \log a + c' \end{aligned}$$

$$\therefore I = \int \frac{dx}{\sqrt{x^2 + a^2}} = \log(x + \sqrt{x^2 + a^2}) + c \text{ where } c = c' - \log a$$

Example 2: Evaluate $\int \frac{dx}{\sqrt{x^2 - a^2}}$

Solution: Let $I = \int \frac{dx}{\sqrt{x^2 - a^2}}$

Put $x = a \sec \theta \Rightarrow dx = a \sec \theta \cdot \tan \theta \cdot d\theta$

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2 - a^2}} &= \int \frac{a \sec \theta \cdot \tan \theta}{\sqrt{a^2 \sec^2 \theta - a^2}} d\theta \\ &= \int \frac{\sec \theta \cdot \tan \theta \cdot d\theta}{\sqrt{\sec^2 \theta - 1}} \\ &= \int \frac{\sec \theta \cdot \tan \theta}{\tan \theta} \cdot d\theta \\ &= \int \sec \theta \cdot d\theta \\ &= \log (\sec \theta + \tan \theta) + c' \\ &= \log (\sec \theta + \sqrt{\sec^2 \theta - 1}) + c' \\ &= \log (x/a + \sqrt{x^2/a^2 - 1}) + c' \end{aligned}$$

$$\begin{aligned}
 &= \log \left(\frac{x + \sqrt{x^2 - a^2}}{a} \right) + c' \\
 &= \log (x + \sqrt{x^2 - a^2}) - \log a + c' \\
 &= \log (x + \sqrt{x^2 - a^2}) + c
 \end{aligned}$$

where $c = c' - \log a$

Hence,
$$I = \int \frac{dx}{\sqrt{x^2 - a^2}} = \log \{x + \sqrt{x^2 - a^2}\} + c$$

Evaluate the following integrals

Example 3: Evaluate $\int \frac{dx}{\sqrt{(9 + 4x^2)}}$

Solution: Let
$$\begin{aligned}
 I &= \int \frac{dx}{\sqrt{(9 + 4x^2)}} \\
 &= \frac{1}{2} \int \frac{dx}{\sqrt{x^2 + (3/2)^2}} \\
 &= \frac{1}{2} \log \{x + \sqrt{x^2 + (3/2)^2}\} + c \\
 &= \frac{1}{2} \log \left\{ x + \sqrt{x^2 + \frac{9}{4}} \right\} + c
 \end{aligned}$$

Example 4: Evaluate $\int \frac{dx}{x \sqrt{(\log x)^2 - 10}}$

Solution: Let
$$I = \int \frac{dx}{x \sqrt{(\log x)^2 - 10}}$$

Put, $\log x = t \Rightarrow \frac{1}{x} \cdot dx = dt$

$$\begin{aligned}
 I &= \int \frac{dt}{\sqrt{t^2 - 10}} = \log (t + \sqrt{t^2 - 10}) + c \\
 &= \log [\log x + \sqrt{(\log x)^2 - 10}] + c
 \end{aligned}$$

Example 5: Evaluate $\int \frac{1}{\sqrt{1 - e^{-2x}}} dx$

Solution: Let
$$I = \int \frac{dx}{\sqrt{1 - e^{-2x}}} = \int \frac{e^x}{\sqrt{e^{2x} - 1}} dx$$

Put $e^x = t \Rightarrow e^x \cdot dx = dt$

$$\int \frac{1}{\sqrt{1 - e^{-2x}}} dx = \int \frac{dt}{\sqrt{t^2 - 1}} = \log (t + \sqrt{t^2 - 1}) + c = \log (e^x + \sqrt{e^{2x} - 1}) + c$$

Example 6: Evaluate $\int \frac{x+1}{\sqrt{x^2+1}} dx$

Solution:

$$\begin{aligned}
 I &= \int \frac{x}{\sqrt{x^2+1}} dx + \int \frac{1}{\sqrt{x^2+1}} dx \\
 &= \int \frac{x}{\sqrt{x^2+1}} dx + \log(x + \sqrt{x^2+1}) + c \quad \dots(1)
 \end{aligned}$$

Now, put $x^2 + 1 = t^2 \Rightarrow 2x \cdot dx = 2t \cdot dt \Rightarrow x \cdot dx = t \cdot dt$

$$\therefore \int \frac{x}{\sqrt{x^2+1}} dx = \int \frac{t \cdot dt}{t} = \int dt = t = \sqrt{x^2+1}$$

From (1) we have $\int \frac{x+1}{\sqrt{x^2+1}} dx = \sqrt{x^2+1} + \log(x + \sqrt{x^2+1}) + c$

Example 7: Evaluate $\int \sqrt{\frac{a+x}{a-x}} dx$

Solution: Let $I = \int \sqrt{\frac{a+x}{a-x}} dx$

Put $x = a \cos 2\theta$

$$\Rightarrow dx = -2a \sin 2\theta \cdot d\theta.$$

$$\begin{aligned}
 \int \sqrt{\frac{a+x}{a-x}} dx &= \int \sqrt{\frac{a(1+\cos 2\theta)}{a(1-\cos 2\theta)}} \cdot (-2a \sin 2\theta) \cdot d\theta \\
 &= -2a \int \sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}} \cdot \sin 2\theta \cdot d\theta \\
 &= -2a \int \sqrt{\frac{2\cos^2 \theta}{2\sin^2 \theta}} \cdot 2 \sin \theta \cdot \cos \theta \cdot d\theta \\
 &= -2a \int 2 \cos^2 \theta \cdot d\theta \\
 &= -2a \int (1 + \cos 2\theta) \cdot d\theta \\
 &= -2a \left[\theta + \frac{\sin 2\theta}{2} \right] + c \\
 &= -a \left[2\theta + \sqrt{1 - \cos^2 2\theta} \right] + c \\
 &= -a \left[\cos^{-1}(x/a) + \sqrt{1 - (x^2/a^2)} \right] + c = -a \cos^{-1}(x/a) - \sqrt{a^2 - x^2} + c
 \end{aligned}$$



Comprehensive Exercise 13.3

Integrate the following functions w.r.t. 'x':

1. (i) $x \sin x^2$ (ii) $x^2 \cos x^3$ (iii) $x^{n-1} \sin x^n$
2. (i) xe^{x^2} (ii) xa^{x^2} (iii) $\frac{x^{P-1}}{2+x^P}$
3. (i) $\sin^2 x \cdot \cos x$ (ii) $\cos^2 x \cdot \sin x$ (iii) $\sin^n x \cdot \cos x$
4. (i) $\frac{4 \sin^{-1} x}{\sqrt{1-x^2}}$ (ii) $\frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}}$ (iii) $\frac{(a+b \sin^{-1} x)^n}{\sqrt{1-x^2}}$
 (iv) $\frac{1}{\sin^{-1} x \sqrt{1-x^2}}$
5. (i) $\frac{x \tan^{-1} x^2}{1+x^4}$ (ii) $\frac{x^2 \tan^{-1} x^3}{1+x^6}$ (iii) $\frac{\sin(2 \tan^{-1} x)}{1+x^2}$

Integration by substitution

6. (i) $\frac{2x}{1+x^2}$ (ii) $\frac{2x+5}{x^2+5x+8}$ (iii) $\frac{e^x}{a+e^x}$
 (iv) $\frac{e^x + e^{-x}}{e^x - e^{-x}}$
7. (i) $\frac{1-\sin x}{x+\cos x}$ (ii) $\frac{e^x - \sin x}{e^x + \cos x}$ (iii) $\frac{\sec^2 x}{1+\tan x}$
 (iv) $\frac{\operatorname{cosec}^2 x}{1+\cot x}$
8. (i) $\int \frac{dx}{3 \sin x - 4 \sin^3 x}$ (ii) $\int \frac{dx}{4 \cos^3 x - 3 \cos x}$ (iii) $\int \frac{1+\tan^2 x}{1-\tan^2 x} dx$
9. (i) $\int \frac{\cot(\log x)}{x} dx$ (ii) $\int \frac{\tan(1+\log x)}{x} dx$ (iii) $\int x^2 \sec x^3 \cdot dx$
 (iv) $\int \frac{x+\sin x}{1+\cos x} dx$

[B.C.A. (Meerut) 2008]



Answers

1.	(i) $-\frac{1}{2} \cos x^2 + c$	(ii) $\frac{1}{3} \sin x^3 + c$	(iii) $-\frac{1}{n} \cos x^n + c$
2.	(i) $\frac{1}{2} e^{x^2} + c$	(ii) $\frac{1}{2} \cdot \frac{a^{x^2}}{\log a} + c$	(iii) $\frac{1}{P} \log(2 + x^P) + c$
3.	(i) $\frac{1}{3} \sin^3 x + c$	(ii) $-\frac{1}{3} \cos^3 x + c$	(iii) $\frac{1}{n+1} \sin^{n+1} x + c$
4.	(i) $2 (\sin^{-1} x)^2 + c$	(ii) $\frac{1}{4} (\sin^{-1} x)^4 + c$	(iii) $\frac{(a + b \sin^{-1} x)^{n+1}}{b(n+1)} + c$
	(iv) $\log(\sin^{-1} x) + c$		
5.	(i) $\frac{1}{4} (\tan^{-1} x^2)^2 + c$	(ii) $\frac{1}{6} (\tan^{-1} x^3)^2 + c$	(iii) $-\frac{1}{2} \cos(2 \tan^{-1} x) + c$
6.	(i) $\log(1 + x^2) + c$	(ii) $\log(x^2 + 5x + 8) + c$	(iii) $\log(a + e^x) + c$
	(iv) $\log(e^x - e^{-x}) + c$		
7.	(i) $\log(x + \cos x) + c$	(ii) $\log(e^x + \cos x) + c$	(iii) $\log(1 + \tan x) + c$
	(iv) $-\log(1 + \cot x) + c$		
8.	(i) $\frac{1}{3} \log \tan \frac{3x}{2} + c$	(ii) $\frac{1}{3} \log(\sec 3x + \tan 3x) + c$	
	(iii) $\frac{1}{2} \log \tan \left(\frac{\pi}{4} + x \right) + c$		
9.	(i) $\log \sin(\log x) + c$	(ii) $\log \sec(1 + \log x) + c$	
	(iii) $\frac{1}{3} \log(\sec^3 x + \tan^3 x) + c$	(iv) $x \tan \frac{x}{2} + c$	

13.6.2 Integration by Parts

Theorem: If u and v are two functions of x :

then; $\int uv \, dx = u \left(\int v \, dx \right) - \int \left(\frac{du}{dx} \right) \int v \, dx \cdot dx$

i.e. The integral of product of two functions

= (first function) $\times$ (Integral of second function)

– Integral of [differential of Ist function $\times$ Integral of IInd function]

Proof: For any two functions $f(x)$ and $g(x)$, we have

$$\frac{d}{dx} \{f(x) \cdot g(x)\} = f(x) \cdot \frac{d}{dx} \{g(x)\} + g(x) \cdot \frac{d}{dx} \{f(x)\}$$

$$\therefore \int \left[(f(x)) \cdot \frac{d}{dx} \{g(x)\} + g(x) \cdot \frac{d}{dx} \{f(x)\} \right] dx = f(x) \cdot g(x)$$

Note: Thus, this result or rule is called as ILATE

Integral of the form $\int f(ax + b) dx$

Theorem: If $\int f(x) dx = \phi(x)$, then $\int f(ax + b) dx = \frac{1}{a} \phi(ax + b)$

Proof: Let $I = \int f(ax + b) dx$. Putting $ax + b = t \Rightarrow adx = dt$ or $dx = \frac{1}{a} dt$

$$\begin{aligned} \therefore I &= \int f(ax + b) dx \\ &= \frac{1}{a} \int f(t) \cdot dt = \frac{1}{a} \phi(t) = \frac{1}{a} \phi(ax + b) \end{aligned}$$

Result 1: Prove that $\int (ax + b)^n \cdot dx = \frac{(ax + b)^{n+1}}{a(n+1)} + c$, $n \neq -1$

Proof: Put $ax + b = t$, we get $adx = dt$ or $dx = \frac{1}{a} dt$

$$\therefore \int (ax + b)^n \cdot dx = \frac{1}{a} \int t^n \cdot dt = \frac{1}{a} \cdot \frac{t^{n+1}}{n+1} + c$$

$$\text{or} \quad \int (ax + b)^n dx = \frac{1}{a} \cdot \frac{(ax + b)^{n+1}}{n+1} + c$$

Result 2: Prove that $\int \frac{1}{ax + b} dx = \frac{1}{a} \log |ax + b| + c$

Proof: $\therefore$ Put $ax + b = t$, we get $adx = dt$ or $dx = \frac{1}{a} dt$

$$\therefore \int \frac{1}{ax + b} dx = \frac{1}{a} \int \frac{1}{t} \cdot dt = \frac{1}{a} \log |t| + c$$

$$\text{or} \quad \int \frac{1}{ax + b} \cdot dx = \frac{1}{a} \log |ax + b| + c$$

Result 3: Prove that $\int \sin(ax + b) dx = -\frac{1}{a} \cdot \cos(ax + b) + c$

Proof: Put $ax + b = t$ so that $dx = \frac{1}{a} dt$

$$\therefore \int \sin(ax + b) dx = \frac{1}{a} \int \sin t \cdot dt = -\frac{1}{a} \cos t + c$$

$$\text{or} \quad \int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + c$$

on comparing these three results with $\int x^n \cdot dx = \frac{x^{n+1}}{n+1} + c$, $n \neq -1$,

$$\int \frac{1}{x} \cdot dx = \log |x| + c \text{ and } \int \sin x \cdot dx = -\cos x + c$$

Trick using here ILATE stand for

I → Inverse trigonometric function

L → Logarithmic function

A → Algebraic function

T → Trigonometric function

E → Exponential function

Illustrative Examples

Example 1: Evaluate:

$$(i) \quad \int x \cos x \, dx$$

$$(ii) \quad \int x \cdot \sec^2 x \, dx$$

$$(iii) \quad \int x \log x \, dx$$

$$(iv) \quad \int x^2 \sin x \, dx$$

$$(v) \quad \int x \cdot \cos^2 x \, dx$$

Solution:

$$(i) \quad \text{Let } I = \int_I x \cdot \cos x \, dx$$

$$\begin{aligned} I &= x \int \cos x \, dx - \int \left[\frac{d}{dx}(x) \cdot \int \cos x \, dx \right] dx \\ &= x \sin x - \int 1 \cdot \sin x \, dx = x \sin x + \cos x + c \end{aligned}$$

$$(ii) \quad \text{Let } I = \int_I x \cdot \sec^2 x \, dx$$

$$\begin{aligned} I &= x \int \sec^2 x \, dx - \int \left[1 \cdot \int (\sec^2 x \, dx) \right] dx \\ &= x \tan x - \int \tan x \, dx = x \tan x - (-\log \cos x) + c \\ I &= x \tan x + \log \cos x + c. \end{aligned}$$

$$(iii) \quad \text{Let } I = \int_I x \log x \, dx$$

$$\begin{aligned} &= (\log x) \int x \, dx - \int \left\{ \frac{d}{dx}(\log x) \cdot \int x \, dx \right\} dx \\ &= (\log x) \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} \, dx = \frac{x^2}{2} \log x - \frac{1}{2} \int x \, dx \\ I &= \frac{x^2}{2} \log x - \frac{1}{2} \cdot \frac{x^2}{2} + c \end{aligned}$$

$$(iv) \quad \text{Let } I = \int_I x^2 \sin x \, dx$$

$$\int x^2 \sin x \, dx = x^2 \int \sin x \, dx - \int \left\{ \frac{d}{dx}(x^2) \int \sin x \, dx \right\} dx$$

$$\begin{aligned}
&= x^2 (-\cos x) - \int 2x (-\cos x) dx \\
&= -x^2 \cos x + 2 \int x \cos x dx \\
&= -x^2 \cos x + 2 [x \int \cos x dx - \int \{1 \int \cos x dx\} dx] \\
&= -x^2 \cos x + 2 [x \sin x - \int \sin x dx] \\
&= -x^2 \cos x + 2 [x \sin x + \cos x] + c
\end{aligned}$$

$$(v) \int x \cos^2 x \cdot dx$$

Let

$$\begin{aligned}
I &= \int x \cdot \cos^2 x \cdot dx \\
&= \frac{1}{2} \int x (2 \cos^2 x) dx = \frac{1}{2} \int x(1 + \cos 2x) dx \\
&= \frac{1}{2} \left[\int x dx + \int x \cos 2x \cdot dx \right] \\
&= \frac{1}{2} \left[\frac{x^2}{2} + x \int \cos 2x \cdot dx - \int \{1 \cdot \int \cos 2x \cdot dx\} dx \right] \\
&= \frac{1}{2} \left[\frac{x^2}{2} + \frac{x \cdot \sin 2x}{2} - \int \frac{\sin 2x}{2} \cdot dx \right] \\
&= \frac{1}{2} \left[\frac{x^2}{2} + \frac{x \sin 2x}{2} - \frac{1}{2} \left(-\frac{\cos 2x}{2} \right) \right] + c \\
I &= \frac{1}{4} \left[x^2 + x \sin 2x + \frac{1}{2} \cos 2x \right] + c
\end{aligned}$$

Example 2: Evaluate:

$$(i) \int x \cdot \tan^{-1} x \cdot dx$$

Let

$$\begin{aligned}
I &= \int x \cdot \tan^{-1} x \cdot dx \\
&= (\tan^{-1} x) \cdot \frac{x^2}{2} - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2} \cdot dx \\
&= \frac{x^2}{2} \cdot \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \cdot dx \\
&= \frac{x^2}{2} \cdot \tan^{-1} x - \frac{1}{2} \int \frac{x^2 + 1 - 1}{x^2 + 1} \cdot dx \\
&= \frac{x^2}{2} \cdot \tan^{-1} x - \frac{1}{2} \left[\int 1 \cdot dx - \int \frac{1}{1+x^2} \cdot dx \right] \\
&= \frac{x^2}{2} \cdot \tan^{-1} x - \frac{1}{2} [x - \tan^{-1} x] + c \\
I &= \frac{1}{2} (x^2 + 1) \tan^{-1} x - \frac{x}{2} + c
\end{aligned}$$

$$\begin{aligned}
 \text{(ii) Let } I &= \int \frac{\log x}{x^2} \cdot dx = \int (\log x) \cdot \frac{1}{x^2} \cdot dx \\
 &= \log x \int \frac{1}{x^2} \cdot dx - \int \left[\frac{1}{x} \cdot \int \frac{1}{x^2} \cdot dx \right] dx \\
 &= \log x \left(-\frac{1}{x} \right) - \int \frac{1}{x} \left(-\frac{1}{x} \right) dx \\
 &= -\frac{1}{x} \log x + \int x^{-2} \cdot dx = -\frac{1}{x} \cdot \log x + \left(-\frac{1}{x} \right) + c \\
 I &= -\frac{1}{x} (1 + \log x) + c
 \end{aligned}$$

Example 3: Evaluate:

$$\text{(i) } \int \log x \, dx$$

$$\text{(ii) } \int \tan^{-1} x \, dx$$

$$\text{(iii) } \int \cos^{-1} \left(\frac{1}{x} \right) \cdot dx$$

Solution:

$$\text{(i) Let } I = \int \log x \, dx$$

$$\begin{aligned}
 I &= \int (\log x) \cdot 1 \, dx \\
 &\quad \text{I} \quad \text{II} \\
 &= (\log x) \int 1 \cdot dx - \int \left\{ \frac{d}{dx} (\log x) \cdot \int 1 \, dx \right\} dx \\
 &= (\log x) \cdot x - \int \frac{1}{x} \cdot x \, dx = x \log x - \int 1 \cdot dx \\
 I &= x \log x - x + c
 \end{aligned}$$

$$\text{(ii) Let}$$

$$\begin{aligned}
 I &= \int (\tan^{-1} x) \cdot 1 \, dx \\
 &\quad \text{I} \quad \text{II} \\
 I &= (\tan^{-1} x) \cdot x - \int \left\{ \frac{d}{dx} (\tan^{-1} x) \cdot x \right\} dx \\
 &= x \cdot \tan^{-1} x - \int \frac{1}{1+x^2} \cdot x \cdot dx \\
 &= x \tan^{-1} x - \frac{1}{2} \int \frac{1}{t} \cdot dt, \text{ where } 1+x^2 = t, \quad 2x \cdot dx = dt, \quad x \, dx = \frac{1}{2} dt \\
 &= x \tan^{-1} x - \frac{1}{2} \log t + c \\
 I &= x \tan^{-1} x - \frac{1}{2} \log (1+x^2) + c
 \end{aligned}$$

$$\text{(iii) Let } I = \int \cos^{-1} \left(\frac{1}{x} \right) \cdot dx$$

$$\text{Let } \cos^{-1} \frac{1}{x} = \theta. \text{ i.e. } \frac{1}{x} = \cos \theta \Rightarrow x = \sec \theta \Rightarrow dx = \sec \theta \cdot \tan \theta \cdot d\theta$$

$$\therefore \int \cos^{-1} \left(\frac{1}{x} \right) dx = \int \theta \cdot \sec \theta \cdot \tan \theta \cdot d\theta$$

$$= \theta \int \sec \theta \cdot \tan \theta \cdot d\theta - \int 1 \cdot \left[\int \sec \theta \cdot \tan \theta \cdot d\theta \right] d\theta$$

$$\begin{aligned}
 &= \theta \cdot \sec \theta - \int \sec \theta \cdot d\theta \\
 &= \theta \cdot \sec \theta - \log (\sec \theta + \tan \theta) + c \\
 &= \theta \cdot \sec \theta - \log (\sec \theta + \sqrt{(\sec^2 \theta - 1)}) + c \\
 &= (\sec^{-1} x) \cdot x - \log (x + \sqrt{(x^2 - 1)}) + c \quad \because (\sec \theta = x) \\
 I &= x \sec^{-1} x - \log (x + \sqrt{x^2 - 1}) + c
 \end{aligned}$$

Example 4: Evaluate $\int \log \{x + \sqrt{x^2 + a^2}\} dx$

Solution: Let $I = \int \log (x + \sqrt{x^2 + a^2}) dx$

$$\begin{aligned}
 &= \int \log (x + \sqrt{a^2 + x^2}) dx \\
 &= \int (\log (x + \sqrt{a^2 + x^2})) 1 \cdot dx \\
 &= \{\log (x + \sqrt{a^2 + x^2})\} x - \int \left\{ \frac{d}{dx} \log (x + \sqrt{a^2 + x^2}) \right\} \cdot x dx \\
 &= x \log (x + \sqrt{a^2 + x^2}) - \int \frac{1}{x + \sqrt{a^2 + x^2}} \times \left\{ 1 + \frac{d}{dx} (a^2 + x^2)^{1/2} \right\} x \cdot dx \\
 &= x \log (x + \sqrt{a^2 + x^2}) - \int \frac{1}{x + \sqrt{a^2 + x^2}} \times \left\{ 1 + \frac{1}{2\sqrt{a^2 + x^2}} \cdot 2x \right\} x \cdot dx \\
 &= x \log (x + \sqrt{a^2 + x^2}) - \int \frac{1}{x + \sqrt{a^2 + x^2}} \cdot \frac{x + \sqrt{a^2 + x^2}}{\sqrt{a^2 + x^2}} \cdot x \cdot dx \\
 &= x \log (x + \sqrt{a^2 + x^2}) - \int \frac{x}{\sqrt{a^2 + x^2}} dx \\
 &= x \log (x + \sqrt{a^2 + x^2}) - \int \frac{t dt}{t} \quad [\text{Putting } a^2 + x^2 = t^2 \Rightarrow 2x \cdot dx = 2t \cdot dt] \\
 &= x \log (x + \sqrt{a^2 + x^2}) - \int dt \\
 &= x \log (x + \sqrt{a^2 + x^2}) - t + c \\
 I &= x \log (x + \sqrt{a^2 + x^2}) - \sqrt{a^2 + x^2} + c
 \end{aligned}$$



Comprehensive Exercise 13.4

Integrate the following functions with respect to x .

1. (i) xe^x (ii) $x^2 e^x$ (iii) $x \cdot \cos 2x$
 (iv) $x \sin ax$ (v) $x \sin x \cos x$ (vi) $x^2 \cos ax$
2. (i) $x^2 \cdot \log x$ (ii) $x^2 \cdot (\log x)^2$ (iii) $x \cdot \sec^2 \cdot 2x$
 (iv) $x \cdot \operatorname{cosec}^2 ax$ (v) $x \cdot \sin^2 \cdot x$

3. (i) $\sin^{-1} x$ (ii) $\cot^{-1} x$ (iii) $\sec^{-1} x$
4. (i) $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ (ii) $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ (iii) $\tan^{-1} \frac{x}{\sqrt{1-x^2}}$
5. (i) $\frac{x^2 \tan^{-1} x}{1+x^2}$ (ii) $\frac{\sin^{-1} x}{(1-x^2)^{3/2}}$ (iii) $\frac{\log \sec^{-1} x}{x\sqrt{x^2-1}}$
- (iv) $x^3 \cos x^2$ (v) $\frac{x}{1+\sin x}$ (vi) $\int \tan^{-1}(\sec x + \tan x) dx$

[B.C.A. (Meerut) 2003, 2004, 2009]



Answers

1.	(i) $e^x(x-1)+c$	(ii) $(x^2-2x+2)e^x+c$
	(iii) $\frac{1}{2}x \sin 2x + \frac{1}{4} \cos 2x + c$	(iv) $\frac{1}{a^2}(\sin ax - ax \cdot \cos ax) + c$
	(v) $-\frac{1}{4}x \cdot \cos 2x + \frac{1}{8} \sin 2x + c$	(vi) $\frac{x^2}{a} \sin ax + \frac{2x}{a^2} \cos ax - \frac{2}{a^3} \sin ax + c$
2.	(i) $\frac{1}{3}x^3 \log x - \frac{1}{9}x^3 + c$	(ii) $\frac{1}{3}x^3 \left[(\log x)^2 - (2/3) \log x + \frac{2}{9} \right] + c$
	(iii) $\frac{1}{2}x \tan 2x - \frac{1}{4} \log \sec 2x + c$	(iv) $\frac{1}{a^2} [\log \sin ax - ax \cdot \cot ax] + c$
	(v) $\frac{1}{4}x^2 - \frac{1}{4}x \sin 2x - \frac{1}{8} \cos 2x + c$	
3.	(i) $x \sin^{-1} x + \sqrt{1-x^2} + c$	(ii) $x \cot^{-1} x + \frac{1}{2} \log(1+x^2) + c$
	(iii) $x \sec^{-1} x - \log(x + \sqrt{x^2-1}) + c$	
4.	(i) $2[x \tan^{-1} x - \log \sqrt{1+x^2}] + c$	(ii) $2[x \cdot \tan^{-1} x - \frac{1}{2} \log(1+x^2)] + c$
	(iii) $x \sin^{-1} x + \sqrt{1-x^2} + c$	
5.	(i) $[x \tan^{-1} x - \log \sqrt{1+x^2}] - \frac{1}{2}(\tan^{-1} x)^2 + c$	
	(ii) $\frac{x \sin^{-1} x}{\sqrt{1-x^2}} + \frac{1}{2} \log(1-x^2) + c$	(iii) $(\sec^{-1} x) \log \left(\frac{\sec^{-1} x}{e} \right) + c$
	(iv) $\frac{1}{2}[x^2 \sin x^2 + \cos x^2] + c$	(v) $x(\tan x - \sec x) + \log(1+\sin x) + c$
	(vi) $[\tan^{-1}(\sec x + \tan x)]^2 + c$	

Illustrative Examples

(Substitution)

Example 1: Evaluate $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$

Solution: Let $I = \int \frac{x \cdot \sin^{-1} x}{\sqrt{1-x^2}} dx$ $\left[\text{Put } \sin^{-1} x = t \Rightarrow x = \sin t, \frac{1}{\sqrt{1-x^2}} dx = dt \right]$

$$\begin{aligned}
 I &= \int (\sin t) \cdot t \cdot dt \\
 &= \int t \cdot \sin t \cdot dt \\
 &= t(-\cos t) - \int 1(-\cos t) dt \\
 &= -t \cdot \cos t + \int \cos t \cdot dt \\
 &= -t \cdot \cos t + \sin t + c \\
 &= \sin t - t \cdot \sqrt{1-\sin^2 t} + c = x - (\sin^{-1} x) \cdot \sqrt{1-x^2} + c \\
 I &= x - (\sqrt{1-x^2}) \sin^{-1} x + c
 \end{aligned}$$

Example 2: Evaluate $\int \frac{\tan^{-1} x}{(1+x^2)^{3/2}} dx$

Solution: Let $I = \int \frac{\tan^{-1} x}{(1+x^2)^{3/2}} dx$ $[\theta = \tan^{-1} x]$

$$\begin{aligned}
 &= \int \frac{\theta}{(1+\tan^2 \theta)^{3/2}} \cdot \sec^2 \theta \cdot d\theta \\
 &= \int \frac{\theta}{\sec^3 \theta} \cdot \sec^2 \theta \cdot d\theta \\
 &= \int \theta \cos \theta \cdot d\theta \\
 &= \theta \cdot \sin \theta - \int 1 \cdot \sin \theta \cdot d\theta \\
 &= \theta \cdot \sin \theta + \cos \theta + c \\
 &= \theta \cdot \tan \theta \cdot \frac{1}{\sec \theta} + \frac{1}{\sec \theta} + c \\
 &= \theta \cdot \tan \theta \cdot \frac{1}{\sqrt{1+\tan^2 \theta}} + \frac{1}{\sqrt{1+\tan^2 \theta}} + c \\
 &= (\tan^{-1} x) \cdot x \cdot \frac{1}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+x^2}} + c \\
 I &= \frac{x}{\sqrt{1+x^2}} \tan^{-1} x + \frac{1}{\sqrt{1+x^2}} + c
 \end{aligned}$$

Example 3: Evaluate:

$$(i) \int \frac{x}{1 + \cos x} dx$$

$$(ii) \int \frac{(x-1)e^x}{(x+1)^3} dx$$

[B.C.A. (Kanpur) 2004, 2008; B.C.A. (Rohilkhand) 2006]

Solution: (i) Let $I = \int \frac{x}{1 + \cos x} dx$

$$I = \int \frac{x}{2 \cos^2 (x/2)} dx$$

$$= \int x \cdot \left(\frac{1}{2} \sec^2 (x/2) \right) dx$$

$$= x \cdot \int \frac{1}{2} \sec^2 \frac{x}{2} \cdot dx - \int \left[1 \cdot \int \left(\frac{1}{2} \sec^2 (x/2) \right) dx \right] dx$$

$$= x \cdot \tan (x/2) - \int \tan (x/2) \cdot dx$$

$$= x \cdot \tan (x/2) - 2 \int \tan t \cdot dt \text{ where } \frac{x}{2} = t = x \cdot \tan (x/2) - 2 \log \sec t + c$$

$$I = x \cdot \tan (x/2) - 2 \log \sec (x/2) + c$$

$$(ii) \quad \text{Let} \quad I = \int \frac{(x-1) e^x \cdot dx}{(x+1)^3}$$

$$I = \int (x-1) e^x \cdot (x+1)^{-3} dx$$

$$= (x-1) e^x \left[\frac{(x+1)^{-3+1}}{-2} \right] - \int \frac{d}{dx} \{ (x-1) e^x \} \cdot \left[\frac{(x+1)^{-2}}{-2} \right] dx$$

$$= -\frac{1}{2} (x-1) e^x \cdot (x+1)^{-2} + \frac{1}{2} \int \{ (x-1) e^x + e^x \cdot 1 \} (x+1)^{-2} dx$$

$$= -\frac{1}{2} (x-1)(x+1)^{-2} e^x + \frac{1}{2} \left[x e^x \frac{(x+1)^{-1}}{-1} - \int (x e^x + e^x) \cdot \frac{(x+1)^{-1}}{-1} \cdot dx \right]$$

$$= -\frac{1}{2} (x-1)(x+1)^{-2} e^x - \frac{1}{2} x e^x \cdot (x+1)^{-1} + \frac{1}{2} \int \frac{e^x (x+1)}{x+1} dx$$

$$= -\frac{(x-1) e^x}{2 (x+1)^2} - \frac{x e^x}{2 (x+1)} + \frac{1}{2} \int e^x \cdot dx$$

$$I = -\frac{(x-1) e^x}{2 (x+1)^2} - \frac{x e^x}{2 (x+1)} + \frac{1}{2} e^x + c$$

Illustrative Examples

(Cancellation of Integrals)

Example 1: Evaluate (i) $\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$ (ii) $\int e^x \sec x (1 + \tan x) dx$

Solution: (i) Let $I = \int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$

$$\begin{aligned}
 &= \int e^x \cdot \frac{1}{x} \cdot dx - \int e^x \cdot \frac{1}{x^2} \cdot dx \\
 &= \int \frac{1}{x} \cdot e^x \cdot dx - \int \frac{1}{x^2} \cdot e^x \cdot dx \\
 &= \frac{1}{x} e^x - \int \left(-\frac{1}{x^2} \right) \cdot e^x \cdot dx - \int \frac{1}{x^2} e^x dx + c \\
 I &= \frac{1}{x} e^x + c
 \end{aligned}$$

(ii) $I = \int e^x \cdot \sec x (1 + \tan x) dx$

$$\begin{aligned}
 &= \int e^x \sec x \cdot dx + \int e^x \cdot \sec x \cdot \tan x \cdot dx \\
 &= (\sec x) e^x - \int \frac{d}{dx} (\sec x) \cdot e^x dx + \int e^x \cdot \sec x \cdot \tan x \cdot dx + c \\
 &= e^x \cdot \sec x - \int (\sec x \tan x) \cdot e^x \cdot dx + \int e^x \cdot \sec x \cdot \tan x \cdot dx + c \\
 I &= e^x \cdot \sec x + c
 \end{aligned}$$

Example 2: Evaluate $\int \frac{x e^x}{(1+x)^2} dx$

Solution: Let

$$\begin{aligned}
 I &= \int \frac{x e^x}{(1+x)^2} \cdot dx \\
 I &= \int \frac{(1+x) - 1}{(1+x)^2} e^x \cdot dx \\
 &= \int \left[\frac{1}{1+x} - \frac{1}{(1+x)^2} \right] \cdot e^x \cdot dx \\
 &= \int e^x \frac{1}{1+x} \cdot dx - \int \frac{1}{(1+x)^2} e^x \cdot dx \\
 &= \int \frac{1}{1+x} \cdot e^x \cdot dx - \int \frac{1}{(1+x)^2} \cdot e^x \cdot dx \\
 &= \frac{1}{1+x} \cdot e^x - \int \frac{d}{dx} \frac{1}{(1+x)} e^x \cdot dx - \int \frac{e^x}{(1+x)^2} dx + c \\
 &= \frac{e^x}{1+x} + \int \frac{1}{(1+x)^2} \cdot e^x \cdot dx - \int \frac{e^x}{(1+x)^2} dx + c \\
 I &= \frac{e^x}{1+x} + c
 \end{aligned}$$

Example 3: Evaluate $\int \frac{x + \sin x}{1 + \cos x} dx$.

[B.C.A. (Meerut) 2008]

Solution: Let

$$\begin{aligned}
 I &= \int \frac{x + \sin x}{1 + \cos x} \cdot dx \\
 &= \int \frac{x + 2 \sin(x/2) \cdot \cos(x/2)}{2 \cos^2(x/2)} \cdot dx \\
 &= \int \left[\frac{x}{2 \cos^2(x/2)} + \frac{2 \sin(x/2) \cdot \cos(x/2)}{2 \cos^2(x/2)} \right] \cdot dx \\
 &= \int x \cdot \frac{1}{2} \sec^2(x/2) dx + \int \tan(x/2) \cdot dx \\
 &= x \cdot \tan(x/2) - \int 1 \cdot \tan(x/2) dx + \int \tan(x/2) dx + c \\
 I &= x \tan(x/2) + c
 \end{aligned}$$

Example 4: Evaluate $\int \frac{(x - \sin x)}{(1 - \cos x)} dx$

Solution: Let

$$\begin{aligned}
 I &= \int \frac{(x - \sin x)}{(1 - \cos x)} \cdot dx \\
 &= \int \frac{x - 2 \sin(x/2) \cdot \cos(x/2)}{2 \sin^2(x/2)} \cdot dx \\
 &= \int \left[\frac{x}{2 \sin^2(x/2)} - \frac{2 \sin(x/2) \cdot \cos(x/2)}{2 \sin^2(x/2)} \right] dx \\
 &= \int x \frac{1}{2} \operatorname{cosec}^2(x/2) dx - \int \cot(x/2) dx \\
 &= x(-\cot x/2) - \int 1 \cdot (-\cot x/2) dx - \int \cot(x/2) \cdot dx + c \\
 &= -x \cdot \cot(x/2) + \int \cot(x/2) \cdot dx - \int \cot(x/2) \cdot dx + c \\
 I &= -x \cdot \cot(x/2) + c
 \end{aligned}$$

Example 5: Evaluate: $\int \sin(\log x) dx$

Solution: Let

$$\begin{aligned}
 I &= \int \sin(\log x) \cdot 1 dx \\
 &= \sin(\log x) \cdot x - \int \cos(\log x) \cdot \frac{1}{x} \cdot x \cdot dx \\
 &= x \sin(\log x) - \int \cos(\log x) \cdot dx \\
 &= x \sin(\log x) - \int \cos(\log x) \cdot 1 dx \\
 &= x \sin(\log x) - \left[\cos(\log x) \cdot x - \int -\sin(\log x) \cdot \frac{1}{x} \cdot x \cdot dx \right] \\
 &= x \sin(\log x) - x \cos(\log x) - \int \sin(\log x) \cdot dx \\
 &= x \sin(\log x) - x \cos(\log x) - I
 \end{aligned}$$

$$\Rightarrow 2I = x \sin(\log x) - x \cos(\log x)$$

$$\Rightarrow I = \frac{x}{2} [\sin(\log x) - \cos(\log x)] + c$$

Example 6: Evaluate: $\int \sec^3 x \cdot dx$

Solution: Let $I = \int \sec^3 x \cdot dx$.

$$= \int \sec^2 x \cdot \sec x \cdot dx$$

$$= \sec x \cdot \tan x - \int \frac{d}{dx} (\sec x) \cdot \tan x \cdot dx$$

$$= \sec x \cdot \tan x - \int \sec x \cdot \tan x \cdot \tan x \cdot dx$$

$$= \sec x \cdot \tan x - \int \sec x (\sec^2 x - 1) dx$$

$$= \sec x \cdot \tan x - \int \sec^3 x dx + \int \sec x dx$$

$$\therefore 2I = \sec x \cdot \tan x + \log(\sec x + \tan x)$$

$$I = \frac{1}{2} \sec x \cdot \tan x + \frac{1}{2} [(\log(\sec x + \tan x))] + c$$

Example 7: Evaluate: $\int \frac{\sin(\log_e x)}{x^3} dx$

Solution: Let $I = \int \frac{\sin(\log_e x)}{x^3} dx$

$$\text{Let } \log_e x = t \Rightarrow x = e^t \Rightarrow dx = e^t \cdot dt$$

$$\therefore I = \int \frac{\sin(\log_e x)}{x^3} \cdot dx$$

$$= \int \frac{\sin t}{e^{3t}} \cdot e^t \cdot dt$$

$$= e^{-2t} (-\cos t) - \int -2e^{-2t} (-\cos t) \cdot dt$$

$$= -e^{-2t} \cos t - 2 \int e^{-2t} \cos t \cdot dt$$

$$= -e^{-2t} \cos t - 2 [e^{-2t} \cdot \sin t - \int -2 e^{-2t} \sin t dt]$$

$$= -e^{-2t} \cos t - 2e^{-2t} \sin t - 4 \int e^{-2t} \cdot \sin t \cdot dt$$

$$= -e^{-2t} (\cos t + 2 \sin t) - 4I$$

$$\Rightarrow 5I = -e^{-2t} (\cos t + 2 \sin t)$$

$$I = -\frac{1}{5} e^{-2t} (\cos t + 2 \sin t)$$

$$I = \frac{-1}{5x^2} \{ \cos (\log_e x) + 2 \sin (\log_e x) \} + c$$

Illustrative Examples

Three Standard Form of Integrals

(i) $\int \sqrt{a^2 - x^2} dx$

We have

$$\begin{aligned} I &= \int \sqrt{a^2 - x^2} \cdot 1 dx \\ &= \sqrt{a^2 - x^2} \cdot x - \int \frac{1}{2\sqrt{a^2 - x^2}} \cdot (-2x) \cdot x \cdot dx \\ &= x\sqrt{a^2 - x^2} + \int \frac{x^2}{\sqrt{a^2 - x^2}} dx \\ &= x\sqrt{a^2 - x^2} + \int \frac{-(a^2 - x^2) + a^2}{\sqrt{a^2 - x^2}} dx \\ &= x\sqrt{a^2 - x^2} - \int \frac{a^2 - x^2}{\sqrt{a^2 - x^2}} dx + a^2 \int \frac{dx}{\sqrt{a^2 - x^2}} \\ &= x\sqrt{a^2 - x^2} - \int \sqrt{a^2 - x^2} dx + a^2 \sin^{-1}(x/a) \end{aligned}$$

$$\Rightarrow 2 \int \sqrt{a^2 - x^2} dx = x\sqrt{a^2 - x^2} + a^2 \sin^{-1}(x/a) + c$$

$$\therefore \int \sqrt{a^2 - x^2} dx = \frac{1}{2} \left[x\sqrt{a^2 - x^2} + a^2 \sin^{-1}(x/a) \right] + c$$

(ii) $\int \sqrt{x^2 + a^2} \cdot dx$

Let

$$\begin{aligned} I &= \int \sqrt{x^2 + a^2} dx \\ &= \int \sqrt{x^2 + a^2} \cdot 1 \cdot dx \\ &= \sqrt{x^2 + a^2} \cdot x - \int \frac{1}{2\sqrt{x^2 + a^2}} 2x \cdot x \cdot dx \\ &= x\sqrt{x^2 + a^2} - \int \frac{x^2}{\sqrt{x^2 + a^2}} dx \\ &= x\sqrt{x^2 + a^2} - \int \frac{(x^2 + a^2) - a^2}{\sqrt{x^2 + a^2}} dx \\ &= x\sqrt{x^2 + a^2} - \int \frac{x^2 + a^2}{\sqrt{x^2 + a^2}} dx + a^2 \int \frac{dx}{\sqrt{x^2 + a^2}} \\ &= x\sqrt{x^2 + a^2} - \int \sqrt{x^2 + a^2} dx + a^2 \log(x + \sqrt{x^2 + a^2}) \end{aligned}$$

$$\Rightarrow 2 \int \sqrt{x^2 + a^2} dx = x\sqrt{x^2 + a^2} + a^2 \log(x + \sqrt{x^2 + a^2}) + c$$

$$\therefore \int \sqrt{x^2 + a^2} dx = \frac{1}{2} \left[x\sqrt{x^2 + a^2} + a^2 \log(x + \sqrt{x^2 + a^2}) \right] + c$$

$$(iii) \int \sqrt{x^2 - a^2} . dx$$

Let

$$I = \int \sqrt{x^2 - a^2} . dx$$

$$= \int \sqrt{x^2 - a^2} . 1 . dx$$

$$= \sqrt{x^2 - a^2} . x - \int \frac{1}{2\sqrt{x^2 - a^2}} . 2x . x . dx$$

$$= x\sqrt{x^2 - a^2} - \int \frac{x^2}{\sqrt{x^2 - a^2}} dx$$

$$= x\sqrt{x^2 - a^2} - \int \frac{(x^2 - a^2) + a^2}{\sqrt{x^2 - a^2}} dx$$

$$= x\sqrt{x^2 - a^2} - \int \sqrt{x^2 - a^2} dx - a^2 \int \frac{dx}{\sqrt{x^2 - a^2}}$$

$$= x\sqrt{x^2 - a^2} - \int \sqrt{x^2 - a^2} dx - a^2 \log(x + \sqrt{x^2 - a^2})$$

$$\Rightarrow 2 \int \sqrt{x^2 - a^2} dx = x\sqrt{x^2 - a^2} - a^2 \log(x + \sqrt{x^2 - a^2}) + c$$

$$\therefore I = \int \sqrt{x^2 - a^2} dx = \frac{1}{2} [x\sqrt{x^2 - a^2} - a^2 \log(x + \sqrt{x^2 - a^2})] + c.$$

Illustrative Examples

Example 1: Evaluate (i) $\int \cos x . \sqrt{4 - \sin^2 x} . dx$ (ii) $\int \frac{x^2 + 1}{\sqrt{x^2 + 3}} dx$

Solution: (i) Let $I = \int \cos x \sqrt{4 - \sin^2 x} dx$.

Putting $\sin x = t \Rightarrow \cos x . dx = dt$

$$I = \int \cos x . \sqrt{4 - \sin^2 x} . dx = \int \sqrt{4 - t^2} dt$$

$$= \int \sqrt{(2)^2 - t^2} . dt$$

$$= \frac{1}{2} \left[t . \sqrt{2^2 - t^2} + 2^2 \sin^{-1}(t/2) \right] + c$$

$$= \frac{1}{2} \left[\sin x \sqrt{4 - \sin^2 x} + 4 \sin^{-1} \left(\frac{\sin x}{2} \right) \right] + c$$

$$(ii) \int \frac{x^2+1}{\sqrt{x^2+3}} dx$$

$$\begin{aligned} \text{Let } I &= \int \frac{x^2+1}{\sqrt{x^2+3}} dx \\ &= \int \frac{(x^2+3)-2}{\sqrt{x^2+3}} dx \\ &= \int \sqrt{x^2+3} dx - 2 \int \frac{dx}{\sqrt{x^2+3}} \\ &= \frac{1}{2} \left[x\sqrt{x^2+3} + 3 \log(x + \sqrt{x^2+3}) \right] - 2 \log(x + \sqrt{x^2+3}) + c \\ &= \frac{1}{2} \left[x\sqrt{x^2+3} - \log(x + \sqrt{x^2+3}) \right] + c \end{aligned}$$

Example 2: Evaluate: $\int \sec x \cdot \tan x \sqrt{\tan^2 x - 4} \cdot dx$

Solution: Let $I = \int \sec x \cdot \tan x \sqrt{\tan^2 x - 4} \cdot dx$

$$\begin{aligned} &= \int \sec x \cdot \tan x \cdot \sqrt{(\sec^2 x - 1) - 4} \cdot dx \\ &= \int \sec x \cdot \tan x \sqrt{\sec^2 x - 5} dx \end{aligned}$$

Put $\sec x = t \Rightarrow \sec x \cdot \tan x \cdot dx = dt$

$$\begin{aligned} I &= \int \sqrt{t^2 - 5} \cdot dt \\ &= \int \sqrt{t^2 - (\sqrt{5})^2} dt \\ &= \frac{1}{2} \left[t \cdot \sqrt{t^2 - (\sqrt{5})^2} - (\sqrt{5})^2 \log(t + \sqrt{t^2 - (\sqrt{5})^2}) \right] + c \\ &= \frac{1}{2} \left[t \cdot \sqrt{t^2 - 5} - 5 \log(t + \sqrt{t^2 - 5}) \right] + c \\ &= \frac{1}{2} \left[\sec x \cdot \sqrt{\sec^2 x - 5} - 5 \log(\sec x + \sqrt{\sec^2 x - 5}) \right] + c \end{aligned}$$



Comprehensive Exercise 13.5

Integration By Parts

Evaluate the following:

1. (i) $\int e^x(x+1) dx$

(ii) $\int e^x(\cos x + \sin x) dx$

(iii) $\int e^x \sin x(\sin x + 2 \cos x) . dx$

2. (i) $\int e^x[e^{\log x} + 1] . dx$

(ii) $\int e^x[e^{2 \log \sec x} + \tan x] . dx$

3. (i) $\int e^x[\log \sin x + \cot x] . dx$

(ii) $\int \frac{e^x(1 - \sin x)}{1 - \cos x} dx$

4. $\int \frac{2x - \sin 2x}{1 - \cos 2x} . dx$

5. $\int \frac{e^x(x^2 + 1)}{(x+1)^2} dx$

6. $\int e^{\sin^{-1} x} dx$

[B.C.A. (Meerut) 2002]

7. Evaluate the following:

(i) $\int e^x . \cos x . dx$

(ii) $\int e^{-x} \cos x . dx$

8. $\int e^x . \sin x . \cos x dx$

9. $\int e^x . \sin^2 x . dx$

10. $\int \sec x . \tan^{-1} x . dx$

11. $\int \operatorname{cosec}^3 x . dx$

12. $\int \sqrt{9 - x^2} dx$

13. (i) $\int \sqrt{2 - 3x^2} dx$

(ii) $\int \left(\sqrt{16x^2 + 25} \right) dx$

14. $\int x^2 \sqrt{a^6 - x^6} dx$

15. $\int \sec x . \tan x . \sqrt{\sec^2 x + 1} dx$

16. $\int e^x . (e^{2x} + 1)^{1/2} . dx$



Answers

1.	(i) $xe^x + c$ (iii) $e^x \cdot \sin^2 x + c$	(ii) $e^x \sin x + c$
2.	(i) $xe^x + c$	(ii) $e^x \tan x + c$
3.	(i) $e^x \log \sin x + c$	(ii) $-e^x \cdot \cot \frac{x}{2} + c$
4.	$-x \cot x + c$	
5.	$\frac{e^x(x-1)}{(x+1)} + c$	
6.	$\frac{e^{\sin^{-1} x}}{2} \sqrt{1-x^2} + x$	
7.	(i) $\frac{1}{2} e^x (\sin x + \cos x) + c$	(ii) $\frac{e^{-x}}{2} (\sin x - \cos x) + c$
8.	$\frac{e^x}{10} (\sin 2x - 2 \cos 2x) + c$	
9.	$\frac{1}{2} e^x - \frac{1}{10} e^x (2 \sin 2x + \cos 2x) + c$	
10.	$\frac{1}{2} [x \sqrt{1+x^2} + \log(x + \sqrt{1+x^2})] + c$	
11.	$-\frac{1}{2} \operatorname{cosec} x \cdot \cot x + \frac{1}{2} \log \tan x/2 + c$	
12.	$\frac{1}{2} x \sqrt{9-x^2} + \frac{9}{2} \sin^{-1}(x/3) + c$	
13.	(i) $\sqrt{3} \left[\frac{1}{2} x \sqrt{(2/3)-x^2} + \frac{1}{3} \sin^{-1}(x\sqrt{3}/2) \right] + c$ (ii) $2x \cdot \sqrt{x^2 + (25/16)} + \frac{25}{8} \log(x - \sqrt{x^2 + (25/16)}) + c$	
14.	$\frac{1}{6} \left[x^3 \sqrt{a^6 - x^6} + \frac{a^6}{6} \sin^{-1} \left(\frac{x^3}{a^3} \right) \right] + c$	
15.	$\frac{1}{2} \left[\sec x + \sqrt{\sec^2 x + x + 1} + \log(\sec x + \sqrt{(\sec^2 x + 1)}) \right] + c$	
16.	$\frac{1}{2} \left[e^x \sqrt{e^{2x} + 1} + \log(e^x + \sqrt{e^{2x} + 1}) \right] + c$	

13.7 Integration (Continued)

13.7.1 Evaluation of $\int \frac{1}{x^2 - a^2} dx$

We have
$$\frac{1}{x^2 - a^2} = \frac{1}{(x+a)(x-a)}$$

Let
$$\frac{1}{x^2 - a^2} = \frac{A}{x+a} + \frac{B}{x-a}. \text{ Then,}$$

$$\frac{1}{(x+a)(x-a)} = \frac{A(x-a) + B(x+a)}{(x+a)(x-a)}$$

$$\Rightarrow A(x-a) + B(x+a) = 1$$

$$\Rightarrow (A+B)x + (B-A)a = 1$$

i.e. Equating the coefficients of x and constant terms on both sides, we have

$$A+B=0. \text{ and } (B-A)a=1.$$

solving these equations; we have;

$$B = \frac{1}{2a}, \text{ and } A = -\frac{1}{2a}.$$

$$\therefore \frac{1}{x^2 - a^2} = \frac{-1}{2a(x+a)} + \frac{1}{2a(x-a)} = \frac{1}{2a} \left[\frac{1}{x-a} - \frac{1}{x+a} \right]$$

$$\begin{aligned} \therefore \int \frac{1}{x^2 - a^2} dx &= \frac{1}{2a} \int \left[\frac{1}{x-a} - \frac{1}{x+a} \right] dx \\ &= \frac{1}{2a} \left[\int \frac{1}{x-a} dx - \int \frac{1}{x+a} dx \right] \\ &= \frac{1}{2a} [\log(x-a) - \log(x+a)] + c \end{aligned}$$

$$\Rightarrow \int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \frac{x-a}{x+a} + c \text{ when } x > a$$

13.7.2 Evaluate $\int \frac{1}{a^2 - x^2} dx$

$\therefore$ We have now;

$$\frac{1}{a^2 - x^2} = \frac{1}{(a+x)(a-x)}$$

Let
$$\frac{1}{a^2 - x^2} = \frac{A}{a+x} + \frac{B}{a-x}$$

$$\Rightarrow 1 = A(a-x) + B(a+x).$$

Putting $x = a$ in the above identity; we have $1 = A(a-a) + B(a+a) \Rightarrow B = \frac{1}{2a}.$

Again putting $x = -a$; we have; $1 = A(a+a) + B(a-a) \Rightarrow A = \frac{1}{2a}$

$$\begin{aligned}
 \therefore \quad \frac{1}{a^2 - x^2} &= \frac{1}{2a(a+x)} + \frac{1}{2a(a-x)} \\
 &= \frac{1}{2a} \left[\frac{1}{a+x} + \frac{1}{a-x} \right] \\
 \therefore \quad \int \frac{1}{a^2 - x^2} dx &= \frac{1}{2a} \int \left[\frac{1}{a+x} + \frac{1}{a-x} \right] dx \\
 &= \frac{1}{2a} \left[\int \frac{dx}{a+x} + \int \frac{dx}{a-x} \right] \\
 &= \frac{1}{2a} [\log(a+x) - \log(a-x)] + c \\
 \therefore \quad \int \frac{1}{a^2 - x^2} dx &= \frac{1}{2a} \log \frac{a+x}{a-x} + c, \text{ when } x < a
 \end{aligned}$$

Example 1: Evaluate: $\int \frac{3x}{(x-1)(x-2)(x-3)} dx$

Solution: Let $\frac{3x}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$

$$\Rightarrow 3x = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$$

Putting $x = 1, 2, 3$, successively on both the sides, we get;

$$3 \times 1 = A(1-2) \cdot (1-3) \Rightarrow A = \frac{3}{2}$$

$$3 \times 2 = B(2-1) \cdot (2-3) \Rightarrow B = -6$$

$$3 \times 3 = C(3-1) \cdot (3-2) \Rightarrow C = \frac{9}{2}$$

$$\therefore \quad \frac{3x}{(x-1)(x-2)(x-3)} = \frac{3}{2(x-1)} - \frac{6}{x-2} + \frac{9}{2(x-3)}$$

$$\therefore \quad \int \frac{3x}{(x-1)(x-2)(x-3)} dx = \frac{3}{2} \int \frac{dx}{x-1} - 6 \int \frac{dx}{x-2} + \frac{9}{2} \int \frac{dx}{x-3}$$

$$= \frac{3}{2} \log(x-1) - 6 \log(x-2) + \frac{9}{2} \log(x-3) + c$$

Example 2: Evaluate $\int \frac{1}{2x^2 + x - 1} dx$

Solution: $2x^2 + x - 1 = 2x^2 + 2x - x - 1 = 2x(x+1) - 1(x+1) = (x+1)(2x-1)$

Let $\frac{1}{2x^2 + x - 1} = \frac{A}{x+1} + \frac{B}{2x-1} = \frac{A(2x-1) + B(x+1)}{(x+1)(2x-1)}$

$$\therefore \quad A(2x-1) + B(x+1) = 1$$

Putting $x = -1$ and $x = \frac{1}{2}$ successively on both sides, we get;

$$A(-2-1) + B(-1+1) = 1 \Rightarrow A = -\frac{1}{3}$$

and $A\left(2 \cdot \frac{1}{2} - 1\right) + B\left(\frac{1}{2} + 1\right) = 1 \Rightarrow B = \frac{2}{3}$

$$\therefore \quad \frac{1}{2x^2 + x - 1} = -\frac{1}{3(x+1)} + \frac{2}{3(2x-1)}$$

$$\begin{aligned}
 \therefore \int \frac{dx}{2x^2 + x - 1} &= \frac{-1}{3} \int \frac{dx}{x+1} + \frac{2}{3} \int \frac{dx}{2x-1} \\
 &= -\frac{1}{3} \log(x+1) + \frac{2}{3} \cdot \frac{1}{2} \log(2x-1) + c \\
 &= \frac{1}{3} \log(2x-1) - \frac{1}{3} \log(x+1) + c \\
 &= \frac{1}{3} [\log(2x-1) - \log(x+1)] + c \\
 \int \frac{1}{2x^2 + x - 1} dx &= \frac{1}{3} \log \frac{(2x-1)}{(x+1)} + c
 \end{aligned}$$

Example 3: Evaluate: $\int \frac{x \cdot dx}{(x^2 - a^2)(x^2 - b^2)}$

$$\therefore \int \frac{x}{(x^2 - a^2)(x^2 - b^2)} dx = \frac{1}{2} \int \frac{1}{(t - a^2)(t - b^2)} dt \quad [\text{Put } x^2 = t, \text{ Then, } 2x \cdot dx = dt]$$

Let
$$\frac{1}{(t - a^2)(t - b^2)} = \frac{A}{t - a^2} + \frac{B}{t - b^2}$$

Putting $t = a^2$ and $t = b^2$ successively on both the sides, we get;

$$1 = A(a^2 - b^2) \Rightarrow A = \frac{1}{a^2 - b^2}$$

and

$$1 = B(b^2 - a^2) \Rightarrow B = \frac{1}{b^2 - a^2}$$

$$\begin{aligned}
 \therefore \int \frac{x}{(x^2 - a^2)(x^2 - b^2)} dx &= \frac{1}{2} \left[\int \frac{1}{(a^2 - b^2)(t - a^2)} + \frac{1}{(b^2 - a^2)(t - b^2)} \right] dt \\
 &= \frac{1}{2(a^2 - b^2)} \int \left[\frac{1}{t - a^2} - \frac{1}{t - b^2} \right] dt. \\
 &= \frac{1}{2(a^2 - b^2)} [\log(t - a^2) - \log(t - b^2)] + c \\
 &= \frac{1}{2(a^2 - b^2)} \log \frac{t - a^2}{t - b^2} + c = \frac{1}{2(a^2 - b^2)} \log \frac{x^2 - a^2}{x^2 - b^2} + c
 \end{aligned}$$

Example 4: Evaluate $\int \frac{x}{(x-2)(x-1)^2} dx$

Solution: Let
$$\frac{x}{(x-2)(x-1)^2} = \frac{A}{x-2} + \frac{B}{x-1} + \frac{C}{(x-1)^2} \quad \dots(1)$$

$$\Rightarrow x = A(x-1)^2 + B(x-1)(x-2) + C(x-2) \quad \dots(2)$$

Putting $x = 1$ in (2) we get;

$$1 = C(1-2) \Rightarrow C = -1$$

Putting $x = 2$ in (2) we get;

$$2 = A(2-1)^2 \Rightarrow A = 2$$

Now, on comparing the coefficients of x^2 on both the sides of the identity (2), we get;

$$0 = A + B \Rightarrow B = -A = -2$$

Substituting the values of A , B , and C in (1), we get

$$\frac{x}{(x-2)(x-1)^2} = \frac{2}{x-2} - \frac{2}{x-1} - \frac{1}{(x-1)^2}$$

$$\begin{aligned} \therefore \int \frac{x}{(x-2)(x-1)^2} dx &= 2 \int \frac{dx}{x-2} - 2 \int \frac{dx}{x-1} - \int \frac{dx}{(x-1)^2} \\ &= 2 \log(x-2) - 2 \log(x-1) + \frac{1}{x-1} + c \\ I &= \int \frac{x \cdot dx}{(x-2)(x-1)^2} = \frac{1}{x-1} + 2 \log \frac{x-2}{x-1} + c \end{aligned}$$

Example 5: Evaluate $\int \frac{x^2 + 1}{(x+1)^3(x-2)} dx$.

Solution: We assume that

$$\frac{x^2 + 1}{(x+1)^3(x-2)} = \frac{A}{x-2} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3}$$

$$\Rightarrow x^2 + 1 = A(x+1)^3 + B(x+1)^2(x-2) + C(x+1)(x-2) + D(x-2) \quad \dots(1)$$

Putting $x = 2$ in (1) we have;

$$4 + 1 = A(2+1)^3 \Rightarrow A = 5/27$$

Putting $x = -1$ in (1), we have

$$(-1)^2 + 1 = D(-1-2) \Rightarrow D = (-2/3)$$

On comparing the coefficients of x^3 and constant terms on both the sides of (1), we get;

$$0 = A + B \Rightarrow B = -A = -(5/27)$$

and

$$1 = A - 2B - 2C - 2D$$

$$\Rightarrow 1 = \frac{5}{27} + \frac{10}{27} - 2C + \frac{4}{3}$$

$$\Rightarrow 2C = \frac{15}{27} + \frac{4}{3} - 1 = \frac{8}{9} \text{ or } C = 4/9$$

$$\begin{aligned} \therefore I &= \int \frac{x^2 + 1}{(x+1)^3(x-2)} dx \\ &= \frac{5}{27} \int \frac{dx}{x-2} - \frac{5}{27} \int \frac{dx}{x+1} + \frac{4}{9} \int \frac{dx}{(x+1)^2} - \frac{2}{3} \int \frac{dx}{(x+1)^3} \\ &= \frac{5}{27} \log(x-2) - \frac{5}{27} \log(x+1) + \frac{4}{9} \int (x+1)^{-2} dx - \frac{2}{3} \int (x+1)^{-3} dx \\ &= \frac{5}{27} \log \left(\frac{x-2}{x+1} \right) - \frac{4}{9} (x+1)^{-1} - \frac{2}{3} \cdot \frac{(x+1)^{-2}}{-2} + C \\ &= \frac{5}{27} \log \left(\frac{x-2}{x+1} \right) - \frac{4}{9(x+1)} + \frac{1}{3(x+1)^2} + c \end{aligned}$$

Example 6: Evaluate $\int \frac{dx}{1+x+x^2+x^3}$.

Solution: $\frac{1}{1+x+x^2+x^3} = \frac{1}{(1+x)(1+x^2)}$

$$\begin{aligned} \text{Let } \frac{1}{1+x+x^2+x^3} &= \frac{A}{1+x} + \frac{Bx+C}{1+x^2} \\ &= \frac{A(1+x^2) + (Bx+C)(1+x)}{(1+x)(1+x^2)} = \frac{x^2(A+B) + x(B+C) + A+C}{(1+x)(1+x^2)} \end{aligned}$$

$$\therefore x^2(A+B) + x(B+C) + (A+C) = 1$$

$$\Rightarrow A+B=0, B+C=0, A+C=1$$

Adding these, we get;

$$2(A+B+C) = 1 \Rightarrow A+B+C = \frac{1}{2}$$

$$\therefore C = \frac{1}{2}, A = \frac{1}{2} \text{ and } B = -\frac{1}{2}$$

$$\begin{aligned} \therefore \frac{1}{1+x+x^2+x^3} &= \frac{\left(\frac{1}{2}\right)}{1+x} + \frac{\left(-\frac{1}{2}x + \frac{1}{2}\right)}{1+x^2} \\ &= \frac{1}{2} \cdot \frac{1}{1+x} + \frac{1}{2} \cdot \frac{(-x+1)}{1+x^2} \\ &= \frac{1}{2} \cdot \frac{1}{1+x} - \frac{1}{2} \cdot \frac{x}{1+x^2} + \frac{1}{2} \cdot \frac{1}{1+x^2} \\ \therefore \int \frac{1}{1+x+x^2+x^3} dx &= \frac{1}{2} \int \frac{1}{1+x} dx - \frac{1}{2} \int \frac{x}{1+x^2} dx + \frac{1}{2} \int \frac{1}{1+x^2} dx \\ &= \frac{1}{2} \log(1+x) - \frac{1}{4} \log(1+x^2) + \frac{1}{2} \tan^{-1} x + c \end{aligned}$$

Example 7: Evaluate $\int \frac{x^4}{(x-1)(x^2+1)} dx$.

$$\begin{aligned} \text{Solution: } \frac{x^4}{(x-1)(x^2+1)} &= \frac{x^4}{x^3-x^2+x-1} \\ &= x+1 + \frac{1}{x^3-x^2+x-1} \quad (\text{By division}) \\ &= x+1 + \frac{1}{(x-1)(x^2+1)} \end{aligned}$$

Now, we assume that:

$$\frac{1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$$

$$\therefore 1 = A(x^2+1) + (Bx+C)(x-1) \quad \dots(1)$$

Putting $x=1$ in both the sides of (1), we get;

$$1 = A(1^2+1) \Rightarrow A = \frac{1}{2}$$

Now, equating the coefficients of x^2 and constant terms on both sides of (1), we have;

$$0 = A + B \Rightarrow B = -A = -\frac{1}{2}$$

and

$$1 = A - C \Rightarrow C = A - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$$

$$\begin{aligned} \therefore \frac{1}{(x-1)(x^2+1)} &= \frac{\left(\frac{1}{2}\right)}{x-1} + \frac{\left(-\frac{1}{2}x - \frac{1}{2}\right)}{x^2+1} \\ &= \frac{1}{2(x+1)} - \frac{1}{2} \cdot \frac{x+1}{x^2+1} \end{aligned}$$

$$\therefore \frac{x^4}{(x-2)(x^2+1)} = x+1 + \frac{1}{2(x-1)} - \frac{1}{2} \cdot \frac{x+1}{x^2+1}$$

$$\begin{aligned} \therefore \int \frac{x^4 \cdot dx}{(x-1)(x^2+1)} &= \int x \cdot dx + \int dx + \frac{1}{2} \int \frac{dx}{x-1} - \frac{1}{2} \int \frac{x+1}{x^2+1} dx \\ &= \frac{x^2}{2} + x + \frac{1}{2} \log(x-1) - \frac{1}{2} \int \frac{x}{x^2+1} dx - \frac{1}{2} \int \frac{dx}{x^2+1} \\ &= \frac{x^2}{2} + x + \frac{1}{2} \log(x-1) - \frac{1}{4} \log(x^2+1) - \frac{1}{2} \tan^{-1} x + c \end{aligned}$$

Example 8: Integrate $\frac{\sin x}{\sin 3x}$ w.r.t. 'x'.

Solution: Let $I = \int \frac{\sin x}{\sin 3x} \cdot dx = \int \frac{\sin x}{3 \sin x - 4 \sin^3 x} dx = \int \frac{dx}{3 - 4 \sin^2 x}$

$$= \int \frac{dx}{3(\cos^2 x + \sin^2 x) - 4 \sin^2 x} = \int \frac{dx}{3 \cos^2 x - \sin^2 x}$$

$$= \int \frac{dx}{\cos^2 x (3 - \tan^2 x)} = \int \frac{1/\cos^2 x}{3 - \tan^2 x} \cdot dx = \int \frac{\sec^2 x}{3 - \tan^2 x} \cdot dx$$

$$= \int \frac{dt}{3 - t^2}, \text{ where } \tan x = t, \sec^2 x \cdot dx = dt$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{3}} \log \frac{\sqrt{3} + t}{\sqrt{3} - t} + c$$

$$I = \int \frac{\sin x}{\sin 3x} dx = \frac{1}{2\sqrt{3}} \log \frac{\sqrt{3} + \tan x}{\sqrt{3} - \tan x} + c$$

Example 9: Evaluate $\int \frac{\sin x \cdot \cos x \cdot dx}{\cos^2 x + 3 \cos x + 2}$.

Solution: Let $\cos x = t \Rightarrow -\sin x \cdot dx = dt$

$$\begin{aligned} \therefore \int \frac{\sin x \cdot \cos x}{\cos^2 x + 3 \cos x + 2} dx &= - \int \frac{t \cdot dt}{t^2 + 3t + 2} \\ &= - \int \frac{t}{(t+1)(t+2)} dt \end{aligned}$$

$$= -\int \left(\frac{2}{t+2} - \frac{1}{t+1} \right) dt$$

(Now after resolving into partial fraction)

$$\begin{aligned} &= \int \frac{dt}{t+1} - 2 \int \frac{dt}{t+2} \\ &= \log(t+1) - 2 \log(t+2) + c \\ &= \log(\cos x + 1) - 2 \log(\cos x + 2) + c \\ &= \log(1 + \cos x) - \log(2 + \cos x)^2 + c \\ I &= \int \frac{\sin x \cdot \cos x \cdot dx}{\cos^2 x + 3 \cos x + 2} = \log \frac{1 + \cos x}{(2 + \cos x)^2} + c \end{aligned}$$

13.7.3 Integral of the Type $\int \frac{1}{ax^2 + bx + c} dx$

We have, $ax^2 + bx + c = a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right)$

$$\begin{aligned} &= a \left[\left(x + \frac{b}{2a} \right)^2 - \left(\frac{b^2}{4a^2} - \frac{c}{a} \right) \right] \\ &= a \left[\left(x + \frac{b}{2a} \right)^2 - \frac{b^2 - 4ac}{4a^2} \right] \end{aligned} \quad \dots(1)$$

Case (I): When the constant term in (1) is negative, we have:

$$\begin{aligned} \int \frac{dx}{ax^2 + bx + c} &= \frac{1}{a} \int \frac{dx}{\left(x + \frac{b}{2a} \right)^2 - \left(\frac{\sqrt{b^2 - 4ac}}{2a} \right)^2} \quad \left[\text{Form } \int \frac{dx}{x^2 - a^2} \right] \\ &= \frac{1}{a} \cdot \frac{1}{2 \frac{\sqrt{b^2 - 4ac}}{2a}} \log \frac{\left(x + \frac{b}{2a} \right) - \frac{\sqrt{b^2 - 4ac}}{2a}}{\left(x + \frac{b}{2a} \right) + \frac{\sqrt{b^2 - 4ac}}{2a}} + c \\ &= \frac{1}{\sqrt{b^2 - 4ac}} \log \cdot \frac{2ax + b - \sqrt{b^2 - 4ac}}{2ax + b + \sqrt{b^2 - 4ac}} + c \end{aligned}$$

Case (II): When constant term be in positive form.

$$\begin{aligned} \text{i.e.} \quad \int \frac{dx}{ax^2 + bx + c} &= \frac{1}{a} \int \frac{dx}{\left(x + \frac{b}{2a} \right)^2 + \left(\frac{\sqrt{4ac - b^2}}{2a} \right)^2} \quad \left[\text{Form } \int \frac{dx}{x^2 + a^2} \right] \\ &= \frac{1}{a} \cdot \frac{1}{\frac{\sqrt{4ac - b^2}}{2a}} \tan^{-1} \frac{\left(x + \frac{b}{2a} \right)}{\frac{\sqrt{4ac - b^2}}{2a}} + c \end{aligned}$$

$$= \frac{2}{\sqrt{4ac - b^2}} \tan^{-1} \frac{2ax + b}{\sqrt{4ac - b^2}} + c$$

13.7.4 Integrals of the Type $\int \frac{px + q}{ax^2 + bx + c} \cdot dx$

$$\therefore \frac{px + q}{ax^2 + bx + c} = \frac{M \frac{d}{dx}(ax^2 + bx + c) + N}{ax^2 + bx + c}$$

Hence the given Integral is

$$\begin{aligned} \int \frac{px + q}{ax^2 + bx + c} dx &= \int \frac{M \frac{d}{dx}(ax^2 + bx + c)}{ax^2 + bx + c} dx + \int \frac{N dx}{ax^2 + bx + c} \\ &= M \log(ax^2 + bx + c) + N \int \frac{dx}{ax^2 + bx + c} \end{aligned}$$

The integral on the right is of the form $\int \frac{dx}{x^2 + a^2}$

Where M and N are constants.

Example 10: Evaluate $\int \frac{dx}{x^2 - x + 1}$.

Solution: Let
$$I = \int \frac{dx}{x^2 - x + 1} = \int \frac{dx}{\left(x^2 - x + \frac{1}{4}\right) + \frac{3}{4}} = \int \frac{dx}{\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}}$$

$$= \int \frac{dx}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \frac{1}{\sqrt{3}/2} \tan^{-1} \left(\frac{x - 1/2}{\sqrt{3}/2} \right) + c$$

$$= \left(\frac{2}{\sqrt{3}} \right) \tan^{-1} \frac{2x - 1}{\sqrt{3}} + c$$

Example 11: Evaluate $\int \frac{dx}{2x^2 + x - 1}$.

Solution: Let
$$I = \int \frac{dx}{2x^2 + x - 1} = \frac{1}{2} \int \frac{dx}{x^2 + x/2 - \frac{1}{2}}$$

$$= \frac{1}{2} \int \frac{dx}{\left\{ \left(x^2 + x/2 + \frac{1}{16}\right) - \frac{1}{16} - \frac{1}{2} \right\}} = \frac{1}{2} \int \frac{dx}{\left(x + \frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2}$$

$$= \frac{1}{2} \cdot \frac{1}{2(3/4)} \log \frac{x + \frac{1}{4} - 3/4}{x + \frac{1}{4} + 3/4} + c. \quad \left[\because \int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \frac{x-a}{x+a} \right]$$

$$= \frac{1}{3} \log \frac{2x - 1}{2(x + 1)} + c$$

Example 12: Evaluate $\int \frac{1}{x [6 (\log x)^2 + 7 \log x + 2]} dx$.

Solution: Putting $\log x = t \Rightarrow \frac{1}{x} \cdot dx = dt$

$$\begin{aligned}
 \text{we have } \int \frac{1}{x [6 (\log x)^2 + 7 \log x + 2]} \cdot dx &= \int \frac{1}{6 t^2 + 7 t + 2} dt \\
 &= \frac{1}{6} \int \frac{1}{t^2 + (7/6)t + \frac{1}{3}} dt \\
 &= \frac{1}{6} \int \frac{1}{(t + 7/12)^2 + \frac{1}{3} - \frac{49}{144}} dt \\
 &= \frac{1}{6} \int \frac{1}{(t + 7/12)^2 - \left(\frac{1}{12}\right)^2} dt \\
 &= \frac{1}{6} \cdot \frac{1}{2 \cdot \frac{1}{12}} \log \left[\frac{t + \frac{7}{12} - \frac{1}{12}}{t + \frac{7}{12} + \frac{1}{12}} \right] + c \\
 &= \log \left(\frac{t + \frac{1}{2}}{t + 2/3} \right) + c \\
 &= \log \left[\frac{3(2t+1)}{2(3t+2)} \right] + c \\
 &= \log \left(\frac{2 \log x + 1}{3 \log x + 2} \right) + \log(3/2) + c \\
 &= \log \left(\frac{2 \log x + 1}{3 \log x + 2} \right) + c_1, \text{ where } c_1 = c + \log \frac{3}{2}
 \end{aligned}$$

Example 13: Evaluate $\int \frac{3x+1}{2x^2-2x+3} dx$

Solution: Let $3x+1 = M \frac{d}{dx}(2x^2-2x+3) + N$

$$\text{i.e.} \quad 3x+1 = M(4x-2) + N$$

$$3x+1 = 4Mx - 2M + N$$

On comparing the coefficients of x and constant terms on both sides; we have;

$$3 = 4M \Rightarrow M = 3/4$$

$$\text{and} \quad N - 2M = 1 \Rightarrow N = 1 + 2 \cdot \frac{3}{4} = \frac{5}{2}$$

$$\therefore \int \frac{3x+1}{2x^2-2x+3} dx = \int \frac{\left[\frac{3}{4}(4x-2) + 5/2 \right]}{2x^2-2x+3} dx$$

$$\begin{aligned}
&= \int \frac{\frac{3}{4}(4x-2)}{2x^2-2x+3} dx + (5/2) \int \frac{dx}{2x^2-2x+3} \\
&= \frac{3}{4} \log(2x^2-2x+3) + (5/2) \int \frac{dx}{2[x^2-x+3/2]} \\
&= \frac{3}{4} \log(2x^2-2x+3) + \frac{5}{4} \int \frac{dx}{x^2-x+\frac{1}{4}+\frac{3}{2}-\frac{1}{4}} \\
&= \frac{3}{4} \cdot \log[2x^2-2x+3] + \frac{5}{4} \cdot \int \frac{dx}{\left(x-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{5}}{2}\right)^2} \\
&= \frac{3}{4} \log(2x^2-2x+3) + \frac{5}{4} \cdot \frac{2}{\sqrt{5}} \tan^{-1} \frac{\left(x-\frac{1}{2}\right)}{\sqrt{5}/2} + c \\
&= \frac{3}{4} \log(2x^2-2x+3) + \frac{\sqrt{5}}{2} \tan^{-1} \frac{2x-1}{\sqrt{5}} + c
\end{aligned}$$

Example 14: Evaluate $\int \frac{4x-3}{3x^2+2x-5} \cdot dx$

Solution: Let $4x-3 = M \frac{d}{dx}(3x^2+2x-5) + N$

$$\text{i.e.,} \quad 4x-3 = M(6x+2) + N$$

$$\Rightarrow \quad 4x-3 = 6Mx + 2M + N$$

On comparing the coefficients of 'x' & constant terms on both the sides, we get,

$$4 = 6M \Rightarrow M = 2/3$$

$$\text{and} \quad -3 = 2M + N \Rightarrow N = -3 - 2M = -3 - 4/3 = \frac{-13}{3}$$

$$\therefore \quad 4x-3 = 2/3 (6x+2) - \frac{13}{3}$$

$$\begin{aligned}
\therefore \int \frac{4x-3}{3x^2+2x-5} dx &= \int \frac{2/3 (6x+2) - 13/3}{3x^2+2x-5} dx \\
&= (2/3) \int \frac{6x+2}{3x^2+2x-5} dx - \frac{13}{3} \int \frac{dx}{3x^2+2x-5} \\
&= (2/3) \log(3x^2+2x-5) - \frac{13}{3} \cdot \frac{1}{3} \int \frac{dx}{x^2 + (2/3)x - 5/3} \\
&= (2/3) \log(3x^2+2x-5) - \frac{13}{9} \int \frac{dx}{\left(x+\frac{1}{3}\right)^2 - \frac{1}{9} - 5/3} \\
&= \frac{2}{3} \log(3x^2+2x-5) - \frac{13}{9} \int \frac{dx}{\left(x+\frac{1}{3}\right)^2 - \frac{16}{9}} \\
&= \frac{2}{3} \log(3x^2+2x-5) - \frac{13}{9} \int \frac{dx}{\left(x+\frac{1}{3}\right)^2 - \left(\frac{4}{3}\right)^2}
\end{aligned}$$

$$\begin{aligned}
 &= \frac{2}{3} \log(3x^2 + 2x - 5) - \frac{13}{9} \frac{1}{2 \times 4/3} \log \frac{x + \frac{1}{3} - \frac{4}{3}}{x + \frac{1}{3} + \frac{4}{3}} + c \\
 &= \frac{2}{3} \log(3x^2 + 2x - 5) - \frac{13}{24} \log \frac{3x - 3}{3x + 5} + c
 \end{aligned}$$

Example 15: Evaluate $\int \frac{2 \sin 2\theta - \cos \theta}{6 - \cos^2 \theta - 4 \sin \theta} d\theta$.

Solution: Let

$$\begin{aligned}
 I &= \int \frac{2 \sin 2\theta - \cos \theta}{6 - \cos^2 \theta - 4 \sin \theta} d\theta \\
 &= \int \frac{4 \sin \theta \cos \theta - \cos \theta}{6 - (1 - \sin^2 \theta) - 4 \sin \theta} d\theta \\
 &= \int \frac{(4 \sin \theta - 1) \cos \theta}{\sin^2 \theta - 4 \sin \theta + 5} d\theta \\
 &= \int \frac{4t - 1}{t^2 - 4t + 5} dt \quad (\text{where } t = \sin \theta)
 \end{aligned}$$

Suppose, $4t - 1 = M \frac{d}{dt}(t^2 - 4t + 5) + N$ then,

$$4t - 1 = M(2t - 4) + N$$

$$\Rightarrow 4t - 1 = 2Mt + N - 4M$$

On comparing the coefficients of t and constant terms on both the sides; we get

$$4 = 2M \Rightarrow M = 2$$

and

$$-1 = N - 4M \Rightarrow N = 4 \times 2 - 1 = 7$$

$$\begin{aligned}
 I &= \int \frac{2(2t - 4) + 7}{t^2 - 4t + 5} dt = 2 \int \frac{2t - 4}{t^2 - 4t + 5} dt + 7 \int \frac{dt}{t^2 - 4t + 5} \\
 &= 2 \log(t^2 - 4t + 5) + 7 \int \frac{dt}{(t - 2)^2 + 1} \\
 &= 2 \log(t^2 - 4t + 5) + 7 \tan^{-1}(t - 2) + c \\
 &= 2 \log(\sin^2 \theta - 4 \sin \theta + 5) + 7 \tan^{-1}(\sin \theta - 2) + c
 \end{aligned}$$

13.7.5 Integral of the Type $\int \frac{1}{\sqrt{ax^2 + bx + c}} dx$

To evaluate these type of Integrals; first we express $\sqrt{ax^2 + bx + c}$ in the form $\sqrt{X^2 + A^2}$ or $\sqrt{X^2 - A^2}$ or $\sqrt{A^2 - X^2}$; where X is a linear function of x , and A is a constant.

After this, we use any one of the following formula according to the case considerations:

$$(i) \int \frac{dx}{\sqrt{x^2 + a^2}} = \log(x + \sqrt{x^2 + a^2}) + c$$

$$(ii) \int \frac{dx}{\sqrt{x^2 - a^2}} = \log(x + \sqrt{x^2 - a^2}) + c$$

$$(iii) \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}(x/a) + c$$

13.7.6 Integral of the Type $\int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx$

To evaluate this integral we express the numerator $(px + q)$ into two parts such that one part is differential coefficient of $ax^2 + bx + c$; as follows:

$$\begin{aligned} \text{Let } px + q &= M \frac{d}{dx}(ax^2 + bx + c) + N \\ &= M(2ax + b) + N \end{aligned}$$

Where M and N are constants to be determined by equating the coefficients of similar terms on both sides, Hence;

$$p = 2aM \Rightarrow M = \frac{p}{2a}$$

$$\text{and } q = bM + N \Rightarrow N = q - \frac{bp}{2a}$$

$$\begin{aligned} \therefore \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx &= \frac{p}{2a} \int \frac{2ax + b}{\sqrt{ax^2 + bx + c}} dx + \left(q - \frac{bp}{2a}\right) \int \frac{1}{\sqrt{ax^2 + bx + c}} dx \\ &= \frac{p}{a} \sqrt{ax^2 + bx + c} + \left(q - \frac{bp}{2a}\right) \int \frac{1}{\sqrt{ax^2 + bx + c}} dx \end{aligned}$$

Illustrative Examples

Example 1: Evaluate $\int \frac{1}{\sqrt{9 + 8x - x^2}} dx$

$$\begin{aligned} \text{Solution: Let } I &= \int \frac{1}{\sqrt{9 + 8x - x^2}} dx = \int \frac{1}{\sqrt{9 - (x^2 - 8x + 16) + 16}} dx \\ &= \int \frac{1}{\sqrt{25 - (x^2 - 8x + 16)}} dx \\ &= \int \frac{1}{\sqrt{(5)^2 - (x - 4)^2}} dx = \sin^{-1}\left(\frac{x - 4}{5}\right) + c \end{aligned}$$

Example 2: Evaluate $\int \frac{1}{\sqrt{1 - 4x - 2x^2}} dx$.

$$\text{Solution: Let } I = \int \frac{1}{\sqrt{1 - 4x - 2x^2}} dx = \frac{1}{\sqrt{2}} \int \frac{1}{\sqrt{\frac{1}{2} - 2x - x^2}} dx$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{2}} \int \frac{1}{\sqrt{\frac{1}{2} - (x^2 + 2x)}} dx \\
 &= \frac{1}{\sqrt{2}} \int \frac{1}{\sqrt{\left(\frac{3}{2}\right) - (x^2 + 2x + 1)}} dx \\
 \Rightarrow &= \frac{1}{\sqrt{2}} \int \frac{1}{\sqrt{\left(\sqrt{\frac{3}{2}}\right)^2 - (x+1)^2}} dx \\
 &= \frac{1}{\sqrt{2}} \int \frac{1}{\sqrt{\left(\sqrt{\frac{3}{2}}\right)^2 - t^2}} dt
 \end{aligned}$$

where, $t = x + 1 \Rightarrow dt = dx$

$$\begin{aligned}
 &= \frac{1}{\sqrt{2}} \sin^{-1} \left(\frac{t}{\sqrt{3/2}} \right) + c \\
 &= \frac{1}{\sqrt{2}} \sin^{-1} \left(\frac{(x+1)\sqrt{2}}{\sqrt{3}} \right) + c
 \end{aligned}$$

Example 3: Evaluate $\int \frac{dx}{\sqrt{4x^2 - x + 4}}$.

Solution: Let

$$\begin{aligned}
 I &= \int \frac{dx}{\sqrt{4x^2 - x + 4}} \\
 &= \frac{1}{2} \int \frac{dx}{\sqrt{x^2 - \frac{1}{4}x + 1}} = \frac{1}{2} \int \frac{dx}{\sqrt{\left(x^2 - \frac{1}{4}x + \frac{1}{64}\right) + 1 - \frac{1}{64}}} \\
 &= \frac{1}{2} \int \frac{dx}{\sqrt{\left(x - \frac{1}{8}\right)^2 + \left(\frac{\sqrt{63}}{8}\right)^2}} \\
 &= \frac{1}{2} \int \frac{dt}{\sqrt{t^2 + \left(\frac{\sqrt{63}}{8}\right)^2}} \text{ where } x - \frac{1}{8} = t \Rightarrow dx = dt \\
 &= \frac{1}{2} \log \left\{ t + \sqrt{t^2 + \left(\frac{\sqrt{63}}{8}\right)^2} \right\} + c \\
 &= \frac{1}{2} \log \left\{ \left(x - \frac{1}{8}\right) + \sqrt{\left(x - \frac{1}{8}\right)^2 + \frac{63}{64}} \right\} + c \\
 &= \frac{1}{2} \log \left\{ \frac{8x-1}{8} + \frac{\sqrt{(8x-1)^2 + 63}}{8} \right\} + c \\
 &= \frac{1}{2} \log \{ (8x-1) + \sqrt{(8x-1)^2 + 63} \} + c - \frac{1}{2} \log 8
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \log (8x-1) + \sqrt{64x^2 - 16x + 64} \} + c \\
 &= \frac{1}{2} \log (8x-1) + 4 \sqrt{4x^2 - x + 4} \} + c
 \end{aligned}$$

Example 4: Evaluate $\int \frac{x+8}{\sqrt{x^2+2x+5}} dx$.

Solution: Let $x+8 = M \frac{d}{dx}(x^2+2x+5) + N$

$$\therefore x+8 = M(2x+2) + N$$

Equating the coefficients of x and constant terms, we get;

$$1 = 2M \Rightarrow M = \frac{1}{2}$$

$$\text{and} \quad 8 = N + 2M \Rightarrow N = 8 - 2 \times \frac{1}{2} = 7$$

$$\begin{aligned}
 \therefore \int \frac{x+8}{\sqrt{x^2+2x+5}} dx &= \frac{1}{2} \int \frac{2x+2}{\sqrt{x^2+2x+5}} dx + 7 \int \frac{1}{\sqrt{x^2+2x+5}} dx \\
 &= I_1 + I_2 \text{ (say)}
 \end{aligned}$$

$$\therefore I_1 = \frac{1}{2} \int \frac{2x+2}{\sqrt{x^2+2x+5}} dx = \frac{1}{2} \int \frac{dt}{\sqrt{t}},$$

$$\text{where } x^2+2x+5 = t \Rightarrow (2x+2) dx = dt$$

$$= \frac{1}{2} \cdot \frac{t^{1/2}}{1/2} = t^{1/2}$$

$$I_1 = \sqrt{x^2+2x+5}$$

$$\begin{aligned}
 \Rightarrow I_2 &= 7 \int \frac{1}{\sqrt{x^2+2x+5}} dx \\
 &= 7 \int \frac{1}{\sqrt{(x+1)^2+2^2}} \cdot dx \\
 &= 7 \log \{(x+1) + \sqrt{(x+1)^2+2^2}\} + c \\
 &= 7 \log \{(x+1) + \sqrt{x^2+2x+5}\} + c
 \end{aligned}$$

$$\therefore I_1 + I_2 = \sqrt{x^2+2x+5} + 7 \log \{(x+1) + \sqrt{x^2+2x+5}\} + c$$

$$\Rightarrow \int \frac{x+8}{\sqrt{x^2+2x+5}} dx = \sqrt{x^2+2x+5} + 7 \log \{(x+1) + \sqrt{x^2+2x+5}\} + c$$

Example 5: Evaluate $\int \frac{2x+1}{\sqrt{2x^2+x-3}} dx$

Solution: Let $2x+1 = M \frac{d}{dx}(2x^2+x-3) + N$

$$\text{Then; } 2x+1 = M(4x+1) + N$$

$$2x+1 = 4Mx + M + N$$

Equating the coefficients of x and constant terms on both the sides, we get;

$$2 = 4M \Rightarrow M = \frac{1}{2}$$

and

$$1 = M + N \Rightarrow N = 1 - M = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\begin{aligned} \therefore \int \frac{2x+1}{\sqrt{2x^2+x-3}} dx &= \int \frac{\frac{1}{2}(4x+1) + \frac{1}{2}}{\sqrt{2x^2+x-3}} dx \\ &= \frac{1}{2} \int \frac{4x+1}{\sqrt{2x^2+x-3}} dx + \frac{1}{2} \int \frac{1}{\sqrt{2x^2+x-3}} dx \\ &= I_1 + I_2 \quad (\text{say}) \end{aligned}$$

Now

$$I_1 = \frac{1}{2} \int \frac{4x+1}{\sqrt{2x^2+x-3}} dx$$

Let

$$2x^2 + x - 3 = t \quad (4x+1) dx = dt$$

$\therefore$

$$I_1 = \frac{1}{2} \int \frac{dt}{\sqrt{t}} = \frac{1}{2} \cdot \frac{t^{1/2}}{1/2} = t^{1/2} = \sqrt{2x^2+x-3}$$

Again,

$$\begin{aligned} I_2 &= \frac{1}{2} \int \frac{1}{\sqrt{2x^2+x-3}} dx \\ &= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \int \frac{1}{\sqrt{x^2 + \frac{1}{2}x - \frac{3}{2}}} dx \\ &= \frac{1}{2\sqrt{2}} \int \frac{dx}{\sqrt{\left(x + \frac{1}{4}\right)^2 - \frac{1}{16} - \frac{3}{2}}} \\ &= \frac{1}{2\sqrt{2}} \int \frac{dx}{\sqrt{\left(x + \frac{1}{4}\right)^2 - (5/4)^2}} \\ &= \frac{1}{2\sqrt{2}} \int \frac{du}{\sqrt{u^2 - \left(\frac{5}{4}\right)^2}} \quad \left[\text{where } x + \frac{1}{4} = u \Rightarrow dx = du\right] \\ &= \frac{1}{2\sqrt{2}} \log \left\{ u + \sqrt{u^2 - \left(\frac{5}{4}\right)^2} \right\} + c \\ &= \frac{1}{2\sqrt{2}} \log \left\{ x + \frac{1}{4} + \sqrt{\left(x + \frac{1}{4}\right)^2 - \left(\frac{5}{4}\right)^2} \right\} + c \\ &= \frac{1}{2\sqrt{2}} \log \left\{ \frac{4x+1 + \sqrt{(4x+1)^2 - 25}}{4} \right\} + c \\ &= \frac{1}{2\sqrt{2}} \log \{4x+1 + \sqrt{(4x+1)^2 - 25}\} - \frac{1}{2\sqrt{2}} \log 4 + c \end{aligned}$$

$$= \frac{1}{2\sqrt{2}} \log\{(4x+1) + \sqrt{16x^2 + 8x - 24}\} + c_1$$

$$\{\text{where } c_1 = c - \frac{1}{2\sqrt{2}} \log 4\}$$

$$= \frac{1}{2\sqrt{2}} \log\{4x+1 + 2\sqrt{2} \sqrt{2x^2 + x - 3}\} + c_1$$

$$\therefore \int \frac{x+1}{\sqrt{x^2 + x - 3}} dx = \sqrt{2x^2 + x - 3} + \frac{1}{2\sqrt{2}} \log\{4x+1 + 2\sqrt{2} \sqrt{2x^2 + x - 3}\} + c_1$$

Example 6: Evaluate: $\int \frac{e^x}{\sqrt{5-4e^x - e^{2x}}} \cdot dx$

Solution: Let $I = \int \frac{e^x}{\sqrt{5-4e^x - e^{2x}}}$

Put $e^x = t \Rightarrow e^x \cdot dx = dt$

$$\begin{aligned} \text{we get } I &= \int \frac{1}{\sqrt{5-4t-t^2}} dt = \int \frac{dt}{5-(t^2+4t)} = \int \frac{dt}{\sqrt{9-(t^2+4t+4)}} \\ &= \int \frac{dt}{\sqrt{3^2-(t+2)^2}} = \sin^{-1}\left(\frac{t+2}{3}\right) + c = \sin^{-1}\left(\frac{e^x+2}{3}\right) + c \end{aligned}$$

13.7.7 Integration of the Type $\int \frac{1}{a+b\cos^2 x} \cdot dx$

and $\int \frac{1}{a+b\sin^2 x} \cdot dx$

Working Rule: To evaluate the integrals involving $\cos^2 x$ or $\sin^2 x$. or both or; $\sin x$. $\cos x$ in the denominator, Proceed as.

(i) Divide the N^r . and D^r . by $\cos^2 x$

(ii) Replace $\sec^2 x$ if any, in the D^r . by $1 + \tan^2 x$

(iii) Put $\tan x = t$ so that $\sec^2 x \cdot dx = dt$

(iv) Now, the integral will reduce to the form $\int \frac{1}{at^2 + bt + c} dt$ which can be evaluated by the method discussed earlier.

Illustrative Examples

Example 1: Evaluation of Integral of the types:

$$(i) \quad \int \frac{dx}{a + b \cos x}$$

$$(ii) \quad \int \frac{dx}{a + b \sin x}$$

Solution:

$$(i) \quad \text{Let} \quad I = \int \frac{dx}{a + b \cos x}$$

$$I = \int \frac{dx}{a (\sin^2 (x/2) + \cos^2 (x/2)) + b (\cos^2 x/2 - \sin^2 (x/2))}$$

$$I = \int \frac{dx}{(a+b) \cos^2 (x/2) + (a-b) \sin^2 (x/2)}$$

$$I = \int \frac{\sec^2 (x/2) dx}{(a+b) + (a-b) \tan^2 (x/2)}$$

$$\text{Now, put } \tan (x/2) = t \Rightarrow \sec^2 (x/2) \cdot \frac{1}{2} \cdot dx = dt$$

$$\text{Then,} \quad I = \int \frac{dx}{a + b \cos x} = \int \frac{2dt}{(a+b) + (a-b) t^2}$$

$$I = \frac{2}{(a-b)} \int \frac{dt}{t^2 + \frac{a+b}{a-b}}$$

Note: Which takes the form of $\int \frac{dt}{t^2 \pm a^2}$ according to $a > b$ or $a < b$ and hence it can be evaluated easily.

$$(ii) \quad \text{Let} \quad I = \int \frac{dx}{a + b \sin x}$$

$$= \int \frac{dx}{a (\cos^2 (x/2) + \sin^2 (x/2)) + 2 b \sin (x/2) \cdot \cos (x/2)}$$

$$I = \int \frac{\sec^2 (x/2) dx}{a (1 + \tan^2 (x/2)) + 2 b \tan (x/2)}$$

$$\text{Now; put } \tan (x/2) = t \Rightarrow \frac{1}{2} \sec^2 (x/2) dx = dt. \text{ we have}$$

$$\begin{aligned} \Rightarrow \quad I &= \int \frac{2 \cdot dt}{a (1 + t^2) + 2bt} \\ &= \frac{2}{a} \int \frac{dt}{1 + t^2 + \frac{2b}{a} \cdot t} \end{aligned}$$

$$\begin{aligned}
 &= \frac{2}{a} \int \frac{dt}{1+t^2 + \frac{2b}{a} \cdot t + \frac{b^2}{a^2} - \frac{b^2}{a^2}} \\
 &= a/2 \int \frac{dt}{1 - \frac{b^2}{a^2} + (t + b/a)^2}
 \end{aligned}$$

It takes the standard form $\int \frac{du}{u^2 \pm a^2}$ according as $a > b$ or $a < b$

13.7.8 Integral of the Type $\int \frac{1}{a \sin x + b \cos x} dx$

To evaluate these type of integrals, we put

$$a = r \cos \theta \text{ and } b = r \sin \theta$$

So, that $r = \sqrt{a^2 + b^2}$ and $\theta = \tan^{-1} b/a$

$$\therefore a \sin x + b \cos x = r \cos \theta \cdot \sin x + r \sin \theta \cdot \cos x = r (\sin x \cdot \cos \theta + \cos x \cdot \sin \theta) = r \sin(x + \theta)$$

$$\begin{aligned}
 \therefore \int \frac{1}{a \sin x + b \cos x} dx &= \frac{1}{r} \int \frac{1}{\sin(x + \theta)} dx \\
 &= \frac{1}{r} \int \operatorname{cosec}(x + \theta) dx = \frac{1}{r} \log \left\{ \tan \left(\frac{x}{2} + \frac{\theta}{2} \right) \right\} + c \\
 &= \frac{1}{\sqrt{a^2 + b^2}} \log \left[\tan \left\{ (x/2) + \frac{1}{2} \tan^{-1} b/a \right\} \right] + c
 \end{aligned}$$

Example 2: Let $I = \int \frac{1}{1 + 3 \sin^2 x} dx$

Solution: Let $I = \int \frac{\sec^2 x}{\sec^2 x + 3 \tan^2 x} dx$ {Dividing N^r and D^r by $\cos^2 x$ }

$$\begin{aligned}
 &= \int \frac{\sec^2 x}{(1 + \tan^2 x) + 3 \tan^2 x} \cdot dx \\
 &= \int \frac{\sec^2 x}{1 + 4 \tan^2 x} \cdot dx \\
 &= \frac{1}{4} \int \frac{\sec^2 x}{\tan^2 x + \left(\frac{1}{2}\right)^2} \cdot dx
 \end{aligned}$$

Now, put $\tan x = t \Rightarrow \sec^2 x \cdot dx = dt$

$$\begin{aligned}
 \therefore I &= \int \frac{1}{1 + 3 \sin^2 x} dx = \frac{1}{4} \int \frac{dt}{t^2 + (1/2)^2} \\
 &= \frac{1}{4} \cdot \frac{1}{1/2} \cdot \tan^{-1} \left(\frac{t}{1/2} \right) + c \\
 &= \frac{1}{2} \tan^{-1} (2 \tan x) + c.
 \end{aligned}$$

Example 3: Evaluate $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} \cdot dx$

Solution: Let $I = \int \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x}$

Now dividing the Nr and Dr by $\cos^2 x$, we get

$$I = \int \frac{\sec^2 x}{a^2 \tan^2 x + b^2} \cdot dx$$

Putting $\tan x = t \Rightarrow \sec^2 x \cdot dx = dt$, we get

$$\begin{aligned} I &= \int \frac{dt}{a^2 t^2 + b^2} = \frac{1}{a^2} \int \frac{dt}{t^2 + \left(\frac{b}{a}\right)^2} \\ &= \frac{1}{a^2} \cdot \frac{1}{\frac{b}{a}} \cdot \tan^{-1} \left(\frac{t}{b/a} \right) + c = \frac{1}{ab} \tan^{-1} \left(\frac{a \tan x}{b} \right) + c \end{aligned}$$

Example 4: Evaluate: $\int \frac{1}{3 + \sin 2x} \cdot dx$

Solution: Let $I = \int \frac{1}{3 + \sin 2x} \cdot dx$

$$= \int \frac{1}{3 + 2 \sin x \cdot \cos x} \cdot dx$$

$$= \int \frac{1}{3 (\cos^2 x + \sin^2 x) + 2 \sin x \cdot \cos x} \cdot dx$$

(Now, dividing Nr. and denominator by $\cos^2 x$)

$$= \int \frac{\sec^2 x}{3 \tan^2 x + 2 \tan x + 3} dx$$

Put, $\tan x = t \Rightarrow \sec^2 x \cdot dx = dt$, then

$$\begin{aligned} I &= \int \frac{1}{3 t^2 + 2 t + 3} dt \\ &= \frac{1}{3} \int \frac{1}{t^2 + \frac{2}{3} t + 1} dt \\ &= \frac{1}{3} \int \frac{1}{\left(t + \frac{1}{3}\right)^2 + \left(\frac{2\sqrt{2}}{3}\right)^2} dt \\ &= \frac{1}{3} \cdot \frac{1}{\left(\frac{2\sqrt{2}}{3}\right)} \tan^{-1} \left\{ \frac{t + \frac{1}{3}}{\frac{2\sqrt{2}}{3}} \right\} + c \\ &= \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{3t + 1}{2\sqrt{2}} \right) + c = \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{3 \cdot \tan x + 1}{2\sqrt{2}} \right) + c \end{aligned}$$

Example 5: Evaluate : $\int \frac{dx}{5+4 \cos x}$.

[B.C.A. (Meerut) 2006, 2007]

Solution: Let $I = \int \frac{dx}{5+4 \cos x}$

$$I = \int \frac{1}{5 (\cos^2 (x/2) + \sin^2 (x/2)) + 4 (\cos^2 (x/2) - \sin^2 (x/2))} dx$$

$$I = \int \frac{dx}{9 \cos^2 (x/2) + \sin^2 (x/2)}$$

$$I = \int \frac{\sec^2 (x/2)}{9 + \tan^2 (x/2)} dx$$

Now, put $\tan x/2 = t \Rightarrow \frac{1}{2} \sec^2 (x/2) dx = dt$ or $\sec^2 (x/2) dx = 2 dt$

$$I = 2 \int \frac{dt}{9 + t^2}$$

$$= 2 \cdot \frac{1}{3} \tan^{-1} \left(\frac{t}{3} \right) + c$$

$$I = \frac{2}{3} \tan^{-1} \left(\frac{1}{3} \cdot \tan (x/2) \right) + c$$

Example 6: Evaluate $\int \frac{1}{4+5 \sin x} dx$

Solution: Let $I = \int \frac{dx}{4+5 \sin x}$

$$= \int \frac{dx}{4 (\cos^2 (x/2) + \sin^2 (x/2)) + 5 \times 2 \sin (x/2) \cdot \cos (x/2)}$$

$$= \int \frac{\sec^2 (x/2)}{4 (1 + \tan^2 (x/2)) + 10 \tan (x/2)} dx$$

Now, put $\tan x/2 = t, \frac{1}{2} \sec^2 (x/2) dx = dt$ or $\sec^2 x/2 \cdot dx = 2 \cdot dt$

We have;

$$I = \int \frac{2 \cdot dt}{4(1+t^2) + 10t}$$

$$= \int \frac{dt}{2t^2 + 5t + 2}$$

$$= \frac{1}{2} \int \frac{dt}{t^2 + (5/2)t + 1}$$

$$= \frac{1}{2} \int \frac{dt}{\left(t + \frac{5}{4}\right)^2 - \frac{9}{16}}$$

Where $u = t + \frac{5}{4} \Rightarrow dt = du$

$$= \frac{1}{2} \int \frac{du}{u^2 - \left(\frac{3}{4}\right)^2} = \frac{1}{2} \cdot \frac{1}{2 \times \frac{3}{4}} \log \frac{u - \frac{3}{4}}{u + \frac{3}{4}} + c$$

$$\begin{aligned}
 &= \frac{1}{3} \log \left(\frac{4u-3}{4u-3} \right) + c = \frac{1}{3} \log \left[\frac{4 \left(t + \frac{5}{4} \right) - 3}{4 \left(t + \frac{5}{4} \right) + 3} \right] + c \\
 &= \frac{1}{3} \log \left(\frac{2t+1}{2t+4} \right) + c = \frac{1}{3} \log \left[\frac{2 \tan(x/2) + 1}{2 \tan(x/2) + 4} \right] + c
 \end{aligned}$$



Comprehensive Exercise 13.6

Evaluate:

1. (i) $\int \frac{dx}{x^2 - 5}$

(ii) $\int \frac{dx}{(x+1)^2 - 4}$

(iii) $\int \frac{x^2 + 1}{x^2 - 1} dx$

(iv) $\int \frac{dx}{(x-1)(x-2)}$

(v) $\int \frac{dx}{(x-1)(x^2 - 4)}$

(vi) $\int \frac{x^2 dx}{(x+1)(x-2)(x-3)}$

(vii) $\int \frac{dx}{2x^2 + 5x + 3}$

(viii) $\int \frac{dx}{x(x+1)^2}$

(ix) $\int \frac{dx}{(x+1)^2(x-1)}$

(x) $\int \frac{dx}{(x+1)^2(x^2 + 1)}$

2. (i) $\int \frac{dx}{1 - x^3}$

(ii) $\int \frac{dx}{x(x^4 + 1)}$

(iii) $\int \frac{dx}{x(x^n + 1)}$

(iv) $\int \frac{dx}{1 + 3e^x + 2e^{2x}}$ [B.C.A. (Delhi) 2005]

[B.C.A. (Meerut) 2002]

3. (i) $\int \frac{1}{x^2 + x + 1} dx$

(ii) $\int \frac{1}{2x^2 + x + 3} dx$

(iii) $\int \frac{4x+3}{2x^2 + 2x+5} dx$

(iv) $\int \frac{1}{2e^{2x} + 3e^x + 1} dx$

4. (i) $\int \frac{dx}{\sqrt{x^2 - 4x + 5}}$

(ii) $\int \frac{dx}{\sqrt{x^2 + 2x}}$

(iii) $\int \frac{dx}{\sqrt{4x^2 - 3x + 5}}$

(iv) $\int \frac{x dx}{\sqrt{x^2 + x + 1}}$

[B.C.A. (Meerut) 2002]

$$(v) \int \frac{(2x+3)}{\sqrt{x^2+4x+1}} dx$$

$$(vi) \int \frac{(3x+1)}{\sqrt{5-2x-x^2}} dx$$

$$(vii) \int_2^3 \frac{dx}{\sqrt{5x-6-x^2}}$$

[B.C.A. (Meerut) 2004]

$$5. (i) \int \frac{1}{1+\cos^2 x} dx$$

$$(ii) \int \frac{1}{1+3\cos^2 x} dx$$

$$(iii) \int \frac{dx}{4\cos^2 x + 9\sin^2 x}$$

$$(iv) \int \frac{1}{2+\cos x} dx$$

$$(v) \int \frac{1}{5+4\sin x} dx$$

$$(vi) \int \frac{1}{\sin x + \cos x} dx$$

[B.C.A. (Meerut) 2000]

$$6. (i) \int \frac{x^2+1}{x^4-x^2+1} dx$$

$$(ii) \int \frac{x^2+1}{x^4+1} dx$$

[B.C.A. (Meerut) 2008]

$$(iii) \int \frac{x^2-1}{x^4+1} dx$$

$$(iv) \int \frac{x^2}{x^4+1} dx$$

[B.C.A. (Delhi) 2007]

$$(v) \int \sqrt{\cot x} dx$$

[B.C.A. (Agra) 2002, 2004;
B.C.A. (Delhi) 2005, 2006]



Answers

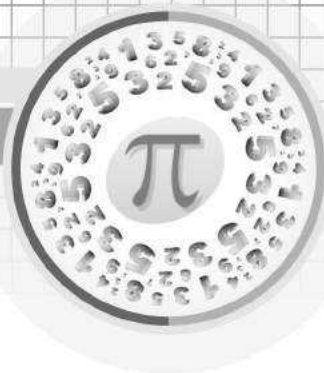
1.	(i) $\frac{1}{2\sqrt{5}} \log \left(\frac{x-\sqrt{5}}{x+\sqrt{5}} \right) + c$ (iii) $x + \log \left(\frac{x-1}{x+1} \right) + c$ (v) $-\frac{1}{3} \log(x-1) + \frac{1}{4} \log(x-2) + \frac{1}{12} \log(x+2) + c$ (vi) $\frac{1}{12} \log(x+1) - \frac{4}{3} \log(x-2) + \frac{9}{4} \log(x-3) + c$ (vii) $\log \left(\frac{x+1}{2x+3} \right) + c$ (ix) $\frac{1}{4} \log \left(\frac{x-1}{x+1} \right) + \frac{1}{2(x+1)} + c$	(ii) $\frac{1}{4} \log \left(\frac{x-1}{x+3} \right) + c$ (iv) $\log \left(\frac{x-2}{x+1} \right) + c$ (viii) $\frac{1}{x+1} + \log \frac{x}{x+1} + c$ (x) $\frac{1}{4} \log \frac{(x+1)^2}{(x^2+1)} - \frac{1}{2(x+1)} + c$
2.	(i) $-\frac{1}{3} \log(1-x) + \frac{1}{6} \log(x^2+x+1) + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + c$ (ii) $\frac{1}{4} \log \left(\frac{x^4}{x^4+1} \right) + c$ (iv) $\log \left\{ \frac{e^x(1+e^x)}{(1+2e^x)^2} \right\} + c$	(iii) $\frac{1}{n} \log \left(\frac{x^n}{x^n+1} \right) + c$
3.	(i) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + c$ (iii) $\log(2x^2+2x+5) + \frac{1}{3} \tan^{-1} \left(\frac{2x+1}{3} \right) + c$ (iv) $-\frac{1}{2} \log(e^{-2x}+3e^{-x}+2) + \frac{3}{2} \log \left(\frac{e^{-x}+1}{e^{-x}+2} \right) + c$	(ii) $\frac{2}{\sqrt{23}} \tan^{-1} \left(\frac{4x+1}{\sqrt{23}} \right) + c$
4.	(i) $\log \{(x-2) + \sqrt{x^2-4x+6}\} + c$ (ii) $\log \{(x+1) + \sqrt{x^2+2x}\} + c$ (iii) $\frac{1}{2} \log [(8x-3)+4\sqrt{4x^2-3x+5}] + c$ (iv) $\sqrt{x^2+x+1} - \frac{1}{2} \log [(2x+1) + 2\sqrt{x^2+x+1}] + c$ (v) $2\sqrt{x^2+4x+1} - \log[(x+2) + \sqrt{x^2+4x+1}] + c$ (vi) $-3\sqrt{5-2x-x^2} - 2 \sin^{-1} \left(\frac{x+1}{\sqrt{6}} \right) + c$	(vii) $-\pi$

5.	(i) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) + c$ (iii) $\frac{1}{6} \tan^{-1} \left(\frac{3 \tan x}{2} \right) + c$ (v) $\frac{2}{3} \tan^{-1} \left[\frac{5 \tan \frac{x}{2} + 4}{3} \right] + c$	(ii) $\frac{1}{2} \tan^{-1} \left(\frac{\tan x}{2} \right) + c$ (iv) $\frac{1}{\sqrt{3}} \log \left[\frac{\sqrt{3} + \tan \frac{x}{2}}{\sqrt{3} - \tan \frac{x}{2}} \right]$ (vi) $\frac{1}{\sqrt{2}} \log \tan \left(\frac{x}{2} + \frac{\pi}{8} \right) + c$
6.	(i) $\tan^{-1} \left(\frac{x^2 - 1}{x} \right) + c$ (iii) $\frac{1}{2\sqrt{2}} \log \left(\frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right) + c$ (iv) $\frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{x^2 - 1}{\sqrt{2}x} \right) + \frac{1}{4\sqrt{2}} \log \left(\frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right) + c$ (v) $-\frac{1}{\sqrt{2}} \left[\int \tan^{-1} \left(\frac{\cot x - 1}{\sqrt{2} \cot x} \right) + \frac{1}{2} \log \left(\frac{\cot x - \sqrt{2} \cot x + 1}{\cot x + \sqrt{2} \cot x + 1} \right) \right] + c$	(ii) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^2 - 1}{\sqrt{2}x} \right) + c$





Unit-4



Chapter

14

Definite Integration

14.1 Fundamental Theorems of Integrals Calculus

If $f(x)$ is continuous on $[a, b]$ and if $f(x)$ is any indefinite integral of $f(x)$, then

$$\int_a^b f(x) \cdot dx = F(b) - F(a)$$

Integral $\int_a^b f(x) dx$ is called the **definite integral** of $f(x)$ between the limits a and b .

Hence a and b are called the lower limit and upper limit respectively of the definite integral. The interval $[a, b]$ is called the range of Integrations.

14.1.1 Properties of Definite Integrals:

Property (i) $\int_a^b f(x) dx = \int_a^b f(t) \cdot dt$

Property (ii) $\int_a^b f(x) dx = - \int_b^a f(x) \cdot dx$

Property (iii) $\int_a^b f(x) dx = \int_a^c f(x) \cdot dx + \int_c^b f(x) \cdot dx$

Where $a < c < b$

Property (iv) $\int_0^a f(x) dx = \int_0^a f(a-x) \cdot dx$

Property (v) $\int_{-a}^a f(x) dx = 0$ or $2 \int_0^a f(x) \cdot dx$

$\Rightarrow$ According as $f(x)$ is an odd or even function of x .

Property (vi) $\int_0^{2a} f(x) \cdot dx = 2 \int_0^a f(x) \cdot dx$, if $f(2a-x) = f(x)$
 $= 0$ if $f(2a-x) = -f(x)$

Illustrative Examples

(On Definite Integration)

Example 1: Evaluate the Integrals:

$$(i) \int_1^2 x^2 dx \quad (ii) \int_1^3 \frac{dx}{x} \quad (iii) \int_0^1 \frac{1}{2x-3} dx$$

Solution:

$$(i) \text{ Let } I = \int_1^2 x^2 \cdot dx = \left[\frac{x^3}{3} \right]_1^2 = \left(\frac{8}{3} - \frac{1}{3} \right) = \frac{7}{3}$$

$$(ii) \text{ Let } I = \int_1^3 \frac{dx}{x} = [\log x]_1^3 = [\log 3 - \log 1] = \log 3$$

$$(iii) \text{ Let } I = \int_0^1 \frac{1}{2x-3} dx = \left[\frac{1}{2} \log(2x-3) \right]_0^1 \\ = \frac{1}{2} [\log(2x-3)]_0^1 \\ = \frac{1}{2} [\log |(-1)| - \log |-3|] = \frac{1}{2} [\log 1 - \log 3] = -\frac{1}{2} \log 3$$

Example 2: Evaluate:

$$(i) \int_0^a \frac{dx}{a^2 + x^2} \quad (ii) \int_0^{\pi/4} \sec x \cdot dx \quad (iii) \int_0^{\pi/4} \tan^2 x \cdot dx$$

$$(iv) \int_0^{\pi/4} \sin^2 x \cdot dx$$

Solution:

$$(i) \text{ Let } I = \int_0^a \frac{dx}{a^2 + x^2} \\ = \frac{1}{a} \left[\tan^{-1} x/a \right]_0^a \\ = \frac{1}{a} \left[\tan^{-1} \frac{a}{a} - \tan^{-1} \frac{0}{a} \right] = \frac{1}{a} [\tan^{-1}(1) - \tan(0)] = \frac{1}{a} \left[\frac{\pi}{4} - 0 \right] = \frac{\pi}{4a}$$

$$(ii) \text{ Let } I = \int_0^{\pi/4} \sec x \cdot dx \\ = [\log (\sec x + \tan x)]_0^{\pi/4}$$

$$\begin{aligned}
 &= \log \left(\sec \frac{\pi}{4} + \tan \frac{\pi}{4} \right) - \log [\sec 0 + \tan^{-1} 0] \\
 &= \{ \log(\sqrt{2} + 1) \} - \{ \log(1 + 0) \} = \log(\sqrt{2} + 1) - \log 1 \\
 &= \log(\sqrt{2} + 1) - 0 = \log(\sqrt{2} + 1)
 \end{aligned}$$

(iii) Let $I = \int_0^{\pi/4} (\tan^2 x \cdot dx)$

$$\begin{aligned}
 &= \int_0^{\pi/4} (\sec^2 x - 1) dx \\
 &= [\tan x]_0^{\pi/4} - [x]_0^{\pi/4} \\
 &= \left(\tan \frac{\pi}{4} - \tan 0 \right) - \left(\frac{\pi}{4} - 0 \right) = (1 - 0) - \left(\frac{\pi}{4} - 0 \right) = \left(1 - \frac{\pi}{4} \right)
 \end{aligned}$$

(iv) Let $I = \int_0^{\pi/4} \sin^2 x \cdot dx = \frac{1}{2} \int_0^{\pi/4} 2 \sin^2 x \cdot dx$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^{\pi/4} (1 - \cos 2x) \cdot dx \\
 &= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/4} \\
 &= \frac{1}{2} \left[\frac{\pi}{4} - \frac{\sin \frac{\pi}{2}}{2} - (0 - 0) \right] = \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} - 0 \right] = \frac{1}{8} [\pi - 2]
 \end{aligned}$$

Example 3: Evaluate $\int_0^{\pi/4} \frac{\sin(\cos 2x)}{\operatorname{cosec} 2x} \cdot dx$

Solution: Let

$$\begin{aligned}
 I &= \int_0^{\pi/4} \frac{\sin(\cos 2x)}{\operatorname{cosec} 2x} \cdot dx \\
 &= \int_0^{\pi/4} \sin 2x \sin(\cos 2x) \cdot dx
 \end{aligned}$$

Put $\cos 2x = t \Rightarrow -2 \sin 2x \cdot dx \Rightarrow dt \Rightarrow \sin 2x \cdot dx = -\frac{1}{2} dt$

When $x = 0$, $t = \cos 0 = 1$ and when $x = \frac{\pi}{4}$, $t = \cos \left(\frac{\pi}{2} \right) = 0$

$$\therefore \int_0^{\pi/4} \sin 2x \cdot \sin(\cos 2x) dx = -\frac{1}{2} \int_1^0 \sin t \cdot dt = \frac{1}{2} [\cos t]_1^0 = \frac{1}{2} [\cos 0 - \cos 1] = \frac{1}{2} (1 - \cos 1)$$

Example 4: Evaluate $\int_0^1 \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

Solution:

$$I = \int_0^1 \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$$

Put $\sin^{-1} x = t$ so that $\frac{1}{\sqrt{1-x^2}} dx = dt$

When $x = 0$, $t = \sin^{-1}(0) = 0$ and when $x = 1$, $t = \sin^{-1}(1) = \frac{\pi}{2}$

$$\therefore \int_0^1 \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = \int_0^{\pi/2} t \cdot dt = \left[\frac{t^2}{2} \right]_0^{\pi/2} = \frac{1}{2} \left[\left(\frac{\pi}{2} \right)^2 - 0 \right] = \frac{\pi^2}{8}$$

Example 5: Evaluate $\int_0^1 \sin^{-1} x \cdot dx$.

[B.C.A.(Meerut) 2005]

Solution:

$$\begin{aligned} I &= \int 1 \cdot \sin^{-1} x \cdot dx \\ &= [\sin^{-1} x \cdot x]_0^1 - \int_0^1 \frac{1}{\sqrt{1-x^2}} \cdot x \cdot dx = \sin^{-1}(1) - \int_0^1 \frac{x \cdot dx}{\sqrt{1-x^2}} \end{aligned}$$

Put

$$1-x^2 = t^2 \Rightarrow -2x \cdot dx = 2t \cdot dt$$

$\therefore$

$$I = \frac{\pi}{2} + \int_1^0 \frac{t \cdot dt}{t} = \frac{\pi}{2} + \int_1^0 dt = \frac{\pi}{2} + [t]_1^0 = \frac{\pi}{2} + 0 - 1 = \left(\frac{\pi}{2} - 1 \right)$$

Example 6: Evaluate: $\int_0^1 \frac{(\tan^{-1} x)^2}{1+x^2} dx$

Solution:

$$I = \int_0^1 \frac{(\tan^{-1} x)^2}{1+x^2} dx$$

Put $\tan^{-1} x = t$, so that $\frac{1}{1+x^2} dx = dt$

Also, when $x = 0$ then $t = \tan^{-1} 0 = 0$ when $x = 1$ then $t = \tan^{-1}(1) = \left(\frac{\pi}{4} \right)$

$$\therefore \int_0^1 \frac{(\tan^{-1} x)^2}{1+x^2} \cdot dx = \int_0^{\pi/4} (t)^2 \cdot dt = \left[\frac{1}{3} t^3 \right]_0^{\pi/4} = \frac{1}{3} \left[\left(\frac{\pi}{4} \right)^3 - 0 \right] = \frac{\pi^3}{192}$$

Examples 7: Evaluate:

$$(i) \int_0^{\pi/4} \sqrt{1+\sin 2x} \cdot dx \qquad (ii) \int_0^{\pi/6} \sqrt{1-\sin 2x} \cdot dx$$

$$(iii) \int_{\pi/4}^{\pi/2} \sqrt{1-\sin 2x} \cdot dx$$

Solution:

$$\begin{aligned} (i) \quad I &= \int_0^{\pi/4} \sqrt{1+\sin 2x} \cdot dx \\ &= \int_0^{\pi/4} \sqrt{\sin^2 x + \cos^2 x + 2 \sin x \cdot \cos x} \cdot dx \\ &= \int_0^{\pi/4} (\sin x + \cos x) \cdot dx \\ &= [\sin x - \cos x]_0^{\pi/4} \\ &= \left(\sin \frac{\pi}{4} - \cos \frac{\pi}{4} \right) - (\sin 0 - \cos 0) = \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) - (0 - 1) = 1 \end{aligned}$$

$$(ii) \quad I = \int_0^{\pi/6} \sqrt{1-\sin 2x} \cdot dx$$

$$\begin{aligned}
 &= \int_0^{\pi/6} \sqrt{\sin^2 x + \cos^2 x - 2 \sin x \cdot \cos x} \cdot dx \\
 &= \int_0^{\pi/6} (\cos x - \sin x) dx. \\
 &= [\sin x + \cos x]_0^{\pi/6} \\
 &= \left(\sin \frac{\pi}{6} + \cos \frac{\pi}{6} \right) - (\sin 0 + \cos 0) \\
 &= \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right) - (0 + 1) = \frac{\sqrt{3} - 1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad I &= \int_{\pi/4}^{\pi/2} \sqrt{1 - \sin 2x} \cdot dx = \int_{\pi/4}^{\pi/2} \sqrt{\sin^2 x + \cos^2 x - 2 \sin x \cdot \cos x} \cdot dx \\
 &= \int_{\pi/4}^{\pi/2} (\sin x - \cos x) \cdot dx \\
 &= [-\cos x - \sin x]_{\pi/4}^{\pi/2} \\
 &= \left(-\cos \frac{\pi}{2} - \sin \frac{\pi}{2} \right) - \left(-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \right) \\
 &= (0 - 1) + \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) = (\sqrt{2} - 1)
 \end{aligned}$$

Example 8: Evaluate $\int_0^1 x \cdot \sqrt{\frac{1-x^2}{1+x^2}} \cdot dx$

Solution: Let $I = \int_0^1 x \cdot \sqrt{\frac{1-x^2}{1+x^2}} dx = \int_0^1 \frac{x\sqrt{1-x^2}}{\sqrt{1+x^2}} \times \frac{\sqrt{1-x^2}}{\sqrt{1-x^2}} dx$

$$\begin{aligned}
 &= \int_0^1 \frac{x(1-x^2)}{\sqrt{1-x^4}} \cdot dx = \int_0^1 \frac{x \cdot dx}{\sqrt{1-x^4}} - \int_0^1 \frac{x^3}{\sqrt{1-x^4}} \cdot dx \\
 &= I_1 - I_2
 \end{aligned}$$

To evaluate I_1 so Put $x^2 = t \Rightarrow 2x \cdot dx = dt \Rightarrow x \cdot dx = \frac{1}{2} dt$

Also when $x=0, t=0$ and when $x=1, t=1$

$$\begin{aligned}
 \therefore I_1 &= \int_0^1 \frac{x \cdot dx}{\sqrt{1-x^4}} = \frac{1}{2} \int_0^1 \frac{dt}{\sqrt{1-t^2}} \\
 &= \frac{1}{2} \left[\sin^{-1} t \right]_0^1 = \frac{1}{2} [\sin^{-1}(1) - \sin^{-1} 0] \\
 &= \frac{1}{2} \left(\frac{\pi}{2} - 0 \right) = \frac{\pi}{4}
 \end{aligned}$$

Again for, I_2 , Put $1-x^4 = u$.

Then $-4x^3 \cdot dx = du \Rightarrow x^3 \cdot dx = -\frac{1}{4} du$

When $x=0, u=1$ and When $x=1, u=0$

∴

$$\begin{aligned}
 I_2 &= \int_0^1 \frac{x^3 \cdot dx}{\sqrt{1-x^4}} = -\frac{1}{4} \int_1^0 \frac{du}{\sqrt{u}} = -\frac{1}{4} \int_1^0 (u)^{-1/2} du \\
 &= -\frac{1}{4} \left[\frac{u^{1/2}}{1/2} \right]_1^0 \\
 &= -\frac{1}{2} (0 - 1) = \frac{1}{2}
 \end{aligned}$$

Hence,

$$I = I_1 - I_2 = \left(\frac{\pi}{4} - \frac{1}{2} \right)$$

Example 9: Evaluate the following integrals

(i) $\int_0^{\pi/2} x \cdot \sin x \cdot dx$

(ii) $\int_1^2 x \cdot \log x \cdot dx$

(iii) $\int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(2 + \sin x)} dx$

(iv) $\int_0^{\pi/2} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$

Solution:

(i) $I = \int_0^{\pi/2} x \cdot \sin x \cdot dx.$

$$\begin{aligned}
 &= [-x \cdot \cos x]_0^{\pi/2} - \int_0^{\pi/2} 1(-\cos x) \cdot dx \\
 &= [-x \cdot \cos x]_0^{\pi/2} + \int_0^{\pi/2} \cos x \cdot dx \\
 &= [-x \cdot \cos x]_0^{\pi/2} + [\sin x]_0^{\pi/2} \\
 &= \left(-\frac{\pi}{2} \cos \frac{\pi}{2} + 0 \cdot \cos 0 \right) + \left(\sin \frac{\pi}{2} - \sin 0 \right) \\
 &= (-\pi/2 \times 0 + 0 \times 1) + (1 - 0) = 1
 \end{aligned}$$

(ii) $I = \int_1^2 x \cdot \log x \cdot dx = \int_1^2 (\log x) \cdot x \cdot dx$

$$\begin{aligned}
 &= \left[(\log x) \cdot \frac{x^2}{2} \right]_1^2 - \int_1^2 \frac{1}{x} \cdot \frac{x^2}{2} \cdot dx \\
 &= \frac{1}{2} \left[x^2 \cdot \log x \right]_1^2 - \frac{1}{2} \int_1^2 x \cdot dx \\
 &= \frac{1}{2} \left[x^2 \log x \right]_1^2 - \frac{1}{2} \left[\frac{x^2}{2} \right]_1^2 \\
 &= \frac{1}{2} [2^2 \log 2 - 1^2 \log 1] - \frac{1}{4} (4 - 1) \\
 &= \frac{1}{2} (4 \log 2 - 0) - \frac{3}{4} = 2 \log 2 - \frac{3}{4}.
 \end{aligned}$$

(iii) $I = \int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(2 + \sin x)} dx.$

Put $\sin x = t \Rightarrow \cos x \cdot dx = dt$

Also, $x = 0, t = \sin 0 = 0$. When $x = \frac{\pi}{2}, t = \sin \frac{\pi}{2} = 1$

$$\therefore \int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(2 + \sin x)} \cdot dx = \int_0^1 \frac{dt}{(1+t)(2+t)}$$

Let
$$\frac{1}{(1+t)(2+t)} = \frac{A}{1+t} + \frac{B}{2+t}$$

$$1 = A(2+t) + B(1+t)$$

Put $t = -1$ in the identity, we get

$$1 = A(2-1) + 0 \Rightarrow A = 1$$

and put $t = -2$, we get

$$1 = 0 + B(1-2) \Rightarrow B = -1$$

$$\therefore \frac{1}{(1+t)(2+t)} = \frac{1}{1+t} - \frac{1}{2+t}$$

$$\begin{aligned} \therefore \int_0^1 \frac{1}{(1+t)(2+t)} dt &= \int_0^1 \frac{1}{1+t} \cdot dt - \int_0^1 \frac{1}{2+t} \cdot dt \\ &= [\log(1+t)]_0^1 - [\log(2+t)]_0^1 \\ &= (\log 2 - \log 1) - (\log 3 - \log 2) \\ &= 2 \log 2 - \log 3 = \log 4 - \log 3 = \log \frac{4}{3} \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad I &= \int_0^{\pi/2} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} \\ &= \int_0^{\pi/2} \frac{dx}{\cos^2 x \left(a^2 + b^2 \frac{\sin^2 x}{\cos^2 x} \right)} \end{aligned}$$

When $b \tan x = t \Rightarrow b \sec^2 x \cdot dx = dt$

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\sec^2 x}{a^2 + b^2 \tan^2 x} \cdot dx = \frac{1}{b} \int \frac{dt}{a^2 + t^2} \\ &= \frac{1}{b} \cdot \frac{1}{a} \tan^{-1} \frac{t}{a} \\ &= \frac{1}{ab} \cdot \tan^{-1} \left(\frac{b \tan x}{a} \right) \end{aligned}$$

$$\begin{aligned} \therefore \int_0^{\pi/2} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} &= \frac{1}{ab} \left[\tan^{-1} \left(\frac{b \tan x}{a} \right) \right]_0^{\pi/2} \\ &= \frac{1}{ab} [\tan^{-1} \infty - \tan^{-1} 0] = \frac{1}{ab} \left(\frac{\pi}{2} - 0 \right) = \frac{\pi}{2ab} \end{aligned}$$

Illustrative Examples

(Properties)

Example 10: Prove that $\int_0^{\pi/2} \log \tan x \cdot dx = 0$

Solution: Let $I = \int_0^{\pi/2} \log \tan x \cdot dx$... (1)

$$I = \int_0^{\pi/2} \log \tan(\pi/2 - x) dx;$$

$$\Rightarrow I = \int_0^{\pi/2} \log \cot x \cdot dx \quad \dots (2)$$

Adding (1) and (2), we find:

$$\begin{aligned} 2I &= \int_0^{\pi/2} \{\log \tan x + \log \cot x\} \cdot dx \\ &= \int_0^{\pi/2} \log(\tan x \cdot \cot x) dx = \int_0^{\pi/2} \log 1 \cdot dx \end{aligned}$$

$$2I = 0 \Rightarrow I = 0$$

$$\Rightarrow \int_0^{\pi/2} \log \tan x \cdot dx = 0$$

Example 11: Prove that $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} = \frac{\pi}{4}$

Solution: $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} \cdot dx.$

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin(\pi/2 - x)}}{\sqrt{\sin(\pi/2 - x)} + \sqrt{\cos(\pi/2 - x)}} \cdot dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad \dots (2)$$

Now, adding (1) and (2), we find

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \\ &= \int_0^{\pi/2} dx = [x]_0^{\pi/2} = \frac{\pi}{2} - 0 \end{aligned}$$

$$\Rightarrow 2I = \frac{\pi}{2} \text{ or } I = \frac{\pi}{4}$$

Example 12: Prove that $\int_{-\pi/2}^{\pi/2} \sin^3 x \cdot \cos^2 x \cdot dx = 0$.

Solution: Let us consider $f(x) = \sin^3 x \cdot \cos^2 x$

$$f(-x) = \sin^3(-x) \cdot \cos^2(-x) = [\sin(-x)]^3 [\cos(-x)]^2 = -\sin^3 x \cdot \cos^2 x$$

$$\Rightarrow f(-x) = -f(x)$$

$$\Rightarrow f(x) \text{ is an odd function}$$

$$\therefore \int_{-\pi/2}^{\pi/2} \sin^3 x \cdot \cos x \cdot dx = 0$$

Example 13: Evaluate $\int_0^{\pi/2} \frac{1}{1 + \cot x} \cdot dx$

Solution:

$$I = \int_0^{\pi/2} \frac{1}{1 + \cot x} \cdot dx$$

$$= \int_0^{\pi/2} \frac{1}{1 + \frac{\cos x}{\sin x}} \cdot dx$$

$$I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} \cdot dx \quad \dots(1)$$

$$I = \int_0^{\pi/2} \frac{\sin\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx$$

$$I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx \quad \dots(2)$$

Adding (1) and (2), we find

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} \cdot dx \\ &= \int_0^{\pi/2} dx = [x]_0^{\pi/2} \end{aligned}$$

$$\Rightarrow 2I = \frac{\pi}{2} \text{ or } I = \frac{\pi}{4}$$

Example 14: Prove that

$$(i) \int_0^{\pi/2} \sin 2x \cdot \log \tan x \cdot dx = 0$$

$$(ii) \int_0^1 \log \frac{(1+x)}{1+x^2} \cdot dx = \frac{\pi}{8} \cdot \log 2$$

or

$$\int_0^{\pi/4} \log (1 + \tan x) \cdot dx = \frac{\pi}{8} \cdot \log 2$$

[B.C.A. (Meerut) 2005, 2009]

Solution: (i) $I = \int_0^{\pi/2} \sin 2x \cdot \log \tan x \cdot dx \quad \dots(1)$

$$I = \int_0^{\pi/2} \sin 2\left(\frac{\pi}{2} - x\right) \log \tan\left(\frac{\pi}{2} - x\right) \cdot dx$$

$$= \int_0^{\pi/2} \sin(\pi - 2x) \cdot \log \cot x \cdot dx$$

$$\Rightarrow I = \int_0^{\pi/2} \sin 2x \cdot \log \cot x \cdot dx \quad \dots(2)$$

Adding (1) and (2), we find

$$\begin{aligned}
 2I &= \int_0^{\pi/2} \sin 2x (\log \tan x + \log \cot x) dx \\
 &= \int_0^{\pi/2} \sin 2x \cdot \log(\tan x \cdot \cot x) dx \\
 &= \int_0^{\pi/2} \sin 2x \cdot \log 1 \cdot dx = \int_0^{\pi/2} \sin(2x) \cdot 0 \cdot dx = 0 \quad (\because \log 1 = 0)
 \end{aligned}$$

$$(ii) \quad I = \int_0^1 \frac{\log(1+x)}{1+x^2} \cdot dx$$

Put $x = \tan \theta$

$$\Rightarrow dx = \sec^2 \theta \cdot d\theta$$

when $x = 0$, $\theta = \tan^{-1} 0 = 0$, when $x = 1$, $\theta = \tan^{-1} (1) = \pi/4$

$$\begin{aligned}
 \therefore \quad I &= \int_0^1 \frac{\log(1+x)}{1+x^2} \cdot dx \\
 &= \int_0^{\pi/4} \frac{\log(1+\tan \theta)}{1+\tan^2 \theta} \cdot \sec^2 \theta d\theta \\
 &= \int_0^{\pi/4} \frac{\log(1+\tan \theta)}{\sec^2 \theta} \cdot \sec^2 \theta \cdot d\theta \\
 \Rightarrow \quad I &= \int_0^{\pi/4} \log(1+\tan \theta) \cdot d\theta \quad \dots(1)
 \end{aligned}$$

$$\begin{aligned}
 I &= \int_0^{\pi/4} \log \left\{ 1 + \tan \left(\frac{\pi}{4} - \theta \right) \right\} \cdot d\theta \\
 &= \int_0^{\pi/4} \log \left(\frac{2}{1+\tan \theta} \right) \cdot d\theta \\
 &= \int_0^{\pi/4} \log 2 \cdot d\theta - \int_0^{\pi/4} \log(1+\tan \theta) \cdot d\theta \\
 &= (\log 2) \int_0^{\pi/4} d\theta - I
 \end{aligned}$$

$$\Rightarrow 2I = (\log 2) \cdot [\theta]_0^{\pi/4}$$

$$\Rightarrow 2I = (\log 2) \cdot \frac{\pi}{4}$$

$$\Rightarrow I = \frac{\pi}{8} \cdot \log 2$$

$$\Rightarrow \int_0^1 \log \frac{(1+x)}{1+x^2} dx = \frac{\pi}{8} \log 2$$

Example 15: Evaluate: (i) $\int_{-\pi/2}^{\pi/2} \sin^2 x \cdot dx$ (ii) $\int_0^{2\pi} \cos^5 x \cdot dx$

Solution: (i) Let $f(x) = \sin^2 x$. then

$$f(-x) = \sin^2(-x) = [\sin(-x)]^2 = \sin^2 x$$

$$\Rightarrow f(-x) = f(x)$$

$$\Rightarrow f(x) \text{ is even function;}$$

$$\therefore \int_{-\pi/2}^{\pi/2} \sin^2 x \cdot dx = 2 \int_0^{\pi/2} \sin^2 x \cdot dx = 2 I \text{ (say)}$$

$$\begin{aligned} \text{Now } I &= \int_0^{\pi/2} \sin^2 x \cdot dx \\ &= \int_0^{\pi/2} \sin^2 (\pi/2 - x) \cdot dx \end{aligned} \quad \dots(1)$$

$$\Rightarrow I = \int_0^{\pi/2} \cos^2 x \cdot dx \quad \dots(2)$$

Adding (1) and (2), we find

$$\begin{aligned} 2 I &= \int_0^{\pi/2} (\sin^2 x + \cos^2 x) dx \\ &= \int_0^{\pi/2} 1 \cdot dx = [x]_0^{\pi/2} = \pi/2 \quad \text{or} \quad I = \frac{\pi}{4} \end{aligned}$$

$$(ii) \quad I = \int_0^{2\pi} \cos^5 x \cdot dx$$

$$\text{Let } f(x) = \cos^5 x$$

$$\Rightarrow f(2\pi - x) = [\cos(2\pi - x)]^5 = [\cos x]^5$$

$$\Rightarrow f(2\pi - x) = \cos^5 x = f(x)$$

$$\therefore \int_0^{2\pi} \cos^5 x \cdot dx = 2 \int_0^{\pi} \cos^5 x \cdot dx$$

$$\begin{aligned} \text{Now, } \int_0^{\pi} \cos^5 x \cdot dx &= \int_0^{\pi} \cos^5 (\pi - x) \cdot dx \\ &= \int_0^{\pi} [\cos(\pi - x)]^5 \cdot dx \\ &= - \int_0^{\pi} \cos^5 x \cdot dx \end{aligned}$$

$$\Rightarrow 2 \int_0^{\pi} \cos^5 x \cdot dx = 0$$

$$\therefore \int_0^{2\pi} \cos^5 x \cdot dx = 0$$

Example 16: Prove that $\int_0^{\pi} \frac{x \cdot dx}{1 + \cos^2 x} = \frac{\pi^2}{2\sqrt{2}}$

Solution:

$$\begin{aligned} I &= \int_0^{\pi} \frac{x}{1 + \cos^2 x} \cdot dx \\ &= \int_0^{\pi} \frac{(\pi - x) \cdot dx}{1 + \cos^2 (\pi - x)} \cdot dx \\ &= \int_0^{\pi} \frac{(\pi - x) \cdot dx}{1 + \cos^2 x} \\ &= \pi \int_0^{\pi} \frac{dx}{1 + \cos^2 x} - \int_0^{\pi} \frac{x \cdot dx}{1 + \cos^2 x} \\ &= \pi \int_0^{\pi} \frac{dx}{1 + \cos^2 x} - I \end{aligned}$$

$$\begin{aligned}
 \Rightarrow 2I &= \pi \int_0^{\pi} \frac{dx}{1 + \cos^2 x} \\
 &= 2\pi \int_0^{\pi/2} \frac{dx}{1 + \cos^2 x} \\
 &= 2\pi \int_0^{\pi/2} \frac{\sec^2 x}{1 + \sec^2 x} \cdot dx \\
 &= 2\pi \int_0^{\pi/2} \frac{\sec^2 x \cdot dx}{\tan^2 x + 2}
 \end{aligned}$$

Put, $\tan x = t \Rightarrow \sec^2 x \cdot dx = dt$

when, $x = 0$, $\tan 0 = 0 \Rightarrow t = 0$; when, $x = \frac{\pi}{2}$, $\tan \frac{\pi}{2} = \infty \Rightarrow t = \infty$

$$\begin{aligned}
 \therefore 2I &= 2\pi \int_0^{\infty} \frac{dt}{t^2 + 2} \\
 &= 2\pi \left[\frac{1}{\sqrt{2}} \tan^{-1} \frac{t}{\sqrt{2}} \right]_0^{\infty} \\
 &= 2 \cdot \frac{\pi}{\sqrt{2}} [\tan^{-1} \infty - \tan^{-1} 0] \\
 &= 2 \cdot \frac{\pi}{\sqrt{2}} \cdot \frac{\pi}{2} = \frac{\pi^2}{\sqrt{2}}
 \end{aligned}$$

$$\Rightarrow I = \frac{\pi^2}{2\sqrt{2}}$$

$$\therefore \int_0^{\pi} \frac{x}{1 + \cos^2 x} \cdot dx = \frac{\pi^2}{2\sqrt{2}}$$

Example 17: Prove that $\int_0^{\pi} \frac{x \cdot dx}{a^2 \cos^2 x + b^2 \sin^2 x} = \frac{\pi^2}{2ab}$

$$\begin{aligned}
 \text{Solution: We have } I &= \int_0^{\pi} \frac{x \cdot dx}{a^2 \cos^2 x + b^2 \sin^2 x} \\
 &= \int_0^{\pi} \frac{(\pi - x) dx}{a^2 \cos^2(\pi - x) + b^2 \sin^2(\pi - x)} \\
 &= \int_0^{\pi} \frac{(\pi - x) \cdot dx}{a^2 \cos^2 x + b^2 \sin^2 x} \\
 &= \int_0^{\pi} \frac{\pi \cdot dx}{a^2 \cos^2 x + b^2 \sin^2 x} - \int_0^{\pi} \frac{x \cdot dx}{a^2 \cos^2 x + b^2 \sin^2 x} \\
 &= \pi \int_0^{\pi} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} - I \\
 \Rightarrow 2I &= \pi \int_0^{\pi} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} \\
 &= 2\pi \int_0^{\pi/2} \frac{\sec^2 x \cdot dx}{a^2 + b^2 \tan^2 x}
 \end{aligned}$$

Now, Put $b \tan x = t \Rightarrow b \sec^2 x \cdot dx = dt$

When $x = 0$, $\tan 0 = 0 \Rightarrow t = 0$, When $x = \frac{\pi}{2}$, $\tan \frac{\pi}{2} = \infty \Rightarrow t = \infty$

$$\begin{aligned}
 \therefore 2I &= \frac{2\pi}{b} \int_0^\infty \frac{dt}{a^2 + t^2} \\
 &= \frac{2\pi}{b} \cdot \frac{1}{a} \left[\tan^{-1} \frac{t}{a} \right]_0^\infty \\
 &= \frac{2\pi}{ab} \left[\tan^{-1}(\infty) - \tan^{-1}(0) \right] = \frac{2\pi}{ab} \cdot \frac{\pi}{2} = \frac{\pi^2}{ab} = \frac{\pi^2}{2ab}
 \end{aligned}$$

$$\text{Hence, } \int_0^\pi \frac{x \cdot dx}{a^2 \cos^2 x + b^2 \sin^2 x} = \frac{\pi^2}{2ab}.$$

Example 18: Prove that $\int_0^\pi \frac{x \cdot \tan x}{\sec x + \tan x} \cdot dx = \pi \left(\frac{\pi}{2} - 1 \right)$.

Solution:

$$\begin{aligned}
 I &= \int_0^\pi \frac{x \cdot \tan x}{\sec x + \tan x} \cdot dx \\
 &= \int_0^\pi \frac{(\pi - x) \tan(\pi - x)}{\sec(\pi - x) + \tan(\pi - x)} \cdot dx \\
 &= \int_0^\pi \frac{(\pi - x) \cdot \tan(-x)}{-\sec x - \tan x} \cdot dx \\
 &= \int_0^\pi \frac{\pi \cdot \tan x}{\sec x + \tan x} \cdot dx - \int_0^\pi \frac{x \tan x}{\sec x + \tan x} \cdot dx \\
 &= \int_0^\pi \frac{\pi \tan x}{\sec x + \tan x} \cdot dx - I
 \end{aligned}$$

$\Rightarrow$

$$\begin{aligned}
 2I &= \pi \int_0^\pi \frac{\tan x}{\sec x + \tan x} \cdot dx \\
 &= \pi \int_0^\pi \frac{\sin x}{1 + \sin x} \cdot dx \\
 &= \pi \int_0^\pi \left(1 - \frac{1}{1 + \sin x} \right) \cdot dx \\
 &= \pi \left\{ \int_0^\pi dx - \int_0^\pi \frac{1}{1 + \sin x} \cdot dx \right\} \\
 &= \pi \left\{ [x]_0^\pi - 2 \int_0^{\pi/2} \frac{1}{1 + \sin x} \cdot dx \right\} \\
 &= \pi \left\{ \pi - 2 \int_0^{\pi/2} \frac{dx}{1 + \sin\left(\frac{\pi}{2} - x\right)} \right\} \\
 &= \pi^2 - 2\pi \int_0^{\pi/2} \frac{1}{1 + \cos x} \cdot dx \\
 &= \pi^2 - 2\pi \int_0^{\pi/2} \frac{1}{2 \cos^2 \frac{x}{2}} \cdot dx \\
 &= \pi^2 - \pi \int_0^{\pi/2} \sec^2 \frac{x}{2} \cdot dx \\
 &= \pi^2 - \pi \left[2 \tan \frac{x}{2} \right]_0^{\pi/2} = \pi^2 - \pi \left[2 \tan \frac{\pi}{4} - 2 \tan 0 \right] = \pi^2 - \pi [2 \cdot 1 - 0]
 \end{aligned}$$

$\Rightarrow$

$$2I = \pi^2 - 2\pi$$

$$\Rightarrow I = \pi \left(\frac{\pi}{2} - 1 \right)$$

Example 19: Evaluate $\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x} \cdot dx$

Solution:

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x} \cdot dx \\ &= \int_0^{\pi/2} \frac{\sin^2 \left(\frac{\pi}{2} - x \right)}{\sin \left(\frac{\pi}{2} - x \right) + \cos \left(\frac{\pi}{2} - x \right)} \cdot dx \end{aligned}$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{\cos^2 x}{\cos x + \sin x} \cdot dx$$

$$\therefore 2I = \int_0^{\pi/2} \frac{\sin^2 x + \cos^2 x}{\sin x + \cos x} \cdot dx$$

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{dx}{\sin x + \cos x} \\ &= \frac{1}{\sqrt{2}} \int_0^{\pi/2} \frac{dx}{\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x} \\ &= \frac{1}{\sqrt{2}} \int_0^{\pi/2} \frac{dx}{\sin \frac{\pi}{4} \sin x + \cos \frac{\pi}{4} \cdot \cos x} \\ &= \frac{1}{\sqrt{2}} \int_0^{\pi/2} \frac{dx}{\cos \left(x - \frac{\pi}{4} \right)} \\ &= \frac{1}{\sqrt{2}} \int_0^{\pi/2} \sec \left(x - \frac{\pi}{4} \right) \cdot dx \\ &= \frac{1}{\sqrt{2}} \left[\log \left\{ \sec \left(x - \frac{\pi}{4} \right) + \tan \left(x - \frac{\pi}{4} \right) \right\} \right]_0^{\pi/2} \\ &= \frac{1}{\sqrt{2}} \left[\log \left\{ \sec \left(\frac{\pi}{2} - \frac{\pi}{4} \right) + \tan \left(\frac{\pi}{2} - \frac{\pi}{4} \right) \right\} - \log \left(\sec \frac{\pi}{4} - \tan \frac{\pi}{4} \right) \right] \\ &= \frac{1}{\sqrt{2}} \log \left\{ \frac{\sec \frac{\pi}{4} + \tan \frac{\pi}{4}}{\sec \frac{\pi}{4} - \tan \frac{\pi}{4}} \right\} \\ &= \frac{1}{\sqrt{2}} \log \left(\frac{\sqrt{2} + 1}{\sqrt{2} - 1} \right) \\ &= \frac{1}{\sqrt{2}} \log \left\{ \frac{(\sqrt{2} + 1)^2}{(\sqrt{2} - 1)(\sqrt{2} + 1)} \right\} = \frac{1}{\sqrt{2}} \log \frac{(\sqrt{2} + 1)^2}{(2 - 1)} \\ 2I &= \frac{1}{\sqrt{2}} \log (\sqrt{2} + 1)^2 \\ \Rightarrow I &= \frac{1}{\sqrt{2}} (\log \sqrt{2} + 1) \end{aligned}$$

Example 20: Prove that, $\int_0^{\pi/2} \log \sin x \cdot dx = -\frac{\pi}{2} \log 2$ or $\frac{\pi}{2} \log \frac{1}{2}$.

[B.C.A. (Meerut) 2002]

Solution: Let $I = \int_0^{\pi/2} \log \sin x \cdot dx$... (1)

$\therefore I = \int_0^{\pi/2} \log \sin \left(\frac{\pi}{2} - x \right) dx$

$\Rightarrow I = \int_0^{\pi/2} \log \cos x \cdot dx$... (2)

Adding (1) and (2), we find

$$\begin{aligned} 2I &= \int_0^{\pi/2} (\log \sin x + \log \cos x) dx. \\ &= \int_0^{\pi/2} \log \sin x \cdot \cos x \cdot dx \\ &= \int_0^{\pi/2} \log \left(\frac{2 \sin x \cdot \cos x}{2} \right) dx \\ &= \int_0^{\pi/2} \log \frac{\sin 2x}{2} \cdot dx \\ &= \int_0^{\pi/2} (\log \sin 2x - \log 2) dx = \int_0^{\pi/2} \log \sin 2x - (\log 2) \frac{\pi}{2} \end{aligned}$$

Putting $2x = t$, $dx = \frac{1}{2} dt$

Also when $x = 0$, $t = 0$ and $x = \frac{\pi}{2}$, $t = \pi$

$\therefore 2I = \int_0^{\pi} (\log \sin t) \frac{1}{2} dt - \frac{\pi}{2} \log 2$

$$= \frac{1}{2} \int_0^{\pi} \log \sin t \cdot dt - \frac{\pi}{2} \cdot \log 2 = \frac{1}{2} \cdot 2 \int_0^{\pi/2} \log \sin t \cdot dt - \frac{\pi}{2} \cdot \log 2$$

$\Rightarrow 2I = I - \frac{\pi}{2} \log 2$

$\Rightarrow I = -\frac{\pi}{2} \log 2 = \frac{\pi}{2} \log(2)^{-1}$

$\Rightarrow I = \frac{\pi}{2} \log \left(\frac{1}{2} \right)$

Example 21: Prove that; $\int_0^{\pi/4} \log \sin 2\theta \cdot d\theta = -\frac{\pi}{4} \log 2$

Solution: $I = \int_0^{\pi/4} \log \sin 2\theta \cdot d\theta$... (1)

$\therefore I = \int_0^{\pi/4} \log \sin 2 \left(\frac{\pi}{4} - \theta \right) \cdot d\theta$

$\Rightarrow I = \int_0^{\pi/4} \log \sin \left(\frac{\pi}{2} - 2\theta \right) \cdot d\theta$

$\Rightarrow I = \int_0^{\pi/4} \log \cos 2\theta \cdot d\theta$... (2)

By Adding (1) and (2), we get

$$\begin{aligned}
 2I &= \int_0^{\pi/4} (\log \sin 2\theta + \log \cos 2\theta) d\theta \\
 &= \int_0^{\pi/4} \log(\sin 2\theta \cos 2\theta) d\theta \\
 &= \int_0^{\pi/4} \log\left(\frac{2 \sin 2\theta \cos 2\theta}{2}\right) d\theta \\
 &= \int_0^{\pi/4} \log \frac{\sin 4\theta}{2} d\theta \\
 &= \int_0^{\pi/4} (\log \sin 4\theta - \log 2) d\theta = \int_0^{\pi/4} \log \sin 4\theta d\theta - \frac{\pi}{4} \log 2
 \end{aligned}$$

on putting $2\theta = t \Rightarrow d\theta = \frac{1}{2} dt$

when $\theta = 0, t = 0$, when $\theta = \pi/4, t = \pi/2$

$$\begin{aligned}
 \therefore 2I &= \frac{1}{2} \int_0^{\pi/2} \log \sin 2t dt - \frac{\pi}{4} \log 2 \\
 &= \frac{1}{2} \int_0^{\pi/2} \log \sin 2\left(\frac{\pi}{2} - t\right) dt - \frac{\pi}{4} \log 2 \\
 &= \frac{1}{2} \int_0^{\pi/2} \log \sin 2t dt - \frac{\pi}{4} \log 2 \\
 &= \int_0^{\pi/4} \log \sin 2t dt - \frac{\pi}{4} \log 2
 \end{aligned}$$

$$\Rightarrow 2I = I - \frac{\pi}{4} \log 2, \text{ from (1), we get}$$

$$I = -\frac{\pi}{4} \log 2$$

$$\text{i.e., } \int_0^{\pi/4} \log \sin 2\theta d\theta = -\frac{\pi}{4} \log 2$$

Example 22: Evaluate

$$(a) \int_{-1}^1 \log\left(\frac{2-x}{2+x}\right) dx \quad (b) \int_0^{\pi/2} \frac{dx}{5 \cos x + 4 \sin x}$$

$$(c) \int_0^6 |x+2| dx$$

[B.C.A. (Meerut) 2009]

Solution: We have

$$\begin{aligned}
 \int_{-1}^1 \log \frac{(2-x)}{(2+x)} dx &= \int_{-1}^1 [\log(2-x) - \log(2+x)] dx \\
 &= \int_{-1}^1 1 \cdot \log(2-x) dx - \int_{-1}^1 1 \cdot \log(2+x) dx \\
 &= [\log(2-x)(x)]_{-1}^1 - \int_{-1}^1 \frac{1}{2-x} \times -1 \times x dx \\
 &\quad - \left\{ [\log(2+x)(x)]_{-1}^1 - \int_{-1}^1 \frac{1}{2+x} \times x dx \right\} \\
 &= [\log(1) + \log 3] + \int_{-1}^1 \frac{xdx}{2-x} - \log 3 + \log 1 + \int_{-1}^1 \frac{xdx}{2+x}
 \end{aligned}$$

$$\begin{aligned}
 &= \log 3 - \log 3 + \int_{-1}^1 x \left[\frac{1}{2-x} + \frac{1}{2+x} \right] dx \\
 &= 0 + \int_{-1}^1 \frac{4x}{4-x^2} = 0 \quad (\text{this is odd function : } f(-x) = -f(x))
 \end{aligned}$$

(b)

$$\begin{aligned}
 I &= \int_0^{\pi/2} \frac{dx}{5 \cos x + 4 \sin x} \\
 I &= \int_0^{\pi/2} \frac{dx}{5 \left(\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} \right) + 8 \sin \frac{x}{2} \cos \frac{x}{2}} \\
 &\quad (\text{divide by } N_r \text{ and } D_r \text{ by } \cos^2 \frac{x}{2})
 \end{aligned}$$

$$= \int_0^{\pi/2} \frac{\sec^2 \frac{x}{2} dx}{5 - 5 \tan^2 \frac{x}{2} + 8 \tan \frac{x}{2}}$$

$$\text{put } \tan \frac{x}{2} = t \Rightarrow \sec^2 \frac{x}{2} dx = 2dt$$

$$\begin{aligned}
 &= \int_0^1 \frac{2dt}{5 - 5t^2 + 8t} \\
 &= \frac{-2}{5} \int_0^1 \frac{dt}{t^2 - \frac{8}{5}t - 1} \\
 &= \frac{-2}{5} \int_0^1 \frac{dt}{\left(t^2 - \frac{8}{5}t + \frac{16}{25}\right) - 1 - \frac{16}{25}} \\
 &= \frac{-2}{5} \int_0^1 \frac{dt}{\left(t - \frac{4}{5}\right)^2 - \frac{41}{25}} \\
 &= \frac{2}{5} \int_0^1 \frac{dt}{\left(\frac{\sqrt{41}}{5}\right)^2 - \left(t - \frac{4}{5}\right)^2} \\
 &= \frac{2}{5} \times \frac{1}{2 \times \frac{\sqrt{41}}{5}} \log \left(\frac{\frac{\sqrt{41}}{5} + t - \frac{4}{5}}{\frac{\sqrt{41}}{5} - t + \frac{4}{5}} \right) \Bigg|_0^1 \\
 &= \frac{1}{\sqrt{41}} \left(\log \left\{ \frac{\left(\frac{\sqrt{41}}{5} + \frac{1}{5}\right)}{\frac{\sqrt{41}}{5} - \frac{1}{5}} \right\} - \log \left\{ \frac{\left(\frac{\sqrt{41}}{5} - \frac{4}{5}\right)}{\frac{\sqrt{41}}{5} + \frac{4}{5}} \right\} \right) \\
 &= \frac{1}{\sqrt{41}} \left\{ \log \frac{\sqrt{41}+1}{\sqrt{41}-1} \times \frac{\sqrt{41}+4}{\sqrt{41}-4} \right\} \\
 &= \frac{1}{\sqrt{41}} \log \frac{(\sqrt{41}+1)(\sqrt{41}+4)}{(\sqrt{41}-1)(\sqrt{41}-4)}
 \end{aligned}$$

$$(c) \quad I = \int_0^6 |x+2| dx = \int_0^6 (x+2) dx = \left[\frac{x^2}{2} + 2x \right]_0^6 = \left(\frac{36}{2} + 12 \right) - (0) = 18 + 12 = 30$$

Evaluate the following:

Example 23: $I = \int_2^8 |x-5| dx$

$$\begin{aligned} &= \int_2^5 |x-5| dx + \int_5^8 |x-5| dx \\ &= -\int_2^5 (x-5) dx + \int_5^8 (x-5) dx \\ &= -\left[\frac{x^2}{2} - 5x \right]_2^5 + \left[\frac{x^2}{2} - 5x \right]_5^8 \\ &= -\frac{1}{2}(25-4) + 5(5-2) + \frac{1}{2}(64-25) - 5(8-5) \\ &= -\frac{21}{2} + 15 + \frac{39}{2} - 15 = \frac{39}{2} - \frac{21}{2} = 9. \end{aligned}$$

Example 24: $I = \int_0^4 |x-1| dx$

$$\begin{aligned} &= \int_0^1 |x-1| dx + \int_1^4 |x-1| dx \\ &= -\int_0^1 (x-1) dx + \int_1^4 (x-1) dx \\ &= -\left[\frac{x^2}{2} - x \right]_0^1 + \left[\frac{x^2}{2} - x \right]_1^4 \\ &= -\left(\frac{1}{2} - 1 \right) + \left(\frac{16}{2} - 4 \right) - \left(\frac{1}{2} - 1 \right) = -\left(-\frac{1}{2} \right) + (8-4) - \left(-\frac{1}{2} \right) = \frac{1}{2} + 4 + \frac{1}{2} = 5. \end{aligned}$$

Example 25: $I = \int_2^5 |x-1| dx = \int_2^5 (x-1) dx$

$$= \left[\frac{x^2}{2} - x \right]_2^5 = \left(\frac{25}{2} - 5 \right) - \left(\frac{4}{2} - 2 \right) = \frac{25}{2} - 5 - 2 + 2 = \frac{15}{2}$$



Comprehensive Exercise

(Define Integral)

Evaluate the following (Definite Integral)

Integral

- | | |
|--|---|
| 1. (i) $\int_4^9 \frac{dx}{\sqrt{x}}$ | (ii) $\int_1^{\sqrt{3}} \frac{1}{1+x^2} dx$ |
| (iii) $\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} dx$ | (iv) $\int_0^{\pi/4} \tan x \cdot dx$ |
| (v) $\int_{\pi/4}^{\pi/2} \cos \theta \cdot \operatorname{cosec}^2 \theta \cdot d\theta$ | (vi) $\int_{-\pi/4}^{\pi/4} \operatorname{cosec}^2 x \cdot dx$ |
| (vii) $\int_0^{\pi/2} \sqrt{1+\sin x} \cdot dx$ | (viii) $\int_0^{\pi} \frac{1}{1+\sin x} \cdot dx$ |
| (ix) $\int_0^{\pi/2} x \cdot \cos x \cdot dx$ | (x) $\int_0^{\pi/2} x^2 \cdot \cos 2x \cdot dx$ |
| 2. (i) $\int_0^{1/2} x \cdot \log(1-x) dx$ | (ii) $\int_{-1}^1 x \cdot \tan^{-1} x \cdot dx$ |
| (iii) $\int_1^2 \frac{dx}{(x+1)(x+2)}$ | (iv) $\int_0^{\infty} \frac{dx}{(x^2+a^2)(x^2+b^2)}$ |
| (v) $\int_0^1 \frac{x^2}{1+x^6} dx$ | (vi) $\int_1^3 \frac{1}{x} \cdot \log x \cdot dx$ |
| (vii) $\int_1^{\sqrt{3}} \frac{\tan^{-1} x}{1+x^2} dx$ | (viii) $\int_0^{\infty} \frac{\sin(\tan^{-1} x)}{1+x^2} dx$ |
| (ix) $\int_{-1}^1 \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$ | (x) $\int_0^1 \sin^{-1} \left(\frac{2x}{1+x^2} \right) dx$ |
| 3. (i) $\int_1^2 \frac{x^2+1}{x^4+1} dx$ | (ii) $\int_1^2 \frac{x^2-1}{x^4+1} dx$ |
| (iii) $\int_1^2 \frac{(1+\log x)^4}{x} \cdot dx$ | (iv) $\int_0^{\pi} \frac{1}{5+4 \cos x} \cdot dx$ |
| (v) $\int_0^{\pi/2} \frac{dx}{5+4 \sin x}$ | (vi) $\int_0^{\pi/2} \frac{d\theta}{4 \cos^2 \theta + 9 \sin^2 \theta}$ |
| (vii) $\int_0^a x \cdot \sqrt{\frac{a^2-x^2}{a^2+x^2}} dx$ | (viii) $\int_0^a \frac{x^4 dx}{\sqrt{a^2-x^2}}$ |

[B.C.A. (Meerut) 2005]

(ix) To prove (i) $\int_0^{\infty} \frac{dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{2ab} \cdot \frac{1}{(a+b)}$

[B.C.A. (Meerut) 2005]
[B.C.A. (Meerut) 2002]

$$(ii) \int_0^{\pi} \frac{dx}{1 + \cos \alpha \sin x} = \frac{\pi \alpha}{\sin \alpha}$$

B.C.A [Meerut, 2010]

$$4. \quad (i) \int_{-5}^5 |x + 2| dx$$

$$(ii) \int_{-1}^1 |2x + 1| dx$$



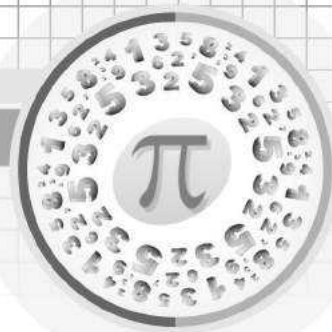
Answers

1.	(i) 2	(ii) $\frac{\pi}{12}$
	(iii) $\frac{\pi}{6}$	(iv) $\frac{1}{2} \log 2$
	(v) $\sqrt{2} - 1$	(vi) -2
	(vii) 2	(viii) 2
	(ix) $\frac{\pi}{2} - 1$	(x) $-\frac{\pi}{4}$
2.	(i) $\frac{3}{8} \log 2 - \frac{5}{16}$	(ii) $\frac{\pi}{2} - 1$
	(iii) $\log \frac{9}{8}$	(iv) $\frac{\pi}{2ab(a+b)}$
	(v) $\frac{\pi}{12}$	(vi) $\frac{1}{2} (\log 3)^2$
	(vii) $\frac{7\pi^2}{288}$	(viii) 1
	(ix) 2	(x) $\frac{\pi}{2} - \log 2$
3.	(i) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3}{2\sqrt{2}} \right)$	(ii) $\frac{1}{2\sqrt{2}} \log \left(\frac{19 + 6\sqrt{2}}{17} \right)$
	(iii) $\frac{1}{5} [(1 + \log 2)^5 - 1]$	(iv) $\frac{\pi}{3}$
	(v) $\frac{2}{3} \tan^{-1} \frac{1}{3}$	(vi) $\frac{\pi}{12}$
	(vii) $\frac{1}{4} a^2 (\pi - 2)$	(viii) $\frac{3\pi a^4}{16}$
4.	(i) 29	(ii) 5 / 2





Unit-4



Chapter

15

Reduction Formulae

15.1 Reduction Formulae

All algebraic linear relations which connects an integral with another integral in which the integrand is of the same form but of lower degree, or a easier to integrate are called **reduction formulae**. In general, integration of such types of functions can be evaluated by applying successively to the reduction formula. This method of integration is called **integration by successive reduction**. Reduction formulae are usually obtained by integration by parts.

15.2 Reduction Formulae for $\int \sin^n x \, dx$ and $\int \cos^n x \, dx$

[B.C.A (Kanpur) 2008, Rohilkhand 2008, 2010]

On integration by parts by taking $\sin x$ as second function, we find

$$\begin{aligned} \int \sin^n x \, dx &= \int \sin^{n-1} x \cdot \sin x \, dx \\ &= \sin^{n-1} x \cdot (-\cos x) - \int (n-1) \sin^{n-2} x \cos x (-\cos x) \, dx \\ &= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) \, dx \\ &= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx. \end{aligned}$$

Transposing the last term in the L.H.S. and then dividing by n , we find

$$\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx \quad \dots(1)$$

which is the required reduction formula for $\sin^n x$.

$$\text{Similarly, } \int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx \quad \dots(2)$$



Illustrative Examples

Example 1: Find the value of

[B.C.A. (Kurukshetra) 2008, 2011]

Solution: We have

$$\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

Put

$$n=6$$

$$\begin{aligned} \therefore \int \sin^6 x \, dx &= -\frac{1}{6} \sin^5 x \cos x + \frac{5}{6} \int \sin^4 x \, dx \\ &= -\frac{1}{6} \sin^5 x \cos x + \frac{5}{6} \left[-\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \int \sin^2 x \, dx \right] \\ &= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x + \frac{5}{8} \int \sin^2 x \, dx \\ &= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x + \frac{5}{8} \left[-\frac{\sin x \cos x}{2} + \frac{1}{2} \int \sin^0 x \, dx \right] \\ &= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x - \frac{5}{16} \sin x \cos x + \frac{5}{16} \int 1 \, dx \\ &= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x - \frac{5}{16} \sin x \cos x + \frac{5}{16} x. \end{aligned}$$

15.3 Walli's Formula

If n is any positive integer, then to find the values of the integrals $\int_0^{\pi/2} \sin^n x \, dx$ and

$$\int_0^{\pi/2} \cos^n x \, dx.$$

[B.C.A. (Delhi) 2004, 2007, 2008, 2009, 2011]

Let $I_n = \int_0^{\pi/2} \sin^n x \, dx$, then from formula (1) of 15.2,

$$I_n = \left[-\frac{\sin^{n-1} x \cos x}{n} \right]_0^{\pi/2} + \frac{n-1}{n} I_{n-2}.$$

If $n \geq 2$, then for both $x=0$ and $x=\frac{\pi}{2}$, $\sin^{n-1} x \cos x = 0$.

$$\begin{aligned} \therefore I_n &= \frac{n-1}{n} I_{n-2} = \frac{n-1}{n} \cdot \frac{n-3}{n-2} I_{n-4} \\ &= \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} I_{n-6} \text{ etc.} \end{aligned}$$

Case I: If n is any positive even integer, then

$$I_n = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \int_0^{\pi/2} x^0 dx$$

$$\text{Hence} \quad \int_0^{\pi/2} \sin^n x dx = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \quad \dots(1)$$

where n is any positive even integer.

Case II: If n is any positive odd integer, then

$$I_n = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{4}{5} \cdot \frac{2}{3} \int_0^{\pi/2} \sin x dx.$$

$$\text{Hence} \quad \int_0^{\pi/2} \sin^n x dx = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \cdot 1 \quad \dots(2)$$

when n is any positive odd integer.

Similarly, we can prove that

$$\int_0^{\pi/2} \cos^n x dx = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \quad \dots(3)$$

where n is any positive even integer and

$$\int_0^{\pi/2} \cos^n x dx = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \cdot 1 \quad \dots(4)$$

where n is any positive odd integer.

$$\text{Illustration:} \quad \int_0^{\pi/2} \sin^9 x dx = \frac{8}{9} \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} = \frac{128}{315}.$$

$$\text{Illustration:} \quad \int_0^{\pi/2} \cos^8 x dx = \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{35}{256} \pi.$$

15.4 Reduction Formula for $\int \sin^m x \cos^n x dx$

This integral can be connected with any of the following integrals:

$$(i) \quad \int \sin^{m-2} x \cos^n x dx \quad (ii) \quad \int \sin^m x \cos^{n-2} x dx$$

$$(iii) \quad \int \sin^{m+2} x \cos^n x dx \quad (iv) \quad \int \sin^m x \cos^{n+2} x dx$$

$$(v) \quad \int \sin^{m+2} x \cos^{n-2} x dx.$$

$$\text{Proof: Let } I(m, n) = \int \sin^m x \cos^n x dx = \int \underbrace{\sin^{m-1} x}_{I} \underbrace{\cos^n x \sin x}_{II} dx.$$

Taking $\sin^{m-1} x$ as the first function and $\cos^n x \sin x$ as the second function and integrating by parts, we get

$$I(m, n) = \sin^{m-1} x \left(-\frac{\cos^{n+1} x}{n+1} \right) - \int (m-1) \sin^{m-2} x \cos x \cdot \left(-\frac{\cos^{n+1} x}{n+1} \right) dx$$

$$\begin{aligned}
&= -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{n+1} \int \sin^{m-2} x \cos^{n+2} x \, dx \\
&= -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{n+1} \int \sin^{m-2} x \cos^n x (1 - \sin^2 x) \, dx \\
&= -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{n+1} I(m-2, n) - \frac{m-1}{n+1} I(m, n).
\end{aligned}$$

Transposing the last term in the L.H.S. and dividing both sides by $1 + \frac{m-1}{n+1}$, we obtain the following reduction formula:

$$I(m, n) = -\frac{\sin^{m-1} x \cos^{n+1} x}{m+n} + \frac{m-1}{m+n} I(m-2, n) \quad \dots(1)$$

Again by writing $\sin^m x \cos^n x = \cos^{n-1} x \cdot \sin^m x \cos x$

and then integrating by parts, we obtain the following reduction formula:

$$I(m, n) = \frac{\sin^{m+1} x \cos^{n-1} x}{m+n} + \frac{n-1}{m+1} I(m, n-2) \quad \dots(2)$$

Formulae (1) and (2) are used to reduce the degree of $\sin x$ and $\cos x$, respectively.

By writing $m+2$ in place of m in formula (1) and then transposing we obtain the following reduction formula:

$$I(m, n) = \frac{\sin^{m+1} x \cos^{n+1} x}{m+1} + \frac{m+n+2}{m+n} I(m+2, n) \quad \dots(3)$$

Again, by writing $n+2$ in place of n in formula (2) and then transposing, we obtain the following reduction formula:

$$I(m, n) = -\frac{\sin^{m+1} x \cos^{n+1} x}{n+1} + \frac{m+n+2}{n+1} I(m, n+2) \quad \dots(4)$$

Alternative Method:

To connect the integral $\int \sin^m x \cos^n x \, dx$ with any one of the following six integrals:

- | | |
|--|---|
| (i) $\int \sin^{m-2} x \cos^n x \, dx$ | (ii) $\int \sin^m x \cos^{n-2} x \, dx$ |
| (iii) $\int \sin^{m+2} x \cos^n x \, dx$ | (iv) $\int \sin^m x \cos^{n+2} x \, dx$ |
| (v) $\int \sin^{m-2} x \cos^{n+2} x \, dx$ | (vi) $\int \sin^{m+2} x \cos^{n-2} x \, dx$ |

To establish the above relations we used the following steps:

Step I: Take $P = \sin^{\lambda+1} x \cos^{\mu+1} x$, where λ and μ are the smaller indices of $\sin x$ and $\cos x$ respectively of the integrals which are to be connected.

Step II: Find $\frac{dP}{dx}$ and arrange it as a linear function of the integrands whose integrals are to be connected.

Step III: Integrate both the sides, transpose and solve to get required relation.

Example 2: To connect $\int \sin^m x \cos^n x \, dx$ with $\int \sin^{m-2} x \cos^{n+2} x \, dx$.

Solution: Let $P = \sin^{(m-2)+1} x \cos^{(n+1)} x$

$$\Rightarrow P = \sin^{m-1} x \cos^{n+1} x.$$

$$\begin{aligned} \therefore \frac{dP}{dx} &= (m-1) \sin^{m-2} x \cos x \cos^{n+1} x + \sin^{m-1} x (n+1) \cos^n x (-\sin x) \\ &= (m-1) \sin^{m-2} x \cos^{n+2} x - (n+1) \sin^m x \cos^n x. \end{aligned}$$

Integrating both sides with respect to x , we get

$$\begin{aligned} P &= (m-1) \int \sin^{m-2} x \cos^{n+2} x \, dx - (n+1) \int \sin^m x \cos^n x \, dx \\ \Rightarrow (n+1) \int \sin^m x \cos^n x \, dx &= -P + (m-1) \int \sin^{m-2} x \cos^{n+2} x \, dx \\ \Rightarrow (n+1) \int \sin^m x \cos^n x \, dx &= -\sin^{m-1} x \cos^{n+1} x \\ &\quad + (m-1) \int \sin^{m-2} x \cos^{n+2} x \, dx \\ \Rightarrow \int \sin^m x \cos^n x \, dx &= -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{m+1} \int \sin^{m-2} x \cos^{n+2} x \, dx. \end{aligned}$$

Example 3: Find the value of $\int \frac{\sin^4 x}{\cos^2 x} \, dx$.

Solution: Here we connect

$$\int \sin^4 x \cos^{-2} x \, dx \quad \text{with} \quad \int \sin^2 x \, dx. \quad \text{So, let}$$

$$P = \sin^{(4-2)+1} x \cos^{(-2+1)} x = \sin^3 x \cos^{-1} x.$$

$$\therefore \frac{dP}{dx} = 3 \sin^2 x + \sin^3 x \cos^{-2} x \sin x = 3 \sin^2 x + \sin^4 x \cos^{-2} x.$$

Integrating both sides w.r. to x , we get

$$\begin{aligned} P &= 3 \int \sin^2 x \, dx + \int \sin^4 x \cos^{-2} x \, dx \\ \Rightarrow \int \sin^4 x \cos^{-2} x \, dx &= P - 3 \int \sin^2 x \, dx \\ \Rightarrow \int \frac{\sin^4 x}{\cos^2 x} \, dx &= \frac{\sin^3 x}{\cos x} - \frac{3}{2} \int 2 \sin^2 x \, dx = \frac{\sin^3 x}{\cos x} - \frac{3}{2} \int (1 - \cos 2x) \, dx \\ &= \frac{\sin^3 x}{\cos x} - \frac{3}{2} \left(x - \frac{\sin 2x}{2} \right) = \frac{\sin^3 x}{\cos x} + \frac{3}{2} \sin x \cos x - \frac{3}{2} x. \end{aligned}$$

15.5 To Find the Value of the Definite Integral $\int_0^{\pi/2} \sin^m x \cos^n x dx$, when m and n are Positive Integers

$$\text{Let } I(m, n) = \int_0^{\pi/2} \sin^m x \cos^n x dx = \left[-\frac{\sin^{m-1} x \cos^{n+1} x}{m+n} \right]_0^{\pi/2} + \frac{m-1}{m+n} I(m-2, n).$$

If $m \geq 2$, then for both $x=0$, $x=\frac{\pi}{2}$, $\sin^{m-1} x \cos^{n+1} x$ is zero.

$$\therefore I(m, n) = \frac{m-1}{m+n} I(m-2, n) \quad \dots(a)$$

Case I: When m and n both are even integer.

Applying formula (a) repeatedly until the degree of $\sin x$ not becomes zero,

$$\begin{aligned} I(m, n) &= \frac{m-1}{m+n} I(m-2, n) = \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} I(m-4, n) \\ &= \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \dots \frac{1}{n+2} I(0, n) \\ &= \frac{(m-1)(m-3)\dots 3 \cdot 1}{(m+n)(m+n-2)\dots (n+2)} \int_0^{\pi/2} \cos^n x dx \\ &= \frac{(m-1)(m-3)\dots 3 \cdot 1}{(m+n)(m+n-2)\dots (n+2)} \cdot \frac{n-1}{n} \cdot \frac{n-3}{n-2} \dots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \quad \dots(1) \end{aligned}$$

Case II: When m is even and n is odd positive integer.

In this case following the above method, we obtain the following formula (remember that here n is odd)

$$I(m, n) = \frac{(m-1)(m-3)\dots 3 \cdot 1}{(m+n)(m+n-2)\dots (n+2)} \cdot \frac{n-1}{n} \cdot \frac{n-3}{n-2} \dots \frac{4}{5} \cdot \frac{2}{3} \quad \dots(2)$$

Case III: When m is odd and n is even positive integer.

This case may be reduces to previous case by writing $\frac{\pi}{2} - x$ in place of x . Hence the formula

(2) is also true in this case.

Case IV: When m and n both are odd positive integers.

As in Case I, we have

$$I(m, n) = \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \dots \frac{2}{n+3} I(1, n)$$

$$\text{But } I(1, n) = \int_0^{\pi/2} \sin x \cos^n x dx = \left[-\frac{\cos^{n+1} x}{n+1} \right]_0^{\pi/2} = \frac{1}{n+1}$$

$$\text{i.e., } I(1, n) = \frac{n-1}{n+1} \cdot \frac{n-3}{n-1} \cdot \frac{n-5}{n-3} \dots \frac{2}{4} \cdot \frac{1}{2}$$

$$\therefore I(m, n) = \frac{(m-1)(m-3)\dots 2}{(m+n)(m+n-2)\dots (n+3)} \cdot \frac{n-1}{n+1} \cdot \frac{n-3}{n-1} \dots \frac{2}{4} \cdot \frac{1}{2} \quad \dots(3)$$

Example 4: Find the value of $\int_0^{\pi/2} \sin^4 x \cos^2 x \, dx$.

Solution: Here $m = 4$ and $n = 2$. Hence by Walli's formula,

$$\int_0^{\pi/2} \sin^4 x \cos^2 x \, dx = \frac{3 \cdot 1 \times 1}{6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2} = \frac{\pi}{32}.$$

Example 5: Find the value of $\int_0^{\pi/2} \cos^3 x \cos 2x \, dx$.

Solution: Since $\cos 2x = \cos^2 x - \sin^2 x$, so by Walli's formula

$$\begin{aligned} \int_0^{\pi/2} \cos^3 x \cos 2x \, dx &= \int_0^{\pi/2} \cos^5 x \, dx - \int_0^{\pi/2} \cos^3 x \sin^2 x \, dx \\ &= \frac{4}{5} \cdot \frac{2}{3} - \frac{2 \times 1}{5 \cdot 3 \cdot 1} = \frac{8}{15} - \frac{2}{15} = \frac{2}{5}. \end{aligned}$$

15.6 Gamma Function

It is easier to express values of some definite integrals in the form of gamma function (the second Eulerian integral). For our present discussion, the following properties of gamma function are just sufficient.

For all positive values of n , $\Gamma(n)$ [read as gamma n]

$$\Gamma(n+1) = n \Gamma(n). \quad \dots(1)$$

Also,
$$\Gamma(1) = 1 \text{ and } \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}. \quad \dots(2)$$

From equations (1) and (2),

$$\begin{aligned} \Gamma(n+1) &= n \Gamma(n) = n(n-1) \Gamma(n-1) \\ &= n(n-1)(n-2) \Gamma(n-2) \\ &= n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 \Gamma(1) \\ &= n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 \end{aligned}$$

Hence
$$\Gamma(n+1) = n! \quad \dots(3)$$

The above relation is just sufficient to define $\Gamma(n)$ if n is either positive integer or half of odd positive integer. For example,

$$\begin{aligned} \Gamma(5) &= 4 \Gamma(4) = 4 \cdot 3 \Gamma(3) = \dots \\ &= 4 \cdot 3 \cdot 2 \cdot 1 \Gamma(1) = 4 \cdot 3 \cdot 2 \cdot 1 = 4! \\ \Gamma\left(\frac{5}{2}\right) &= \frac{3}{2} \Gamma\left(\frac{3}{2}\right) = \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}. \end{aligned}$$

15.7 For any Positive Integer

We have

$$\int_0^{\pi/2} \sin^m x \cos^n x \, dx = \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2 \Gamma\left(\frac{m+n+2}{2}\right)}.$$

Example 6: Find the value of $\int_0^{\pi/2} \sin^4 x \cos^2 x \, dx$.

Solution: We have by gamma function

$$\int_0^{\pi/2} \sin^4 x \cos^2 x \, dx = \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{3}{2}\right)}{2 \Gamma(4)} = \frac{\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{2 \cdot 3 \cdot 2 \cdot 1} \cdot \frac{1}{2} \sqrt{\pi} = \frac{\pi}{32}.$$

Example 7: Find the value of $\int_0^{\pi/2} \cos^3 x \cos 2x \, dx$.

Solution: The given integral

$$\begin{aligned} &= \int_0^{\pi/2} \cos^3 x (\cos^2 x - \sin^2 x) \, dx \\ &= \int_0^{\pi/2} \cos^5 x \, dx - \int_0^{\pi/2} \sin^2 x \cos^3 x \, dx \\ &= \int_0^{\pi/2} \sin^0 x \cos^5 x \, dx - \int_0^{\pi/2} \sin^2 x \cos^3 x \, dx \\ &= \frac{\Gamma\left(\frac{1}{2}\right) \Gamma(3)}{2 \Gamma\left(\frac{7}{2}\right)} - \frac{\Gamma\left(\frac{3}{2}\right) \Gamma(2)}{2 \Gamma\left(\frac{7}{2}\right)} \\ &= \frac{\frac{\sqrt{\pi}}{2} \cdot 2 \cdot 1}{2 \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}} - \frac{\frac{1}{2} \sqrt{\pi} \cdot 1}{2 \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}} = \frac{2}{5}. \end{aligned}$$

Example 8: Evaluate:

(i) $\int_0^{\pi/4} \sin^4 2\theta \, d\theta$

(ii) $\int_0^{\pi/4} \sin^4 \theta \, d\theta.$

Solution: (i) The given integral

$$\begin{aligned} &= \int_0^{\pi/4} \sin^4 2\theta \, d\theta \\ &= \frac{1}{2} \int_0^{\pi/2} \sin^4 t \, dt, \text{ put } 2\theta = t \Rightarrow d\theta = dt/2 \\ &= \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \\ &= \frac{3\pi}{32}. \end{aligned}$$

[From Walli's Formula]

$$\begin{aligned}
 \text{(ii) The given integral} &= \int_0^{\pi/4} \sin^4 \theta \, d\theta = \frac{1}{4} \int_0^{\pi/4} (2 \sin^2 \theta)^2 \, d\theta \\
 &= \frac{1}{4} \int_0^{\pi/4} (1 - \cos 2\theta)^2 \, d\theta = \frac{1}{4} \int_0^{\pi/4} (1 - 2 \cos 2\theta + \cos^2 2\theta) \, d\theta \\
 &= \frac{1}{4} \int_0^{\pi/4} \left[1 - 2 \cos 2\theta + \frac{1}{2} (2 \cos^2 2\theta) \right] d\theta \\
 &= \frac{1}{4} \int_0^{\pi/4} \left[1 - 2 \cos 2\theta + \frac{1}{2} (1 + \cos 4\theta) \right] d\theta \\
 &= \frac{3}{8} \int_0^{\pi/4} d\theta - \frac{1}{2} \int_0^{\pi/4} \cos 2\theta \, d\theta + \frac{1}{8} \int_0^{\pi/4} \cos 4\theta \, d\theta \\
 &= \frac{3}{8} [\theta]_0^{\pi/4} - \frac{1}{4} \int_0^{\pi/2} \cos t \, dt + \frac{1}{32} \int_0^{\pi} \cos u \, du, \\
 &\quad \text{put } 2\theta = t \Rightarrow d\theta = \frac{dt}{2} \text{ and } 4\theta = u \Rightarrow d\theta = \frac{du}{4}] \\
 &= \frac{3\pi}{32} - \frac{1}{4} [\sin t]_0^{\pi/2} + 0 \quad [\text{By property of definite integral}] \\
 &= \frac{3\pi}{32} - \frac{1}{4} = \frac{3\pi - 8}{32}.
 \end{aligned}$$

Example 9: Prove that

$$\begin{aligned}
 \text{(i)} \quad &\int_0^1 x^2 (1-x^2)^{3/2} \, dx = \frac{\pi}{32}. \\
 \text{(ii)} \quad &\int_0^{2a} x^{9/2} (2a-x)^{-1/2} \, dx = \frac{63\pi a^5}{8}.
 \end{aligned}$$

Solution: (i) Here we substituting $x = \sin \theta \Rightarrow dx = \cos \theta \, d\theta$. Also, from $x = \sin \theta$, $\theta = 0$ when $x = 0$ and $\theta = \frac{\pi}{2}$ when $x = 1$.

$$\begin{aligned}
 \therefore \int_0^1 x^2 (1-x^2)^{3/2} \, dx &= \int_0^{\pi/2} \sin^2 \theta (\cos^3 \theta) \cdot \cos \theta \, d\theta \\
 &= \int_0^{\pi/2} \sin^2 \theta \cos^4 \theta \, d\theta = \frac{\Gamma\left(\frac{3}{2}\right) \Gamma\left(\frac{5}{2}\right)}{2\Gamma(4)} = \frac{\frac{1}{2} \sqrt{\pi} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{2 \cdot 3 \cdot 2 \cdot 1} = \frac{\pi}{32}.
 \end{aligned}$$

$$\text{(ii)} \quad I = \int_0^{2a} x^{9/2} (2a-x)^{-1/2} \, dx.$$

$$\text{Let} \quad x = 2a \sin^2 \theta, \Rightarrow dx = 4a \sin \theta \cos \theta \, d\theta.$$

$$\begin{aligned}
 \therefore I &= \int_0^{\pi/2} (2a)^{9/2} \sin^9 \theta (2a - 2a \sin^2 \theta)^{-1/2} \cdot 4a \sin \theta \cos \theta \, d\theta \\
 &= (2a)^4 \cdot 4a \int_0^{\pi/2} \sin^9 \theta \cos^{-1} \theta \cdot \sin \theta \cos \theta \, d\theta \\
 &= 64a^5 \int_0^{\pi/2} \sin^{10} \theta \, d\theta
 \end{aligned}$$

$$= 64a^5 \cdot \frac{9}{10} \cdot \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= \frac{63\pi a^5}{8}$$

[From Walli's Formula]

Example 10: Prove that

$$(i) \int_0^1 x^{3/2} (1-x)^{3/2} dx = \frac{3\pi}{128}$$

$$(ii) \int_0^{2a} x^3 \sqrt{2ax - x^2} dx = \frac{7\pi a^5}{8}$$

$$(iii) \int_0^{2a} x^3 (2ax - x^2)^{3/2} dx = \frac{9\pi a^7}{16}$$

Solution: (i) Let $I = \int_0^1 x^{3/2} (1-x)^{3/2} dx$.Put $x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta d\theta$, we have

$$I = \int_0^{\pi/2} \sin^3 \theta \cos^3 \theta \cdot 2 \sin \theta \cos \theta d\theta$$

$$= 2 \int_0^{\pi/2} \sin^4 \theta \cos^4 \theta d\theta = 2 \frac{\Gamma\left(\frac{4+1}{2}\right) \Gamma\left(\frac{4+1}{2}\right)}{2\Gamma\left(\frac{4+4+2}{2}\right)}$$

$$= \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{5}{2}\right)}{\Gamma(5)} = \frac{\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{3\pi}{128}$$

$$(ii) \text{ Let } I = \int_0^{2a} x^3 \sqrt{2ax - x^2} dx = \int_0^{2a} x^{7/2} \sqrt{2a - x} dx.$$

Put $x = 2a \sin^2 \theta \Rightarrow dx = 4a \sin \theta \cos \theta d\theta$, we have

$$I = \int_0^{\pi/2} 2^{7/2} a^{7/2} \sin^7 \theta \cdot 2^{1/2} a^{1/2} \cos \theta \cdot 4a \sin \theta \cos \theta d\theta$$

$$= 64a^4 \int_0^{\pi/2} \sin^8 \theta \cos^2 \theta d\theta = 64a^5 \frac{\Gamma\left(\frac{8+1}{2}\right) \Gamma\left(\frac{2+1}{2}\right)}{2\Gamma\left(\frac{8+2+2}{2}\right)}$$

$$= 32a^5 \frac{\Gamma\left(\frac{9}{2}\right) \Gamma\left(\frac{3}{2}\right)}{\Gamma(6)} = 32a^5 \frac{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \cdot \frac{1}{2} \sqrt{\pi}}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{7\pi a^5}{8}$$

$$(iii) \text{ Let } I = \int_0^{2a} x^3 (2ax - x^2)^{3/2} dx = \int_0^{2a} x^{9/2} (2a - x)^{3/2} dx$$

Put $x = 2a \sin^2 \theta, \Rightarrow dx = 4a \sin \theta \cos \theta d\theta$. We find

$$I = \int_0^{\pi/2} (2a)^{9/2} \sin^9 \theta (2a)^{3/2} \cos^3 \theta \cdot 4a \sin \theta \cos \theta d\theta$$

$$\begin{aligned}
 &= 256a^7 \int_0^{\pi/2} \sin^{10} \theta \cos^4 \theta \, d\theta = 256a^7 \frac{\Gamma\left(\frac{10+1}{2}\right) \Gamma\left(\frac{4+1}{2}\right)}{2\Gamma\left(\frac{10+4+2}{2}\right)} \\
 &= 256a^7 \frac{\Gamma\left(\frac{11}{2}\right) \Gamma\left(\frac{5}{2}\right)}{2\Gamma(8)} = 256a^7 \frac{\frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}}{2 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{9\pi a^7}{16}
 \end{aligned}$$

Example 11: Evaluate: $\int_0^a \frac{x^4 \, dx}{\sqrt{a^2 - x^2}}$.

Solution: Let $I = \int_0^a \frac{x^4 \, dx}{\sqrt{a^2 - x^2}}$

$$\begin{aligned}
 &= \int_0^{\pi/2} \frac{a^4 \sin^4 \theta \cdot a \cos \theta \, d\theta}{\sqrt{a^2 - a^2 \sin^2 \theta}}, \text{ on putting } x = a \sin \theta, \Rightarrow dx = a \cos \theta \, d\theta \\
 &= a^4 \int_0^{\pi/2} \sin^4 \theta \, d\theta = a^4 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \quad \quad \quad [\text{From Walli's Formula}] \\
 &= \frac{3\pi a^4}{16}
 \end{aligned}$$

Example 12: Evaluate:

$$\begin{aligned}
 (i) \quad & \int_0^a x^4 \sqrt{a^2 - x^2} \, dx & (ii) \quad & \int_0^a x^2 (a^2 - x^2)^{3/2} \, dx \\
 (iii) \quad & \int_0^1 x^4 (1 - x^2)^{3/2} \, dx & (iv) \quad & \int_0^2 x (8 - x^3)^{1/3} \, dx = \frac{16\pi}{9\sqrt{3}}.
 \end{aligned}$$

Solution: (i) Let $I = \int_0^a x^4 \sqrt{a^2 - x^2} \, dx$.

Put $x = a \sin \theta \Rightarrow dx = a \cos \theta \, d\theta$, we have

$$\begin{aligned}
 I &= \int_0^{\pi/2} a^4 \sin^4 \theta \cdot a \cos \theta \cdot a \cos \theta \, d\theta \\
 &= a^6 \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta \, d\theta = a^6 \frac{\Gamma\left(\frac{4+1}{2}\right) \Gamma\left(\frac{2+1}{2}\right)}{2\Gamma\left(\frac{4+2+2}{2}\right)} \\
 &= a^6 \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{3}{2}\right)}{2\Gamma(4)} = a^6 \frac{\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \cdot \frac{1}{2} \sqrt{\pi}}{2 \cdot 3 \cdot 2 \cdot 1} = \frac{\pi a^6}{32}
 \end{aligned}$$

(ii) Let $I = \int_0^a x^2 (a^2 - x^2)^{3/2} \, dx$

Put $x = a \sin \theta \Rightarrow dx = a \cos \theta \, d\theta$, we have

$$\begin{aligned}
 I &= \int_0^{\pi/2} a^2 \sin^2 \theta \cdot a^3 \cos^3 \theta \cdot a \cos \theta d\theta \\
 &= a^6 \int_0^{\pi/2} \sin^2 \theta \cos^4 \theta d\theta = a^6 \frac{\Gamma\left(\frac{2+1}{2}\right) \Gamma\left(\frac{4+1}{2}\right)}{2 \Gamma\left(\frac{2+4+2}{2}\right)} \\
 &= a^6 \frac{\Gamma\left(\frac{3}{2}\right) \Gamma\left(\frac{5}{2}\right)}{2\Gamma(4)} = \frac{\pi a^6}{32}
 \end{aligned}$$

(iii) Let $I = \int_0^1 x^4 (1-x^2)^{3/2} dx$.

Put $x = \sin \theta \Rightarrow dx = \cos \theta d\theta$, we have

$$\begin{aligned}
 I &= \int_0^{\pi/2} \sin^4 \theta \cdot \cos^3 \theta \cdot \cos \theta d\theta \\
 &= \int_0^{\pi/2} \sin^4 \theta \cos^4 \theta d\theta = \frac{\Gamma\left(\frac{4+1}{2}\right) \Gamma\left(\frac{4+1}{2}\right)}{2 \Gamma\left(\frac{4+4+2}{2}\right)} \\
 &= \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{5}{2}\right)}{2\Gamma(5)} = \frac{\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{2 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{3\pi}{256}
 \end{aligned}$$

Example 13: If m and n are positive integer, then prove that

$$\int_0^1 x^{m-1} (1-x)^{n-1} dx = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}.$$

Solution: Let $I = \int_0^1 x^{m-1} (1-x)^{n-1} dx$.

Put $x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta d\theta$ and when $x=0, \theta=0$ and when $x=1, \theta = \frac{\pi}{2}$.

$$\begin{aligned}
 \therefore I &= \int_0^{\pi/2} \sin^{2m-2} \theta \cdot \cos^{2n-2} \theta \cdot 2 \sin \theta \cos \theta d\theta \\
 &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta \\
 &= 2 \frac{\Gamma\left(\frac{2m-1+1}{2}\right) \Gamma\left(\frac{2n-1+1}{2}\right)}{2 \Gamma\left(\frac{2m-1+2n-1+2}{2}\right)} = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}.
 \end{aligned}$$

Example 14: If $I_n = \int_0^a (a^2 - x^2)^n dx$, and $n > 0$, then show that $I_n = \frac{2na^2}{2n+1} I_{n-1}$.

Solution: $I_n = \int_0^a (a^2 - x^2)^n \cdot 1 dx$

$$= \left[(a^2 - x^2)^n \cdot x \right]_0^a - \int_0^a n (a^2 - x^2)^{n-1} \cdot (-2x) \cdot x \, dx,$$

Integrate by parts, we find

$$\begin{aligned} &= 0 - 2n \int_0^a (a^2 - x^2)^{n-1} \cdot \{(a^2 - x^2) - a^2\} \, dx \\ &= -2n \int_0^a (a^2 - x^2)^n \, dx + 2na^2 \int_0^a (a^2 - x^2)^{n-1} \, dx \\ &= -2nI_n + 2na^2 I_{n-1} \end{aligned}$$

$$\text{or} \quad (2n+1) I_n = 2na^2 I_{n-1}$$

$$\therefore I = \frac{2na^2}{2n+1} I_{n-1}$$

Example 15: (i) If $I_n = \int \sin^n x \, dx$, then prove that

$$nI_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2}$$

(ii) If $I_n = \int \cos^n x \, dx$, then prove that $nI_n = \cos^{n-1} x \sin x + (n-1) I_{n-2}$

Solution: (i) From the reduction formula (1) of 15.2, we have

$$\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$\text{or} \quad I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}.$$

$$\therefore nI_n = -\sin^{n-1} x \cos x + (n-1)I_{n-2}$$

(ii) From the reduction formula (2) of 15.2, we have

$$\int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$$\text{or} \quad I_n = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} I_{n-2}$$

$$\therefore n I_n = \cos^{n-1} x \sin x + (n-1)I_{n-2}$$

Example 16: (i) Connect $\int \sin^m x \cos^n x \, dx$ with $\int \sin^{m+2} x \cos^{n-2} x \, dx$.

(ii) Connect $\int \sin^m x \cos^n x \, dx$ with $\int \sin^{m-2} x \cos^n x \, dx$ by reduction formula.

Solution: Let $I(m, n) = \int \sin^m x \cos^n x \, dx$.

Then from 15.4

$$I(m, n) = -\frac{\sin^{m-1} x \cos^{n+1} x}{m+n} + \frac{m-1}{m+n} I(m-2, n) \quad \dots(1)$$

$$I(m, n) = \frac{\sin^{m+1} x \cos^{n-1} x}{m+n} + \frac{n-1}{m+n} I(m, n-2) \quad \dots(2)$$

$$I(m, n) = \frac{\sin^{m+1} x \cos^{n+1} x}{m+1} + \frac{m+n+2}{m+1} I(m+2, n) \quad \dots(3)$$

$$I(m, n) = -\frac{\sin^{m+1} x \cos^{n+1} x}{n+1} + \frac{m+n+2}{n+1} I(m, n+2) \quad \dots(4)$$

(i) Using (2) in reduction formula (3), we get

$$\begin{aligned} I(m, n) &= \frac{\sin^{m+1} x \cos^{n+1} x}{m+1} + \frac{m+n+2}{m+1} \\ &\quad \cdot \left[\frac{\sin^{m+3} x \cos^{n-1} x}{m+n+2} + \frac{n-1}{m+n+2} I(m+2, n-2) \right] \\ &= \frac{\sin^{m+1} x \cos^{n+1} x}{m+1} + \frac{\sin^{m+3} x \cos^{n-1} x}{m+1} + \frac{n-1}{m+1} I(m+2, n-2) \\ \therefore \int \sin^m x \cos^n x \, dx &= \frac{\sin^{m+1} x \cos^{n+1} x}{m+1} + \frac{\sin^{m+3} x \cos^{n-1} x}{m+1} \\ &\quad + \frac{n-1}{m+1} \int \sin^{m+2} x \cos^{n-2} x \, dx. \end{aligned}$$

(ii) From the reduction formula (1), we have

$$\int \sin^m x \cos^n x \, dx = -\frac{\sin^{m-1} x \cos^{n-1} x}{m+n} + \frac{m-1}{m+n} \int \sin^{m-2} x \cos^n x \, dx.$$

Example 17: Show that $\int_0^{\pi/6} \cos^6 3\theta \sin^2 6\theta \, d\theta = \frac{7\pi}{384}$.

Solution:

$$\begin{aligned} I &= \int_0^{\pi/6} \cos^6 3\theta \sin^2 6\theta \, d\theta = \int_0^{\pi/6} \cos^6 3\theta (2 \sin 3\theta \cos 3\theta)^2 \, d\theta \\ &= 4 \int_0^{\pi/6} \sin^2 3\theta \cos^8 3\theta \, d\theta \\ &= \frac{4}{3} \int_0^{\pi/2} \sin^2 t \cos^8 t \, dt, \text{ by putting } 3\theta = t, \Rightarrow d\theta = \frac{dt}{3} \\ &= \frac{4}{3} \cdot \frac{\Gamma\left(\frac{2+1}{2}\right) \Gamma\left(\frac{8+1}{2}\right)}{2\Gamma\left(\frac{2+8+2}{2}\right)} = \frac{2\Gamma\left(\frac{3}{2}\right) \Gamma\left(\frac{9}{2}\right)}{3\Gamma(6)} \quad [\text{From Gamma Function}] \\ &= \frac{2 \cdot \frac{1}{2} \sqrt{\pi} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{3 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{7\pi}{384}. \end{aligned}$$

Example 18: Evaluate: $\int_0^{\pi/6} \cos^4 3\theta \sin^2 6\theta \, d\theta$.

Solution:

$$\begin{aligned} \text{Here } I &= \int_0^{\pi/6} \cos^4 3\theta \sin^2 6\theta \, d\theta \\ &= \int_0^{\pi/2} \cos^4 t \sin^2 2t \, \frac{dt}{3}, \text{ by putting } 3\theta = t \Rightarrow d\theta = \frac{dt}{3} \end{aligned}$$

$$\begin{aligned}
 &= \frac{4}{3} \int_0^{\pi/2} \cos^6 t \sin^2 t \, dt = \frac{4}{3} \frac{\Gamma\left(\frac{6+1}{2}\right) \Gamma\left(\frac{2+1}{2}\right)}{2\Gamma\left(\frac{6+2+2}{2}\right)} \\
 &= \frac{4}{3} \frac{\Gamma\left(\frac{7}{2}\right) \Gamma\left(\frac{3}{2}\right)}{2\Gamma(5)} = \frac{2}{3} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \frac{\sqrt{\pi} \cdot \frac{1}{2} \sqrt{\pi}}{4!} = \frac{5\pi}{192}
 \end{aligned}$$

Example 19: Prove that

$$\int_0^{\pi/2} \sin^m \theta \, d\theta = \frac{m-1}{m} \int_0^{\pi/2} \sin^{m-2} \theta \, d\theta.$$

Solution: From the reduction formula (1) of 15.2, we have

$$\begin{aligned}
 \int_0^{\pi/2} \sin^m \theta \, d\theta &= \left[-\frac{\sin^{m-1} \theta \cos \theta}{m} \right]_0^{\pi/2} + \frac{m-1}{m} \int_0^{\pi/2} \sin^{m-2} \theta \, d\theta \\
 &= 0 + \frac{m-1}{m} \int_0^{\pi/2} \sin^{m-2} \theta \, d\theta = \frac{m-1}{m} \int_0^{\pi/2} \sin^{m-2} \theta \, d\theta
 \end{aligned}$$



Comprehensive Exercise 15.1

(Reduction Formula)

Evaluate:

- $\int_0^{\pi/2} \cos^4 \theta \, d\theta.$
- $\int_0^{\pi/8} \cos^3 4x \, dx$
- $\int_0^{\pi/2} \sin^2 x \cos^4 x \, dx$
- $\int_0^{\pi/2} \sin^5 x \cos^6 x \, dx$

Show that:

- $\int_0^1 x^4 (1-x^2)^{5/2} \, dx = \frac{3\pi}{512}.$
- $\int_0^1 x^6 (1-x^2)^{1/2} \, dx = \frac{5\pi}{256}.$
- $\int_0^1 x^{3/2} \sqrt{1-x} \, dx = \frac{\pi}{16}.$
- $\int_0^a x^2 (ax-x^2)^{1/2} \, dx = \frac{5\pi a^4}{128}.$
- $\int_0^{2a} x^{1/2} (2a-x)^{1/2} \, dx = \frac{63\pi a^5}{8}.$
- If $I_n = \int \cos^n x \, dx$, then prove that $nI_n = \cos^{n-1} x \sin x + (n-1)I_{n-2}.$

Evaluate:

- $\int \sin^2 x \cos^4 x \, dx.$
- $\int \sin^2 x \cos^6 x \, dx.$
- Prove that $\int_a^b (x-a)^m (b-x)^n \, dx = (b-a)^{m+n+1} \frac{\Gamma(m+1) \Gamma(n+1)}{\Gamma(m+n+2)}.$



Answers

1.	$3\pi/16$	2.	$\pi/32$
5.	$1/6$	6.	$8/693$
11.	$\frac{1}{6} \left[-\cos^5 x \sin x \sin x + \frac{1}{4} \cos^3 x \sin x + \frac{3}{8} \cos x \sin x \right] + \frac{x}{16}$		
12.	$\frac{1}{8} \left(-\cos^7 x + \frac{1}{6} \cos^5 x + \frac{5}{24} \cos^3 x + \frac{5}{16} \cos x \right) \sin x + \frac{5x}{128}$		

15.8 Reduction Formulae for $\int \tan^n x \, dx$ and $\int \cot^n x \, dx$

We can write $\tan^n x = \tan^{n-2} x \cdot \tan^2 x = \tan^{n-2} x \cdot (\sec^2 x - 1)$,

Then integrate we get $\int \tan^n x \, dx = \int \tan^{n-2} x \sec^2 x \, dx - \int \tan^{n-2} x \, dx$
 $= \int \tan^{n-2} x \cdot d(\tan x) - \int \tan^{n-2} x \, dx.$

Hence, $\int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx \quad \dots(1)$

Similarly, $\int \cot^n x \, dx = \int \cot^{n-2} x (\operatorname{cosec}^2 x - 1) \, dx$
 $= - \int \cot^{n-2} x \cdot d(\cot x) - \int \cot^{n-2} x \, dx$

Hence, $\int \cot^n x \, dx = -\frac{\cot^{n-1} x}{n-1} - \int \cot^{n-2} x \, dx \quad \dots(2)$

15.9 Reduction Formulae for $\int \sec^n x \, dx$ and $\int \operatorname{cosec}^n x \, dx$

We can write $\sec^n x = \sec^{n-2} x \cdot \sec^2 x$ and integrating by parts, we get

$$\begin{aligned}
 I_n &= \int \sec^n x \, dx = \int \sec^{n-2} x \cdot \sec^2 x \, dx \\
 &= \sec^{n-2} x \tan x - \int (n-2) \sec^{n-3} x \sec x \tan x \cdot \tan x \, dx \\
 &= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) \, dx \\
 &= \sec^{n-2} x \tan x + (n-2) \int \sec^{n-2} x \, dx - (n-2) \int \sec^n x \, dx.
 \end{aligned}$$

Transposing the last term in the L.H.S. and dividing by $(n-1)$, we find

$$\int \sec^n x \, dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx. \quad \dots(1)$$

Hence $I_n = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}.$

Similarly, we can write as

$$\operatorname{cosec}^n x = \operatorname{cosec}^{n-2} x \cdot \operatorname{cosec}^2 x$$

and integrating by parts or by putting $x + \frac{\pi}{2}$ in place of x in the above formula we can prove that

$$\int \operatorname{cosec}^n x \, dx = -\frac{\operatorname{cosec}^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1} \int \operatorname{cosec}^{n-2} x \, dx \quad \dots(2)$$

Example 20: Evaluate: $\int \cot^4 x \, dx$.

Solution: From reduction formula (2) of 15.8, we get

$$\begin{aligned} \int \cot^4 x \, dx &= -\frac{\cot^3 x}{4-1} - \int \cot^2 x \, dx \\ &= -\frac{1}{3} \cot^3 x - \left[-\frac{\cot x}{2-1} - \int \cot^0 x \, dx \right] \\ &= -\frac{1}{3} \cot^3 x + \cot x + \int 1 \, dx = -\frac{1}{3} \cot^3 x + \cot x + x. \end{aligned}$$

Example 21: Evaluate: $\int \tan^4 x \, dx$.

Solution: Putting $n=4, 2$ respectively in formula (1) of 15.8, we get

$$\int \tan^4 x \, dx = \frac{\tan^3 x}{3} - \int \tan^2 x \, dx$$

$$\text{and} \quad \int \tan^2 x \, dx = \frac{\tan x}{1} - \int \tan^0 x \, dx = \tan x - \int 1 \, dx = \tan x - x$$

$$\text{Hence} \quad \int \tan^4 x \, dx = \frac{1}{3} \tan^3 x - (\tan x - x).$$

Example 22: Evaluate: $\int_0^{\pi/4} \sec^3 x \, dx$.

Solution: Putting $n=3$ in formula (1) of 15.9, we find

$$\begin{aligned} \int_0^{\pi/4} \sec^3 x \, dx &= \left[\frac{\sec x \tan x}{2} \right]_0^{\pi/4} + \frac{1}{2} \int_0^{\pi/4} \sec x \, dx \\ &= \frac{1}{2} [\sqrt{2} - 0] + \frac{1}{2} \left[\log \tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right]_0^{\pi/4} \\ &= \frac{1}{\sqrt{2}} + \frac{1}{2} \log \tan \frac{3\pi}{8}. \end{aligned}$$

Example 23: If $I_n = \int \sec^n x \, dx$, then prove that

$$I_n = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

Solution: From reduction formula (1) of 15.9, we get

$$\int \sec^n x \, dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$$

$$\therefore I_n = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

Example 24: Evaluate: $\int_0^{\pi/4} \tan^5 x \, dx$.

Solution: From reduction formula (1) of 15.8, we get

$$\begin{aligned} \int_0^{\pi/4} \tan^5 x \, dx &= \left[\frac{\tan^4 x}{4} \right]_0^{\pi/4} - \int_0^{\pi/4} \tan^3 x \, dx \\ &= \frac{1}{4} - \left\{ \left[\frac{\tan^2 x}{2} \right]_0^{\pi/4} - \int_0^{\pi/4} \tan x \, dx \right\} \\ &= \frac{1}{4} - \frac{1}{2} + [\log \sec x]_0^{\pi/4} = -\frac{1}{4} + \left[\log \sec \frac{\pi}{4} - \log \sec 0 \right] \\ &= -\frac{1}{4} + \log \sqrt{2} = \frac{1}{2} \left[\log 2 - \frac{1}{2} \right] \end{aligned}$$

Example 25: If $\phi(n) = \int_0^{\pi/4} \tan^n x \, dx$, then prove that

$$\phi(n) + \phi(n-2) = \frac{1}{n-1}$$

Hence find the value of $\phi(5)$.

Solution: From reduction formula (1) of 15.8, we have

$$\begin{aligned} \phi(n) &= \int_0^{\pi/4} \tan^n x \, dx = \left[\frac{\tan^{n-1} x}{n-1} \right]_0^{\pi/4} - \int_0^{\pi/4} \tan^{n-2} x \, dx \\ &= \frac{1}{n-1} - \phi(n-2). \end{aligned}$$

$$\therefore \phi(n) + \phi(n-2) = \frac{1}{n-1} \quad \dots(1)$$

Putting $n=5$ in equation (1), we get

$$\begin{aligned} \phi(5) &= \frac{1}{4} - \phi(3) = \frac{1}{4} - \left[\frac{1}{2} - \phi(1) \right], & [\text{From (1)}] \\ &= -\frac{1}{4} + \phi(1) = -\frac{1}{4} + \int_0^{\pi/4} \tan x \, dx \\ &= -\frac{1}{4} + [\log \sec x]_0^{\pi/4} = -\frac{1}{4} + \log \sqrt{2} = \frac{1}{2} \left[\log 2 - \frac{1}{2} \right]. \end{aligned}$$



Comprehensive Exercise 15.2

$(\tan^n \theta, \cot^n \theta)$

Integrate with respect to x :

1. $\tan^3 x$ 2. $\cot^5 x$ 3. $\sec^3 x$ 4. $\operatorname{cosec}^5 x$

Evaluate:

5. $\int \frac{d\theta}{\sin^3 \theta}$ 6. $\int_0^{\pi/4} \sec^5 \theta \, d\theta$.
7. $\int_0^a (a^2 + x^2)^{5/2} \, dx$.
8. If $I_n = \int \operatorname{cosec}^n x \, dx$, then prove that $I_n = -\frac{\operatorname{cosec}^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1} I_{n-2}$.
9. $\int (1+x^2)^{3/2} \, dx$.



Answers

- | | |
|----|--|
| 1. | $\frac{1}{2} \tan^2 x + \log \cos x$ |
| 2. | $-\frac{1}{4} \cot^4 x + \frac{1}{2} \cot^2 x + \log \sin x$ |
| 3. | $\frac{1}{2} \sec x \tan x + \frac{1}{2} \log (\sec x + \tan x)$ |
| 4. | $-\frac{1}{4} \operatorname{cosec}^3 x \cot x - (3/8) \operatorname{cosec} x \cot x + (3/8) \log \tan (x/2)$ |
| 5. | $-\frac{1}{2} \operatorname{cosec} \theta \cot \theta + \frac{1}{2} \log \tan \theta/2$ |
| 6. | $(7/4\sqrt{2}) + (3/8) \log (\sqrt{2} + 1)$ |
| 7. | $(a^6/48) \{67\sqrt{2} + 15 \log (\sqrt{2} + 1)\}$ |
| 9. | $\frac{1}{4} x (1+x^2)^{1/2} + \frac{3}{8} x (1+x^2)^{1/2} + \frac{3}{8} [x + \sqrt{1+x^2}]$ |

15.10 Reduction Formulae for $x^n \sin mx$ and $x^n \cos mx$

On integration by parts, we get

$$\begin{aligned}
 \int x^n \sin mx \, dx &= x^n \left(-\frac{\cos mx}{m} \right) - \int nx^{n-1} \left(\frac{-\cos mx}{m} \right) dx \\
 &= -\frac{x^n \cos mx}{m} + \frac{n}{m} \int x^{n-1} \cos mx \, dx \\
 &= -\frac{x^n \cos mx}{m} + \frac{n}{m} \left\{ \frac{x^{n-1} \sin mx}{m} - \frac{n-1}{m} \int x^{n-2} \sin mx \, dx \right\}
 \end{aligned}$$

$$\text{or} \quad \int x^n \sin mx \, dx = -\frac{x^n \cos mx}{m} + \frac{nx^{n-1} \sin mx}{m^2} - \frac{n(n-1)}{m^2} \cdot \int x^{n-2} \sin mx \, dx$$

which is the required **Reduction Formula**.

Similarly, we can prove

$$\int x^n \cos mx \, dx = \frac{x^n \sin mx}{m} + \frac{nx^{n-1} \cos mx}{m^2} - \frac{n(n-1)}{m^2} \int x^{n-2} \cos mx \, dx$$

15.11 Reduction Formulae for $x \sin^n x$ and $x \cos^n x$

$$\begin{aligned} \text{Let } I_n &= \int x \sin^n x \, dx = \int (x \sin^{n-1} x) \sin x \, dx \\ &= (x \sin^{n-1} x) (-\cos x) + \int \{ \sin^{n-1} x + x(n-1) \sin^{n-2} x \cos x \} \cdot \cos x \, dx \\ &= -x \cos x \sin^{n-1} x + \int \sin^{n-1} x \cos x \, dx + (n-1) \int x \sin^{n-2} x \cos^2 x \, dx \\ &= -x \cos x \sin^{n-1} x + \left\{ \frac{\sin^n x}{n} \right\} + (n-1) \int x \sin^{n-2} x (1 - \sin^2 x) \, dx \\ &= -x \cos x \sin^{n-1} x + \frac{\sin^n x}{n} + (n-1) \int x \sin^{n-2} x \, dx - (n-1) \int x \sin^n x \, dx \\ &= -x \cos x \sin^{n-1} x + \frac{\sin^n x}{n} + (n-1) I_{n-2} - (n-1) I_n \end{aligned}$$

$$\Rightarrow \quad nI_n = -x \cos x \sin^{n-1} x + \frac{\sin^n x}{n} + (n-1) I_{n-2}$$

$$\therefore \quad \int x \sin^n x \, dx = -\frac{x \cos x \sin^{n-1} x}{n} + \frac{\sin^n x}{n^2} + \frac{n-1}{n} \int x \sin^{n-2} x \, dx$$

which is the required **reduction formula**.

Similarly, the reduction formula for $\int x \cos^n x \, dx$ can be obtained as:

$$\int x \cos^n x \, dx = \frac{x \sin x \cos^{n-1} x}{n} + \frac{\cos^n x}{n^2} - \frac{n-1}{n} \int x \cos^{n-2} x \, dx.$$

15.12 Reduction Formulae for $\int \cos^m x \sin nx \, dx$

$$\text{Let } I(m, n) = \int \cos^m x \sin nx \, dx.$$

Taking $\sin nx$ as the second function integrating by parts, we find

$$\begin{aligned} I(m, n) &= \cos^m x \cdot \left[\frac{\cos nx}{(-n)} \right] - \int m \cos^{m-1} x (-\sin x) \cdot \frac{\cos nx}{-n} \, dx \\ &= -\frac{\cos^m x \cos nx}{n} - \frac{m}{n} \int \cos^{m-1} x (\sin x \cos nx) \, dx. \end{aligned}$$

But $\sin x \cos nx = \sin nx \cos x - \sin (n-1)x$. By this relation in the above equation, we find

$$\begin{aligned} I(m, n) &= -\frac{\cos^m x \cos nx}{n} - \frac{m}{n} \int \cos^{m-1} x [\sin nx \cos x - \sin (n-1)x] dx \\ &= -\frac{\cos^m x \cos nx}{n} + \frac{m}{n} \int \cos^{m-1} x \sin (n-1)x dx - \frac{m}{n} I(m, n). \end{aligned}$$

Transposing the last term in the L.H.S. and then dividing both the sides by $\frac{m+n}{n}$, we find

$$\int \cos^m x \sin nx dx = -\frac{\cos^m x \cos nx}{m+n} + \frac{m}{m+n} \int \cos^{m-1} x \sin (n-1)x dx$$

or
$$I(m, n) = -\frac{\cos^m x \cos nx}{m+n} + \frac{m}{m+n} I(m-1, n-1)$$

Similarly,
$$I(m, n) = \int \cos^m x \cos nx dx = \frac{\cos^m x \sin nx}{m+n} + \frac{m}{m+n} I(m-1, n-1)$$

and
$$I(m, n) = \int \sin^m x \cos nx dx$$

$$= -\frac{n \sin^m x \sin nx + m \sin^{m-1} x \cos x \cos nx}{m^2 - n^2} + \frac{m(m-1)}{m^2 - n^2} I(m-2, n)$$

Example 26: Evaluate: $\int_0^1 (\sin^{-1} x)^4 dx$.

Solution: Let $I = \int_0^1 (\sin^{-1} x)^4 dx$. On putting $x = \sin t$, so that $dx = \cos t dx$

$$\begin{aligned} I &= \int_0^{\pi/2} t^4 \cos t dt, \\ &= \left[\frac{t^4 \cdot \sin t}{1} \right]_0^{\pi/2} + \left[\frac{4t^3 \cos t}{1^2} \right]_0^{\pi/2} - \frac{4 \cdot 3}{1^2} \int_0^{\pi/2} t^2 \cos t dt, \text{ on putting} \end{aligned}$$

$$\begin{aligned} & \quad n=4, m=1 \text{ in the second reduction formula of 15.10} \\ &= \left[\left(\frac{\pi}{2} \right)^2 \cdot 1 - 0 \right] + 0 - 12 \left\{ \left[\frac{t^2 \sin t}{1} \right]_0^{\pi/2} + \left[\frac{2t \cos t}{1^2} \right]_0^{\pi/2} - \frac{2 \cdot 1}{1^2} \int_0^{\pi/2} 1^2 \cos t dt \right\} \\ &= \frac{\pi^4}{16} - 12 \left\{ \left[\left(\frac{\pi}{2} \right)^2 - 0 \right] + 0 - 2 [\sin t]_0^{\pi/2} \right\} = \frac{\pi^4}{16} - 12 \left[\frac{\pi^2}{4} - 2(1-0) \right] \\ &= \frac{\pi^4}{16} - 12 \left[\frac{\pi^2}{4} - 2 \right] = \frac{1}{16} \pi^4 - 3\pi^2 + 24 \end{aligned}$$

Example 27: If n is any positive integer, then prove that

$$\int_0^{\pi/2} \cos^n x \cos nx dx = \frac{\pi}{2^{n+1}}.$$

Solution: Let $I_n = \int_0^{\pi/2} \cos^n x \cos nx dx$. Then on integration by parts, we get

$$I_n = \left[\cos^n x \frac{\sin nx}{n} \right]_0^{\pi/2} - \int_0^{\pi/2} n \cos^{n-1} x (-\sin x) \cdot \frac{\sin nx}{n} dx$$

$$\begin{aligned}
 &= 0 + \int_0^{\pi/2} \cos^{n-1} x \cdot (\sin nx \cdot \sin x) dx \\
 &= \int_0^{\pi/2} \cos^{n-1} x [\cos (n-1)x - \cos nx \cos x] dx = I_{n-1} - I_n.
 \end{aligned}$$

Hence $\int_0^{\pi/2} \cos^n x \cos nx dx = \frac{1}{2} \int_0^{\pi/2} \cos^{n-1} x \cos (n-1)x dx.$

Applying the above formula successively, we obtain

$$I_n = \frac{1}{2} I_{n-1} = \frac{1}{2^2} I_{n-2} = \frac{1}{2^3} I_{n-3} = \dots$$

$$\therefore I_n = \frac{1}{2^n} I_0 = \frac{1}{2^n} \int_0^{\pi/2} \cos^0 x \cos 0x dx = \frac{1}{2^n} \int_0^{\pi/2} dx.$$

Hence $\int_0^{\pi/2} \cos^n x \cos nx dx = \frac{1}{2^n} [x]_0^{\pi/2} = \frac{\pi}{2^{n+1}}$

Example 28: Prove that $\int_0^{\pi/2} \cos^m x \sin mx dx = \frac{1}{2^{m+1}} \left\{ 2 + \frac{2^2}{2} + \frac{2^3}{3} + \dots + \frac{2^m}{m} \right\}.$

Solution: We have

$$\int_0^{\pi/2} \cos^m x \sin mx dx = \left[-\frac{\cos^m x \cos mx}{m+m} \right]_0^{\pi/2} + \frac{m}{m+m} \int_0^{\pi/2} \cos^{m-1} x \sin (m-1)x dx$$

$$\Rightarrow I(m, m) = \frac{1}{2m} + \frac{1}{2} I(m-1, m-1). \quad \dots(1)$$

On substituting $m-1$ in place of m in equation (1), we get

$$I(m-1, m-1) = \frac{1}{2(m-1)} + \frac{1}{2} I(m-2, m-2) \quad \dots(2)$$

Now, from equations (1) and (2),

$$\begin{aligned}
 I(m, m) &= \frac{1}{2m} + \frac{1}{2} \left[\frac{1}{2(m-1)} + \frac{1}{2} I(m-2, m-2) \right] \\
 &= \frac{1}{2m} + \frac{1}{2^2(m-1)} + \frac{1}{2^2} I(m-2, m-2) \\
 &= \frac{1}{2m} + \frac{1}{2^2(m-1)} + \frac{1}{2^3(m-2)} + \frac{1}{2^3} I(m-3, m-3) \\
 &= \frac{1}{2m} + \frac{1}{2^2(m-1)} + \frac{1}{2^3(m-2)} + \dots + \frac{1}{2^m \{m-(m-1)\}} \\
 &\quad + \frac{1}{2^m} I(m-m, m-m), \text{ on carrying over the above process} \\
 &= \frac{1}{2m} + \frac{1}{2^2(m-1)} + \frac{1}{2^3(m-2)} + \dots + \frac{1}{2^m \cdot 1} + \frac{1}{2^m} I(0, 0) \\
 &= \frac{1}{2m} + \frac{1}{2^2(m-1)} + \frac{1}{2^3(m-2)} + \dots + \frac{1}{2^m \cdot 1} + \frac{1}{2^m} \int_0^{\pi/2} 0 dx
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2^m} + \frac{1}{2^2 (m-1)} + \frac{1}{2^3 (m-2)} + \dots + \frac{1}{2^m \cdot 1} \\
 &\quad \left[\because \int_0^{\pi/2} 0 \, dx = [c]_0^{\pi/2} = c - c = 0 \right] \\
 &= \frac{1}{2^m \cdot 1} + \frac{1}{2^{m-1} \cdot 2} + \frac{1}{2^{m-2} \cdot 3} + \dots + \frac{1}{2^m},
 \end{aligned}$$

on writing the series in reverse order

$$\begin{aligned}
 &= \frac{1}{2^{m+1}} \left[\frac{2^{m+1}}{2^m \cdot 1} + \frac{2^{m+1}}{2^{m-1} \cdot 2} + \frac{2^{m+1}}{2^{m-2} \cdot 3} + \dots + \frac{2^{m+1}}{2^m} \right] \\
 \therefore \quad I(m, m) &= \frac{1}{2^{m+1}} \left[\frac{2}{1} + \frac{2^2}{2} + \frac{2^3}{3} + \dots + \frac{2^m}{m} \right].
 \end{aligned}$$

Example 29: If $U_n = \int_0^{\pi/2} x^n \sin x \, dx$ and $n > 1$, then show that

$$U_n + n(n-1)U_{n-2} = n\left(\frac{\pi}{2}\right)^{n-1}.$$

Hence find $\int_0^{\pi/2} x^5 \sin x \, dx$

Solution: On two successive integration by parts, we obtain

$$\begin{aligned}
 U_n &= \left[x^n \cdot (-\cos x) \right]_0^{\pi/2} - \int_0^{\pi/2} nx^{n-1} (-\cos x) \, dx \\
 &= 0 + n \left[x^{n-1} \cdot \sin x - \int (n-1)x^{n-2} \sin x \, dx \right]_0^{\pi/2} \\
 &= n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1) \int_0^{\pi/2} x^{n-2} \sin x \, dx. \quad \dots(1)
 \end{aligned}$$

Hence $U_n + n(n-1)U_{n-2} = n\left(\frac{\pi}{2}\right)^{n-1}.$

Second Part: On putting $n = 5, 3$ respectively in (1), we get

$$\begin{aligned}
 \int_0^{\pi/2} x^5 \sin x \, dx &= 5\left(\frac{\pi}{2}\right)^4 - 5 \cdot 4 \int_0^{\pi/2} x^3 \sin x \, dx \\
 &= 5\left(\frac{\pi}{2}\right)^4 - 20 \left\{ 3\left(\frac{\pi}{2}\right)^2 - 3 \cdot 2 \int_0^{\pi/2} x \sin x \, dx \right\} \\
 &= 5\left(\frac{\pi}{2}\right)^4 - 60\left(\frac{\pi}{2}\right)^2 + 120 \int_0^{\pi/2} x \sin x \, dx.
 \end{aligned}$$

But $\int_0^{\pi/2} x \sin x \, dx = \left[x \cdot (-\cos x) - 1 \cdot (1 - \cos x) \right]_0^{\pi/2} = \left[-x \cos x + \sin x \right]_0^{\pi/2} = 1.$

Hence $\int_0^{\pi/2} x^5 \sin x \, dx = 5\left(\frac{\pi}{2}\right)^4 - 60\left(\frac{\pi}{2}\right)^2 + 120.$

Example 30: Evaluate:

$$(i) \int_0^{\pi/2} x^3 \sin 3x \, dx.$$

$$(ii) \int_0^{\pi/2} x \cos^3 x \, dx.$$

Solution: (i) Let $I = \int_0^{\pi/2} x^3 \sin 3x \, dx$.

On integration by parts, we get

$$\begin{aligned} I &= \left[x^3 \cdot \frac{(-\cos 3x)}{3} \right]_0^{\pi/2} - \int_0^{\pi/2} 3x^2 \cdot \frac{(-\cos 3x)}{3} \, dx \\ &= 0 + \int_0^{\pi/2} x^2 \cos 3x \, dx = \left[x^2 \cdot \frac{\sin 3x}{3} \right]_0^{\pi/2} - \int_0^{\pi/2} 2x \cdot \frac{\sin 3x}{3} \, dx \\ &= \frac{1}{3} \left[\left(\frac{\pi}{2} \right)^2 \cdot (-1) - 0 \right] - \frac{2}{3} \int_0^{\pi/2} x \sin 3x \, dx \\ &= -\frac{\pi^2}{12} - \frac{2}{3} \left\{ \left[x \cdot \frac{(-\cos 3x)}{3} \right]_0^{\pi/2} - \int_0^{\pi/2} 1 \cdot \frac{(-\cos 3x)}{3} \, dx \right\} \\ &= -\frac{\pi^2}{12} - \frac{2}{3} \left\{ 0 + \frac{1}{3} \int_0^{\pi/2} \cos 3x \, dx \right\} = -\frac{\pi^2}{12} - \frac{2}{9} \left[\frac{\sin 3x}{3} \right]_0^{\pi/2} \\ &= -\frac{\pi^2}{12} - \frac{2}{27} [(-1) - 0] = \frac{2}{27} - \frac{\pi^2}{12}. \end{aligned}$$

$$(ii) \text{ Let } I = \int_0^{\pi/2} x \cos^3 x \, dx.$$

On integration by parts, we get

$$I = \left[x \cdot \frac{\sin x \cos^2 x}{3} \right]_0^{\pi/2} + \left[\frac{\cos^3 x}{3^2} \right]_0^{\pi/2} + \frac{2}{3} \int_0^{\pi/2} x \cos x \, dx,$$

on putting $n=3$ in the second reduction formula of 15.11

$$\begin{aligned} &= 0 + \frac{1}{9} [0 - 1] + \frac{2}{3} \left\{ \left[x \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} 1 \cdot \sin x \, dx \right\} \\ &= -\frac{1}{9} + \frac{2}{3} \left\{ \left[\frac{\pi}{2} \cdot 1 - 0 \right] - \left[(-\cos x) \right]_0^{\pi/2} \right\} \\ &= -\frac{1}{9} + \frac{\pi}{3} + \frac{2}{3} [0 - 1] = -\frac{1}{9} + \frac{\pi}{3} - \frac{2}{3} = \frac{1}{3} \left(\pi - \frac{7}{3} \right). \end{aligned}$$

Example 31: If $U_n = \int_0^{\pi/2} \theta \sin^n \theta \, d\theta$ and $n > 1$, then show that $U_n = \frac{1}{n^2} + \frac{n-1}{n} U_{n-2}$

and deduce that $U_5 = \frac{149}{225}$.

Solution: In 15.11, we have

$$\int \theta \sin^n \theta \, d\theta = -\frac{\theta \sin^{n-1} \theta \cos \theta}{n} + \frac{\sin^n \theta}{n^2} + \frac{n-1}{n} \int \theta \sin^{n-2} \theta \, d\theta$$

$$\therefore \int_0^{\pi/2} \theta \sin^n \theta \, d\theta = \left\{ -\frac{\theta \sin^{n-1} \theta \cos \theta}{n} + \frac{\sin^n \theta}{n^2} \right\}_0^{\pi/2} + \frac{n-1}{n} \int_0^{\pi/2} \theta \sin^{n-2} \theta \, d\theta.$$

$$\Rightarrow U_n = \left\{ \frac{1}{n^2} - 0 \right\} + \frac{n-1}{n} U_{n-2}.$$

$$\text{Hence } U_n = \frac{1}{n^2} + \frac{n-1}{n} U_{n-2}.$$

Putting $n=5, 3$ respectively in equation (1), we get

$$U_5 = \frac{1}{25} + \frac{4}{5} U_3, \quad U_3 = \frac{1}{9} + \frac{2}{3} U_1.$$

$$\begin{aligned} \therefore U_5 &= \left(\frac{1}{25} + \frac{4}{45} \right) + \frac{8}{15} \int_0^{\pi/2} \theta \sin \theta \, d\theta \\ &= \frac{29}{225} + \frac{8}{15} \left[\left\{ -\theta \cos \theta \right\}_0^{\pi/2} + \int_0^{\pi/2} \cos \theta \, d\theta \right] \\ &= \frac{29}{225} + \frac{8}{15} \left[0 + \left\{ \sin \theta \right\}_0^{\pi/2} \right] = \frac{29}{225} + \frac{8}{15} = \frac{149}{225}. \end{aligned}$$

Example 32: Show that $\int_0^{\pi/2} \cos^{n-2} x \sin nx \, dx = \frac{1}{n-1}$ where n is a positive integer greater than 1.

$$\begin{aligned} \text{Solution: Let } I &= \int_0^{\pi/2} \cos^{n-2} x \sin nx \, dx = \int_0^{\pi/2} \cos^{n-2} x \sin \{(n-1)x + x\} \, dx \\ &= \int_0^{\pi/2} \cos^{n-2} x \{ \sin(n-1)x \cos x + \cos(n-1)x \sin x \} \, dx \\ &= \int_0^{\pi/2} \cos^{n-1} x \sin(n-1)x \, dx + \int_0^{\pi/2} \cos^{n-2} x \cos(n-1)x \sin x \, dx \\ &= I_1 + I_2, \text{ say.} \end{aligned}$$

$$\begin{aligned} \text{Now } I_1 &= \int_0^{\pi/2} \cos^{n-1} x \sin(n-1)x \, dx \\ &= \left[\cos^{n-1} x \cdot \left\{ -\frac{\cos(n-1)x}{n-1} \right\} \right]_0^{\pi/2} \\ &\quad - \int_0^{\pi/2} (n-1) \cos^{n-2} x \cdot (-\sin x) \cdot \left\{ -\frac{\cos(n-1)x}{n-1} \right\} dx \\ &= \frac{1}{n-1} - \int_0^{\pi/2} \cos^{n-2} x \cos(n-1)x \sin x \, dx = \frac{1}{n-1} - I_2 \end{aligned}$$

$$\therefore I = \frac{1}{n-1} - I_2 + I_2 = \frac{1}{n-1}$$

$$\text{Hence } \int_0^{\pi/2} \cos^{n-2} x \sin nx \, dx = \frac{1}{n-1}.$$



Comprehensive Exercise 15.3

$$(x^m \sin \theta, x^n \cos m \theta)$$

Evaluate:

1. $\int x^2 \sin 2x \, dx.$
2. $\int x^2 \cos 2x \, dx.$
3. $\int x \sin^3 x \, dx.$
4. $\int x \cos^5 x \, dx.$

Prove that:

5. $\int_0^{\pi/2} \cos^5 x \sin 3x \, dx = \frac{1}{3}.$
6. If $u_n = \int_0^{\pi/2} x^n \sin mx \, dx$, then prove that $u_n = \frac{n}{m^2} \left(\frac{\pi}{2}\right)^{n-1} - \frac{n^{(n-1)}}{m^2} u_{n-2}$ where m is of the form $4r + 1$.
7. If $I(m, n) = \int_0^{\pi/2} \cos^m x \cos nx \, dx$, then prove that $I(m, n) = \frac{m(m-1)}{m^2 - n^2} I(m-2, n).$
8. If $I(m, n) = \int_0^{\pi/2} \cos^m x \sin nx \, dx$, then prove that

$$I(m, n) = \frac{1}{m+n} + \frac{m}{m+n} I(m-1, n-1).$$
9. If $f(m, n) = \int_0^{\pi/2} \cos^m x \cos nx \, dx$, then prove that

$$f(m, n) = \frac{m}{m-n} f(m-1, n+1) = \frac{m}{m+n} f(m-1, n-1).$$



Answers

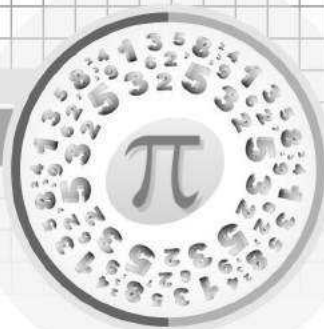
$$(x^m \sin \theta, x^n \cos m \theta)$$

- | | |
|----|--|
| 1. | $\left(\frac{1}{4} - \frac{1}{2} x^2\right) \cos 2x + \frac{1}{2} x \sin 2x$ |
| 2. | $\frac{1}{2} x^2 \sin 2x + \frac{1}{2} x \cos 2x - \frac{1}{4} \sin 2x$ |
| 3. | $\frac{-x \sin^2 x \cos x}{3} + \frac{1}{9} \sin^3 x + \frac{2}{3} (\sin x - x \cos x)$ |
| 4. | $I_5 = \frac{1}{5} \left(x \sin x \cos^4 x + \frac{1}{5} \cos^5 x + 4I_3 \right) = \text{etc.}$ |





Unit-4



Chapter 16

Gamma and Beta Functions

16.1 Beta Function

Definite integral $\int_0^1 x^{m-1}(1-x)^{n-1} dx$, for $m > 0, n > 0$ is called the Beta Function and is denoted by $B(m, n)$ or $\beta(m, n)$.

Thus $B(m, n)$ or $\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$, $m > 0, n > 0$

where m, n are any positive numbers, integral or fractional. Beta function also called the Eulerian integral of the first kind.

16.2 An Important Property of Beta Function

Theorem 1: Beta function is symmetrical in m and n

i.e., $B(m, n) = B(n, m)$

Proof: We have by definition

$$\begin{aligned}
 B(m, n) &= \int_0^1 x^{m-1}(1-x)^{n-1} dx, \quad m > 0, n > 0 \\
 &= \int_0^1 (1-x)^{m-1} \{1-(1-x)\}^{n-1} dx \quad [\because \int_0^a f(x) dx = \int_0^a f(a-x) dx] \\
 &= \int_0^1 (1-x)^{m-1} x^{n-1} dx \\
 &= \int_0^1 x^{n-1} (1-x)^{m-1} dx = B(n, m)
 \end{aligned}$$

$\therefore B(m, n) = B(n, m)$

16.3 Other Forms of Beta Function

Theorem 2: Show that

$$(i) \quad B(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$(ii) \quad B(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$(iii) \quad B(m, n) = 2 \int_0^{\infty} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

Proof: We know that

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \dots(1)$$

put $x = \frac{1}{(1+y)}$ in (1) and $dx = -\frac{1}{(1+y)^2} dy$

when $x = 0$, $y = \infty$ and when $x = 1$, $y = 0$

$$\begin{aligned} \therefore B(m, n) &= \int_{\infty}^0 \frac{1}{(1+y)^{m-1}} \cdot \left[1 - \frac{1}{1+y}\right]^{n-1} \left[-\frac{1}{(1+y)^2}\right] dy \\ &= \int_{\infty}^0 \frac{1}{(1+y)^{m-1+2}} \cdot \frac{y^{n-1}}{(1+y)^{n-1}} dy \\ &= \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy \quad \left[\because \int_0^a f(x) dx = \int_0^a f(t) dt \right] \\ B(m, n) &= \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}} dx \quad \dots(2) \end{aligned}$$

This is the (ii) part of theorem.

Since Beta function is symmetrical in m and n , we have

$$B(m, n) = \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

This is the (i) part of theorem

Put $x = \sin^2 \theta$ in (1) and $dx = 2 \sin \theta \cos \theta d\theta$

Now
$$B(m, n) = \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (\cos^2 \theta)^{n-1} (2 \sin \theta \cos \theta) d\theta$$

Hence
$$B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

This is (iii) part of theorem.

16.4 Evaluation of Beta Function

There are three cases

(i) **Case I:** When n is positive integer, then

$$B(m, n) = \frac{(n-1)!}{m(m+1)(m+2)\dots(m+n-1)}$$

(ii) **Case II:** When m is positive integer, then

$$B(m, n) = \frac{(m-1)!}{n(n+1)(n+2)\dots(n+m-1)}$$

(iii) **Case III:** When m and n both are positive integers, then

$$B(m, n) = \frac{(m-1)!(n-1)!}{(m+n-1)!}$$

Proof: Case I: We know that

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \dots(1)$$

(II) (I)

Integrate (1) by parts when x^{m-1} as the second function

$$\begin{aligned} B(m, n) &= [(1-x)^{n-1} \int x^{m-1} dx]_0^1 - \int_0^1 \left[\frac{d}{dx} \{ (1-x)^{n-1} \} \int x^{m-1} dx \right] \\ &= \left[(1-x)^{n-1} \frac{x^m}{m} \right]_0^1 + \int_0^1 (n-1) \left[(1-x)^{n-2} \frac{x^m}{m} \right] dx \\ &= 0 + \frac{n-1}{m} \int_0^1 x^m (1-x)^{n-2} dx \\ &= \frac{n-1}{m} \int_0^1 x^{(m+1)-1} (1-x)^{(n-1)-1} dx \\ &= \frac{n-1}{m} B(m+1, n-1) \end{aligned}$$

By repeated application of this process, we get

$$\begin{aligned} B(m, n) &= \frac{(n-1)}{m} \cdot \frac{(n-2)}{m+1} \cdot \frac{(n-3)}{m+2} \dots \frac{1}{(m+n-2)} B(m+n-1, 1) \\ &= \frac{n-1}{m} \cdot \frac{n-2}{m+1} \cdot \frac{n-3}{m+2} \dots \frac{1}{(m+n-2)} \int_0^1 x^{m+n-2} (1-x)^0 dx \\ &= \frac{n-1}{m} \cdot \frac{n-2}{m+1} \cdot \frac{n-3}{m+2} \dots \frac{1}{m+n-2} \int_0^1 x^{m+n-2} dx \\ &= \frac{(n-1)!}{m(m+1)(m+2)\dots(m+n-2)} \left[\frac{x^{m+n-1}}{m+n-1} \right]_0^1 \end{aligned}$$

$$\therefore B(m, n) = \frac{(n-1)!}{m(m+1)(m+2)\dots(m+n-1)}$$

Case II: Since the Beta function is symmetrical in m and n

$$i.e., B(m, n) = B(n, m)$$

$$\therefore B(m, n) = \frac{(m-1)!}{n(n+1)(n+2)\dots(n+m-1)}$$

Case III: When m and n both are positive integers then from case I, we get

$$\begin{aligned} B(m, n) &= \frac{(n-1)!}{m(m+1)(m+2)\dots(m+n-1)} \\ &= \frac{(n-1)!(m-1)(m-2)\dots(2)(1)}{m(m+1)\dots(m+n-1) \times (m-1)(m-2)\dots 2 \cdot 1} \\ &\quad [\text{Multiplying the } N_r \text{ and } D_r \text{ by } (m-1)!] \\ &= \frac{(n-1)!(m-1)!}{(m+n-1)!} \end{aligned}$$

16.5 Gamma Function

Second Integral of Euler

The definite integral $\int_0^\infty e^{-x} x^{n-1} dx$, where $n > 0$ is called (Gamma function of n), it is denoted by

16.5.1 (Properties of Gamma Function)

Theorem 3: Show that

- (i) $\Gamma 1 = 1$
- (ii) $\Gamma n + 1 = n\Gamma n, n > 0$
- (iii) $\Gamma(n+1) = n!, n = 1, 2, 3$

Proof:(i) We have, by definition of gamma function

$$\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx \quad \dots(1)$$

Put $n = 1$ in (1), we get

$$\Gamma 1 = \int_0^\infty e^{-x} dx = [-e^{-x}]_0^\infty = [-e^{-\infty} - e^0] = -[0 - 1] = 1$$

(ii) We have, by definition of gamma function

$$\begin{aligned} \Gamma(n+1) &= \int_0^\infty e^{-x} x^{(n+1)-1} dx \\ &= \int_0^\infty e^{-x} x^n dx \end{aligned}$$

Integration by parts taking e^{-x} as second function.

$$\Gamma(n+1) = \left\{ -x^n e^{-x} \right\}_0^\infty - \int_0^\infty nx^{n-1}(-e^{-x}) dx$$

Now
$$\lim_{x \rightarrow \infty} \frac{x^n}{e^x} = \lim_{x \rightarrow \infty} \frac{x^n}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\dots+\frac{x^n}{n!}} = 0$$

$\therefore \Gamma(n+1) = n \int_0^\infty e^{-x} x^{n-1} dx$

$$\Gamma(n+1) = n\Gamma n$$

(iii) $\Gamma(n+1) = n\Gamma n$

Put $n=1, 2, 3, 4, \dots$ in $\Gamma n+1 = n\Gamma n$, we get

$$\Gamma 2 = 1\Gamma 1 = 1 = 1!$$

$$\Gamma 3 = 2\Gamma 2 = 2.2 = 2!$$

$$\Gamma 4 = 3\Gamma 3 = 3.2.\Gamma 2 = 3.2.1 = 3!$$

In general form $\Gamma(n+1) = n!$, if n is positive integer

Remark: $\Gamma 0 = \infty, \Gamma(-n) = \infty, n$ is positive integer

Theorem 4: Prove that

(i) $\frac{\Gamma n}{a^n} = \int_0^\infty e^{-ay} y^{n-1} dy$ (ii) $\Gamma(1/2) = \sqrt{\pi}$ (iii) $\Gamma n = \int_0^1 \left[\log\left(\frac{1}{y}\right) \right]^{n-1} dy$

Proof: (i) We have by definition of gamma function

$$\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$$

Put $x = ay$ and $dx = a dy$ when $x=0 \Rightarrow y=0$ and when $x \rightarrow \infty \Rightarrow y \rightarrow \infty$

$\therefore \Gamma n = \int_0^\infty e^{-ay} a^n y^{n-1} dy$

Hence $\int_0^\infty e^{-ay} y^{n-1} dy = \frac{\Gamma n}{a^n}$

(ii) We know

$$\Gamma n = \int_0^\infty e^{-t} t^{n-1} dt \quad \dots(1)$$

Put (1) $n=1/2$ in (1)

$$\Gamma 1/2 = \int_0^\infty e^{-t} t^{-1/2} dt$$

Put $t = x^2$ and $dt = 2x dx$ in (1)

$$\Gamma 1/2 = 2 \int_0^\infty e^{-x^2} dx \quad \dots(2)$$

Replace x by y

$$\Gamma 1/2 = 2 \int_0^\infty e^{-y^2} dy \quad \dots(3)$$

Multiplying (2) and (3)

$$\begin{aligned}\{\Gamma(1/2)\}^2 &= 4 \int_0^\infty e^{-x^2} dx \int_0^\infty e^{-y^2} dy \\ &= 4 \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy \quad \dots(4)\end{aligned}$$

Change in (Polar coordinates)

$$x = r \cos \theta, y = r \sin \theta, dx dy = r dr d\theta$$

Put these values in (4)

$$\begin{aligned}(\Gamma(1/2))^2 &= 4 \int_0^{\pi/2} \int_0^\infty e^{-r^2} r dr d\theta \\ &= 2 \int_0^{\pi/2} [-e^{-r^2}]_0^\infty d\theta = 2 \int_0^{\pi/2} d\theta = 2 \cdot \frac{\pi}{2} = \pi\end{aligned}$$

$$\therefore \Gamma(1/2) = \sqrt{\pi}$$

(iii) Let $e^{-x} = y$ then $-e^{-x} dx = dy$ and $-x = \log y$ or $x = \log \left(\frac{1}{y} \right)$

Put these values in $\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$

(when $x=0$ then $y=1$ and $y=0$ then $x=\infty$)

$$\Gamma n = - \int_1^0 \left[\log \left(\frac{1}{y} \right) \right]^{n-1} dy = \int_0^1 \left(\log \frac{1}{y} \right)^{n-1} dy$$

16.6 Relation between Beta and Gamma Function

Theorem 5: Prove that

$$B(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}, \quad m > 0, n > 0$$

Proof: We know that

$$\frac{\Gamma m}{z^m} = \int_0^\infty e^{-zx} x^{m-1} dx \quad \dots(1)$$

$$\therefore \Gamma m = z^m \int_0^\infty e^{-zx} x^{m-1} dx = \int_0^\infty z^m e^{-zx} x^{m-1} dx$$

Multiplying both sides by $e^{-z} z^{n-1}$, we get

$$\Gamma m e^{-z} z^{n-1} = \int_0^\infty e^{-z(1+x)} z^{m+n-1} x^{m-1} dx \quad \dots(2)$$

Now integrate both sides of (1) with respect to z from 0 to ∞ , we get

$$\Gamma m \int_0^\infty e^{-z} z^{n-1} dz = \int_0^\infty \left[\int_0^\infty e^{-z(1+x)} z^{m+n-1} x^{m-1} dx \right] dz$$

or
$$\Gamma m \Gamma n = \int_0^\infty \left[\int_0^\infty e^{-z(1+x)} z^{m+n-1} dz \right] x^{m-1} dx$$

Using (1)

$$\begin{aligned}\Gamma m \Gamma n &= \int_0^\infty \frac{\Gamma(m+n)}{(1+x)^{m+n}} x^{m-1} dx = \Gamma(m+n) \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx \\ &= \Gamma(m+n) B(m, n)\end{aligned}$$

$$\therefore B(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

Theorem 6: Prove that

$$\Gamma n \Gamma(1-n) = \frac{\pi}{\sin n\pi}, \text{ where } 0 < n < 1$$

Proof: We know that

$$B(m, n) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx \quad \dots(1)$$

$$\text{and} \quad B(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)} \quad \dots(2)$$

From (1) and (2)

$$\frac{\Gamma m \Gamma n}{\Gamma(m+n)} = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

Put $m+n=1$ in above equation or $m=1-n$, then

$$\Gamma n \Gamma(1-n) = \Gamma 1 \int_0^\infty \frac{x^{n-1}}{(1+x)} dx = \int_0^\infty \frac{x^{n-1}}{(1+x)} dx \quad [\Gamma 1 = 1]$$

$$\text{But} \quad \int_0^\infty \frac{x^{n-1}}{(1+x)} dx = \frac{\pi}{\sin n\pi}$$

$$\therefore \Gamma n \Gamma(1-n) = \frac{\pi}{\sin n\pi}, \text{ where } 0 < n < 1$$

Theorem 7: Prove that

$$\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{\Gamma \frac{(p+1)}{2} \Gamma \frac{(q+1)}{2}}{2 \Gamma \frac{(p+q+2)}{2}}$$

Proof: The Beta function, we know from

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \dots(1)$$

$$\text{Let} \quad x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta d\theta$$

$$\text{and} \quad 1-x = 1 - \sin^2 \theta = \cos^2 \theta, x=0 \text{ but } \theta=0 \text{ and } x=1 \text{ then } \theta = \frac{\pi}{2}$$

$$\text{Hence new limit is } \theta=0 \text{ to } \theta = \frac{\pi}{2}$$

Put these values in (1)

$$B(m, n) = \int_0^{\pi/2} \sin^{2m-2} \theta \cos^{2n-2} \theta \cdot 2 \sin \theta \cos \theta d\theta$$

$$\text{or } \frac{\Gamma m \Gamma n}{\Gamma(m+n)} = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta \quad \dots(2)$$

$$\text{From (2) } 2m-1 = p \text{ or } m = \frac{p+1}{2}$$

$$\text{and put } 2n-1 = q \text{ or } n = \frac{q+1}{2}$$

$$\frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)} = 2 \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta$$

$$\text{or } \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)} \quad \dots(3)$$

Deductions: put $p = q = 0$ in (3)

$$\int_0^{\pi/2} d\theta = \frac{\Gamma_{1/2} \Gamma_{1/2}}{2\Gamma_1} = \frac{\left(\Gamma \frac{1}{2}\right)^2}{2}$$

$$\text{or } [\theta]_0^{\pi/2} = \frac{1}{2} \left(\Gamma \frac{1}{2}\right)^2$$

$$\text{or } \left(\frac{\pi}{2} - 0\right) = \frac{1}{2} \left(\Gamma \frac{1}{2}\right)^2$$

$$\text{or } \left(\Gamma \frac{1}{2}\right)^2 = \pi$$

$$\text{or } \Gamma \frac{1}{2} = \sqrt{\pi}$$

16.7 Duplication Formula

Show that: $\Gamma m \Gamma\left(m + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}} \Gamma(2m)$, where $m > 0$

Proof: We know that

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}, \text{ where } m > 0, n > 0.$$

Put $n = m$, then

$$B(m, m) = \frac{[\Gamma(m)]^2}{\Gamma(2m)} \quad \dots(1)$$

By the definition of Beta function, we have

$$B(m, m) = \int_0^1 x^{m-1} (1-x)^{m-1} dx$$

Let

$$x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta d\theta$$

when $x = 0$ then $\theta = 0$ and when $x = 1$ then $\theta = \frac{1}{2}\pi$

Then

$$\begin{aligned} B(m, m) &= \int_0^{\pi/2} \sin^{2(m-1)} \theta \cdot \cos^{2(m-1)} \theta \cdot 2 \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2m-1} \theta d\theta = 2 \int_0^{\pi/2} (\sin \theta \cos \theta)^{2m-1} d\theta \\ &= 2 \int_0^{\pi/2} \left(\frac{\sin 2\theta}{2} \right)^{2m-1} d\theta = \frac{1}{2^{2m-2}} \int_0^{\pi/2} \sin^{2m-1} 2\theta d\theta \\ &= \frac{1}{2^{2m-2}} \int_0^{\pi} \sin^{2m-1} \phi \cdot \frac{d\phi}{2}, \text{ Put } 2\theta = \phi \Rightarrow d\theta = \frac{1}{2} d\phi \\ &= \frac{1}{2^{2m-1}} \int_0^{\pi} \sin^{2m-1} \phi d\phi = \frac{1}{2^{2m-1}} \cdot 2 \int_0^{\pi/2} \sin^{2m-1} \phi d\phi \\ &= \frac{1}{2^{2m-2}} \int_0^{\pi/2} \sin^{2m-1} \phi \cdot \cos^0 \phi d\phi \\ &= \frac{1}{2^{2m-2}} \cdot \frac{\Gamma \frac{1}{2} (2m-1+1) \Gamma \frac{1}{2} (0+1)}{2 \Gamma \frac{1}{2} (2m-1+0+2)} = \frac{1}{2^{2m-1}} \cdot \frac{\Gamma(m) \Gamma \left(\frac{1}{2} \right)}{\Gamma \left(m + \frac{1}{2} \right)} \\ &= \frac{1}{2^{2m-1}} \cdot \frac{\Gamma(m) \sqrt{\pi}}{\Gamma \left(m + \frac{1}{2} \right)} \quad \dots (2) \end{aligned}$$

$$\left[\because \Gamma \left(\frac{1}{2} \right) = \sqrt{\pi} \right]$$

Comparing $B(m, n)$ in equation (1) and (2), we get

$$\frac{[\Gamma(m)]^2}{\Gamma(2m)} = \frac{1}{2^{2m-1}} \cdot \frac{\Gamma(m) \cdot \sqrt{\pi}}{\Gamma \left(m + \frac{1}{2} \right)}$$

or
$$\Gamma(m) \Gamma \left(m + \frac{1}{2} \right) = \frac{\sqrt{\pi}}{2^{2m-1}} \Gamma(2m).$$

Theorem 8: Solve the following

(i) $\int_0^\infty e^{-ax} \cos bx \cdot x^{m-1} dx$ (ii) $\int_0^\infty e^{-ax} \sin bx \cdot x^{m-1} dx.$

Proof: We know that

$$\int_0^\infty e^{-ax} e^{ibx} x^{m-1} dx = \int_0^\infty e^{-(a-ib)x} x^{m-1} dx = \frac{\Gamma(m)}{(a-ib)^m} = (a-ib)^{-m} \Gamma(m) \quad \dots (1)$$

Let $a = k \cos \alpha$ and $b = k \sin \alpha$, then

$$\alpha = \tan^{-1}(b/a) \text{ and } k = \sqrt{a^2 + b^2}$$

Then

$$(a-ib)^{-m} = [k(\cos \alpha - i \sin \alpha)]^{-m} = k^{-m} (\cos \alpha - i \sin \alpha)^{-m}$$

$$= k^{-m} (\cos m\alpha + i \sin m\alpha), \quad [\text{By Demovier's theorem}]$$

Now, from eqn. (1)

$$\int_0^{\infty} e^{-ax} e^{ibx} x^{m-1} dx = k^{-m} (\cos m\alpha + i \sin m\alpha) \Gamma(m)$$

or
$$\int_0^{\infty} e^{-ax} (\cos bx + i \sin bx) x^{m-1} dx = \frac{\Gamma(m)}{k^m} (\cos m\alpha + i \sin m\alpha),$$

$$[\because e^{i\theta} = \cos \theta + i \sin \theta, \text{ by Euler theorem}]$$

or
$$\int_0^{\infty} e^{-ax} \cos bx \cdot x^{m-1} dx + i \int_0^{\infty} e^{-ax} \sin bx \cdot x^{m-1} dx$$

$$= \frac{\Gamma(m)}{k^m} \cos m\alpha + i \frac{\Gamma(m)}{k^m} \sin m\alpha. \quad \dots(2)$$

Now, equating real and imaginary parts on both sides, we find

$$\int_0^{\infty} e^{-ax} \cos bx \cdot x^{m-1} dx = \frac{\Gamma(m)}{k^m} \cos m\alpha,$$

and
$$\int_0^{\infty} e^{-ax} \sin bx \cdot x^{m-1} dx = \frac{\Gamma(m)}{k^m} \sin m\alpha,$$

where $k = \sqrt{a^2 + b^2}$ and $\alpha = \tan^{-1}(b/a),$

Remark: $\Gamma\left(\frac{1}{n}\right)\Gamma\left(\frac{2}{n}\right)\Gamma\left(\frac{3}{n}\right)\dots\Gamma\left(\frac{n-1}{n}\right) = \frac{(2\pi)^{(n-1)/2}}{(n)^{1/2}}$

Illustrative Examples

Example 1: Evaluate $\int_0^1 x^4 (1 - \sqrt{x})^5 dx$

Solution: Let us consider

$$\sqrt{x} = t \quad \text{or} \quad x = t^2 \quad \text{or} \quad dx = 2t dt$$

Hence
$$\int_0^1 x^4 (1 - \sqrt{x})^5 dx = \int_0^1 (t^2)^4 (1 - t)^5 (2t) dt$$

$$= 2 \int_0^1 t^9 (1 - t)^5 dt$$

$$= 2 \int_0^1 t^{10-1} (1 - t)^{6-1} dt$$

$$= 2 B(10, 6)$$

$$= 2 \frac{(10-1)!(6-1)!}{(10+6-1)!} = 2 \frac{9!5!}{15!} = \frac{2 \times 5.4.3.2.1}{15.14.13.12.11.10} = \frac{1}{15015}$$

Example 2: Prove that $B(m, n) = B(m+1, n) + B(m, n+1)$ where $m, n > 0$

Solution: We know that

$$B(m+1, n) = \frac{\Gamma(m+1)\Gamma n}{\Gamma(m+n+1)} \quad \dots(1)$$

and
$$B(m, n+1) = \frac{\Gamma(m)\Gamma(n+1)}{\Gamma(m+n+1)} \quad \dots(2)$$

Hence
$$\begin{aligned} \text{R.H.S} &= B(m+1, n) + B(m, n+1) \\ &= \frac{\Gamma(m+1)\Gamma(n)}{\Gamma(m+n+1)} + \frac{\Gamma m \Gamma(n+1)}{\Gamma(m+n+1)} \\ &= \frac{m\Gamma(m)\Gamma n}{\Gamma(m+n+1)} + \frac{\Gamma m n \Gamma n}{\Gamma(m+n+1)} = \frac{\Gamma m \Gamma n (m+n)}{(m+n)\Gamma(m+n)} \\ &= \frac{\Gamma(m)\Gamma n}{\Gamma(m+n)} = B(m, n) = \text{L.H.S} \end{aligned}$$

Example 3: Prove that

(i)
$$\frac{B(p, q+1)}{q} = \frac{B(p+1, q)}{p} = \frac{B(p, q)}{p+q}$$

(ii)
$$\frac{B(m+1, n)}{B(m, n)} = \frac{m}{m+n}$$

Solution: (i) We have

$$\begin{aligned} \frac{B(p, q+1)}{q} &= \frac{1}{q} B(p, q+1) \\ &= \frac{1}{q} \int_0^1 x^{p-1} (1-x)^{(q+1)-1} dx = \frac{1}{q} \int_0^1 (1-x)^q x^{p-1} dx \\ &= \frac{1}{q} \left[\left\{ (1-x)^q \frac{x^p}{p} \right\}_0^1 - \int_0^1 q(1-x)^{q-1} (-1) \frac{x^p}{p} dx \right] \\ &\quad \text{(Integrating by parts)} \\ &= \frac{1}{q} \left[0 + \frac{q}{p} \int_0^1 x^{(p+1)-1} (1-x)^{q-1} dx \right] = \frac{1}{p} B(p+1, q) \quad \dots(1) \end{aligned}$$

Again
$$\begin{aligned} \frac{B(p+1, q)}{p} &= \frac{1}{p} B(p+1, q) \\ &= \frac{1}{p} \int_0^1 x^{p+1-1} (1-x)^{q-1} dx \\ &= \frac{1}{p} \int_0^1 x^p (1-x)^{q-1} dx \\ &= \frac{1}{p} \int_0^1 x^{(p-1)} x (1-x)^{q-1} dx \\ &= \frac{1}{p} \int_0^1 x^{p-1} \{1 - (1-x)\} (1-x)^{q-1} dx \\ &= \frac{1}{p} \int_0^1 x^{p-1} (1-x)^{q-1} dx - \frac{1}{p} \int_0^1 x^{p-1} (1-x)^q dx \\ &= \frac{1}{p} B(p, q) - \frac{1}{p} B(p, q+1) \end{aligned}$$

$$\therefore \frac{B(p, q+1)}{p} + \frac{B(p+1, q)}{p} = \frac{B(p, q)}{p}$$

$$\text{or} \quad \frac{B(p, q+1)}{q} = \frac{B(p, q)}{p+q} \quad \dots(2)$$

Now, from equation (1) and (2)

$$\frac{B(p, q+1)}{q} = \frac{B(p+1, q)}{p} = \frac{B(p, q)}{p+q} \quad \dots(3)$$

We have from equation (3)

$$B(p, q+1) = \frac{q}{p+q} B(p, q) \text{ and } B(p+1, q) = \frac{p}{p+q} B(p, q)$$

$$\begin{aligned} \therefore B(p+1, q) + B(p, q+1) &= \frac{p}{p+q} B(p, q) + \frac{q}{p+q} B(p, q) \\ &= \frac{p+q}{p+q} B(p, q) = B(p, q) \end{aligned}$$

(ii) We have

$$B(m+1, n) = B(n, m+1) \quad [\text{Beta function is symmetrical in } m+1, n]$$

$$= \int_0^1 x^{n-1} (1-x)^{(m+1)-1} dx = \int_0^1 x^m (1-x)^{n-1} dx$$

$$= \left[(1-x)^n \cdot \frac{x^{m+1}}{n+1} \right]_0^1 - \int_0^1 m (1-x)^{m-1} (-1) \frac{x^n}{n} dx$$

[Integrating by part]

$$= 0 + \frac{m}{n} \int_0^1 x^{n-1} x (1-x)^{m-1} dx \quad [x = 1 - (1-x)]$$

$$= \frac{m}{n} \left[\int_0^1 x^{n-1} (1-x)^{m-1} dx - \int_0^1 x^{n-1} (1-x)^m dx \right]$$

$$= \frac{m}{n} [B(n, m) - B(n, m+1)] = \frac{m}{n} (m, n) - \frac{m}{n} B(m+1, n)$$

$$\text{or} \left(1 + \frac{m}{n}\right) B(m+1, n) = \frac{m}{n} B(m, n) \quad (\text{by transform})$$

$$\text{or } (n+m) B(m+1, n) = m B(m, n)$$

$$\text{or} \quad \frac{B(m+1, n)}{B(m, n)} = \frac{m}{m+n}$$

Example 4: Evaluate the following integrals:

$$(i) \int_0^\infty \frac{x^8 (1-x^6)}{(1+x)^{24}} dx$$

$$(ii) \int_0^\infty \frac{xdx}{1+x^6}$$

Solution: (i) We have

$$\int_0^\infty \frac{x^8 (1-x^6)}{(1+x)^{24}} dx = \int_0^\infty \frac{x^8}{(1+x)^{24}} dx - \int_0^\infty \frac{x^{14}}{(1+x)^{24}} dx$$

$$\begin{aligned}
&= \int_0^\infty \frac{x^{9-1}}{(1+x)^{9+15}} dx - \int_0^\infty \frac{x^{15-1}}{(1+x)^{15+9}} dx \\
&= B(9, 15) - B(15, 9) \\
&= B(9, 15) - B(9, 15) \quad [\because B(9, 15) = B(15, 9)] \\
&= 0
\end{aligned}$$

(ii) Let

$$I = \int_0^\infty \frac{x dx}{1+x^6},$$

Let

$$x^6 = y \quad \text{or} \quad x = y^{1/6} \Rightarrow dx = \frac{1}{6} y^{-5/6} dy$$

 $\therefore$

$$\begin{aligned}
I &= \frac{1}{6} \int_0^\infty \frac{y^{1/6} y^{-5/6}}{1+y} dy = \frac{1}{6} \int_0^\infty \frac{y^{-2/3}}{1+y} dy = \frac{1}{6} \int_0^\infty \frac{y^{(1/3-1)}}{(1+y)^{(1/3+2/3)}} dy \\
&= \frac{1}{6} B\left(\frac{1}{3}, \frac{2}{3}\right) = \frac{1}{6} \frac{\Gamma \frac{1}{3} \Gamma \frac{2}{3}}{\Gamma\left(\frac{1}{3} + \frac{2}{3}\right)} = \frac{1}{6} \frac{\Gamma \frac{1}{3} \Gamma\left(1 - \frac{1}{3}\right)}{\Gamma 1} = \frac{1}{6} \cdot \frac{\pi}{\sin \frac{\pi}{3}} \\
&= \frac{1}{6} \cdot \frac{\pi}{(\sqrt{3}/2)} = \frac{2\pi}{6\sqrt{3}} = \frac{\pi}{3\sqrt{3}}
\end{aligned}$$

Example 5: Show that $\int_0^{\pi/2} (\tan x)^n dx = \frac{\pi}{2} \sec \frac{n\pi}{2}$, where $-1 < n < 1$.

Solution: We have $\int_0^{\pi/2} (\tan x)^n dx = \int_0^{\pi/2} \sin^n x \cdot \cos^{-n} x dx$

$$\begin{aligned}
&= \frac{\Gamma \frac{1}{2}(n+1) \cdot \Gamma \frac{1}{2}(-n+1)}{2\Gamma \frac{1}{2}(n-n+2)}
\end{aligned}$$

where $-n+1 > 0$ or $n < 1$ or $n+1 > 0$ or $n > -1$

or

$$\begin{aligned}
&= \frac{1}{2} \Gamma \frac{1}{2}(n+1) \Gamma \frac{1}{2}(1-n) = \frac{1}{2} \Gamma \frac{1}{2}(n+1) \Gamma \left[1 - \frac{1}{2}(n+1)\right] \\
&= \frac{1}{2} \frac{\pi}{\sin \frac{1}{2}(n+1)\pi}, \quad \left[\because \Gamma(n) \Gamma(1-n) = \frac{\pi}{\sin n\pi} \right] \\
&= \frac{\pi}{2} \cdot \frac{1}{\sin \left(\frac{1}{2}\pi + \frac{n\pi}{2}\right)} = \frac{\pi}{2} \cdot \frac{1}{\cos \left(\frac{1}{2}n\pi\right)} = \frac{\pi}{2} \sec \frac{n\pi}{2},
\end{aligned}$$

where $-1 < n < 1$.

Example 6: Prove that

$$(i) \int_0^\infty x^{2n-1} e^{-ax^2} dx = \frac{\Gamma(n)}{2a^n}$$

$$(ii) \int_0^\infty x^m e^{-ax^n} dx = \frac{\Gamma[(m+1)/n]}{na^{(m+1)/n}}$$

$$(iii) \int_0^1 \frac{dx}{\sqrt{-\log x}} = \sqrt{\pi}$$

Solution: (i) Let $I = \int_0^\infty x^{2n-1} e^{-ax^2} dx = \int_0^\infty x^{2n-2} e^{-ax^2} x dx$

Now, $ax^2 = z \Rightarrow 2ax dx = dz$. when $x = 0, z = 0$ and then $x \rightarrow \infty, z \rightarrow \infty$.

$$\begin{aligned} \therefore I &= \int_0^\infty \left(\frac{z}{a}\right)^{n-1} e^{-z} \frac{1}{2a} dz = \frac{1}{2a^n} \int_0^\infty e^{-z} z^{n-1} dz \\ &= \frac{1}{2a^n} \Gamma(n), \quad [\text{by definition of Gamma function}] \end{aligned}$$

(ii) Let $I = \int_0^\infty x^m e^{-ax^n} dx = \int_0^\infty \frac{x^m}{x^{n-1}} e^{-ax^n} x^{n-1} dx$

$$= \int_0^\infty x^{m-n+1} e^{-ax^n} x^{n-1} dx$$

Again let $ax^n = t \Rightarrow na x^{n-1} dx = dt$, also $x = 0, t = 0$ and when $x \rightarrow \infty, t \rightarrow \infty$.

$$\begin{aligned} \therefore I &= \int_0^\infty \left(\frac{t}{a}\right)^{(m-n+1)/n} e^{-t} \cdot \frac{1}{na} dt, \quad \left[\because ax^n = t \Rightarrow x = \left(\frac{t}{a}\right)^{1/n} \right] \\ &= \frac{1}{na \cdot a^{(m-n+1)/n}} \int_0^\infty t^{\{(m+1)/n\}-1} e^{-t} dt \\ &= \frac{1}{na^{(m+1)/n}} \Gamma\{(m+1)/n\} \end{aligned}$$

(iii) Let $I = \int_0^1 \frac{dx}{\sqrt{(-\log x)}} = \int_0^1 \frac{dx}{\sqrt{\{\log(1/x)\}}} = \int_0^1 \left(\log \frac{1}{x}\right)^{-1/2} dx$.

Again let $\log(1/x) = y$ i.e., $1/x = e^y$ i.e., $x = e^{-y} \Rightarrow dx = -e^{-y} dy$.

Also $x \rightarrow \infty, y \rightarrow \infty$ and when $x = 1, y = 0$.

$$\therefore I = -\int_\infty^0 y^{-1/2} e^{-y} dy = \int_0^\infty e^{-y} y^{1/2-1} dy = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Example 7. Evaluate the following

(i) $\int_0^{\pi/2} \sqrt{\tan \theta} d\theta$ (ii) $\int_0^{\pi/2} \sqrt{\cot \theta} d\theta$ (iii) $\int_0^\infty \frac{dy}{1+y^4}$

Solution: (i) We have

$$\begin{aligned} I &= \int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \int_0^{\pi/2} \frac{\sin^{1/2} \theta}{\cos^{1/2} \theta} d\theta \\ &= \int_0^{\pi/2} (\sin \theta)^{1/2} (\cos \theta)^{-1/2} d\theta = \frac{\Gamma\left(\frac{1}{2}+1\right) \Gamma\left(\frac{-\frac{1}{2}+1}{2}\right)}{2\Gamma\left(\frac{1}{2}-\frac{1}{2}+2\right)} = \frac{\Gamma\frac{3}{4} \Gamma\frac{1}{4}}{2\Gamma 1} \\ &= \frac{\Gamma\frac{1}{4} \Gamma\left(1-\frac{1}{4}\right)}{2} = \frac{1}{2} \cdot \frac{\pi}{\sin \frac{\pi}{4}} = \frac{\pi}{2\left(\frac{1}{\sqrt{2}}\right)} = \frac{\pi}{\sqrt{2}} \end{aligned}$$

(iii) Let $I = \int_0^\infty \frac{dy}{1+y^4}$

Let $y^2 = \tan \theta$ or $y = \sqrt{\tan \theta} \Rightarrow dy = \frac{1}{2} (\tan \theta)^{-1/2} \sec^2 \theta d\theta$

when $y=0$ then $\theta=0$ and when $y \rightarrow \infty$ then $\theta = \frac{\pi}{2}$

$$\begin{aligned} \therefore I &= \int_0^{\pi/2} \frac{1}{1+\tan^2 \theta} \times \frac{1}{2} (\tan \theta)^{-1/2} \sec^2 \theta \, d\theta \\ &= \frac{1}{2} \int_0^{\pi/2} (\tan \theta)^{-1/2} d\theta = \frac{1}{2} \int_0^{\pi/2} (\sin \theta)^{-1/2} (\cos \theta)^{1/2} d\theta \\ &= \frac{1}{2} \cdot \frac{\Gamma\left(\frac{-1}{2}+1\right) \Gamma\left(\frac{1}{2}+1\right)}{\frac{2}{\left(-\frac{1}{2}+\frac{1}{2}+2\right)}} = \frac{1}{4} \Gamma\left(\frac{1}{4}\right) \Gamma\left(1-\frac{1}{4}\right) = \frac{1}{4} \frac{\pi}{\sin \frac{\pi}{4}} = \frac{\pi}{2\sqrt{2}} \end{aligned}$$

$$\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \pi$$
$$\begin{aligned}
 I &= \int_0^{\pi/2} (\sin \theta)^{1/2} d\theta \times \int_0^{\pi/2} (\sin \theta)^{-1/2} d\theta \\
 &= \int_0^{\pi/2} (\sin \theta)^{-1/2} (\cos \theta)^0 d\theta \times \int_0^{\pi/2} (\sin \theta)^{1/2} (\cos \theta)^0 d\theta \\
 &= \frac{\Gamma\left(\frac{-1}{2}+1\right) \Gamma\left(\frac{0+1}{2}\right)}{2\Gamma\left(\frac{-1}{2}+0+2\right)} \times \frac{\Gamma\left(\frac{1}{2}+1\right) \Gamma\left(\frac{0+1}{2}\right)}{2\Gamma\left(\frac{1}{2}+0+2\right)} \\
 &= \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{1}{2}\right)}{2\Gamma\left(\frac{3}{4}\right)} \times \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)}{2\Gamma\left(\frac{5}{4}\right)} = \frac{\Gamma\left(\frac{1}{4}\right) \sqrt{\pi} \times \Gamma\pi}{2.2 \cdot \frac{1}{4} \Gamma\left(\frac{1}{4}\right)} = \pi
 \end{aligned}$$
$$\int_0^1 \frac{x^2 dx}{(1-x^4)^{1/2}} \times \int_0^1 \frac{dx}{(1+x^4)^{1/2}} = \frac{\pi}{4\sqrt{2}} \quad \dots(1)$$

Solution: Let $I_1 = \int_0^1 \frac{x^2 dx}{(1-x^4)^{1/2}}$

Let

$$x^2 = \sin \theta \text{ or } x = (\sin \theta)^{1/2} \Rightarrow dx = \frac{1}{2} (\sin \theta)^{-1/2} \cos \theta d\theta$$

$$\left[\text{when } x = 0 \Rightarrow \theta = 0, \text{ when } x = 1 \Rightarrow \theta = \frac{\pi}{2} \right]$$

 $\therefore$

$$\begin{aligned} I_1 &= \int_0^{\pi/2} \frac{\sin \theta}{(1 - \sin^2 \theta)^{1/2}} \times \frac{1}{2} (\sin \theta)^{-1/2} \cos \theta d\theta \\ &= \frac{1}{2} \int_0^{\pi/2} \sqrt{\sin \theta} d\theta \end{aligned}$$

Let

$$I_2 = \int_0^1 \frac{dx}{(1 + x^4)^{1/2}}$$

$$x^2 = \tan \theta \text{ or } x = (\tan \theta)^{1/2} \Rightarrow dx = \frac{1}{2} (\tan \theta)^{-1/2} \sec^2 \theta d\theta$$

Put in I_2 , we get

$$\begin{aligned} I_2 &= \int_0^1 \frac{dx}{(1 + x^4)^{1/2}} = \frac{1}{2} \int_0^{\pi/4} \frac{\frac{1}{2} (\tan \theta)^{-1/2} \sec^2 \theta d\theta}{(1 + \tan^2 \theta)^{1/2}} \\ &= \frac{1}{2} \int_0^{\pi/4} \frac{d\theta}{\sqrt{\sin \theta} \cos \theta} = \frac{1}{\sqrt{2}} \int_0^{\pi/4} \frac{d\theta}{\sqrt{\sin 2\theta}} \end{aligned}$$

$$[\text{Put } 2\theta = \phi \text{ and } d\theta = \frac{d\phi}{2}]$$

Then

$$I_2 = \frac{1}{2\sqrt{2}} \int_0^{\pi/2} \frac{d\phi}{\sqrt{\sin \phi}} = \frac{1}{2\sqrt{2}} \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \left[\because \int_0^a g(x) dx = \int_0^a f(t) dt \right]$$

Put I_1, I_2 in (1)

$$\begin{aligned} \int_0^1 \frac{x^2 dx}{(1 - x^4)^{1/2}} \times \int_0^1 \frac{dx}{(1 + x^4)^{1/2}} &= I_1 \times I_2 \\ &= \frac{1}{2} \int_0^{\pi/2} \sqrt{\sin \theta} d\theta \times \frac{1}{2\sqrt{2}} \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \\ &= \frac{1}{4\sqrt{2}} \left[\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \right] \\ &= \frac{1}{4\sqrt{2}} \times \pi = \frac{\pi}{4\sqrt{2}} \end{aligned}$$

(from example 8)

Example 10: Show that

$$\int_0^a (a-x)^{m-1} x^{n-1} dx = a^{m+n-1} B(m, n) = \frac{a^{m+n-1} \Gamma m \Gamma n}{\Gamma(m+n)}$$

Solution: We have

$$\begin{aligned} \int_0^a (a-x)^{m-1} x^{n-1} dx &= \int_0^1 (a-ay)^{m-1} (ay)^{n-1} a dy && [\text{where } x = ay \Rightarrow dx = a dy] \\ &= \int_0^1 a^{(m-1)+(n-1)+1} (1-y)^{m-1} y^{n-1} dy \\ &= a^{m+n-1} \int_0^1 y^{n-1} (1-y)^{m-1} dy = a^{m+n-1} B(m, n) \\ &= \frac{a^{m+n-1} \Gamma m \Gamma n}{\Gamma(m+n)} && \left[\because B(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)} \right] \end{aligned}$$

Example 11: Show that

$$\int_0^{\infty} \frac{\sin bz}{z} dz = \frac{\pi}{2}.$$

Solution: We have

$$\begin{aligned} I &= \int_0^{\infty} \int_0^{\infty} e^{-xz} \sin bz \, dx \, dz \\ &= \int_0^{\infty} \left[\frac{e^{-xz}}{-z} \right]_0^{\infty} \sin bz \, dz, \text{ integrate w.r.t } x \\ &= \int_0^{\infty} \frac{\sin bz}{z} dz \end{aligned} \quad \dots(1)$$

Again integrate w.r.t z

$$\begin{aligned} I &= \int_0^{\infty} \int_0^{\infty} e^{-xz} \sin bz \, dx \, dz \\ &= \int_0^{\infty} \left[\int_0^{\infty} e^{-xz} \sin bz \, dz \right] dx \\ &= \int_0^{\infty} \frac{b}{b^2 + x^2} dx, \\ &= \left[\tan^{-1} \frac{x}{b} \right]_0^{\infty} = \frac{\pi}{2} \end{aligned} \quad \dots(2)$$

Comparing eqn. (1) and (2), we get

$$\int_0^{\infty} \frac{\sin bz}{z} dz = \frac{\pi}{2}.$$

Example 12: Show that

$$\int_0^{\infty} \cos(bz^{1/n}) dz = \frac{1}{b^n} \Gamma(n+1) \cdot \cos \frac{n\pi}{2}.$$

Solution: Let $z^{1/n} = x$ i.e., $z = x^n \Rightarrow dz = nx^{n-1} dx$

$$\begin{aligned} \therefore \int_0^{\infty} \cos(bz^{1/n}) dz &= \int_0^{\infty} \cos(bx) \cdot nx^{n-1} dx \\ &= n \int_0^{\infty} x^{n-1} \cos(bx) dx \\ &= n \text{ real part of } \int_0^{\infty} e^{-ibx} x^{n-1} dx \\ &= n \text{ real part of } \frac{\Gamma(n)}{(ib)^n} \\ &= n \text{ real part of } \frac{\Gamma(n)}{b^n} \cdot \left(\cos \frac{1}{2} \pi + i \sin \frac{1}{2} \pi \right)^{-n} \\ &= \frac{\Gamma(n+1)}{b^n} \left(\cos \frac{n\pi}{2} - i \sin \frac{n\pi}{2} \right) \text{ real part of } = \frac{1}{b^n} \Gamma(n+1) \cos \left(\frac{n\pi}{2} \right) \end{aligned}$$



Comprehensive Exercise

(Beta and Gamma)

1. Show that:

$$(i) \int_0^a \frac{dx}{(a^n - x^n)^{1/n}} = \frac{1}{n} \cdot \frac{\pi}{\sin\left(\frac{\pi}{n}\right)}$$

$$(ii) \int_0^2 (8 - x^3)^{-1/3} dx = \frac{2\pi}{3\Gamma 3}$$

$$(iii) \int_a^b (x-a)^{m-1} (b-x)^{n-1} dx = (b-a)^{m+n-1} \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

$$(iv) \int_0^\infty x^m e^{-n^2 x^2} dx = \frac{1}{2n^{m+1}} \Gamma\left(\frac{m+1}{2}\right)$$

$$(v) \int_0^1 x^m (1-x^n)^p dx = \frac{1}{n} B\left(\frac{m+1}{n}, p+1\right)$$

$$(vi) \int_0^1 (1-x^n)^{1/n} dx = \frac{1}{n} \frac{(\Gamma(1/n))^2}{2\Gamma(2/n)}$$

$$(vii) \int_0^{\pi/2} (\sin \theta)^p d\theta = \frac{1}{2} B\left(\frac{1}{2}, \frac{p+1}{2}\right)$$

$$(viii) \int_{-\infty}^\infty \cos\left(\frac{\pi x^2}{2}\right) dx = 1$$

2. Show that $\int_0^{\pi/2} \sin^6 \theta d\theta = \frac{5\pi}{32}$

3. Show that

$$(i) \int_0^{\pi/2} \sin^2 \theta \cos^4 \theta d\theta = \frac{5\pi}{256}$$

$$(ii) \int_0^{\pi/2} \frac{d\theta}{\sqrt{a \cos^4 \theta + b \sin^4 \theta}} = \frac{(\Gamma 1/4)^2}{4(ab)^{1/4} \sqrt{\pi}}$$

4. Establish the relation between Beta and Gamma function.

5. Show that (i) $B(m, m) \cdot B\left(m + \frac{1}{2}, m + \frac{1}{2}\right) = \frac{\pi m^{-1}}{2^{4m-1}}$

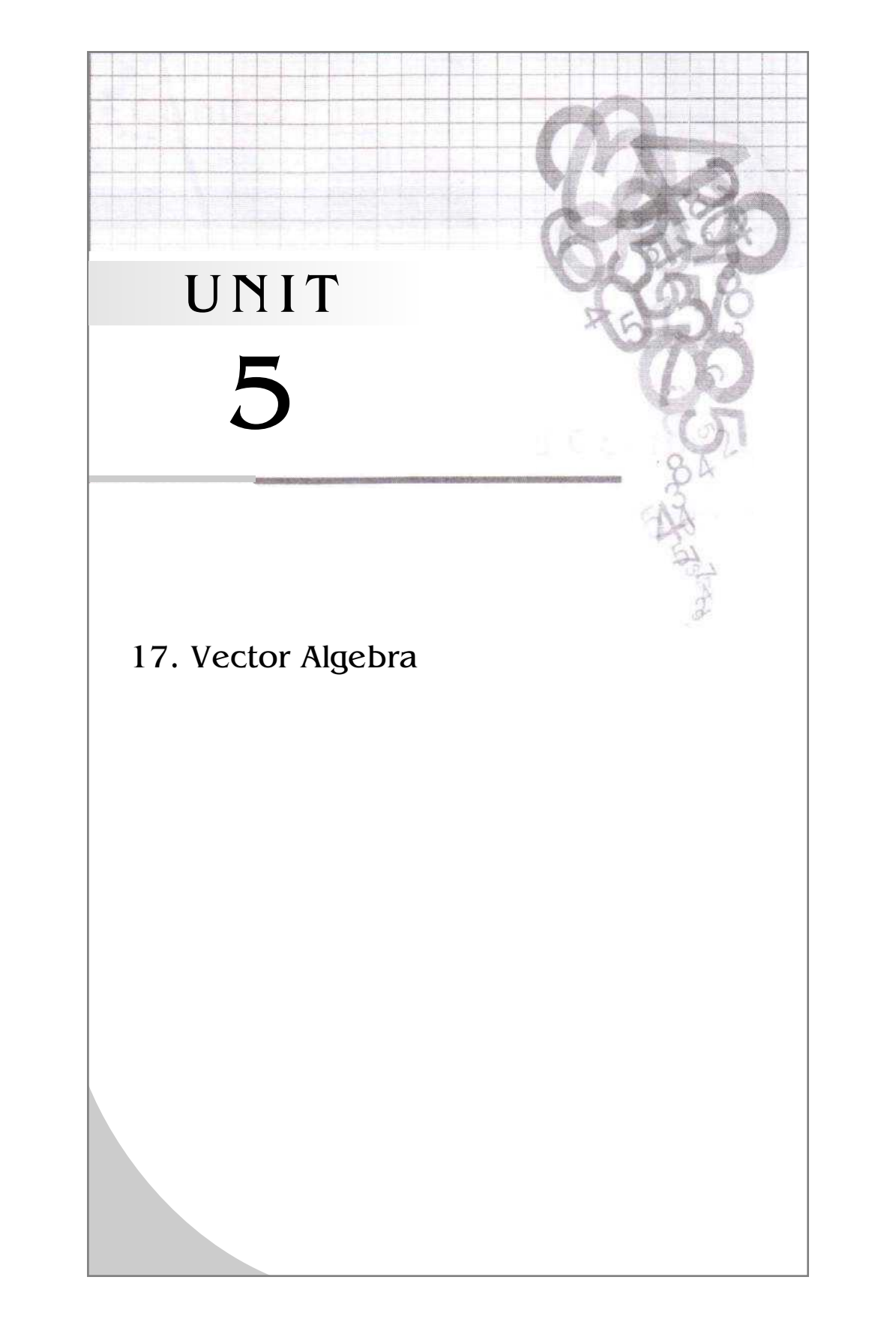
$$(ii) \Gamma \frac{1}{9} \Gamma \frac{2}{9} \Gamma \frac{3}{9} \dots \Gamma \frac{8}{9} = \frac{16}{3} \pi^4$$

$$(iii) \Gamma 0.1 \Gamma 0.2 \Gamma 0.3 \dots \Gamma 0.9 = \frac{(2\pi)^{9/2}}{\sqrt{10}}$$

$$(iv) \int_0^n \left(1 - \frac{x}{n}\right)^n x^{p-1} dx = n^p B(p, n+1)$$

[Hint: $x/n = y$]





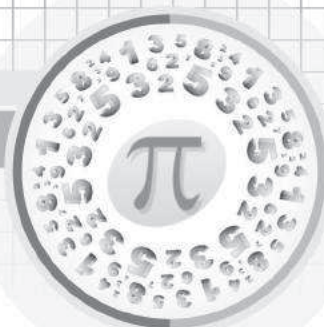
UNIT

5

17. Vector Algebra



Unit-5



Chapter

17

Vector Algebra

17.1 Vector Algebra

17.1.1 Vector

A directed line segment is called a **vector**. A directed line segment with initial point A , and the terminal point B , is the vector denoted by $\vec{AB}$.



The magnitude of $\vec{AB} = |\vec{AB}|$

We shall denote a vector by single letter with an arrow on its and its magnitude is denoted by this letter only.

Thus, $\vec{AB} = \vec{a}$, $|\vec{AB}| = |\vec{a}| = a$

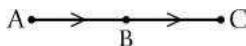
17.1.2 Unit Vector

A vector $\vec{a}$ of length unit is called **unit vector**. It is denoted by $\hat{a}$ and defined as

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

17.1.3 Collinear Vectors

Vector having the same or parallel supports are known as **collinear vectors**.



Hence, $\vec{AB}$, $\vec{BC}$ and $\vec{AC}$ are **like vectors**.

17.1.4 Like Vectors

Collinear vector having the same direction are called like vectors.

Thus, $\vec{AB}$, $\vec{BC}$ and $\vec{AC}$ are like vectors.

17.1.5 Unlike Vectors

Collinear vectors having opposite direction are known as unlike vectors.

17.1.6 Localized Vectors

A vector drawn parallel to a given vector through a specified point as the initial point is called a **localized vector**.

17.1.7 Coplanar Vector

Three or more non-zero vectors lying in the same plane or parallel to the same plane are said to be coplanar, otherwise they are called non-coplanar.

17.1.8 Position Vector

The position vector of point P with respect to its origin is denoted by

$$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$$

Where $\hat{i}$, $\hat{j}$, $\hat{k}$ be unit vectors along x , y , z axes

17.2 Laws of Addition of Vector

17.2.1 Vector Addition

Let $\vec{a}$ and $\vec{b}$ be any two vectors. Take any point O and draw segment $\vec{OA}$ and $\vec{AB}$ such that $\vec{OA} = \vec{a}$ and $\vec{AB} = \vec{b}$. Join OB . Then $\vec{OB}$ is called the sum of resultant of $\vec{a}$ and $\vec{b}$.

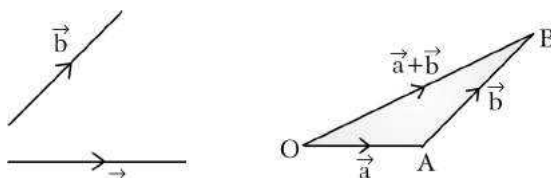


Fig. 17.1

17.2.2 Triangle Law of Addition of Forces

In $\triangle OAB$, if $\vec{OA}$ and $\vec{AB}$ represent $\vec{a}$ and $\vec{b}$ respectively, then $\vec{OB}$ represents $(\vec{a} + \vec{b})$.

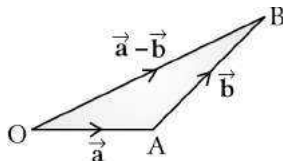


Fig. 17.2

This is known as triangle law of addition of forces.

17.2.3 Parallelogram Law of Addition of Forces

In a parallelogram $OACB$ if $\vec{OA}$ and $\vec{OB}$ represent $\vec{a}$ and $\vec{b}$ respectively, then $\vec{OC}$ represents $(\vec{a} + \vec{b})$. This is known as parallelogram law of addition of forces.

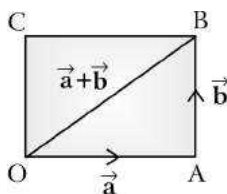


Fig. 17.3

Theorems

Theorem 1: (*Commutative law*) vector addition is commutative

i.e., $\vec{a} + \vec{b} = \vec{b} + \vec{a}$

[B.C.A. (Garhwal) 2011]

Proof: Let $\vec{a}$ and $\vec{b}$ be given vectors represented by $\vec{OA}$ and $\vec{OB}$ respectively then complete the parallelogram $OACB$.

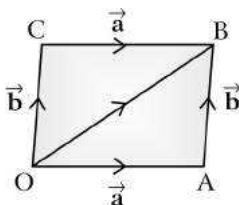


Fig. 17.4

Then

$$\vec{OC} = \vec{AB} = \vec{b}, \quad \vec{CB} = \vec{OA} = \vec{a}, \quad \vec{b} + \vec{a} = \vec{OC} + \vec{CB} = \vec{OB}$$

Hence,

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

17.3 Components of a Vector

Let O be the origin and let $P(x, y, z)$ be any point in space. Let $\hat{i}, \hat{j}, \hat{k}$ be unit vectors along the x, y, z axis respectively. Let position vector of P be $\vec{r}$, then

$$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$$

This form of vector is called its component form. Here x, y, z are called the scalar of $\vec{r}$ and $x \hat{i} + y \hat{j} + z \hat{k}$ are called its vector components.

Also
$$|\vec{r}| = |x \hat{i} + y \hat{j} + z \hat{k}| = \sqrt{x^2 + y^2 + z^2}$$

17.3.1 Direction Ratios and Direction Cosines of a Vector

Consider a vector $\vec{r} = a \hat{i} + b \hat{j} + c \hat{k}$

Then, the number a, b, c are called the **direction ratios** of $\vec{r}$. Direction cosines of $\vec{r}$ are given by

$$\frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

If l, m, n are the direction cosines of a vector then $l^2 + m^2 + n^2 = 1$.

If $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be two points in space, then direction ratios of

$$\vec{AB} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

The direction cosine of $\vec{AB} = \frac{x_2 - x_1}{r}, \frac{y_2 - y_1}{r}, \frac{z_2 - z_1}{r}$

where
$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Theorem 2: (Associative law) Vector addition is associative

i.e.,
$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$$

Proof: Let $\vec{OA} = \vec{a}$, $\vec{AB} = \vec{b}$ and $\vec{BC} = \vec{c}$

Join, OB, OC and AC

$$(\vec{a} + \vec{b}) + \vec{c} = (\vec{OA} + \vec{AB}) + \vec{BC} = (\vec{OB} + \vec{BC}) = \vec{OC}$$

$$\vec{a} + (\vec{b} + \vec{c}) = \vec{OA} + (\vec{AB} + \vec{BC}) = \vec{OA} + \vec{AC} = \vec{OC}$$

$\therefore (\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$

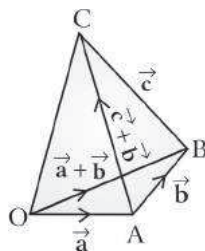


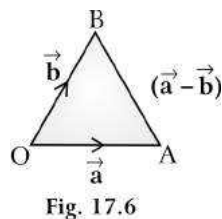
Fig. 17.5

17.3.2 Difference of Two Vectors

For any two vectors $\vec{a}$ and $\vec{b}$, we define $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$

Let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ then $\vec{BO} = -\vec{b}$

$\therefore (\vec{a} - \vec{b}) = \vec{a} + (-\vec{b}) = \vec{OA} + \vec{BO} = \vec{BO} + \vec{OA} = \vec{BA}$



Thus $(\vec{OA} - \vec{OB}) = \vec{BA}$

Similarly $(\vec{OB} - \vec{OA}) = \vec{AB}$

17.4 Scalar Multiplication

The scalar multiplication of $\vec{a}$ by a scalar λ is the vector $\lambda \vec{a}$ such that

(i) $|\lambda \vec{a}| = |\lambda| |\vec{a}|$

(ii) Direction of $|\lambda \vec{a}|$ is the same as that of $\vec{a}$, when $\lambda > 0$ and opposite to that of.

Illustrative Examples

(Vectors)

Example 1: Find the unit vector in the direction of the vector $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$

[B.C.A. (Rohilkhand) 2004, 2006]

Solution: We have, $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$

$$\Rightarrow |\vec{a}| = \sqrt{(1)^2 + (2)^2 + (3)^2} = \sqrt{14}$$

Unit vector in the direction of $\vec{a}$ is given by

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} + 2\hat{j} + 3\hat{k}}{\sqrt{1+4+9}} = \left(\frac{1}{\sqrt{14}} \hat{i} + \frac{2}{\sqrt{14}} \hat{j} + \frac{3}{\sqrt{14}} \hat{k} \right)$$

Example 2: Find vector in the direction of the vector $\vec{a} = 3\hat{i} + \hat{j}$ that has magnitude of 5 units.

Solution: We have $\vec{a} = 3\hat{i} + \hat{j}$

$$\Rightarrow |\vec{a}| = \sqrt{(3)^2 + (1)^2} = \sqrt{10}$$

Unit vector in the direction of $\vec{a}$ is given by

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{(3\hat{i} + \hat{j})}{\sqrt{10}} = \left(\frac{3}{\sqrt{10}}\hat{i} + \frac{1}{\sqrt{10}}\hat{j} \right)$$

Hence, the required vector, $5\vec{a} = \left(\frac{15}{\sqrt{10}}\hat{i} + \frac{5}{\sqrt{10}}\hat{j} \right)$

Example 3: Let $\vec{a} = 3\hat{i} + 2\hat{j}$ and $\vec{b} = 2\hat{i} + 3\hat{j}$ is $|\vec{a}| = |\vec{b}|$. Is $\vec{a} = \vec{b}$?

[B.C.A. (Delhi) 2000, 2009]

Solution: We have $|\vec{a}| = \sqrt{(3)^2 + (2)^2} = \sqrt{13}$

and $|\vec{b}| = \sqrt{(2)^2 + (3)^2} = \sqrt{13}$

$\therefore |\vec{a}| = |\vec{b}|$, but $3\hat{i} + 2\hat{j} \neq 2\hat{i} + 3\hat{j}$

$\therefore \vec{a} \neq \vec{b}$

Example 4: If $P(1, 2, -3)$ and $Q(-1, -2, 1)$ are two points in space then find

(i) The direction ratios of $\vec{PQ}$ and (ii) The direction cosines of $\vec{PQ}$

Solution: (i) Direction ratios of PQ are

$$(-1-1), (-2-2), (1+3), \text{ i.e., } -2, -4, 4$$

(ii) Direction cosines of PQ are

$$\frac{-2}{\sqrt{(-2)^2 + (-4)^2 + (4)^2}}, \frac{-4}{\sqrt{(-2)^2 + (-4)^2 + (4)^2}}, \frac{4}{\sqrt{(-2)^2 + (-4)^2 + (4)^2}}$$

i.e., $\frac{-2}{6}, \frac{-4}{6}, \frac{4}{6}$

i.e., $\frac{-1}{3}, \frac{-2}{3}, \frac{2}{3}$

Example 5: Find a unit vector in the direction of $\vec{AB}$, where $A(1, 2, 3)$ and $B(4, 5, 6)$ are the given points.

Solution: We have

Position vector (P.V.) of $A = \hat{i} + 2\hat{j} + 3\hat{k}$

Position vector (P.V.) of $B = 4\hat{i} + 5\hat{j} + 6\hat{k}$

$$\begin{aligned} \therefore \vec{AB} &= (\text{P.V. of } B) - (\text{P.V. of } A) \\ &= (4\hat{i} + 5\hat{j} + 6\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) = (3\hat{i} + 3\hat{j} + 3\hat{k}) \end{aligned}$$

$$|\vec{AB}| = \sqrt{(3)^2 + (3)^2 + (3)^2} = \sqrt{27}$$

$$\therefore \quad \text{Unit vector in the direction of } \vec{AB} = \frac{\vec{AB}}{|\vec{AB}|}$$

$$= \frac{3\hat{i} + 3\hat{j} + 3\hat{k}}{\sqrt{27}} = \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}} = \left(\frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k} \right)$$

Example 6: Find the position vector of the mid-point of the vector joining the points.

$$\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k} \quad \text{and} \quad \vec{B} = (4\hat{i} + \hat{j} - 2\hat{k})$$

Solution: The position vector of A and B are given by

$$\vec{a} = (2\hat{i} + 3\hat{j} + 4\hat{k}) \quad \text{and} \quad \vec{b} = (4\hat{i} + \hat{j} - 2\hat{k})$$

$$\therefore \text{ P.V. of mid-point of } AB = \frac{1}{2}(\vec{a} + \vec{b})$$

$$= \frac{1}{2} [\{2\hat{i} + 3\hat{j} + 4\hat{k}\} + \{4\hat{i} + \hat{j} - 2\hat{k}\}] = 3\hat{i} + 2\hat{j} + \hat{k}$$

Example 7: Show that the points A, B, C with position vector $\vec{a} = (3\hat{i} - 4\hat{j} - 4\hat{k})$, $\vec{b} = (2\hat{i} - \hat{j} + \hat{k})$ and $\vec{c} = (\hat{i} - 3\hat{j} - 5\hat{k})$ respectively, form the vertices of a right-angled triangle. [B.C.A. (Agra) 2008]

Solution: We have $\vec{AB} = (\text{P.V. of } \vec{B}) - (\text{P.V. of } \vec{A}) = \vec{b} - \vec{a}$

$$= (2\hat{i} - \hat{j} + \hat{k}) - (3\hat{i} - 4\hat{j} - 4\hat{k}) = -\hat{i} + 3\hat{j} + 5\hat{k}$$

Similarly, $\vec{BC} = (\text{P.V. of } \vec{C}) - (\text{P.V. of } \vec{B})$

$$= (\vec{c} - \vec{b}) = (\hat{i} - 3\hat{j} - 5\hat{k}) - (2\hat{i} - \hat{j} + \hat{k}) = -\hat{i} - 2\hat{j} - 6\hat{k}$$

$$\vec{CA} = (\text{P.V. of } \vec{A}) - \text{P.V. of } \vec{C} = \vec{a} - \vec{c}$$

$$= (3\hat{i} - 4\hat{j} - 4\hat{k}) - (\hat{i} - 3\hat{j} - 5\hat{k}) = (2\hat{i} - \hat{j} + \hat{k})$$

$$\therefore |\vec{AB}|^2 = (-1)^2 + (3)^2 + (5)^2 = 35,$$

$$|\vec{BC}|^2 = (-1)^2 + (-2)^2 + (-6)^2 = 41$$

$$\text{and} \quad |\vec{CA}|^2 = (2)^2 + (-1)^2 + (1)^2 = 6$$

$$\text{Thus} \quad |\vec{AB}|^2 + |\vec{CA}|^2 = |\vec{BC}|^2$$

$$\therefore (AB)^2 + (CA)^2 = (BC)^2$$

Hence, ΔABC is right angle at A .

Example 8: If $\vec{a} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$, and $\vec{c} = \hat{i} + \hat{j} + \hat{k}$ then find $|\vec{a} + 2\vec{b} + 3\vec{c}|$.

Solution: We have $-\vec{a} + 2\vec{b} + 3\vec{c} = -(2\hat{i} + \hat{j} - 3\hat{k}) + 2(3\hat{i} - 2\hat{j} + \hat{k}) + 3(\hat{i} + \hat{j} + \hat{k})$
 $= 7\hat{i} - 2\hat{j} + 8\hat{k}$

$$\therefore |\vec{a} + 2\vec{b} + 3\vec{c}| = \sqrt{(7)^2 + (-2)^2 + (8)^2} = \sqrt{49 + 4 + 64} = \sqrt{117}$$

17.4.1 Scalar Product of Two Vectors

Let $\vec{a}$ and $\vec{b}$ be two vectors and let θ be the angle between them. Then, the **scalar product** or dot product of $\vec{a}$ and $\vec{b}$ denoted by $\vec{a} \cdot \vec{b}$ is defined as $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = ab \cos \theta$.

(i) **Angle between two vectors in terms of scalar product:** Let θ be the angle between two non-zero vectors $\vec{a}$ and $\vec{b}$. Then $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$.

$$\therefore \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \Rightarrow \theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \right)$$

(ii) **Length of a vector:** Let $\vec{a}$ be any vector.

$$\text{Then, } \vec{a} \cdot \vec{a} = |\vec{a}| |\vec{a}| \cos 0^\circ = |\vec{a}|^2$$

$$\therefore |\vec{a}| = \sqrt{(\vec{a} \cdot \vec{a})}$$

Note:

(i) If $\vec{a} = 0$ or $\vec{b} = 0$ we define $\vec{a} \cdot \vec{b} = 0$

(ii) If $\vec{a}$ and $\vec{b}$ is like or parallel vectors then $\theta = 0$.

$$\vec{a} \cdot \vec{b} = ab \cos 0 = ab$$

(iii) If $\vec{a}$ and $\vec{b}$ is unlike vector then $\theta = \pi$

$$\vec{a} \cdot \vec{b} = ab \cos \pi = -ab.$$

(iv) If $\vec{a}$ and $\vec{b}$ are orthogonal vectors then $\theta = \frac{\pi}{2}$

$$\vec{a} \cdot \vec{b} = ab \cos \frac{\pi}{2} = 0$$

17.4.2 Orthonormal Vector Triad

Let $\hat{i}$, $\hat{j}$, $\hat{k}$ be unit vectors along three mutually perpendicular co-ordinate axis i.e., x , y , z axes respectively. Then these vectors are said to form an **orthonormal triad** of vectors.

We have

$$(i) \quad \hat{i} \cdot \hat{i} = |\hat{i}| |\hat{i}| \cos 0 = 1$$

$$\text{Similarly,} \quad \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$$

$$\text{Thus,} \quad \hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$$

$$(ii) \quad \hat{i} \cdot \hat{j} = |\hat{i}| |\hat{j}| \cos 90 = 0$$

$$\text{Thus,} \quad \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

17.4.3 Geometrical Interpretation of Scalar Product

Let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ and $\angle AOB = \theta$ and draw $BM \perp OA$, then

$$\begin{aligned} \vec{a} \cdot \vec{b} &= |\vec{a}| |\vec{b}| \cos \theta \\ &= a (b \cos \theta) \\ &= a (OB \cos \theta) \\ &= a (OM) \\ &= a (\text{projection of } \vec{b} \text{ on } \vec{a}) \end{aligned}$$

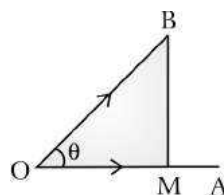


Fig. 17.7

$$\text{Thus,} \quad \vec{a} \cdot \vec{b} = a (\text{projection of } \vec{b} \text{ on } \vec{a})$$

$$\text{Similarly,} \quad \vec{b} \cdot \vec{a} = b (\text{projection of } \vec{a} \text{ on } \vec{b})$$

Hence we conclude that

$$(i) \quad \text{Projection of } \vec{a} \text{ on } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

$$(ii) \quad \text{Projection of } \vec{b} \text{ on } \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

17.4.4 Physical Significance of Scalar Product

Let O be a particle and let $\vec{OA} = \vec{a}$ be force magnitude a . Let $\vec{OB} = \vec{b}$ be the displacement due to this force. Draw $BM \perp$ to OA .

$$\text{Then, } OM = OB \cos \theta = b \cos \theta$$

$$\text{The displacement in the direction of force} = OM = b \cos \theta$$

$$\therefore \quad \text{work done} = a (b \cos \theta) = \vec{a} \cdot \vec{b}$$

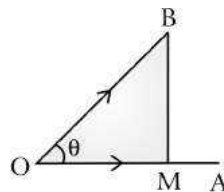


Fig. 17.8

17.5 Properties of Scalar Product

Theorem 1: (Commutative law): For any two vectors $\vec{a}$ and $\vec{b}$

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

[B.C.A. (Bundelkhand) 2007]

Proof: Let $\vec{a}$ and $\vec{b}$ be any two non-zero vectors, and let θ be the angle between them.

Then $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = ab \cos \theta$, and

$$\vec{b} \cdot \vec{a} = |\vec{b}| |\vec{a}| \cos(-\theta) = ba \cos \theta = ab \cos \theta$$

Thus,

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

Theorem 2: For any two vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \cdot \vec{b} = 0$ or $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$

or

$$\vec{b} \perp \vec{a}$$

Proof: Let $\vec{a} \cdot \vec{b} = 0$

Then, $\vec{a} \cdot \vec{b} = 0 \Rightarrow ab \cos \theta = 0$

$$\Rightarrow a = 0 \text{ or } b = 0 \text{ or } \cos \theta = 0$$

$$\Rightarrow a = 0 \text{ or } b = 0 \text{ or } \theta = \frac{\pi}{2}$$

$$\Rightarrow \vec{a} = \vec{0} \text{ or } \vec{b} = \vec{0} \text{ or } \vec{a} \perp \vec{b}$$

Conversely, Let $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{a} \perp \vec{b}$

If $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ then $\vec{a} \cdot \vec{b} = 0$

If $\vec{a} \perp \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = ab \cos \frac{\pi}{2} = 0$

Theorem 3: For any two vectors $\vec{a}$ and $\vec{b}$

$$(i) \quad \vec{a} \cdot (-\vec{b}) = -(\vec{a} \cdot \vec{b}) = (-\vec{a}) \cdot \vec{b}$$

$$(ii) \quad (-\vec{a}) \cdot (-\vec{b}) = \vec{a} \cdot \vec{b}$$

Proof: Let $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$. Increase AO to A' and BO to B' such that $OA' = OA$ and $OB' = OB$.

Then $\vec{OA'} = -\vec{a}$ and $\vec{OB'} = -\vec{b}$

Let $\angle AOB = \theta$ then $\angle AOB' = \pi - \theta$

$$(i) \quad \vec{a} \cdot (-\vec{b}) = |\vec{a}| |-\vec{b}| \cos(\pi - \theta)$$

$$= |\vec{a}| |\vec{b}| (-\cos \theta) = -ab \cos \theta = -(\vec{a} \cdot \vec{b})$$

$$\text{Similarly, } (-\vec{a}) \cdot \vec{b} = -(\vec{a} \cdot \vec{b})$$

$$\text{Hence, } \vec{a} \cdot (-\vec{b}) = (-\vec{a}) \cdot \vec{b} = -(\vec{a} \cdot \vec{b})$$

$$(ii) \quad (-\vec{a}) \cdot (-\vec{b}) = |-\vec{a}| |-\vec{b}| \cos(\angle A'OB') = |\vec{a}| |\vec{b}| \cos \theta = ab \cos \theta = \vec{a} \cdot \vec{b}$$

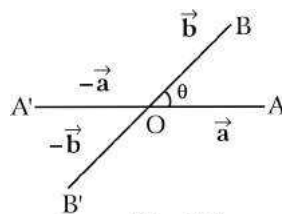


Fig. 17.9

Theorem 4: (Distribution law): For any three vectors $\vec{a}, \vec{b}, \vec{c}$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

[B.C.A. (Bundelkhand) 2006]

Proof: Let O be the origin. Let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$, $\vec{BC} = \vec{c}$

$$\text{Then } \vec{OC} = \vec{OB} + \vec{BC} = (\vec{b} + \vec{c})$$

Draw $BM \perp OA$ and $CN \perp OA$

Now, $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{OC} = |\vec{a}| (OC \cos \theta)$, where $\angle AOC = \theta$

$$= a(ON) [(OC) \cos \theta = ON]$$

$$= a(OM + MN)$$

$$= a(OM) + a(MN)$$

$$= a(\text{projection of } \vec{b} \text{ on } \vec{a})$$

$$+ a(\text{projection of } \vec{c} \text{ on } \vec{a})$$

$$= \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

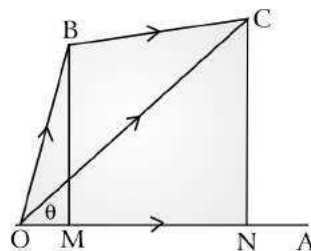


Fig. 17.10

$$\text{Hence, } \vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

Theorem 5: $\vec{a} \cdot (\vec{b} - \vec{c}) = (\vec{a} \cdot \vec{b}) - (\vec{a} \cdot \vec{c})$

$$\text{Proof: } \vec{a} \cdot (\vec{b} - \vec{c}) = \vec{a} \cdot [\vec{b} + (-\vec{c})]$$

$$= \vec{a} \cdot \vec{b} + \vec{a} \cdot (-\vec{c})$$

[By distribution law]

$$= \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c},$$

$$(\vec{a} \cdot (-\vec{c})) = (-\vec{a} \cdot \vec{c})$$

$$\text{Hence, } \vec{a} \cdot (\vec{b} - \vec{c}) = \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c}$$

Remark: If m is a real number, and $\vec{a}$ and $\vec{b}$ be any two vectors, then

$$(\vec{ma}) \cdot \vec{b} = m(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (m\vec{b}).$$

Theorems 6: For any two vectors $\vec{a}$ and $\vec{b}$

$$(i) \quad (\vec{a} + \vec{b})^2 = \vec{a}^2 + \vec{b}^2 + 2\vec{a} \cdot \vec{b}$$

[B.C.A. (Meerut) 2003, 2005]

$$(ii) \quad (\vec{a} - \vec{b})^2 = \vec{a}^2 + \vec{b}^2 - 2\vec{a} \cdot \vec{b}$$

[B.C.A. (Delhi) 2002, 2008]

$$(iii) \quad (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = \vec{a}^2 - \vec{b}^2$$

$$(iv) \quad \vec{a} \cdot \vec{b} = \frac{1}{4} \left[(\vec{a} + \vec{b})^2 - (\vec{a} - \vec{b})^2 \right]$$

Proof: (i) $(\vec{a} + \vec{b})^2 = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b})$

$$= (\vec{a} + \vec{b}) \cdot \vec{a} + (\vec{a} + \vec{b}) \cdot \vec{b}$$

[by the distribution law]

$$= \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b}$$

[by the distribution law]

$$= \vec{a}^2 + \vec{b}^2 + 2\vec{a} \cdot \vec{b},$$

$$[\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}]$$

(ii) $(\vec{a} - \vec{b})^2 = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})$

$$= (\vec{a} - \vec{b}) \cdot \vec{a} + (\vec{a} - \vec{b}) \cdot (-\vec{b})$$

$$= \vec{a} \cdot \vec{a} - \vec{b} \cdot \vec{a} - (\vec{a} - \vec{b}) \cdot \vec{b}$$

$$= \vec{a} \cdot \vec{a} - \vec{b} \cdot \vec{a} - \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b}$$

$$[\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}]$$

$$= \vec{a} \cdot \vec{a} - 2\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b}$$

$$= \vec{a}^2 + \vec{b}^2 - 2\vec{a} \cdot \vec{b}$$

(iii) $(\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) = (\vec{a} + \vec{b}) \cdot \vec{a} - (\vec{a} + \vec{b}) \cdot \vec{b}$

$$= \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{b}$$

$$= \vec{a}^2 - \vec{b}^2$$

$$(\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a})$$

(iv) We have

$$(\vec{a} + \vec{b})^2 - (\vec{a} - \vec{b})^2 = (\vec{a}^2 + \vec{b}^2 + 2\vec{a} \cdot \vec{b}) - (\vec{a}^2 + \vec{b}^2 - 2\vec{a} \cdot \vec{b})$$

[from (1) and (2) parts]

$$= 4(\vec{a} \cdot \vec{b})$$

Hence,
$$\vec{a} \cdot \vec{b} = \frac{1}{4} \left[(\vec{a} + \vec{b})^2 - (\vec{a} - \vec{b})^2 \right]$$

Theorem 7: State and prove triangle inequality.

[B.C.A. (Purvanchal) 2008]

Proof: Triangle inequality is

$$|x + y| \leq |x| + |y|$$

$$\begin{aligned} \therefore |x + y|^2 &= (x + y)^2 \\ &= x^2 + y^2 + 2xy \\ &\leq x^2 + y^2 + 2|xy| \\ &= |x|^2 + |y|^2 + 2|x||y| \\ &= (|x| + |y|)^2 \end{aligned}$$

$$\text{Thus, } |x + y|^2 \leq (|x| + |y|)^2$$

$$\Rightarrow |x + y| \leq |x| + |y|$$

Theorem 8: State and prove Cauchy's inequality.

[B.C.A. (Meerut) 2001; B.C.A. (Bhopal) 2011]

or

Using vectors, prove Cauchy's inequality.

$$(a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

Proof: Let θ be the angle between $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$$\text{Then, } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\leq |\vec{a}| |\vec{b}| \quad (\cos \theta \leq 1)$$

$$\Rightarrow (a_1b_1 + a_2b_2 + a_3b_3) \leq \sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}$$

$$\Rightarrow (a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

17.5.1 Components of $\vec{b}$ Along and Perpendicular to $\vec{a}$

$$\text{Let } \vec{OA} = \vec{a}, \quad \vec{OB} = \vec{b}$$

$$\text{and } \angle AOB = \theta$$

$$\therefore \vec{b} = \vec{OB} = \vec{OM} + \vec{MB}$$

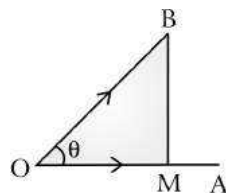


Fig. 17.11

Thus, $\vec{OM}$ and $\vec{MB}$ are components of $\vec{b}$ along $\vec{a}$ and perpendicular to $\vec{a}$ respectively.

$$\begin{aligned}\therefore \vec{OM} &= (OM) \hat{a} = (OB \cos \theta) \hat{a} = (|\vec{b}| \cos \theta) \left(\frac{\vec{a}}{|\vec{a}|} \right) \\ &= \frac{\vec{a} |\vec{b}| \cos \theta}{|\vec{a}|^2} = \frac{(\vec{a} \cdot \vec{b}) \vec{a}}{|\vec{a}|^2}\end{aligned}$$

and
$$\vec{MB} = \vec{OB} - \vec{OM} = \vec{b} - \frac{(\vec{a} \cdot \vec{b}) \vec{a}}{|\vec{a}|^2}$$

Hence, the components of $\vec{b}$ along and perpendicular to $\vec{a}$ are $\frac{(\vec{a} \cdot \vec{b}) \vec{a}}{|\vec{a}|^2}$ and $\vec{b} - \frac{(\vec{a} \cdot \vec{b}) \vec{a}}{|\vec{a}|^2}$ respectively.

17.6 Scalar Product in Terms of Rectangular Components

Let
$$\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} \text{ and } \vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}.$$

Then,
$$\begin{aligned}\vec{a} \cdot \vec{b} &= (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \cdot (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \\ &= a_1 b_1 + a_2 b_2 + a_3 b_3 \\ (\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0)\end{aligned}$$

(i) **Condition of perpendicularity:**

Let
$$\begin{aligned}\vec{a} &= a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} \text{ and} \\ \vec{b} &= b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}\end{aligned}$$

Then,
$$\begin{aligned}\vec{a} \perp \vec{b} &\Leftrightarrow \vec{a} \cdot \vec{b} = 0 \\ \Leftrightarrow a_1 b_1 + a_2 b_2 + a_3 b_3 &= 0\end{aligned}$$

(ii) **Angle between two vectors:** Let $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ and $\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$ and θ is the angle between them, then $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$.

$$\Leftrightarrow a_1 b_1 + a_2 b_2 + a_3 b_3 = |\vec{a}| |\vec{b}| \cos \theta$$

$$\Leftrightarrow \cos \theta = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{|\vec{a}| |\vec{b}|}$$

$$\Leftrightarrow \theta = \cos^{-1} \left\{ \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}} \right\}$$

Illustrative Examples

Example 1: If $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} + 2\hat{j} - \hat{k}$, find the value of

$$(\vec{a} + 3\vec{b}) \cdot (2\vec{a} - \vec{b})$$

[B.C.A. (Delhi) 2005, 2007; B.C.A. (Kanpur) 2007]

Solution: We have $(\vec{a} + 3\vec{b}) = (\hat{i} + \hat{j} + 2\hat{k}) + 3(3\hat{i} + 2\hat{j} - \hat{k})$

$$= (\hat{i} + \hat{j} + 2\hat{k}) + 9\hat{i} + 6\hat{j} - 3\hat{k}$$

$$= 10\hat{i} + 7\hat{j} - \hat{k}$$

and $2\vec{a} - \vec{b} = 2(\hat{i} + \hat{j} + 2\hat{k}) - (3\hat{i} + 2\hat{j} - \hat{k})$

$$= (2\hat{i} + 2\hat{j} + 4\hat{k}) - (3\hat{i} + 2\hat{j} - \hat{k}) = -\hat{i} + 0\hat{j} + 5\hat{k}$$

$\therefore (\vec{a} + 3\vec{b}) \cdot (2\vec{a} - \vec{b}) = (10\hat{i} + 7\hat{j} - \hat{k}) \cdot (-\hat{i} + 0\hat{j} + 5\hat{k})$

$$= -10 + 0 - 5 = -15$$

Example 2: Find the projection of $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$

[B.C.A. (Meerut) 2003]

Solution: Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

$$= \frac{(2\hat{i} - \hat{j} + \hat{k}) \cdot (\hat{i} - 2\hat{j} + \hat{k})}{\sqrt{(1)^2 + (-2)^2 + (1)^2}} = \frac{2 + 2 + 1}{\sqrt{6}} = \frac{5}{\sqrt{6}}$$

Example 3: Find the value of a which makes the vectors $a\hat{i} + 2\hat{j} + 3\hat{k}$ and $-\hat{i} + 5\hat{j} + a\hat{k}$ perpendicular.

[B.C.A. (Meerut) 2007]

Solution: For two vectors $\vec{A} = a\hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{B} = -\hat{i} + 5\hat{j} + a\hat{k}$ to be perpendicular

Then, $\vec{A} \cdot \vec{B} = 0$

$$\Rightarrow (a\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (-\hat{i} + 5\hat{j} + a\hat{k}) = 0$$

$$\Rightarrow -a + 10 + 3a = 0$$

$$\Rightarrow 2a + 10 = 0$$

or $a = -5$

Example 4: Find a unit vector perpendicular to both of the vectors.

$$\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$$

$$\vec{B} = 7\hat{i} - 5\hat{j} + \hat{k}$$

[B.C.A. (Meerut) 2008]

Solution: Let the required unit vector is $\vec{C} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$.

Such that $|\vec{C}| = 1 \Rightarrow \sqrt{c_1^2 + c_2^2 + c_3^2} = 1$

$$\text{or } c_1^2 + c_2^2 + c_3^2 = 1 \quad \dots(1)$$

Also $\vec{C}$ is $\perp$ to $\vec{A}$ and $\vec{B}$ then

$$\vec{C} \cdot \vec{A} = 0 \text{ and } \vec{C} \cdot \vec{B} = 0$$

$$\text{i.e., } \vec{C} \cdot \vec{A} = 0 \Rightarrow 2c_1 - 3c_2 + c_3 = 0 \quad \dots(2)$$

$$\vec{C} \cdot \vec{B} = 0 \Rightarrow 7c_1 - 5c_2 + c_3 = 0 \quad \dots(3)$$

Solution (2) and (3) by cross multiplication.

$$\frac{c_1}{-3+5} = \frac{c_2}{7-2} = \frac{c_3}{-10+21} = k$$

$$\text{or } \frac{c_1}{2} = \frac{c_2}{5} = \frac{c_3}{11} = k$$

$$c_1 = 2k, c_2 = 5k, c_3 = 11k$$

Put these values in (1), we get

$$4k^2 + 25k^2 + 121k^2 = 1$$

$$150k^2 = 1, k = \frac{1}{\sqrt{150}} = \frac{1}{5\sqrt{6}}$$

$$\therefore c_1 = \frac{2}{5\sqrt{6}}, c_2 = \frac{5}{5\sqrt{6}}, c_3 = \frac{11}{5\sqrt{6}}$$

Then

$$C = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$$

$$C = \frac{2}{5\sqrt{6}} \hat{i} + \frac{5}{5\sqrt{6}} \hat{j} + \frac{11}{5\sqrt{6}} \hat{k} \quad \text{or } C = \frac{2\hat{i} + 5\hat{j} + 11\hat{k}}{5\sqrt{6}}$$

Example 5: Find a vector whose magnitude is 3 units and which is perpendicular to each of the vectors.

$$\vec{a} = 3\hat{i} + \hat{j} - 4\hat{k} \quad \text{and} \quad \vec{b} = 6\hat{i} + 5\hat{j} - 2\hat{k}$$

Solution: Let required vector $\vec{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$

$$\text{then } |\vec{c}| = \sqrt{c_1^2 + c_2^2 + c_3^2} = 3 \Rightarrow \sqrt{c_1^2 + c_2^2 + c_3^2}$$

$$\Rightarrow c_1^2 + c_2^2 + c_3^2 = 9 \quad \dots(1)$$

$$\text{Also } \vec{c} \perp \vec{a} \Rightarrow \vec{c} \cdot \vec{a} = 0$$

$$\Rightarrow (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}) \cdot (3\hat{i} + \hat{j} - 4\hat{k}) = 0$$

$$\Rightarrow 3c_1 + c_2 - 4c_3 = 0 \quad \dots(2)$$

Also $\vec{c} \perp \vec{b} \Rightarrow \vec{c} \cdot \vec{b} = 0$

$$\Rightarrow (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}) \cdot (6\hat{i} + 5\hat{j} - 2\hat{k}) = 0$$

$$\Rightarrow 6c_1 + 5c_2 - 2c_3 = 0 \quad \dots(3)$$

From (2) and (3)

$$\frac{c_1}{(-2+20)} = \frac{c_2}{(-24+6)} = \frac{c_3}{(15-6)} = k$$

$$\Rightarrow \frac{c_1}{18} = \frac{c_2}{-18} = \frac{c_3}{9} = k$$

$$\Rightarrow \frac{c_1}{2} = \frac{c_2}{-2} = \frac{c_3}{1} = k$$

$$\Rightarrow c_1 = 2k, c_2 = -2k, c_3 = k$$

Put these values in (1) we find

$$4k^2 + 4k^2 + k^2 = 9 \Rightarrow k^2 = 1 \Rightarrow k = 1$$

$$\therefore c_1 = 2, c_2 = -2 \text{ and } c_3 = 1$$

Hence, $\vec{c} = (2\hat{i} - 2\hat{j} + \hat{k})$ is the required vector.

Example 6: If $\vec{AB} = 2\vec{i} + \vec{j} - 3\vec{k}$ and the coordinates of A are (1, 2, -1), find the coordinates of B.

Solution: Let O be the origin. Then $\vec{OA} = \hat{i} + 2\hat{j} - \hat{k}$

$$\therefore \vec{AB} = \vec{OB} - \vec{OA}$$

$$\begin{aligned} \Rightarrow \vec{OB} &= \vec{AB} + \vec{OA} \\ &= (2\hat{i} + \hat{j} - 3\hat{k}) + (\hat{i} + 2\hat{j} - \hat{k}) = (3\hat{i} + 3\hat{j} - 4\hat{k}) \end{aligned}$$

Hence, the coordinates of B are (3, 3, -4)

Example 7: If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = |\vec{b}| = \sqrt{2}$ and $\vec{a} \cdot \vec{b} = -1$ find the angle between $\vec{a}$ and $\vec{b}$. [B.C.A (Rohilkhand) 2004]

Solution: Let θ be the angle between $\vec{a}$ and $\vec{b}$. Then

$$\vec{a} \cdot \vec{b} = -1$$

$$\Leftrightarrow |\vec{a}| |\vec{b}| \cos \theta = -1$$

$$\Leftrightarrow \sqrt{2} \times \sqrt{2} \cos \theta = -1$$

$$\Leftrightarrow \cos \theta = -\frac{1}{2}$$

$$\Leftrightarrow \theta = \frac{2\pi}{3} \quad [\because 0 < \theta \leq 2\pi]$$

Hence, the angle between $\vec{a}$ and $\vec{b}$ is $\frac{2\pi}{3}$

Example 8: Find the angle between the vector

$$\vec{a} = \hat{i} + \hat{j} + \hat{k} \text{ and } \vec{b} = \hat{i} - \hat{j} + \hat{k}$$

Also the projection of $\vec{a}$ and $\vec{b}$.

[B.C.A. (Meerut) 2003, 2005]

Solution: We have $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$

Let θ is the angle between $\vec{a}$ and $\vec{b}$, then $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$

$$\therefore \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} - \hat{j} + \hat{k})}{\sqrt{3} \sqrt{3}} = \frac{1-1+1}{3} = 1/3$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{3}\right) = 70.52^\circ$$

$$\text{We know, the projection of } \vec{a} \text{ or } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} - \hat{j} + \hat{k})}{|\hat{i} - \hat{j} + \hat{k}|} = \frac{1-1+1}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

Example 9: If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 4$ find $|\vec{a} - \vec{b}|$.

[B.C.A. (Kanpur) 2007]

Solution: $|\vec{a} - \vec{b}|^2 = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})$

$$= \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = |\vec{a}|^2 - 2(\vec{a} \cdot \vec{b}) + |\vec{b}|^2 = (4 - 2 \times 4 + 9) = 5$$

$$\therefore |\vec{a} - \vec{b}| = \sqrt{5}$$

Example 10: If $\vec{a}, \vec{b}, \vec{c}$ are vectors such that $|\vec{a}| = 5$, $|\vec{b}| = 4$ and $|\vec{c}| = 3$ and each is perpendicular to the sum of the other, two, find $|\vec{a} + \vec{b} + \vec{c}|$.

Solution: We have

$$\vec{a} \perp (\vec{b} + \vec{c}), \vec{b} \perp (\vec{c} + \vec{a}), \text{ and } \vec{c} \perp (\vec{a} + \vec{b})$$

$$\Rightarrow \vec{a} \cdot (\vec{b} + \vec{c}) = 0, \vec{b} \cdot (\vec{c} + \vec{a}) = 0 \text{ and } \vec{c} \cdot (\vec{a} + \vec{b}) = 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = 0, \vec{b} \cdot \vec{c} + \vec{b} \cdot \vec{a} = 0 \text{ and } \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} = 0$$

[by the distributive law]

$$\Rightarrow 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0 \quad [\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}]$$

$$\begin{aligned} \therefore |\vec{a} + \vec{b} + \vec{c}|^2 &= (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) \\ &= \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} + \vec{c} \cdot \vec{c} \\ &= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \quad [\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}] \\ &= 25 + 16 + 9 + 0 = 50 \end{aligned}$$

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{50} = 5\sqrt{2}$$

Example 11: If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, prove that $\vec{a}$ and $\vec{b}$ are perpendicular to each other.

[B.C.A. (Meerut) 2008, 2009]

Solution: Let $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$. Then

$$|\vec{a} + \vec{b}|^2 = |\vec{a} - \vec{b}|^2 \Rightarrow (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})$$

$$\Rightarrow \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b}$$

$$\Rightarrow 4\vec{a} \cdot \vec{b} = 0, 4 \neq 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} = 0$$

$$\Rightarrow \vec{a} \perp \vec{b}$$

Hence, $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| \Rightarrow \vec{a} \perp \vec{b}$

Example 12: If $\hat{a}$ and $\hat{b}$ are unit vectors and θ is the angle between them, prove that

$$\sin\left(\frac{\theta}{2}\right) = \frac{1}{2}|\hat{a} - \hat{b}|.$$

[B.C.A. (Meerut) 2001, B.C.A. (Kanpur) 2006]

Solution: We have

$$\begin{aligned} |\hat{a} - \hat{b}|^2 &= (\hat{a} - \hat{b}) \cdot (\hat{a} - \hat{b}) \\ &= \hat{a} \cdot \hat{a} + \hat{b} \cdot \hat{b} - 2\hat{a} \cdot \hat{b} \quad [\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{a}] \end{aligned}$$

$$= [1 + 1 - 2 \times 1 \cos \theta] = 2(1 - \cos \theta)$$

$$|\hat{a} - \hat{b}|^2 = 4 \sin^2 \frac{\theta}{2}$$

$$\Rightarrow |\hat{a} - \hat{b}| = 2 \sin \frac{\theta}{2}$$

Example 13: For any vector $\vec{a}$ in space, show that

$$(\vec{a} \cdot \hat{i})\hat{i} + (\vec{a} \cdot \hat{j})\hat{j} + (\vec{a} \cdot \hat{k})\hat{k} = \vec{a}$$

Solution: Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, then

$$\vec{a} \cdot \hat{i} = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot \hat{i} = a_1$$

$$[\hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0]$$

$$\vec{a} \cdot \hat{j} = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot \hat{j} = a_2$$

$$\vec{a} \cdot \hat{k} = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot \hat{k} = a_3$$

$$(\vec{a} \cdot \hat{i})\hat{i} + (\vec{a} \cdot \hat{j})\hat{j} + (\vec{a} \cdot \hat{k})\hat{k} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} = \vec{a}$$

Example 14: If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$ then find the value of $(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$.

Solution: Since $\vec{a}, \vec{b}, \vec{c}$ are unit vectors, then

$$|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$$

Now,

$$\vec{a} + \vec{b} + \vec{c} = 0$$

$$\Rightarrow (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$$

$$\Rightarrow \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} + \vec{c} \cdot \vec{c} = 0$$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$$

Example 15: If $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$ and $|\vec{a}| = 3, |\vec{b}| = 5, |\vec{c}| = 7$, find angle between $\vec{a}$ and $\vec{b}$. [B.C.A (Meerut) 2009]

Solution: Let θ be the angle between $\vec{a}$ and $\vec{b}$

Now,

$$\vec{a} + \vec{b} + \vec{c} = 0$$

$$\Rightarrow (\vec{a} + \vec{b}) = -\vec{c}$$

$$\Rightarrow (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = (-\vec{c}) \cdot (-\vec{c})$$

$$\Rightarrow \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} + 2\vec{a} \cdot \vec{b} = \vec{c} \cdot \vec{c} \quad [\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}]$$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}||\vec{b}|\cos\theta = |\vec{c}|^2$$

$$\Rightarrow 9 + 25 + 2 \cdot 3 \cdot 5 \cdot \cos\theta = 49$$

$$\Rightarrow \cos\theta = \frac{1}{2}$$

$$\Rightarrow \theta = 60^\circ$$

Hence, the angle between $\vec{a}$ and $\vec{b}$ is 60° .

Example 16: If $\vec{a}, \vec{b}, \vec{c}$ are three mutually perpendicular vectors of the same magnitude prove that $(\vec{a} + \vec{b} + \vec{c})$ is equally inclined to the vectors $\vec{a}, \vec{b}$ and $\vec{c}$.

Solution: We have $|\vec{a}| = |\vec{b}| = |\vec{c}| = x$... (1)

Since $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors, then

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0 \quad \dots (2)$$

Now,

$$\begin{aligned} |\vec{a} + \vec{b} + \vec{c}|^2 &= (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) \\ &= \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{c} + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \\ &= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 \quad [\text{From (1)}] \\ &= 3x^2 \end{aligned}$$

$$\therefore |\vec{a} + \vec{b} + \vec{c}| = x\sqrt{3}$$

Suppose $(\vec{a} + \vec{b} + \vec{c})$ makes angles $\theta_1, \theta_2, \theta_3$, with $\vec{a}, \vec{b}, \vec{c}$ respectively.

$$\text{Then, } (\vec{a} + \vec{b} + \vec{c}) \cdot \vec{a} = |\vec{a} + \vec{b} + \vec{c}| |\vec{a}| \cos\theta_1 = (\sqrt{3}x) \cdot x \cdot \cos\theta_1$$

$$\Rightarrow \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a} = \sqrt{3}x^2 \cos\theta_1 \quad [\text{From (1) and (2)}]$$

$$\Rightarrow x^2 = (\sqrt{3}x^2) \cos\theta_1$$

$$\Rightarrow \cos\theta_1 = \frac{1}{\sqrt{3}} \Rightarrow \theta_1 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\text{Similarly, } \theta_2 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \text{ and } \theta_3 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\therefore \theta_1 = \theta_2 = \theta_3 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

Hence, $(\vec{a} + \vec{b} + \vec{c})$ is equally inclined to $\vec{a}, \vec{b}$ and $\vec{c}$

Example 17: If $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$ show that $\vec{a} = 0$ or $\vec{b} = \vec{c}$ or $\vec{a} \perp (\vec{b} - \vec{c})$

Solution: We have $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$

$$\Rightarrow \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} = 0$$

$$\Rightarrow \vec{a} \cdot (\vec{b} - \vec{c}) = 0$$

$$\Rightarrow \vec{a} = 0 \text{ or } \vec{b} - \vec{c} = 0 \text{ or } \vec{a} \perp (\vec{b} - \vec{c})$$

$$\Rightarrow \vec{a} = 0 \text{ or } \vec{b} = \vec{c} \text{ or } \vec{a} \perp (\vec{b} - \vec{c})$$

Example 18: Find the angle between $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{B} = 6\hat{i} - 3\hat{j} + 2\hat{k}$

[B.C.A (Meerut) 2003, 2008]

Solution: We have $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$, $\vec{B} = 6\hat{i} - 3\hat{j} + 2\hat{k}$

Let θ be an angle between $\vec{A}$ and $\vec{B}$ then

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} = \frac{(2\hat{i} + 2\hat{j} - \hat{k}) \cdot (6\hat{i} - 3\hat{j} + 2\hat{k})}{\sqrt{(2)^2 + (2)^2 + (-1)^2} \sqrt{(6)^2 + (-3)^2 + (2)^2}}$$

$$\cos \theta = \frac{12 - 6 - 2}{\sqrt{9} \sqrt{49}} = \frac{4}{3 \times 7} = \frac{4}{21}$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{4}{21} \right)$$

Example 19: If $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. Show that $(\vec{a} + \vec{b})$ is perpendicular to $(\vec{a} - \vec{b})$.

[B.C.A. (Meerut) 2002, 2005, 2010]

Solution: We have $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$

$$\text{Then } \vec{a} + \vec{b} = 4\hat{i} + \hat{j} - \hat{k}$$

$$\vec{a} - \vec{b} = -2\hat{i} + 3\hat{j} - 5\hat{k}$$

If $(\vec{a} + \vec{b})$ is perpendicular to $(\vec{a} - \vec{b})$ then show $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0$

$$\text{i.e., } (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = (4\hat{i} + \hat{j} - \hat{k}) \cdot (-2\hat{i} + 3\hat{j} - 5\hat{k}) = -8 + 3 + 5 = 0$$

Example 20: Using vectors, show that the angle between the two diagonals of a cube is

$$\cos^{-1} \left(\frac{1}{3} \right).$$

[B.C.A. (Delhi) 2002, 2006]

Solution: Let OA , OB and OC be the coterminous edges of a cube taken, along the axes in such a way that $OA = OB = OC = a$ (Let).

Then, the coordinates of the vertices of the cube are

$$O(0,0,0), A(a,0,0), B(0,a,0), C(0,0,a), P(a,a,a)$$

$$L(0,a,a), M(a,0,a) \text{ and } N(a,a,0)$$

The four diagonals are OP, AL, BM and CN

$$\therefore \vec{OP} = a\hat{i} + a\hat{j} + a\hat{k}$$

$$\begin{aligned} \vec{AL} &= (P.V. \text{ of } \vec{L}) - (P.V. \text{ of } \vec{A}) \\ &= (0\hat{i} + a\hat{j} + a\hat{k}) - (a\hat{i} + 0\hat{j} + 0\hat{k}) \\ &= (-a\hat{i} + a\hat{j} + a\hat{k}) \end{aligned}$$

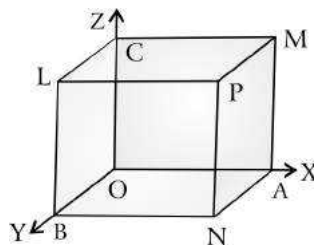


Fig. 17.12

Let θ be angle between OP and AL , then

$$\begin{aligned} \cos \theta &= \frac{\vec{OP} \cdot \vec{AL}}{|\vec{OP}| |\vec{AL}|} = \frac{(a\hat{i} + a\hat{j} + a\hat{k}) \cdot (-a\hat{i} + a\hat{j} + a\hat{k})}{\sqrt{a^2 + a^2 + a^2} \sqrt{(-a)^2 + a^2 + a^2}} \\ \theta &= \cos^{-1} \left(\frac{1}{3} \right) \end{aligned}$$

Similarly, the angle between the other pairs of diagonals are each equal to $\cos^{-1} \left(\frac{1}{3} \right)$.

Example 21: Let $\vec{a}$ and $\vec{b}$ be two non zero vectors, prove that

$$\vec{a} \perp \vec{b} \Leftrightarrow |\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|.$$

Solution: Let $\vec{a} \perp \vec{b}$, then $(\vec{a} \cdot \vec{b}) = 0$

$$\begin{aligned} \text{Now, } |\vec{a} + \vec{b}|^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\ &= \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = |\vec{a}|^2 + |\vec{b}|^2 \quad [\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a} = 0] \end{aligned}$$

$$\begin{aligned} \text{Also, } |\vec{a} - \vec{b}|^2 &= (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) \\ &= \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} \\ &= |\vec{a}|^2 + |\vec{b}|^2 \quad [\because \vec{a} \cdot \vec{b} = 0, \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}] \end{aligned}$$

$$\text{Thus, } |\vec{a} + \vec{b}|^2 = |\vec{a} - \vec{b}|^2$$

$$\Rightarrow |\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$$

$$\vec{a} \perp \vec{b} \Rightarrow |\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$$

Conversely, suppose that $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then

$$|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| \Rightarrow |\vec{a} + \vec{b}|^2 = |\vec{a} - \vec{b}|^2$$

$$\Rightarrow (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})$$

$$\Rightarrow \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b}$$

$$\Rightarrow 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a}) = 0 \quad (\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a})$$

$$\Rightarrow 4(\vec{a} \cdot \vec{b}) = 0 \quad (4 \neq 0)$$

$$\Rightarrow \vec{a} \cdot \vec{b} = 0$$

$$\Rightarrow \vec{a} \perp \vec{b}$$

$$\text{Thus, } |\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| \Rightarrow \vec{a} \perp \vec{b}$$

Example 22: Show that the points $A(2, -1, 1)$, $B(1, -3, -5)$, $C(3, -4, -4)$ are the vertices of a right-angle triangle. Also, find the remaining angles of the triangle.

Solution: Let O be the origin, then

$$\vec{OA} = (2\hat{i} - \hat{j} + \hat{k}), \vec{OB} = (\hat{i} - 3\hat{i} - 5\hat{k})$$

$$\text{and } \vec{OC} = (3\hat{i} - 4\hat{j} - 4\hat{k})$$

$$\therefore \vec{AB} = (\vec{OB} - \vec{OA}) = (-\hat{i} - 2\hat{i} - 6\hat{k})$$

$$\vec{BC} = (\vec{OC} - \vec{OB}) = (2\hat{i} - \hat{j} + \hat{k})$$

$$\vec{CA} = (\vec{OA} - \vec{OC}) = (-\hat{i} + 3\hat{j} + 5\hat{k})$$

$$\therefore \vec{AB} + \vec{BC} + \vec{CA} = 0$$

This shows that A, B, C are the vertices of a triangle

$$\vec{BC} \cdot \vec{CA} = (2\hat{i} - \hat{j} + \hat{k}) \cdot (-\hat{i} + 3\hat{j} + 5\hat{k}) = -2 - 3 + 5 = 0$$

$$\therefore \vec{BC} \perp \vec{CA} \text{ and therefore } \angle C = 90^\circ$$

Hence, $\triangle ABC$ is right angled at C

Now, $\angle A$ is the angle between $\vec{AB}$ and $\vec{AC}$

$$\therefore \vec{AB} \cdot \vec{AC} = |\vec{AB}| |\vec{AC}| \cos A$$

$$\cos A = \frac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}| |\vec{AC}|} = \frac{(-\hat{i} - 2\hat{j} - 6\hat{k}) \cdot (\hat{i} - 3\hat{j} - 5\hat{k})}{\sqrt{(-1)^2 + (-2)^2 + (-6)^2} \sqrt{(1)^2 + (-3)^2 + (-5)^2}} = \frac{\sqrt{35}}{\sqrt{41}}$$

$$\Rightarrow A = \cos^{-1} \left(\sqrt{\frac{35}{41}} \right)$$

[B.C.A. (Meerut) 2009]

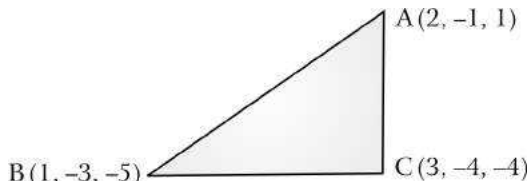


Fig. 17.13

Further, $\angle B$ is the angle between $\vec{BC}$ and $\vec{BA}$

$$\therefore \vec{BC} \cdot \vec{BA} = |\vec{BC}| |\vec{BA}| \cos B$$

$$\Rightarrow \cos B = \frac{\vec{BC} \cdot \vec{BA}}{|\vec{BC}| |\vec{BA}|} = \frac{(2\hat{i} - \hat{j} + \hat{k}) \cdot (\hat{i} + 2\hat{j} + 6\hat{k})}{\sqrt{4+1+1}\sqrt{1+4+36}} = \frac{6}{\sqrt{41}}$$

$$\Rightarrow B = \cos^{-1} \left(\sqrt{\frac{6}{41}} \right)$$

Hence, $A = \cos^{-1} \left(\sqrt{\frac{35}{41}} \right)$ and $B = \cos^{-1} \left(\sqrt{\frac{6}{41}} \right)$

Example 23: Find the area of triangle formed by the points $A(1, 2, 3)$, $B(-2, 1, -4)$, and $C(3, 4, -2)$.

Solution: We know that

Area of $\Delta ABC = \frac{1}{2} ac \sin B$, where $a = BC$ and $c = BA$. Let O be the origin, then

$$\vec{OA} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{OB} = -2\hat{i} + \hat{j} - 4\hat{k}$$

and

$$\vec{OC} = 3\hat{i} + 4\hat{j} - 2\hat{k}$$

$\therefore$

$$\vec{BC} = (\vec{OC} - \vec{OB}) = (5\hat{i} + 3\hat{j} + 2\hat{k})$$

$$\vec{BA} = (\vec{OA} - \vec{OB}) = (3\hat{i} + \hat{j} + 7\hat{k})$$

$$c = |\vec{BA}| = \sqrt{3^2 + 1^2 + 7^2} = \sqrt{59}$$

$$a = |\vec{BC}| = \sqrt{5^2 + 3^2 + 2^2} = \sqrt{38}$$

$$\cos B = \frac{\vec{BC} \cdot \vec{BA}}{|\vec{BC}| |\vec{BA}|} = \frac{(5\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (3\hat{i} + \hat{j} + 7\hat{k})}{\sqrt{38} \sqrt{59}} = \frac{32}{\sqrt{2242}}$$

$$\therefore \sin B = \sqrt{1 - \cos^2 B} = \sqrt{1 - \frac{1024}{2242}} = \sqrt{\frac{1218}{2242}}$$

Area of $\Delta ABC = \frac{1}{2} ac \sin B$

$$= \frac{1}{2} \times \sqrt{38} \times \sqrt{59} \times \sqrt{\frac{1218}{2242}} = \frac{1}{2} \sqrt{1218} \text{ square units.}$$

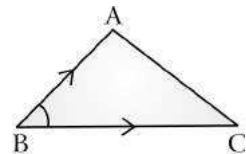


Fig. 17.14

17.6.1 Application of Scalar Product in Mechanics

The work done by a force is a scalar quantity. Suppose that a force $\vec{F}$, acting on a particle at O , displaces it to a point A such that $\vec{OA} = \vec{d}$ then.

Work done by $\vec{F}$ to displace the particle by a distance $\vec{d}$

$$W = \vec{F} \cdot \vec{d} = |\vec{F}| |\vec{d}| \cos \theta.$$

Illustrative Examples

Example 1: Find the work done by a force $\vec{F} = \hat{i} + 3\hat{j} - \hat{k}$ in moving a particle from the point A with position vector $(2\hat{i} - 6\hat{j} + 7\hat{k})$ to the point B with position vector $(3\hat{i} - \hat{j} - 5\hat{k})$.

[B.C.A. (Kanpur) 2001, 2007]

Solution: Let $\vec{d}$ be the displacement of the particle in moving from A to B , then

$$\begin{aligned}\vec{d} = \vec{AB} &= (P.V. \text{ of } B) - (P.V. \text{ of } A) \\ &= (3\hat{i} - \hat{j} - 5\hat{k}) - (2\hat{i} - 6\hat{j} + 7\hat{k}) = \hat{i} + 5\hat{j} - 12\hat{k}\end{aligned}$$

And $\vec{F} = \hat{i} + 3\hat{j} - \hat{k}$

$\therefore$ Work done $W = \vec{F} \cdot \vec{d} = (\hat{i} + 3\hat{j} - \hat{k}) \cdot (\hat{i} + 5\hat{j} - 12\hat{k}) = 1 + 15 + 12 = 28 \text{ units.}$

Example 2: Forces of magnitudes 5 and 3 units, acting in the direction $(6\hat{i} + 2\hat{j} + 3\hat{k})$ and $(3\hat{i} - 2\hat{j} + 6\hat{k})$ respectively, acts on particle, which is displaced from the point $A(2, -3, -1)$ to $B(4, 3, 1)$. Find the work done by the forces.

Solution: Let $\vec{F}$ = The resultant force

$\vec{d}$ = The displacement

$$\begin{aligned}\therefore \vec{F} &= \frac{5(6\hat{i} + 2\hat{j} + 3\hat{k})}{|6\hat{i} + 2\hat{j} + 3\hat{k}|} + \frac{3(3\hat{i} - 2\hat{j} + 6\hat{k})}{|3\hat{i} - 2\hat{j} + 6\hat{k}|} \\ &= \frac{30\hat{i} + 10\hat{j} + 15\hat{k}}{\sqrt{36 + 4 + 9}} + \frac{9\hat{i} - 6\hat{j} + 18\hat{k}}{\sqrt{9 + 4 + 36}} = \frac{1}{7}(39\hat{i} + 4\hat{j} + 33\hat{k})\end{aligned}$$

$$\begin{aligned}\vec{d} = \vec{AB} &= (P.V. \text{ of } B) - (P.V. \text{ of } A) \\ &= (4\hat{i} + 3\hat{j} + \hat{k}) - (2\hat{i} - 3\hat{j} - \hat{k}) = (2\hat{i} + 6\hat{j} + 2\hat{k})\end{aligned}$$

$$\begin{aligned}\therefore \text{Total work done} &= \vec{F} \cdot \vec{d} \\ &= \frac{1}{7}(39\hat{i} + 4\hat{j} + 33\hat{k}) \cdot (2\hat{i} + 6\hat{j} + 2\hat{k}) \\ &= \frac{1}{7}(78 + 24 + 66) = 24 \text{ units.}\end{aligned}$$

Example 3: Find work done by the force $\vec{F} = \hat{i} + \hat{j} + \hat{k}$ acting on a particle, if it is displaced from the point $A(3,3,3)$ and the point $B(4,4,4)$.

Solution: We have

$$\begin{aligned}\text{Displacement } \vec{d} = \vec{AB} &= (P.V. \text{ of } B) - (P.V. \text{ of } A) \\ &= (4\hat{i} + 4\hat{j} + 4\hat{k}) - (3\hat{i} + 3\hat{j} + 3\hat{k}) = \hat{i} + \hat{j} + \hat{k} \\ \vec{F} &= (\hat{i} + \hat{j} + \hat{k})\end{aligned}$$

$$\begin{aligned}\therefore \text{Work done} &= \vec{F} \cdot \vec{d} \\ &= (\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} + \hat{j} + \hat{k}) = 1 + 1 + 1 = 3 \text{ units.}\end{aligned}$$

Example 4: A force of 3 units, acting parallel to $(\hat{i} + \hat{j} + \hat{k})$, displaces a particle from point $A(-2, -3, 4)$ to the point $B(2, 3, 1)$. Find the work done.

Solution: The given force is

$$\vec{F} = \frac{3(\hat{i} + \hat{j} + \hat{k})}{|\hat{i} + \hat{j} + \hat{k}|} = \frac{3\hat{i} + 3\hat{j} + 3\hat{k}}{\sqrt{3}} = \sqrt{3}(\hat{i} + \hat{j} + \hat{k})$$

$$\begin{aligned}\vec{d} = \vec{AB} &= P.V. \text{ of } B - P.V. \text{ of } A \\ &= (2\hat{i} + 3\hat{j} + \hat{k}) - (-2\hat{i} - 3\hat{j} + 4\hat{k}) = 4\hat{i} + 6\hat{j} - 3\hat{k}\end{aligned}$$

$$\begin{aligned}\therefore \text{Work done } w &= \vec{F} \cdot \vec{d} \\ &= \sqrt{3}(\hat{i} + \hat{j} + \hat{k}) \cdot (4\hat{i} + 6\hat{j} - 3\hat{k}) = \sqrt{3}(4 + 6 - 3) = 7\sqrt{3} \text{ units}\end{aligned}$$

Example 5: A particle acted upon by constant force

$$\vec{P} = 2\hat{i} + 5\hat{j} + 6\hat{k}, Q = -\hat{i} - 2\hat{j} - \hat{k}$$

is displaced from the point A with position vector $(4\hat{i} - 3\hat{j} - 2\hat{k})$ to the point B with position vector $(6\hat{i} + \hat{j} - 3\hat{k})$. Find the total work done by the forces.

Solution: Let $\vec{F} = \vec{P} + \vec{Q}$ = Resultant forces

$$\Rightarrow \vec{F} = (2\hat{i} + 5\hat{j} + 6\hat{k}) + (-\hat{i} - 2\hat{j} - \hat{k}) = \hat{i} + 3\hat{j} + 5\hat{k}$$

$$\begin{aligned}\vec{d} = \vec{AB} &= P.V. \text{ of } B - P.V. \text{ of } A \\ &= (6\hat{i} + \hat{j} - 3\hat{k}) - (4\hat{i} - 3\hat{j} - 2\hat{k}) = 2\hat{i} + 4\hat{j} - \hat{k}\end{aligned}$$

$$\begin{aligned}\therefore \text{Total work done } W &= \vec{F} \cdot \vec{d} \\ &= (\hat{i} + 3\hat{j} + 5\hat{k}) \cdot (2\hat{i} + 4\hat{j} - \hat{k}) = 2 + 12 - 5 = 9 \text{ units}\end{aligned}$$

Theorems

Theorem 1: (Cosine formula) In $\triangle ABC$, if $BC = a$, $CA = b$ and $AB = c$, prove the following using vector method:

$$(i) \quad \cos A = \frac{(b^2 + c^2 - a^2)}{2bc} \quad (ii) \quad \cos B = \frac{(c^2 + a^2 - b^2)}{2ca}$$

$$(iii) \quad \cos C = \frac{(a^2 + b^2 - c^2)}{2ab}$$

Proof: In $\triangle ABC$, Let $\vec{BC} = \vec{a}$, $\vec{CA} = \vec{b}$ and $\vec{AB} = \vec{c}$.

Then, $|\vec{BC}| = |\vec{a}| = a$, $|\vec{CA}| = |\vec{b}| = b$ and $|\vec{AB}| = |\vec{c}| = c$

$$(i) \quad \vec{BC} + \vec{CA} + \vec{AB} = \vec{0}$$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \vec{b} + \vec{c} = -\vec{a}$$

$$\Rightarrow (\vec{b} + \vec{c}) \cdot (\vec{b} + \vec{c}) = (-\vec{a}) \cdot (-\vec{a})$$

$$\Rightarrow \vec{b} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{b} + \vec{c} \cdot \vec{c} = \vec{a} \cdot \vec{a}$$

$$\Rightarrow |\vec{b}|^2 + 2\vec{b} \cdot \vec{c} + |\vec{c}|^2 = |\vec{a}|^2$$

$$\Rightarrow |\vec{b}|^2 + 2|\vec{b}||\vec{c}|\cos(\pi - A) + |\vec{c}|^2 = |\vec{a}|^2 \quad [\because \text{angle between } \vec{b} \text{ and } \vec{c} \text{ is } (\pi - A)]$$

$$\Rightarrow b^2 - 2bc \cos A + c^2 = a^2$$

$$\Rightarrow \cos A = \frac{(b^2 + c^2 - a^2)}{2bc}$$

$$(ii) \quad \vec{BC} + \vec{CA} + \vec{AB} = \vec{0}$$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \vec{a} + \vec{c} = -\vec{b}$$

$$\Rightarrow (\vec{a} + \vec{c}) \cdot (\vec{a} + \vec{c}) = (-\vec{b}) \cdot (-\vec{b})$$

$$\Rightarrow \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{c} + \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{c} = \vec{b} \cdot \vec{b}$$

$$\Rightarrow |\vec{a}|^2 + 2\vec{a} \cdot \vec{c} + |\vec{c}|^2 = |\vec{b}|^2$$

$$\Rightarrow |\vec{a}|^2 + 2|\vec{a}||\vec{c}|\cos(\pi - B) + |\vec{c}|^2 = |\vec{b}|^2 \quad [\because \text{angle between } \vec{a} \text{ and } \vec{c} \text{ is } (\pi - B)]$$

$$\Rightarrow a^2 - 2ac \cos B + c^2 = b^2$$

$$\Rightarrow \cos B = \frac{(a^2 + c^2 - b^2)}{2ac}$$

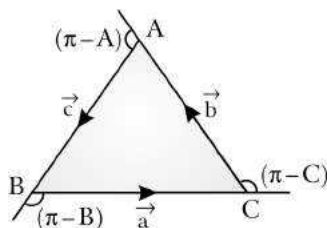


Fig. 17.15

$$[\because \vec{c} \cdot \vec{b} = \vec{b} \cdot \vec{c}]$$

$$[\because \vec{c} \cdot \vec{a} = \vec{a} \cdot \vec{c}]$$

$$(iii) \quad \vec{BC} + \vec{CA} + \vec{AB} = \vec{0}$$

$$\Rightarrow \quad \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \quad \vec{a} + \vec{b} = -\vec{c}$$

$$\Rightarrow \quad (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = (-\vec{c}) \cdot (-\vec{c})$$

$$\Rightarrow \quad \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} = \vec{c} \cdot \vec{c}$$

$$\Rightarrow \quad |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 = |\vec{c}|^2 \quad [\because \vec{b} \cdot \vec{a} = \vec{a} \cdot \vec{b}]$$

$$\Rightarrow \quad |\vec{a}|^2 + 2|\vec{a}||\vec{b}|\cos(\pi - C) + |\vec{b}|^2 = |\vec{c}|^2 \quad [\because \text{angle between } \vec{a} \text{ and } \vec{b} \text{ is } (\pi - C)]$$

$$\Rightarrow \quad a^2 - 2ab \cos C + b^2 = c^2 \Rightarrow \cos C = \frac{(a^2 + b^2 - c^2)}{2ab}$$

Theorem 2: (Projection formula) in $\triangle ABC$, if $BC = \vec{a}$, $CA = \vec{b}$ and $AB = \vec{c}$, prove the following, using vector method:

$$(i) \quad a = b \cos C + c \cos B$$

$$(ii) \quad b = c \cos A + a \cos C$$

$$(iii) \quad c = a \cos B + b \cos A$$

Proof: In $\triangle ABC$, Let $\vec{BC} = \vec{a}$, $\vec{CA} = \vec{b}$ and $\vec{AB} = \vec{c}$. Then

$$|\vec{BC}| = |\vec{a}| = a, |\vec{CA}| = |\vec{b}| = b \text{ and } |\vec{AB}| = |\vec{c}| = c$$

$$(i) \quad \vec{BC} + \vec{CA} + \vec{AB} = \vec{0}$$

$$\Rightarrow \quad \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \quad \vec{a} = -(\vec{b} + \vec{c})$$

$$\Rightarrow \quad \vec{a} \cdot \vec{a} = -\vec{a} \cdot (\vec{b} + \vec{c})$$

$$\Rightarrow \quad |\vec{a}|^2 = -(\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c})$$

$$\Rightarrow \quad |\vec{a}|^2 = -\{|\vec{a}| \cdot |\vec{b}| \cos(\pi - C) + |\vec{a}| \cdot |\vec{c}| \cos(\pi - B)\}$$

$$\Rightarrow \quad |\vec{a}|^2 = -\{|\vec{a}| \cdot |\vec{b}| \cos(\pi - C) + |\vec{a}| \cdot |\vec{c}| \cos(\pi - B)\}$$

$$\Rightarrow \quad a^2 = ab \cos C + ac \cos B$$

$$\Rightarrow \quad a = b \cos C + c \cos B$$

$$(ii) \quad \vec{BC} + \vec{CA} + \vec{AB} = \vec{0}$$

$$\Rightarrow \quad \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \quad \vec{b} = -(\vec{a} + \vec{c})$$

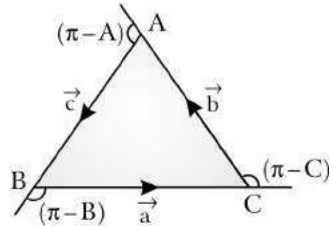


Fig. 17.16

$$\begin{aligned}
\Rightarrow \quad \vec{b} \cdot \vec{b} &= -\vec{b} \cdot (\vec{a} + \vec{c}) \\
\Rightarrow \quad \vec{b} \cdot \vec{b} &= -(\vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{c}) \\
\Rightarrow \quad \vec{b} \cdot \vec{b} &= -(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c}) \\
\Rightarrow \quad |\vec{b}|^2 &= -\{|\vec{a}| \cdot |\vec{b}| \cos(\pi - C) + |\vec{b}| \cdot |\vec{c}| \cos(\pi - A)\} \\
\Rightarrow \quad b^2 &= ab \cos C + bc \cos A \\
\Rightarrow \quad b &= a \cos C + c \cos A
\end{aligned}$$

(iii) Left as an exercise

Theorem 3: *An angle in a semicircle is a right angle.*

Proof: Let O be the centre of a semicircle with BOA as its diameter.

Let P be a point on the semicircle. Then, $\angle BPA$ is an angle in a semicircle.

Take O as the origin.

Let the position vectors of A and P be $\vec{a}$ and $\vec{r}$ respectively.

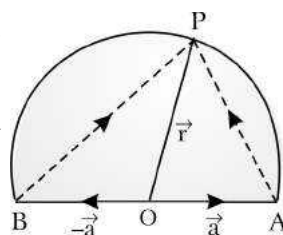


Fig. 17.17 (a)

Then, the position vector of B is $-\vec{a}$.

$$\vec{AP} = (\text{position vector of } P) - (\text{position vector of } A) = (\vec{OP} - \vec{OA}) = (\vec{r} - \vec{a})$$

$$\begin{aligned}
\vec{BP} &= (\text{position vector of } P) - (\text{Position vector of } B) \\
&= (\vec{OP} - \vec{OB}) = [\vec{r} - (-\vec{a})] = (\vec{r} + \vec{a})
\end{aligned}$$

$$\begin{aligned}
\therefore \quad \vec{AP} \cdot \vec{BP} &= (\vec{r} - \vec{a}) \cdot (\vec{r} + \vec{a}) \\
&= \vec{r} \cdot \vec{r} + \vec{r} \cdot \vec{a} - \vec{a} \cdot \vec{r} - \vec{a} \cdot \vec{a} \\
&= |\vec{r}|^2 - |\vec{a}|^2 && [\because \vec{a} \cdot \vec{r} = \vec{r} \cdot \vec{a}] \\
&= 0 && [\because |\vec{r}| = |\vec{a}| = \text{radius}]
\end{aligned}$$

$$\Rightarrow \quad \vec{BP} \perp \vec{AP} \Rightarrow \angle BPA = 90^\circ$$

Hence, an angle in a semicircle is a right angle.

Theorem 4: (Pythagoras theorem) *Prove by vector method that in a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.*

Proof: Consider $\triangle OAB$, in which $\angle AOB = 90^\circ$

Then, we have to prove that $OA^2 + OB^2 = AB^2$

Take O as the origin.

Let the position vectors of A and B be $\vec{a}$ and $\vec{b}$ respectively. Then,

$$\vec{OA} = \vec{a} \text{ and } \vec{OB} = \vec{b}.$$

$$\therefore \vec{AB} = (\vec{OB} - \vec{OA}) = (\vec{b} - \vec{a})$$

$$\vec{OA} \perp \vec{OB} \Rightarrow \vec{a} \cdot \vec{b} = 0 \quad \dots(1)$$

Now,

$$AB^2 = |\vec{AB}|^2$$

$$= \vec{AB} \cdot \vec{AB}$$

$$= (\vec{b} - \vec{a}) \cdot (\vec{b} - \vec{a})$$

$$= \vec{b} \cdot \vec{b} - \vec{b} \cdot \vec{a} - \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{a} \quad [\text{by distributive law}]$$

$$= |\vec{b}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{a}|^2 \quad [\because \vec{a} \cdot \vec{a} = \vec{a} \cdot \vec{a}]$$

$$= |\vec{a}|^2 + |\vec{b}|^2 \quad [\text{using (1)}]$$

$$= OA^2 + OB^2$$

$$\text{Hence,} \quad OA^2 + OB^2 = AB^2$$

Theorem 5: Prove by vector method that in a right-angled triangle, the midpoint of the hypotenuse is equidistant from its vertices.

[B.C.A. (Indore) 2010]

Proof: Consider $\triangle AOB$, in which $\angle AOB = 90^\circ$. Let D be the midpoint of the hypotenuse AB . Then, we have to show that $OD = DA = DB$.

Take O as the origin.

Let the position vectors of A and B be $\vec{a}$ and $\vec{b}$ respectively.

$$\text{Then,} \quad \vec{a} = \vec{OA} = \vec{OD} + \vec{DA}$$

$$\text{and} \quad \vec{b} = \vec{OB} = \vec{OD} + \vec{DB} = \vec{OD} - \vec{DA} \quad [\because \vec{DB} = -\vec{DA}]$$

$$\therefore OA \perp OB \Rightarrow \vec{a} \cdot \vec{b} = 0$$

$$\Rightarrow (\vec{OD} + \vec{DA}) \cdot (\vec{OD} - \vec{DA}) = 0$$

$$\Rightarrow \vec{OD} \cdot \vec{OD} - \vec{DA} \cdot \vec{DA} = 0$$

$$\Rightarrow |\vec{OD}|^2 = |\vec{DA}|^2$$

$$\Rightarrow |\vec{OD}| = |\vec{DA}|$$

$$\Rightarrow OD = DA$$

$$\Rightarrow OD = DA = DB$$

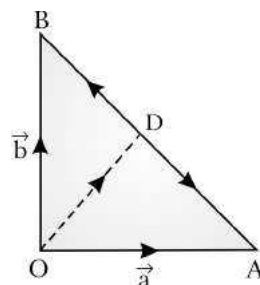


Fig. 17. 18

[$\because D$ is the midpoint of AB]

Hence, D is equidistant from O , A and B .

Theorem 6: In an isosceles triangle, the median to the base is perpendicular to the base.

[B.C.A. (Delhi) 2011]

Proof: Consider a ΔABC in which $BA = CA$

Let O be the midpoint of BC .

Then, we have to show that $OA \perp BC$.

Take O as the origin.

Let the position vectors of A and C be $\vec{a}$ and $\vec{c}$ respectively.

Then, P.V. of B is $-\vec{c}$

$$\vec{BA} = (\text{P.V. of } A) - (\text{P.V. of } B) = [\vec{a} - (-\vec{c})] = (\vec{a} + \vec{c})$$

$$\vec{CA} = (\text{P.V. of } A) - (\text{P.V. of } C) = (\vec{a} - \vec{c})$$

$$\therefore BA = CA \Rightarrow BA^2 = CA^2$$

$$\Rightarrow |\vec{BA}|^2 = |\vec{CA}|^2$$

$$\Rightarrow \vec{BA} \cdot \vec{BA} = \vec{CA} \cdot \vec{CA}$$

$$\Rightarrow (\vec{a} + \vec{c}) \cdot (\vec{a} + \vec{c}) = (\vec{a} - \vec{c}) \cdot (\vec{a} - \vec{c})$$

$$\Rightarrow \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{c} + \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{c} = \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{c} - \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{c}$$

$$\Rightarrow 4\vec{a} \cdot \vec{c} = 0 \Rightarrow \vec{a} \cdot \vec{c} = 0$$

$$\Rightarrow OA \perp OC, \text{ and therefore, } OA \perp BC$$

Hence, $OA \perp BC$

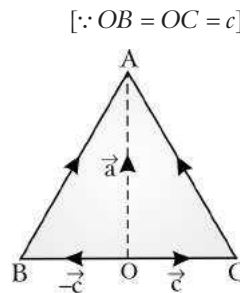


Fig. 17.19

$$[\because \vec{c} \cdot \vec{a} = \vec{a} \cdot \vec{c}]$$

Theorem 7: If two medians of a triangle are equal then the triangle is isosceles.

[B.C.A. (Lucknow) 2008]

Proof: Let AC and BD be two equal medians of ΔOAB .

Then, we have to prove that $OA = OB$.

Take O as the origin.

Let the position vectors of A and B be $\vec{a}$ and $\vec{b}$ respectively. Then,

$$\text{P.V. of } C = \vec{OC} = \frac{1}{2}\vec{OB} = \frac{1}{2}\vec{b}, \text{ and}$$

$$\text{P.V. of } D = \vec{OD} = \frac{1}{2}\vec{OA} = \frac{1}{2}\vec{a}$$

$$\therefore \quad \vec{AC} = (P.V. \text{ of } C) - (P.V. \text{ of } A) = \left(\frac{1}{2} \vec{b} - \vec{a} \right)$$

$$\vec{BD} = (P.V. \text{ of } D) - (P.V. \text{ of } B) = \left(\frac{1}{2} \vec{a} - \vec{b} \right)$$

$$\therefore \quad AC = BD$$

$$\Rightarrow \quad AC^2 = BD^2$$

$$\Rightarrow \quad \vec{AC} \cdot \vec{AC} = \vec{BD} \cdot \vec{BD}$$

$$\Rightarrow \quad \left(\frac{1}{2} \vec{b} - \vec{a} \right) \cdot \left(\frac{1}{2} \vec{b} - \vec{a} \right) = \left(\frac{1}{2} \vec{a} - \vec{b} \right) \cdot \left(\frac{1}{2} \vec{a} - \vec{b} \right)$$

$$\Rightarrow \quad \frac{1}{4} (\vec{b} \cdot \vec{b}) + (\vec{a} \cdot \vec{a}) - \frac{1}{2} (\vec{b} \cdot \vec{a}) - \frac{1}{2} (\vec{a} \cdot \vec{b})$$

$$= \frac{1}{4} (\vec{a} \cdot \vec{a}) + (\vec{b} \cdot \vec{b}) - \frac{1}{2} (\vec{a} \cdot \vec{b}) - \frac{1}{2} (\vec{b} \cdot \vec{a})$$

$$\Rightarrow \quad \frac{1}{4} |\vec{b}|^2 + |\vec{a}|^2 = \frac{1}{4} |\vec{a}|^2 + |\vec{b}|^2$$

$$\Rightarrow \quad |\vec{a}|^2 = |\vec{b}|^2$$

$$\Rightarrow \quad OA^2 = OB^2 \Rightarrow OA = OB$$

Hence, $\triangle OAB$ is isosceles.

Theorem 8: Prove by vector method that the altitudes of a triangle are concurrent.

Proof: Let $\triangle ABC$ be a triangle in which $AD \perp BC$ and $BE \perp CA$

Let AD and BE intersect at a point O .

Join CO and produce it to meet AB at F .

Then, we have to show that $CF \perp AB$.

Take O as the origin.

Let $\vec{a}, \vec{b}, \vec{c}$ be the position vectors of A, B, C respectively. Then,

$$\vec{OA} = \vec{a}, \vec{OB} = \vec{b} \text{ and } \vec{OC} = \vec{c}$$

$$\therefore \quad \vec{BC} = (P.V. \text{ of } C) - (P.V. \text{ of } B) = (\vec{c} - \vec{b}) \text{ and}$$

$$\vec{CA} = (P.V. \text{ of } A) - (P.V. \text{ of } C) = (\vec{a} - \vec{c})$$

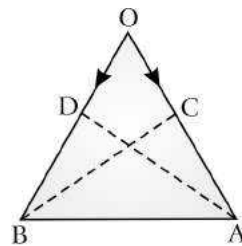


Fig. 17.20

Now, $AD \perp BC \Rightarrow \vec{OA} \perp \vec{BC}$

$$\Rightarrow \vec{OA} \cdot \vec{BC} = 0$$

$$\Rightarrow \vec{a} \cdot (\vec{c} - \vec{b}) = 0$$

...(1)

And, $BE \perp CA \Rightarrow \vec{OB} \perp \vec{CA}$

$$\Rightarrow \vec{OB} \cdot \vec{CA} = 0$$

$$\Rightarrow \vec{b} \cdot (\vec{a} - \vec{c}) = 0$$

$$\Rightarrow \vec{b} \cdot \vec{a} - \vec{b} \cdot \vec{c} = 0$$

$$\Rightarrow \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} = 0 \quad [\because \vec{b} \cdot \vec{a} = \vec{a} \cdot \vec{b}] \quad \dots(2)$$

Adding (1) and (2), we get

$$\vec{a} \cdot \vec{c} - \vec{b} \cdot \vec{c} = 0 \Rightarrow (\vec{a} - \vec{b}) \cdot \vec{c} = 0$$

$$\Rightarrow \vec{BA} \cdot \vec{OC} = 0$$

$$\Rightarrow \vec{OC} \perp \vec{AB} \Rightarrow CF \perp AB$$

Thus, the altitudes of $\triangle ABC$ meet at a point O .

Hence, the altitudes of a triangle are concurrent.

Theorem 9: Prove by vector method that the right bisectors of the sides of a triangle are concurrent.

Proof: Consider a $\triangle ABC$ in which D, E, F are the midpoints of BC, CA and AB respectively. Let DO and EO be the perpendicular bisectors of BC and CA respectively, intersecting at O . Join OF .

Then, we have to show that $OF \perp AB$. Take O as the origin. Join OA, OB and OC .

Let the position vectors of A, B, C be $\vec{a}, \vec{b}, \vec{c}$ respectively. Then, the position vectors of D, E, F are

$$\frac{1}{2}(\vec{b} + \vec{c}), \frac{1}{2}(\vec{c} + \vec{a}) \text{ and } \frac{1}{2}(\vec{a} + \vec{b}) \text{ respectively.}$$

$$\therefore \vec{OD} = \frac{1}{2}(\vec{b} + \vec{c}), \vec{OE} = \frac{1}{2}(\vec{c} + \vec{a}) \text{ and } \vec{OF} = \frac{1}{2}(\vec{a} + \vec{b})$$

$$\text{Also, } \vec{BC} = (\text{P.V. of } C) - (\text{P.V. of } B) = (\vec{c} - \vec{b})$$

$$\vec{CA} = (\text{P.V. of } A) - (\text{P.V. of } C) = (\vec{a} - \vec{c})$$

$$\vec{OD} \perp \vec{BC} \Rightarrow \vec{OD} \cdot \vec{BC} = 0$$

$$\Rightarrow \frac{1}{2}(\vec{b} + \vec{c}) \cdot (\vec{c} - \vec{b}) = 0$$

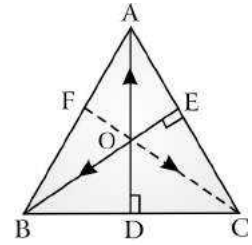


Fig. 17.21

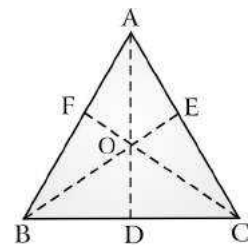


Fig. 17.22

$$\begin{aligned}
 &\Rightarrow (\vec{c} + \vec{b}) \cdot (\vec{c} - \vec{b}) = 0 \\
 &\Rightarrow \vec{c} \cdot \vec{c} - \vec{b} \cdot \vec{b} = 0 \\
 &\Rightarrow |\vec{c}|^2 - |\vec{b}|^2 = 0 \quad \dots(1)
 \end{aligned}$$

$$\begin{aligned}
 \vec{OE} \perp \vec{CA} &\Rightarrow \vec{OE} \cdot \vec{CA} = 0 \\
 &\Rightarrow \frac{1}{2}(\vec{c} + \vec{a}) \cdot (\vec{a} - \vec{c}) = 0 \\
 &\Rightarrow (\vec{a} + \vec{c}) \cdot (\vec{a} - \vec{c}) = 0 \\
 &\Rightarrow \vec{a} \cdot \vec{a} - \vec{c} \cdot \vec{c} = 0 \\
 &\Rightarrow |\vec{a}|^2 - |\vec{c}|^2 = 0 \quad \dots(2)
 \end{aligned}$$

Adding (1) and (2), we get

$$\begin{aligned}
 |\vec{a}|^2 - |\vec{b}|^2 = 0 &\Rightarrow (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0 \\
 &\Rightarrow \frac{1}{2}(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0 \\
 &\Rightarrow \vec{OF} \cdot \vec{BA} = 0 \Rightarrow OF \perp BA, \text{ i.e., } OF \perp AB
 \end{aligned}$$

This shows that OF is also the right bisector of AB .

Hence, the right bisectors of the sides of a triangle are concurrent.

Theorem 10: Prove by vector method that the diagonals of a rhombus bisect each other at right angles.

Proof: Let $OABC$ be a rhombus whose diagonals intersect at a point D .

Let O be the origin.

Let the position vectors of A and C be $\vec{a}$ and $\vec{c}$ respectively. Then,

$$\vec{OA} = \vec{a} \quad \text{and} \quad \vec{OC} = \vec{c}$$

$$\therefore \vec{OB} = \vec{OA} + \vec{AB} = \vec{OA} + \vec{OC} = (\vec{a} + \vec{c}) \quad [\because \vec{AB} = \vec{OC}]$$

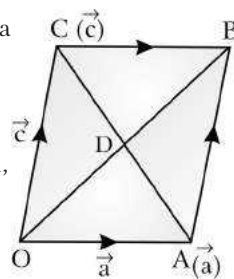


Fig. 17.23

$$\text{Position vector of midpoint of } OB = \frac{1}{2}(\vec{a} + \vec{c})$$

$$\text{Position vector of midpoint of } AC = \frac{1}{2}(\vec{a} + \vec{c})$$

Thus, the midpoints of OB and AC coincide.

So, the diagonals OB and AC bisect each other.

$$\begin{aligned}
 \text{Now, } \vec{OB} \cdot \vec{AC} &= (\vec{a} + \vec{c}) \cdot (\vec{c} - \vec{a}) \quad [\because \vec{AC} = (P.V. \text{ of } C) - (P.V. \text{ of } A) = (\vec{c} - \vec{a})] \\
 &= (\vec{c} + \vec{a}) \cdot (\vec{c} - \vec{a}) \\
 &= \vec{c} \cdot \vec{c} - \vec{a} \cdot \vec{a} = |\vec{c}|^2 - |\vec{a}|^2 = OC^2 - OA^2 = 0 \\
 &\quad [\because OC = OA, \text{ sides of the same rhombus}]
 \end{aligned}$$

$$\Rightarrow \vec{OB} \perp \vec{AC}$$

Hence, the diagonals OB and AC bisect each other at right angles.

Theorem 11: If the diagonal of a parallelogram are equal, prove by vector method that it is a rectangle. [B.C.A. (Agra) 2010 (Dec.)]

Proof: Let $OABC$ be a parallelogram in which diagonal $OB = \text{diagonal } AC$.

Then, we have to prove that $OABC$ is a rectangle.

Take O as the origin.

Let the position vectors of A and C be $\vec{a}$ and $\vec{c}$ respectively. Then,

$$\vec{OA} = \vec{a} \text{ and } \vec{OC} = \vec{c}$$

$$\therefore \vec{AC} = (P.V. \text{ of } C) - (P.V. \text{ of } A) = (\vec{c} - \vec{a})$$

$$\vec{OB} = \vec{OA} + \vec{AB} = \vec{OA} + \vec{OC} = (\vec{a} + \vec{c}) = (\vec{c} + \vec{a})$$

$$\therefore OB = AC \Rightarrow OB^2 = AC^2$$

$$\Rightarrow \vec{OB} \cdot \vec{OB} = \vec{AC} \cdot \vec{AC}$$

$$\Rightarrow (\vec{c} + \vec{a}) \cdot (\vec{c} + \vec{a}) = (\vec{c} - \vec{a}) \cdot (\vec{c} - \vec{a})$$

$$\Rightarrow \vec{c} \cdot \vec{c} + 2\vec{c} \cdot \vec{a} + \vec{a} \cdot \vec{a} = \vec{c} \cdot \vec{c} - 2\vec{c} \cdot \vec{a} + \vec{a} \cdot \vec{a}$$

$$\Rightarrow 4\vec{c} \cdot \vec{a} = 0 \Rightarrow \vec{c} \cdot \vec{a} = 0$$

$$\Rightarrow \vec{OC} \perp \vec{OA}.$$

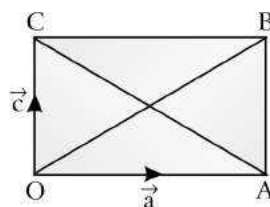


Fig. 17.24

Hence $OABC$ is a rectangle.



Comprehensive Exercise 17.1

1. Find $\vec{a} \cdot \vec{b}$ when:

(i) $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = 3\hat{i} - 4\hat{j} - 2\hat{k}$

(ii) $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = -2\hat{j} + 4\hat{k}$

(iii) $\vec{a} = \hat{i} + 5\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{k}$

2. Find the value of λ for which $\vec{a}$ and $\vec{b}$ are perpendicular where:

(i) $\vec{a} = 3\hat{i} - \hat{j} + 4\hat{k}$ and $\vec{b} = -\lambda\hat{i} + 3\hat{j} + 3\hat{k}$

(ii) $\vec{a} = 3\hat{i} + 2\hat{j} - 5\hat{k}$ and $\vec{b} = -5\hat{j} + \lambda\hat{k}$

3. Show that the vectors:

$$\frac{1}{7}(2\hat{i}+3\hat{j}+6\hat{k}), \frac{1}{7}(3\hat{i}-6\hat{j}+2\hat{k})$$

and $\frac{1}{7}(6\hat{i}+2\hat{j}-3\hat{k})$

are mutually perpendicular unit vectors.

4. Let $\vec{a} = 4\hat{i} + 5\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j} - \hat{k}$, find a vector $\vec{d}$ which is perpendicular to both $\vec{a}$ and $\vec{b}$, and is such that $\vec{d} \cdot \vec{c} = 21$.
5. The dot product of a vector with the vectors $(\hat{i} + \hat{j} - 3\hat{k})$, $(\hat{i} + 3\hat{j} - 2\hat{k})$ and $(2\hat{i} + \hat{j} + 4\hat{k})$ are 0, 5, 8 respectively. Find the vector.
6. Find the angle between the vectors $\vec{a}$ and $\vec{b}$, when:
- (i) $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} - \hat{k}$
- (ii) $\vec{a} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + 4\hat{k}$
7. If $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 4\hat{k}$ then:
- (i) Calculate the angle between $(2\vec{a} + \vec{b})$ and $(\vec{a} + 2\vec{b})$
- (ii) Show that $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ are orthogonal.
8. Find the angle between $\vec{a}$ and $\vec{b}$ when:
- (i) $|\vec{a}| = 2, |\vec{b}| = 1$ and $\vec{a} \cdot \vec{b} = \sqrt{3}$
- (ii) $|\vec{a}| = \sqrt{3}, |\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = \sqrt{6}$
9. Let $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$.
Find the projection of (i) $\vec{a}$ on $\vec{b}$ and (ii) $\vec{b}$ on $\vec{a}$
10. Find projection of $(8\hat{i} + \hat{j})$ in the direction of $(\hat{i} + 2\hat{j} - 2\hat{k})$
11. If $\vec{a}$ is a unit of vector such that $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 8$ find $|\vec{x}|$.
12. If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a}| = 2, |\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 4$, find $|\vec{a} - \vec{b}|$
13. For any two vectors $\vec{a}$ and $\vec{b}$, prove that $(\vec{a} \cdot \vec{b})^2 \leq |\vec{a}|^2 |\vec{b}|^2$
14. If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 3, |\vec{b}| = 5$ and $|\vec{c}| = 7$, find the angle between $\vec{a}$ and $\vec{b}$
15. If $A(2, 3, 4)$, $B(5, 4, -1)$, $C(3, 6, 2)$ and $D(1, 2, 0)$ be four points, show that $\vec{AB}$ is perpendicular to CD .

16. If $\vec{AB} = 3\hat{i} - \hat{j} + 2\hat{k}$ and the coordinates of A are $(0, -2, -1)$, find the coordinates of B .
17. Find the work done by the force $F = 2\hat{i} + \hat{j} + \hat{k}$ in moving a particle from the point A with position vector $(2\hat{i} + 2\hat{j} + 2\hat{k})$ to point B with position vector $(3\hat{i} + 4\hat{j} + 5\hat{k})$.

[B.C.A. (Kanpur) 2005]



Answers

1.	(i) 9	(ii) 8	(iii) -7
2.	(i) $\lambda = 3$	(ii) $\lambda = -2$	
4.	$\vec{d} = 7(\hat{i} - \hat{j} - \hat{k})$		
5.	$\hat{i} + 2\hat{j} + \hat{k}$		
6.	(i) $\cos^{-1}\left(\frac{5}{7}\right)$	(ii) $\cos^{-1}\left(\sqrt{\frac{3}{7}}\right)$	
7.	(i) $\cos^{-1}\left(\frac{31}{50}\right)$		
8.	30°	45°	
9.	(i) $\frac{5\sqrt{6}}{3}$	(ii) $\frac{10\sqrt{17}}{17}$	
10.	$\frac{10}{3}$		
11.	$ \vec{x} = 3$		
12.	$\sqrt{5}$		
14.	60°		
16.	$B(3, -3, 1)$		
17.	7 units		

17.7 Vector (or Cross) Product of Vectors

Let $\vec{a}$ and $\vec{b}$ be two non-zero, non-parallel vectors and let θ be the angle between them such that $0 < \theta < \pi$, then, the vector product of $\vec{a}$ and $\vec{b}$ is defined as $\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$

Where $\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

Note: 1. If $\vec{a}$ and $\vec{b}$ are parallel or collinear i.e., when $\theta=0$ or $\theta=\pi$ then we define, $\vec{a} \times \vec{b} = 0$,

in particular, $\vec{a} \times \vec{b} = 0$.

2. If $\vec{a} = 0$ or $\vec{b} = 0$, we define $\vec{a} \times \vec{b} = 0$.

3. For any vector $\vec{a}$, we define $\vec{a} \times \vec{a} = |a| |a| \sin \theta \hat{n} = 0$.

(i) **Angle between two vectors:** Let θ be the angle between $\vec{a}$ and $\vec{b}$ then,

$$\vec{a} \times \vec{b} = (ab \sin \theta) \hat{n}, \text{ where } |\vec{a}| = a, |\vec{b}| = b$$

$$\Rightarrow |\vec{a} \times \vec{b}| = (ab \sin \theta) |\hat{n}|$$

$$\Rightarrow |\vec{a} \times \vec{b}| = (ab \sin \theta) |\hat{n}|$$

$$\Rightarrow \sin \theta = \frac{|\vec{a} \times \vec{b}|}{ab} = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|} \right)$$

(ii) **Unit vector perpendicular to two given vectors:** We have, $(\vec{a} \times \vec{b})$ is a vector perpendicular to both $\vec{a}$ and $\vec{b}$. So, a unit vector $\hat{n}$, perpendicular to both $\vec{a}$ and $\vec{b}$ is given by $\hat{n} = \frac{(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$.

Note: There are two vectors, each of magnitude λ and perpendicular to each of the vector $\vec{a}$ and $\vec{b}$, namely.

$$\frac{\pm \lambda (\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$$

17.8 Properties of Vector Product

Theorems

Theorem 1: Vector product is not commutative.

Proof: Let $\vec{a}$ and $\vec{b}$ be two non-zero, non-parallel vectors, and let θ be the angle between them, where $0 < \theta < \pi$.

Let $\hat{n}$ be a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ then, $\vec{a}, \vec{b}, \hat{n}$ form right-handed system.

$$\therefore \vec{a} \times \vec{b} = (|\vec{a}| |\vec{b}| \sin \theta) \hat{n}$$

$$\Rightarrow \vec{a} \times \vec{b} = (ab \sin \theta) \hat{n}$$

Again $\vec{b}, \vec{a}$ and $-\hat{n}$ form a right handed system

$$\therefore \vec{b} \times \vec{a} = (|\vec{b}| |\vec{a}| \sin \theta) (-\hat{n})$$

$$\Rightarrow \vec{a} \times \vec{b} = -(ba \sin \theta) \hat{n} = -(ab \sin \theta) \hat{n}$$

$$\therefore \vec{b} \times \vec{a} = -\vec{a} \times \vec{b}.$$

Note: (1) For any vectors $\vec{a}$ and $\vec{b}$

$$(-\vec{b}) \times \vec{a} = -(\vec{b} \times \vec{a}) = \vec{b} \times (-\vec{a})$$

$$(2) \text{ For any scalar } m, (m\vec{a} \times \vec{b}) = m(\vec{a} \times \vec{b}) = \vec{a} \times (m\vec{b})$$

Theorem 2: (Distributive law) for any vectors $\vec{a}, \vec{b}, \vec{c}$

$$\vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})$$

Proof: Let $\vec{P} = [\vec{a} \times (\vec{b} + \vec{c}) - (\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c})]$ and let $\vec{r}$ be any arbitrary vector.

$$\begin{aligned} \text{Then, } \vec{r} \cdot \vec{P} &= \vec{r} \cdot [\vec{a} \times (\vec{b} + \vec{c}) - (\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c})] \\ &= \vec{r} \cdot [\vec{a} \times (\vec{b} + \vec{c})] - \vec{r} \cdot (\vec{a} \times \vec{b}) - \vec{r} \cdot (\vec{a} \times \vec{c}) \\ &= (\vec{r} \times \vec{a}) \cdot (\vec{b} + \vec{c}) - (\vec{r} \times \vec{a}) \cdot \vec{b} - (\vec{r} \times \vec{a}) \cdot \vec{c} \\ &= (\vec{r} \times \vec{a}) \cdot \vec{b} + (\vec{r} \times \vec{a}) \cdot \vec{c} - (\vec{r} \times \vec{a}) \cdot \vec{b} - (\vec{r} \times \vec{a}) \cdot \vec{c} \\ &= 0. \end{aligned}$$

$$\text{Now, } \vec{r} \cdot \vec{P} = 0 \Rightarrow \vec{r} = \vec{0} \text{ or } \vec{P} = \vec{0} \text{ or } (\vec{r} \perp \vec{P}).$$

Since $\vec{r}$ is an arbitrary vector, we may choose it in such a way that $\vec{r} \neq 0$ and $\vec{r}$ is not perpendicular to $\vec{P}$. Then, $\vec{P} = 0$.

$$\Rightarrow \vec{a} \times (\vec{b} + \vec{c}) - (\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c}) = 0$$

$$\Rightarrow \vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})$$

$$\text{Hence, } \vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c}).$$

Theorem 3: For any three vectors $\vec{a}, \vec{b}, \vec{c}$

$$\vec{a} \times (\vec{b} - \vec{c}) = (\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c})$$

Proof: We have

$$\begin{aligned} \vec{a} \times (\vec{b} - \vec{c}) &= \vec{a} \times [\vec{b} + (-\vec{c})] \\ &= (\vec{a} \times \vec{b}) + \vec{a} \times (-\vec{c}) && \text{[by the distributive law]} \\ &= (\vec{a} \times \vec{b}) + [-(\vec{a} \times \vec{c})] \\ &= (\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c}) && [\vec{a} \times (-\vec{c}) = -(\vec{a} \times \vec{c})] \end{aligned}$$

Hence, $\vec{a} \times (\vec{b} - \vec{c}) = \vec{a} \times \vec{b} - \vec{a} \times \vec{c}$

Theorem 4: Two non-zero vectors $\vec{a}$ and $\vec{b}$ are parallel or collinear iff $(\vec{a} \times \vec{b}) = 0$.

Proof: Let $\vec{a} \neq 0$ and $\vec{b} \neq 0$, θ be the angle between them.

Let us suppose that $\vec{a}$ and $\vec{b}$ are parallel or collinear then, $\theta = 0$ or $\theta = \pi$

$$\therefore \sin \theta = 0$$

$$\therefore \vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n} = 0$$

Thus, whenever two non-zero vectors $\vec{a}$ and $\vec{b}$ are parallel or collinear, then

$$\vec{a} \times \vec{b} = 0$$

Conversely, let $\vec{a}$ and $\vec{b}$ be non-zero vectors such that

$$\vec{a} \times \vec{b} = 0$$

$$\text{Then, } \vec{a} \times \vec{b} = 0 \Rightarrow |\vec{a} \times \vec{b}| = 0$$

$$\Rightarrow |\vec{a}| |\vec{b}| \sin \theta = 0$$

$$\Rightarrow \sin \theta = 0 \quad [as \ \vec{a} \neq 0, \ \vec{b} \neq 0]$$

$$\Rightarrow \theta = 0 \text{ or } \pi$$

$$\Rightarrow \vec{a} \text{ and } \vec{b} \text{ are parallel or collinear.}$$

Thus, $(\vec{a} \neq 0, \vec{b} \neq 0 \text{ and } \vec{a} \times \vec{b} = 0) \Rightarrow (\vec{a} \text{ and } \vec{b} \text{ are parallel or collinear}).$

Theorem 5: For any two vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b} = 0 \Rightarrow \vec{a} = 0$ or $\vec{b} = 0$ or $\vec{a}$ and $\vec{b}$ are parallel or collinear.

Proof: Let θ be the angle between $\vec{a}$ and $\vec{b}$, then

$$\begin{aligned}\vec{a} \times \vec{b} = 0 &\Rightarrow (ab \sin \theta) \hat{n} = 0 \\ &\Rightarrow ab \sin \theta = 0 \\ &\Rightarrow a = 0 \text{ or } b = 0 \text{ or } \sin \theta = 0 \\ &\Rightarrow a = 0 \text{ or } b = 0 \text{ or } \theta = 0 \text{ or } \pi \\ &\Rightarrow a = 0 \text{ or } b = 0 \text{ or } \vec{a} \text{ and } \vec{b} \text{ are parallel or collinear.}\end{aligned}$$

Theorem 6: Let $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$ then it does not imply that $\vec{b} = \vec{c}$.

Proof:

$$\begin{aligned}\vec{a} \times \vec{b} &= \vec{a} \times \vec{c} \\ \Rightarrow (\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c}) &= 0 \\ \Rightarrow \vec{a} \times (\vec{b} - \vec{c}) &= 0 \\ \Rightarrow \vec{a} = 0 \text{ or } (\vec{b} - \vec{c}) = 0 \text{ or } \vec{a} \parallel (\vec{b} - \vec{c}) \\ \Rightarrow \vec{a} = 0 \text{ or } \vec{b} = \vec{c} \text{ or } \vec{a} \parallel (\vec{b} - \vec{c})\end{aligned}$$

$\therefore (\vec{a} \times \vec{b}) = \vec{a} \times \vec{c}$ does not always means that $\vec{b} = \vec{c}$.

17.8.1 Geometrical Interpretation of Vector Product

Theorem 1: The area of a parallelogram whose adjacent sides are represented by $\vec{a}$ and $\vec{b}$ is $|\vec{a} \times \vec{b}|$.

Proof: Let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ and $\angle AOB = \theta$.

Complete the parallelogram $OACB$. Join AB , Draw $BD \perp OA$.

In $\triangle OAB$, we have

$$\text{base} = OA = |\vec{a}|$$

$$\text{altitude} = BD = (OB) \sin \theta = |\vec{b}| \sin \theta$$

$$\therefore \text{area of } \triangle OAB = \frac{1}{2} \times \text{base} \times \text{altitude}$$

$$= \frac{1}{2} \cdot |\vec{a}| \cdot |\vec{b}| \sin \theta = \frac{1}{2} ab \sin \theta = \frac{1}{2} |\vec{a} \times \vec{b}|$$

$$\therefore \text{area of parallelogram } OACB = 2 \times (\text{area of } \triangle OAB)$$

$$= |\vec{a} \times \vec{b}|.$$

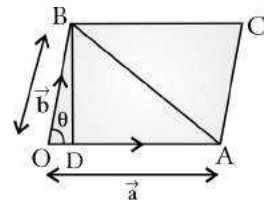


Fig. 17.25

Hence, the area of a parallelogram whose adjacent sides are represented by $\vec{a}$ and $\vec{b}$ is given by $|\vec{a} \times \vec{b}|$.

Note: $(\vec{a} \times \vec{b})$ is called the vector area of the parallelogram.

Theorem 2: The area of ΔOAB , where $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$ is $\frac{1}{2} |\vec{a} \times \vec{b}|$.

Proof: In ΔOAB , $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ and let $\angle AOB = \theta$

Draw $BL \perp OA$

In ΔOAB , we have

base = $OA = |\vec{a}|$, and altitude = $BL = (OB) \sin \theta = |\vec{b}| \sin \theta$

$$\therefore \text{area of } \Delta OAB = \frac{1}{2} \times \text{base} \times \text{altitude} = \frac{1}{2} |\vec{a}| \cdot |\vec{b}| \sin \theta = \frac{1}{2} |\vec{a} \times \vec{b}|$$

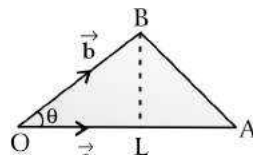


Fig. 17.26

Note: 1. $\frac{1}{2} (\vec{a} \times \vec{b})$ is called the vector area of ΔOAB

2. Area of ΔAOB is given by

$$\frac{1}{2} |\vec{OA} \times \vec{OB}| = \frac{1}{2} |\vec{BO} \times \vec{BA}| = \frac{1}{2} |\vec{AO} \times \vec{AB}|$$

Theorem 3: The vector area of quadrilateral $ABCD$ with diagonal AC and BD is $\frac{1}{2} (\vec{AC} \times \vec{BD})$.

Proof: Vector area of a quadrilateral $ABCD$

$$= (\text{vector area of } \Delta ABC) + (\text{vector area of } \Delta ACD)$$

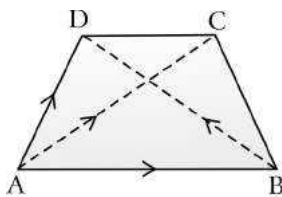


Fig. 17.27

$$= \frac{1}{2} (\vec{AB} \times \vec{AC}) + \frac{1}{2} (\vec{AC} \times \vec{AD})$$

$$= -\frac{1}{2} (\vec{AC} \times \vec{AB}) + \frac{1}{2} (\vec{AC} \times \vec{AD}) = \frac{1}{2} [\vec{AC} \times (\vec{AD} - \vec{AB})]$$

$$= \frac{1}{2} (\vec{AC} \times \vec{BD}) \quad [\because \vec{AB} + \vec{BD} = \vec{AD}]$$

17.8.2 Vector Product of an Orthonormal Vector Triad

For mutually perpendicular unit vector $\hat{i}, \hat{j}, \hat{k}$, we have

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$$

$$\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$$

and

$$\hat{j} \times \hat{i} = -\hat{k}, \hat{k} \times \hat{j} = -\hat{i}, \hat{i} \times \hat{k} = -\hat{j}.$$

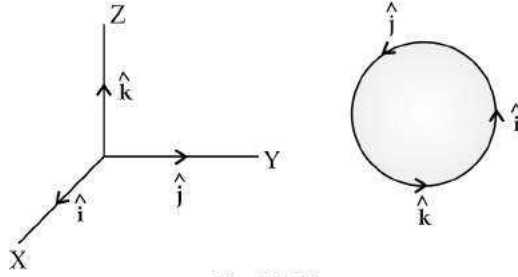


Fig. 17.28

17.8.3 Vector Product in Terms of Components

Theorem 1: Let $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ and $\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$, then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Proof: We have $\vec{a} \times \vec{b} = (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \times (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k})$

$$= a_1 b_1 (\hat{i} \times \hat{i}) + a_1 b_2 (\hat{i} \times \hat{j}) + a_1 b_3 (\hat{i} \times \hat{k}) + a_2 b_1 (\hat{j} \times \hat{i}) + a_2 b_2 (\hat{j} \times \hat{j})$$

$$+ a_2 b_3 (\hat{j} \times \hat{k}) + a_3 b_1 (\hat{k} \times \hat{i}) + a_3 b_2 (\hat{k} \times \hat{j}) + a_3 b_3 (\hat{k} \times \hat{k})$$

$$= (a_2 b_3 - a_3 b_2) \hat{i} + (a_3 b_1 - a_1 b_3) \hat{j} + (a_1 b_2 - a_2 b_1) \hat{k}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Theorem 2: If θ is the angle between the vectors

$$\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} \text{ and } \vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k} \text{ then}$$

$$\sin \theta = \frac{\sqrt{(a_1 b_2 - a_2 b_1)^2 + (a_2 b_3 - a_3 b_2)^2 + (a_3 b_1 - a_1 b_3)^2}}{(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)}$$

Proof: We have

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= (a_2 b_3 - a_3 b_2) \hat{i} + (a_3 b_1 - a_1 b_3) \hat{j} + (a_1 b_2 - a_2 b_1) \hat{k}$$

$$\begin{aligned}
\Rightarrow \quad |\vec{a} \times \vec{b}|^2 &= (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 \\
\Rightarrow \quad (ab \sin \theta)^2 &= (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 \\
\Rightarrow \quad \sin^2 \theta &= \frac{(a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2}{|\vec{a}|^2 |\vec{b}|^2} \\
&= \frac{(a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2}{(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)} \\
\Rightarrow \quad \sin \theta &= \sqrt{\frac{(a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2}{(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)}}
\end{aligned}$$

Theorem 3: (Lagrange's identity) For any vector $\vec{a}$ and $\vec{b}$ prove that

$$|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} \end{vmatrix}$$

[B.C.A (Meerut) 2002, 2004]

Proof: Let θ be the angle between $\vec{a}$ and $\vec{b}$, then

$$\begin{aligned}
(\vec{a} \times \vec{b})^2 &= (\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) \\
&= (|\vec{a}| |\vec{b}| \sin \theta) \hat{n} \cdot (|\vec{a}| |\vec{b}| \sin \theta) \hat{n} \\
&= |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta (\hat{n} \cdot \hat{n}) & [\hat{n} \cdot \hat{n} = 1] \\
&= |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta \\
&= |\vec{a}|^2 |\vec{b}|^2 (1 - \cos^2 \theta) \\
&= |\vec{a}|^2 |\vec{b}|^2 - |\vec{a}|^2 |\vec{b}|^2 \cos^2 \theta \\
&= |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 & \dots(1) \\
&= (\vec{a} \cdot \vec{a})(\vec{b} \cdot \vec{b}) - (\vec{a} \cdot \vec{b})^2 \\
&= \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} \end{vmatrix} & \dots(2)
\end{aligned}$$

Hence, $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} \end{vmatrix}$ [From (1) and (2)]

Theorem 4: For any three vectors $\vec{a}, \vec{b}, \vec{c}$

$$\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b}) = 0$$

Proof: We have

$$\begin{aligned} & \vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b}) \\ &= \vec{a} \times \vec{b} + \vec{a} \times \vec{c} + \vec{b} \times \vec{c} + \vec{b} \times \vec{a} + \vec{c} \times \vec{a} + \vec{c} \times \vec{b} \quad [\text{by distributive law}] \\ &= \vec{a} \times \vec{b} + (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) - (\vec{a} \times \vec{b}) - (\vec{b} \times \vec{c}) - (\vec{c} \times \vec{a}) = 0 \end{aligned}$$

Theorem 5: If $\vec{a}, \vec{b}, \vec{c}$ are vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$, then prove that

$$(\vec{a} \times \vec{b}) = (\vec{b} \times \vec{c}) = (\vec{c} \times \vec{a})$$

Proof: We have $\vec{a} + \vec{b} + \vec{c} = 0 \Rightarrow \vec{a} + \vec{b} = -\vec{c}$

$$\Rightarrow (\vec{a} + \vec{b}) \times \vec{b} = (-\vec{c}) \times (\vec{b})$$

$$\Rightarrow (\vec{a} \times \vec{b}) + \vec{b} \times \vec{b} = (-\vec{c}) \times (\vec{b})$$

$$\Rightarrow \vec{a} \times \vec{b} + 0 = \vec{b} \times \vec{c}$$

$$\Rightarrow \vec{a} \times \vec{b} = \vec{b} \times \vec{c} \quad \dots(1)$$

Also, $\vec{a} + \vec{b} + \vec{c} = 0 \Rightarrow \vec{c} + \vec{b} = -\vec{a}$

$$\Rightarrow (\vec{b} + \vec{c}) \times \vec{c} = (-\vec{a}) \times \vec{c}$$

$$\Rightarrow \vec{b} \times \vec{c} + \vec{c} \times \vec{c} = -(\vec{a} \times \vec{c}) \quad [(\vec{c} \times \vec{c} = 0)]$$

$$\Rightarrow \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \quad \dots(2)$$

Hence from (1) and (2), we get

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

Theorem 6: For any two vectors $\vec{a}$ and $\vec{b}$

$$(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = 2(\vec{a} \times \vec{b})$$

Proof: We have $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})$

$$= \vec{a} \times \vec{a} + \vec{a} \times \vec{b} - \vec{b} \times \vec{a} - \vec{b} \times \vec{b} \quad [\text{by distributive law}]$$

$$= (\vec{a} \times \vec{b} - \vec{b} \times \vec{a})$$

$$= (\vec{a} \times \vec{b} + \vec{a} \times \vec{b}) \quad (\vec{a} \times \vec{b} = -\vec{b} \times \vec{a})$$

$$= 2(\vec{a} \times \vec{b})$$

$$= 2 \text{ (vector area of parallelogram with adjacent sides } \vec{a} \text{ and } \vec{b}).$$

Theorem 7: The points A, B, C with position vector $\vec{a}, \vec{b}$ and $\vec{c}$ respectively are collinear if and only if $(\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) + (\vec{a} \times \vec{b}) = 0$.

Proof: We have

$$\vec{AB} = (P.V. \text{ of } B) - (P.V. \text{ of } A) = (\vec{b} - \vec{a})$$

$$\vec{BC} = (P.V. \text{ of } C) - (P.V. \text{ of } B) = (\vec{c} - \vec{b})$$

Now, A, B, C are collinear

$$\Leftrightarrow \vec{AB} \text{ and } \vec{BC} \text{ are parallel}$$

$$\Leftrightarrow (\vec{b} - \vec{a}) \times (\vec{c} - \vec{b}) = 0$$

$$\Leftrightarrow (\vec{b} - \vec{a}) \times \vec{c} - (\vec{b} - \vec{a}) \times \vec{b} = 0 \quad [\text{by distributive law}]$$

$$\Leftrightarrow (\vec{b} \times \vec{c}) - (\vec{a} \times \vec{c}) - (\vec{b} \times \vec{b}) + (\vec{a} \times \vec{b}) = 0 \quad [\text{by distributive law}]$$

$$\Leftrightarrow (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) + (\vec{a} \times \vec{b}) = 0.$$

Theorem 8: When $\vec{a} \cdot \vec{b} = 0$ and $\vec{a} \times \vec{b} = 0$, then $\vec{a} = 0$ or $\vec{b} = 0$

Proof: As $\vec{a} \cdot \vec{b} = 0$ and $\vec{a} \times \vec{b} = 0$,

$$\Rightarrow \vec{a} = 0 \text{ or } \vec{b} = 0 \text{ or } \vec{a} \perp \vec{b} \text{ and } \vec{a} = 0 \text{ or } \vec{b} = 0 \text{ or } \vec{a} \parallel \vec{b}$$

$$\Rightarrow \vec{a} = 0 \text{ or } \vec{b} = 0 \quad [\because \vec{a} \perp \vec{b} \text{ and } \vec{a} \parallel \vec{b} \text{ can never so hold together}].$$

Theorem 9: If $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$, $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$ and $\vec{a} \neq 0$ then $\vec{b} = \vec{c}$

[B.C.A. (Rohilkhand) 2004]

Proof: We have $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$ and $\vec{a} \neq 0$

$$\Rightarrow \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} = 0$$

$$\Rightarrow \vec{a} \cdot (\vec{b} - \vec{c}) = 0 \text{ and } \vec{a} \neq 0$$

$$\Rightarrow (\vec{b} - \vec{c}) = 0 \text{ or } \vec{a} \perp (\vec{b} - \vec{c})$$

$$\Rightarrow \vec{b} = \vec{c} \text{ or } \vec{a} \perp (\vec{b} - \vec{c}) \quad \dots(1)$$

Again $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$ and $\vec{a} \neq 0$

$$\Rightarrow (\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c}) = 0 \text{ and } \vec{a} \neq 0$$

$$\Rightarrow \vec{a} \times (\vec{b} - \vec{c}) = 0 \text{ and } \vec{a} \neq 0$$

$$\Rightarrow \quad \vec{b} - \vec{c} = 0 \quad \text{or} \quad \vec{a} \parallel (\vec{b} - \vec{c})$$

$$\Rightarrow \quad \vec{b} = \vec{c} \quad \text{or} \quad \vec{a} \parallel (\vec{b} - \vec{c}) \quad \dots(2)$$

Hence from (1) and (2) we conclude that either $\vec{b} = \vec{c}$ or $\vec{a} \perp (\vec{b} - \vec{c})$ and $\vec{a} \parallel (\vec{b} - \vec{c})$

$\therefore \vec{b} = \vec{c}$ [since $\vec{a}$ cannot be both parallel and perpendicular to $(\vec{b} - \vec{c})$].

Theorem 10: If $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$, show that $(\vec{a} - \vec{d})$ is parallel to $(\vec{b} - \vec{c})$, it is given that $a \neq d$ and $b \neq c$. [B.C.A (Delhi) 2003]

Proof: $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$

$$\Rightarrow \quad \vec{a} \times \vec{d} - \vec{a} \times \vec{c} = \vec{c} \times \vec{d} - \vec{b} \times \vec{d}$$

$$\Rightarrow \quad \vec{a} \times (\vec{b} - \vec{c}) + (\vec{b} - \vec{c}) \times \vec{d} = 0$$

$$\Rightarrow \quad \vec{a} \times (\vec{b} - \vec{c}) - \vec{d} \times (\vec{b} - \vec{c}) = 0$$

$$\Rightarrow \quad (\vec{a} - \vec{d}) \times (\vec{b} - \vec{c}) = 0$$

$$\Rightarrow \quad (\vec{a} - \vec{d}) \parallel (\vec{b} - \vec{c})$$

Hence, $(\vec{a} - \vec{d})$ is parallel to $(\vec{b} - \vec{c})$.

Theorem 11: If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of A, B, C of a triangle ABC then the area of triangle ABC is

$$\frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$$

Also, the condition for points $\vec{a}, \vec{b}, \vec{c}$ to be collinear is

$$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = 0$$

Proof: Let O be the origin, then

$$\vec{OA} = \vec{a}, \vec{OB} = \vec{b} \text{ and } \vec{OC} = \vec{c}$$

$$\therefore \quad \vec{AB} = (\vec{b} - \vec{a}) \text{ and } \vec{AC} = (\vec{c} - \vec{a})$$

So, the vector area of the triangle ABC

$$\begin{aligned} &= \frac{1}{2} \vec{AB} \times \vec{AC} = \frac{1}{2} (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) \\ &= \frac{1}{2} [\vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{a} \times \vec{c} + \vec{a} \times \vec{a}] \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} [\vec{b} \times \vec{c} + \vec{a} \times \vec{b} + \vec{c} \times \vec{a}] \quad [\vec{a} \times \vec{a} = 0] \\
 &= \frac{1}{2} [\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}]
 \end{aligned}$$

$$\therefore \text{area of the triangle } ABC = \frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$$

Further, let A, B, C be collinear.

Then, the vector area of triangle ABC is $\vec{0}$.

Hence, $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = 0$.

Theorem 12: Let A, B, C be three non-collinear points with position vectors $\vec{a}, \vec{b}, \vec{c}$ respectively then the length of the perpendicular from C on AB is

$$\frac{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}{|\vec{b} - \vec{a}|}$$

Proof: Let P be the length of the perpendicular from C to AB .

$$\text{Then, area of the triangle } ABC = \frac{1}{2} P(AB) = \frac{1}{2} p |\vec{AB}| = \frac{1}{2} p |\vec{b} - \vec{a}|$$

$$\text{Also, area of the triangle } ABC = \frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$$

$$\therefore \frac{1}{2} p |\vec{b} - \vec{a}| = \frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$$

$$\text{or } p = \frac{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}{|\vec{b} - \vec{a}|}$$



Illustrative Examples

(Cross Product)

Example 1: Find the unit vector perpendicular to both the vectors $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = (2\hat{i} + 2\hat{j} - \hat{k})$. [B.C.A (Meerut) 2008, B.C.A (Kanpur) 2002]

Solution: We know that a unit vector is perpendicular to both $\vec{a}$ and $\vec{b}$

$$= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

Now,

$$\begin{aligned}\vec{a} \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 3 \\ 2 & 2 & -1 \end{vmatrix} \\ &= (1-6)\hat{i} + \hat{j}(6+4) + (8+2)\hat{k} \\ &= -5\hat{i} + 10\hat{j} + 10\hat{k} \\ |\vec{a} \times \vec{b}| &= \sqrt{(-5)^2 + (10)^2 + (10)^2} = 15\end{aligned}$$

$\therefore$ a unit vector is perpendicular to both $\vec{a}$ and $\vec{b}$

$$= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{(-5\hat{i} + 10\hat{j} + 10\hat{k})}{15} = \frac{(-\hat{i} + 2\hat{j} + 2\hat{k})}{3}$$

Note: Another unit vector perpendicular to each of the given vectors $\vec{a}$ and $\vec{b}$

$$= -\frac{(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|} = \frac{1}{3}(\hat{i} - 2\hat{j} - 2\hat{k}).$$

Example 2: Find the unit vector perpendicular to the plane of vectors $\vec{a}$ and $\vec{b}$ where

$$\vec{a} = \hat{i} - \hat{j} + 2\hat{k} \text{ and } \vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$$

[B.C.A.(Meerut) 2009]

Solution: We know that a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

$$= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

Now

$$\begin{aligned}\vec{a} \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 2 & 3 & -1 \end{vmatrix} \\ &= \hat{i}(1-6) + \hat{j}(4+1) + \hat{k}(3+2) \\ &= -5\hat{i} + 5\hat{j} + 5\hat{k} \\ |\vec{a} \times \vec{b}| &= \sqrt{(-5)^2 + (5)^2 + (5)^2} = 5\sqrt{3}\end{aligned}$$

$\therefore$ a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

$$\begin{aligned}&= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{-5\hat{i} + 5\hat{j} + 5\hat{k}}{5\sqrt{3}} \\ &= \frac{-\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}\end{aligned}$$

Example 3: If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ then find $\vec{a} \times \vec{b}$ verify that $\vec{a}$ and $(\vec{a} \times \vec{b})$ are perpendicular to each other.

[B.C.A. (Indra Prastha) 2006, 2009]

Solution: We have
$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & 3 & -5 \end{vmatrix}$$

$$= \hat{i}(10 - 9) + \hat{j}(6 + 5) + \hat{k}(3 + 4)$$

Thus,
$$\vec{a} \times \vec{b} = (\hat{i} + 11\hat{j} + 7\hat{k})$$

Now
$$\vec{a} \cdot (\vec{a} \times \vec{b}) = (\hat{i} - 2\hat{j} + 3\hat{k}) \cdot (\hat{i} + 11\hat{j} + 7\hat{k})$$

$$= 1 - 22 + 21 = 0$$

$$\Rightarrow \vec{a} \perp (\vec{a} \times \vec{b})$$

Example 4: If $\vec{a} = 4\hat{i} + 3\hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} + 2\hat{k}$ find $|\vec{b} \times 2\vec{a}|$

Solution: We have $\vec{b} = (3\hat{i} + 2\hat{k})$ and $2\vec{a} = (8\hat{i} + 6\hat{j} + 4\hat{k})$

$$\therefore \vec{b} \times 2\vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & 2 \\ 8 & 6 & 4 \end{vmatrix} = \hat{i}(0 - 12) + \hat{j}(16 - 12) + \hat{k}(18 - 0)$$

$$= -12\hat{i} + 4\hat{j} + 18\hat{k}$$

$$\therefore |\vec{b} \times 2\vec{a}| = \sqrt{(-12)^2 + (4)^2 + (18)^2} = \sqrt{484} = 22$$

Hence,
$$|\vec{b} \times 2\vec{a}| = 22$$

Example 5: If $|\vec{a}| = 2$, $|\vec{b}| = 7$ and $(\vec{a} \times \vec{b}) = (3\hat{i} + 2\hat{j} + 6\hat{k})$, find the angle between $\vec{a}$ and $\vec{b}$.

Solution: Let θ be the angle between $\vec{a}$ and $\vec{b}$, then

$$\vec{a} \times \vec{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$$

$$\Rightarrow |\vec{a} \times \vec{b}| = \sqrt{3^2 + 2^2 + 6^2} = \sqrt{49} = 7$$

$$\Rightarrow |\vec{a}||\vec{b}|\sin\theta = 7$$

$$\Rightarrow \sin\theta = \frac{7}{|\vec{a}||\vec{b}|} = \frac{7}{2 \times 7} = \frac{1}{2}$$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{1}{2}\right) \Rightarrow \theta = \frac{\pi}{6}$$

Hence, the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$.

Example 6: Find the value of λ for which the vectors $\vec{a} = \hat{i} + \lambda\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} + 2\hat{j} + 9\hat{k}$ are parallel. [B.C.A. (Rohilkhand) 2007]

Solution: Since $\vec{a}$ and $\vec{b}$ are parallel then $\vec{a} \times \vec{b} = 0$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \lambda & 3 \\ 3 & 2 & 9 \end{vmatrix} = 0$$

$$\hat{i}(9\lambda - 6) + \hat{j}(9 - 9) + \hat{k}(2 - 3\lambda) = 0$$

$$\Rightarrow 9\lambda - 6 = 0, 2 - 3\lambda = 0$$

$$\Rightarrow \lambda = \frac{2}{3}.$$

Hence, the required value of λ is $\frac{2}{3}$.

Example 7: Find a vector of magnitude 15, which is perpendicular to both the vectors $(4\hat{i} - \hat{j} + 8\hat{k})$ and $(-\hat{j} + \hat{k})$.

Solution: Let $\vec{a} = (4\hat{i} - \hat{j} + 8\hat{k})$ and $\vec{b} = (-\hat{j} + \hat{k})$

A unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

$$\begin{aligned} &= \frac{(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|} \\ \vec{a} \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 8 \\ 0 & -1 & 1 \end{vmatrix} = 7\hat{i} - 4\hat{j} - 4\hat{k} \end{aligned}$$

$$|\vec{a} \times \vec{b}| = \sqrt{49 + 16 + 16} = \sqrt{81} = 9$$

$\therefore$ a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

$$= \frac{7\hat{i} - 4\hat{j} - 4\hat{k}}{9}$$

$\therefore$ a vector of magnitudes 15 and perpendicular to both $\vec{a}$ and $\vec{b}$

$$= \frac{15(7\hat{i} - 4\hat{j} - 4\hat{k})}{9} = \frac{5}{3}(7\hat{i} - 4\hat{j} - 4\hat{k})$$

Example 8: Let $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = 3\hat{j} - \hat{k}$ and $\vec{c} = 7\hat{i} - \hat{k}$. Find a vector $\vec{d}$ such that it is perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{c} \cdot \vec{d} = 1$.

Solution: We have $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 0 & 3 & -1 \end{vmatrix} = (\hat{i} + \hat{j} + 3\hat{k})$

$$|\vec{a} \times \vec{b}| = \sqrt{(1)^2 + (1)^2 + (3)^2} = \sqrt{11}.$$

$\therefore$ a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

$$= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{(\hat{i} + \hat{j} + 3\hat{k})}{\sqrt{11}}$$

The required vector

$$\vec{d} = \frac{x(\hat{i} + \hat{j} + 3\hat{k})}{\sqrt{11}}$$

Then

$$\vec{c} \cdot \vec{d} = 1 \Rightarrow (7\hat{i} - \hat{k}) \cdot \frac{x(\hat{i} + \hat{j} + 3\hat{k})}{\sqrt{11}} = 1$$

$\Rightarrow$

$$(7 + 0 - 3)x = \sqrt{11}$$

$\Rightarrow$

$$x = \frac{\sqrt{11}}{4}$$

$\therefore$ the required vector is

$$\begin{aligned} \vec{d} &= \frac{\sqrt{11}}{4} \cdot \frac{(\hat{i} + \hat{j} + 3\hat{k})}{\sqrt{11}} \\ &= \frac{1}{4}(\hat{i} + \hat{j} + 3\hat{k}). \end{aligned}$$

Example 9: In $\triangle OAB$, $\vec{OA} = 3\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{OB} = \hat{i} + 3\hat{j} + \hat{k}$ find the area of the triangle.

Solution: Area of $\triangle OAB = \left| \frac{1}{2} (\vec{OA} \times \vec{OB}) \right|$

Now,

$$\begin{aligned} \vec{OA} \times \vec{OB} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & -1 \\ 1 & 3 & 1 \end{vmatrix} \\ &= (2+3)\hat{i} - (3+1)\hat{j} + (9-2)\hat{k} \\ &= 5\hat{i} - 4\hat{j} + 7\hat{k} \end{aligned}$$

$\therefore$

$$\begin{aligned} \text{area of } \triangle OAB &= \left| \frac{1}{2} (\vec{OA} \times \vec{OB}) \right| = \left| \frac{5}{2}\hat{i} - 2\hat{j} + \frac{7}{2}\hat{k} \right| \\ &= \sqrt{\frac{25}{4} + 4 + \frac{49}{4}} = \frac{\sqrt{90}}{2} \\ &= \frac{3}{2}\sqrt{10} \text{ units} \end{aligned}$$

Example 10: Find the area of triangle whose vertices are $A(1, 1, 1)$, $B(1, 2, 3)$ and $C(2, 3, 1)$.

Solution: We have

$$\vec{A} = (\hat{i} + \hat{j} + \hat{k}), \vec{B} = (\hat{i} + 2\hat{j} + 3\hat{k}) \text{ and}$$

$$\vec{C} = (2\hat{i} + 3\hat{j} + \hat{k})$$

$$\begin{aligned}\vec{AB} &= P.V. \text{ of } B - P.V. \text{ of } A = (\hat{i} + 2\hat{j} + 3\hat{k}) - (\hat{i} + \hat{j} + \hat{k}) \\ &= 0\hat{i} + \hat{j} + 2\hat{k}\end{aligned}$$

$$\begin{aligned}\vec{AC} &= P.V. \text{ of } C - P.V. \text{ of } A = (2\hat{i} + 3\hat{j} + \hat{k}) - (\hat{i} + \hat{j} + \hat{k}) \\ &= \hat{i} + 2\hat{j} + 0\hat{k}\end{aligned}$$

$$\text{Area of triangle } ABC = \left| \frac{1}{2} (\vec{AB} \times \vec{AC}) \right|$$

$$= \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{vmatrix}$$

$$= \frac{1}{2} \{ (0-4) \} \hat{i} + \hat{j} \{ 2-0 \} + \hat{k} \{ 0-1 \}$$

$$= \left| -2\hat{i} + \hat{j} - \frac{1}{2}\hat{k} \right|$$

$$= \frac{\sqrt{21}}{2} \text{ Square units}$$

Example 11: Show that the three points $5\hat{i} + 6\hat{j} + 7\hat{k}$, $7\hat{i} - 8\hat{j} + 9\hat{k}$ and $3\hat{i} + 20\hat{j} + 5\hat{k}$ are collinear. [B.C.A. (Meerut) 2006]

Solution: Let

$$\vec{a} = 5\hat{i} + 6\hat{j} + 7\hat{k}$$

$$\vec{b} = 7\hat{i} - 8\hat{j} + 9\hat{k}$$

$$\vec{c} = 3\hat{i} + 20\hat{j} + 5\hat{k}$$

We know that the point A, B, C with position vectors $\vec{a}, \vec{b}, \vec{c}$ respectively are collinear if and only if $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = 0$.

$$\text{Now, } \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 6 & 7 \\ 7 & -8 & 9 \end{vmatrix} = 110\hat{i} + 4\hat{j} - 82\hat{k} \quad \dots(1)$$

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & -8 & 9 \\ 3 & 20 & 5 \end{vmatrix} = -220\hat{i} - 8\hat{j} + 164\hat{k} \quad \dots(2)$$

$$\vec{c} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 20 & 5 \\ 5 & 6 & 7 \end{vmatrix} = 110\hat{i} + 4\hat{j} - 82\hat{k} \quad \dots(3)$$

$$(1) + (2) + (3) \Rightarrow \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = 0$$

Hence, given points are collinear.

Example 12: Using vector method, show that the points $A(2, -1, 3)$, $B(4, 3, 1)$ and $C(3, 1, 2)$ are collinear.

Solution: We know, the points A, B, C are collinear $\Leftrightarrow$ area of a triangle is zero

Now,

$$\vec{A} = 2\hat{i} - \hat{j} + 3\hat{k}, \quad \vec{B} = 4\hat{i} + 3\hat{j} + \hat{k}, \quad \vec{C} = 3\hat{i} + \hat{j} + 2\hat{k}$$

$$\vec{AB} = P.V. \text{ of } B - P.V. \text{ of } A = (4\hat{i} + 3\hat{j} + \hat{k}) - (2\hat{i} - \hat{j} + 3\hat{k})$$

$$= 2\hat{i} + 4\hat{j} - 2\hat{k}$$

$$\vec{AC} = P.V. \text{ of } C - P.V. \text{ of } A = (3\hat{i} + \hat{j} + 2\hat{k}) - (2\hat{i} - \hat{j} + 3\hat{k})$$

$$= (\hat{i} + 2\hat{j} - \hat{k})$$

$$\therefore \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 4 & -2 \\ 1 & 2 & -1 \end{vmatrix} = 2 \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 1 & 2 & -1 \end{vmatrix} = 2 \times 0 = \vec{0}$$

Thus, $\vec{AB}$ and $\vec{AC}$ are parallel vectors, having a common point, A .

Hence, the points A, B, C are collinear.

Example 13: Show that the three points $\hat{i} - 2\hat{j} + 3\hat{k}$, $2\hat{i} + 3\hat{j} - 4\hat{k}$ and $-7\hat{j} + 10\hat{k}$ are collinear. [B.C.A. (Meerut) 2008]

Solution: Let

$$\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$$

$$\vec{b} = 2\hat{i} + 3\hat{j} - 4\hat{k}$$

$$\vec{c} = 0\hat{i} - 7\hat{j} + 10\hat{k}$$

We know that the points A, B, C with position vectors, $\vec{a}, \vec{b}, \vec{c}$ respectively are collinear iff

$$(\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) = 0$$

Now,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & 3 & -4 \end{vmatrix} = \hat{i}(8-9) + \hat{j}(6+4) + \hat{k}(3+4)$$

$$= -\hat{i} + 10\hat{j} + 7\hat{k} \quad \dots(1)$$

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -4 \\ 0 & -7 & 10 \end{vmatrix} = 2\hat{i} - 20\hat{j} - 14\hat{k} \quad \dots(2)$$

$$\begin{aligned} \vec{c} \times \vec{a} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -7 & 10 \\ 1 & -2 & 3 \end{vmatrix} = \hat{i}(-21+20) + \hat{j}(10-0) + \hat{k}(0+7) \\ &= -\hat{i} + 10\hat{j} + 7\hat{k} \quad \dots(3) \end{aligned}$$

$$(1) + (2) + (3) \Rightarrow (\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) = 0.$$

Hence given points are collinear.

Example 14: Show that the points having position vectors

$$(\vec{a} - 2\vec{b} + 3\vec{c}), (-2\vec{a} + 3\vec{b} + 2\vec{c}), (-8\vec{a} + 13\vec{b})$$

are collinear, whatever be $\vec{a}, \vec{b}, \vec{c}$.

[B.C.A. (Meerut) 2003]

Solution: Let A, B, C be the given points whose position vectors are

$$\vec{A} = \vec{a} - 2\vec{b} + 3\vec{c}$$

$$\vec{B} = -2\vec{a} + 3\vec{b} + 2\vec{c}$$

$$\vec{C} = -8\vec{a} + 13\vec{b} + 0\vec{c}$$

$$\text{Then } \vec{AB} = (P.V. \text{ of } B) - (P.V. \text{ of } A) = (-2\vec{a} + 3\vec{b} + 2\vec{c}) - (\vec{a} - 2\vec{b} + 3\vec{c})$$

$$= (-3\vec{a} + 5\vec{b} - \vec{c})$$

$$\vec{AC} = (P.V. \text{ of } C) - (P.V. \text{ of } A)$$

$$= (-8\vec{a} + 13\vec{b}) - (\vec{a} - 2\vec{b} + 3\vec{c}) = -9\vec{a} + 15\vec{b} - 3\vec{c}$$

$$\therefore \vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ -3 & 5 & -1 \\ -9 & 15 & -3 \end{vmatrix}$$

$$= \vec{a}(-15+15) + \vec{b}(9-9) + \vec{c}(-45+45)$$

$$= 0 + 0 + 0 = 0$$

$\therefore \vec{AB}$ and $\vec{AC}$ are parallel vectors, having a common point A , hence the points A, B, C are collinear.

Example 15: Find the scalar components of a unit vector which is perpendicular to each of the vectors $\hat{i} + 2\hat{j} - \hat{k}$ and $(3\hat{i} - \hat{j} + 2\hat{k})$.

Solution: Let $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$

Then, the vector perpendicular to each of $\vec{a}$ and $\vec{b}$ are

$$\frac{\pm(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$$

Now,

$$(\vec{a} \times \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 3 & -1 & 2 \end{vmatrix} = 3\hat{i} - 5\hat{j} - 7\hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{9 + 25 + 49} = \sqrt{83}$$

$\therefore$ each of the vectors $\pm \frac{(3\hat{i} - 5\hat{j} - 7\hat{k})}{\sqrt{83}}$ is perpendicular to both $\vec{a}$ and $\vec{b}$. Therefore, the required components are $\frac{3\hat{i} - 5\hat{j} - 7\hat{k}}{\sqrt{83}}$ and $-\frac{3\hat{i} - 5\hat{j} - 7\hat{k}}{\sqrt{83}}$.

Example 16: Find the area of the parallelogram whose adjacent sides are represented by the vectors $(\hat{i} - 3\hat{j} + \hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$.

Solution: Let $ABCD$ be the parallelogram whose adjacent sides be given by the vectors.

$$\vec{AB} = (\hat{i} - 3\hat{j} + \hat{k}) \text{ and } \vec{AD} = (\hat{i} + \hat{j} + \hat{k})$$

$$\therefore \text{ area of parallelogram } ABCD = |\vec{AB} \times \vec{AD}|$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= |-4\hat{i} + 4\hat{k}|$$

$$= \sqrt{32} \text{ square units}$$

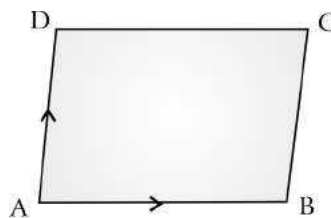


Fig. 17.29

Example 17: Find the area of the parallelogram whose diagonals are represented by the vectors

$$\vec{d}_1 = (4\hat{i} - \hat{j} - 3\hat{k}) \text{ and } \vec{d}_2 = (-2\hat{i} + \hat{j} - 2\hat{k})$$

Solution: Area of parallelogram $= \frac{1}{2} |\vec{d}_1 \times \vec{d}_2|$

Now,

$$\vec{d}_1 \times \vec{d}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & -3 \\ -2 & 1 & -2 \end{vmatrix} = 5\hat{i} + 14\hat{j} + 2\hat{k}$$

$$\begin{aligned}\text{Area of the parallelogram} &= \frac{1}{2} |\vec{d}_1 \times \vec{d}_2| \\ &= \frac{1}{2} \sqrt{25 + (14)^2 + 4} = \frac{1}{2} \sqrt{225} = \frac{15}{2} \text{ square units.}\end{aligned}$$

Vector moment (or torque)– Let $\vec{F}$ be the force acting at a point P , and let A be a given point outside the line of action of $\vec{F}$. Then we define moment of $\vec{F}$ applied at P about point $A = \vec{AP} \times \vec{F}$.

Example 18: Find the moment of $\vec{F}$ about the point $(2, -1, 3)$, when the force $\vec{F} = 3\hat{i} + 2\hat{j} - 4\hat{k}$ is acting at the point $(1, -1, 2)$.

Solution: We know that moment of $\vec{F}$ applied at P about $A = (\vec{AP} \times \vec{F})$. Now, the force $\vec{F} = 3\hat{i} + 2\hat{j} - 4\hat{k}$ is applied at the point $P(1, -1, 2)$ we have to find the moment of $\vec{F}$ about the point $A(2, -1, 3)$. Now,

$$\begin{aligned}\vec{AP} &= (P.V. \text{ of } P) - (P.V. \text{ of } A) = (\hat{i} - \hat{j} + 2\hat{k}) - (2\hat{i} - \hat{j} + 3\hat{k}) \\ &= (-\hat{i} - \hat{k})\end{aligned}$$

$$\begin{aligned}\text{Moment of } \vec{F} \text{ applied at } P \text{ about } A &= (\vec{AP} \times \vec{F}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 0 & -1 \\ 3 & 2 & -4 \end{vmatrix} \\ &= (0 + 2)\hat{i} - (4 + 3)\hat{j} + (-2 - 0)\hat{k} = (2\hat{i} - 7\hat{j} - 2\hat{k})\end{aligned}$$

Example 19: Find the vector moment of three forces $(\hat{i} + 2\hat{j} - 3\hat{k})$, $(2\hat{i} + 3\hat{j} + 4\hat{k})$ and $(-\hat{i} - \hat{j} + \hat{k})$, acting on a particle at a point $P(0, 1, 2)$, about the point $A(1, -2, 0)$.

Solution: Let $\vec{F}$ be the resultant of the given forces. Then

$$\begin{aligned}\vec{F} &= (\hat{i} + 2\hat{j} - 3\hat{k}) + (2\hat{i} + 3\hat{j} + 4\hat{k}) + (-\hat{i} - \hat{j} + \hat{k}) \\ &= (2\hat{i} + 4\hat{j} + 2\hat{k})\end{aligned}$$

$$\text{Moment of } \vec{F} \text{ applied at } P \text{ about } A = (\vec{AP} \times \vec{F})$$

$$\begin{aligned}\text{where } \vec{AP} &= (P.V. \text{ of } p) - (P.V. \text{ of } A) \\ &= (0\hat{i} + \hat{j} + 2\hat{k}) - (\hat{i} - 2\hat{j} + 0\hat{k}) = (-\hat{i} + 3\hat{j} + 2\hat{k})\end{aligned}$$

$\therefore$ the required vector moment

$$\begin{aligned}(\vec{AP} \times \vec{F}) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & 2 \\ 2 & 4 & 2 \end{vmatrix} = (6 - 8)\hat{i} + \hat{j}(4 + 2) + \hat{k}(-4 - 6) \\ &= -2\hat{i} + 6\hat{j} - 10\hat{k}.\end{aligned}$$

Example 20: Find the moment of the couple consisting of the forces $\vec{F} = 3\hat{i} + 2\hat{j} - \hat{k}$ acting through the point $(\hat{i} - \hat{j} + \hat{k})$ and $-\vec{F}$ acting through the point $(2\hat{i} - 3\hat{j} - \hat{k})$.

Solution: Let A and B be the points with position vector $(\hat{i} - \hat{j} + \hat{k})$ and $(2\hat{i} - 3\hat{j} - \hat{k})$ respectively.

Then,

$$\begin{aligned}\vec{BA} &= (P.V. \text{ of } A) - (P.V. \text{ of } B) \\ &= (\hat{i} - \hat{j} + \hat{k}) - (2\hat{i} - 3\hat{j} - \hat{k}) = (-\hat{i} + 2\hat{j} + 2\hat{k})\end{aligned}$$

Also $\vec{F} = 3\hat{i} + 2\hat{j} - \hat{k}$

$\therefore$ Moment of the couple $= \vec{BA} \times \vec{F}$

$$\begin{aligned}&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 2 \\ 3 & 2 & -1 \end{vmatrix} \\ &= (-2 - 4)\hat{i} - (1 - 6)\hat{j} + (-2 - 6)\hat{k} = (-6\hat{i} + 5\hat{j} - 8\hat{k})\end{aligned}$$

Magnitude of this moment $= \sqrt{36 + 25 + 64} = \sqrt{125} = 5\sqrt{5}$.

17.8.4 Solving Geometrical Problems Using Vectors Method

Theorem 1: (Sine rule) In a triangle ABC , prove by vector method that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}; \text{ where } BC = a, CA = b \text{ and } AB = c$$

Proof: Let $\vec{BC} = \vec{a}$, $\vec{CA} = \vec{b}$ and $\vec{AB} = \vec{c}$. Then

$$\vec{a} + \vec{b} + \vec{c} = \vec{BC} + \vec{CA} + \vec{AB} = \vec{0}$$

Now, $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

$$\Rightarrow \vec{a} \times (\vec{a} + \vec{b} + \vec{c}) = \vec{a} \times \vec{0}$$

$$\Rightarrow \vec{a} \times \vec{a} + \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = \vec{0}$$

$$\Rightarrow \vec{a} \times \vec{b} = -\vec{a} \times \vec{c}$$

$$\Rightarrow \vec{a} \times \vec{b} = \vec{c} \times \vec{a}$$

Again, $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

$$\Rightarrow \vec{b} \times (\vec{a} + \vec{b} + \vec{c}) = \vec{b} \times \vec{0}$$

$$\Rightarrow \vec{b} \times \vec{a} + \vec{b} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0}$$

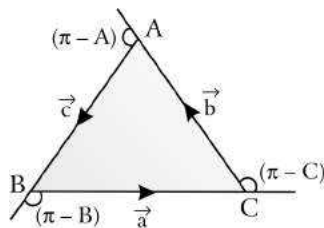


Fig. 17.30

$$[\because \vec{a} \times \vec{a} = \vec{0}]$$

...(1)

$$\Rightarrow -\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0} \quad [\because \vec{b} \times \vec{b} = \vec{0} \text{ and } \vec{b} \times \vec{a} = -\vec{a} \times \vec{b}]$$

$$\Rightarrow \vec{b} \times \vec{c} = \vec{a} \times \vec{b} \quad \dots(2)$$

$$\therefore \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \quad [\text{From (1) and (2)}]$$

$$\Rightarrow |\vec{a} \times \vec{b}| = |\vec{b} \times \vec{c}| = |\vec{c} \times \vec{a}|$$

$$\Rightarrow ab \sin(\pi - C) = bc \sin(\pi - A) = ca \sin(\pi - B)$$

$$\Rightarrow ab \sin C = bc \sin A = ca \sin B$$

$$\Rightarrow \frac{\sin C}{c} = \frac{\sin A}{a} = \frac{\sin B}{b} \quad [\text{dividing each by } abc]$$

$$\Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Theorem 2: The parallelograms on the same base and between the same parallels are equal in area.

Proof: Let $ABCD$ and $ABEF$ be two parallelograms on the same base AB , and between the same parallels AB and DE . Then,

vector area of parallelogram $ABEF$

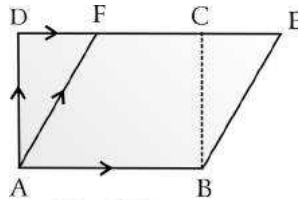


Fig. 17.31

$$= \vec{AB} \times \vec{AF}$$

$$= \vec{AB} \times (\vec{AD} + \vec{DF})$$

$$= (\vec{AB} \times \vec{AD}) + (\vec{AB} \times \vec{DF}) \quad [\text{by the distributive law}]$$

$$= \vec{AB} \times \vec{AD} \quad [\because \vec{AB} \parallel \vec{DF} \Rightarrow \vec{AB} \times \vec{DF} = \vec{0}]$$

$$= \text{vector area of } ||gm\ ABCD$$

Hence, area $(||gm\ ABEF) = \text{area}(||gm\ ABCD)$

Theorem 3: If D, E, F are the midpoints of the sides BC, CA and AB respectively of ΔABC then

$$\text{area}(\Delta DEF) = \frac{1}{4} \text{area}(\Delta ABC).$$

Proof: Let A be the origin, and let the position vectors of B and C be $\vec{b}$ and $\vec{c}$ respectively.

Then, the position vectors of D, E, F are $\frac{1}{2}(\vec{b} + \vec{c})$, $\frac{1}{2}\vec{c}$ and $\frac{1}{2}\vec{b}$ respectively.

$$\therefore \vec{DE} = (P.V. \text{ of } E) - (P.V. \text{ of } D)$$

$$= \frac{1}{2}\vec{c} - \frac{1}{2}(\vec{b} + \vec{c}) = -\frac{1}{2}\vec{b}$$

$$\vec{DF} = (P.V. \text{ of } F) - (P.V. \text{ of } D)$$

$$= \frac{1}{2}\vec{b} - \frac{1}{2}(\vec{b} + \vec{c}) = -\frac{1}{2}\vec{c}$$

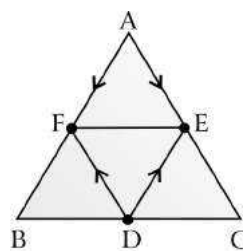


Fig. 17.32

$$\text{Vector area of } \triangle DEF = \frac{1}{2}(\vec{DE} \times \vec{DF})$$

$$= \frac{1}{2} \left[\left(-\frac{1}{2}\vec{b} \right) \times \left(-\frac{1}{2}\vec{c} \right) \right] = \frac{1}{4} \left[\frac{1}{2}(\vec{b} \times \vec{c}) \right]$$

$$= \frac{1}{4} (\text{vector area of } \triangle ABC)$$

$$\therefore \text{area}(\triangle DEF) = \frac{1}{4} \text{area}(\triangle ABC)$$

Theorem 4: For any two angles α and β , $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$.

[B.C.A. (Indra Prastha) 2011]

Proof: Draw a circle of unit radius with its center at the origin O . Let P and Q be two points on the circle such that $\angle POX = \alpha$, $\angle QOX = \beta$ and $\angle POQ = (\alpha - \beta)$, as shown in the figure 7.33.

Let $\hat{i}$ and $\hat{j}$ be unit vectors along OX and OY respectively, and let $\hat{k}$ be a unit vector perpendicular to the XOY -plane. Then $\hat{i}, \hat{j}, \hat{k}$ form a right-handed triad.

$$\text{Also, } OP = |\vec{OP}| = 1 \text{ and } OQ = |\vec{OQ}| = 1$$

Draw $PM \perp OX$ and $QN \perp OX$

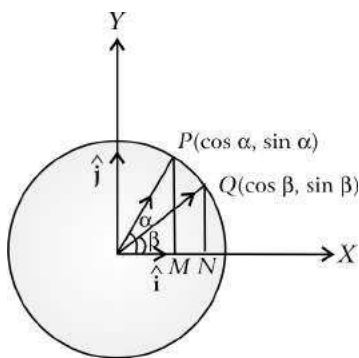


Fig. 17.33

$$\begin{aligned} \vec{OP} &= \vec{OM} + \vec{MP} = (OM)\hat{i} + (MP)\hat{j} \\ &= \{(OP) \cos \alpha\}\hat{i} + \{(OP) \sin \alpha\}\hat{j} = (\cos \alpha)\hat{i} + (\sin \alpha)\hat{j} \end{aligned}$$

$$\begin{aligned} \vec{OQ} &= \vec{ON} + \vec{NQ} = (ON)\hat{i} + (NQ)\hat{j} \\ &= \{(OQ) \cos \beta\}\hat{i} + \{(OQ) \sin \beta\}\hat{j} = (\cos \beta)\hat{i} + (\sin \beta)\hat{j} \end{aligned}$$

$$\begin{aligned} \therefore \vec{OQ} \times \vec{OP} &= |\vec{OQ}| |\vec{OP}| \sin(\alpha - \beta) \hat{k} = 1 \cdot 1 \cdot \sin(\alpha - \beta) \hat{k} \\ &= \sin(\alpha - \beta) \hat{k} \end{aligned}$$

...(1)

Also,

$$\vec{OQ} \times \vec{OP} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \beta \sin \beta & 0 & \\ \cos \alpha \sin \alpha & 0 & \end{vmatrix}$$

$$= (\sin \alpha \cos \beta - \cos \alpha \sin \beta) \hat{k} \quad \dots(2)$$

From (1) and (2) we get

$$\begin{aligned} \sin(\alpha - \beta) \hat{k} &= (\sin \alpha \cos \beta - \cos \alpha \sin \beta) \hat{k} \\ \Rightarrow |\sin(\alpha - \beta) \hat{k}| &= |(\sin \alpha \cos \beta - \cos \alpha \sin \beta) \hat{k}| \\ \Rightarrow \sin(\alpha - \beta) &= (\sin \alpha \cos \beta - \cos \alpha \sin \beta) \end{aligned}$$

Theorem 5: For any two angles α and β , $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$.

Proof: Draw a circle of unit radius with its centre at the origin O . Let OP and OQ make angles α and β with OX , α in the anticlockwise direction and β in the clockwise direction. Then, $\angle POQ = (\alpha + \beta)$.

Also,

$$OP = |\vec{OP}| = 1 \quad \text{and} \quad OQ = |\vec{OQ}| = 1$$

Draw $PM \perp OX$ and $QN \perp OY$

Let $\hat{i}$ and $\hat{j}$ be unit vectors along OX and OY respectively, and let $\hat{k}$ be a unit vector perpendicular to the XOY -plane.

Then, $\hat{i}, \hat{j}, \hat{k}$ form a right-handed triad.

Now, $\vec{OP} = \vec{OM} + \vec{MP} = (OM) \hat{i} + (MP) \hat{j}$

$$\begin{aligned} &= \{(OP) \cos \alpha\} \hat{i} + \{(OP) \sin \alpha\} \hat{j} \\ &= (\cos \alpha) \hat{i} + (\sin \alpha) \hat{j} \quad [\because OP = 1] \end{aligned}$$

And, $\vec{OQ} = \vec{ON} + \vec{NQ} = (ON) \hat{i} + (NQ)(-\hat{j})$

$$\begin{aligned} &= \{(OQ) \cos \beta\} \hat{i} - \{(OQ) \sin \beta\} \hat{j} \\ &= (\cos \beta) \hat{i} - (\sin \beta) \hat{j} \quad -[\because OQ = 1] \end{aligned}$$

$$\begin{aligned} \therefore \vec{OQ} \times \vec{OP} &= \{(\cos \beta) \hat{i} - (\sin \beta) \hat{j}\} \times \{(\cos \alpha) \hat{i} + (\sin \alpha) \hat{j}\} \\ &= (\cos \alpha \cos \beta) (\hat{i} \times \hat{i}) + (\sin \alpha \cos \beta) (\hat{i} \times \hat{j}) \\ &\quad - (\cos \alpha \sin \beta) (\hat{j} \times \hat{i}) - (\sin \alpha \sin \beta) (\hat{j} \times \hat{j}) \\ &= (\sin \alpha \cos \beta) \hat{k} + (\cos \alpha \cos \beta) \hat{k} \\ &\quad [\because \hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \vec{0}, \hat{i} \times \hat{j} = \hat{k} \text{ and } \hat{j} \times \hat{i} = -\hat{k}] \\ &= (\sin \alpha \cos \beta + \cos \alpha \sin \beta) \hat{k} \end{aligned}$$

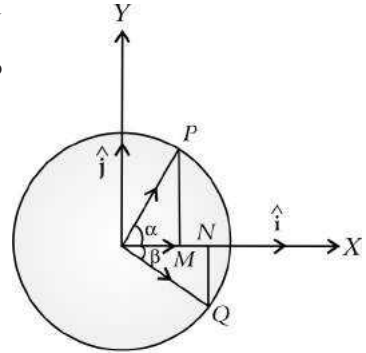


Fig. 17.34

$$\Rightarrow \{|\vec{OQ}||\vec{OP}|\sin \angle POQ\} \hat{k} = (\sin \alpha \cos \beta + \cos \alpha \sin \beta) \hat{k}$$

$$\Rightarrow \sin(\alpha + \beta) \hat{k} = (\sin \alpha \sin \beta + \cos \alpha \sin \beta) \hat{k}$$

$$\Rightarrow |\sin(\alpha + \beta) \hat{k}| = |\sin \alpha \sin \beta + \cos \alpha \sin \beta| |\hat{k}|$$

$$\Rightarrow \sin(\alpha + \beta) = (\sin \alpha \cos \beta + \cos \alpha \sin \beta)$$



Comprehensive Exercise 17.2

- Find $(\vec{a} \times \vec{b})$ and $|\vec{a} \times \vec{b}|$, when:
 - $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} + 5\hat{j} - 2\hat{k}$
 - $\vec{a} = \hat{i} - 7\hat{j} + 7\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} + 2\hat{k}$
 - $\vec{a} = 3\hat{i} + 4\hat{j}$ and $\vec{b} = \hat{i} + \hat{j} + \hat{k}$
- If $\vec{a} = 3\hat{i} + 4\hat{j} - 7\hat{k}$ and $\vec{b} = 6\hat{i} + 2\hat{j} - 3\hat{k}$ then find $(\vec{a} \times \vec{b})$. Verify that $\vec{b}$ and $\vec{a} \times \vec{b}$ are perpendicular to each other.
- Find the unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$, when:
 - $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ [B.C.A. (Meerut) 2000]
 - $\vec{a} = 4\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 4\hat{j} - \hat{k}$ [B.C.A. (Kanpur) 2001]
 - $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + \hat{k}$ [B.C.A (Rohilkhand) 2007]
- Find the unit vectors perpendicular to the plane of the vectors $\vec{a} = 2\hat{i} - 6\hat{j} - 3\hat{k}$ and $\vec{b} = 4\hat{i} + 3\hat{j} - \hat{k}$
- Find a vector of magnitude 6 which is perpendicular to both the vectors $\vec{a} = 4\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = -2\hat{i} + \hat{j} - 2\hat{k}$
- If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} - 3\hat{k}$ then find $|\vec{b} \times 2\vec{a}|$
- If $|\vec{a}| = 5$, $|\vec{b}| = 13$ and $|\vec{a} \times \vec{b}| = 25$, find $\vec{a} \cdot \vec{b}$
- Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find a vector $\vec{d}$ such that it is perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{c} \cdot \vec{d} = 18$.
- Find the value of λ and μ which $(2\hat{i} + 6\hat{j} + 27\hat{k}) \times (\hat{i} + \lambda\hat{j} + \mu\hat{k}) = \vec{0}$.

10. Find the area of the parallelogram whose adjacent sides are respectively by the vectors:
- (i) $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} + 4\hat{k}$
- (ii) $\vec{a} = 2\hat{i}$ and $\vec{b} = 3\hat{j}$
11. Find the area of the parallelogram whose diagonals are represented by the vectors:
- (i) $\vec{d}_1 = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{d}_2 = \hat{i} - 3\hat{j} + 4\hat{k}$
- (ii) $\vec{d}_1 = \hat{i} - 3\hat{j} + 2\hat{k}$ and $\vec{d}_2 = -\hat{i} + 2\hat{j}$
12. Using vectors, find the area of ΔABC , whose vertices are
- (i) $A(3, -1, 2)$, $B(1, -1, -3)$ and $C(4, -3, 1)$
- (ii) $A(1, 2, 3)$, $B(2, 5, -1)$ and $C(-1, 1, 2)$
13. Find the unit vector perpendicular to the plane ABC , where the points A, B, C are $(3, -1, 2)$, $(1, -1, -3)$ and $(4, -3, 1)$ respectively.
14. Show that the points having position vector $(-2\vec{a} + 3\vec{b} + 5\vec{c})$, $(\vec{a} + 2\vec{b} + 3\vec{c})$ and $(7\vec{a} - \vec{c})$ are collinear, whatever be $\vec{a}, \vec{b}, \vec{c}$.
15. Using vector method, show that the given points A, B, C are collinear where:
- (i) $A(3, -5, 1)$, $B(-1, 0, 8)$ and $C(7, -10, -6)$
- (ii) $A(6, -7, -1)$, $B(2, -3, 1)$ and $C(4, -5, 0)$. [B.C.A. (Delhi) 2001, 2005]
16. Show that the points A, B, C with position vectors $(3\hat{i} - 2\hat{j} + 4\hat{k})$, $(\hat{i} + \hat{j} + \hat{k})$ and $(-\hat{i} + 4\hat{j} - 2\hat{k})$ respectively are collinear. [B.C.A. (Meerut) 2004]
17. Find the moments of the force $\vec{F} = \hat{i} - \hat{j} + \hat{k}$, acting at a point P with position vector $(4\hat{i} + 3\hat{j} - 2\hat{k})$ about the point $A(1, 0, 1)$.
18. A force $\vec{F} = 2\hat{i} + 3\hat{j} - \hat{k}$ is acting at the point $(2, -1, 4)$. Find its moment about the point $(1, -2, 1)$.
19. Two unlike forces $(3\hat{i} + \hat{j})$ and $(-3\hat{i} - \hat{k})$ of equal magnitude are acting on a body at points $(\hat{i} + 2\hat{j} - \hat{k})$ and $(2\hat{i} - \hat{j} + 3\hat{k})$ respectively. Find the moment of couple formed by these force. Also, find the magnitude of moment of the couple.



Answers

1.	(i) $\vec{a} \times \vec{b} = (-17i + 13j + 7k)$ and $ \vec{a} \times \vec{b} = 13\sqrt{3}$
	(ii) $\vec{a} \times \vec{b} = 19(\hat{j} + \hat{k})$ and $ \vec{a} \times \vec{b} = 19\sqrt{2}$
	(iii) $\vec{a} \times \vec{b} = (4\hat{i} - 3\hat{j} - \hat{k})$ and $ \vec{a} \times \vec{b} = \sqrt{26}$
2.	$\vec{a} \times \vec{b} = (2\hat{i} - 5\hat{j} - 30\hat{k})$
3.	(i) $\pm \frac{1}{\sqrt{3}}(-\hat{i} + \hat{j} + \hat{k})$ (ii) $\pm \frac{1}{\sqrt{209}}(2\hat{i} + 3\hat{j} + 14\hat{k})$
	(iii) $\pm \frac{1}{3\sqrt{10}}(\hat{i} + 8\hat{j} + 5\hat{k})$
4.	$\pm \frac{1}{7}(3\hat{i} - 2\hat{j} + 6\hat{k})$ 5. $\pm 2(-\hat{i} + 2\hat{j} + 2\hat{k})$
6.	$4\sqrt{19}$ 7. 60
8.	$\vec{d} = 64\hat{i} - 2\hat{j} - 28\hat{k}$ 9. $\lambda = 3, \mu = \frac{27}{2}$
10.	(i) $10\sqrt{3}$ square units (ii) 6 square units.
11.	(i) $5\sqrt{3}$ square units (ii) $\frac{1}{2}\sqrt{21}$ square units.
12.	(i) $\frac{1}{2}\sqrt{165}$ square units (ii) $\frac{1}{2}\sqrt{155}$ square units.
13.	$\frac{-\hat{i} - 2\hat{j} + 4\hat{k}}{\sqrt{21}}$ 17. $(-6\hat{j} - 6\hat{k})$
18.	$-10\hat{i} + 7\hat{j} + \hat{k}$ 19. $(3\hat{i} - 11\hat{j} - 9\hat{k}), \sqrt{211}$ units.

17.9 Product of Three Vectors

17.9.1 Scalar Triple Product

Scalar triple product of three vectors $\vec{a}, \vec{b}, \vec{c}$ is defined as $(\vec{a} \times \vec{b}) \cdot \vec{c}$ and is denoted by $[\vec{a}, \vec{b}, \vec{c}]$ or $[\vec{a} \ \vec{b} \ \vec{c}]$ i.e. $[\vec{a}, \vec{b}, \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) =$ scalar quantity.

Geometrically it represents the volume of a parallelepiped whose coterminal edges are represented by $\vec{a}, \vec{b}, \vec{c}$.

Theorems

(Scalar Triple Product)

Theorem 1: For any three vectors $\vec{a}, \vec{b}, \vec{c}$

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{a} \cdot (\vec{b} \times \vec{c})$$

i.e. dot and cross are interchanges.

Proof: Let $\vec{a}, \vec{b}, \vec{c}$ form a right-handed system, representing the coterminous edges of a parallelepiped of volume V .

Then $V = (\vec{a} \times \vec{b}) \cdot \vec{c}$.

Again, $\vec{b}, \vec{c}, \vec{a}$ form a right-handed system representing the coterminous edges of the same parallelepiped.

$$\therefore V = (\vec{b} \times \vec{c}) \cdot \vec{a} = \vec{a} \cdot (\vec{b} \times \vec{c}) \quad [\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}]$$

Hence, $(\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{a} \cdot (\vec{b} \times \vec{c})$

Theorem 2: To show that

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{b} \times \vec{c}) \cdot \vec{a} = (\vec{c} \times \vec{a}) \cdot \vec{b}$$

The scalar triple product of three vectors remains unchanged as long as their cyclic order remains unchanged.

Proof: Let $\vec{a}, \vec{b}, \vec{c}$ form a right-handed system, representing the coterminous edges of a rectangular parallelepiped of volume V .

Then $V = (\vec{a} \times \vec{b}) \cdot \vec{c}$

Now, $\vec{b}, \vec{c}, \vec{a}$ as well as $\vec{c}, \vec{a}, \vec{b}$ form a right-handed system representing the coterminous edges of the same parallelepiped

$$\therefore V = (\vec{b} \times \vec{c}) \cdot \vec{a} \text{ and } V = (\vec{c} \times \vec{a}) \cdot \vec{b}$$

Thus $(\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{b} \times \vec{c}) \cdot \vec{a} = (\vec{c} \times \vec{a}) \cdot \vec{b}$

Hence, $[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$

Theorem 3: The scalar triple product changes in sign but not in magnitude when the cyclic order of vectors is changed.

Proof: For any three vectors $\vec{a}, \vec{b}, \vec{c}$ we have

$$[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$$

$$\therefore [\vec{c} \ \vec{b} \ \vec{a}] = (\vec{c} \times \vec{b}) \cdot \vec{a} = -(\vec{b} \times \vec{c}) \cdot \vec{a} = -[\vec{b} \ \vec{c} \ \vec{a}] = -[\vec{a} \ \vec{b} \ \vec{c}]$$

And $[\vec{a} \ \vec{c} \ \vec{b}] = (\vec{a} \times \vec{c}) \cdot \vec{b} = -(\vec{c} \times \vec{a}) \cdot \vec{b} = -[\vec{c} \ \vec{a} \ \vec{b}] = -[\vec{a} \ \vec{b} \ \vec{c}]$

$$\therefore [\vec{c} \ \vec{b} \ \vec{a}] = -[\vec{a} \ \vec{b} \ \vec{c}] \text{ and } [\vec{a} \ \vec{c} \ \vec{b}] = -[\vec{a} \ \vec{b} \ \vec{c}]$$

Theorem 4: The scalar triple product vanished if any two of its vectors are equal

i.e. $[\vec{a} \ \vec{a} \ \vec{b}] = 0, [\vec{a} \ \vec{b} \ \vec{a}] = 0 \text{ and } [\vec{b} \ \vec{a} \ \vec{a}] = 0$

Proof: We have $[\vec{a} \ \vec{a} \ \vec{b}] = (\vec{a} \times \vec{a}) \cdot \vec{b} = 0 \cdot \vec{b} = 0$

$$\therefore (\vec{a} \ \vec{b} \ \vec{a}) = [\vec{a} \ \vec{a} \ \vec{b}] = (\vec{a} \times \vec{a}) \cdot \vec{b} = 0 \cdot \vec{b} = 0$$

And $[\vec{b} \ \vec{a} \ \vec{a}] = [\vec{a} \ \vec{b} \ \vec{a}] = 0$

Hence, $[\vec{a} \ \vec{a} \ \vec{b}] = [\vec{a} \ \vec{b} \ \vec{a}] = [\vec{b} \ \vec{a} \ \vec{a}] = 0$

Theorem 5: The scalar triple product vanishes if any two of its vectors are parallel or collinear.

Proof: Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} \parallel \vec{b}$ or $\vec{a}$ and $\vec{b}$ are collinear.

Then $\vec{a} = m \vec{b}$ for some scalar m

$$\therefore [\vec{a} \ \vec{b} \ \vec{c}] = [m \vec{b}, \vec{b}, \vec{c}] = (m \vec{b} \times \vec{b}) \cdot \vec{c} = 0 \cdot \vec{c} = 0.$$

Theorem 6: The necessary and sufficient condition for three non-zero, non-collinear vectors $\vec{a}, \vec{b}, \vec{c}$ to be coplanar is that $[\vec{a} \ \vec{b} \ \vec{c}] = 0$

Proof: Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero, non-parallel vectors. Then $\vec{b} \times \vec{c}$ is perpendicular to the plane of $\vec{b}$ and $\vec{c}$

$$\Rightarrow (\vec{b} \times \vec{c}) \perp \vec{a} \quad [\therefore \vec{a}, \vec{b}, \vec{c} \text{ are coplanar}]$$

$$\Rightarrow \vec{a} \cdot (\vec{b} \times \vec{c}) = 0$$

$$\Rightarrow [\vec{a} \ \vec{b} \ \vec{c}] = 0.$$

Conversely, Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero, non-collinear vectors such that

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0.$$

Then, $[\vec{a} \ \vec{b} \ \vec{c}] = 0 \Rightarrow \vec{a} \cdot (\vec{b} \times \vec{c}) = 0$

$$\Rightarrow \vec{a} = 0 \text{ or } \vec{b} \times \vec{c} = 0 \text{ or } \vec{b} \times \vec{c} \perp \vec{a}$$

$$\Rightarrow \vec{b} \times \vec{c} = 0 \text{ or } \vec{b} \times \vec{c} \perp \vec{a} \quad [\vec{a} \neq 0]$$

$$\Rightarrow (\vec{b} \times \vec{c}) \perp \vec{a}$$

Thus, $(\vec{b} \times \vec{c})$ is perpendicular to $\vec{a}$.

But, $(\vec{b} \times \vec{c})$ is perpendicular to the plane of $\vec{b}$ and $\vec{c}$

$\therefore \vec{a}$ must lie in the plane of $\vec{b}$ and $\vec{c}$.

Hence, $\vec{a}, \vec{b}, \vec{c}$ must be coplanar.

Theorem 7: (Scale triple product in terms of components)

Let $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}, \vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$

and $\vec{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$.

Then $[\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

Proof: We have $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \hat{i}(a_2 b_3 - a_3 b_2) + \hat{j}(a_3 b_1 - a_1 b_3) + \hat{k}(a_1 b_2 - a_2 b_1)$

$\therefore [\vec{a}, \vec{b}, \vec{c}] = (\vec{a} \times \vec{b}) \cdot \vec{c}$

$$= (a_2 b_3 - a_3 b_2) c_1 + (a_3 b_1 - a_1 b_3) c_2 + (a_1 b_2 - a_2 b_1) c_3$$

$$= a_1 (b_2 c_3 - b_3 c_2) + a_2 (b_3 c_1 - b_1 c_3) + a_3 (b_1 c_2 - b_2 c_1)$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Note: For three vectors $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}, \vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$ and $\vec{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$ to be coplanar we have to show

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

Illustrative Examples

(Scalar Triple Product)

Example 1: Show that the vectors $\hat{i} - 3\hat{j} + 4\hat{k}, 2\hat{i} - \hat{j} + 2\hat{k}$ and $4\hat{i} - 7\hat{j} + 10\hat{k}$ are coplanar.

[B.C.A. (Meerut) 2008]

Solution: Let $\vec{a} = \hat{i} - 3\hat{j} + 4\hat{k}$

$$\vec{b} = 2\hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{c} = 4\hat{i} - 7\hat{j} + 10\hat{k}$$

$$\begin{aligned} \therefore [\vec{a} \ \vec{b} \ \vec{c}] &= \begin{vmatrix} 1 & -3 & 4 \\ 2 & -1 & 2 \\ 4 & -7 & 10 \end{vmatrix} = 1(-10+14) + 3(20-8) + 4(-14+4) \\ &= 4 + 36 - 40 = 0 \end{aligned}$$

Hence, the given vectors are coplanar.

Example 2: Show that the vectors $5\hat{i} + 6\hat{j} + 7\hat{k}$, $7\hat{i} - 8\hat{j} + 9\hat{k}$ and $3\hat{i} + 20\hat{j} + 5\hat{k}$ are coplanar.

[B.C.A. (Meerut) 2007]

Solution: Let

$$\vec{a} = 5\hat{i} + 6\hat{j} + 7\hat{k}$$

$$\vec{b} = 7\hat{i} - 8\hat{j} + 9\hat{k}$$

$$\vec{c} = 3\hat{i} + 20\hat{j} + 5\hat{k}$$

$$\begin{aligned} \therefore [\vec{a} \ \vec{b} \ \vec{c}] &= \begin{vmatrix} 5 & 6 & 7 \\ 7 & -8 & 9 \\ 3 & 20 & 5 \end{vmatrix} = 5(-40-180) + 6(27-35) + 7(140+24) \\ &= 5 \times -220 + 6 \times -8 + 7 \times 164 \\ &= -1100 - 48 + 1148 = 0 \end{aligned}$$

Example 3: Show that the four points with position vectors

$$(4\hat{i} + 5\hat{j} + \hat{k}), (-\hat{j} - \hat{k}), (3\hat{i} + 9\hat{j} + 4\hat{k}) \text{ and } 4(-\hat{i} + \hat{j} + \hat{k}) \text{ are coplanar.}$$

[B.C.A. (Delhi) 2003, 2007]

Solution: Let the given vector are the points A, B, C, D then points A, B, C, D are coplanar

$$\Leftrightarrow \vec{AB}, \vec{AC}, \vec{AD} \text{ are coplanar}$$

$$\Leftrightarrow [\vec{AB}, \vec{AC}, \vec{AD}] = 0$$

Now,

$$\begin{aligned} \vec{AB} &= (P.V. \text{ of } B) - (P.V. \text{ of } A) \\ &= (-\hat{j} - \hat{k}) - (4\hat{i} + 5\hat{j} + \hat{k}) = (-4\hat{i} - 6\hat{j} - 2\hat{k}) \end{aligned}$$

$$\begin{aligned} \vec{AC} &= P.V. \text{ of } C - P.V. \text{ of } A \\ &= (3\hat{i} + 9\hat{j} + 4\hat{k}) - (4\hat{i} + 5\hat{j} + \hat{k}) = (-\hat{i} + 4\hat{j} + 3\hat{k}) \end{aligned}$$

$$\begin{aligned} \vec{AD} &= (P.V. \text{ of } D) - (P.V. \text{ of } A) \\ &= (-4\hat{i} + 4\hat{j} + 4\hat{k}) - (4\hat{i} + 5\hat{j} + \hat{k}) = (-8\hat{i} - \hat{j} + 3\hat{k}) \end{aligned}$$

$$\begin{aligned} \therefore [\vec{AB}, \vec{AC}, \vec{AD}] &= \begin{vmatrix} -4 & -6 & -2 \\ -1 & 4 & 3 \\ -8 & -1 & 3 \end{vmatrix} \\ &= -4(12+3) + 6(-3+24) - 2(1+32) = 0 \end{aligned}$$

$$\therefore \vec{AB}, \vec{AC}, \vec{AD} \text{ are coplanar.}$$

Hence, the points A, B, C, D are coplanar.

Example 4: Find the value of λ so that the vectors

$$\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}, \vec{b} = \hat{i} + 2\hat{j} - 3\hat{k} \text{ and } \vec{c} = \hat{j} + \hat{k} \text{ are coplanar.}$$

[B.C.A. (Rohilkhand) 2005]

Solution: The given vectors will be coplanar if $[\vec{a}, \vec{b}, \vec{c}] = 0$

Now, $[\vec{a} \ \vec{b} \ \vec{c}] = 0$

$$\Rightarrow \begin{vmatrix} 2 & -3 & 1 \\ 1 & 2 & -3 \\ 0 & 1 & \lambda \end{vmatrix} = 0$$

$$2(2\lambda + 3) + 3(\lambda + 0) + 1(1 - 0) = 0$$

$$\Rightarrow 7\lambda + 7 = 0 \Rightarrow \lambda = -1.$$

Example 5: Prove that $[\hat{i} \ \hat{j} \ \hat{k}] = 1$

Solution: $\therefore [\hat{i}, \hat{j}, \hat{k}] = (\hat{i} \times \hat{j}) \cdot \hat{k} = \hat{k} \cdot \hat{k} = 1$

Example 6: If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = -3\hat{i} + \hat{j} + 2\hat{k}$, find $[\vec{a} \ \vec{b} \ \vec{c}]$.

Solution: We have

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 2 & 1 & 3 \\ -1 & 2 & 1 \\ -3 & 1 & 2 \end{vmatrix}$$

$$= 2(4 - 1) + 1(-3 + 2) + 3(-1 + 6) = 6 - 1 + 15 = 20.$$

Example 7: Find the value of λ so that four points with position vectors $(-6\hat{i} + 3\hat{j} + 2\hat{k})$, $(3\hat{i} + \lambda\hat{j} + 4\hat{k})$, $(5\hat{i} + 7\hat{j} + 3\hat{k})$ and $(-13\hat{i} + 17\hat{j} - \hat{k})$ are coplanar.

Solution: Let the given points be A, B, C, D respectively, then

$$\vec{AB} = P.V. \text{ of } B - P.V. \text{ of } A$$

$$= (3\hat{i} + \lambda\hat{j} + 4\hat{k}) - (-6\hat{i} + 3\hat{j} + 2\hat{k}) = 9\hat{i} + (\lambda - 3)\hat{j} + 2\hat{k}.$$

$$\vec{AC} = P.V. \text{ of } C - P.V. \text{ of } A = (5\hat{i} + 7\hat{j} + 3\hat{k}) - (-6\hat{i} + 3\hat{j} + 2\hat{k})$$

$$= 11\hat{i} + 4\hat{j} + \hat{k}$$

$$\vec{AD} = P.V. \text{ of } D - P.V. \text{ of } A = (-13\hat{i} + 17\hat{j} - \hat{k}) - (-6\hat{i} + 3\hat{j} + 2\hat{k})$$

$$= -7\hat{i} + 14\hat{j} - 3\hat{k}$$

Now A, B, C, D are coplanar $\Leftrightarrow [\vec{AB}, \vec{AC}, \vec{AD}] = 0$

$$\Leftrightarrow \begin{vmatrix} 9 & \lambda - 3 & 2 \\ 11 & 4 & 1 \\ -7 & 14 & -3 \end{vmatrix} = 0$$

$$\Leftrightarrow 9(-12-14) - (\lambda-3)(-33+7) + 2(11 \times 14 + 28) = 0$$

$$\Leftrightarrow 26\lambda + 52 = 0$$

$$\Leftrightarrow \lambda = -2$$

Example 8: Find the volume of parallelepiped whose edges are represented by

$$\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$$

$$\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$$

and

$$\vec{c} = 2\hat{i} + \hat{j} - \hat{k}.$$

Solution: Here, volume of parallelepiped is

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 2 & -3 & 1 \\ 1 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix}$$

$$= 2(1-2) + 3(-1-4) + 1(1+2) = -2 - 15 + 3 = -14.$$

Example 9: Show that the points $A(-1, 4, -3)$, $B(3, 2, -5)$, $C(-3, 8, -5)$ and $D(-3, 2, 1)$ are coplanar.

Solution: Let the position vectors of $A = -\hat{i} + 4\hat{j} - 3\hat{k}$, $B = 3\hat{i} + 2\hat{j} - 5\hat{k}$, $C = (-3\hat{i} + 8\hat{j} - 5\hat{k})$ and $D = (-3\hat{i} + 2\hat{j} + \hat{k})$.

$$\begin{aligned} \therefore \vec{AB} &= P.V. \text{ of } B - P.V. \text{ of } A = (3\hat{i} + 2\hat{j} - 5\hat{k}) - (-\hat{i} + 4\hat{j} - 3\hat{k}) = 4\hat{i} - 2\hat{j} - 2\hat{k} \\ \vec{AC} &= P.V. \text{ of } C - P.V. \text{ of } A = (-3\hat{i} + 8\hat{j} - 5\hat{k}) - (-\hat{i} + 4\hat{j} - 3\hat{k}) = -2\hat{i} + 4\hat{j} - 2\hat{k} \\ \vec{AD} &= P.V. \text{ of } D - P.V. \text{ of } A = (-3\hat{i} + 2\hat{j} + \hat{k}) - (-\hat{i} + 4\hat{j} - 3\hat{k}) = -2\hat{i} - 2\hat{j} + 4\hat{k} \end{aligned}$$

$$\therefore [\vec{AB}, \vec{AC}, \vec{AD}] = \begin{vmatrix} 4 & -2 & -2 \\ -2 & 4 & -2 \\ -2 & -2 & 4 \end{vmatrix} = 4(16-4) + 2(-8-4) - 2(4+8) = 48 - 24 - 24 = 0$$

$\Rightarrow A, B, C, D$ are coplanar.

Example 10: If $\vec{a}, \vec{b}, \vec{c}$ be any three vectors, then prove that

$$[\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}] = 2[\vec{a}, \vec{b}, \vec{c}] \quad [\text{B.C.A. (Meerut) 2004}]$$

Solution: L.H.S. $= [\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}]$

$$\begin{aligned} &= (\vec{a} + \vec{b}) \cdot \{(\vec{b} + \vec{c}) \times (\vec{c} + \vec{a})\} \\ &= (\vec{a} + \vec{b}) \cdot \{\vec{b} \times \vec{c} + \vec{b} \times \vec{a} + \vec{c} \times \vec{c} + \vec{c} \times \vec{a}\} \\ &= (\vec{a} + \vec{b}) \cdot \{\vec{b} \times \vec{c} + \vec{b} \times \vec{a} + 0 + \vec{c} \times \vec{a}\} \quad [\vec{c} \times \vec{c} = 0] \\ &= \vec{a} \cdot (\vec{b} \times \vec{c}) - \vec{a} \cdot (\vec{a} \times \vec{b}) + \vec{b} \cdot (\vec{c} \times \vec{a}) \\ &\quad + \vec{b} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{a}) + \vec{b} \cdot (\vec{c} \times \vec{a}) \end{aligned}$$

$$\begin{aligned}
 &= [\vec{a} \vec{b} \vec{c}] - [\vec{a} \vec{a} \vec{b}] + [\vec{a} \vec{c} \vec{a}] + [\vec{b} \vec{b} \vec{c}] + [\vec{b} \vec{b} \vec{a}] + [\vec{b} \vec{c} \vec{a}] \\
 &= [\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{c}] \quad [\vec{a} \vec{a} \vec{b}] = 0 \\
 &= 2[\vec{a} \vec{b} \vec{c}] \\
 &= \text{R.H.S.}
 \end{aligned}$$

Example 11: Show that $\vec{a}, \vec{b}, \vec{c}$ are coplanar if and only if $(\vec{a} + \vec{b}), (\vec{b} + \vec{c}), (\vec{c} + \vec{a})$ are coplanar.

Solution: Let $(\vec{a} + \vec{b}), (\vec{b} + \vec{c}), (\vec{c} + \vec{a})$ are coplanar

$$\begin{aligned}
 &\Leftrightarrow [\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}] = 0 \\
 &\Leftrightarrow (\vec{a} + \vec{b}) \cdot [(\vec{b} + \vec{c}) \times (\vec{c} + \vec{a})] = 0 \\
 &\Leftrightarrow (\vec{a} + \vec{b}) \cdot [\vec{b} \times \vec{c} + \vec{b} \times \vec{a} + \vec{c} \times \vec{c} + \vec{c} \times \vec{a}] = 0 \\
 &\Leftrightarrow \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{a} \cdot (\vec{b} \times \vec{a}) + 0 + \vec{a} \cdot (\vec{c} \times \vec{a}) \\
 &\quad + \vec{b} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{a}) + \vec{b} \cdot (\vec{c} \times \vec{a}) = 0 \\
 &\Leftrightarrow [\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{a}] + [\vec{a} \vec{c} \vec{a}] + [\vec{b} \vec{b} \vec{c}] + [\vec{b} \vec{b} \vec{a}] + [\vec{b} \vec{c} \vec{a}] = 0 \\
 &\Leftrightarrow [\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{c}] = 0 \\
 &\Leftrightarrow 2[\vec{a} \vec{b} \vec{c}] = 0, \quad 2 \neq 0 \\
 &\Leftrightarrow [\vec{a} \vec{b} \vec{c}] = 0
 \end{aligned}$$

Example 12: For any three vectors $\vec{a}, \vec{b}, \vec{c}$ show that the vectors $(\vec{a} - \vec{b}), (\vec{b} - \vec{c}), (\vec{c} - \vec{a})$ are coplanar.

Solution: For $(\vec{a} - \vec{b}), (\vec{b} - \vec{c}), (\vec{c} - \vec{a})$ to be coplanar we have to show

$$\begin{aligned}
 &[\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}] = 0 \\
 \therefore &(\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}) = (\vec{a} - \vec{b}) \cdot [(\vec{b} - \vec{c}) \times (\vec{c} - \vec{a})] \\
 &= (\vec{a} - \vec{b}) \cdot [\vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{c} \times \vec{c} + \vec{c} \times \vec{a}] \\
 &= (\vec{a} - \vec{b}) \cdot [\vec{b} \times \vec{c} - \vec{b} \times \vec{a} - 0 + \vec{c} \times \vec{a}] \\
 &= \vec{a} \cdot (\vec{b} \times \vec{c}) - \vec{a} \cdot (\vec{b} \times \vec{a}) - \vec{b} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{a}) - \vec{b} \cdot (\vec{c} \times \vec{a}) \\
 &= [\vec{a} \vec{b} \vec{c}] - 0 + 0 + 0 - [\vec{b} \vec{c} \vec{a}] \\
 &= [\vec{a} \vec{b} \vec{c}] - [\vec{a} \vec{b} \vec{c}] = 0
 \end{aligned}$$

Hence, $(\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a})$ are coplanar.

Example 13: Show that $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar if and only if $\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}$ are non-coplanar.

Solution: Let $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar

$$\Leftrightarrow [\vec{a} \ \vec{b} \ \vec{c}] \neq 0$$

$$\Leftrightarrow 2[\vec{a} \ \vec{b} \ \vec{c}] \neq 0$$

$$\Leftrightarrow [\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}] \neq 0$$

$$\Leftrightarrow \vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a} \text{ are non-coplanar.}$$

Example 14: If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of points A, B, C , prove that $(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$ is a vector perpendicular to the plane of triangle ABC .

Solution: We have to show that $(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$ is perpendicular to each of the vectors $\vec{AB}, \vec{BC}$ and $\vec{CA}$.

$$\therefore \vec{AB} = (\vec{b} - \vec{a}), \vec{BC} = (\vec{c} - \vec{b}) \text{ and } \vec{CA} = (\vec{a} - \vec{c})$$

$$\begin{aligned} \text{Now, } & (\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}) \cdot (\vec{b} - \vec{a}) \\ &= (\vec{a} \times \vec{b}) \cdot (\vec{b} - \vec{a}) + (\vec{b} \times \vec{c}) \cdot (\vec{b} - \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{b} - \vec{a}) \\ &= (\vec{a} \times \vec{b}) \cdot \vec{b} - (\vec{a} \times \vec{b}) \cdot \vec{a} + (\vec{b} \times \vec{c}) \cdot \vec{b} - (\vec{b} \times \vec{c}) \cdot \vec{a} + (\vec{c} \times \vec{a}) \cdot \vec{b} - (\vec{c} \times \vec{a}) \cdot \vec{a} \\ &= 0 - 0 - 0 - [\vec{b} \ \vec{c} \ \vec{a}] + [\vec{c} \ \vec{a} \ \vec{b}] = 0 \end{aligned} \quad [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$$

$$\text{Similarly } (\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}) \cdot (\vec{c} - \vec{b}) = 0$$

$$\text{and } (\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}) \cdot (\vec{a} - \vec{c}) = 0.$$

Thus, $(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$ is perpendicular to each one of the vector $\vec{AB}, \vec{BC}$ and $\vec{CA}$ and therefore, it is perpendicular to the plane of ΔABC .

Example 15: For any three vectors $\vec{a}, \vec{b}, \vec{c}$ prove that $[\vec{a}, \vec{b} + \vec{c}, \vec{a} + \vec{b} + \vec{c}] = 0$.

Solution: We have $[\vec{a}, \vec{b} + \vec{c}, \vec{a} + \vec{b} + \vec{c}]$

$$\begin{aligned} &= [\vec{a} \times (\vec{b} + \vec{c})] \cdot (\vec{a} + \vec{b} + \vec{c}) \\ &= [(\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})] \cdot (\vec{a} + \vec{b} + \vec{c}) \\ &= (\vec{a} \times \vec{b}) \cdot \vec{a} + (\vec{a} \times \vec{c}) \cdot \vec{a} + (\vec{a} \times \vec{b}) \cdot \vec{b} + (\vec{a} \times \vec{c}) \cdot \vec{b} + (\vec{a} \times \vec{b}) \cdot \vec{c} + (\vec{a} \times \vec{c}) \cdot \vec{c} \\ &= [\vec{a} \ \vec{b} \ \vec{a}] + [\vec{a} \ \vec{c} \ \vec{a}] + [\vec{a} \ \vec{b} \ \vec{b}] + [\vec{a} \ \vec{c} \ \vec{b}] + [\vec{a} \ \vec{b} \ \vec{c}] + [\vec{a} \ \vec{c} \ \vec{c}] \end{aligned}$$

$$\begin{aligned}
 &= 0 + 0 + 0 + [\vec{a} \vec{c} \vec{b}] + [\vec{a} \vec{b} \vec{c}] + 0 \\
 &= [\vec{a} \vec{b} \vec{c}] - [\vec{a} \vec{b} \vec{c}] \\
 &= 0
 \end{aligned}$$

Hence, $[\vec{a}, \vec{b} + \vec{c}, \vec{a} + \vec{b} + \vec{c}] = 0$



Comprehensive Exercise 17.3

(Vectors)

- If $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} - 5\hat{k}$, $\vec{c} = 3\hat{i} - 4\hat{j} - 4\hat{k}$, then find $\vec{a} \cdot (\vec{b} \times \vec{c})$.
 - If $\vec{a} = 5\hat{i} - 4\hat{j} + \hat{k}$, $\vec{b} = -4\hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{c} = \hat{i} - 2\hat{j} - 7\hat{k}$ then find $\vec{a} \cdot (\vec{b} \times \vec{c})$.
- Find the volume of the parallelopiped whose edges are:
 - $\hat{i} - 2\hat{j} + 3\hat{k}$, $2\hat{i} + \hat{j} - \hat{k}$, $\hat{j} + \hat{k}$
 - $(2\hat{i} - 3\hat{j} + \hat{k})$, $(\hat{i} - \hat{j} + 2\hat{k})$, $(2\hat{i} + \hat{j} - \hat{k})$.
- Show that the vector $\vec{a} = \hat{i} + 3\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} - \hat{k}$, $\vec{c} = 7\hat{j} + 3\hat{k}$ are parallel to the same plane.
- Find the scalar P such that the vector $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} - 3\hat{k}$, $3\hat{i} + p\hat{j} + 5\hat{k}$ are coplanar.
- Prove following four points are coplanar:
 - $-\vec{a} + 4\vec{b} - 3\vec{c}$, $3\vec{a} + 2\vec{b} - 5\vec{c}$, $-3\vec{a} + 8\vec{b} - 5\vec{c}$, $-3\vec{a} + 2\vec{b} + \vec{c}$
 - $2\vec{a} - 3\vec{b} - \vec{c}$, $\vec{a} - 2\vec{b} + 3\vec{c}$, $3\vec{a} + 4\vec{b} - 2\vec{c}$, $\vec{a} - 6\vec{b} + 6\vec{c}$
 - $A(4, 5, 1)$, $B(0, -1, -1)$, $C(3, 9, 4)$ and $D(-4, 4, 4)$.
- Find the value of λ for which the four points with position vectors $(\hat{i} + 2\hat{j} + 3\hat{k})$, $(3\hat{i} - \hat{j} + 2\hat{k})$, $(-2\hat{i} + \hat{j} + \hat{k})$ and $(6\hat{i} - 4\hat{j} + 2\hat{k})$ are coplanar.



Answers

1.	(i) 0 (ii) 0
2.	(i) 12 (ii) -14
4.	-4
6.	$\lambda = 3$

17.9.2 Vector Triple Product of Vectors

If $\vec{a}, \vec{b}, \vec{c}$ be any three vectors then $\vec{a} \times (\vec{b} \times \vec{c})$ and $(\vec{a} \times \vec{b}) \times \vec{c}$ etc. are known as **vector triple products**.

Theorem 1: For any three vectors $\vec{a}, \vec{b}, \vec{c}$

$$(i) \quad \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$$

$$(ii) \quad (\vec{b} \times \vec{c}) \times \vec{a} = (\vec{b} \cdot \vec{a}) \vec{c} - (\vec{c} \cdot \vec{a}) \vec{b}$$

Proof: Let $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$, $\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$ and $\vec{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$

$$(i) \quad \vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = (b_2 c_3 - b_3 c_2) \hat{i} + \hat{j} (b_3 c_1 - c_3 b_1) + \hat{k} (b_1 c_2 - b_2 c_1)$$

$$= d_1 \hat{i} + d_2 \hat{j} + d_3 \hat{k}$$

Where $d_1 = b_2 c_3 - b_3 c_2$, $d_2 = b_3 c_1 - b_1 c_3$, $d_3 = b_1 c_2 - b_2 c_1$

$$\vec{a} \times (\vec{b} \times \vec{c}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ d_1 & d_2 & d_3 \end{vmatrix}$$

$$= (a_2 d_3 - a_3 d_2) \hat{i} + (a_3 d_1 - a_1 d_3) \hat{j} + (a_1 d_2 - a_2 d_1) \hat{k}$$

$$= P_1 \hat{i} + P_2 \hat{j} + P_3 \hat{k}$$

Where $P_1 = (a_2 d_3 - a_3 d_2)$, $P_2 = (a_3 d_1 - a_1 d_3)$, $P_3 = (a_1 d_2 - a_2 d_1)$

Now,

$$P_1 = (a_2 d_3 - a_3 d_2)$$

$$= a_2 (b_1 c_2 - b_2 c_1) - a_3 (b_3 c_1 - b_1 c_3)$$

$$= (a_2 c_2 + a_3 c_3) b_1 - (a_2 b_2 + a_3 b_3) c_1$$

$$= (a_1 c_1 + a_2 c_2 + a_3 c_3) b_1 - (a_1 b_1 + a_2 b_2 + a_3 b_3) c_1$$

$$= (\vec{a} \cdot \vec{c}) b_1 - (\vec{a} \cdot \vec{b}) c_1$$

Thus, $P_1 = (\vec{a} \cdot \vec{c}) b_1 - (\vec{a} \cdot \vec{b}) c_1$

Similarly, $P_2 = (\vec{a} \cdot \vec{c}) b_2 - (\vec{a} \cdot \vec{b}) c_2$

$$P_3 = (\vec{a} \cdot \vec{c}) b_3 - (\vec{a} \cdot \vec{b}) c_3$$

$$\therefore \vec{a} \times (\vec{b} \times \vec{c}) = [(\vec{a} \cdot \vec{c}) b_1 - (\vec{a} \cdot \vec{b}) c_1] \hat{i} + [(\vec{a} \cdot \vec{c}) b_2 - (\vec{a} \cdot \vec{b}) c_2] \hat{j}$$

$$+ [(\vec{a} \cdot \vec{c}) b_3 - (\vec{a} \cdot \vec{b}) c_3] \hat{k}$$

$$= (\vec{a} \cdot \vec{c}) (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) - (\vec{a} \cdot \vec{b}) (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k})$$

$$= (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$$

Hence, $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$

$$\begin{aligned}
 \text{(ii)} \quad (\vec{b} \times \vec{c}) \times \vec{a} &= -\vec{a} \times (\vec{b} \times \vec{c}) \\
 &= -[(\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}] = (\vec{a} \cdot \vec{b}) \vec{c} - (\vec{a} \cdot \vec{c}) \vec{b} \\
 &= (\vec{b} \cdot \vec{a}) \vec{c} - (\vec{c} \cdot \vec{a}) \vec{b}
 \end{aligned}$$

$$\text{Hence, } (\vec{b} \times \vec{c}) \times \vec{a} = (\vec{b} \cdot \vec{a}) \vec{c} - (\vec{c} \cdot \vec{a}) \vec{b}$$

Illustrative Examples

(Vector Triple Product)

Example 1: Verify that $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$ given $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + 2\hat{j} - 5\hat{k}$.

Solution: $\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 3 & 2 & -5 \end{vmatrix} = \hat{i}(5-2) + \hat{j}(3+10) + \hat{k}(4+3)$

$$= 3\hat{i} + 13\hat{j} + 7\hat{k}$$

$$\text{L.H.S} = \vec{a} \times (\vec{b} \times \vec{c}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & 13 & 7 \end{vmatrix} = \hat{i}(-25) + 2\hat{j} + 7\hat{k}$$

$$\vec{a} \cdot \vec{c} = 3 + 4 - 15 = -8$$

$$\vec{a} \cdot \vec{b} = 2 - 2 + 3 = 3$$

$$\begin{aligned}
 \text{R.H.S} &= (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} \\
 &= -8(2\hat{i} - \hat{j} + \hat{k}) - 3(3\hat{i} + 2\hat{j} - 5\hat{k}) \\
 &= -25\hat{i} + 2\hat{j} + 7\hat{k}
 \end{aligned}$$

$$\text{L.H.S} = \text{R.H.S}$$

Example 2: Show that $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b}) = 0$.

[B.C.A. (Meerut) 2005]

Solution: We have $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b})$

$$= \left[(\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} \right] + \left[(\vec{b} \cdot \vec{a}) \vec{c} - (\vec{b} \cdot \vec{c}) \vec{a} \right] + \left[(\vec{c} \cdot \vec{b}) \vec{a} - (\vec{c} \cdot \vec{a}) \vec{b} \right] = 0$$

Example 3: Show that $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$ if and only if $(\vec{a} \times \vec{c}) \times \vec{b} = 0$ or if $\vec{a}$ and $\vec{c}$ are collinear.

Solution: $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$ if $-\vec{c} \times (\vec{a} \times \vec{b}) = \vec{a} \times (\vec{b} \times \vec{c})$

$$\text{if} \quad -\left[(\vec{c} \cdot \vec{b}) \vec{a} - (\vec{c} \cdot \vec{a}) \vec{b}\right] = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$$

$$\text{if} \quad (\vec{c} \cdot \vec{a}) \vec{b} - (\vec{c} \cdot \vec{b}) \vec{a} = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$$

$$\text{if} \quad (\vec{c} \cdot \vec{b}) \vec{a} = (\vec{a} \cdot \vec{b}) \vec{c} \quad \therefore \vec{c} \cdot \vec{a} = \vec{a} \cdot \vec{c}$$

$$\text{if} \quad (\vec{b} \cdot \vec{c}) \vec{a} - (\vec{b} \cdot \vec{a}) \vec{c} = 0. \quad \text{Since } \vec{c} \cdot \vec{b} = \vec{b} \cdot \vec{c} \text{ and } \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

$$\text{if} \quad \vec{b} \times (\vec{a} \times \vec{c}) = 0 \quad \text{i.e. if } (\vec{a} \times \vec{c}) \times \vec{b} = 0$$

$$\text{if either} \quad \vec{b} = 0 \quad \text{or} \quad \vec{a} \times \vec{c} = 0.$$

But $\vec{b} \neq 0$, therefore $\vec{a} \times \vec{c} = 0$ which is possible only if $\vec{a} = l \vec{c}$

$$\text{since} \quad \vec{a} \times \vec{c} = l \vec{c} \times \vec{c} = l (\vec{c} \times \vec{c}) = 0$$

Hence, $\vec{a}$ and $\vec{c}$ are collinear.

Example 4: Prove that $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k}) = 2 \vec{a}$

[B.C.A. (Meerut) 2000]

Solution: We have $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k})$

$$\begin{aligned} &= (\hat{i} \cdot \hat{i}) \vec{a} - (\hat{i} \cdot \vec{a}) \hat{i} + (\hat{j} \cdot \hat{j}) \vec{a} - (\hat{j} \cdot \vec{a}) \hat{j} + (\hat{k} \cdot \hat{k}) \vec{a} - (\hat{k} \cdot \vec{a}) \hat{k} \\ &= \vec{a} - (\vec{a} \cdot \hat{i}) \hat{i} + \vec{a} - (\vec{a} \cdot \hat{j}) \hat{j} + \vec{a} - (\vec{a} \cdot \hat{k}) \hat{k} \\ &= 3 \vec{a} - \vec{a} \quad [\hat{i} \cdot \vec{a} = a_1] \\ &= 2 \vec{a} \end{aligned}$$

Example 5: If $\hat{a}, \hat{b}, \hat{c}$ be three non-parallel unit vectors such that $\hat{a} \times (\hat{a} \times \hat{c}) = \frac{1}{2} \hat{b}$, find the angle which $\hat{a}$ makes with $\hat{b}$ and $\hat{c}$ respectively.

[B.C.A. (Meerut) 2002, 2005, 2008]

Solution: Let θ_1 be the angle between $\hat{a}$ and $\hat{b}$ and let θ_2 be the angle between $\hat{a}$ and $\hat{c}$, then

$$\begin{aligned} \hat{a} \times (\hat{b} \times \hat{c}) &= \frac{1}{2} \hat{b} \\ \Rightarrow (\hat{a} \cdot \hat{c}) \hat{b} - (\hat{a} \cdot \hat{b}) \hat{c} &= \frac{1}{2} \hat{b} \\ \Rightarrow \left(\hat{a} \cdot \hat{c} - \frac{1}{2} \right) \hat{b} &= (\hat{a} \cdot \hat{b}) \hat{c} \\ \Rightarrow \hat{a} \cdot \hat{b} &= 0 \text{ and } \hat{a} \cdot \hat{c} - \frac{1}{2} = 0 \\ \Rightarrow \hat{a} \cdot \hat{b} &= 0 \text{ and } \hat{a} \cdot \hat{c} = \frac{1}{2} \\ \Rightarrow |\hat{a}| |\hat{b}| \cos \theta_1 &= 0 \text{ and } |\hat{a}| |\hat{c}| \cos \theta_2 = \frac{1}{2} \end{aligned}$$

$$\Rightarrow \cos \theta_1 = 0 \text{ and } \cos \theta_2 = 1/2 \quad [\because |\hat{a}| = |\hat{b}| = |\hat{c}| = 1]$$

$$\Rightarrow \theta_1 = \pi/2 \text{ and } \theta_2 = \pi/3$$

Hence, $\theta_1 = \pi/2$ and $\theta_2 = \pi/3$

Example 6: Prove that $[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}] = [\vec{a}, \vec{b}, \vec{c}]^2$

Solution: We have $(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) = \vec{d} \times (\vec{c} \times \vec{a})$, where $\vec{d} = (\vec{b} \times \vec{c})$

$$\begin{aligned} &= (\vec{d} \cdot \vec{a}) \vec{c} - (\vec{d} \cdot \vec{c}) \vec{a} \\ &= (\vec{b} \times \vec{c} \cdot \vec{a}) \vec{c} - (\vec{b} \times \vec{c} \cdot \vec{c}) \vec{a} \\ &= [\vec{b} \vec{c} \vec{a}] \vec{c} - [\vec{b} \vec{c} \vec{c}] \vec{a} \quad [\vec{b} \vec{c} \vec{c}] = 0 \\ &= [\vec{b} \vec{c} \vec{a}] \vec{c} \end{aligned}$$

$$\begin{aligned} \Rightarrow (\vec{a} \times \vec{b}) \cdot \{(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})\} &= (\vec{b} \vec{c} \vec{a}) (\vec{a} \times \vec{b}) \cdot \vec{c} \\ &= [\vec{b} \vec{c} \vec{a}] [\vec{a} \vec{b} \vec{c}] \\ &= [\vec{a} \vec{b} \vec{c}]^2. \end{aligned}$$

Example 7: For any three vectors $\vec{a}, \vec{b}, \vec{c}$ prove that

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c}) \Leftrightarrow \vec{b} \times (\vec{c} \times \vec{a}) = \vec{0}.$$

Solution:

$$\begin{aligned} &(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c}) \\ \Leftrightarrow &-\vec{c} \times (\vec{a} \times \vec{b}) = \vec{a} \times (\vec{b} \times \vec{c}) \\ \Leftrightarrow &-[(\vec{c} \cdot \vec{b}) \vec{a} - (\vec{c} \cdot \vec{a}) \vec{b}] = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} \\ \Leftrightarrow &(\vec{c} \cdot \vec{a}) \vec{b} - (\vec{c} \cdot \vec{b}) \vec{a} = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} \\ \Leftrightarrow &(\vec{a} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{c}) \vec{a} = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{a}) \vec{c} \\ \Leftrightarrow &(\vec{b} \cdot \vec{a}) \vec{c} - (\vec{b} \cdot \vec{c}) \vec{a} = \vec{0} \\ \Leftrightarrow &\vec{b} \times (\vec{c} \times \vec{a}) = \vec{0} \end{aligned}$$

Hence, $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c}) \Leftrightarrow \vec{b} \times (\vec{c} \times \vec{a}) = \vec{0}$

Example 8: For any two vectors $\vec{a}$ and $\vec{b}$, prove that

$$\vec{a} \times [\vec{a} \times (\vec{a} \times \vec{b})] = (\vec{a} \cdot \vec{a}) (\vec{b} \times \vec{a}).$$

Solution: We have

$$\begin{aligned} \vec{a} \times [\vec{a} \times (\vec{a} \times \vec{b})] &= \vec{a} \times [(\vec{a} \cdot \vec{b}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{b}] \\ &= (\vec{a} \cdot \vec{b}) (\vec{a} \times \vec{a}) - (\vec{a} \cdot \vec{a}) (\vec{a} \times \vec{b}) \end{aligned}$$

$$\begin{aligned}
 &= -(\vec{a} \cdot \vec{a})(\vec{a} \times \vec{b}) \quad [\because \vec{a} \times \vec{a} = \vec{0}] \\
 &= (\vec{a} \cdot \vec{a})(\vec{b} \times \vec{a})
 \end{aligned}$$

Hence, $\vec{a} \times [\vec{a} \times (\vec{a} \times \vec{b})] = (\vec{a} \cdot \vec{a})(\vec{b} \times \vec{a})$

Example 9: For any four vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ prove that

$$\vec{a} \times [\vec{b} \times (\vec{c} \times \vec{d})] = (\vec{b} \cdot \vec{d})(\vec{a} \times \vec{c}) - (\vec{b} \cdot \vec{c})(\vec{a} \times \vec{d}).$$

Solution: We have $\vec{a} \times [\vec{b} \times (\vec{c} \times \vec{d})] = \vec{a} \times [(\vec{b} \cdot \vec{d})\vec{c} - (\vec{b} \cdot \vec{c})\vec{d}]$

$$= (\vec{b} \cdot \vec{d})(\vec{a} \times \vec{c}) - (\vec{b} \cdot \vec{c})(\vec{a} \times \vec{d})$$

Hence, $\vec{a} \times [\vec{b} \times (\vec{c} \times \vec{d})] = (\vec{b} \cdot \vec{d})(\vec{a} \times \vec{c}) - (\vec{b} \cdot \vec{c})(\vec{a} \times \vec{d})$

Example 10: For any vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ prove that

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = \begin{vmatrix} \vec{a} \cdot \vec{c} & \vec{a} \cdot \vec{d} \\ \vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{d} \end{vmatrix}$$

Solution: We have $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \times \vec{b}) \cdot \vec{r}$, where $(\vec{c} \times \vec{d}) = \vec{r}$

$$= \vec{a} \cdot (\vec{b} \times \vec{r})$$

$$= \vec{a} \cdot [\vec{b} \times (\vec{c} \times \vec{d})] \quad [\because \vec{r} = \vec{c} \times \vec{d}]$$

$$= \vec{a} \cdot [(\vec{b} \cdot \vec{d})\vec{c} - (\vec{b} \cdot \vec{c})\vec{d}]$$

$$= (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{b} \cdot \vec{c})(\vec{a} \cdot \vec{d})$$

$$= \begin{vmatrix} \vec{a} \cdot \vec{c} & \vec{a} \cdot \vec{d} \\ \vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{d} \end{vmatrix}$$

Example 11: For any four vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ prove that

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \vec{b} \vec{d}]\vec{c} - [\vec{a} \vec{b} \vec{c}]\vec{d}$$

Solution: We have $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{r} \times (\vec{c} \times \vec{d})$, where $\vec{r} = \vec{a} \times \vec{b}$

$$= (\vec{r} \cdot \vec{d})\vec{c} - (\vec{r} \cdot \vec{c})\vec{d}$$

$$= [(\vec{a} \times \vec{b}) \cdot \vec{d}]\vec{c} - [(\vec{a} \times \vec{b}) \cdot \vec{c}]\vec{d}$$

$$= [\vec{a} \vec{b} \vec{d}]\vec{c} - [\vec{a} \vec{b} \vec{c}]\vec{d}.$$

Example 12: If $\vec{a}$ and $\vec{b}$ are vectors such that

$$(\vec{a} \times \vec{b}) \times \vec{b} = (\vec{b} \times \vec{a}) \times \vec{a}, \text{ then prove that } \vec{a} \times \vec{b} = \vec{0}.$$

Solution: We have $(\vec{a} \times \vec{b}) \times \vec{b} = (\vec{b} \times \vec{a}) \times \vec{a}$

$$\Rightarrow (\vec{a} \times \vec{b}) \times \vec{b} = -(\vec{a} \times \vec{b}) \times \vec{a}$$

$$\Rightarrow (\vec{a} \times \vec{b}) \times \vec{b} + (\vec{a} \times \vec{b}) \times \vec{a} = \vec{0}$$

$$\Rightarrow (\vec{a} \times \vec{b}) \times (\vec{a} + \vec{b}) = \vec{0}$$

$$\Rightarrow (\vec{a} \times \vec{b}) = \vec{0} \text{ or } (\vec{a} + \vec{b}) = \vec{0} \text{ or } (\vec{a} \times \vec{b}) \parallel (\vec{a} + \vec{b}) \quad \dots(1)$$

$$\text{Now, } \vec{a} + \vec{b} = \vec{0} \Rightarrow \vec{a} = -\vec{b} \Rightarrow \vec{a} \times \vec{b} = -\vec{b} \times \vec{b} = \vec{0}.$$

Also, $(\vec{a} \times \vec{b})$ being a vector perpendicular to both $\vec{a}$ and $\vec{b}$, it is perpendicular to $(\vec{a} + \vec{b})$.

Thus, $(\vec{a} \times \vec{b})$ can not be parallel to $(\vec{a} + \vec{b})$

$$\text{Hence, } \vec{a} \times \vec{b} = \vec{0}$$

